Elementary Category Theory

A First Course Built from the Ground Up

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Volume I

The Architecture of Categories

Preface

This book is an experiment.

Category theory has a reputation: it is something you learn *after* years of mathematics, as a kind of universal solvent that dissolves the walls between algebra, geometry, logic, and the rest. Every existing textbook on the subject assumes you already live in those rooms.

This book tries something different. It treats category theory not as a retrospective unifier of mathematics but as a *starting point*. We build from nothing. The only raw material is the idea that there are **things**, and between things there are **connections**. Everything else grows from there. The goal is a full formal understanding of category theory — its definitions, its theorems, its classical results — and eventually its connections to other branches of mathematics. Those connections appear only after we have built the language to appreciate them.

Who this book is for. If you can follow a recipe, read a transit map, or understand why two different roads can take you to the same place, you have all the prerequisites. We will not assume familiarity with algebra, calculus, set theory, or any other branch of mathematics. When those subjects appear later as examples, they will be introduced fresh.

How each section works. Each section follows a cycle. The cycle has a direction: it opens outward from an intuitive starting point, and then folds back inward to a precise formal statement. Different sections need different numbers of steps, but the vocabulary of steps is always drawn from the same small set:

- **Informally** — The intuitive, everyday-language version of the idea. Start here every time.
- **Expanding** — Worked examples, diagrams, and the kind of out-loud thinking you should practice when you encounter these ideas in the wild.
- **Formalizing** — Precise definitions and notation are introduced. This is where the informal idea gets its mathematical clothing.
- **Reorganizing** — We go back to the examples from the Expanding step and restate them *using* the formal vocabulary we just learned. The facts are the same; the language is new.
- **Simplifying** — We compress. Informal sentences are replaced, one by one, with their formal equivalents, until the statement fits in a line or two of notation.
- **Formally** — The CT-idiomatic form. What a working category theorist would write, and what a reader who has completed the cycle can now read fluently.

The cycle is not a secret. We use it openly, and we label each step so you always know where you are. Not every section needs all six steps. What every section does is complete a round trip: from plain language to formal notation and back to a compressed formal statement that carries the full informal meaning inside it.

The internal-dialog boxes. Throughout the book you will find boxes labeled “Internal Dialog.” These model the running commentary that a skilled reader maintains while working with diagrams and definitions. Learning to produce that commentary yourself—in your own head, in pencil in the margin—is one of the main skills this book tries to teach.

Chapter bookends. When a chapter covers several distinct ideas, it opens with an **Overview** and closes with a **Formal Restatement**. The Overview is where you are *before* the chapter: a plain-language description of what is coming. The Formal Restatement is where you are *after*: the same ideas, in CT-idiomatic notation. A reader who finishes the chapter and reads the Formal Restatement should be able to hear the Overview behind every line.

Restarting is part of reading this book. The cycle structure is designed for repeated passes. On a first reading, you will use the Informal and Expanding steps to build intuition. On a second pass, the Reorganizing and Simplifying steps will feel different — you will recognize the formal language as yours. On a third pass, you may find yourself reading the

Formal Restatement first and working backwards. All of these are the right way to read. The book does not penalize you for returning to earlier sections; it rewards it.

A note on what we avoid. We will not say things like "consider the category of groups" or "the fundamental groupoid of a topological space" until we have built those ideas from scratch. The temptation to borrow ready-made examples from other fields is real, but it defeats the purpose. Our examples come from everyday experience: maps and routes, instructions and steps, classifications and comparisons. The mathematics, when it appears, will be mathematics we have built together.

Let's begin.

Chapter 1

Things and Arrows

Chapter Overview

Category theory is the study of *things* and the *connections between things*. In this chapter we give those words precise names, examine several everyday examples, and establish three rules that connections must follow: they can be chained end-to-end; every thing connects to itself by a special "do nothing" connection; and when a diagram shows multiple paths between the same two points, all paths must agree.

Each of those sentences is informal. By the end of the chapter each one appears again in the compact notation a category theorist would write. The Formal Restatement at the close of the chapter collects them together. A reader returning to this chapter a second time may find it useful to read the Formal Restatement first, then work through the chapter as a commentary on it.

1.1 What Category Theory Is About

1.1.1 Informally: The Central Idea

Category theory is the study of **things** and **arrows between things**.

That's it. That's the whole subject.

Of course, "things" and "arrows" are deliberately vague words. The power of category theory comes precisely from that vagueness. A thing can be a recipe step, a city, a sentence, a number, a proof, or a process. An arrow can be a road, a translation, a logical implication, a comparison, or a transformation of any kind.

What category theory cares about is not *what things are made of inside*, but *how they connect to other things*. Two things that connect to everything else in exactly the same way are, for the purposes of category theory, *the same thing* — even if they look completely different up close.

This is a radical idea, and we will return to it many times.

1.1.2 Expanding: Three Everyday Pictures

Let's build the intuition with three examples that have nothing to do with mathematics.

Example: The Transit Map

A city has train stations. Between stations there are train lines. If you want to get from Station A to Station C, and there is a line from A to B and a line from B to C, you can travel A → B → C.

$$A \xrightarrow{\text{line 1}} B \xrightarrow{\text{line 2}} C$$

The *things* are the stations. The *arrows* are the train lines. Notice: we don't care what is inside a station — its architecture, its size, whether it has a coffee shop. We only care how it connects.

Internal Dialog

When I see a diagram like the one above, I ask myself three questions:

1. *What are the dots?* Here: train stations.
2. *What are the arrows?* Here: train lines you can ride.
3. *Can I follow a path?* Here: yes, A to B to C.

These three questions apply to *every* diagram in this book. Make them into a habit.

Example: The Recipe Steps

A recipe is a sequence of steps. "Chop onions" must happen before "sauté onions." "Sauté onions" must happen before "add to pot."

$$\text{chop} \xrightarrow{\text{before}} \text{sauté} \xrightarrow{\text{before}} \text{add to pot}$$

The *things* are the steps. The *arrows* mean "must happen before." If chopping must happen before sautéing, and sautéing before adding, then chopping must happen before adding — even though there is no single arrow drawn directly from "chop" to "add to pot."

This indirect conclusion is not magic. It follows from the arrows we have. In category theory we will give this a formal name: **composition**.

Example: The Approval Chain

A company requires that budget requests be approved by a manager before going to the director, and by the director before going to the board.

$$\text{request} \xrightarrow{\text{manager}} \text{approved} \xrightarrow{\text{director}} \text{final approval}$$

Again: *things* (states of the request), *arrows* (approvals that move the request forward). The pattern is identical to the transit map and the recipe. This is the first sign that something deeper is going on: very different situations share the same shape.

Category theory is, among other things, the science of recognizing when two situations have the same shape.

1.1.3 Formalizing: Inside Versus Outside

There is a habit of mind that category theory requires, and it is worth practicing from the very beginning.

Stop asking what things are made of.

When you look at a train station on a map, you do not ask how many platforms it has or which architect designed it. You ask: which lines does it connect to? Category theory extends this discipline to *everything*. When we work with a thing in category theory, we are forbidden from looking "inside" it. We can only look at what arrows arrive at it and what arrows leave it.

This sounds limiting. In practice it is liberating.

Suppose you have two completely different-looking objects — a subway station and an airport terminal, say — but every route that goes through one also goes through the other, and vice versa. For the purposes of your journey, they are *interchangeable*. Category theory gives you a way to say that precisely and work with it rigorously.

Reader's Annotation

Try this: pick any two things in your life that are functionally interchangeable but look different inside. A paperback and an e-book of the same novel. An email and a printed letter carrying the same message. A physical key and a key card opening the same door.

In each case: the *inside* differs, but the *connections* are the same. In category theory we say these are **isomorphic** — a word we will define carefully later.

The discipline of "look only at connections, not at insides" is called working **externally** or working **up to isomorphism**. Start training the habit now: when you look at a thing, ask not *what is it?* but *what does it connect to?*

A historical note. Category theory was introduced in 1945 by Samuel Eilenberg and Saunders Mac Lane. Their language turned out to carry a philosophical viewpoint: the question *"what is X?"* is replaced by *"what role does X play?"* This is sometimes called **structuralism** in the philosophy of mathematics. This book does not require you to adopt a philosophical position, but it is worth knowing the viewpoint is there — it shapes every definition we write.

Where the Name Comes From

Morphism is from Greek *morphē* ("form, shape") plus *-ism*. A morphism is a "form-preserving thing" — a map between mathematical objects that respects the structure they carry. The word was standard in 19th-century mathematics (homomorphism, isomorphism); Eilenberg and Mac Lane (1945) adopted it as the generic term for an arrow in a category. **Hom-set** is from algebraic "set of homomorphisms"; the notation $\mathrm{Hom}(A, B)$ predates category theory and was kept.

1.2 Objects and Morphisms

1.2.1 Informally: Naming the Pieces

The *things* in category theory have an official name: **objects**.

The *arrows* have an official name: **morphisms**.

Every morphism goes *from* one object *to* another. We write

$$f : A \to B$$

to mean "there is a morphism named f going from object A to object B." We call A the **domain** of f and B the **codomain** of f.

Four words: objects, morphisms, domain, codomain. Everything else in category theory is built from these four words.

1.2.2 Expanding: Reading Morphism Notation

Let's slow down and practice reading $f : A \to B$, because it will appear on almost every page of this book.

Internal Dialog

I see the expression $f : A \to B$. Here is what I say to myself:

- "f is the name of a morphism."
- "A is where f starts — its domain."
- "B is where f ends — its codomain."
- "The arrow $\to$ tells me this is a one-directional connection."
- "I do *not* assume I can go from B back to A just because I can go from A to B. Arrows have directions."

If someone asks me to draw this, I draw a dot labeled A, a dot labeled B, and an arrow from A to B labeled f:

$$A \xrightarrow{f} B$$

This picture and the expression $f : A \to B$ carry exactly the same information.

Example: Morphisms as Temperature Comparisons

Suppose our objects are temperature readings: 0řC, 20řC, 37řC, 100řC (freezing, room temperature, body temperature, boiling). A morphism f : 0řC → 100řC could mean "is colder than." A morphism g : 20řC → 37řC could also mean "is colder than."

Both f and g are instances of the same *kind* of arrow ("is colder than"), but they are different morphisms because they connect different objects. Two morphisms are the same only if they have the same name *and* the same domain *and* the same codomain.

Example: Morphisms as Translations

Suppose our objects are the phrases "hello," "hola," "bonjour," "ciao." A morphism $t : \text{hello} \to \text{hola}$ could mean "is a Spanish translation of." A morphism $u : \text{hello} \to \text{bonjour}$ could mean "is a French translation of."

Both start at the same object ("hello") but end at different objects. They are different morphisms. The object "hello" has *two outgoing morphisms* in this setup.

Reader's Annotation

When you encounter any new example, before doing anything else, identify:

1. What are the *objects*? (the dots in the diagram)
2. What are the *morphisms*? (the arrows)
3. For each morphism: what is its *domain*? what is its *codomain*?

Write these down, in words, every time. This routine prevents most of the confusion that comes from reading diagrams too quickly.

1.2.3 Formalizing: What Morphisms Are Not

Several natural assumptions about morphisms turn out to be wrong. Let's clear them now.

Morphisms are not always functions. In many examples, morphisms will be functions from one set to another. But "morphism" does not mean "function." A morphism is anything that fits the pattern: domain, codomain, goes from one to the other consistently. We will see morphisms that are proofs, comparisons, paths, and transformations.

Multiple morphisms can share the same domain and codomain. If $f : A \to B$ and $g : A \to B$, that does *not* mean $f = g$. There can be many different morphisms between the same two objects.

A morphism $f : A \to B$ does not imply a morphism $B \to A$. Going from A to B is not the same as going from B to A. In the recipe example, "chop before sauté" does not imply "sauté before chop" — that would be a contradiction.

There may be no morphism between two objects. Not every pair of objects is connected. Not every city has a direct train to every other city.

1.2.4 Reorganizing: The Earlier Examples in Formal Terms

Now that we have the vocabulary, let's restate the examples from the opening section using it.

Transit map. The stations are *objects*. The line from Station A to Station B is a *morphism* with domain A and codomain B. "There is a train line from A to B" becomes: there exists a morphism $f : A \to B$. "There is no direct route from B to A" becomes: there is no morphism with domain B and codomain A in this example.

Temperature. The readings 0řC, 20řC, 37řC, 100řC are *objects*. The relation "is colder than" provides a *morphism* f : 0řC $\to$ 100řC. The domain is the colder reading; the codomain is the warmer one.

Translations. "Hello," "hola," "bonjour" are *objects*. "Is a Spanish translation of" is a *morphism* $t : \text{hello} \to \text{hola}$. A second Spanish translation (formal versus informal) would be a second, distinct morphism $t' : \text{hello} \to \text{hola}$ — same domain, same codomain, but $t \neq t'$.

In every case, the informal description maps directly onto the formal vocabulary. No facts changed; only the language did.

1.2.5 Simplifying: Four Phrases, Four Terms

The informal sentences we used have formal counterparts:

Informal phrase	*Formal term*
"there is a connection from X to Y"	$f : X \to Y$
"where the connection starts"	domain
"where the connection ends"	codomain
"all connections from X to Y"	$\mathrm{Hom}(X, Y)$

The last entry, $\mathrm{Hom}(X, Y)$, is the **hom-set**: the collection of all morphisms with domain X and codomain Y. We have replaced four English phrases with four mathematical terms. The terms carry the same meaning in any context.

1.2.6 Formally: Morphisms and Hom-Sets

A **morphism** $f : A \to B$ is determined by three data: its domain A, its codomain B, and its identity as a morphism. Two morphisms are equal if and only if they have equal domains, equal codomains, and are the same arrow.

The **hom-set** $\mathrm{Hom}(A, B)$ is the collection of all morphisms with domain A and codomain B. When we need to specify the category $\mathcal{C}$, we write $\mathrm{Hom}_{\mathcal{C}}(A, B)$.

1.2.6.0.1 Alternative notations

In some literature: $f \in \mathrm{Hom}(A, B)$; or inline as $A \xrightarrow{f} B$; or in diagrams without labels when the morphism is clear from context. This book defaults to $f : A \to B$.

1.2.6.0.2 Hom-sets may be empty

$\mathrm{Hom}(A, B) = \varnothing$ is permitted — there need not be a morphism between every pair of objects. The definition requires only that *when* morphisms exist, they obey the rules of composition and identity (introduced in the next two sections).

Where the Name Comes From

Composition of functions has been central to mathematics since the 17th century (Leibniz). The symbol $\circ$ is 19th-century standardization. Right-to-left reading order is universal: $g \circ f$ means "apply f, then g," preserving the substitution rule $(g \circ f)(x) = g(f(x))$. Some CS traditions reverse (write $f\,;\,g$); mathematical category theory uses $g \circ f$ throughout.

1.3 Composition

1.3.1 Informally: Chaining Arrows

If there is a morphism from A to B, and a morphism from B to C, then there must also be a morphism from A to C.

This is the rule of **composition**. Whenever two arrows line up end-to-end, you get a third arrow for free — the one that goes from start to finish in one step.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & B \arrow[r, "g"] & C \end{tikzcd}$$

We write the composed morphism as $g \circ f$, read "g after f." It goes from A to C. The order reads right to left: first f, then g.

1.3.2 Expanding: Practicing Composition

Internal Dialog

I see: $f : A \to B$ and $g : B \to C$. I ask: "Do these line up?" The codomain of f is B. The domain of g is B. They match. So composition is possible.
I write: $g \circ f : A \to C$.
I read this as: "first do f, then do g." I always verify: domain of composite = domain of $f = A$; codomain of composite = codomain of $g = C$.

Example: Composition in the Recipe

Recall:

$$f : \text{chop} \to \text{sauté}, \qquad g : \text{sauté} \to \text{add to pot}$$

Composition gives:

$$g \circ f : \text{chop} \to \text{add to pot}$$

"Chopping must happen before adding to pot" — a fact we knew intuitively. Composition is the formal way to record it.

Example: A Chain of Three

Suppose $f : A \to B$, $g : B \to C$, $h : C \to D$. We can compose in steps:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rrr, bend right, "h \circ g \circ f"'] & B \arrow[r, "g"] \arrow[rr, bend left, "h \circ g"] & C \arrow[r, "h"] & D \end{tikzcd}$$

Both $(h \circ g) \circ f$ and $h \circ (g \circ f)$ give the same morphism $A \to D$. This is not a coincidence — it is a law called **associativity**.

Where the Name Comes From

Associative (Latin *associare*, "to join with as a companion") describes operations where grouping doesn't matter: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. The term was coined by William Rowan Hamilton (1843) in his work on quaternions, where associativity holds but commutativity fails. Composition in any category is associative by axiom.

1.3.3 Formalizing: Associativity

Associativity says that when composing a chain of three morphisms, the grouping of intermediate steps doesn't matter:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

Why does this matter? Because it means we can write $h \circ g \circ f$ without parentheses — the chain has a unique composite regardless of how we group the steps.

Example: Associativity in Transit

- f: travel Airport → Downtown
- g: travel Downtown → Stadium
- h: travel Stadium → Hotel

Whether you plan the trip as (Airport→Stadium) then Hotel, or Airport then (Downtown→Hotel), the journey is the same. Grouping the *planning* differently does not change the *journey*. Associativity just says this obvious fact always holds for morphisms.

Associativity has a useful consequence: any finite chain of composable arrows has a unique composite. We don't need to specify parenthesization. The composite of $f_1, f_2, \ldots, f_n$ (when they chain in order) is well-defined.

Composition is also a *partial* operation: $g \circ f$ is defined *only* when $\mathrm{cod}(f) = \mathrm{dom}(g)$. The domain and codomain are the type-checking mechanism that tells you which compositions are legal.

1.3.4 Reorganizing: Recipe Composition in Formal Terms

In the recipe example, $f : \text{chop} \to \text{sauté}$ and $g : \text{sauté} \to \text{add}$. These are composable because $\mathrm{cod}(f) = \mathrm{dom}(g) = \text{sauté}$.

Their composite $g \circ f : \text{chop} \to \text{add}$ is the formal encoding of "chopping must happen before adding." The intermediate step (sautéing) is *inside* the composite; it doesn't appear in the domain or codomain of $g \circ f$.

For the transit chain — Airport, Downtown, Stadium, Hotel — we have morphisms f, g, h and composites $g \circ f$, $h \circ g$, and $h \circ g \circ f$. Associativity says the last one is unambiguous: $(h \circ g) \circ f = h \circ (g \circ f) = h \circ g \circ f$.

1.3.5 Simplifying: The Composition Rule

The entire informal discussion of chaining compresses to two lines:

$$f : A \to B, \quad g : B \to C \implies g \circ f : A \to C$$

$$(h \circ g) \circ f = h \circ (g \circ f)$$

The first line says when composition is defined and what it produces. The second line says that the order of grouping doesn't matter. Everything else we said was commentary on these two lines.

1.3.6 Formally: Composition as a Partial Operation

Composition is a family of functions

$$\circ : \mathrm{Hom}(B, C) \times \mathrm{Hom}(A, B) \;\to\; \mathrm{Hom}(A, C)$$

one for each triple of objects A, B, C. It satisfies **associativity**: for all $f \in \mathrm{Hom}(A, B)$, $g \in \mathrm{Hom}(B, C)$, $h \in \mathrm{Hom}(C, D)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

This is equivalently expressed as a commutative diagram: the two composite paths from A to D in the chain $A \to B \to C \to D$ are equal.

1.3.6.0.1 Why composition precedes identity in this presentation

Composition is the operative structure of a category. Identity morphisms (next section) are a *constraint* on composition: they are the neutral elements that leave composites unchanged. Logically you must understand the operation before you can understand its neutral element.

1.4 Identity Morphisms

1.4.1 Informally: Every Object Has a "Stay Put" Arrow

Every object has a special morphism that goes from *itself to itself* and does nothing. We call it the **identity morphism** on that object and write $\mathrm{id}_A : A \to A$.

"Does nothing" has a precise meaning: composing any morphism with the identity gives back the original morphism unchanged. For any $f : A \to B$:

$$f \circ \mathrm{id}_A = f \qquad \text{and} \qquad \mathrm{id}_B \circ f = f$$

$$\begin{tikzcd} A \arrow[loop left, "\mathrm{id}_A"] \arrow[r, "f"] & B \arrow[loop right, "\mathrm{id}_B"] \end{tikzcd}$$

1.4.2 Expanding: The "Do Nothing" Connection

Example: The Optional Rest Step

Suppose a recipe optionally rests dough for an hour. On a day when you're in a hurry, you skip the step: the dough is the same dough it was before. This "do nothing" is an identity morphism: it takes the dough as input and returns it unchanged.

Example: The Zero-Distance Route

In the transit example, "stay at Station A" is a valid route: you start at A and end at A. It composes correctly with other routes:

$$\text{(stay at A) then (A to B)} = \text{(A to B)}$$

$$\text{(A to B) then (stay at B)} = \text{(A to B)}$$

The "stay put" route is the identity morphism on Station A.

Internal Dialog

When I see a loop arrow on an object — an arrow that starts and ends at the same dot — my first question is: "Is this the identity, or a non-trivial self-loop?"

I can tell them apart by testing: does composing with this arrow change any other morphism? If no, it's the identity. If yes, it's something else.

Uniqueness of identity. Each object has *exactly one* identity morphism. Suppose id_A and id'_A both satisfy the neutrality law. Then:

$$\mathrm{id}_A = \mathrm{id}_A \circ \mathrm{id}'_A = \mathrm{id}'_A$$

The first equality uses id'_A as a right identity; the second uses id_A as a left identity. They must be the same morphism.

Reader's Annotation

This is the first categorical proof in the book. Notice its structure: we used the identity *laws* (equations that must hold) rather than looking inside any object. That is the categorical style. If a proof ever asks you to look inside an object, something has gone wrong.

1.4.3 Formalizing: Neutrality

The identity morphism is characterized entirely by the equations it satisfies. For any $f : A \to B$:

$$f \circ \mathrm{id}_A = f \qquad \text{(right unit law)}$$

$$\mathrm{id}_B \circ f = f \qquad \text{(left unit law)}$$

These are called the **unit laws**. Together they say: composing with the identity on the appropriate side leaves any morphism unchanged.

1.4.4 Reorganizing: Restating the Examples Formally

Transit. The "stay at Station A" route is $\mathrm{id}_A \in \mathrm{Hom}(A, A)$. For any route $f : A \to B$: $f \circ \mathrm{id}_A = f$ (riding the zero-distance route at A, then taking route f, is just route f). And $\mathrm{id}_B \circ f = f$ (taking route f and then staying at B is just route f).

Recipe. The "skip the rest step" action is $\mathrm{id}_{\mathrm{dough}} \in \mathrm{Hom}(\mathrm{dough}, \mathrm{dough})$. Doing nothing before or after any step leaves that step unchanged.

In both cases, the informal description maps exactly onto the unit laws.

1.4.5 Simplifying: Two Laws, One Line

The two unit laws collapse to a single display:

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f \qquad \text{for all } f : A \to B$$

This single line captures everything the identity morphism does. The uniqueness proof shows that id_A is the *only* morphism satisfying this condition.

1.4.6 Formally: Identity as Part of the Data

The identity is not derived — it is specified. Part of the data of a category is an assignment: to each object A, a distinguished morphism $\mathrm{id}_A \in \mathrm{Hom}(A, A)$, satisfying:

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f \qquad \text{for all } A, B, f : A \to B.$$

The full category definition — collecting objects, morphisms, composition, and identity together — is the subject of Chapter 2.

1.4.6.0.1 Displayed as triangles

The unit laws are often displayed as commutative triangles:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "f"'] & B \arrow[d, "\mathrm{id}_B"] \\ & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "\mathrm{id}_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \\ & B \end{tikzcd}$$

Each diagram says: the two paths from the top-left object to the bottom-right object are equal. That is precisely $\mathrm{id}_B \circ f = f$ and $f \circ \mathrm{id}_A = f$.

1.5 Commutative Diagrams

1.5.1 Informally: When All Paths Agree

A **commutative diagram** is a diagram of objects and morphisms in which every path between the same two endpoints gives the same composite morphism.

The simplest example is a triangle:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "h"'] & B \arrow[d, "g"] \\ & C \end{tikzcd}$$

This diagram *commutes* when $g \circ f = h$. Two paths from A to C; one composite result.

1.5.2 Expanding: Reading Commutative Diagrams

Internal Dialog

I see a diagram with multiple paths between two objects. My procedure:

1. **Identify** all paths from each object to each other object. I trace every possible route, like reading a map.
2. **Write** the composite morphism for each path using $\circ$ notation.
3. **Ask**: are these composites equal? If yes, the diagram (or those paths) commutes.
4. **Note**: a triangle asserts one equation, a square asserts one, a more complex diagram may assert many simultaneously.

Example: A Commutative Square

The most common diagram in category theory:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "p"'] & B \arrow[d, "q"] \\ C \arrow[r, "g"'] & D \end{tikzcd}$$

This commutes when $q \circ f = g \circ p$: going right-then-down equals going down-then-right.

In everyday terms: suppose A is a sentence in English, B a formal logical statement, C the same sentence in French, D the same formal statement. Then f = formalize English, p = translate English to French, q = formalize French, g = match formalizations. The square commutes if "translate then formalize" gives the same result as "formalize then translate." This is a genuine question, not a foregone conclusion.

Example: The Recipe as a Commutative Triangle

We have f : chop $\to$ sauté, g : sauté $\to$ add, and their composite $g \circ f$: chop $\to$ add. If we also draw h : chop $\to$ add meaning "chop must happen before add," then the triangle

$$\begin{tikzcd} \text{chop} \arrow[r, "f"] \arrow[dr, "h"'] & \text{sauté} \arrow[d, "g"] \\ & \text{add} \end{tikzcd}$$

commutes when $h = g \circ f$: the direct "chop before add" relation is the same as the two-step route via sautéing. Commutativity here says these two descriptions of the same ordering constraint are equal.

Reader's Annotation

Whenever you encounter a diagram, write next to it: "This commutes because __________." If you can't fill in the blank, you don't yet understand what the diagram is asserting. Diagrams are *claims* — they say that two composed morphisms are equal. Read them as equations, not just pictures.

1.5.3 Formalizing: Diagrams as Equations

Every commutative diagram is shorthand for a collection of equations between composed morphisms. The diagram is often easier to parse than the equations; the two are entirely equivalent.

A triangle asserts one equation. A square asserts one equation. A more complex diagram with five or six objects may assert many equations simultaneously, one for each pair of objects with multiple paths between them.

There is also a converse skill: given a collection of equations relating composed morphisms, you can often *draw* them as a commutative diagram. Practicing this translation in both directions builds fluency.

Not every diagram in category theory is required to commute. Sometimes a diagram merely depicts objects and morphisms without asserting equations. Convention: when a diagram is stated to commute (explicitly or by context), it does. When in doubt, check whether the author says "the following diagram commutes" or just "consider the following diagram."

1.5.4 Reorganizing: Diagrams and Their Equations

Triangle. The triangle with $f : A \to B$, $g : B \to C$, $h : A \to C$ commutes iff $h = g \circ f$. Equivalently: $h \in \mathrm{Hom}(A, C)$ and $g \circ f \in \mathrm{Hom}(A, C)$ and they are equal.

Square. The square with $f : A \to B$, $q : B \to D$, $p : A \to C$, $g : C \to D$ commutes iff $q \circ f = g \circ p$ in $\mathrm{Hom}(A, D)$.

Pasting. If we have two commutative squares sharing an edge:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "p"'] & B \arrow[r, "g"] \arrow[d, "q"] & C \arrow[d, "r"] \\ D \arrow[r, "s"'] & E \arrow[r, "t"'] & F \end{tikzcd}$$

Left square commutes: $q \circ f = s \circ p$. Right square commutes: $r \circ g = t \circ q$. The outer rectangle commutes: $r \circ g \circ f = t \circ s \circ p$. This is the **pasting lemma**.

1.5.5 Simplifying: Commutativity in One Line

A diagram commutes if and only if: for every pair of objects X, Y and every two paths p_1, p_2 of morphisms from X to Y, the composites are equal.

$$\text{diagram commutes} \iff \forall\, X, Y,\ \forall\,\text{paths } p_1, p_2 : X \rightsquigarrow Y, \quad \mathrm{comp}(p_1) = \mathrm{comp}(p_2)$$

All of our examples are instances of this single condition.

1.5.6 Formally: The Pasting Lemma

1.5.6.0.1 Pasting lemma

Let two commutative squares share a common edge. If $q \circ f = s \circ p$ and $r \circ g = t \circ q$, then $r \circ g \circ f = t \circ s \circ p$.

Proof. $r \circ g \circ f = (r \circ g) \circ f = (t \circ q) \circ f = t \circ (q \circ f) = t \circ (s \circ p) = (t \circ s) \circ p = t \circ s \circ p$. □

The pasting lemma is used constantly in categorical proofs: large commutative diagrams are built from smaller confirmed pieces, like tiling a floor. Once you know it, you can decompose any complex diagram into its square or triangle sub-diagrams and confirm each in turn.

1.5.6.0.2 Diagrams without commutativity

A diagram that is merely depicted (not asserted to commute) is called a **diagram in $\mathcal{C}$** or a **graph of shape J in $\mathcal{C}$**. The formal definition uses the concept of a functor from an index category J, which we will develop in Chapter 3.

Exercises

What are the two ingredients of any category, before laws?	Objects and morphisms.
Is a morphism a function?	Not necessarily. It is an arrow with a source and a target. Functions are one example.
Can two different morphisms have the same source and target?	Yes. The hom-set may contain many morphisms.
Can a morphism have two different sources?	No. The source is part of the morphism's data.
What does $f : A \to B$ tell you?	The name of the morphism, its source A, and its target B.
If $f : A \to B$ and $g : B \to C$, is $g \circ f$ defined?	Yes. The target of f matches the source of g.
If $f : A \to B$ and $g : C \to D$ with $B \neq C$, is $g \circ f$ defined?	No. Composition requires the matching condition.
What is the source and target of $g \circ f$ when $f : A \to B$ and $g : B \to C$?	Source A, target C.
How is $g \circ f$ pronounced?	"g after f," or "g composed with f."
Why is composition written right-to-left?	So that the morphism applied first sits closest to the input. Like $g(f(x))$.

What does the identity morphism id_A do under composition?	It leaves morphisms unchanged: $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$.
How many identity morphisms does an object have?	Exactly one.
State the associativity law in symbols.	$(h \circ g) \circ f = h \circ (g \circ f)$.
Why does associativity let us write $h \circ g \circ f$ without parentheses?	Because the grouping doesn't change the result.
When we say a diagram *commutes*, what do we mean?	Every path between two given objects gives the same composite morphism.
In a triangle with $f : A \to B$, $g : B \to C$, and $h : A \to C$, what equation makes it commute?	$h = g \circ f$.
Is a single morphism $f : A \to B$ a (trivial) commutative diagram?	Yes. There is only one path, so it commutes vacuously.
Can a diagram fail to commute?	Yes. Two paths might compose to different morphisms.
What does an empty hom-set $\mathrm{Hom}(A, B) = \varnothing$ mean?	There are no morphisms from A to B at all.
Must every pair of objects have a morphism between them?	No.
What is the smallest category with a non-identity morphism?	**2**: two objects with one arrow between them, plus the two identities.
In the recipe category, what is the composite of "chop" and "sauté"?	A single morphism Chop $\to$ Add.
Why is the discrete category on a set S called "discrete"?	It has no morphisms besides identities.
In a discrete category on $\{A, B\}$, is $\mathrm{Hom}(A, B)$ empty?	Yes. There are no non-identity morphisms.
Why does the unit law require both $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$?	Because f has both a source and a target, and identity must be neutral on both sides.
What goes wrong if you drop associativity?	Composites of three or more morphisms become ambiguous.
What goes wrong if you drop identities?	Composition has no neutral element; many constructions break.
What is the most reliable test that a triangle commutes?	Compute both paths and check they are equal as morphisms.
Can a morphism be its own inverse without being the identity?	Yes; for example, a swap in a permutation.
If f and g are both identities on A, what can you conclude?	$f = g$. There is only one identity per object.

Chapter Summary

Chapter Summary

Objects are the things in a category. We work with them only through their connections, never by looking inside.
Morphisms ($f : A \to B$) are the directed connections between objects. Each has a domain and a codomain.
Composition: if $f : A \to B$ and $g : B \to C$, then $g \circ f : A \to C$. Composition is associative: $(h \circ g) \circ f = h \circ (g \circ f)$.
Identity: every object A has a morphism $\mathrm{id}_A : A \to A$ satisfying $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$ for any $f : A \to B$. It is unique.
Commutative diagrams assert that all paths between the same two objects compose to the same morphism. They are equations in picture form.

The reader's three questions, to be asked of every new example:

1. What are the objects?

2. What are the morphisms (domain, codomain, name)?
3. Which diagrams commute, and what equations do they assert?

Formal Restatement

Formal Restatement

The following collects the formal content of this chapter. A reader returning here after completing the chapter should be able to read each item and hear the informal discussion behind it.

Morphisms. A morphism $f : A \to B$ has domain A and codomain B. The hom-set $\mathrm{Hom}(A, B)$ is the collection of all morphisms from A to B. Two morphisms are equal iff they have equal domains, equal codomains, and are the same arrow.

Composition. There is a family of functions $\circ : \mathrm{Hom}(B, C) \times \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, C)$ satisfying associativity:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

for all $f \in \mathrm{Hom}(A, B)$, $g \in \mathrm{Hom}(B, C)$, $h \in \mathrm{Hom}(C, D)$.

Identity. Each object A carries a distinguished morphism $\mathrm{id}_A \in \mathrm{Hom}(A, A)$ satisfying the unit laws:

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f \qquad \text{for all } f \in \mathrm{Hom}(A, B).$$

The identity morphism is unique.

Commutativity. A diagram commutes iff for every pair of objects X, Y and every two paths of morphisms from X to Y, the two composites are equal.

Pasting lemma. If two commutative squares share a common edge, the outer rectangle commutes.

Note: Objects, morphisms, composition, and identity together form the data of a **category** — to be defined precisely in Chapter 2, where the axioms above become the definition.

In Chapter 2, we assemble all these pieces into a single package: a **category**.

Chapter 2

Categories

Chapter Overview

Chapter 1 introduced four ideas: objects, morphisms, composition, and identity. Each came with rules — composition is associative, identity is neutral — but we never packaged all four ideas and their rules into a single named thing.

This chapter does that. A *category* is the name for the package. Once we have the name and the definition, we can look at many different situations and ask: "is this a category?" The answer is often yes — and recognizing that two very different-looking situations are both categories is the first payoff of the theory.

The chapter also introduces *isomorphism* — the precise meaning of "these two objects are the same for our purposes" — and the *opposite category*, which reverses all arrows and will recur throughout the book.

The Formal Restatement at the end collects the definitions in compact notation.

2.1 The Definition of a Category

2.1.1 Informally: Packaging What We Have

We have been working with objects and morphisms for an entire chapter. We have two rules: composition is associative, and identity morphisms are neutral. Now we bundle everything together and give the bundle a name.

A **category** is a collection of objects and morphisms, equipped with composition and identity morphisms, satisfying the associativity and unit laws.

That's it. A category is not a complicated new idea. It is a name for the structure we have already been using.

2.1.2 Expanding: Checking Our Examples

The three running examples from Chapter 1 are all categories. Let's verify this for one of them in detail.

Example: The Recipe as a Category

Recall the recipe with three steps: Chop, Sauté, Add.

$$\begin{tikzcd} \text{Chop} \arrow[loop left, "\mathrm{id}_C"] \arrow[r, "f"] & \text{Sauté} \arrow[loop above, "\mathrm{id}_S"] \arrow[r, "g"] & \text{Add} \arrow[loop right, "\mathrm{id}_A"] \end{tikzcd}$$

We check each requirement:

- **Objects?** Yes: Chop, Sauté, Add.
- **Morphisms?** Yes: f, g, $g \circ f$, and the three identity morphisms.
- **Composition?** Yes: $g \circ f$ exists and goes from Chop to Add, as required.
- **Identity?** Yes: each step has a "do nothing at this step" morphism.
- **Associativity?** There are only two non-identity, non-composed morphisms (f and g), so the only chain of three is degenerate — associativity holds vacuously.
- **Unit laws?** $f \circ \mathrm{id}_C = f$, $\mathrm{id}_S \circ f = f$. Checked.

The recipe is a category.

Internal Dialog

When I want to verify that something is a category, I run through a checklist:

1. Can I name the objects?
2. Can I name the morphisms, with domains and codomains?

3. Is composition defined wherever it should be, and associative?
4. Does every object have an identity morphism that is neutral?

If all four answers are yes, I have a category. This checklist applies to every example in this book and every example I'll encounter elsewhere.

2.1.3 Formalizing: The Four-Part Definition

Where the Name Comes From

Category as a mathematical term was introduced by *Samuel Eilenberg* (1913–1998, Polish-American mathematician) and *Saunders Mac Lane* (1909–2005, American mathematician) in their 1945 paper *General Theory of Natural Equivalences*. The original motivation: they needed a precise framework to formulate the notion of "natural transformation," which in turn required "functor," which required *the things functors go between* — i.e., *categories*. The word *category* predates the math: it's from Greek *katēgoria* ("predicate, statement, accusation"), via Aristotle's *Categories* (4th c. BCE) and Kant's table of categories (1781). Eilenberg and Mac Lane chose the word for its sense of "a class of mathematical structures of the same kind" — a class one can speak *about*.

A **category** $\mathcal{C}$ consists of four pieces of data:

1. A collection $\mathrm{Ob}(\mathcal{C})$, whose elements are called **objects**.
2. For each pair of objects $A, B \in \mathrm{Ob}(\mathcal{C})$, a collection $\mathrm{Hom}_{\mathcal{C}}(A, B)$, whose elements are called **morphisms** from A to B.
3. For each triple of objects A, B, C, a **composition** function
$$\circ \;:\; \mathrm{Hom}(B, C) \times \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(A, C).$$
4. For each object A, a distinguished **identity morphism** $\mathrm{id}_A \in \mathrm{Hom}(A, A)$.

This data must satisfy two **axioms**:

- **Associativity**: For all $f \in \mathrm{Hom}(A, B)$, $g \in \mathrm{Hom}(B, C)$, $h \in \mathrm{Hom}(C, D)$:
$$(h \circ g) \circ f = h \circ (g \circ f).$$
- **Unit laws**: For all $f \in \mathrm{Hom}(A, B)$:
$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

Reader's Annotation

When you meet a definition with this structure — "consists of" followed by "satisfying" — read the "consists of" part as a list of *what you must supply*, and the "satisfying" part as a list of *what your supply must pass*. The data is what you bring to the table; the axioms are the tests your data must survive.

2.1.4 Reorganizing: The Examples Pass the Tests

Recipe category. We supply: objects $\{\mathrm{Chop}, \mathrm{Sauté}, \mathrm{Add}\}$; morphisms f, g, $g \circ f$, and three identities; composition by the rule of chaining steps; identity by the "stay at this step" morphism. The associativity test passes (checked above). The unit test passes.

Transit category. The stations are objects. The train lines are morphisms. The "stay at station" routes are identities. Any journey along lines in sequence is a composite. Since real transit routes are unambiguous regardless of how you mentally group the legs, associativity holds. The transit map is a category.

Temperature ordering. The readings {0řC, 20řC, 37řC, 100řC} are objects. A morphism $A \to B$ exists exactly when $A \leq B$ ("is no warmer than"). Identity: every temperature is no warmer than itself. Composition: if $A \leq B$ and $B \leq C$ then $A \leq C$. Associativity holds because $\leq$ is transitive, and there is at most one morphism between any two objects so associativity is automatic. This is a category.

2.1.5 Simplifying: The Definition in Brief

A category packages four things: objects, morphisms, composition, identity. It satisfies two laws:
$$(h \circ g) \circ f = h \circ (g \circ f) \qquad \text{and} \qquad f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

Every example we have seen passes both laws. These two equations are the entire content of the definition.

2.1.6 Formally: The Standard Formulation

2.1.6.0.1 Notation

We write $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$ for categories and use bold letters $\mathbf{C}$, $\mathbf{D}$ in some contexts. The notation $f \in \mathcal{C}(A, B)$ is an alternative to $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. When the category is clear from context we write $\mathrm{Hom}(A, B)$.

2.1.6.0.2 Size

In the definition above we said "collection" rather than "set." The distinction matters: if $\mathrm{Ob}(\mathcal{C})$ is a set, $\mathcal{C}$ is called **small**; if the hom-sets $\mathrm{Hom}(A, B)$ are all sets, $\mathcal{C}$ is called **locally small**. Most categories we encounter are at least locally small. We will not dwell on set-theoretic foundations here; the interested reader may consult Appendix A (to be written).

2.1.6.0.3 Uniqueness of identities revisited

The proof from Chapter 1 that identity morphisms are unique works uniformly in any category: if e, e' both satisfy the unit laws at A, then $e = e \circ e' = e'$. No additional hypotheses are needed.

2.2 A Catalogue of Small Categories

2.2.1 Informally: Categories Can Be Tiny

Not all categories are large. Some have very few objects and morphisms. These tiny categories are not toys — they are the building blocks used throughout the rest of category theory to state definitions and theorems in a compact, diagram-based way.

We build five small categories from scratch.

2.2.2 Expanding: Five Small Categories

Example: 0: The Empty Category

Objects: none. Morphisms: none.
Does it satisfy the axioms? Every axiom has the form "for all objects $A\ldots$" or "for all morphisms $f\ldots$". With nothing to quantify over, every such statement is true *vacuously* — it holds because there is nothing that could fail it.
The empty category is valid. It is the categorical equivalent of a blank page: a place where nothing has been written, not a place where something is wrong.

Example: 1: One Object, One Morphism

Objects: one object, call it $*$. Morphisms: one morphism, id_*.
The only composition is $\mathrm{id}_* \circ \mathrm{id}_* = \mathrm{id}_*$. The only unit law is $\mathrm{id}_* \circ \mathrm{id}_* = \mathrm{id}_*$. Both are satisfied. The category **1** is the simplest non-empty category: one dot, one loop.

Example: 2: Two Objects, One Arrow

Objects: 0 and 1. Morphisms: id_0, id_1, and one morphism $\iota : 0 \to 1$.

$$\begin{tikzcd} 0 \arrow[loop left, "\mathrm{id}_0"] \arrow[r, "\iota"] & 1 \arrow[loop right, "\mathrm{id}_1"] \end{tikzcd}$$

There is no morphism from 1 back to 0. The only non-trivial composition would require a morphism starting at 1, but there is none (besides id_1 which goes $1 \to 1$). So composition is complete. The unit laws hold for ι because $\iota \circ \mathrm{id}_0 = \iota = \mathrm{id}_1 \circ \iota$.
The category **2** is sometimes called the **walking morphism**: it contains exactly one non-identity morphism and nothing else. It is used later to "pick out" a single morphism from a category.

Example: An Ordering Category

Take any collection of objects with a relation $\leq$ that is:

- **Reflexive**: $A \leq A$ for all A (gives identity),
- **Transitive**: $A \leq B$ and $B \leq C$ implies $A \leq C$ (gives composition).

Define $\mathrm{Hom}(A, B)$ to contain exactly one morphism when $A \leq B$, and to be empty otherwise. Since there is at most one morphism between any two objects, associativity is automatic. Identity comes from reflexivity.
Such a category is called a **preorder category**. Our temperature example is a preorder category on $\{0\text{řC}, 20\text{řC}, 37\text{řC}, 100\text{řC}\}$ with the usual $\leq$ on temperatures.
A preorder category where $A \leq B$ and $B \leq A$ implies $A = B$ is called a **poset category**.

Example: A One-Object Category

Take a single object $*$. Let the morphisms from $*$ to $*$ be some collection M of "actions" that can be composed in sequence, where composition is associative and there is a "do nothing" action.
For concreteness: suppose $*$ is a light panel, and the morphisms are sequences of button presses. Pressing button A then button B is a composite action. The "do nothing" action is the identity. If button presses compose associatively (which they do, since doing A then B then C is the same regardless of grouping), this is a category.
A one-object category is precisely a **monoid**: a set with an associative binary operation and an identity element.
We will return to this connection when we discuss functors.

Internal Dialog

When I encounter a small category, I draw it as a directed graph and check two things:

1. Is the graph *closed under composition*? For every two composable edges, is there an edge representing their composite?
2. Does every vertex have a *loop* for its identity?

If both hold and composition is associative, I have a category. For tiny categories, associativity is often automatic because there simply are not enough morphisms to chain in three different ways.

2.2.3 Formalizing: What Makes a Category Small?

In the example above we used the word "small" informally. Let's be precise.

- $\mathcal{C}$ is **small** if $\mathrm{Ob}(\mathcal{C})$ is a set (not a proper class).
- $\mathcal{C}$ is **finite** if $\mathrm{Ob}(\mathcal{C})$ and all hom-sets are finite sets.
- $\mathcal{C}$ is **discrete** if the only morphisms are identity morphisms: $\mathrm{Hom}(A, B) = \varnothing$ for $A \neq B$ and $\mathrm{Hom}(A, A) = \{\mathrm{id}_A\}$.
- $\mathcal{C}$ is **thin** (or a **preorder**) if there is at most one morphism between any two objects.

The categories $\mathbf{0}$, $\mathbf{1}$, $\mathbf{2}$ are all small and finite. The temperature example is small, finite, and thin. A discrete category on n objects has n objects and n morphisms (just identities).

2.2.4 Reorganizing: Naming What We Already Know

Our examples from Chapter 1 now have names:

- The recipe (Chop $\to$ Sauté $\to$ Add) is a small, finite, *thin* category — there is exactly one morphism from any step to any step that comes later, and none in the other direction.
- The temperature ordering is a small, finite, *poset* category.
- The approval chain is a small, finite, thin category, identical in shape to the recipe.

The single-object category $\mathbf{1}$ is both a category and a monoid. We will see later that any monoid can be viewed as a one-object category, and vice versa.

2.2.5 Simplifying: A Table of Small Categories

Name	Objects	Non-id morphisms	Thin?	Finite?
$\mathbf{0}$	0	0	yes	yes
$\mathbf{1}$	1	0	yes	yes
$\mathbf{2}$	2	1	yes	yes
Preorder	any	varies	yes	varies
One-object (monoid)	1	varies	no	varies

2.2.6 Formally: Discrete and Indiscrete Categories

2.2.6.0.1 Discrete category on a collection S

Given any collection S, the **discrete category** $\mathrm{Disc}(S)$ has $\mathrm{Ob} = S$ and $\mathrm{Hom}(A, B) = \varnothing$ for $A \neq B$, $\mathrm{Hom}(A, A) = \{\mathrm{id}_A\}$. Composition and identity are trivially defined. This is the "objects only" category: all structure has been stripped away except the objects themselves.

2.2.6.0.2 Indiscrete category on S

The **indiscrete category** $\mathrm{Indis}(S)$ has $\mathrm{Ob} = S$ and $\mathrm{Hom}(A, B) = \{*_{AB}\}$ for every pair A, B (including $A = B$, where $*_{AA} = \mathrm{id}_A$). Composition is forced: the only possible composite of any two composable morphisms is the unique morphism between the appropriate objects. All axioms hold automatically.

Discrete and indiscrete categories are the two extremes: one has no morphisms beyond identities, the other has exactly one morphism between every pair.

Where the Name Comes From

Isomorphism is from Greek *isos* ("equal") + *morphē* ("form") — "same form." The term was coined by the 19th-century algebraists studying group theory; it generalized from groups to all algebraic and categorical settings. An isomorphism is, in the category-theoretic sense, a morphism with a two-sided inverse — the strongest natural notion of "sameness" available in any category.

2.3 Isomorphism

2.3.1 Informally: Going There and Back

Two objects are "the same for our purposes" when you can travel from one to the other and then back, and both round trips (there-and-back, or back-and-there) leave you exactly where you started — not approximately where you started, but *exactly*.

This is the categorical meaning of "these two things are interchangeable." We call this an **isomorphism**.

2.3.2 Expanding: Round Trips

Example: The Two-Road Town

Suppose two stations, A and B, have a road from A to B and a road from B to A. These roads are an isomorphism if: driving from A to B and then immediately back puts you exactly at A with nothing changed, and driving from B to A and then immediately back puts you exactly at B.
This is stronger than just saying the roads exist. It requires that the round trips are identity journeys.

Example: A Recipe Step You Can Undo

Suppose a recipe step "chill the dough" has a corresponding step "bring dough to room temperature." If doing both in either order leaves the dough in its original state, the two steps are inverse morphisms. Chilling and then warming = identity. Warming and then chilling = identity.
Note: in reality, chilling and warming might not be perfect inverses (some moisture is lost, some structure changes). Whether they are isomorphisms is a question about your specific model of the process, not about the real-world chemistry.

Internal Dialog

I see two morphisms $f : A \to B$ and $g : B \to A$. I ask:

- Does $g \circ f = \mathrm{id}_A$? (Going from A to B then back returns to A as if nothing happened.)
- Does $f \circ g = \mathrm{id}_B$? (Going from B to A then back returns to B as if nothing happened.)

If both hold, f and g are inverse to each other, and both are isomorphisms.

2.3.3 Formalizing: The Definition

A morphism $f : A \to B$ is an **isomorphism** if there exists a morphism $g : B \to A$ such that:

$$g \circ f = \mathrm{id}_A \qquad \text{and} \qquad f \circ g = \mathrm{id}_B.$$

The morphism g is called the **inverse** of f, written f^{-1}.

Two objects A and B are **isomorphic**, written $A \cong B$, if there exists an isomorphism $f : A \to B$.

Inverses are unique. If g and g' are both inverses of f, then:

$$g = g \circ \mathrm{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_A \circ g' = g'.$$

Every isomorphism has exactly one inverse.

2.3.4 Reorganizing: Which Objects Are Isomorphic?

In 2: the objects 0 and 1 are *not* isomorphic. The only morphism $0 \to 1$ is ι, and there is no morphism $1 \to 0$. Without a morphism in both directions, there is no isomorphism.

In the temperature ordering: 20řC and 37řC are not isomorphic. A morphism 20řC $\to$ 37řC exists (the "is colder than" relation), but no morphism goes in the other direction. In a poset category, two distinct objects are *never* isomorphic.

In 1: the single object $*$ is isomorphic to itself, via id_*. Every object is always isomorphic to itself, via its identity morphism.

In the indiscrete category on $\{A, B\}$: $A \cong B$. There is a morphism $*_{AB} : A \to B$ and a morphism $*_{BA} : B \to A$, and their composites are the unique morphisms $*_{AA} = \mathrm{id}_A$ and $*_{BB} = \mathrm{id}_B$.

2.3.5 Simplifying: Isomorphism in One Line

$f : A \to B$ is an isomorphism iff $\exists\, g : B \to A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

The inverse g is then unique and denoted f^{-1}.

$A \cong B$ means an isomorphism between them exists.

2.3.6 Formally: Properties of Isomorphism

2.3.6.0.1 Isomorphism is an equivalence relation on objects

- **Reflexive**: $A \cong A$ via id_A (its own inverse).
- **Symmetric**: if $f : A \xrightarrow{\sim} B$ then $f^{-1} : B \xrightarrow{\sim} A$.
- **Transitive**: if $f : A \xrightarrow{\sim} B$ and $g : B \xrightarrow{\sim} C$, then $g \circ f : A \xrightarrow{\sim} C$ with inverse $f^{-1} \circ g^{-1}$.

The notation $\xrightarrow{\sim}$ over an arrow marks it as an isomorphism.

2.3.6.0.2 Isomorphisms in special categories

In a thin (preorder) category, the only isomorphisms are identity morphisms — since $f : A \to B$ and $g : B \to A$ existing simultaneously forces $A = B$ (by antisymmetry, if the preorder is a poset). In a groupoid (a category where *every* morphism is an isomorphism), all morphisms are invertible.

Where the Name Comes From

The **opposite category** $\mathcal{C}^{\mathrm{op}}$ reverses the direction of every morphism. The notation $^{\mathrm{op}}$ (also written $\mathcal{C}^{\circ}$ or $\mathcal{C}^*$) is universal; the superscript "op" is short for *opposite*. The construction is one of the most useful — and most underused-by-beginners — operations in category theory. Every theorem in $\mathcal{C}$ has a dual theorem in $\mathcal{C}^{\mathrm{op}}$; this is the principle behind the dichotomies limit/colimit, product/coproduct, terminal/initial.

2.4 The Opposite Category

2.4.1 Informally: Turning All Arrows Around

Every category has a mirror image. In the mirror, every arrow points the other way. An arrow that went from A to B now goes from B to A. Composition still works — you just have to track the reversal.

This mirror is called the **opposite category**.

2.4.2 Expanding: Reversing Familiar Examples

Example: The Recipe Reversed

The original recipe category has $f : \text{Chop} \to \text{Sauté}$ and $g : \text{Sauté} \to \text{Add}$. The opposite category has $f^{\mathrm{op}} : \text{Sauté} \to \text{Chop}$ and $g^{\mathrm{op}} : \text{Add} \to \text{Sauté}$.
The original morphism $g \circ f : \text{Chop} \to \text{Add}$ becomes $(g \circ f)^{\mathrm{op}} : \text{Add} \to \text{Chop}$.
Importantly, $(g \circ f)^{\mathrm{op}} = f^{\mathrm{op}} \circ g^{\mathrm{op}}$. The order of composition reverses when you reverse arrows. This is the key bookkeeping fact.

Example: The Opposite of a Preorder

The opposite of the temperature preorder has morphisms going from warmer to colder: 100řC $\to$ 37řC $\to$ 20řC $\to$ 0řC. This is the "is no colder than" relation.
In general, the opposite of a preorder category is another preorder category, with the order reversed.

Internal Dialog

When I reverse a category, I keep two things straight:

1. Every morphism $f : A \to B$ becomes $f^{\mathrm{op}} : B \to A$. Domain and codomain swap.
2. Composition reverses order: if $h = g \circ f$ in $\mathcal{C}$, then $h^{\mathrm{op}} = f^{\mathrm{op}} \circ g^{\mathrm{op}}$ in $\mathcal{C}^{\mathrm{op}}$.

The objects stay the same. Only the arrows and their order change.

2.4.3 Formalizing: The Definition

Given a category $\mathcal{C}$, its **opposite category** $\mathcal{C}^{\mathrm{op}}$ is defined as follows:

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$ (same objects).
- $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$ (morphisms are reversed).
- Composition in $\mathcal{C}^{\mathrm{op}}$: given $f^{\mathrm{op}} \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ and $g^{\mathrm{op}} \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C)$, their composite is $(f \circ g)^{\mathrm{op}}$, where $\circ$ is composition in $\mathcal{C}$.
- Identity: $\mathrm{id}_A^{\mathrm{op}} = \mathrm{id}_A$.

$\mathcal{C}^{\mathrm{op}}$ **is a category.** We verify:

- *Associativity*: inherited from $\mathcal{C}$, with order reversed.
- *Unit laws*: $f^{\mathrm{op}} \circ \mathrm{id}_A^{\mathrm{op}} = (\mathrm{id}_A \circ f)^{\mathrm{op}} = f^{\mathrm{op}}$, and similarly on the other side.

2.4.4 Reorganizing: The Opposite of the Opposite

Applying the opposite construction twice returns to the original: $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$. Reversing arrows twice gives back the original arrows.

2.4.5 Simplifying: Opposite in Notation

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$$
$$(g \circ_{\mathcal{C}^{\mathrm{op}}} f) = (f \circ_{\mathcal{C}} g)^{\mathrm{op}}$$
$$(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$$

2.4.6 Formally: Why the Opposite Category Matters

2.4.6.0.1 Duality

The opposite category is the engine of **categorical duality**. Any statement that is true of all categories remains true when you replace every $\mathcal{C}$ with $\mathcal{C}^{\mathrm{op}}$ — but the statement it becomes may sound quite different. Every theorem has a *dual theorem*, obtained by reversing all arrows. This doubles the value of every proof.

2.4.6.0.2 Contravariant functors

In Chapter 3 we will define functors. A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ is simply a (covariant) functor from $\mathcal{C}^{\mathrm{op}}$ to $\mathcal{D}$. The opposite category allows us to treat contravariance as a special case of covariance rather than a separate concept.

Exercises

What four pieces of data specify a category?	Objects, morphisms, composition, and identities (one per object).
What two laws do those pieces obey?	Associativity of composition, and the unit law for identities.
Is “a set with extra structure” enough to be a category?	No. A category is data plus laws; the morphisms and composition are part of the structure.
In **0**, how many morphisms are there?	Zero. No objects, no morphisms.
In **1**, how many morphisms?	One: the identity on the unique object.
In **2**, how many morphisms in total?	Three: two identities and one non-identity arrow.
Why does **2** have no other arrows?	There is nothing to compose the existing arrow with that would create a new one.
What is a preorder, viewed as a category?	A category where every hom-set has at most one element.
In a preorder-category, when is $A \to B$ inhabited?	When $A \leq B$ in the preorder.
What does antisymmetry give you on top of a preorder?	Iso-equality: $A \cong B$ implies $A = B$ on the nose.
What is a monoid, viewed as a category?	A category with one object, whose morphisms compose like the monoid operation.

In a one-object category, what is the identity morphism?	The unit of the monoid.
What is an isomorphism in a category?	A morphism with a two-sided inverse.
Why must inverses be two-sided?	Because a one-sided inverse need not be unique or behave well under composition.
If $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$, what is g called?	The inverse of f (and vice versa).
Are inverses unique when they exist?	Yes. If g and g' are both inverses of f, then $g = g'$.
Why is uniqueness of inverses important?	It lets us speak of *the* inverse, written f^{-1}.
What does $A \cong B$ mean?	There exists an isomorphism between A and B.
Is $\cong$ reflexive? Symmetric? Transitive?	Yes to all three.
What is the opposite category $\mathcal{C}^{\mathrm{op}}$?	Same objects as $\mathcal{C}$, but every arrow reversed; composition is reversed too.
If $f : A \to B$ in $\mathcal{C}$, what is it in $\mathcal{C}^{\mathrm{op}}$?	An arrow $B \to A$.
Why is $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$?	Reversing arrows twice puts them back the way they were.
What is the principle of duality?	Every theorem about $\mathcal{C}$ yields a theorem about $\mathcal{C}^{\mathrm{op}}$ by reversing arrows.
What is a contravariant operation, in opposite-category terms?	An ordinary (covariant) operation from $\mathcal{C}^{\mathrm{op}}$.
Why is a small category called "small"?	Its collection of objects (and morphisms) is a set, not a proper class.
Is **Set** small?	No. The collection of all sets is too large to be a set.
Can a category have a proper class of objects?	Yes; such categories are called *large*.
If f is an iso and $f \circ g$ is defined, can you cancel f from $f \circ g = f \circ h$?	Yes. Compose with f^{-1} on the left.
If f is *not* an iso, can you still cancel?	Not in general.
Where does the cycle "Informally $\to$ Expanding $\to \ldots \to$ Formally" end up in a Chapter?	In the Formal Restatement at the chapter's end.

Chapter Summary

Chapter Summary

A **category** $\mathcal{C}$ packages: objects, hom-sets, composition, and identity morphisms, satisfying associativity and the unit laws.
Small categories include **0** (empty), **1** (one object), **2** (the walking morphism), preorder categories (at most one morphism between any two objects), and one-object categories (monoids).
An **isomorphism** $f : A \to B$ has an inverse $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. Isomorphic objects are interchangeable. The inverse is unique.
The **opposite category** $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$ but all morphisms reversed. Every theorem about categories has a dual obtained by replacing $\mathcal{C}$ with $\mathcal{C}^{\mathrm{op}}$.

The category checklist:

1. Name the objects.
2. Name the morphisms (with domains and codomains).
3. Verify composition is defined and associative.
4. Verify every object has a neutral identity morphism.

Formal Restatement

Formal Restatement

Category. A category $\mathcal{C}$ consists of $\mathrm{Ob}(\mathcal{C})$, hom-sets $\mathrm{Hom}(A, B)$ for all A, B, composition $\circ$, and identities id_A, satisfying:

$$(h \circ g) \circ f = h \circ (g \circ f), \qquad f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

Special classes. $\mathcal{C}$ is *small* if $\mathrm{Ob}(\mathcal{C})$ is a set; *thin* if $|\mathrm{Hom}(A, B)| \leq 1$ for all A, B; *discrete* if $\mathrm{Hom}(A, B) = \varnothing$ for $A \neq B$.

Isomorphism. $f : A \to B$ is an isomorphism iff $\exists\, f^{-1} : B \to A$ with $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$. $A \cong B$ means such f exists. The inverse is unique.

Opposite category. $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$; $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$; composition $(g^{\mathrm{op}} \circ_{\mathrm{op}} f^{\mathrm{op}}) = (f \circ g)^{\mathrm{op}}$; $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Chapter 3 introduces functors: structure-preserving maps between categories.

Chapter 3

Functors

Chapter Overview

A category is a world of objects connected by morphisms. A *functor* is a map between two such worlds that respects their structure. It sends objects to objects and morphisms to morphisms, but not arbitrarily: it must preserve composition and identity. A functor that ignores the structure — sending everything to a single point — is still a functor; one that scrambles the composition rules is not.
By the end of this chapter we will have seen that categories and functors themselves form a category. This is one of the first signs that category theory is "self-referential" in a productive way: the tools we build can be applied to the framework that built them.

Where the Name Comes From

Functor was coined by Eilenberg and Mac Lane (1945) as a portmanteau of *function* + the agent suffix *-or* ("operator," "actuator"). A functor is a "function-like thing performing a mapping between categories," operating on objects and morphisms simultaneously and preserving composition and identity. The word was deliberately new — distinct from "function" (which operates on sets) — and has stuck unchanged. **Covariant / contravariant** are from tensor calculus (covariant tensors transform with the coordinate change; contravariant against it); Eilenberg-Mac Lane adopted the terminology for functors that preserve vs. reverse the direction of morphisms.

3.1 Structure-Preserving Maps

3.1.1 Informally: Translating Between Categories

Suppose you have two categories. A functor from the first to the second is a systematic translation: every object in the first world is sent to some object in the second, and every arrow in the first world is sent to an arrow in the second — in a way that respects the structure.

"Respects the structure" has two precise requirements:

- Composites translate to composites.
- Identity morphisms translate to identity morphisms.

If these two conditions hold, the translation is a functor.

3.1.2 Expanding: A Functor from Recipes to Approvals

We have seen the recipe category (Chop $\to$ Sauté $\to$ Add) and the approval chain category (Request $\to$ Approved $\to$ Final). Both are thin categories with three objects in a line. A functor between them is a systematic renaming that preserves the ordering structure.

Example: The Status Functor

Define F as follows:

Recipe	*Approval chain*
Chop	$\mapsto$ Request
Sauté	$\mapsto$ Approved
Add	$\mapsto$ Final

On objects: $F(\text{Chop}) = \text{Request}$, and so on. On morphisms: $F(f) =$ the unique morphism from Request to Approved (the manager's approval). $F(g) =$ the unique morphism from Approved to Final. $F(g \circ f) =$ the unique morphism from Request to Final.

Is this a functor? We check:

- **Preserves composition**: $F(g \circ f) = F(g) \circ F(f)$. Both sides are the unique morphism from Request to Final. ✓
- **Preserves identity**: $F(\mathrm{id}_{\mathrm{Chop}}) = \mathrm{id}_{\mathrm{Request}}$. In a thin category, identity morphisms are the only morphisms from an object to itself, so this is automatic. ✓

F is a functor.

Example: A Functor Between Orderings

Consider the temperature ordering $\mathcal{T}$:

$$0\text{ř}\mathrm{C} \to 20\text{ř}\mathrm{C} \to 37\text{ř}\mathrm{C} \to 100\text{ř}\mathrm{C}$$

and the size ordering $\mathcal{S}$: Small $\to$ Medium $\to$ Large.
Define $G : \mathcal{T} \to \mathcal{S}$ by:

$$G(0\text{ř}\mathrm{C}) = \text{Small}, \quad G(20\text{ř}\mathrm{C}) = \text{Small}, \quad G(37\text{ř}\mathrm{C}) = \text{Medium}, \quad G(100\text{ř}\mathrm{C}) = \text{Large}.$$

Two temperature values (freezing and room temperature) both map to Small. On morphisms: if $A \leq B$ in $\mathcal{T}$, then $G(A) \leq G(B)$ in $\mathcal{S}$ (check: $G(0\text{ř}) = G(20\text{ř}) = \text{Small} \leq \text{Small}$, and so on). Composition and identity are preserved because all hom-sets have at most one element.
G is a functor. It "collapses" two temperature values into one size value without breaking the ordering structure.

Internal Dialog

When I want to check that F is a functor from $\mathcal{C}$ to $\mathcal{D}$, I run two checks:

1. **Composition**: For every composable pair $f : A \to B$ and $g : B \to C$ in $\mathcal{C}$: is $F(g \circ f) = F(g) \circ F(f)$?
2. **Identity**: For every object A in $\mathcal{C}$: is $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$?

If both hold for all morphisms and objects, F is a functor. Note that the composition on the left of each equation is in $\mathcal{C}$, and on the right it is in $\mathcal{D}$.

3.1.3 Formalizing: The Two Conditions

A **functor** $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. An **object assignment**: for each $A \in \mathrm{Ob}(\mathcal{C})$, an object $F(A) \in \mathrm{Ob}(\mathcal{D})$.
2. A **morphism assignment**: for each $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, a morphism $F(f) \in \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$.

Satisfying:

- **Functoriality (composition)**: $F(g \circ f) = F(g) \circ F(f)$ for all composable f, g in $\mathcal{C}$.
- **Functoriality (identity)**: $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$ for all $A \in \mathrm{Ob}(\mathcal{C})$.

We say F is a functor *from $\mathcal{C}$ to $\mathcal{D}$*, or a **covariant functor**, or just a functor. ($\mathcal{C}$ is the **source** or **domain**; $\mathcal{D}$ is the **target** or **codomain**.)

3.1.4 Reorganizing: What the Conditions Enforce

The two conditions enforce that a functor does not invent or destroy structure:

Composition condition. If, in $\mathcal{C}$, the path $A \xrightarrow{f} B \xrightarrow{g} C$ exists and composes to $h : A \to C$, then in $\mathcal{D}$ the path $F(A) \xrightarrow{F(f)} F(B) \xrightarrow{F(g)} F(C)$ exists and composes to $F(h) = F(g) \circ F(f)$. Every composable triangle in $\mathcal{C}$ maps to a composable triangle in $\mathcal{D}$, and the triangles commute.

Identity condition. Every identity loop $A \xrightarrow{\mathrm{id}_A} A$ in $\mathcal{C}$ maps to the identity loop $F(A) \xrightarrow{\mathrm{id}_{F(A)}} F(A)$ in $\mathcal{D}$. Functors do not turn identity morphisms into non-trivial morphisms.

Together: *a functor sends commutative diagrams to commutative diagrams.*

3.1.5 Simplifying: The Functoriality Conditions

$$F(g \circ f) = F(g) \circ F(f)$$
$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$$

The first equation says composition commutes with F. The second says identity commutes with F. These two equations are the complete definition of a functor (given the object and morphism assignments).

3.1.6 Formally: Functors and Commutative Diagrams

3.1.6.0.1 Functors preserve commutativity

If a diagram in $\mathcal{C}$ commutes, its image under any functor $F : \mathcal{C} \to \mathcal{D}$ commutes in $\mathcal{D}$. This follows directly from the functoriality conditions: every path in the original diagram corresponds to an equal composite, and F maps that equality to an equality of composites in $\mathcal{D}$.

3.1.6.0.2 Functors and isomorphisms

If $f : A \to B$ is an isomorphism in $\mathcal{C}$, then $F(f) : F(A) \to F(B)$ is an isomorphism in $\mathcal{D}$, with inverse $F(f^{-1})$. Proof: $F(f^{-1}) \circ F(f) = F(f^{-1} \circ f) = F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$, and similarly on the other side.

Functors *preserve* isomorphisms. They may also *create* them: two non-isomorphic objects in $\mathcal{C}$ can have isomorphic images in $\mathcal{D}$.

3.2 More Examples of Functors

3.2.1 Informally: Functors Are Everywhere

Once you have the definition, you start seeing functors in many places. Some translate faithfully, preserving all distinctions. Some collapse distinctions. Some ignore most of the source and map everything to a single point. All of these are functors, provided the two conditions hold.

3.2.2 Expanding: Four Functors

Example: The Identity Functor

For any category $\mathcal{C}$, define $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ by:

$$\mathrm{Id}_{\mathcal{C}}(A) = A, \qquad \mathrm{Id}_{\mathcal{C}}(f) = f.$$

Everything goes to itself. Both functoriality conditions hold trivially. This is the "do nothing" functor on $\mathcal{C}$.

Example: A Constant Functor

For any categories $\mathcal{C}$ and $\mathcal{D}$, and any fixed object $D \in \mathcal{D}$, define $\Delta_D : \mathcal{C} \to \mathcal{D}$ by:

$$\Delta_D(A) = D \text{ for all } A, \qquad \Delta_D(f) = \mathrm{id}_D \text{ for all } f.$$

Every object maps to D, every morphism maps to id_D.
Check: $\Delta_D(g \circ f) = \mathrm{id}_D = \mathrm{id}_D \circ \mathrm{id}_D = \Delta_D(g) \circ \Delta_D(f)$. ✓ $\Delta_D(\mathrm{id}_A) = \mathrm{id}_D = \mathrm{id}_{\Delta_D(A)}$. ✓
The constant functor collapses all of $\mathcal{C}$ to a single point in $\mathcal{D}$. It is the most forgetful functor possible.

Example: A Functor from 1 to $\mathcal{C}$

A functor $F : \mathbf{1} \to \mathcal{C}$ (from the one-object category to $\mathcal{C}$) picks out one object in $\mathcal{C}$: whatever $F(*)$ is. The only morphism in $\mathbf{1}$ is id_*, and it must map to $\mathrm{id}_{F(*)}$.
This means: *functors from* $\mathbf{1}$ *are in exact correspondence with objects of* $\mathcal{C}$. The category $\mathbf{1}$ is the "object selector." A functor from $\mathbf{1}$ is a way of pointing at a single object.

Example: A Functor from 2 to $\mathcal{C}$

A functor $F : \mathbf{2} \to \mathcal{C}$ picks out a single morphism in $\mathcal{C}$: the image of $\iota : 0 \to 1$. The two identity morphisms of $\mathbf{2}$ must map to identities, and $F(\iota)$ can be any morphism $F(0) \to F(1)$.
Functors from $\mathbf{2}$ *are in exact correspondence with morphisms of* $\mathcal{C}$. The category $\mathbf{2}$ is the "morphism selector." This is why $\mathbf{2}$ is called the "walking morphism": it is the minimal category containing a non-trivial morphism, and any morphism in any category can be captured by a functor from it.

3.2.3 Formalizing: Forgetful and Inclusion Functors

Some functors appear so often they deserve names.

A **forgetful functor** is one that "forgets" some of the structure of its source. We have not yet built enough structure to give a formal example, but the constant functor Δ_D is an extreme case: it forgets everything. The temperature-to-size functor G from the previous section is a mild forgetfulness: it forgets which of 0řC and 20řC you started at.

An **inclusion functor** is one that embeds a subcategory into a larger category without changing any object or morphism. For example, any category $\mathcal{C}$ that contains the recipe category as a subcategory has an inclusion functor $\mathrm{Recipe} \hookrightarrow \mathcal{C}$ that is the identity on all objects and morphisms of Recipe.

3.2.4 Reorganizing: What Functors See and Ignore

Functor	What it sees	What it ignores
$\mathrm{Id}_{\mathcal{C}}$	everything	nothing
Δ_D	nothing	everything
Temperature $\to$ Size	ordering	which of two temps is cooler
$\mathbf{1} \to \mathcal{C}$	one object	all other objects and morphisms
$\mathbf{2} \to \mathcal{C}$	one morphism	all other morphisms

Every functor lies on a spectrum from "sees everything" to "sees nothing." The two conditions (preserve composition, preserve identity) constrain what it can ignore, but leave a great deal of freedom.

3.2.5 Simplifying: Functors as Translations

A functor $F : \mathcal{C} \to \mathcal{D}$ is a translation of $\mathcal{C}$- language into $\mathcal{D}$-language. The two conditions guarantee that every sentence that is true in $\mathcal{C}$ (every commutative diagram) translates to a true sentence in $\mathcal{D}$. The translation may collapse distinctions but cannot introduce falsehoods.

3.2.6 Formally: Faithful, Full, and Essentially Surjective

3.2.6.0.1 Faithfulness and fullness

A functor $F : \mathcal{C} \to \mathcal{D}$ is:

- **faithful** if for all $A, B \in \mathcal{C}$, the function $F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$ is injective (no two distinct morphisms get identified).
- **full** if that function is surjective (every morphism between the images is hit).
- **fully faithful** if it is both.

A faithful functor "doesn't identify" morphisms; a full one "doesn't miss" morphisms between images. The identity functor is fully faithful. The constant functor is neither (if $\mathcal{C}$ has more than one morphism).

3.2.6.0.2 Essentially surjective

F is **essentially surjective** if for every $D \in \mathcal{D}$, there exists $C \in \mathcal{C}$ with $F(C) \cong D$. Combined with full faithfulness, this gives an **equivalence of categories** — the categorical substitute for isomorphism between categories, which we will develop in a later chapter.

3.3 Composition of Functors

3.3.1 Informally: Translating Twice

If F translates from $\mathcal{C}$ to $\mathcal{D}$, and G translates from $\mathcal{D}$ to $\mathcal{E}$, then following F then G gives a translation directly from $\mathcal{C}$ to $\mathcal{E}$. This composite translation is a functor, called the **composite functor** $G \circ F$.

3.3.2 Expanding: Composing the Examples

We have $F : \text{Recipe} \to \text{Approval}$ (the status functor from §3.1) and suppose $H : \text{Approval} \to \mathcal{S}$ maps each approval stage to a project phase (Request $\mapsto$ Planning, Approved $\mapsto$ Building, Final $\mapsto$ Done).

The composite $H \circ F : \text{Recipe} \to \mathcal{S}$ maps:

$$\text{Chop} \mapsto \text{Planning}, \quad \text{Sauté} \mapsto \text{Building}, \quad \text{Add} \mapsto \text{Done}.$$

Internal Dialog

I check that $H \circ F$ is a functor:

- $(H \circ F)(g \circ f) = H(F(g \circ f)) = H(F(g) \circ F(f)) = H(F(g)) \circ H(F(f)) = (H \circ F)(g) \circ (H \circ F)(f)$. ✓
- $(H \circ F)(\mathrm{id}_A) = H(F(\mathrm{id}_A)) = H(\mathrm{id}_{F(A)}) = \mathrm{id}_{H(F(A))} = \mathrm{id}_{(H \circ F)(A)}$. ✓

Each step uses the functoriality of F (first) and G (second). The argument works for any two composable functors.

3.3.3 Formalizing: The Composite Functor

Given functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{E}$, their **composite** $G \circ F : \mathcal{C} \to \mathcal{E}$ is defined by:

$$(G \circ F)(A) = G(F(A)), \qquad (G \circ F)(f) = G(F(f)).$$

$G \circ F$ is a functor. Functoriality for composition:

$$(G \circ F)(g \circ f) = G(F(g \circ f)) = G(F(g) \circ F(f)) = G(F(g)) \circ G(F(f)) = (G \circ F)(g) \circ (G \circ F)(f).$$

Functoriality for identity: $(G \circ F)(\mathrm{id}_A) = G(F(\mathrm{id}_A)) = G(\mathrm{id}_{F(A)}) = \mathrm{id}_{G(F(A))} = \mathrm{id}_{(G\circ F)(A)}$.

Composition of functors is associative. For $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{E}$, $H : \mathcal{E} \to \mathcal{F}$:

$$(H \circ G) \circ F = H \circ (G \circ F).$$

This holds because it holds pointwise on objects and morphisms (function composition is associative).

Identity functors are neutral. $\mathrm{Id}_{\mathcal{D}} \circ F = F = F \circ \mathrm{Id}_{\mathcal{C}}$.

3.3.4 Reorganizing: Functors Are Morphisms

We now have:

- **Objects**: categories $\mathcal{C}$, $\mathcal{D}$, ...
- **Morphisms**: functors $F : \mathcal{C} \to \mathcal{D}$
- **Composition**: $G \circ F : \mathcal{C} \to \mathcal{E}$, associative
- **Identity**: $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$, neutral

This matches the pattern for a category exactly. Categories and functors form a category.

3.3.5 Simplifying: Functor Composition

$$(G \circ F)(A) = G(F(A)), \qquad (G \circ F)(f) = G(F(f))$$
$$(H \circ G) \circ F = H \circ (G \circ F), \qquad \mathrm{Id}_{\mathcal{D}} \circ F = F = F \circ \mathrm{Id}_{\mathcal{C}}$$

These are the same associativity and unit equations as for morphisms in any category — because functors *are* morphisms, in the category of categories.

3.3.6 Formally: The Category Cat

3.3.6.0.1 Definition

The **category of categories** **Cat** has:

- Objects: (small) categories.
- Morphisms: functors between them.
- Composition: composite functor $G \circ F$.
- Identity: identity functor $\mathrm{Id}_{\mathcal{C}}$.

The axioms are verified above. **Cat** is a category.

3.3.6.0.2 Size remark

To avoid set-theoretic paradoxes (a "category of all categories" including itself), **Cat** is typically defined to have *small* categories as objects. The category of all locally small categories is sometimes called **CAT**. We will not belabor this point, but the reader should be aware that size considerations are lurking.

3.3.6.0.3 Endofunctors

A functor $F : \mathcal{C} \to \mathcal{C}$ from a category to itself is called an **endofunctor**. Endofunctors can be composed repeatedly: $F^n = F \circ F \circ \cdots \circ F$ (n times). Endofunctors on a category $\mathcal{C}$ form a category $[\mathcal{C}, \mathcal{C}]$ (a **functor category**, to be developed in Chapter 4).

3.4 Contravariant Functors

3.4.1 Informally: Going Backwards

Some translations between categories send morphisms in the *opposite* direction. An arrow $f : A \to B$ in the source gets sent to an arrow $F(f) : F(B) \to F(A)$ in the target — the direction is reversed.

Such a translation still has to respect composition and identity, but the composition rule looks like a reversal. This is a **contravariant functor**.

3.4.2 Expanding: The Reversal Pattern

Example: An Opposite Ordering

Consider the temperature ordering $\mathcal{T}$ (colder to warmer) and the size ordering $\mathcal{S}^{\text{op}}$ (larger to smaller: Large $\to$ Medium $\to$ Small).
Define $F : \mathcal{T} \to \mathcal{S}^{\text{op}}$ by: colder things are *larger* in $\mathcal{S}^{\text{op}}$:

$$F(0řC) = \text{Large}, \quad F(20řC) = \text{Medium}, \quad F(37řC) = \text{Small}, \quad F(100řC) = \text{Small}.$$

A morphism $0řC \to 37řC$ in $\mathcal{T}$ (colder to warmer) maps to Large $\to$ Small in $\mathcal{S}^{\text{op}}$ (larger to smaller). The direction of the ordering is preserved in the opposite sense.
This is a covariant functor $\mathcal{T} \to \mathcal{S}^{\text{op}}$, or equivalently a contravariant functor $\mathcal{T} \to \mathcal{S}$.

Internal Dialog

I see a map F that sends morphisms "backwards": $f : A \to B$ maps to $F(f) : F(B) \to F(A)$. I ask:
- Does $F(g \circ f) = F(f) \circ F(g)$? (Note the reversed order.)
- Does $F(\text{id}_A) = \text{id}_{F(A)}$?

If yes to both, F is a contravariant functor. The reversal in composition is not a mistake — it is forced by the direction flip.

3.4.3 Formalizing: Contravariance as a Functor from the Opposite

A **contravariant functor** $F : \mathcal{C} \to \mathcal{D}$ is exactly a (covariant) functor $F : \mathcal{C}^{\text{op}} \to \mathcal{D}$.

On objects: F assigns an object $F(A) \in \mathcal{D}$ to each $A \in \mathcal{C}$. On morphisms: for $f : A \to B$ in $\mathcal{C}$, F assigns $F(f) : F(B) \to F(A)$ in $\mathcal{D}$.

Functoriality:

$$F(g \circ f) = F(f) \circ F(g) \qquad \text{(note reversed order)}$$
$$F(\text{id}_A) = \text{id}_{F(A)}$$

This is the covariant functoriality applied to $\mathcal{C}^{\text{op}}$: in $\mathcal{C}^{\text{op}}$, composition of f^{op} and g^{op} gives $(f \circ g)^{\text{op}}$, and F applied to $(f \circ g)^{\text{op}}$ gives $F(f \circ g) = F(g) \circ F(f)$... wait, applying the covariant rule to $\mathcal{C}^{\text{op}}$ correctly gives $F(f^{\text{op}} \circ_{\text{op}} g^{\text{op}}) = F(f^{\text{op}}) \circ F(g^{\text{op}})$, which translates back to $F(g \circ f) = F(f) \circ F(g)$ in $\mathcal{D}$.

3.4.4 Reorganizing: Contravariance = Covariance on the Opposite

We do not need a new concept for contravariant functors. Every contravariant functor $\mathcal{C} \to \mathcal{D}$ is a covariant functor $\mathcal{C}^{\text{op}} \to \mathcal{D}$. The opposite category absorbs the reversal.

This is the first example of the *duality principle* in practice: instead of developing a parallel theory for "backwards" maps, we build the reversal into the objects (via $\mathcal{C}^{\text{op}}$) and reuse the existing theory.

3.4.5 Simplifying: Contravariance in One Line

A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor:

$$F : \mathcal{C}^{\text{op}} \longrightarrow \mathcal{D}.$$

3.4.6 Formally: Bifunctors

3.4.6.0.1 Bifunctors

A **bifunctor** is a functor of two arguments: $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$, where $\mathcal{C} \times \mathcal{D}$ is the **product category** whose objects are pairs (C, D) and whose morphisms are pairs (f, g). A mixed-variance bifunctor has the form $F : \mathcal{C}^{\text{op}} \times \mathcal{D} \to \mathcal{E}$: contravariant in the first argument, covariant in the second. The hom-functor $\text{Hom}(-, -)$ (to be studied in Chapter 5) is the canonical example.

Exercises

What is a functor, in one sentence?	A structure-preserving map between categories.
What two pieces of data does a functor consist of?	An action on objects, and an action on morphisms.
What two laws does a functor obey?	It preserves identities, and it preserves composition.

Write the identity law for a functor F.	$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.
Write the composition law.	$F(g \circ f) = F(g) \circ F(f)$.
What goes wrong if a "functor" fails to preserve identities?	It fails to be a functor. There is no rescue.
Is every function on objects extendable to a functor?	No. You must also have a well-defined action on morphisms that respects the laws.
What is the identity functor $\mathrm{Id}_\mathcal{C}$?	The functor that sends every object and every morphism to itself.
What is a constant functor?	One that sends every object to a fixed X and every morphism to id_X.
Is a constant functor really a functor?	Yes. Check: it sends id_A to id_X and $g \circ f$ to $\mathrm{id}_X \circ \mathrm{id}_X = \mathrm{id}_X$.
What does a functor out of **1** pick out?	A single object in the target category.
What does a functor out of **2** pick out?	A single morphism (with chosen source and target) in the target category.
What does a functor out of a discrete category on S pick out?	A choice of one object for each element of S.
What is a faithful functor?	One that is injective on each hom-set.
What is a full functor?	One that is surjective on each hom-set.
What is a fully faithful functor?	One that is bijective on each hom-set — both full and faithful.
Can a fully faithful functor be non-injective on objects?	Yes. Equivalences allow that.
What is a contravariant functor?	One that reverses the direction of morphisms: $F(g \circ f) = F(f) \circ F(g)$.
How can we recast a contravariant functor as a covariant one?	As a (covariant) functor out of $\mathcal{C}^{\mathrm{op}}$.
What is the opposite functor F^{op}?	The same functor regarded as $F^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$.
If F is a functor and f is an isomorphism, is $F(f)$ an isomorphism?	Yes. Apply F to both $g \circ f = \mathrm{id}$ and $f \circ g = \mathrm{id}$.
What is the formula for $F(f^{-1})$?	$F(f^{-1}) = F(f)^{-1}$. Functors preserve inverses.
Why do functors preserve isomorphisms?	Because they preserve identities and composition — and isomorphism is defined using both.
What is the category **Cat**?	The category whose objects are (small) categories and whose morphisms are functors.
In **Cat**, what is composition of functors?	Sequential application: $(G \circ F)(X) = G(F(X))$, and similarly on morphisms.
What is the identity morphism on $\mathcal{C}$ in **Cat**?	The identity functor $\mathrm{Id}_\mathcal{C}$.
Why must **Cat** restrict to small categories?	To avoid set-theoretic foundation problems with collecting all categories.
What is a bifunctor?	A functor of two arguments, often $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$.
If F is faithful, can F send distinct objects to the same image?	Yes. Faithfulness is about morphisms, not objects.
Why is the functor-axiom $F(g \circ f) = F(g) \circ F(f)$ the heart of the definition?	It is what makes diagrams in $\mathcal{C}$ become diagrams in $\mathcal{D}$, preserving commutativity.

Chapter Summary

Chapter Summary

A **functor** $F : \mathcal{C} \to \mathcal{D}$ assigns objects to objects and morphisms to morphisms, satisfying:

$$F(g \circ f) = F(g) \circ F(f), \qquad F(\mathrm{id}_A) = \mathrm{id}_{F(A)}.$$

Functors **preserve commutativity** of diagrams and **preserve isomorphisms**.
Composition of functors $(G \circ F)(A) = G(F(A))$ is associative and has the identity functor as neutral element.
Categories and functors form a category, called **Cat**.
A **contravariant functor** $\mathcal{C} \to \mathcal{D}$ is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.
Special functors: identity ($\mathrm{Id}_{\mathcal{C}}$), constant ($\Delta_D$), faithful (injective on hom-sets), full (surjective on hom-sets).
Functors from **1** pick out objects; functors from **2** pick out morphisms.

Formal Restatement

Formal Restatement

Functor. $F : \mathcal{C} \to \mathcal{D}$ assigns $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ and $F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$, satisfying:

$$F(g \circ f) = F(g) \circ F(f), \qquad F(\mathrm{id}_A) = \mathrm{id}_{F(A)}.$$

Composition. $(G \circ F)(A) = G(F(A))$, $(G \circ F)(f) = G(F(f))$; associative; $\mathrm{Id}_{\mathcal{D}} \circ F = F = F \circ \mathrm{Id}_{\mathcal{C}}$.

Properties. F is *faithful* iff injective on each hom-set; *full* iff surjective on each hom-set; *essentially surjective* iff every $D \in \mathcal{D}$ satisfies $D \cong F(C)$ for some C.

Contravariant. A contravariant functor $\mathcal{C} \to \mathcal{D}$ is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$; sends $f : A \to B$ to $F(f) : F(B) \to F(A)$; satisfies $F(g \circ f) = F(f) \circ F(g)$.

Cat. The category of small categories and functors: objects are small categories, morphisms are functors, composition and identity as above.

Chapter 4 introduces natural transformations: morphisms between functors, which complete the three-level hierarchy of category theory.

Chapter 4

Natural Transformations

Chapter Overview

Chapter 3 built one layer of structure on top of categories: between any two categories there are functors, and these functors themselves form a category.

This chapter adds a third layer. Between any two functors there are *natural transformations* — morphisms between morphisms. A natural transformation is a coordinated way of moving from one functor to another, where "coordinated" has a very precise meaning: a single commutative square must hold for every morphism in the source category.

By the end of this chapter we will have built a three-level hierarchy: objects, morphisms, transformations between morphisms. Together with the functor categories that hold them, these three levels are the structural backbone of every result in the rest of the book.

Where the Name Comes From

Natural transformation was the *original motivation* for category theory. Eilenberg and Mac Lane (1945) had a clear sense of what "natural" meant in algebra — a construction is natural if it doesn't depend on arbitrary choices (no coordinates, no basis, etc.) — but no formal definition. To make "natural" precise, they needed "a map between maps that commutes with everything else," which led to defining functors, which led to defining the things functors go between (categories), which led to category theory itself.

The word *natural* echoes the long-standing mathematical usage: "a natural transformation" is one that arises "in nature" (= without arbitrary choices). The naturality square (commutative square in the definition) is the formalization: the transformation commutes with *every* morphism in the source category.

4.1 What a Natural Transformation Is

4.1.1 Informally: A Coordinated Family of Morphisms

Suppose we have two functors F and G, both going from category $\mathcal{C}$ to category $\mathcal{D}$. They send each object $A \in \mathcal{C}$ to a (possibly different) object in $\mathcal{D}$. A natural transformation from F to G is a recipe that, at every object A of $\mathcal{C}$, gives a morphism in $\mathcal{D}$ from $F(A)$ to $G(A)$.

But these morphisms cannot be chosen independently. They must be *coordinated* so that they agree with how F and G act on morphisms. That coordination requirement is the entire content of the definition.

4.1.2 Expanding: Two Functors, a Family of Bridges

Example: Two Recipes, One Translation

Suppose the recipe category $\mathcal{R}$ (Chop $\to$ Sauté $\to$ Add) maps to the approval chain $\mathcal{A}$ (Request $\to$ Approved $\to$ Final) in two different ways:

- Functor F: every step maps to Request.
- Functor G: Chop $\mapsto$ Request, Sauté $\mapsto$ Approved, Add $\mapsto$ Final.

A natural transformation $\alpha : F \Rightarrow G$ supplies, for each recipe step A, a morphism in $\mathcal{A}$ from $F(A)$ to $G(A)$:

$$\begin{aligned} \alpha_{\text{Chop}} &: \text{Request} \to \text{Request} \\ \alpha_{\text{Sauté}} &: \text{Request} \to \text{Approved} \\ \alpha_{\text{Add}} &: \text{Request} \to \text{Final} \end{aligned}$$

In a thin category like $\mathcal{A}$, each of these is the unique morphism between its endpoints. The natural transformation α provides "advancement arrows" that move from F's constant Request view to G's finer-grained view, one step at a time.

Internal Dialog

When I encounter $\alpha : F \Rightarrow G$, I picture two parallel functor arrows from $\mathcal{C}$ to $\mathcal{D}$ and a sheaf of small morphisms in $\mathcal{D}$ connecting their object-by-object outputs:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & \mathcal{D} \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$$

The double arrow $\Rightarrow$ is the visual signature of a natural transformation. It lives "between" two functors.

4.1.3 Formalizing: The Naturality Condition

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A **natural transformation** $\alpha : F \Rightarrow G$ consists of:

- For each object $A \in \mathrm{Ob}(\mathcal{C})$, a morphism $\alpha_A : F(A) \to G(A)$ in $\mathcal{D}$, called the **component of α at A**.

Subject to the **naturality condition**: for every morphism $f : A \to B$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "\alpha_A"] \arrow[d, "F(f)"'] & G(A) \arrow[d, "G(f)"] \\ F(B) \arrow[r, "\alpha_B"'] & G(B) \end{tikzcd}$$

That is:

$$G(f) \circ \alpha_A \;=\; \alpha_B \circ F(f).$$

This square is called the **naturality square** at f.

How to Say This

- $\alpha : F \Rightarrow G$ is read "alpha, a natural transformation from F to G" or "F goes to G via α." The double arrow is often called *double-arrow*, *nat-arrow*, or *2-arrow*.
- α_A is "the component of α at A," or "alpha sub-A," or simply "alpha at A."
- Some authors write $\alpha : F \to G$ when the context is clearly functors. Either notation is standard.
- The naturality square "commutes" or, less formally, "is natural in A" (when we want to emphasize that this commutativity must hold as A varies).

4.1.4 Reorganizing: Verifying Naturality on Our Example

Return to $\alpha : F \Rightarrow G$ from the recipe example. For naturality to hold, every morphism $f : A \to B$ in $\mathcal{R}$ must produce a commutative square. Take $f : \text{Chop} \to \text{Sauté}$. The square is:

$$\begin{tikzcd} F(\text{Chop}) = \text{Request} \arrow[r, "\alpha_{\text{Chop}}"] \arrow[d, "F(f)=\mathrm{id}_{\text{Request}}"'] & G(\text{Chop}) = \text{Request} \arrow[d, "G(f)"] \\ F(\text{Sauté}) = \text{Request} \arrow[r, "\alpha_{\text{Sauté}}"'] & G(\text{Sauté}) = \text{Approved} \end{tikzcd}$$

Both paths from Request (top-left) to Approved (bottom-right) compose to the unique morphism between those two objects in $\mathcal{A}$. The square commutes because $\mathcal{A}$ is thin — there is only one possible composite, so there is nothing to fail. The other naturality squares (for g, $g \circ f$, and the identities) commute for the same reason.

The natural transformation α is valid.

4.1.5 Simplifying: Naturality as One Equation

For every $f : A \to B$ in $\mathcal{C}$:

$$G(f) \circ \alpha_A \;=\; \alpha_B \circ F(f).$$

That equation, quantified over all f, is the entire definition of naturality. Everything else — the components, the diagrams, the names — is bookkeeping around this single equation.

4.1.6 Formally: Notation and Conventions

4.1.6.0.1 Notational summary

For $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$:

$$\begin{aligned} \alpha_A &: F(A) \to G(A) && \text{(component at } A\text{)} \\ G(f) \circ \alpha_A &= \alpha_B \circ F(f) && \text{(naturality at } f : A \to B\text{)} \end{aligned}$$

4.1.6.0.2 Identity natural transformation

For any functor $F : \mathcal{C} \to \mathcal{D}$, there is an identity natural transformation $\mathrm{id}_F : F \Rightarrow F$ whose components are the identities of $\mathcal{D}$:

$$(\mathrm{id}_F)_A \;=\; \mathrm{id}_{F(A)}.$$

Naturality is automatic: $F(f) \circ \mathrm{id}_{F(A)} = F(f) = \mathrm{id}_{F(B)} \circ F(f)$ by the unit laws.

4.1.6.0.3 Components determine the transformation

A natural transformation is fully specified by its components together with the verification of naturality. Two natural transformations $\alpha, \beta : F \Rightarrow G$ are equal iff $\alpha_A = \beta_A$ for every A. There is no extra data beyond the components.

4.2 Vertical Composition

4.2.1 Informally: Stacking Transformations

If $\alpha : F \Rightarrow G$ takes us from F to G, and $\beta : G \Rightarrow H$ takes us from G to H, then doing α followed by β should take us from F to H. This composed transformation is denoted $\beta \cdot \alpha$ (or sometimes just $\beta\alpha$) and is called the **vertical composition**.

It is called *vertical* because in the standard diagram, we picture the two transformations stacked top-to-bottom, with the composed transformation running the full height.

4.2.2 Expanding: Stacking Components

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F"{name=F}] \arrow[r, "G"{description, name=G}] \arrow[r, bend right=60, "H"'{name=H}] & \mathcal{D} \arrow[from=F, to=G, Rightarrow, shorten=2pt, "\alpha"] \arrow[from=G, to=H, Rightarrow, shorten=2pt, "\beta"] \end{tikzcd}$$

At each object A of $\mathcal{C}$, we have

$$\begin{aligned} \alpha_A &: F(A) \to G(A), \\ \beta_A &: G(A) \to H(A). \end{aligned}$$

These two morphisms compose in $\mathcal{D}$:

$$(\beta \cdot \alpha)_A \;\coloneqq\; \beta_A \circ \alpha_A \;:\; F(A) \to H(A).$$

The component of the composed transformation at A is the composite of the two original components at A.

4.2.3 Formalizing: Naturality of the Composite

We must check that $\beta \cdot \alpha$ is itself a natural transformation — i.e., that its components $(\beta \cdot \alpha)_A$ satisfy the naturality condition. For any $f : A \to B$ in $\mathcal{C}$:

$$\begin{aligned} H(f) \circ (\beta \cdot \alpha)_A &= H(f) \circ \beta_A \circ \alpha_A && \text{(definition of composite)} \\ &= \beta_B \circ G(f) \circ \alpha_A && \text{(naturality of } \beta \text{ at } f\text{)} \\ &= \beta_B \circ \alpha_B \circ F(f) && \text{(naturality of } \alpha \text{ at } f\text{)} \\ &= (\beta \cdot \alpha)_B \circ F(f) && \text{(definition of composite).} \end{aligned}$$

The two outer expressions equal, so the naturality square for $\beta \cdot \alpha$ at f commutes. $\beta \cdot \alpha$ is a natural transformation.

How to Say This

- $\beta \cdot \alpha$ is "beta vertical alpha" or "beta after alpha" or simply "beta composed with alpha."
- The dot $\cdot$ is the conventional symbol for vertical composition. Some authors drop it entirely and write $\beta\alpha$.
- Order: $\beta \cdot \alpha$ does α first, then β — right to left, matching morphism composition.

4.2.4 Reorganizing: Associativity and Units

Vertical composition is associative: for $\alpha : F \Rightarrow G$, $\beta : G \Rightarrow H$, $\gamma : H \Rightarrow K$:

$$(\gamma \cdot \beta) \cdot \alpha \;=\; \gamma \cdot (\beta \cdot \alpha).$$

This follows componentwise from the associativity of composition in $\mathcal{D}$:

$$\begin{aligned}
((\gamma \cdot \beta) \cdot \alpha)_A &= (\gamma \cdot \beta)_A \circ \alpha_A \\
&= (\gamma_A \circ \beta_A) \circ \alpha_A \\
&= \gamma_A \circ (\beta_A \circ \alpha_A) && \text{(associativity in } \mathcal{D}) \\
&= \gamma_A \circ (\beta \cdot \alpha)_A \\
&= (\gamma \cdot (\beta \cdot \alpha))_A.
\end{aligned}$$

Identity transformations are units for vertical composition:

$$\alpha \cdot \mathrm{id}_F \;=\; \alpha \;=\; \mathrm{id}_G \cdot \alpha \qquad \text{for } \alpha : F \Rightarrow G.$$

4.2.5 Simplifying: A Familiar Pattern

We have:

- Objects: functors $F, G, H, \ldots$ all $\mathcal{C} \to \mathcal{D}$.
- Morphisms: natural transformations $\alpha : F \Rightarrow G$.
- Composition: vertical composition $\beta \cdot \alpha$, associative.
- Identity: $\mathrm{id}_F : F \Rightarrow F$, neutral.

The pattern is exactly that of a category. We name the resulting category in the next section.

4.2.6 Formally: Componentwise Verification Pattern

4.2.6.0.1 The componentwise pattern

Most facts about natural transformations are proved by "checking componentwise": reduce an equation between natural transformations to an equation between their components at each object, then verify the latter using the rules of the target category $\mathcal{D}$. Naturality of the candidate composite or identity is then verified separately.

The proof above of associativity is a paradigmatic example. The reader should expect to recognize this pattern often.

Where the Name Comes From

The **functor category** $[\mathcal{C}, \mathcal{D}]$ (also written $\mathcal{D}^{\mathcal{C}}$ or $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$) is a fundamental construction: a category whose objects are functors and morphisms are natural transformations. The exponential notation $\mathcal{D}^{\mathcal{C}}$ generalizes set-theoretic exponentiation; the bracket $[\mathcal{C}, \mathcal{D}]$ is from enriched-category literature. Functor categories are the testbed for many higher-categorical constructions (Vol II Ch. 22: 2-category structure; Vol IV Ch. 43: quasi-categories).

4.3 Functor Categories

4.3.1 Informally: Functors as Objects

Take any two categories $\mathcal{C}$ and $\mathcal{D}$. The functors from $\mathcal{C}$ to $\mathcal{D}$ are objects in a new category, whose morphisms are the natural transformations between them. This new category is called the **functor category** from $\mathcal{C}$ to $\mathcal{D}$.

4.3.2 Expanding: Putting the Pieces Together

We have already collected the ingredients in the previous section. We name them once more, now framed as a category:

- **Objects**: functors $F : \mathcal{C} \to \mathcal{D}$.
- **Morphisms**: natural transformations $\alpha : F \Rightarrow G$ between such functors.
- **Composition**: vertical composition $\beta \cdot \alpha$.
- **Identity**: identity natural transformation id_F.

The axioms have been verified:

$$\begin{aligned}
(\gamma \cdot \beta) \cdot \alpha &= \gamma \cdot (\beta \cdot \alpha) && \text{(associativity)} \\
\alpha \cdot \mathrm{id}_F &= \alpha = \mathrm{id}_G \cdot \alpha && \text{(unit laws).}
\end{aligned}$$

This is a category.

4.3.3 Formalizing: Notation

The functor category from $\mathcal{C}$ to $\mathcal{D}$ is denoted in several equivalent ways:

$$[\mathcal{C}, \mathcal{D}] \;=\; \mathcal{D}^{\mathcal{C}} \;=\; \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

How to Say This

- $[\mathcal{C}, \mathcal{D}]$ — "the functor category from $\mathcal{C}$ to $\mathcal{D}$," or "brackets $\mathcal{C}$, $\mathcal{D}$." Common in modern texts.
- $\mathcal{D}^{\mathcal{C}}$ — "$\mathcal{D}$ to the $\mathcal{C}$" or "$\mathcal{D}$ exponential $\mathcal{C}$." This notation suggests an analogy with exponentiation that becomes precise once we have products.
- $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ — "Fun of $\mathcal{C}$, $\mathcal{D}$" or simply "the functors from $\mathcal{C}$ to $\mathcal{D}$."

This book uses $[\mathcal{C}, \mathcal{D}]$ by default and falls back to $\mathcal{D}^{\mathcal{C}}$ when the exponential analogy is the point.

4.3.4 Reorganizing: A Familiar Example

The category $[\mathbf{1}, \mathcal{C}]$ has:

- Objects: functors from $\mathbf{1}$ to $\mathcal{C}$, which (we showed in §3.2) correspond exactly to objects of $\mathcal{C}$.
- Morphisms: natural transformations between such functors. A natural transformation between two such functors F and G, picking out objects $A = F(*)$ and $B = G(*)$, has one component $\alpha_* : A \to B$. Naturality at id_* is automatic.

So $[\mathbf{1}, \mathcal{C}]$ is essentially the same as $\mathcal{C}$ itself — objects are objects of $\mathcal{C}$, morphisms are morphisms of $\mathcal{C}$. This is our first "equivalence of categories," though we have not yet defined that term precisely. It will become precise in a later chapter.

4.3.5 Simplifying: The Exponential Notation

$$\mathcal{D}^{\mathcal{C}} \;=\; \text{the category whose objects are functors } \mathcal{C} \to \mathcal{D}.$$

4.3.6 Formally: Functor Categories Are Categories

4.3.6.0.1 Existence

For any (small) $\mathcal{C}$ and any $\mathcal{D}$, the functor category $[\mathcal{C}, \mathcal{D}]$ exists and is itself a category.

4.3.6.0.2 Size considerations

If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, then $[\mathcal{C}, \mathcal{D}]$ is locally small. If both are small, $[\mathcal{C}, \mathcal{D}]$ is small.

4.3.6.0.3 Why this matters

Functor categories are the natural setting for many results that follow. The Yoneda lemma, presheaves, sheaves, and most universal constructions either live inside a functor category or are characterized in terms of one.

4.4 Natural Isomorphism

4.4.1 Informally: When Two Functors Are "the Same"

In Chapter 2 we said two objects A and B of a category are *isomorphic* when there is an invertible morphism between them. The same idea, applied to the functor category $[\mathcal{C}, \mathcal{D}]$, gives a notion of two *functors* being isomorphic. Two functors F and G are *naturally isomorphic* when there is an invertible natural transformation between them.

A natural isomorphism is just an isomorphism in a functor category, no more, no less. But because functors carry so much structure, this is one of the most useful notions in the entire subject.

4.4.2 Expanding: Componentwise Invertibility

A natural transformation $\alpha : F \Rightarrow G$ has an inverse $\alpha^{-1} : G \Rightarrow F$ exactly when there is a natural transformation in the other direction such that the two vertical composites are identity natural transformations:

$$\begin{aligned} \alpha^{-1} \cdot \alpha &= \mathrm{id}_F, \\ \alpha \cdot \alpha^{-1} &= \mathrm{id}_G. \end{aligned}$$

By the componentwise pattern, these equations hold iff for every A:

$$\begin{aligned} (\alpha^{-1})_A \circ \alpha_A &= \mathrm{id}_{F(A)}, \\ \alpha_A \circ (\alpha^{-1})_A &= \mathrm{id}_{G(A)}. \end{aligned}$$

So α is a natural isomorphism iff every component α_A is an isomorphism in $\mathcal{D}$, with $(\alpha^{-1})_A = (\alpha_A)^{-1}$.

4.4.3 Formalizing: Definition and Notation

A natural transformation $\alpha : F \Rightarrow G$ is a **natural isomorphism** if there exists a natural transformation $\alpha^{-1} : G \Rightarrow F$ with $\alpha^{-1} \cdot \alpha = \mathrm{id}_F$ and $\alpha \cdot \alpha^{-1} = \mathrm{id}_G$.

Equivalently: α is a natural isomorphism iff every component $\alpha_A : F(A) \to G(A)$ is an isomorphism in $\mathcal{D}$.

Two functors F and G are **naturally isomorphic**, written $F \cong G$, if there is a natural isomorphism between them.

How to Say This

- $F \cong G$ — "F is naturally isomorphic to G," or "F and G are naturally isomorphic," or (in informal speech) "F and G are the same up to natural isomorphism."
- The double arrow with a tilde, $F \overset{\sim}{\Longrightarrow} G$ or $F \overset{\sim}{\Rightarrow} G$, indicates that the natural transformation α is a natural isomorphism. Reads as "alpha is a natural iso from F to G."
- Some authors say "isomorphic" (without "naturally") when the context is clear that the isomorphism is in a functor category.

4.4.4 Reorganizing: Naturality of the Inverse Is Automatic

It is worth checking explicitly that if α has componentwise inverses $(\alpha_A)^{-1}$ in $\mathcal{D}$, then the family of inverses is itself a natural transformation. We verify the naturality square for the candidate inverse:

$$\begin{aligned} F(f) \circ (\alpha_A)^{-1} &= (\alpha_B)^{-1} \circ \alpha_B \circ F(f) \circ (\alpha_A)^{-1} && \text{(inserting identity)} \\ &= (\alpha_B)^{-1} \circ G(f) \circ \alpha_A \circ (\alpha_A)^{-1} && \text{(naturality of } \alpha\text{)} \\ &= (\alpha_B)^{-1} \circ G(f) && \text{(inverse cancels).} \end{aligned}$$

The outer equality is precisely the naturality square for the inverse at $f : A \to B$. So having componentwise inverses suffices — the naturality of the inverse comes for free.

4.4.5 Simplifying: Three Equivalent Statements

The following are equivalent for $\alpha : F \Rightarrow G$:

(i) α is an isomorphism in $[\mathcal{C}, \mathcal{D}]$.
(ii) there exists α^{-1} with $\alpha^{-1} \cdot \alpha = \mathrm{id}_F$ and $\alpha \cdot \alpha^{-1} = \mathrm{id}_G$.
(iii) every component α_A is an isomorphism in $\mathcal{D}$.

The third is by far the most useful test: to check a natural isomorphism, check that every component is an isomorphism in the target category.

4.4.6 Formally: Natural Isomorphism in Practice

4.4.6.0.1 Natural vs. unnatural isomorphism

Sometimes one can find isomorphisms $F(A) \cong G(A)$ for every object A, but no single choice of isomorphisms that is natural in A. In that case F and G are *not* naturally isomorphic, even though they are "pointwise isomorphic." Naturality forces the family of isomorphisms to cohere with morphisms.

This distinction is one of the historical motivations for the entire subject: Eilenberg and Mac Lane introduced natural transformations precisely to formalize "natural" versus "arbitrary" choices.

4.4.6.0.2 Functor categories have isomorphism

Just as in any category, "naturally isomorphic" is an equivalence relation on the objects of $[\mathcal{C}, \mathcal{D}]$ (the functors). The proof is the same proof as for ordinary isomorphism — it works in any category, and the functor category is no exception.

Exercises

Functors are maps between categories. What is a map between functors?	A natural transformation.
If $F, G : \mathcal{C} \to \mathcal{D}$, what does $\alpha : F \Rightarrow G$ consist of?	A morphism $\alpha_A : F(A) \to G(A)$ in $\mathcal{D}$ for every object A of $\mathcal{C}$.
What is α_A called?	The component of α at A.
What constraint do the components satisfy?	The naturality square: for every $f : A \to B$, $G(f) \circ \alpha_A = \alpha_B \circ F(f)$.
What does the naturality square look like as a commuting diagram?	A square with $F(A), G(A), F(B), G(B)$ at the corners, and $F(f), G(f), \alpha_A, \alpha_B$ on the sides.

Why is it called *natural*?	Because the components are chosen in a way that respects all morphisms, not just one.
What is the identity natural transformation on F?	id_F, with $(\mathrm{id}_F)_A = \mathrm{id}_{F(A)}$.
Does id_F satisfy naturality?	Yes, trivially: both paths around the square reduce to $F(f)$.
What is vertical composition?	Composing $\alpha : F \Rightarrow G$ with $\beta : G \Rightarrow H$ to get $\beta \cdot \alpha : F \Rightarrow H$.
What is the component formula for $\beta \cdot \alpha$?	$(\beta \cdot \alpha)_A = \beta_A \circ \alpha_A$.
Is vertical composition associative?	Yes.
What is the functor category $[\mathcal{C}, \mathcal{D}]$?	The category whose objects are functors $\mathcal{C} \to \mathcal{D}$ and whose morphisms are natural transformations between them.
What is the identity morphism in $[\mathcal{C}, \mathcal{D}]$?	The identity natural transformation id_F.
What are two other notations for $[\mathcal{C}, \mathcal{D}]$?	$\mathcal{D}^{\mathcal{C}}$ and $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$.
When is $\alpha : F \Rightarrow G$ called a natural isomorphism?	When every component α_A is an isomorphism in $\mathcal{D}$.
If every component is an iso, is the inverse natural too?	Yes. The pointwise inverses assemble into a natural transformation.
What does $F \cong G$ mean in $[\mathcal{C}, \mathcal{D}]$?	There exists a natural isomorphism $F \Rightarrow G$.
Can F and G act differently on objects yet be naturally isomorphic?	Yes; their object-values need only be isomorphic, not equal.
Why is naturality the right notion of "map of functors"?	Without it, components could behave incoherently across morphisms; naturality enforces coherence.
What is horizontal composition?	Composing $\alpha : F \Rightarrow G$ in $[\mathcal{C}, \mathcal{D}]$ with $\beta : H \Rightarrow K$ in $[\mathcal{D}, \mathcal{E}]$ to get $\beta * \alpha : H \circ F \Rightarrow K \circ G$.
What is the whiskering βF?	Horizontal composition $\beta * \mathrm{id}_F$: precomposing each component of β with F.
What is the component of βF at A?	$\beta_{F(A)}$.
What is $\alpha_A : F(A) \to G(A)$ best read as?	"α component at A," a single morphism in $\mathcal{D}$.
Is naturality a property or extra data?	A property: the components are the data, and naturality is a condition they must satisfy.
Do natural transformations between functors $\mathcal{C} \to \mathcal{D}$ form a set?	They form a hom-set in $[\mathcal{C}, \mathcal{D}]$.
Why do we need naturality if pointwise isomorphism would do?	Pointwise iso ignores coherence; many "isomorphic" functors fail to be naturally isomorphic.
Where does the symbol $\Rightarrow$ come from?	It distinguishes a 2-cell (between 1-cells) from a 1-cell (between 0-cells).
Is $F \Rightarrow G$ a different thing from $F \to G$?	Conventionally yes: $\to$ is a morphism, $\Rightarrow$ is a natural transformation between functors.
What is the simplest non-trivial natural transformation example?	The identity-to-identity isomorphism, or the inclusion of a constant functor into a non-constant one.
Why did we want functor categories at all?	So that we can apply categorical reasoning *to functors themselves* — the same machinery, one level up.

Chapter Summary

Chapter Summary

A **natural transformation** $\alpha : F \Rightarrow G$ between two parallel functors $F, G : \mathcal{C} \to \mathcal{D}$ consists of a component $\alpha_A : F(A) \to G(A)$ for each object A, such that for every morphism $f : A \to B$ in $\mathcal{C}$, the naturality square commutes:

$$G(f) \circ \alpha_A \;=\; \alpha_B \circ F(f).$$

Vertical composition: $(\beta \cdot \alpha)_A = \beta_A \circ \alpha_A$. Associative; identity natural transformations are units.
Functor categories: $[\mathcal{C}, \mathcal{D}]$ (also written $\mathcal{D}^{\mathcal{C}}$) has functors as objects and natural transformations as morphisms.
Natural isomorphism $F \cong G$: an isomorphism in $[\mathcal{C}, \mathcal{D}]$, equivalently a natural transformation whose every component is an isomorphism in $\mathcal{D}$.

The three levels:

1. objects (live in a category),
2. morphisms (between objects in the same category),
3. natural transformations (between functors with the same source and target categories).

Formal Restatement

Formal Restatement

Natural transformation. For $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\alpha : F \Rightarrow G$ assigns to each $A \in \mathcal{C}$ a morphism $\alpha_A : F(A) \to G(A)$ in $\mathcal{D}$, subject to:

$$G(f) \circ \alpha_A \;=\; \alpha_B \circ F(f) \qquad \text{for all } f : A \to B \text{ in } \mathcal{C}.$$

Identity. $\mathrm{id}_F : F \Rightarrow F$ has components $(\mathrm{id}_F)_A = \mathrm{id}_{F(A)}$.

Vertical composition. For $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$:

$$(\beta \cdot \alpha)_A \;=\; \beta_A \circ \alpha_A.$$

Associative; $\mathrm{id}_G \cdot \alpha = \alpha = \alpha \cdot \mathrm{id}_F$.

Functor category. $[\mathcal{C}, \mathcal{D}] = \mathcal{D}^{\mathcal{C}}$ has functors $\mathcal{C} \to \mathcal{D}$ as objects and natural transformations as morphisms.

Natural isomorphism. $\alpha : F \Rightarrow G$ is a natural isomorphism iff α is invertible in $[\mathcal{C}, \mathcal{D}]$ iff every component α_A is an isomorphism in $\mathcal{D}$. Two functors F and G are naturally isomorphic, $F \cong G$, when such an α exists.

Chapter 5 introduces universal properties, beginning with initial and terminal objects: the first examples of objects characterized by their relationship to everything else.

Chapter 5

Universal Properties

Chapter Overview

Category theory characterizes objects by their relationships to other objects, not by what they are made of inside. The cleanest expression of this principle is the *universal property*: a description of an object that says, in effect, "this object is the unique one (up to unique isomorphism) that participates in such-and-such a way in such-and-such a situation."

This chapter introduces the universal-property pattern through its two simplest cases: *initial objects* (those with a unique map out to every other object) and *terminal objects* (those with a unique map in from every other object). We then state the general pattern, prove that universal objects are unique up to unique isomorphism, and use the pattern to preview *products* and *coproducts* — the first universal constructions involving more than one object at a time.

Universal properties are everywhere from this chapter onward. The investment made here in understanding the pattern pays off repeatedly.

Where the Name Comes From

Universal property captures one of the deepest patterns in mathematics: characterizing an object not by what it *is* but by what it *does* — its universal mapping behavior. The terminology is borrowed from physics and analysis (universal cover, universal constant, etc.) where "universal" means "applies in every case." In category theory, a universal property is an object that "solves a problem uniquely up to unique isomorphism" — it is the universal solution to a mapping question. The pattern was formalized by Bourbaki (mid-20th century) and made into the central technique of category theory by Mac Lane and Eilenberg.

Initial and **terminal** objects are the universal-property basics: initial = unique map *out* to everything; terminal = unique map *in* from everything. "Initial" and "terminal" are from the structure of arrows — first/last in any path.

5.1 Initial Objects

5.1.1 Informally: One Way Out

An **initial object** in a category is an object from which there is exactly one morphism to every other object — including itself. "Exactly one" is the key phrase: not zero, not two, not many. The initial object has a unique outbound connection to every destination.

If you imagine the category as a transit map, an initial object is a station from which there is one and only one route to every station in the city. There may be many such routes from other stations, but starting at the initial object, your route is forced.

5.1.2 Expanding: Looking for Initial Objects

Example: 0 Is Initial in Itself

The empty category $\mathbf{0}$ has no objects. The statement "for every object A, there is exactly one morphism from X to A" is vacuously true for any X, but $\mathbf{0}$ has no objects to play the role of X either. Vacuously, the empty category is its own (trivial) example.

Example: 1 Is Initial in 2 Read Backwards?

Recall **2**: two objects $0, 1$ with one non-identity morphism $\iota : 0 \to 1$. The object 0 has:

$$\mathrm{Hom}(0, 0) = \{\mathrm{id}_0\},$$
$$\mathrm{Hom}(0, 1) = \{\iota\}.$$

Exactly one morphism to each object. So 0 is the initial object of **2**. The object 1, by contrast, has $\mathrm{Hom}(1, 0) = \varnothing$ — no morphism to 0 at all. So 1 is not initial.

Example: The Recipe Category

The recipe category has Chop, Sauté, Add with morphisms Chop → Sauté → Add (and their composite). From Chop:

$$\mathrm{Hom}(\mathrm{Chop}, \mathrm{Chop}) = \{\mathrm{id}_{\mathrm{Chop}}\},$$
$$\mathrm{Hom}(\mathrm{Chop}, \mathrm{Saut\acute{e}}) = \{f\},$$
$$\mathrm{Hom}(\mathrm{Chop}, \mathrm{Add}) = \{g \circ f\}.$$

Exactly one morphism to each. So Chop is initial. It is the unique step from which a unique route reaches every other step.

Example: No Initial Object

Take the indiscrete category on two objects $\{A, B\}$: hom-sets are all singletons including $\mathrm{Hom}(A, B)$, $\mathrm{Hom}(B, A)$, $\mathrm{Hom}(A, A) = \{\mathrm{id}_A\}$, $\mathrm{Hom}(B, B) = \{\mathrm{id}_B\}$. Both A and B have exactly one morphism to each object, so both are initial. We will see in the next section that this forces $A \cong B$, but here we note: *there can be more than one initial object* in a category, but they must be isomorphic.

Internal Dialog

To check whether an object X is initial, I ask, for each object A in the category: how many morphisms $X \to A$ are there?

- If exactly one for every A: X is initial.
- If zero for some A: X is not initial.
- If two or more for some A: X is not initial.

The check is local at X but quantifies over every A in the category. In small categories, this is a finite check. In large categories, it is usually a structural argument.

5.1.3 Formalizing: The Definition

An object $X \in \mathrm{Ob}(\mathcal{C})$ is **initial** if for every object $A \in \mathrm{Ob}(\mathcal{C})$, the hom-set $\mathrm{Hom}(X, A)$ is a singleton:

$$|\mathrm{Hom}_{\mathcal{C}}(X, A)| = 1 \quad \text{for every } A \in \mathrm{Ob}(\mathcal{C}).$$

We write $\mathbf{0}$ for an initial object (when one exists), and $!_A$ or simply ! for the unique morphism $\mathbf{0} \to A$.

How to Say This

- "initial object" is sometimes called the *universal repelling object*, or the *strict initial* in some refined contexts.
- The notation $\mathbf{0}$ (boldface zero) is conventional. Some authors use $\perp$ (read "bottom"), particularly when the category comes from logic.
- The unique morphism is often written $!_A$ ("shriek to A") or $!_{\mathbf{0},A}$. The exclamation point itself is read "shriek" or "bang."
- Drawn as a dashed arrow when emphasizing uniqueness: $\mathbf{0} \dashrightarrow A$.

5.1.4 Reorganizing: The Examples Restated

$\mathbf{0}$ in **2**:	$\lvert\mathrm{Hom}(0, 0)\rvert = 1$, $\lvert\mathrm{Hom}(0, 1)\rvert = 1$. Initial.
Chop in Recipe:	$\lvert\mathrm{Hom}(\mathrm{Chop}, X)\rvert = 1$ for $X \in \{\mathrm{Chop}, \mathrm{Saut\acute{e}}, \mathrm{Add}\}$.
Indiscrete on $\{A, B\}$:	both A and B initial; both isomorphic.
$\mathbf{0}$ (empty category):	vacuously its own (trivial) initial object.

5.1.5 Simplifying: Initial in One Line

$$X \text{ is initial in } \mathcal{C} \iff \forall A \in \mathcal{C},\ \exists! f \in \mathrm{Hom}(X, A).$$

The symbol $\exists!$ is read "there exists a unique."

5.1.6 Formally: Uniqueness Sketch

5.1.6.0.1 Initial objects are unique up to unique isomorphism

Suppose X and Y are both initial in $\mathcal{C}$. There is a unique $f : X \to Y$ (because X is initial), and a unique $g : Y \to X$ (because Y is initial). Consider $g \circ f : X \to X$. There is exactly one morphism $X \to X$ (because X is initial), and id_X is such a morphism. So $g \circ f = \mathrm{id}_X$. By the same argument, $f \circ g = \mathrm{id}_Y$. Hence f is an isomorphism, and uniqueness of f as a morphism gives uniqueness of the isomorphism.

We will state and reuse this argument in its general form in §5.3. Notice that no inspection of what X or Y are "made of" is involved — the entire proof uses only the universal property.

5.2 Terminal Objects

5.2.1 Informally: One Way In

A **terminal object** is the mirror image of an initial object: an object to which there is exactly one morphism from every other object. Instead of "one way out," it is "one way in."

5.2.2 Expanding: A Categorical Mirror

Example: 1 Is Terminal in 2

In $\mathbf{2}$, the object 1 has:

$$\mathrm{Hom}(0, 1) = \{\iota\},$$
$$\mathrm{Hom}(1, 1) = \{\mathrm{id}_1\}.$$

Exactly one morphism into 1 from every object. So 1 is terminal. (The object 0 is not terminal, since $\mathrm{Hom}(1, 0) = \varnothing$.)

Example: Add in the Recipe Category

In the recipe category, Add receives the morphisms $\mathrm{id}_{\mathrm{Add}}$, g, and $g \circ f$. One from each step. So Add is terminal.

Example: An Object That Is Both

In $\mathbf{1}$ (one object, one morphism), the unique object $*$ has exactly one morphism out (to itself) and exactly one morphism in (from itself). It is simultaneously initial and terminal. Such an object is called a **zero object**.

5.2.3 Formalizing: Duality

An object $X \in \mathrm{Ob}(\mathcal{C})$ is **terminal** if for every object $A \in \mathrm{Ob}(\mathcal{C})$, the hom-set $\mathrm{Hom}(A, X)$ is a singleton:

$$|\mathrm{Hom}_{\mathcal{C}}(A, X)| = 1 \quad \text{for every } A \in \mathrm{Ob}(\mathcal{C}).$$

Duality. An object is terminal in $\mathcal{C}$ iff it is initial in $\mathcal{C}^{\mathrm{op}}$. The two notions are dual to each other in the formal sense of §2.4: any statement about initial objects becomes a statement about terminal objects when you replace $\mathcal{C}$ by $\mathcal{C}^{\mathrm{op}}$, and vice versa.

How to Say This

- "terminal object" is also called the *universal attracting object*, or in some texts the *final object*. "Terminal" is most common in modern usage.
- Notation: $\mathbf{1}$ (boldface one) is standard. Some authors use $\top$ (read "top"), especially in logic contexts.
- The unique morphism into a terminal object is also written $!$ or $!_A : A \to \mathbf{1}$.
- A *zero object*, which is both initial and terminal, is sometimes denoted 0, $\mathbf{0}$, or $*$.

5.2.4 Reorganizing: The Same Theorem, Twice

The uniqueness-up-to-unique-isomorphism result we proved for initial objects holds, by duality, for terminal objects: any two terminal objects are isomorphic via a unique isomorphism. This is not a second proof to write; it is the same proof, transported across the opposite-category construction.

This is the first concrete payoff of §2.4. We proved one result and got two theorems. From here on, every result about a universal property comes in two flavors (covariant and contravariant), and we will prove only one of them.

5.2.5 Simplifying: Initial and Terminal

$$X \text{ initial} \iff \forall A,\ \exists!\, f : X \to A.$$
$$X \text{ terminal} \iff \forall A,\ \exists!\, f : A \to X.$$

5.2.6 Formally: Zero Morphisms

5.2.6.0.1 Zero objects and zero morphisms

If $\mathcal{C}$ has a zero object $\mathbf{0}$ (an object that is both initial and terminal), then for any pair A, B there is a distinguished morphism

$$0_{A,B} : A \to B,$$

obtained by composing the unique $A \to \mathbf{0}$ with the unique $\mathbf{0} \to B$. This is the **zero morphism** from A to B. Zero morphisms have the property that any composite involving a zero morphism is again a zero morphism. They behave like the algebraic zero in contexts where one applies.

5.2.6.0.2 Existence is not guaranteed

Many categories have no initial or terminal object at all. The empty category $\mathbf{0}$ has no terminal object (no object exists to receive the unique morphism from elsewhere). The category of nonempty preorders under monotone maps has neither. Existence of universal objects is context-dependent.

5.3 The Universal Property Pattern

5.3.1 Informally: "Unique So That..."

The defining feature of every universal property is a phrase of the form:

> *for every* object X with property P, *there exists a unique* morphism $u : X \to U$ (or $U \to X$) *such that* some diagram commutes.

The object U is the universal one. The phrase "unique so that" is the signature.

Initial and terminal objects fit this template in the simplest possible way: P is trivial ("X is any object"), and the diagram has only one path. More involved universal properties add data and conditions, but the shape "for every ..., there exists a unique ... such that ..." is preserved.

5.3.2 Expanding: A Worked-Out Template

Example: Initial Object as a Universal Property

> An object $\mathbf{0}$ is *initial* if:
> *for every* object A,
> *there exists a unique* morphism $!_A : \mathbf{0} \to A$.

No diagram is needed beyond the morphism itself, because there is no other data to compare against.

Example: Terminal Object as a Universal Property

> An object $\mathbf{1}$ is *terminal* if:
> *for every* object A,
> *there exists a unique* morphism $!_A : A \to \mathbf{1}$.

Example: The Universal Property of a Product (Preview)

For objects A and B in $\mathcal{C}$, a *product* is an object P equipped with two morphisms $\pi_1 : P \to A$ and $\pi_2 : P \to B$ such that:

> *for every* object X and every pair of morphisms $f : X \to A$, $g : X \to B$,
> *there exists a unique* morphism $\langle f, g\rangle : X \to P$ *such that* the diagram
>
> $$\begin{tikzcd} & X \arrow[dl, bend right, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, bend left, "g"] & \\ A & P \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B \end{tikzcd}$$
>
> commutes.

Here the universal property has additional data (π_1, π_2) and a diagram with multiple paths. But the signature phrase is the same.

5.3.3 Formalizing: The Pattern

A **universal property** for a class of objects is a specification of:

1. A category $\mathcal{C}$ in which we look for the universal object.
2. Some **data** D that the universal object must carry.
3. A class of **competitors**: pairs (X, d_X) where X is an object and d_X is the same kind of data attached to X.
4. A **uniqueness requirement**: for every competitor (X, d_X), there is exactly one morphism from X to the universal object (or from the universal object to X) compatible with the data.

The universal object is the "optimal" member of the class of competitors in a precise sense made by the uniqueness requirement.

5.3.4 Reorganizing: Recognizing the Pattern

Whenever you see a definition of the form

> *For every X with ..., there exists a unique $u : X \to U$ such that ... commutes,*

you are looking at a universal property. The next few chapters introduce many such definitions:

- products and coproducts;
- equalizers and coequalizers;
- pullbacks and pushouts;
- limits and colimits;
- the Yoneda embedding;
- adjunctions.

Each is built from the same template. Recognizing the template is half the work of understanding the definition.

5.3.5 Simplifying: The Uniqueness Schema

$$\forall\, X \in \mathcal{P},\ \exists!\, u : X \to U \text{ such that } \Phi(u).$$

Here $\mathcal{P}$ is the class of competitors, U is the universal object, and $\Phi(u)$ is the commutativity condition. Reading this as a sentence in English gives the prose form of any universal property.

5.3.6 Formally: Universal Elements via Representable Functors

5.3.6.0.1 Universal properties as natural isomorphisms

Every universal property can be restated as the assertion that a certain functor $\mathcal{C} \to \mathbf{Set}$ is **representable**: that is, naturally isomorphic to a hom-functor $\mathrm{Hom}_{\mathcal{C}}(U, -)$ or $\mathrm{Hom}_{\mathcal{C}}(-, U)$. We will not develop **Set** or representability here — both come in a later chapter — but it is worth mentioning that the pattern of this section has a clean abstract formulation: a universal property is a representation of a certain functor.

5.4 Uniqueness Up to Unique Isomorphism

5.4.1 Informally: "The" Universal Object

Universal objects are not, in general, literally unique. There can be many initial objects in a category. What *is* unique is the *isomorphism* between any two of them: not just that they are isomorphic, but that there is exactly one isomorphism connecting them, given by the universal property itself.

This is the precise sense in which a universal object is "the" universal object. "The" here is shorthand for "unique up to unique isomorphism."

5.4.2 Expanding: Counting Initial Objects

Example: Two Initial Objects Must Be Uniquely Isomorphic

Recall the indiscrete category on $\{A, B\}$. Both A and B are initial. The unique morphism $A \to B$ (call it α) and the unique morphism $B \to A$ (call it β) must satisfy:

$$\begin{aligned} \beta \circ \alpha &= \mathrm{id}_A && \text{(only one morphism } A \to A\text{, must be the identity)} \\ \alpha \circ \beta &= \mathrm{id}_B && \text{(only one morphism } B \to B\text{, must be the identity).} \end{aligned}$$

So α is an isomorphism with inverse β. The isomorphism is unique because α itself was unique (only one morphism

$A \to B$ exists).

5.4.3 Formalizing: The Theorem

Theorem (Uniqueness of Initial Objects). If X and Y are both initial in $\mathcal{C}$, then there is a unique isomorphism $X \xrightarrow{\sim} Y$.

Proof. Let $f : X \to Y$ and $g : Y \to X$ be the unique morphisms guaranteed by the initiality of X and Y respectively. Compute:

$$g \circ f : X \to X.$$

There is only one morphism $X \to X$ (by initiality of X), and id_X is one. So:

$$g \circ f = \mathrm{id}_X.$$

Similarly:

$$\begin{aligned} f \circ g &: Y \to Y, \\ f \circ g &= \mathrm{id}_Y. \end{aligned}$$

So f is an isomorphism, with inverse g. The isomorphism f is unique among morphisms $X \to Y$ because $\mathrm{Hom}(X, Y)$ has exactly one element. □

By duality (apply this theorem in $\mathcal{C}^{\mathrm{op}}$):

Theorem (Uniqueness of Terminal Objects). If X and Y are both terminal in $\mathcal{C}$, then there is a unique isomorphism $X \xrightarrow{\sim} Y$.

5.4.4 Reorganizing: The General Principle

The argument generalizes. For any universal property, suppose U and U' both satisfy it with the same data. Then:

- The universal property of U' applied to U gives a unique morphism $u : U \to U'$ compatible with the data.
- The universal property of U applied to U' gives a unique morphism $u' : U' \to U$.
- The composite $u' \circ u : U \to U$ is forced by the universal property to be id_U (uniqueness applied to U compared against itself).
- Similarly $u \circ u' = \mathrm{id}_{U'}$.

So u is the unique isomorphism between U and U'. This works for every universal property in this book; the argument is the same in each case.

5.4.5 Simplifying: The Slogan

A universal object is **unique up to unique isomorphism**. Or, in notation:

$$U, U' \text{ both satisfy the same universal property} \implies \exists!\, u : U \xrightarrow{\sim} U'.$$

5.4.6 Formally: Treating Universal Objects as Concrete

5.4.6.0.1 The convention

We routinely speak of "the" initial object, "the" terminal object, "the" product of A and B, and so on, even though strictly speaking there is a whole equivalence class of such objects. The justification is exactly the theorem of this section: the choice of representative within the equivalence class is harmless because any two representatives are related by a unique canonical isomorphism.

5.4.6.0.2 When uniqueness fails

In some weaker contexts (e.g., bicategories, weak limits), the uniqueness of the isomorphism relaxes to "unique up to higher coherent isomorphism." We will encounter such weakened versions in later chapters. For ordinary (1-)category theory, the strict uniqueness above holds without modification.

Where the Name Comes From

Product and **coproduct** take their names from set theory: the Cartesian product $A \times B$ in **Set** is the prototype. The categorical *product* generalizes Cartesian product via the universal mapping property (maps into $A \times B$ correspond to pairs of maps into A and B). The *coproduct* (also called *sum*) is the dual: in **Set** it's the disjoint union; in **Vect** it's the direct sum; in **Top** it's the disjoint union with topology. The "co-" prefix marks duality (cross-ref Vol I Ch. 2's opposite categories).

5.5 Toward Products and Coproducts

5.5.1 Informally: Two Objects Combined

Suppose we have two objects A and B in a category, and we want to talk about a single object that "contains" both of them, in some categorical sense. There are two natural ways to do this:

- Combine them so that you can *extract* either one: this leads to *products*.
- Combine them so that you can *embed* either one: this leads to *coproducts*.

Each combination has a universal property. This section sketches both as previews; the full development comes in Chapter 6.

5.5.2 Expanding: Pairing and Choice

Example: Products as "Both at Once"

Imagine you have a Chop step and a Sauté step. A combined task "Chop and Sauté" should let you extract either the chop part or the sauté part. That is, there should be two "projections" $\pi_1 : \text{Chop} \wedge \text{Sauté} \to \text{Chop}$ and $\pi_2 : \text{Chop} \wedge \text{Sauté} \to \text{Sauté}$. Given any other step X that can independently be sent to both Chop and Sauté, there should be a unique way to send X to the combined task.

Example: Coproducts as "Either-Or"

Conversely, a step "Chop or Sauté" should accept a chop or a sauté input and route it through. So there are two "inclusions" $\iota_1 : \text{Chop} \to \text{Chop} \vee \text{Sauté}$ and $\iota_2 : \text{Sauté} \to \text{Chop} \vee \text{Sauté}$. Given any other step Y that can receive a chop and can receive a sauté, there should be a unique way to receive from the disjoint union.

5.5.3 Formalizing: The Two Universal Properties

A **product** of A and B in $\mathcal{C}$ is an object $A \times B$ together with morphisms $\pi_1 : A \times B \to A$ and $\pi_2 : A \times B \to B$, such that:

> *for every* object X with morphisms $f : X \to A$ and $g : X \to B$,
> *there exists a unique* morphism $\langle f, g\rangle : X \to A \times B$ *such that*
> $\pi_1 \circ \langle f, g\rangle = f$ *and* $\pi_2 \circ \langle f, g\rangle = g$.

$$\begin{tikzcd} & X \arrow[dl, bend right, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, bend left, "g"] & \\ A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B \end{tikzcd}$$

A **coproduct** of A and B is an object $A + B$ (also written $A \sqcup B$) with morphisms $\iota_1 : A \to A + B$ and $\iota_2 : B \to A + B$, such that:

> *for every* object Y with morphisms $f : A \to Y$ and $g : B \to Y$,
> *there exists a unique* morphism $[f, g] : A + B \to Y$ *such that*
> $[f, g] \circ \iota_1 = f$ *and* $[f, g] \circ \iota_2 = g$.

$$\begin{tikzcd} A \arrow[r, "\iota_1"] \arrow[dr, bend right, "f"'] & A + B \arrow[d, dashed, "{[f,g]}"] & B \arrow[l, "\iota_2"'] \arrow[dl, bend left, "g"] \\ & Y & \end{tikzcd}$$

How to Say This

- $A \times B$ — "A cross B," "the product of A and B," or sometimes "A product B." Some authors write $A \wedge B$ (read "A meet B" or "A and B"), especially in lattice-theoretic contexts.
- $A + B$ — "A plus B," "the coproduct of A and B," or "A sum B." The notation $A \sqcup B$ (read "A disjoint union B") is also common; $A \vee B$ ("A join B," "A or B") appears in lattice-theoretic contexts.
- π_1, π_2 — "pi sub one" and "pi sub two," or "first/second projection," or "left/right projection."
- ι_1, ι_2 — "iota sub one," "iota sub two," or "first/second injection" or "left/right inclusion."
- $\langle f, g\rangle$ — "angle f comma g," "the pairing of f and g," or "f paired with g."
- $[f, g]$ — "bracket f comma g," "the cotuple of f and g," or "case analysis on f and g."

5.5.4 Reorganizing: Duality Again

A coproduct in $\mathcal{C}$ is a product in $\mathcal{C}^{\text{op}}$. Every theorem about products has a corresponding theorem about coproducts obtained by reversing arrows. The pairing $\langle f, g\rangle$ in a product corresponds to the cotuple $[f, g]$ in a coproduct. The projections π_i correspond to the inclusions ι_i.

We will use this duality without comment for the rest of the book.

5.5.5 Simplifying: The Product and Coproduct Schema

$$\text{Product:} \quad \forall (X, f, g),\ \exists!\, \langle f, g\rangle : X \to A \times B,\ \pi_i \circ \langle f, g\rangle = (f, g)_i.$$
$$\text{Coproduct:} \quad \forall (Y, f, g),\ \exists!\, [f, g] : A + B \to Y,\ [f, g] \circ \iota_i = (f, g)_i.$$

5.5.6 Formally: Existence and Notation

5.5.6.0.1 When products exist

Not every category has products. The categories from Chapter 1 (recipe, approval chain) generally do not, because there are not enough morphisms to support pairings. We will exhibit categories that *do* have products in Chapter 6.

5.5.6.0.2 Binary, finite, and infinite

The product of two objects is the *binary* product. One can also speak of products of finitely many objects, indexed products $\prod_{i \in I} A_i$, and infinite products. The universal-property template extends uniformly to all of these. The same goes for coproducts.

5.5.6.0.3 Uniqueness up to unique isomorphism, restated

By the theorem of §5.4, any two products (P, π_1, π_2) and (P', π_1', π_2') of the same pair A, B are related by a unique isomorphism $P \xrightarrow{\sim} P'$ commuting with the projections. Likewise for coproducts. We may therefore speak of "the" product and "the" coproduct.

Exercises

What does it mean for an object X to be initial?	There is exactly one morphism from X to every object.
What does it mean for X to be terminal?	There is exactly one morphism from every object to X.
Is the empty category $\mathbf{0}$ initial or terminal in itself?	Vacuously its own (trivial) initial object.
Which object is initial in $\mathbf{2}$?	The object 0 (source of the unique non-identity arrow).
Which is terminal in $\mathbf{2}$?	The object 1.
Can a category have no initial object?	Yes.
Can a category have more than one initial object?	Yes, but any two are uniquely isomorphic.
Why "up to unique isomorphism"?	The isomorphism between two initial objects is itself uniquely determined.
What is the standard notation for the unique morphism $\mathbf{0} \to A$?	$!_A$, sometimes called "the shriek to A."
What is a zero object?	An object that is both initial and terminal.
Given a zero object, what is the zero morphism $A \to B$?	The composite $A \to \mathbf{0} \to B$ through the zero object.
What is the signature phrase for a universal property?	"For every X with such-and-such, there exists a unique u such that some diagram commutes."
What four pieces specify a universal property?	The ambient category, the data, the competitors, and the uniqueness requirement.
When two objects satisfy the same universal property, how are they related?	By a unique isomorphism.
Sketch the uniqueness argument.	Get morphisms $u : U \to U'$ and $u' : U' \to U$ from each side's universal property; argue $u' \circ u$ and $u \circ u'$ must be identities.

Why does "unique morphism $X \to X$" force $u' \circ u = \mathrm{id}_X$?	Because id_X also has the property, and there is only one such morphism.
What is a product $A \times B$?	An object equipped with projections π_1, π_2 such that any (f, g) pair factors uniquely through it.
What is the unique map $X \to A \times B$ induced by $f : X \to A$ and $g : X \to B$?	$\langle f, g\rangle$, the pairing of f and g.
Which equation does $\langle f, g\rangle$ satisfy?	$\pi_1 \circ \langle f, g\rangle = f$ and $\pi_2 \circ \langle f, g\rangle = g$.
What is the coproduct $A + B$?	An object equipped with inclusions ι_1, ι_2 such that any (f, g) pair factors uniquely through it.
What is the unique map $A + B \to Y$ induced by $f : A \to Y$ and $g : B \to Y$?	$[f, g]$, the cotuple of f and g.
Which equation does $[f, g]$ satisfy?	$[f, g] \circ \iota_1 = f$ and $[f, g] \circ \iota_2 = g$.
What is the relation between products and coproducts under duality?	A coproduct in $\mathcal{C}$ is a product in $\mathcal{C}^{\mathrm{op}}$.
Why might a category fail to have products?	Not enough morphisms to support pairings; or no candidate object exists.
In the recipe category, does Chop × Sauté exist?	Generally not; there's no object with the required pairing.
If a product exists, is it literally unique?	No; unique up to unique isomorphism.
Why is "unique up to unique iso" good enough for everyday work?	We pick a representative and use it; any other representative is interchangeable via the canonical iso.
What word captures all of products, coproducts, initial, and terminal?	Universal property (or universal construction).
How is a universal property reframed in terms of representable functors?	As a natural isomorphism between some functor and a hom-functor.
Why bother with universal properties when concrete constructions exist?	They identify the *essence* of the construction, free of arbitrary internal details.

Chapter Summary

Chapter Summary

An **initial object 0** has a unique morphism to every object. A **terminal object 1** has a unique morphism from every object. They are dual to each other.
Uniqueness up to unique isomorphism: any two objects satisfying the same universal property are connected by a unique isomorphism.
The **universal property pattern**:

$$\forall\, X \text{ with data, } \exists!\, u : X \to U \text{ or } U \to X \text{ compatible with the data.}$$

Products ($A \times B$) and **coproducts** ($A + B$) are the first universal constructions involving two objects: a product comes with projections π_1, π_2 and a pairing operation $\langle f, g\rangle$; a coproduct comes with inclusions ι_1, ι_2 and a cotuple $[f, g]$.

Recognizing a universal property: look for "for every . . . , there exists a unique . . . such that some diagram commutes."

Formal Restatement

Formal Restatement

Initial object. $\mathbf{0} \in \mathcal{C}$ is initial iff $|\mathrm{Hom}(\mathbf{0}, A)| = 1$ for every $A \in \mathcal{C}$.

Terminal object. $\mathbf{1} \in \mathcal{C}$ is terminal iff $|\mathrm{Hom}(A, \mathbf{1})| = 1$ for every $A \in \mathcal{C}$.

Duality. X is terminal in $\mathcal{C}$ iff X is initial in $\mathcal{C}^{\mathrm{op}}$.

Uniqueness theorem. Any two objects satisfying the same universal property are related by a unique isomorphism:

$$U, U' \text{ both satisfy the property} \implies \exists!\, u : U \xrightarrow{\sim} U'.$$

Product. $(A \times B, \pi_1, \pi_2)$ is a product of A, B iff for every (X, f, g) with $f : X \to A$, $g : X \to B$:

$$\exists!\, \langle f, g\rangle : X \to A \times B \text{ with } \pi_1 \circ \langle f, g\rangle = f,\ \pi_2 \circ \langle f, g\rangle = g.$$

Coproduct. $(A + B, \iota_1, \iota_2)$ is a coproduct of A, B iff for every (Y, f, g) with $f : A \to Y$, $g : B \to Y$:

$$\exists!\, [f, g] : A + B \to Y \text{ with } [f, g] \circ \iota_1 = f,\ [f, g] \circ \iota_2 = g.$$

Chapter 6 develops the full theory of limits and colimits, of which initial and terminal objects, products and coproducts, are the first examples.

Chapter 6

Limits and Colimits

Chapter Overview

Chapter 5 introduced the universal-property pattern through its two simplest cases (initial and terminal objects) and previewed a richer case (products and coproducts). This chapter develops the systematic generalization: *limits* and *colimits*.

The picture is this. A **diagram** in a category $\mathcal{C}$ is just a functor $D : \mathcal{J} \to \mathcal{C}$ from some small "shape" category $\mathcal{J}$. A **limit** of D is a universal way to choose a single object of $\mathcal{C}$ that "sees the whole diagram through the projections"; a **colimit** is the dual — a universal way to "glue the whole diagram together along the inclusions." Once this template is in place, products, coproducts, equalizers, coequalizers, pullbacks and pushouts all become instances of the same construction with different choices of $\mathcal{J}$. Every theorem we prove about limits will instantly give a theorem about each of these specific cases.

We end with the slogan: *limits in a functor category are computed pointwise*. This is the first taste of why functor categories are so well-behaved.

Where the Name Comes From

Limit and **colimit** were introduced by *Saunders Mac Lane* as the general categorical pattern subsuming products, equalizers, pullbacks, and their duals. The word *limit* echoes classical analysis (limit of a sequence, limit of a function): a single object "approached" by the whole diagram. The dual *colimit* ("co-limit") generalizes coproduct, coequalizer, and pushout — the "glued-together" constructions. The pattern is one of the central unifications of category theory: dozens of seemingly different constructions across mathematics become instances of one universal-property template.

Diagram as a functor $D : \mathcal{J} \to \mathcal{C}$ formalizes "the shape" of a drawing: $\mathcal{J}$ is the shape-category whose objects are "slots" and whose morphisms are "required arrows." The viewpoint is due to Grothendieck and Mac Lane.

6.1 Diagrams as Functors

6.1.1 Informally: Picture-Then-Functor

When we draw a diagram in $\mathcal{C}$, we are picking some objects and some morphisms in $\mathcal{C}$ and arranging them. We do not draw every object and every morphism; we draw a finite (or small) selection that "has a shape." The shape itself is a category — a category of *slots* or *positions* for the objects, with arrows between slots in the appropriate places.

A functor $D : \mathcal{J} \to \mathcal{C}$ then fills the slots: each object of $\mathcal{J}$ becomes an object of $\mathcal{C}$, each arrow becomes a morphism of $\mathcal{C}$, and the functoriality conditions guarantee that the picture in $\mathcal{C}$ has the right relationships between its parts.

6.1.2 Expanding: The Shape Categories

Example: Two-Object Diagram for a Product

The shape is the discrete category on two objects, $\bullet \quad \bullet$. A functor from this category to $\mathcal{C}$ is a choice of two objects of $\mathcal{C}$ — nothing more. This is the shape we use for binary products.

Example: Parallel Pair for an Equalizer

The shape is a category with two objects and two parallel non-identity arrows:

$$\bullet \rightrightarrows \bullet$$

A functor from this category to $\mathcal{C}$ picks two objects and two parallel morphisms $f, g : A \to B$ between them. This is the shape we use for equalizers.

Example: Cospan for a Pullback

The shape is a category with three objects arranged so that two arrows meet at a common target:

$$\bullet \longrightarrow \bullet \longleftarrow \bullet$$

A functor from this category to $\mathcal{C}$ picks a cospan $A \to C \leftarrow B$ in $\mathcal{C}$. This is the shape we use for pullbacks.

Internal Dialog

The standard moves are: pick a shape $\mathcal{J}$, then study *all* $\mathcal{J}$-shaped diagrams in $\mathcal{C}$ at once. The functor formalism gives us a clean language for "a $\mathcal{J}$-shaped diagram in $\mathcal{C}$": it is exactly an object of $[\mathcal{J}, \mathcal{C}]$.

6.1.3 Formalizing: Indexing Category and Diagram

A **shape** or **index category** is a small category $\mathcal{J}$. A **diagram of shape $\mathcal{J}$ in $\mathcal{C}$** is a functor

$$D : \mathcal{J} \to \mathcal{C}.$$

Two such diagrams are equivalent in the sense of the functor category $[\mathcal{J}, \mathcal{C}]$: a morphism between them is a natural transformation.

How to Say This

- "diagram" refers either to the picture or to the underlying functor D. We use both interchangeably.
- Common shapes have names: *discrete* (no arrows), *parallel pair*, *cospan*, *span*, *walking arrow* (just $\bullet \to \bullet$), the natural numbers $\mathbb{N}$ viewed as a category, etc.
- The phrase "D-shaped diagram" usually means a diagram whose index category equals (or resembles) the shape used by D.

6.1.4 Simplifying: From Pictures to Functors and Back

Every commutative-diagram picture we draw can be encoded as a functor $D : \mathcal{J} \to \mathcal{C}$ for some $\mathcal{J}$. Going from a picture to a functor is mechanical: the slots and arrows of the picture *are* $\mathcal{J}$, and the labels of the slots and arrows give the action of D. Going from a functor to a picture is also mechanical: draw $\mathcal{J}$ on a chalkboard, then write D's values inside each slot and on each arrow.

6.2 Cones and Limits

6.2.1 Informally: An Apex Watching the Diagram

A **cone over** D is a single object X of $\mathcal{C}$, together with a family of morphisms — one $\lambda_j : X \to D(j)$ for each object j of $\mathcal{J}$ — that respects all the arrows of $\mathcal{J}$. The word "cone" comes from the picture: X sits as an apex above the diagram, with rays λ_j reaching down to each slot.

A **limit** of D is a *universal* cone: a cone L such that every other cone factors through L in a unique way.

6.2.2 Expanding: The Naturality Condition for Cones

Example: Cone over a Two-Object Diagram (Product)

A diagram of shape "two objects, no arrows" picks A and B in $\mathcal{C}$. A cone over it is an object X together with two morphisms $X \to A$ and $X \to B$. There is no naturality constraint because there are no non-identity arrows in the shape. So a cone is just a pair (f, g) with $f : X \to A$, $g : X \to B$. The limit, when it exists, is the product $A \times B$ together with its projections.

Example: Cone over a Parallel Pair (Equalizer)

A diagram of shape "parallel pair" picks $f, g : A \to B$. A cone over it is an object X with morphisms $\lambda_A : X \to A$ and $\lambda_B : X \to B$ such that $f \circ \lambda_A = \lambda_B$ and $g \circ \lambda_A = \lambda_B$. Combining: $f \circ \lambda_A = g \circ \lambda_A$. So λ_A is a morphism into A that "equalizes" f and g, and λ_B is then forced.

Example: Cone over a Cospan (Pullback)

A diagram of shape $\bullet \to \bullet \leftarrow \bullet$ picks $f : A \to C$ and $g : B \to C$. A cone over it is an object X with morphisms $\lambda_A : X \to A$, $\lambda_B : X \to B$, and $\lambda_C : X \to C$ such that $f \circ \lambda_A = \lambda_C$ and $g \circ \lambda_B = \lambda_C$. The morphism λ_C is redundant (forced by either of the others), so a pullback cone is really a pair (λ_A, λ_B) with $f \circ \lambda_A = g \circ \lambda_B$.

Internal Dialog

A cone over D packages a candidate limiting object X with all the "projection-like" morphisms going down to each $D(j)$. The naturality constraint is what makes the rays compatible with the structure of the diagram.

6.2.3 Formalizing: Cones and the Limit

Let $\Delta_X : \mathcal{J} \to \mathcal{C}$ denote the **constant functor** at X: every object goes to X, every morphism goes to id_X.

A **cone over D with apex X** is a natural transformation

$$\lambda : \Delta_X \Rightarrow D.$$

Unpacking: a family $\lambda_j : X \to D(j)$ for each object j, such that for every $\alpha : j \to k$ in $\mathcal{J}$,

$$D(\alpha) \circ \lambda_j = \lambda_k.$$

This is the naturality square for λ: it commutes because the Δ_X side is just id_X.

A **limit of D** is a cone $\lambda : \Delta_L \Rightarrow D$ with the universal property: for every cone $\mu : \Delta_X \Rightarrow D$, there is a *unique* morphism $u : X \to L$ in $\mathcal{C}$ such that $\mu_j = \lambda_j \circ u$ for every j.

We write $\lim D$ for the limit object (when one exists), and call the λ_j the **limit projections** or **limit legs**.

How to Say This

- $\lim D$ — "limit of D," sometimes written $\lim_{\mathcal{J}} D$ to emphasize the indexing category.
- Δ_X — "Delta sub X" or "the constant functor at X."
- $\lambda_j : L \to D(j)$ — the *j-th leg* or *projection* of the limit cone.
- Limits are sometimes written $\lim_{\leftarrow}$ or with an underset arrow: the convention $\varprojlim$ also appears.

6.2.4 Reorganizing: Products, Equalizers, Pullbacks Restated

$$\begin{aligned}
\text{Product of } A, B\text{:} \quad & \lim D \text{ where } D : \text{discrete on } 2 \to \mathcal{C}, \\
& D(1) = A,\ D(2) = B. \\
\text{Equalizer of } f, g\text{:} \quad & \lim D \text{ where } D : (\bullet \rightrightarrows \bullet) \to \mathcal{C}, \\
& D \text{ sends the two arrows to } f \text{ and } g. \\
\text{Pullback of } f, g \text{ over } C\text{:} \quad & \lim D \text{ where } D : (\bullet \to \bullet \leftarrow \bullet) \to \mathcal{C}, \\
& D \text{ encodes the cospan } f, g.
\end{aligned}$$

Each of these is the same universal-property pattern of §5.3, with a specific shape for $\mathcal{J}$.

6.2.5 Simplifying: The Limit Schema

$$\boxed{\forall (X, \mu),\ \exists!\, u : X \to L,\ \mu_j = \lambda_j \circ u \ \text{ for every } j.}$$

6.2.6 Formally: Uniqueness and Functoriality of Limits

6.2.6.0.1 Uniqueness up to unique isomorphism

By the general theorem of §5.4, any two limits of the same diagram are related by a unique isomorphism compatible with the legs. We may speak of *the* limit of D.

6.2.6.0.2 The limit functor

When $\mathcal{C}$ has all $\mathcal{J}$-shaped limits, the assignment $D \mapsto \lim D$ extends to a functor

$$\lim : [\mathcal{J}, \mathcal{C}] \to \mathcal{C}.$$

Its action on a morphism of diagrams (a natural transformation $\eta : D \Rightarrow D'$) sends $\lim D$ to $\lim D'$ via the unique mediating morphism guaranteed by the limit universal property.

Where the Name Comes From

The names of the small-limit constructions tell their geometric stories:

- **Equalizer**: where two parallel arrows "become equal" — the universal sub-object on which $f = g$.
- **Pullback**: from algebraic geometry, where one "pulls back" a function along a map of varieties; the fibre product $X \times_Z Y$.
- **Pushout** (dual of pullback): from topology / gluing constructions; "pushes out" along inclusions.
- **Coequalizer**: the dual of equalizer; the universal quotient on which $f = g$.

The terminology is mostly from algebraic topology (Eilenberg-Steenrod 1952) and algebraic geometry (Grothendieck school). The category-theoretic unification subsumes them all under the limit/colimit umbrella.

6.3 Equalizers, Pullbacks, and the Zoo of Limits

6.3.1 Informally: One Pattern, Many Names

We have already seen products, equalizers, and pullbacks emerge as specific limits. This section catalogues the standard limit shapes explicitly, so the names become familiar.

6.3.2 Expanding: Each Shape, Its Universal Property

Example: Equalizer of $f, g : A \to B$

The equalizer is an object E together with a morphism $e : E \to A$ satisfying $f \circ e = g \circ e$, universal among such pairs:

$$E \xrightarrow{e} A \overset{f}{\underset{g}{\rightrightarrows}} B$$

For every $h : X \to A$ with $f \circ h = g \circ h$, there is a unique $u : X \to E$ with $e \circ u = h$.

Example: Pullback of f, g over C

The pullback (sometimes called *fibered product*) is an object P with morphisms $p_A : P \to A$ and $p_B : P \to B$ satisfying $f \circ p_A = g \circ p_B$, universal among such squares:

$$\begin{tikzcd} P \arrow[r, "p_B"] \arrow[d, "p_A"'] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & C \end{tikzcd}$$

For every commuting square (h, k) with $f \circ h = g \circ k$, there is a unique $u : X \to P$ with $p_A \circ u = h$ and $p_B \circ u = k$.

Example: The Limit of 1-shaped Diagrams

The shape $\mathbf{1}$ (one object, no non-identity morphisms) sends to just one object A. A cone is a morphism $X \to A$. The limit is A itself with $\lambda = \mathrm{id}_A$.

Example: The Limit of 0-shaped Diagrams (Terminal Object)

The shape $\mathbf{0}$ (empty category) has no objects and no morphisms. The functor $D : \mathbf{0} \to \mathcal{C}$ picks nothing. A cone over D is an object X with no constraints — just an object. The limit is the *terminal object* $\mathbf{1}$: the universal object with a unique map from every other object. So terminal = limit over the empty diagram.

6.3.3 Formalizing: A Single Universal Property, Five Names

For a diagram D of shape $\mathcal{J}$:

$$\begin{aligned} \mathcal{J} \text{ empty:} \quad & \lim D = \mathbf{1} \text{ (terminal object)} \\ \mathcal{J} = \mathbf{1}: \quad & \lim D = D(\text{the only object}) \\ \mathcal{J} = \text{discrete on } n: \quad & \lim D = \prod_{i=1}^{n} D(i) \text{ (n-ary product)} \\ \mathcal{J} = \bullet \rightrightarrows \bullet: \quad & \lim D = \mathrm{Eq}(f, g) \text{ (equalizer)} \\ \mathcal{J} = \bullet \to \bullet \leftarrow \bullet: \quad & \lim D = A \times_C B \text{ (pullback)} \end{aligned}$$

6.3.4 Reorganizing: The Limit Construction in Set

When $\mathcal{C} = \mathbf{Set}$, the limit of $D : \mathcal{J} \to \mathbf{Set}$ has a concrete description. Define

$$\lim D = \left\{ (x_j)_{j \in \mathrm{Ob}(\mathcal{J})} \;\middle|\; x_j \in D(j),\ D(\alpha)(x_j) = x_k \text{ for every } \alpha : j \to k \right\}.$$

The legs λ_j are the obvious projections. One can check this satisfies the universal property. So in **Set** "limit" is "compatible tuples."

6.3.5 Simplifying: When Do Limits Exist?

A category has *finite limits* if it has limits for every diagram whose shape $\mathcal{J}$ has finitely many objects and morphisms. Equivalently (the limit zoo collapses): a category has finite limits iff it has a terminal object, all binary products, and all equalizers.

A category has *all (small) limits* if it has limits for every small-shaped diagram. Such a category is called **(small-)complete**.

6.3.6 Formally: The Equalizer Construction in Set

6.3.6.0.1 Equalizers in Set

For $f, g : A \to B$ in **Set**:

$$\mathrm{Eq}(f, g) = \{a \in A \mid f(a) = g(a)\}, \quad e : \mathrm{Eq}(f, g) \hookrightarrow A.$$

Pullbacks have a similar concrete description:

$$A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\}.$$

6.4 Colimits as Dual Limits

6.4.1 Informally: Glue Instead of Project

The dual of a limit is a colimit. Where a limit is an apex *above* the diagram with rays going *down*, a colimit is a vertex *below* the diagram with rays going *up*. Where a limit's universal property is about *mapping in*, a colimit's is about *mapping out*.

Roughly, a colimit *glues together* the objects in the diagram along the morphisms in the diagram. In **Set**, this gluing is realized as a quotient of a disjoint union by an equivalence relation.

6.4.2 Expanding: Standard Colimit Shapes

Example: Coproducts

For the discrete two-object shape: the colimit is the coproduct $A + B$ together with inclusions ι_1, ι_2.

Example: Coequalizers

For the parallel-pair shape with $f, g : A \to B$: the colimit is the **coequalizer** of f and g — the universal object Q with a morphism $q : B \to Q$ such that $q \circ f = q \circ g$. In **Set**, Q is B modulo the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$.

Example: Pushouts

For the span shape $\bullet \leftarrow \bullet \to \bullet$ with $f : C \to A$ and $g : C \to B$: the colimit is the **pushout**, written $A +_C B$ or $A \sqcup_C B$. In **Set**, it is $(A \sqcup B)/\sim$, where $f(c) \sim g(c)$ for all $c \in C$.

6.4.3 Formalizing: Cocones and Colimits

A **cocone under D with co-apex X** is a natural transformation

$$\mu : D \Rightarrow \Delta_X.$$

Unpacking: morphisms $\mu_j : D(j) \to X$ for each j, with $\mu_k \circ D(\alpha) = \mu_j$ for every $\alpha : j \to k$.

A **colimit of D** is a universal cocone: $\mu : D \Rightarrow \Delta_C$ such that every cocone $\nu : D \Rightarrow \Delta_X$ factors uniquely as $\nu_j = u \circ \mu_j$ for some unique $u : C \to X$.

We write $\mathrm{colim}\, D$ (also $\lim_{\to}$ or $\varinjlim D$) for the colimit.

How to Say This

- $\operatorname{colim} D$ — "colimit of D"
- $\varinjlim D$ or $\lim_{\to}$ — the arrow-direction notation
- "cocone" is sometimes written "co-cone"
- Specific names: coproduct, coequalizer, pushout, initial object (colimit over empty)

6.4.4 Reorganizing: Duality, Once Stated

A colimit of $D : \mathcal{J} \to \mathcal{C}$ is a limit of $D^{\text{op}} : \mathcal{J}^{\text{op}} \to \mathcal{C}^{\text{op}}$. Therefore every theorem we prove about limits has a dual for colimits, by reversing arrows. We will prove things about limits and rely on duality for the colimit side without comment.

6.4.5 Simplifying: The Colimit Schema

$$\boxed{\forall (X, \nu),\ \exists!\, u : C \to X,\ \nu_j = u \circ \mu_j \text{ for every } j.}$$

6.4.6 Formally: Cocomplete Categories

6.4.6.0.1 Cocompleteness

A category is **cocomplete** if it has all small colimits. Equivalently, it has an initial object, all small coproducts, and all coequalizers.

6.4.6.0.2 The free constructions

Many "free" constructions in mathematics — free monoid on a set, free group on a set, free R-module on a set — are colimits in disguise. Free constructions go hand-in-hand with adjunctions (Chapter 7).

6.5 Preservation and Reflection of Limits

6.5.1 Informally: Which Functors Behave Nicely?

Once we have limits and colimits, the natural question is: which functors *respect* them? A functor that respects limits is invaluable, because we can compute the limit in the source and then apply the functor, rather than computing inside the target.

Three properties matter: *preservation* (the functor sends limits to limits), *reflection* (the functor turns limits-in-target back into limits-in-source), and *creation* (the strongest, which we discuss formally).

6.5.2 Expanding: A Functor That Preserves Products

Example: The Forgetful Functor U : Mon → Set

U sends a monoid to its underlying set, and a monoid homomorphism to the underlying function. It preserves products: $U(M_1 \times M_2) = U(M_1) \times U(M_2)$, with projections sent to projections. We will see in Chapter 7 that U is the right adjoint of a free-monoid functor, and right adjoints preserve all limits.

Example: A Functor That Does Not Preserve Limits

The functor **Set** → **Set** that sends X to the two-element set $\{0, 1\}$ regardless of X does not preserve products: $\{0, 1\} = F(A \times B) \neq F(A) \times F(B) = \{0, 1\} \times \{0, 1\}$ which has four elements. Constant functors typically do not preserve limits.

6.5.3 Formalizing: Preservation, Reflection, Creation

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor, $D : \mathcal{J} \to \mathcal{C}$ a diagram, and $\lambda : \Delta_L \Rightarrow D$ a limiting cone over D.

- F **preserves** the limit of D if $F\lambda : \Delta_{F(L)} \Rightarrow F \circ D$ is a limiting cone over $F \circ D$.
- F **reflects** the limit of D if: whenever $\lambda' : \Delta_X \Rightarrow D$ is a cone over D in $\mathcal{C}$ and $F\lambda'$ is limiting over $F \circ D$, then λ' was limiting over D already.
- F **creates** the limit of D if every limiting cone over $F \circ D$ in $\mathcal{D}$ uniquely lifts to a cone over D in $\mathcal{C}$, and the lifted cone is itself a limit of D.

Preservation says "limits in $\mathcal{C}$ map to limits in $\mathcal{D}$." Reflection says "limits in $\mathcal{D}$ come from limits in $\mathcal{C}$." Creation says "the limit in $\mathcal{C}$ is determined by the limit in $\mathcal{D}$."

6.5.4 Reorganizing: A Slogan and Two Anchors

Anchor 1. Right adjoints preserve limits (Chapter 7).

Anchor 2. Fully faithful functors reflect limits.

These two facts together account for an enormous fraction of the limit-preservation phenomena one meets in practice.

6.5.5 Simplifying: The Preservation Schema

$$F \text{ preserves } \lim D \iff F(\lim D) \cong \lim(F \circ D),$$

naturally and via the canonical mediating morphism, not merely up to abstract isomorphism.

6.5.6 Formally: Limits in Functor Categories

6.5.6.0.1 Limits are computed pointwise

For a small category $\mathcal{J}$ and any category $\mathcal{D}$ with $\mathcal{J}$-limits, the functor category $[\mathcal{C}, \mathcal{D}]$ has $\mathcal{J}$-limits, and they are computed pointwise:

$$(\lim D)(C) \cong \lim_j D(j)(C)$$

for each object C of $\mathcal{C}$. The evaluation functors $\mathrm{ev}_C : [\mathcal{C}, \mathcal{D}] \to \mathcal{D}$ jointly create all limits. This is one of the most-used facts in category theory: calculations in functor categories reduce to calculations object-by-object in the target.

Exercises

What is a diagram in $\mathcal{C}$, formally?	A functor $D : \mathcal{J} \to \mathcal{C}$ from a small category $\mathcal{J}$.
What plays the role of "the shape"?	The index category $\mathcal{J}$.
What is the constant functor Δ_X?	The functor sending every object to X and every arrow to id_X.
What is a cone over D with apex X?	A natural transformation $\Delta_X \Rightarrow D$.
Unpacked, what data does a cone consist of?	A morphism $\lambda_j : X \to D(j)$ for every j, compatible with the arrows of $\mathcal{J}$.
What is a limit of D?	A universal cone over D.
Why is the limit unique up to unique iso?	It satisfies a universal property; §5.4 applies.
Empty index category: what is the limit?	The terminal object.
Singleton index category: what is the limit?	The unique object that D picks out.
Discrete two-object index: what is the limit?	The (binary) product.
Parallel pair index: what is the limit?	The equalizer.
Cospan index: what is the limit?	The pullback.
What is a cocone under D?	A natural transformation $D \Rightarrow \Delta_X$.
What is a colimit?	A universal cocone.
Empty index: what is the colimit?	The initial object.
Discrete two-object index: colimit?	The coproduct.
Parallel pair: colimit?	The coequalizer.
Span index: colimit?	The pushout.
Why call the colimit "a glue"?	It glues the diagram's objects together along the diagram's arrows.
In **Set**, how is the pullback realized?	As $\{(a, b) \in A \times B \mid f(a) = g(b)\}$.

In **Set**, how is the equalizer realized?	As $\{a \in A \mid f(a) = g(a)\}$ with the inclusion into A.
In **Set**, how is the coequalizer realized?	As B modulo the equivalence relation generated by $f(a) \sim g(a)$.
What does "$\mathcal{C}$ is complete" mean?	$\mathcal{C}$ has all small limits.
What is the minimal checklist for completeness?	Terminal object, all binary products, all equalizers.
What is "$\mathcal{C}$ is cocomplete"?	Has all small colimits.
Minimal checklist for cocompleteness?	Initial object, all binary coproducts, all coequalizers.
What does "F preserves limits" mean?	F sends limiting cones to limiting cones.
What is "F reflects limits"?	If $F\lambda$ is a limit, then λ was already a limit.
Where are limits computed in $[\mathcal{C}, \mathcal{D}]$?	Pointwise, in $\mathcal{D}$.
What is the formula for $(\lim D)(C)$ pointwise?	$\lim_j D(j)(C)$.
Why is "pointwise" so useful?	It reduces hard calculations in functor categories to easier ones in the target category.
Are products limits over what shape?	The discrete category on two (or more) objects.
Why is duality a labour-saver here?	We prove a theorem about limits, and get a colimit theorem free by reversing arrows.

Chapter Summary

Chapter Summary

A **diagram** in $\mathcal{C}$ is a functor $D : \mathcal{J} \to \mathcal{C}$ from a small shape category. A **cone** over D is a natural transformation $\Delta_X \Rightarrow D$; a **cocone** under D is $D \Rightarrow \Delta_X$.
A **limit** of D is a universal cone; a **colimit** is a universal cocone. Products, equalizers, pullbacks (and their duals) are all limits/colimits of specific shapes.
Limits and colimits are unique up to unique isomorphism, and they are **computed pointwise** in functor categories.
A category is **complete** if it has all small limits, and **cocomplete** if it has all small colimits. Equivalently: terminal/initial + binary products/coproducts + equalizers/coequalizers.
A functor **preserves** a limit when it sends limiting cones to limiting cones; it **reflects** when target-limits descend to source-limits.

Formal Restatement

Formal Restatement

Diagram. A diagram of shape $\mathcal{J}$ in $\mathcal{C}$ is a functor $D : \mathcal{J} \to \mathcal{C}$.

Cone. A cone over D with apex X is a natural transformation $\lambda : \Delta_X \Rightarrow D$:

$$\lambda_j : X \to D(j), \quad D(\alpha) \circ \lambda_j = \lambda_k \text{ for every } \alpha : j \to k.$$

Limit. L is a limit of D, with cone λ, iff for every cone $\mu : \Delta_X \Rightarrow D$:

$$\exists!\, u : X \to L \text{ with } \mu_j = \lambda_j \circ u \text{ for all } j.$$

Colimit. C is a colimit of D, with cocone μ, iff for every cocone $\nu : D \Rightarrow \Delta_X$:

$$\exists!\, u : C \to X \text{ with } \nu_j = u \circ \mu_j \text{ for all } j.$$

Completeness. $\mathcal{C}$ is (small-)complete iff it has all small limits, iff it has a terminal object, all binary products, and all equalizers. Dually for cocompleteness.

Pointwise limits. $[\mathcal{C}, \mathcal{D}]$ has $\mathcal{J}$-limits whenever $\mathcal{D}$ does, and:

$$(\lim_j D_j)(C) = \lim_j (D_j(C)).$$

Chapter 7 develops adjunctions, the most important source of limit-preserving and colimit-preserving functors.

Chapter 7

Adjunctions

Chapter Overview

We have now built a substantial toolkit: categories, functors, natural transformations, universal properties, limits, and colimits. The next step is to study how categories themselves *relate*.

An **adjunction** is the fundamental relationship between two categories. It is the categorical formalization of a recurring informal pattern: *constructing* something on one side "corresponds to" *checking* something on the other. The construction goes *left* (a functor L) and the checking goes *right* (a functor R), and the two are linked by a bijection between hom-sets:

$$\mathrm{Hom}_{\mathcal{D}}(L(A), B) \;\cong\; \mathrm{Hom}_{\mathcal{C}}(A, R(B)).$$

This bijection, when it holds naturally in both variables, is called an **adjunction** and is written $L \dashv R$.

The reason adjunctions matter is that almost every important construction in mathematics is part of an adjunction: free monoid/forgetful, free group/forgetful, tensor/hom, $\exists$/pullback, abelianization/inclusion, ... The list is essentially endless.

The deep fact of this chapter: left adjoints preserve colimits, right adjoints preserve limits. This is what makes adjunctions the principal tool for proving preservation theorems.

Where the Name Comes From

Adjunction (or *adjoint pair*) was introduced by *Daniel Kan* in his 1958 paper "Adjoint Functors." The name is borrowed from linear algebra: in a Hilbert space, the *adjoint* T^* of an operator T is the unique operator satisfying the pairing $\langle Tx, y\rangle = \langle x, T^*y\rangle$. Kan's categorical adjoint follows the same template: the hom-set bijection $\mathcal{D}(LA, B) \cong \mathcal{C}(A, RB)$ is the categorical version of the inner-product pairing, with Hom playing the role of $\langle -, -\rangle$. Kan's insight was that this pattern is *everywhere*: free/forgetful pairs, tensor/hom pairs, base-change pairs in geometry, modal operators in logic — all are adjunctions.

The symbol $\dashv$ ("left tack" or "turnstile") is read "left adjoint to." It evokes the asymmetry: L on the left of $\dashv$ is the left adjoint; R on the right is the right adjoint.

7.1 The Hom-Set Bijection

7.1.1 Informally: Free and Forgetful

Here is the prototypical pattern. Consider an object A in some category $\mathcal{C}$ of "raw stuff" (say a set). We want to "freely build a richer structure" on A — a monoid, a group, a module — landing in some richer category $\mathcal{D}$. The free construction is $L(A)$, an object of $\mathcal{D}$.

Conversely, given an object B in $\mathcal{D}$, we can *forget* its extra structure, landing back in $\mathcal{C}$. The forgetful is $R(B)$, an object of $\mathcal{C}$.

The magic: a structured map $L(A) \to B$ (i.e., a homomorphism in $\mathcal{D}$) corresponds to an unstructured map $A \to R(B)$ (just a function on raw stuff). Once you fix where each generator of $L(A)$ goes in B, the rest of the homomorphism is forced. This correspondence is a bijection $\mathrm{Hom}_{\mathcal{D}}(L(A), B) \cong \mathrm{Hom}_{\mathcal{C}}(A, R(B))$.

7.1.2 Expanding: The Pattern in Concrete Settings

Example: Free Monoid and Forgetful

Take $L =$ free monoid functor, $R =$ forgetful from monoids to sets. For a set A and a monoid B:

- A monoid homomorphism $L(A) \to B$ is determined by where it sends each $a \in A$ (the generators).
- A function $A \to R(B)$ is exactly a choice of an element of B for each $a \in A$.

These data are the same. The bijection is the formal statement of this informal fact.

Example: Discrete and Forgetful for Topology

Let L send a set X to "X with the discrete topology," and R send a topological space to its underlying set. A continuous map $L(X) \to Y$ is just any function $X \to R(Y)$, because every function from a discrete space is continuous. So we get a bijection

$$\mathrm{Hom}_{\mathbf{Top}}(L(X), Y) \cong \mathrm{Hom}_{\mathbf{Set}}(X, R(Y)).$$

L is left adjoint to R.

Example: Tensor and Hom

For modules over a commutative ring R, the tensor product with a fixed module M is left adjoint to the hom-functor out of M:

$$\mathrm{Hom}(N \otimes M, P) \cong \mathrm{Hom}(N, \mathrm{Hom}(M, P)).$$

This bijection is so important it has a special name: *tensor-hom adjunction*. It generalizes the "currying" of functions in **Set**: $Z^{X \times Y} = (Z^Y)^X$.

Internal Dialog

The pattern in each case: one functor *builds something* and the other *forgets or evaluates*. The bijection says that the building in one direction matches the using in the other.

7.1.3 Formalizing: Adjoint Functors

Let $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ be functors. We say L is **left adjoint to** R (equivalently R is **right adjoint to** L), and we write $L \dashv R$, if there is a *natural* bijection

$$\varphi_{A,B} : \mathrm{Hom}_{\mathcal{D}}(L(A), B) \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(A, R(B))$$

for every $A \in \mathcal{C}$ and $B \in \mathcal{D}$. The naturality requires that φ commute with precomposition by $L(f)$ in $\mathcal{D}$ and with f in $\mathcal{C}$, and similarly with $R(g)$.

If φ exists, we say L and R form an **adjoint pair**, or that (L, R, φ) is an **adjunction**.

How to Say This

- $L \dashv R$ — "L is left adjoint to R," or "the adjunction L-R."
- The symbol $\dashv$ is read "turnstile," written `\dashv`. The reflected $\vdash$ goes the other direction.
- φ is sometimes called the *adjunction iso* or just *the natural bijection*.
- Naturality is in *both* variables: A contravariantly (from $\mathcal{C}^{\mathrm{op}}$) and B covariantly.

7.1.4 Reorganizing: Unit and Counit

Plug $B = L(A)$ into the bijection: the right-hand side becomes $\mathrm{Hom}_{\mathcal{C}}(A, R(L(A)))$. The identity $\mathrm{id}_{L(A)}$ on the left corresponds to a morphism on the right:

$$\eta_A : A \to R(L(A)).$$

This morphism is called the **unit of the adjunction at** A.

Dually, plug $A = R(B)$: the left-hand side becomes $\mathrm{Hom}_{\mathcal{D}}(L(R(B)), B)$. The identity $\mathrm{id}_{R(B)}$ on the right corresponds to:

$$\varepsilon_B : L(R(B)) \to B,$$

the **counit at** B.

The collections (η_A) and (ε_B) assemble into natural transformations

$$\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow R \circ L, \qquad \varepsilon : L \circ R \Rightarrow \mathrm{Id}_{\mathcal{D}}.$$

7.1.5 Simplifying: The Triangle Identities

$$(R\varepsilon) \circ (\eta R) = \mathrm{id}_R,$$
$$(\varepsilon L) \circ (L\eta) = \mathrm{id}_L.$$

These are the **triangle identities** (or *counit-unit laws*). They encode the requirement that the bijection φ is the natural bijection of an adjunction, expressed entirely in terms of L, R, η, and ε.

7.1.6 Formally: Three Equivalent Definitions

7.1.6.0.1 Adjunctions, three ways

The following data are interchangeable:

1. A natural bijection φ as above.
2. Natural transformations η, ε satisfying the triangle identities.
3. For every A, a universal arrow $\eta_A : A \to R(L(A))$ from A to R (or, dually, for every B, a universal arrow from L to B).

Each formulation is convenient in different proofs. The hom-set bijection is best for computing examples; the unit/counit form is best for proofs in 2-category theory; the universal-arrow form is best for the Adjoint Functor Theorems (Chapter 10).

7.2 Adjoint Pairs in the Wild

7.2.1 Informally: A Tour

The same hom-set bijection $\mathrm{Hom}(LA, B) \cong \mathrm{Hom}(A, RB)$ appears all over mathematics. Recognizing it is half the battle.

7.2.2 Expanding: A Catalogue

Example: Free Group — Forgetful

L sends a set to the free group on it; R sends a group to its underlying set. The unit $\eta_A : A \to RL(A)$ is the inclusion of generators.

Example: Abelianization — Inclusion

$L : \mathbf{Grp} \to \mathbf{Ab}$ sends a group G to its abelianization $G/[G, G]$. R is the inclusion of abelian groups into all groups. The unit is the quotient map $G \to G^{\mathrm{ab}}$.

Example: Discrete — Forgetful for Posets

L sends a set X to the discrete poset (no nontrivial order). R sends a poset to its underlying set. $L \dashv R$.

Example: Quantifiers as Adjoints

For a function $f : X \to Y$, the pullback functor $f^* : \mathbf{Set}/Y \to \mathbf{Set}/X$ has both a left adjoint $\exists_f$ and a right adjoint $\forall_f$. The left adjoint is "image" and the right adjoint is "Frobenius's all-fiber condition." This makes existential and universal quantification adjoints to substitution.

Example: Curry: Product — Function-Space

In $\mathbf{Set}$ (and any cartesian closed category):

$$\mathrm{Hom}(A \times B, C) \;\cong\; \mathrm{Hom}(A, C^B).$$

$(- \times B)$ is left adjoint to $(-)^B$.

7.2.3 Formalizing: A Recipe for Identifying Adjunctions

To spot an adjunction $L \dashv R$:

1. Identify the two categories $\mathcal{C}$ and $\mathcal{D}$.
2. Identify the candidate functors L and R.
3. Write out a typical hom-set $\mathrm{Hom}_{\mathcal{D}}(LA, B)$ in concrete terms.
4. Write out $\mathrm{Hom}_{\mathcal{C}}(A, RB)$ in concrete terms.
5. Look for a natural bijection between the two descriptions.

If you can carry out steps 3–5, you have your adjunction.

7.2.4 Reorganizing: Adjoints Are Unique

Theorem (Uniqueness of adjoints). If R has a left adjoint, that left adjoint is unique up to natural isomorphism. Dually for right adjoints.

The proof is the standard universal-property uniqueness argument from §5.4, applied to the unit (or counit).

7.2.5 Simplifying: "Adjoints" Are Functorial Pairs

We will routinely write things like "the left adjoint to R is L" or "$\exists_f$ is left adjoint to f^*." This is a statement about a *pair* of functors and a natural bijection, but the bijection is canonically determined by either functor and need not be named.

7.2.6 Formally: Composability of Adjunctions

7.2.6.0.1 Adjunctions compose

If $L \dashv R$ between $\mathcal{C}$ and $\mathcal{D}$, and $L' \dashv R'$ between $\mathcal{D}$ and $\mathcal{E}$, then $L' \circ L \dashv R \circ R'$ between $\mathcal{C}$ and $\mathcal{E}$. The bijection composes:

$$\begin{aligned} \mathrm{Hom}_{\mathcal{E}}(L'L(A), B) &\cong \mathrm{Hom}_{\mathcal{D}}(L(A), R'(B)) \\ &\cong \mathrm{Hom}_{\mathcal{C}}(A, RR'(B)). \end{aligned}$$

Where the Name Comes From

The **unit** η and **counit** ϵ of an adjunction are named for the role they play. The *unit* $\eta : \mathrm{id} \Rightarrow RL$ is the "coming-in" transformation (analogous to the multiplicative unit "1" in a monoid); the *counit* $\epsilon : LR \Rightarrow \mathrm{id}$ is the "going-out" transformation. The triangle identities ($\epsilon L \circ L\eta = \mathrm{id}_L$ and $R\epsilon \circ \eta R = \mathrm{id}_R$) say that going around the unit-counit-cycle on either side returns to the identity — a coherence condition. The Greek letters η (eta) and ϵ (epsilon) are conventional.
RAPL and **LAPC** are mnemonic acronyms: *Right Adjoints Preserve Limits*, *Left Adjoints Preserve Colimits*. They are universally taught in category theory courses and are perhaps the most useful one-line summaries of adjunction theory.

7.3 Adjunctions and Limits

7.3.1 Informally: Right Adjoints Preserve Limits

The most useful theorem about adjunctions is this:

> *If R has a left adjoint, then R preserves all limits that exist in its source.*

Dually, left adjoints preserve all colimits.

Why does this matter? Because limits are everywhere (Chapter 6) and checking that a functor preserves them is, in general, painful. But once we know our functor is a right adjoint, preservation is automatic.

7.3.2 Expanding: Sketch of the Proof

The proof is short. Suppose $L \dashv R$, $D : \mathcal{J} \to \mathcal{D}$ is a diagram, and $\lim D$ exists. We want to show $R(\lim D) = \lim(R \circ D)$.

For any object A:

$$\begin{aligned} \mathrm{Hom}(A, R(\lim D)) &\cong \mathrm{Hom}(L(A), \lim D) && \text{(adjunction)} \\ &\cong \lim_j \mathrm{Hom}(L(A), D(j)) && \text{(limit/hom)} \\ &\cong \lim_j \mathrm{Hom}(A, R(D(j))) && \text{(adjunction, pointwise)} \\ &\cong \mathrm{Hom}(A, \lim_j R(D(j))). && \text{(limit/hom)} \end{aligned}$$

This natural isomorphism for every A identifies $R(\lim D)$ with $\lim(R \circ D)$ up to unique iso. Hence R preserves the limit.

7.3.3 Formalizing: The Statement, Precisely

Theorem (RAPL: Right Adjoints Preserve Limits). Let $L \dashv R$ with $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$. For any diagram $D : \mathcal{J} \to \mathcal{D}$ whose limit exists, the canonical morphism

$$R(\lim D) \xrightarrow{\cong} \lim(R \circ D)$$

is an isomorphism.

Dually: **LAPC (Left Adjoints Preserve Colimits).**

7.3.4 Reorganizing: A Test for "No Left Adjoint"

The contrapositive is useful: if R *fails* to preserve some limit, then R has no left adjoint. For example:

- The functor $\pi_0 : \mathbf{Top} \to \mathbf{Set}$ (path components) fails to preserve products (cf. the line $\times$ line example), so it has no left adjoint (in the unenriched sense).

- The free-group functor $L : \mathbf{Set} \to \mathbf{Grp}$ does not preserve products, so it has no right adjoint either. (It is itself a left adjoint, of course — of the forgetful functor.)

7.3.5 Simplifying: The Adjunction Slogan

$$\boxed{L \dashv R \;\Rightarrow\; R \text{ preserves limits}, \quad L \text{ preserves colimits.}}$$

7.3.6 Formally: Hom-Functor Preservation

7.3.6.0.1 Hom-functors and limits

For any object A of $\mathcal{C}$, the hom-functor $\mathrm{Hom}(A, -)$ preserves limits. This is sometimes called "co-Yoneda preservation" and it is the limit/hom step used in the proof of RAPL. Dually, $\mathrm{Hom}(-, A)$ (contravariant) sends colimits to limits.

7.4 Equivalences as Adjunctions

7.4.1 Informally: When the Two Sides Are Almost the Same

A special and important case: when both the unit η and the counit ε are natural isomorphisms. Then L and R are inverses up to natural isomorphism, and the two categories $\mathcal{C}$ and $\mathcal{D}$ are essentially the same. We call this an **equivalence** of categories.

7.4.2 Expanding: Equivalence vs. Isomorphism

An *isomorphism* of categories would require $R \circ L = \mathrm{Id}_{\mathcal{C}}$ and $L \circ R = \mathrm{Id}_{\mathcal{D}}$ *on the nose*. This is too strict — and usually irrelevant. What we really want is that the two sides are the same *up to natural isomorphism*, which is exactly what equivalence demands.

Example: Finite Sets and Their Cardinality Skeletons

Let $\mathcal{C}$ be **FinSet** (all finite sets and functions), and $\mathcal{D}$ be the skeleton $\mathbf{FinSet}_{\mathrm{sk}}$ with objects $\underline{n} = \{1, 2, \ldots, n\}$ for $n \geq 0$. The inclusion $\mathcal{D} \hookrightarrow \mathcal{C}$ is one half of an equivalence; the other functor sends a finite set X to $\underline{|X|}$. Every finite set is isomorphic to some $\underline{n}$, so the unit and counit are natural isomorphisms.

7.4.3 Formalizing: Equivalence of Categories

An **equivalence of categories** between $\mathcal{C}$ and $\mathcal{D}$ is a pair of functors $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms

$$R \circ L \;\cong\; \mathrm{Id}_{\mathcal{C}}, \qquad L \circ R \;\cong\; \mathrm{Id}_{\mathcal{D}}.$$

We write $\mathcal{C} \simeq \mathcal{D}$.

Equivalently: $L \dashv R$, both unit and counit are natural isomorphisms.

Theorem (Recognition of Equivalences). A functor $F : \mathcal{C} \to \mathcal{D}$ is one half of an equivalence iff F is fully faithful and **essentially surjective** (every object of $\mathcal{D}$ is isomorphic to some $F(C)$).

7.4.4 Reorganizing: The Strict-Equivalence Slogan

Isomorphism of categories is the strict version; equivalence is the *right* version. Whenever we say "two categories are the same" we almost always mean equivalence.

7.4.5 Simplifying: The Equivalence Schema

$$F \text{ is an equivalence} \quad \Longleftrightarrow \quad F \text{ is fully faithful and essentially surjective.}$$

7.4.6 Formally: Working with Skeletons

7.4.6.0.1 Skeletons and equivalence

Every category $\mathcal{C}$ is equivalent to its **skeleton**: a subcategory containing one object per isomorphism class. Working with a skeleton is often convenient, because it eliminates redundant objects without changing the categorical content.

Exercises

What is an adjunction, in one sentence?	A natural bijection between $\mathrm{Hom}_{\mathcal{D}}(LA, B)$ and $\mathrm{Hom}_{\mathcal{C}}(A, RB)$.

Which is the left adjoint: L or R?	L. It appears on the left of the hom inside $\mathcal{D}$.
What does $L \dashv R$ mean?	L is left adjoint to R.
In "free monoid — forgetful," which is left?	Free monoid is left; forgetful is right.
What is the unit of an adjunction?	A natural transformation $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow RL$.
What is the counit?	A natural transformation $\varepsilon : LR \Rightarrow \mathrm{Id}_{\mathcal{D}}$.
How does η_A relate to $\mathrm{id}_{L(A)}$?	Under the bijection, η_A corresponds to $\mathrm{id}_{L(A)}$.
How does ε_B relate to $\mathrm{id}_{R(B)}$?	Under the inverse bijection, ε_B corresponds to $\mathrm{id}_{R(B)}$.
What are the triangle identities?	$(R\varepsilon)(\eta R) = \mathrm{id}_R$ and $(\varepsilon L)(L\eta) = \mathrm{id}_L$.
Why do we care about the triangle identities?	They are the data-only form of the adjunction (no bijection needed).
Are adjoints unique?	Yes, up to natural isomorphism.
What is the slogan for limits and adjoints?	Right adjoints preserve limits; left adjoints preserve colimits. (RAPL/LAPC.)
If R fails to preserve some limit, what can you conclude?	R has no left adjoint.
Why does the hom-set bijection imply RAPL?	Apply $\mathrm{Hom}(A, -)$, which preserves limits; chase the bijection.
What is the prototype adjunction in **Set**?	Currying: $(- \times B) \dashv (-)^B$.
What is the tensor-hom adjunction in modules?	$(- \otimes M) \dashv \mathrm{Hom}(M, -)$.
Quantifiers as adjoints to what?	Adjoints to the pullback functor f^*: $\exists_f \dashv f^* \dashv \forall_f$.
What is an equivalence of categories?	A pair $L \dashv R$ where both unit and counit are natural isomorphisms.
Why prefer equivalence to isomorphism of categories?	Isomorphism is too strict; almost all "sameness" in CT is up to natural iso.
What is essential surjectivity?	Every object in $\mathcal{D}$ is isomorphic to some $F(C)$.
When is F an equivalence?	When F is fully faithful and essentially surjective.
What is a skeleton of a category?	A subcategory with one object per isomorphism class.
Is every category equivalent to its skeleton?	Yes.
Do adjunctions compose?	Yes: $L \dashv R$ and $L' \dashv R'$ give $L'L \dashv RR'$.
Does the identity functor have an adjoint?	Yes; it is adjoint to itself.
Can a functor have more than one left adjoint?	Only up to natural isomorphism; there is essentially one.
Why is the unit "the obvious" inclusion in many examples?	Because the unit corresponds to $\mathrm{id}_{L(A)}$, the most basic morphism on the freely built object.
What kind of functor is $\pi_0 : \mathbf{Top} \to \mathbf{Set}$?	Not a right adjoint in the unenriched sense (it fails to preserve some products).
What is the practical use of recognizing an adjunction?	You instantly get a wealth of preservation theorems and natural bijections.
What's the deeper meaning of $L \dashv R$?	The construction L *represents* the same data as the checking R, viewed dually.

Chapter Summary

Chapter Summary

An **adjunction** $L \dashv R$ is a natural bijection $\mathrm{Hom}(LA, B) \cong \mathrm{Hom}(A, RB)$. Equivalently, it is a pair of natural transformations $\eta : \mathrm{Id} \Rightarrow RL$ (**unit**) and $\varepsilon : LR \Rightarrow \mathrm{Id}$ (**counit**) satisfying the **triangle identities**.
Adjoints are unique up to natural isomorphism. They compose. They are ubiquitous: free/forgetful pairs, tensor-hom, quantifiers, abelianization, discrete topology, etc.
The major theorem: **Right Adjoints Preserve Limits** (RAPL) and **Left Adjoints Preserve Colimits** (LAPC).
An **equivalence of categories** is an adjunction with invertible unit and counit. It is recognized by being **fully faithful and essentially surjective**.

Formal Restatement

Formal Restatement

Adjunction. $L : \mathcal{C} \to \mathcal{D}$ is left adjoint to $R : \mathcal{D} \to \mathcal{C}$ ($L \dashv R$) iff there is a natural bijection

$$\varphi_{A,B} : \mathrm{Hom}_{\mathcal{D}}(LA, B) \cong \mathrm{Hom}_{\mathcal{C}}(A, RB).$$

Unit and counit. $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow RL$, $\varepsilon : LR \Rightarrow \mathrm{Id}_{\mathcal{D}}$, satisfying:

$$(R\varepsilon)(\eta R) = \mathrm{id}_R, \qquad (\varepsilon L)(L\eta) = \mathrm{id}_L.$$

Limit preservation. $L \dashv R \;\Rightarrow\; R$ preserves limits, L preserves colimits.

Equivalence. $\mathcal{C} \simeq \mathcal{D}$ iff there exist $L \dashv R$ with η, ε both natural isomorphisms; equivalently, there exists a fully faithful, essentially surjective functor.

Chapter 8 develops monads, which arise from adjunctions and capture algebraic structure relative to a base category.

Chapter 8

Monads

Chapter Overview

In Chapter 7 we saw that an adjunction $L \dashv R$ packages a "construction" (the left adjoint) together with a "forgetful" (the right adjoint). What does this construction *leave behind* on the source category? The endofunctor $R \circ L : \mathcal{C} \to \mathcal{C}$, together with the unit $\eta : \mathrm{Id} \Rightarrow RL$ and a "multiplication" $\mu = R\varepsilon L : RLRL \Rightarrow RL$ arising from the counit.

This package — an endofunctor with a unit and a compatible multiplication — is called a **monad**. Monads are the categorical formalization of *algebraic structure* carried by a functor. They appear in algebra (the free-monoid monad, the free-group monad), in topology (free topological monoids), in logic and computer science (the list monad, the Maybe monad, the state monad), and in geometry (sheaf constructions).

The story of this chapter: define monads in their own right, show how every adjunction gives rise to a monad, study *algebras for a monad* (the Eilenberg–Moore category), and conclude with the relationship to the original adjunction. Along the way we develop the Kleisli construction, the categorical home of effectful computation.

Where the Name Comes From

Monad is from Greek *monas* ("unit, single thing"). The term was introduced by *Saunders Mac Lane* in 1971 (*Categories for the Working Mathematician*), displacing two earlier names: *triple* (Eilenberg-Moore 1965; "triple" because a monad is the triple (T, η, μ)) and *standard construction* (Godement 1958). Mac Lane chose "monad" partly for its connection to Leibniz's philosophical *monads* (single, indivisible substances) and partly because the categorical structure really is a "single algebraic object" on the endofunctor category.

Eilenberg-Moore category of T-algebras is named for *Eilenberg* (cross-ref Vol I Ch. 2) and *John Moore*; their 1965 paper formalized the notion. **Kleisli category** is named for *Heinrich Kleisli* (1930–2011, Swiss mathematician), whose 1965 note introduced the construction.

8.1 Monads as Algebraic Endofunctors

8.1.1 Informally: Endofunctor + Unit + Multiplication

A **monad** on a category $\mathcal{C}$ consists of three things:

- An endofunctor $T : \mathcal{C} \to \mathcal{C}$. Think: "a way of building richer objects from $\mathcal{C}$-objects."
- A unit $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow T$. Think: "the inclusion of A into $T(A)$ as the generators."
- A multiplication $\mu : T \circ T \Rightarrow T$. Think: "compose nested T-structures into one."

These satisfy three laws: a unit law (twice) and an associativity law, to be made precise below.

The classic example: $T(X) = X^*$, the free-monoid (list) functor on **Set**. The unit $\eta_X : X \to X^*$ sends x to the singleton list $[x]$. The multiplication $\mu_X : (X^*)^* \to X^*$ concatenates a list of lists into a single list.

8.1.2 Expanding: Examples That Matter

Example: The List Monad on Set

$T(X) = X^*$ (finite sequences from X).

- Unit $\eta_X(x) = [x]$.
- Multiplication $\mu_X([[a_1, \ldots], [b_1, \ldots], \ldots]) = [a_1, \ldots, b_1, \ldots, \ldots]$.

The laws say: the unit at the front or back of a list does nothing ($\mu \circ \eta T = \mathrm{id} = \mu \circ T\eta$), and concatenating nested lists is associative ($\mu \circ T\mu = \mu \circ \mu T$).

Example: The Maybe Monad

$T(X) = X + \{*\}$ (a copy of X plus a distinguished "failure" value).

- Unit $\eta_X(x) = x \in X \subset X + \{*\}$.
- Multiplication μ_X flattens nested options: an outer $*$ becomes $*$; an outer value that is itself $*$ becomes $*$; an outer value that is an inner real value is unwrapped.

The Maybe monad is the standard formalism for "computations that may fail."

Example: The Powerset Monad

$T(X) = \mathcal{P}(X)$, the powerset.

- Unit $\eta_X(x) = \{x\}$.
- Multiplication μ_X takes a set of sets to its union.

The Kleisli morphisms (§8.4) for this monad are exactly relations: functions $X \to \mathcal{P}(Y)$ are relations $X \rightharpoonup Y$.

Internal Dialog

The three examples share a common shape: build something on X, "flatten" when you build it twice, and have a way to include X into $T(X)$ as the trivial case.

8.1.3 Formalizing: The Definition

A **monad on $\mathcal{C}$** is a triple (T, η, μ) where:

- $T : \mathcal{C} \to \mathcal{C}$ is a functor.
- $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow T$ is a natural transformation.
- $\mu : T \circ T \Rightarrow T$ is a natural transformation.

And satisfying:

$$\mu \circ \mu T = \mu \circ T\mu \qquad \text{(associativity)}$$
$$\mu \circ \eta T = \mathrm{id}_T \qquad \text{(left unit)}$$
$$\mu \circ T\eta = \mathrm{id}_T \qquad \text{(right unit).}$$

The first equation lives in $T^3 \Rightarrow T$; the others in $T \Rightarrow T$.

How to Say This

- T — "the monad" (loosely; really it's the whole triple).
- η — "eta," the **unit** or **return**.
- μ — "mu," the **multiplication** or **join** (the latter in CS).
- "return" and "join" are the names used in functional programming for η and μ.
- The composition $T \circ T$ is sometimes written T^2, and the n-fold composition T^n for any n.

8.1.4 Reorganizing: The Three Laws as Diagrams

Associativity.

$$\begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd}$$

Unit laws.

$$\begin{tikzcd} T \arrow[r, "\eta T"] \arrow[dr, "\mathrm{id}_T"'] & T^2 \arrow[d, "\mu"] & T \arrow[l, "T\eta"'] \arrow[dl, "\mathrm{id}_T"] \\ & T & \end{tikzcd}$$

8.1.5 Simplifying: The Monad Schema

A monad is a *monoid object* in the category of endofunctors of $\mathcal{C}$ under composition. The unit object of this monoidal category is Id, the tensor product is functor composition, and the associativity/unitality of (T, η, μ) are exactly the associativity/unit laws of a monoid in this monoidal category.

This is why one sometimes hears the slogan "a monad is a monoid in the category of endofunctors." The slogan is true. Whether it is helpful at first reading is a matter of taste.

8.1.6 Formally: Naturality of η and μ

8.1.6.0.1 Naturality is automatic if you write down the right diagrams

The naturality of η says that for $f : A \to B$, $T(f) \circ \eta_A = \eta_B \circ f$. The naturality of μ says that for $f : A \to B$, $T(f) \circ \mu_A = \mu_B \circ T^2(f)$. Both follow from η, μ being natural transformations between functors of the appropriate type.

8.2 Adjunctions Give Monads

8.2.1 Informally: RL Is Always a Monad

Take any adjunction $L \dashv R$ between $\mathcal{C}$ and $\mathcal{D}$. Set $T = R \circ L$. Then T is an endofunctor of $\mathcal{C}$, and the adjunction's unit $\eta : \mathrm{Id} \Rightarrow RL = T$ is the unit of a monad. The multiplication is the whiskering of the counit:

$$\mu := R\varepsilon L : RLRL = T^2 \Rightarrow RL = T.$$

The triangle identities of the adjunction translate directly to the monad laws.

8.2.2 Expanding: The Free-Monoid Adjunction $\rightsquigarrow$ The List Monad

The classical example: $L : \mathbf{Set} \to \mathbf{Mon}$, $R : \mathbf{Mon} \to \mathbf{Set}$, $L \dashv R$. Then RL on **Set** sends a set X to the underlying set of the free monoid on X, which is X^*. The induced monad is the list monad.

In general, whenever you see a free/forgetful adjunction, you get a monad whose algebras (in a sense to be defined below) are the original structured objects.

8.2.3 Formalizing: The Theorem

Theorem. Let $L \dashv R$ be an adjunction with unit η and counit ε. Define $T = RL$ and $\mu = R\varepsilon L$. Then (T, η, μ) is a monad on $\mathcal{C}$.

Proof sketch. The unit law $\mu \circ \eta T = \mathrm{id}_T$ expands to $R\varepsilon L \circ \eta RL$, which by the triangle identity for R equals id_{RL}. Similarly for $\mu \circ T\eta$. Associativity uses the naturality of ε together with the fact that L and R are functors. $\square$

8.2.4 Reorganizing: Every Adjunction Has a Monad

- Free monoid $\dashv$ Forgetful $\rightsquigarrow$ List monad.
- Free group $\dashv$ Forgetful $\rightsquigarrow$ Free-group monad.
- Discrete $\dashv$ Forgetful $\rightsquigarrow$ Identity monad (essentially trivial).
- Stone–Čech $\dashv$ Forgetful (for compact Hausdorff spaces) $\rightsquigarrow$ Stone–Čech monad on **Top**.

8.2.5 Simplifying: $T = RL$, $\mu = R\varepsilon L$, η as Given

That is the recipe. Memorize the recipe.

8.2.6 Formally: Comonads from Adjunctions

8.2.6.0.1 The dual: comonads

Just as RL is a monad on $\mathcal{C}$, the composite LR is a *comonad* on $\mathcal{D}$: a functor with a **counit** $\varepsilon : LR \Rightarrow \mathrm{Id}$ and a **comultiplication** $\delta : LR \Rightarrow LRLR$ satisfying the dual laws. Comonads formalize "contexts" or "streams" in computer science.

8.3 Algebras and the Eilenberg–Moore Category

8.3.1 Informally: A Monad Is an Algebraic Theory; Algebras Are Its Models

A monad on $\mathcal{C}$ encodes a kind of algebraic structure. The **algebras for the monad** are the objects of $\mathcal{C}$ carrying that structure.

Concretely: an algebra for the list monad on **Set** is an object X equipped with a structure map $\alpha : X^* \to X$ "compatible with list operations." This compatibility makes α a way of *multiplying* a list of X-elements into a single X-element — i.e., X becomes a monoid. The algebras for the list monad are *monoids*.

More generally: the algebras for the monad induced by an adjunction recover the right-hand category, up to equivalence (when nice conditions hold). This is the *monadicity* story.

8.3.2 Expanding: What Is an Algebra?

Example: A List-Monad Algebra Is a Monoid

Given $(X, \alpha : X^* \to X)$ satisfying the laws:

- $\alpha \circ \eta_X = \mathrm{id}_X$ (unit): the singleton list $[x]$ multiplies to x.
- $\alpha \circ \mu_X = \alpha \circ T\alpha$ (associativity): flattening and then multiplying equals multiplying each list and then multiplying the results.

From α we recover the monoid operations: the identity is $\alpha([\,])$ (the empty list), and the binary operation is $x \cdot y = \alpha([x, y])$. These operations satisfy the monoid axioms; conversely any monoid yields a list-algebra structure.

Example: A Powerset-Monad Algebra Is a Complete Sup-Semilattice

For the powerset monad $T(X) = \mathcal{P}(X)$ on **Set**, an algebra structure $\alpha : \mathcal{P}(X) \to X$ assigns to each subset a supremum, in a way satisfying the laws — exactly the structure of a complete sup-semilattice.

8.3.3 Formalizing: T-Algebras and the EM Category

For a monad (T, η, μ) on $\mathcal{C}$, a **T-algebra** is a pair (X, α) where:

- X is an object of $\mathcal{C}$.
- $\alpha : T(X) \to X$ is a morphism, the **structure map**.

Subject to:

$$\begin{aligned} \alpha \circ \eta_X &= \mathrm{id}_X && \text{(unit)} \\ \alpha \circ \mu_X &= \alpha \circ T\alpha && \text{(associativity).} \end{aligned}$$

A morphism of T-algebras $f : (X, \alpha) \to (Y, \beta)$ is a morphism $f : X \to Y$ of $\mathcal{C}$ such that $f \circ \alpha = \beta \circ T(f)$.

The category of T-algebras and their morphisms is the **Eilenberg–Moore category** of T, written $\mathcal{C}^T$ or T-**Alg**.

8.3.4 Reorganizing: The EM Adjunction

The Eilenberg–Moore category comes with a canonical adjunction $F^T \dashv U^T$:

- $U^T : \mathcal{C}^T \to \mathcal{C}$ sends (X, α) to X and a morphism to its underlying morphism.
- $F^T : \mathcal{C} \to \mathcal{C}^T$ sends X to $(T(X), \mu_X)$, the *free T-algebra* on X.

The monad induced by $F^T \dashv U^T$ is exactly the original T. This is the *universal* adjunction with this monad: any adjunction giving rise to T factors uniquely through the EM adjunction.

8.3.5 Simplifying: The Algebra Slogan

Algebras for a monad are the objects on which the monad's structure can be "evaluated."

8.3.6 Formally: Monadicity

8.3.6.0.1 Monadicity

An adjunction $L \dashv R : \mathcal{D} \to \mathcal{C}$ is **monadic** if the canonical comparison functor $K : \mathcal{D} \to \mathcal{C}^T$ (sending B to $(R(B), R\varepsilon_B)$) is an equivalence of categories. Many "algebraic" adjunctions are monadic: monoids over sets, groups over sets, rings over sets, R-modules over abelian groups. The Beck monadicity theorem gives a precise criterion (which we will not prove here).

8.4 The Kleisli Category

8.4.1 Informally: Effectful Maps as a New Category

Many "computations with effects" have the type $X \to T(Y)$ rather than $X \to Y$. A list-valued function $X \to Y^*$. A partial function $X \to Y_*$. A relation $X \to \mathcal{P}(Y)$. These should be the morphisms of *some* category. They are: the **Kleisli category** of T.

8.4.2 Expanding: Composition of Effectful Maps

If $f : X \to T(Y)$ and $g : Y \to T(Z)$, how do we compose them? Not naively, because the target of f is $T(Y)$, not Y. We need to "lift" g to $T(Y) \to T^2(Z)$, then flatten via μ:

$$g *_K f := \mu_Z \circ T(g) \circ f \quad : \quad X \to T(Z).$$

This is **Kleisli composition**. The identity at X is the unit $\eta_X : X \to T(X)$.

8.4.3 Formalizing: The Kleisli Category

For a monad (T, η, μ) on $\mathcal{C}$, the **Kleisli category** $\mathcal{C}_T$ has:

- Objects: same as $\mathcal{C}$.
- Morphisms $X \to Y$: morphisms $X \to T(Y)$ in $\mathcal{C}$.
- Composition: Kleisli composition as above.
- Identities: $\eta_X : X \to T(X)$ as the identity at X.

Theorem. $\mathcal{C}_T$ is a category. The Kleisli adjunction $F_T \dashv U_T$ between $\mathcal{C}$ and $\mathcal{C}_T$ gives rise to the same monad T. Among all adjunctions giving rise to T, the Kleisli adjunction is the *initial* one (every other such adjunction factors uniquely through it), while the EM adjunction is the *terminal* one.

8.4.4 Reorganizing: EM vs. Kleisli

Both categories give the same monad T. They sit at opposite ends:

- Kleisli $\mathcal{C}_T$: "the smallest category whose adjunction yields T."
- Eilenberg–Moore $\mathcal{C}^T$: "the largest such category."

Between them sit all adjunctions giving rise to T. The original adjunction $L \dashv R$ that we started from sits somewhere in this sandwich.

8.4.5 Simplifying: Kleisli for Programmers

In Haskell-flavoured pseudocode:

$$g *_K f = \lambda x.\ \text{“run } f \text{ on } x\text{, then run } g \text{ on the result, flattened.”}$$

This is the `>>=` ("bind") operator. The Kleisli category formalizes "programs with T-effects."

8.4.6 Formally: Kleisli for the List Monad

8.4.6.0.1 Kleisli for lists

For the list monad, $\mathcal{C}_T = \mathbf{Set}_T$ has objects sets and morphisms $X \to Y$ given by functions $X \to Y^*$. Kleisli composition of $f : X \to Y^*$ and $g : Y \to Z^*$ is: apply f to $x \in X$, yielding a list $[y_1, \dots, y_n]$; apply g to each y_i, yielding lists $[z_{i1}, \dots]$; concatenate them all into one list in Z^*. This is the categorical version of nondeterministic computation.

Exercises

What is a monad on $\mathcal{C}$?	A triple (T, η, μ) with T an endofunctor, $\eta : \mathrm{Id} \Rightarrow T$, $\mu : T^2 \Rightarrow T$, satisfying associativity and unit laws.
What is η called?	The unit (or "return").
What is μ called?	The multiplication (or "join").
State the associativity law.	$\mu \circ \mu T = \mu \circ T\mu$.
State the two unit laws.	$\mu \circ \eta T = \mathrm{id}_T = \mu \circ T\eta$.
Which functor on **Set** gives the list monad?	The free-monoid functor $X \mapsto X^*$.
What is the unit for the list monad?	$x \mapsto [x]$.
What is the multiplication for the list monad?	Concatenation of lists of lists.
What is the Maybe monad?	$T(X) = X + \{*\}$ with η the inclusion and μ the flattening of nested options.
What is the powerset monad?	$T(X) = \mathcal{P}(X)$ with $\eta = \{-\}$ and $\mu = \bigcup$.
How does every adjunction give a monad?	Set $T = RL$, $\eta =$ adjunction unit, $\mu = R\varepsilon L$.

Which adjunction gives the list monad?	Free monoid (left) $\dashv$ Forgetful (right) on **Set** $\rightleftarrows$ **Mon**.
What is a T-algebra?	A pair $(X, \alpha : TX \to X)$ satisfying $\alpha \circ \eta = \mathrm{id}$ and $\alpha \circ \mu = \alpha \circ T\alpha$.
What are the algebras for the list monad?	Monoids.
What are the algebras for the powerset monad?	Complete sup-semilattices.
What is a morphism of T-algebras?	A morphism f of underlying objects such that $f \circ \alpha = \beta \circ Tf$.
What is the Eilenberg–Moore category?	The category of T-algebras and algebra morphisms, denoted $\mathcal{C}^T$.
What is the free T-algebra on X?	$(T(X), \mu_X)$, the algebra given by the monad's own multiplication.
What adjunction does EM provide?	$F^T \dashv U^T : \mathcal{C}^T \to \mathcal{C}$, where U^T forgets the structure and F^T is the free algebra.
What monad does $F^T \dashv U^T$ produce?	The original T.
What is the Kleisli category $\mathcal{C}_T$?	Same objects as $\mathcal{C}$; morphisms $X \to Y$ are morphisms $X \to T(Y)$.
What is Kleisli composition?	$g *_K f = \mu \circ T(g) \circ f$.
What is the identity in the Kleisli category?	$\eta_X : X \to T(X)$.
What is the role of η in the Kleisli category?	The identity morphism on each object.
How does EM compare to Kleisli?	EM is the terminal (largest) adjunction giving T; Kleisli is the initial (smallest).
Is every monad induced by some adjunction?	Yes, via either EM or Kleisli.
What is a comonad?	The dual of a monad: (W, ε, δ) with $\varepsilon : W \Rightarrow \mathrm{Id}$, $\delta : W \Rightarrow W^2$.
Slogan: "A monad is a monoid in ..."?	"... the category of endofunctors with composition as tensor."
What does monadicity mean?	The original adjunction is equivalent to its EM adjunction.
Why care about Kleisli in computer science?	Kleisli composition formalizes "programs with effects"; bind in Haskell is Kleisli composition.
What is the trivial monad?	The identity functor with $\eta = \mu = \mathrm{id}$.

Chapter Summary

Chapter Summary

A **monad** on $\mathcal{C}$ is a triple (T, η, μ): an endofunctor with a unit and a multiplication satisfying associativity and unit laws. "A monoid in the category of endofunctors."
Every adjunction $L \dashv R$ gives a monad $T = RL$ with multiplication $\mu = R\varepsilon L$. Conversely, every monad is the monad of some adjunction — two canonical choices: **Eilenberg–Moore** (terminal) and **Kleisli** (initial).
A T**-algebra** is a structure map $\alpha : TX \to X$ subject to the monad laws. The category of T-algebras is the Eilenberg–Moore category $\mathcal{C}^T$. The **Kleisli category** $\mathcal{C}_T$ has morphisms $X \to T(Y)$ with Kleisli composition.
For the list monad: algebras are monoids, Kleisli morphisms are list-valued (nondeterministic) functions.

Formal Restatement

Formal Restatement

Monad. (T, η, μ) on $\mathcal{C}$ with $T : \mathcal{C} \to \mathcal{C}$, $\eta : \mathrm{Id} \Rightarrow T$, $\mu : T^2 \Rightarrow T$, satisfying

$$\mu \circ \mu T = \mu \circ T\mu, \qquad \mu \circ \eta T = \mathrm{id}_T = \mu \circ T\eta.$$

Adjunction $\Rightarrow$ monad. $L \dashv R$ yields $(RL, \eta, R\varepsilon L)$.

T-algebra. $(X, \alpha : TX \to X)$ with $\alpha \circ \eta_X = \mathrm{id}_X$ and $\alpha \circ \mu_X = \alpha \circ T\alpha$.

EM category. $\mathcal{C}^T$, the category of T-algebras and algebra morphisms; canonically adjoint to $\mathcal{C}$, producing T.

Kleisli category. $\mathcal{C}_T$, with $\mathrm{Hom}_{\mathcal{C}_T}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(X, TY)$ and Kleisli composition $g *_K f = \mu \circ T(g) \circ f$.

Slogan. "A monad is a monoid in the category of endofunctors."

Chapter 9 develops the Yoneda Lemma — the master theorem that ties together representable functors, naturality, and universal properties.

Chapter 9

The Yoneda Lemma

Chapter Overview

We come now to the most quoted single theorem of category theory: the **Yoneda Lemma**. The statement is short. The consequences are vast.

The starting point is the family of **hom-functors**. For each object A of $\mathcal{C}$, the functor $\mathrm{Hom}(A, -) : \mathcal{C} \to \mathbf{Set}$ records "how the rest of $\mathcal{C}$ maps out of A." This is a functor that captures A *from outside*: not what A is made of, but how it sits in the web of morphisms.

The Yoneda Lemma says, in one line:

$$\mathrm{Nat}(\mathrm{Hom}(A, -), F) \;\cong\; F(A).$$

Read: "Natural transformations from $\mathrm{Hom}(A, -)$ into any functor F are in bijection with elements of $F(A)$." The bijection is given by "evaluate the transformation at the identity id_A."

From this short statement flow many fundamental facts:

- An object is completely determined by its hom-functor (up to iso). This is the **Yoneda embedding** of $\mathcal{C}$ into $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.
- Universal properties can be cleanly recast as **representability**: a functor is representable iff it is isomorphic to some $\mathrm{Hom}(A, -)$.
- Many "natural" constructions in mathematics are, after Yoneda, just *evaluation at the right object*.

We give the lemma, prove it, and then walk through its three signature applications.

Where the Name Comes From

Yoneda lemma is named for *Nobuo Yoneda* (1930–1996, Japanese mathematician). The most-told story: Yoneda communicated the result to Saunders Mac Lane on a Paris train platform during a 1954 mathematics conference; Mac Lane carried it back to the West. Yoneda himself may never have published the lemma in its now-standard form — he was primarily an algebraic topologist working on Ext groups, and the lemma was for him a tool, not a centerpiece. The lemma has since become *the* central organizing theorem of category theory; appears in Vol I Ch. 9 (this), Vol II Ch. 15 (representability), Vol III Ch. 28 (formal via ends), Vol IV Ch. 48 (∞-categorical).

Representable functor: a functor isomorphic to some $\mathrm{Hom}(A, -)$. The term "representable" is from Eilenberg-Mac Lane; "representing object" is the object A doing the representing. The Yoneda philosophy: an object is determined by *what maps into it* or *out of it*.

9.1 Hom-Functors and Representability

9.1.1 Informally: An Object Seen Through Its Outgoing Maps

Fix an object A in $\mathcal{C}$. For each object B, the hom-set $\mathrm{Hom}(A, B)$ tells us "how many ways there are to map A into B, and what they look like." As B varies, these hom-sets fit together into a functor $\mathrm{Hom}(A, -) : \mathcal{C} \to \mathbf{Set}$, with action on a morphism $g : B \to C$ given by postcomposition: $g_*(f) = g \circ f$.

The functor $\mathrm{Hom}(A, -)$ *is* A, seen from outside. Everything about how A relates to other objects is recorded there.

9.1.2 Expanding: The Two Hom-Functors

Example: Covariant Hom-Functor

$\mathrm{Hom}(A, -) : \mathcal{C} \to \mathbf{Set}$ sends:

- Object B to the set $\mathrm{Hom}(A, B)$.
- Morphism $g : B \to C$ to the function $g_* = (g \circ -) : \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, C)$.

Functoriality is immediate: $(g \circ f)_* = g_* \circ f_*$ and $(\mathrm{id})_* = \mathrm{id}$, both by associativity and identity laws.

Example: Contravariant Hom-Functor

$\mathrm{Hom}(-, A) : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ sends:

- Object B to the set $\mathrm{Hom}(B, A)$.
- Morphism $g : B \to C$ (in $\mathcal{C}$, reversed in $\mathcal{C}^{\mathrm{op}}$) to the function $g^* = (- \circ g) : \mathrm{Hom}(C, A) \to \mathrm{Hom}(B, A)$, given by *pre*composition.

Internal Dialog

Two hom-functors per object, one covariant and one contravariant. The covariant one is “maps out of A”; the contravariant is “maps into A.”

9.1.3 Formalizing: Representable Functors

A functor $F : \mathcal{C} \to \mathbf{Set}$ is **representable** if there exists an object A of $\mathcal{C}$ and a natural isomorphism

$$F \;\cong\; \mathrm{Hom}(A, -).$$

The object A is called a **representing object** for F (it is unique up to isomorphism when it exists).

Dually, $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is representable if $F \cong \mathrm{Hom}(-, A)$ for some A.

How to Say This

- “representable” — the functor is “representable by A.”
- $\mathrm{Hom}(A, -)$ — “the covariant hom out of A,” “A’s functor of points,” or simply “the representable at A.”
- $\mathrm{Hom}(-, A)$ — “the contravariant hom into A,” or “the representable presheaf at A.”

9.1.4 Reorganizing: Universal Properties as Representability

Most universal properties can be repackaged as “the functor of X with such-and-such property is representable.” For example: a product $A \times B$ represents the functor $X \mapsto \mathrm{Hom}(X, A) \times \mathrm{Hom}(X, B)$. A pullback represents the functor of pairs of morphisms agreeing under composition. This packaging is uniform, and it is the door to the Yoneda Lemma.

9.1.5 Simplifying: The Hom-Functor Recipe

To check whether $F : \mathcal{C} \to \mathbf{Set}$ is representable, look for an A such that $F(B) \cong \mathrm{Hom}(A, B)$ in a way natural in B. If such an A exists, F is representable; the universal element of F is the image of id_A in $F(A)$ under the corresponding isomorphism.

9.1.6 Formally: Naturality of the Hom-Functor

9.1.6.0.1 Hom as a bifunctor

The two-sided $\mathrm{Hom}(-, -) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$ is a bifunctor: contravariant in the first slot, covariant in the second. The covariant and contravariant hom-functors are obtained by fixing the appropriate slot.

9.2 The Yoneda Lemma

9.2.1 Informally: A Natural Transformation Is Just an Element

Suppose we are given a functor $F : \mathcal{C} \to \mathbf{Set}$ and an object A of $\mathcal{C}$. How many natural transformations $\alpha : \mathrm{Hom}(A, -) \Rightarrow F$ are there?

Pick any natural transformation α. Its component at A is a function $\alpha_A : \mathrm{Hom}(A, A) \to F(A)$. Apply it to the special element $\mathrm{id}_A \in \mathrm{Hom}(A, A)$:

$$x = \alpha_A(\mathrm{id}_A) \in F(A).$$

The Yoneda Lemma says: this single element x determines α *completely*, and conversely every $x \in F(A)$ arises this way from a unique α.

9.2.2 Expanding: The Bijection

Given $x \in F(A)$, define $\alpha^x : \mathrm{Hom}(A, -) \Rightarrow F$ by:

$$\alpha^x_B(f : A \to B) := F(f)(x) \in F(B).$$

That is, transport x from $F(A)$ to $F(B)$ along $F(f)$.

The two assignments

$$\alpha \;\mapsto\; \alpha_A(\mathrm{id}_A) \quad \text{and} \quad x \;\mapsto\; \alpha^x$$

are inverse to each other. They establish a bijection between natural transformations $\mathrm{Hom}(A, -) \Rightarrow F$ and elements of $F(A)$.

9.2.3 Formalizing: The Statement

The Yoneda Lemma. For any functor $F : \mathcal{C} \to \mathbf{Set}$ and any object A of $\mathcal{C}$, there is a bijection

$$\boxed{\mathrm{Nat}(\mathrm{Hom}(A, -),\ F) \xrightarrow{\cong} F(A), \quad \alpha \;\mapsto\; \alpha_A(\mathrm{id}_A).}$$

This bijection is natural in both A (contravariantly) and F (covariantly).

Proof. The map "$\alpha \mapsto \alpha_A(\mathrm{id}_A)$" is well-defined. We exhibit an inverse: given $x \in F(A)$, define $\alpha^x_B(f) := F(f)(x)$. Naturality of α^x comes for free from functoriality of F:

$$\begin{aligned} \alpha^x_C(g \circ f) &= F(g \circ f)(x) && \text{(definition)} \\ &= F(g)(F(f)(x)) && \text{(functoriality)} \\ &= F(g)(\alpha^x_B(f)) && \text{(definition).} \end{aligned}$$

This is the naturality square at the morphism $g : B \to C$ applied to f.

To see "$\alpha \mapsto \alpha_A(\mathrm{id}_A) \mapsto \alpha^{\alpha_A(\mathrm{id}_A)}$" is the identity, fix any $f : A \to B$. We need $\alpha_B^{\alpha_A(\mathrm{id}_A)}(f) = \alpha_B(f)$:

$$\begin{aligned} \alpha_B^{\alpha_A(\mathrm{id}_A)}(f) &= F(f)(\alpha_A(\mathrm{id}_A)) && \text{(definition of } \alpha^{(-)}) \\ &= \alpha_B(f \circ \mathrm{id}_A) && \text{(naturality of } \alpha \text{ at } f) \\ &= \alpha_B(f). \end{aligned}$$

The reverse composite "$x \mapsto \alpha^x \mapsto \alpha^x_A(\mathrm{id}_A)$" gives $F(\mathrm{id}_A)(x) = x$. So we have inverse bijections. □

9.2.4 Reorganizing: The Contravariant Version

Dually, for a contravariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$:

$$\mathrm{Nat}(\mathrm{Hom}(-, A),\ F) \;\cong\; F(A).$$

The bijection again sends α to $\alpha_A(\mathrm{id}_A)$.

9.2.5 Simplifying: The One-Line Slogan

$$\boxed{\mathrm{Nat}(\mathrm{Hom}(A, -), F) \;\cong\; F(A) \;\; (\text{natural in } A,\ F).}$$

9.2.6 Formally: Naturality of the Yoneda Isomorphism

9.2.6.0.1 Naturality in F and A

The bijection of the Yoneda Lemma is natural in two variables. For a morphism $f : A' \to A$ in $\mathcal{C}$, we get a natural transformation $f^* : \mathrm{Hom}(A, -) \Rightarrow \mathrm{Hom}(A', -)$ given by precomposition, and the Yoneda bijection respects this. For a natural transformation $\theta : F \Rightarrow G$, we similarly get compatible maps. These coherences are what allow Yoneda to upgrade to the Yoneda *embedding*.

9.3 The Yoneda Embedding

9.3.1 Informally: A Category Inside a Bigger Category

Apply Yoneda with $F = \mathrm{Hom}(B, -)$ for a second object B:

$$\mathrm{Nat}(\mathrm{Hom}(A, -),\ \mathrm{Hom}(B, -)) \;\cong\; \mathrm{Hom}(B, A).$$

So natural transformations between the hom-functors of A and B are exactly the morphisms $B \to A$ in $\mathcal{C}$. The hom-functors faithfully encode $\mathcal{C}$.

This gives us a functor: take each object A to its (contravariant) hom-functor $\mathrm{Hom}(-, A)$. The Yoneda Lemma says this functor is fully faithful. We have embedded $\mathcal{C}$ inside the much larger category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ of presheaves on $\mathcal{C}$.

9.3.2 Expanding: The Embedding in Detail

Define
$$Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$$
by $Y(A) = \mathrm{Hom}(-, A)$ on objects, and $Y(g : A \to A') = (- \circ g)_*$ on morphisms (postcomposition with g in each component).

By the Yoneda Lemma applied to $F = \mathrm{Hom}(-, A')$:
$$\mathrm{Hom}_{[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]}(Y(A), Y(A')) \;\cong\; \mathrm{Hom}(-, A')(A) = \mathrm{Hom}(A, A').$$

So Y is fully faithful: it gives a bijection between morphisms in $\mathcal{C}$ and natural transformations between the representable presheaves. Hence $\mathcal{C}$ sits inside $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ as a full subcategory, via the Yoneda embedding.

9.3.3 Formalizing: The Embedding Theorem

Theorem (Yoneda Embedding). The functor
$$Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}], \quad A \mapsto \mathrm{Hom}(-, A)$$
is fully faithful. It identifies $\mathcal{C}$ with the full subcategory of representable presheaves.

9.3.4 Reorganizing: Three Consequences

1. Objects are determined by their hom-functors. If $\mathrm{Hom}(-, A) \cong \mathrm{Hom}(-, B)$ as presheaves, then $A \cong B$. (A fully faithful functor reflects isomorphism.)

2. Universal properties name objects up to canonical iso. A universal property describes an object as the representing object of a functor. By Yoneda, this object is unique up to canonical iso.

3. "Set is the universe of presheaves." $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the "free completion" of $\mathcal{C}$ under colimits, in a precise sense. Every presheaf is a colimit of representables.

9.3.5 Simplifying: The Yoneda Slogan

$$\boxed{\mathcal{C} \hookrightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}], \quad A \mapsto \mathrm{Hom}(-, A).}$$

9.3.6 Formally: Presheaves and Probing

9.3.6.0.1 Presheaves

A **presheaf on $\mathcal{C}$** is a functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$. Yoneda says $\mathcal{C}$ embeds in its category of presheaves. The full subcategory of representable presheaves is $Y(\mathcal{C})$, equivalent to $\mathcal{C}$ itself.

9.3.6.0.2 Probing with representables

For any presheaf F, the Yoneda Lemma says $F(A) \cong \mathrm{Nat}(YA, F)$. So evaluating F at A is the same as "probing" F with the representable at A. This is the perspective that motivates the slogan "points of a sheaf are maps from a representable."

9.4 Applications of Yoneda

9.4.1 Informally: Three Patterns, One Theorem

We have three signature uses of Yoneda. Each is a translation: "a property of an object" becomes "a property of the corresponding representable."

9.4.2 Expanding: Three Applications

Example: Application 1: Adjunctions Restated

$L \dashv R$ iff $\mathrm{Hom}(LA, B) \cong \mathrm{Hom}(A, RB)$ naturally. By Yoneda, this is the same as a natural isomorphism $\mathrm{Hom}(LA, -) \cong \mathrm{Hom}(A, R-)$, or alternatively $\mathrm{Hom}(-, RB) \cong \mathrm{Hom}(L-, B)$. So adjunction is a relation between hom-functors, exposed by Yoneda.

Example: Application 2: Equality of Functors

To show two functors $F, G : \mathcal{C} \to \mathcal{D}$ are isomorphic, it suffices to show $\mathrm{Hom}(-, F(A)) \cong \mathrm{Hom}(-, G(A))$ naturally in A. By Yoneda, this is equivalent to $F(A) \cong G(A)$ canonically, hence to $F \cong G$ as functors. Many "$F = G$" arguments in practice are Yoneda arguments in disguise.

Example: Application 3: Representability Lifts

If F is representable, then F preserves all (small) limits that exist in $\mathcal{C}$. (This is a special case of the fact from Chapter 6 that $\mathrm{Hom}(A, -)$ preserves limits.) So representability is a strong constraint: it forces a functor to be limit-preserving.

9.4.3 Formalizing: Density of Representables

Theorem (Density). Every presheaf $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is canonically a colimit of representables. More specifically, F is the colimit of the diagram

$$\mathrm{el}(F) \to \mathcal{C} \xrightarrow{Y} [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}],$$

where $\mathrm{el}(F)$ is the category of elements of F (pairs (A, x) with $x \in F(A)$).

9.4.4 Reorganizing: The Yoneda Toolkit

To use Yoneda effectively:

1. Translate the problem about objects of $\mathcal{C}$ into a problem about representable presheaves.
2. Use the embedding's full-faithfulness or the Yoneda bijection.
3. Translate back if needed.

This is the move "check on representables." It is sometimes called *the Yoneda manoeuvre* or simply *by Yoneda*.

9.4.5 Simplifying: The Yoneda Maxim

> *To know an object, know its hom-functor. To know a functor, know how it acts on representables. To compare two things, compare them on representables.*

9.4.6 Formally: The Co-Yoneda Lemma

9.4.6.0.1 Co-Yoneda

The dual statement: for a presheaf F, the canonical map from the colimit $\int^A F(A) \times \mathrm{Hom}(-, A)$ to F is an isomorphism. This is the formal statement of "every presheaf is a colimit of representables." The notation $\int^A$ here is a coend, a kind of co-limit indexed over a "twisted" diagram. Coends are beyond the scope of this chapter.

Exercises

What is $\mathrm{Hom}(A, -)$?	The functor $\mathcal{C} \to \mathbf{Set}$ sending B to the set of morphisms $A \to B$.
Is $\mathrm{Hom}(A, -)$ covariant or contravariant?	Covariant.
What is $\mathrm{Hom}(-, A)$?	The functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ sending B to morphisms $B \to A$.
How does $\mathrm{Hom}(A, -)$ act on $g : B \to C$?	Postcomposition: $f \mapsto g \circ f$.
How does $\mathrm{Hom}(-, A)$ act on $g : B \to C$?	Precomposition: $f \mapsto f \circ g$.
What does it mean for F to be representable?	$F \cong \mathrm{Hom}(A, -)$ for some A.
Is the representing object unique?	Yes, up to isomorphism.
State the Yoneda Lemma.	$\mathrm{Nat}(\mathrm{Hom}(A, -), F) \cong F(A)$.
What is the explicit bijection?	$\alpha \mapsto \alpha_A(\mathrm{id}_A)$.
What is the inverse map?	$x \mapsto \alpha^x$ where $\alpha^x_B(f) = F(f)(x)$.

Why is the inverse α^x natural?	Because F is a functor: $F(g \circ f) = F(g)F(f)$.
What does $\alpha_A(\mathrm{id}_A)$ represent?	The "universal element" that determines α.
Yoneda for contravariant functors: state it.	$\mathrm{Nat}(\mathrm{Hom}(-, A), F) \cong F(A)$ for $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.
What is the Yoneda embedding?	The functor $A \mapsto \mathrm{Hom}(-, A)$ from $\mathcal{C}$ into $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.
Is the Yoneda embedding faithful?	Yes.
Is it full?	Yes (this is what the Yoneda Lemma supplies).
So the Yoneda embedding is what?	Fully faithful.
What is a presheaf on $\mathcal{C}$?	A functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.
What is a representable presheaf?	One of the form $\mathrm{Hom}(-, A)$.
Consequence: when are two objects isomorphic?	Iff their representables are isomorphic as presheaves.
Hom-functors preserve what?	All small limits.
Therefore representable functors preserve what?	All small limits.
What does the density theorem say?	Every presheaf is canonically a colimit of representables.
What is the slogan for using Yoneda?	"Check on representables."
Yoneda restatement of adjunction $L \dashv R$?	$\mathrm{Hom}(LA, -) \cong \mathrm{Hom}(A, R-)$ as natural transformations.
Yoneda restatement of $A \cong B$?	$\mathrm{Hom}(A, -) \cong \mathrm{Hom}(B, -)$ naturally.
What is the "category of elements" of a presheaf F?	The category with objects (A, x), $x \in F(A)$, and morphisms $f : A \to A'$ with $F(f)(x') = x$.
Why is the Yoneda Lemma considered "deep"?	It links naturality, representability, and elements in a single short statement.
What was the most surprising consequence to you?	Often: that universal properties identify objects up to unique iso *via Yoneda*.
Does Yoneda hold for general $\mathcal{V}$-enriched categories?	Yes; it generalizes essentially verbatim, with **Set** replaced by $\mathcal{V}$.

Chapter Summary

Chapter Summary

The **hom-functors** $\mathrm{Hom}(A, -)$ and $\mathrm{Hom}(-, A)$ record how an object A relates to all others. A functor is **representable** if it is naturally isomorphic to such a hom-functor.
The **Yoneda Lemma**: natural transformations from $\mathrm{Hom}(A, -)$ to any functor F are in bijection with elements of $F(A)$. Explicitly: $\alpha \mapsto \alpha_A(\mathrm{id}_A)$.
The **Yoneda Embedding** $A \mapsto \mathrm{Hom}(-, A)$ is fully faithful, embedding $\mathcal{C}$ into the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.
Consequence: objects are determined by their hom-functors (up to iso), representable functors preserve limits, and every presheaf is a colimit of representables (**density**).
The slogan: **check on representables**.

Formal Restatement

Formal Restatement

Hom-functor. $\mathrm{Hom}(A, -) : \mathcal{C} \to \mathbf{Set}$ sends B to $\mathrm{Hom}(A, B)$ and $g : B \to C$ to $g_* = (g \circ -)$.

Representability. $F : \mathcal{C} \to \mathbf{Set}$ is representable iff $F \cong \mathrm{Hom}(A, -)$ for some A.

Yoneda Lemma. For $F : \mathcal{C} \to \mathbf{Set}$ and $A \in \mathcal{C}$:

$$\mathrm{Nat}(\mathrm{Hom}(A, -),\ F) \ \cong\ F(A), \quad \alpha \ \mapsto\ \alpha_A(\mathrm{id}_A).$$

The bijection is natural in A and F.

Yoneda Embedding. $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$, $A \mapsto \mathrm{Hom}(-, A)$. Fully faithful.

Density. Every presheaf F is a colimit of representables, indexed by $\mathrm{el}(F)$.

Chapter 10 uses representability and Yoneda to develop the Adjoint Functor Theorems — general existence theorems for left and right adjoints.

Chapter 10

Adjoint Functor Theorems

Chapter Overview

Chapter 7 explained *what* adjunctions are and showed that they are ubiquitous. But it left open a hard question: given a functor R, how do we know whether it has a left adjoint? In examples we often guess the left adjoint and verify the bijection. But for many naturally occurring functors there is no obvious candidate.

This chapter develops the **Adjoint Functor Theorems** (AFTs): existence theorems that say, under suitable conditions, the answer is yes. The two main statements are:

- **The General Adjoint Functor Theorem (GAFT).** If R preserves all small limits and satisfies a smallness condition (the *solution-set condition*), then R has a left adjoint.
- **The Special Adjoint Functor Theorem (SAFT).** If R preserves all small limits, and its source category is *well-powered, complete, and has a small cogenerating set*, then R has a left adjoint.

The dual statements give criteria for right adjoints. Behind both theorems lies the same idea: a left adjoint LA would have to be the "initial object" in the category of A-pointed targets; the theorems provide conditions guaranteeing this initial object exists.

This is the natural climax of the book to date. We have built the cycle all the way from objects and morphisms through limits, adjunctions, and Yoneda. The Adjoint Functor Theorems use *all* of this machinery.

Where the Name Comes From

The **Adjoint Functor Theorem** (AFT) was proved by *Peter Freyd* in his 1964 book *Abelian Categories*. Freyd's theorem characterized when a functor between locally small categories admits a left/right adjoint, in terms of two conditions: (1) preserves the appropriate limits/colimits; (2) satisfies a smallness condition (the *solution-set condition*). The two main variants are:

- **GAFT** (General AFT): solution-set condition.
- **SAFT** (Special AFT): well-powered + cogenerating set.

The **solution-set condition** is named for the technique it uses: to find a left adjoint LA, one needs a "solution set" — a small collection of arrows out of A that captures every possible target. The condition trades size for existence, in the standard set-theoretic manner. Modern formulations (Vol IV Ch. 52, presentable case) replace solution-set by accessibility, but the spirit is unchanged.

10.1 The Adjoint Functor Question

10.1.1 Informally: Existence Without Construction

When we have an explicit construction of a left adjoint (free monoid, free group, free abelian group, abelianization, ...), we can check the hom-set bijection directly. But the categorical perspective sometimes gives us a functor R *first*, and asks whether a left adjoint *exists*, without telling us how to build one.

For example: "Does the inclusion of compact Hausdorff spaces into all topological spaces have a left adjoint?" Yes — it is the Stone–Čech compactification. But this is a hard theorem to prove directly; one might want a general criterion that gives existence for free.

The Adjoint Functor Theorems are exactly such criteria.

10.1.2 Expanding: A Reformulation via Initial Objects

Suppose we want a left adjoint LA for a fixed object A of $\mathcal{C}$. Define the **comma category** $(A \downarrow R)$:

- Objects: pairs (B, f) with $B \in \mathcal{D}$ and $f : A \to RB$.
- Morphisms $(B, f) \to (B', f')$: morphisms $g : B \to B'$ in $\mathcal{D}$ such that $R(g) \circ f = f'$.

The object LA together with the unit $\eta_A : A \to R(LA)$ is an *initial object* of $(A \downarrow R)$: every other pair (B, f) admits a unique morphism from (LA, η_A) to itself. So:

Existence of $LA \iff$ *existence of an initial object in the comma category* $(A \downarrow R)$.

So the existence of a left adjoint reduces to: do certain comma categories all have initial objects?

10.1.3 Formalizing: The Reduction

Theorem. A functor $R : \mathcal{D} \to \mathcal{C}$ has a left adjoint L iff for every $A \in \mathcal{C}$, the comma category $(A \downarrow R)$ has an initial object. In that case, the initial object of $(A \downarrow R)$ is (LA, η_A), where η is the unit.

This is the universal-arrow form of the adjunction (§7.1, the third equivalent definition).

10.1.4 Reorganizing: When Does $(A \downarrow R)$ Have an Initial Object?

In general, having an initial object is a *strong* condition — many categories have none. But $(A \downarrow R)$ inherits structure from $\mathcal{C}$ and $\mathcal{D}$. The strategy of the AFTs is:

- **Get limits in $(A \downarrow R)$.** If R preserves limits in $\mathcal{D}$ and $\mathcal{C}$ has the right properties, then $(A \downarrow R)$ has small limits.
- **Take a weakly initial set.** A *set* of objects from which every other object has a morphism. Limits over the weakly initial set yield an initial object.

The smallness condition — existence of a weakly initial set — is exactly the **solution-set condition** of GAFT.

10.1.5 Simplifying: The Strategy in One Sentence

To show R has a left adjoint, show R preserves small limits and that for each A the "cone of targets" over A is small.

10.1.6 Formally: Comma Categories Live Above Both Sides

10.1.6.0.1 Comma categories

The comma category $(A \downarrow R)$ has projection functors to both $\mathcal{C}$ (constant at A, essentially) and $\mathcal{D}$ (sending (B, f) to B). Limits in $(A \downarrow R)$ are computed by taking limits of the underlying objects and morphisms; the universal property of the projection makes this work.

10.2 The General Adjoint Functor Theorem

10.2.1 Informally: Limits + Smallness $\Rightarrow$ Adjoint

GAFT is the most general of the adjoint-functor theorems.

Theorem (Freyd's General Adjoint Functor Theorem). Let $\mathcal{D}$ be a complete category, and $R : \mathcal{D} \to \mathcal{C}$ a functor. Then R has a left adjoint iff:

1. R preserves all small limits, and
2. R satisfies the **solution-set condition**: for every $A \in \mathcal{C}$, there is a set I of objects $\{B_i\}_{i \in I}$ in $\mathcal{D}$ and morphisms $f_i : A \to R(B_i)$ such that every morphism $A \to RB$ in $\mathcal{C}$ factors as $A \xrightarrow{f_i} RB_i \xrightarrow{R(g)} RB$ for some i and some $g : B_i \to B$ in $\mathcal{D}$.

10.2.2 Expanding: Reading the Solution-Set Condition

The solution-set condition says: for each A, there is a *set* (not a proper class) of "candidate units" $f_i : A \to R(B_i)$ such that every actual unit factors through one of them. This is a *smallness* condition — it tames a potentially proper-class-sized collection of candidates down to a set.

In practice, the solution-set condition holds in any reasonable category. The condition is automatic if $\mathcal{D}$ has, for instance, a small dense subcategory or a small cogenerating set.

10.2.3 Formalizing: The Statement, Again

For clarity, restate. Given $R : \mathcal{D} \to \mathcal{C}$ with $\mathcal{D}$ complete:

$$\begin{aligned} & R \text{ has a left adjoint } L \\ \iff & R \text{ preserves small limits} \\ & \quad \text{and for each } A \in \mathcal{C},\ \exists \text{ set } \{(B_i, f_i)\}_{i \in I} \\ & \quad \text{such that any } (B, f) \text{ factors through some } (B_i, f_i). \end{aligned}$$

Proof sketch. ($\Leftarrow$) Build LA as the limit of the diagram of all factorizations in the comma category through the solution set. Use limit-preservation of R to verify the universal property. The solution-set condition is what guarantees this limit is indexed by a small enough diagram.

($\Rightarrow$) If L exists, then (LA, η_A) is the initial object of $(A \downarrow R)$; take $\{LA\}$ to be the solution set. Limit preservation is RAPL. □

10.2.4 Reorganizing: When Each Hypothesis Bites

Completeness of $\mathcal{D}$ is what lets us construct LA as a limit.

R **preserves limits** is what makes the constructed LA have the right universal property.

Solution-set condition is what makes the construction *size-feasible*: without it, the relevant "limit" would be over a proper class.

10.2.5 Simplifying: GAFT as a Recipe

To verify GAFT:

1. Check $\mathcal{D}$ has all small limits.
2. Check R preserves them.
3. Check the solution-set condition (often the trickiest step).

10.2.6 Formally: The Dual for Right Adjoints

10.2.6.0.1 GAFT for right adjoints

By duality: a functor $L : \mathcal{C} \to \mathcal{D}$ with $\mathcal{D}$ cocomplete has a right adjoint iff L preserves all small colimits and satisfies the dual solution-set condition.

10.3 The Special Adjoint Functor Theorem

10.3.1 Informally: A Cleaner Statement Under Extra Conditions

GAFT is general but requires checking the solution-set condition, which is sometimes awkward. SAFT replaces the solution-set condition with two cleaner structural assumptions on $\mathcal{D}$: *well-poweredness* and *a small cogenerating set*.

10.3.2 Expanding: Well-Powered and Cogenerating

Well-powered. A category $\mathcal{D}$ is **well-powered** if for every object B, the collection of subobjects of B (equivalence classes of monomorphisms into B) is a set, not a proper class. This holds in almost every category one meets in practice.

Cogenerating set. A set $\{S_j\}_{j \in J}$ in $\mathcal{D}$ is **cogenerating** (or a "set of cogenerators") if the functor

$$\mathcal{D} \to \prod_j \mathbf{Set}, \quad B \mapsto (\mathrm{Hom}(B, S_j))_{j \in J}$$

is faithful; equivalently, morphisms $f, g : B \to B'$ are equal iff $h \circ f = h \circ g$ for every $h : B' \to S_j$ for every j.

10.3.3 Formalizing: SAFT

Theorem (Special Adjoint Functor Theorem). Let $\mathcal{D}$ be complete, well-powered, and have a small cogenerating set. Let $R : \mathcal{D} \to \mathcal{C}$ be a functor. Then R has a left adjoint iff R preserves all small limits.

10.3.4 Reorganizing: Why SAFT Is Easier

The two structural conditions on $\mathcal{D}$ (well-powered and cogenerating set) together imply the solution-set condition of GAFT for every R. So under SAFT's hypotheses, you do not need to verify solution-sets separately; you only check limit-preservation.

In categories like **CHaus** (compact Hausdorff spaces) or any locally presentable category, SAFT's hypotheses are automatic, so the existence of an adjoint reduces to checking limit-preservation.

10.3.5 Simplifying: SAFT in Practice

For "well-behaved" target categories: limit-preservation is enough.

10.3.6 Formally: Locally Presentable Categories

10.3.6.0.1 A modern reformulation

The most powerful adjoint-functor theorem in current use is the AFT for **locally presentable categories**: in a locally presentable category, a functor between locally presentable categories has a left adjoint iff it preserves limits. This subsumes both GAFT and SAFT for an enormous class of practical examples.

10.4 Worked Examples

10.4.1 Informally: Watching the Theorems Work

We apply the theorems to three classical adjoints that we have seen existing for "other" reasons, and observe the AFT machinery at work.

10.4.2 Expanding: Three Workouts

Example: Stone–Čech Compactification, via SAFT

The inclusion i : **CHaus** $\hookrightarrow$ **Top** has a left adjoint β (the Stone–Čech compactification). To prove existence by SAFT, we need:

- **CHaus** is complete (yes; limits of compact Hausdorff spaces are compact Hausdorff).
- **CHaus** is well-powered (yes; subobjects of a compact Hausdorff space are closed subsets, which form a set).
- **CHaus** has a small cogenerating set: $\{[0,1]\}$ is one. (Two morphisms into a CHaus space agree iff they agree after composition with every map to $[0,1]$, which is Urysohn-like.)
- i preserves limits (yes; it preserves the underlying topology).

SAFT delivers the adjoint β without our needing to construct it explicitly.

Example: Free R-Module Functor, via GAFT

Let R be a ring. The forgetful U : R-**Mod** $\to$ **Set** has a left adjoint (the free R-module functor). R-**Mod** is complete, U preserves limits, and the solution-set condition holds with a singleton set: $\{R^X\}$, the free module on the set X. GAFT delivers existence.

Example: Sheafification, via SAFT in $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$

The inclusion of sheaves into presheaves is a fully faithful functor with a left adjoint, called **sheafification**. SAFT applies: the presheaf category is complete, well-powered, and the family of representables $\{Y(C)\}_{C\in\mathcal{C}}$ is jointly cogenerating. The inclusion of sheaves preserves limits because limits of sheaves are computed pointwise. SAFT gives sheafification as a left adjoint.

10.4.3 Formalizing: A General Recipe

When confronted with "does R have a left adjoint?", try SAFT first (it requires less work to verify). If $\mathcal{D}$ is not known to be well-powered or cogenerated, fall back to GAFT and check the solution-set condition by hand.

If even GAFT fails, there is a fundamental obstruction: either R does not preserve limits (and so no left adjoint exists at all), or the candidate left adjoint involves a proper-class-sized colimit and is not representable.

10.4.4 Reorganizing: When Adjoints Fail to Exist

- The inclusion **Set** $\hookrightarrow$ **Class** has no left adjoint, because **Class** is not a category in the usual sense (size issues).
- The forgetful **TopGrp** $\to$ **Grp** (forgetting the topology of a topological group) has no left adjoint, because the candidate would be a free topological group on a discrete group, which exists but is hard to verify without further smallness.

10.4.5 Simplifying: The Final Slogan

$$\boxed{\text{Preserves all limits} + \text{Smallness} \;\Rightarrow\; \text{Has a left adjoint.}}$$

10.4.6 Formally: Connection to Yoneda

10.4.6.0.1 AFT and representability

The Adjoint Functor Theorems are really representability theorems in disguise. Asking whether R has a left adjoint is asking whether each of the presheaves $\mathrm{Hom}_{\mathcal{C}}(A, R(-))$ is representable. The theorems give conditions under which a presheaf is representable, and left-adjoint-existence is a special case.

Exercises

What question does an Adjoint Functor Theorem answer?	Given a functor R, does it have a left adjoint?
When does R certainly *not* have a left adjoint?	When R fails to preserve some small limit.
Why?	By RAPL: right adjoints must preserve all small limits.
What is the comma category $(A \downarrow R)$?	Pairs $(B, f : A \to RB)$, with the obvious morphisms.
Why does $(A \downarrow R)$ matter for adjunctions?	LA exists iff $(A \downarrow R)$ has an initial object.
What initial object does it have?	(LA, η_A), the universal arrow from A to R.
State GAFT in slogan form.	Complete target, R preserves limits, plus solution-set $\Rightarrow R$ has a left adjoint.
What is the solution-set condition?	For each A, a set of $f_i : A \to RB_i$ such that every $f : A \to RB$ factors through one.
Why does the condition need a *set*?	To avoid proper-class-sized constructions.
In the SAFT, which two extra assumptions replace the solution-set condition?	Well-poweredness and a small cogenerating set.
What is a well-powered category?	One in which each object has only a set's worth of subobjects.
What is a cogenerating set?	A set $\{S_j\}$ such that hom-into-cogenerators is jointly faithful.
Give an example of a cogenerator in **CHaus**.	$[0, 1]$.
Where does SAFT apply that GAFT doesn't need to be invoked?	When the target category is locally presentable; SAFT often suffices.
What is the connection to Yoneda?	Both GAFT and SAFT are representability theorems for the presheaves $\mathrm{Hom}(A, R(-))$.
What strategy underlies both theorems?	Construct LA as a limit over the comma category, controlled by a smallness hypothesis.
Why does completeness of $\mathcal{D}$ enter?	Because we construct LA as a limit of objects in $\mathcal{D}$.
Why does limit-preservation of R enter?	Because LA's universal property must hold after applying R.
Where does the solution-set condition enter the proof?	To ensure the limit we take is over a set-sized diagram.
Can a functor have neither a left nor a right adjoint?	Yes; many do.
Can a functor have both?	Yes; e.g., the identity functor has itself on both sides.
What does "f^* has both adjoints" say in logic?	There are both existential and universal quantifiers, given by left and right adjoints.
Is the forgetful functor **Mon** $\to$ **Set** adjoint-bearing?	Yes; it has a left adjoint (free monoid), no right adjoint.
Why no right adjoint to forgetful **Mon** $\to$ **Set**?	The forgetful doesn't preserve coproducts on the nose.
In a locally presentable category, what's the criterion for an adjoint?	Limit preservation suffices.
What's the dual of GAFT for right adjoints?	Cocomplete target, L preserves colimits, plus dual solution-set condition.
Sheafification is an example of which theorem applied?	SAFT.

Why is the AFT considered a deep theorem?	It turns existence-of-adjoint into a checkable list of conditions, often the only practical route.
What's the central tool common to all the theorems?	Comma categories with initial objects.
Final slogan?	"Preserves all limits + smallness $\Rightarrow$ has a left adjoint."

Chapter Summary

Chapter Summary

The **Adjoint Functor Theorems** give criteria for a functor R to have a left adjoint.
GAFT: If $\mathcal{D}$ is complete, R preserves small limits, and R satisfies the **solution-set condition**, then R has a left adjoint.
SAFT: If additionally $\mathcal{D}$ is well-powered and has a small cogenerating set, the solution-set condition is automatic; limit preservation suffices.
The underlying mechanism: LA exists iff the comma category $(A \downarrow R)$ has an initial object, which it does under the hypotheses, constructed as a limit over a solution-set-indexed diagram.
These theorems supply existence of Stone–Čech compactification, sheafification, free module functors, and many other classical left adjoints — without requiring an explicit construction in each case.

Formal Restatement

Formal Restatement

Comma category. $(A \downarrow R)$ has objects (B, f) with $f : A \to RB$, and morphisms $g : B \to B'$ with $R(g) \circ f = f'$.

Adjoint criterion. R has a left adjoint L iff $(A \downarrow R)$ has an initial object for every A. That initial object is (LA, η_A).

GAFT. If $\mathcal{D}$ is complete, then $R : \mathcal{D} \to \mathcal{C}$ has a left adjoint iff:

- R preserves all small limits, and
- for every A, there is a set $\{(B_i, f_i)\}$ such that every $f : A \to RB$ factors as $R(g) \circ f_i$ for some i, $g : B_i \to B$.

SAFT. If $\mathcal{D}$ is complete, well-powered, and has a small cogenerating set, then $R : \mathcal{D} \to \mathcal{C}$ has a left adjoint iff R preserves all small limits.

Dual. Apply to $\mathcal{D}^{\mathrm{op}}$ for right-adjoint existence.

This concludes the foundational arc of the book. Chapter 11 onward will turn to connections with other areas of mathematics, where the toolkit of categories, functors, natural transformations, limits, adjunctions, monads, Yoneda, and AFT becomes the working vocabulary.

Chapter 11

Kan Extensions

Chapter Overview

You have learned how to move objects between categories (functors), how to compare functors (natural transformations), how to build objects universally (limits and colimits), and how to relate whole categories (adjunctions). There is one construction that quietly underlies all of these: the **Kan extension**.

When a functor $F : \mathcal{C} \to \mathcal{D}$ cannot be defined on a larger category $\mathcal{E}$ without additional choices, a Kan extension asks: what is the *best possible approximation* to extending F along a functor $p : \mathcal{C} \to \mathcal{E}$? The answer turns out to subsume limits, colimits, and adjunctions as special cases. Saunders Mac Lane wrote that Kan extensions are "perhaps the most important concept of category theory."

This chapter builds Kan extensions from that intuition outward: we start with the idea of approximation, develop the colimit formula for left Kan extensions, the limit formula for right Kan extensions, and then reveal how Kan extensions themselves are adjoint functors between functor categories.

Where the Name Comes From

Kan extension is named for *Daniel M. Kan* (1927–2013, Israeli-American mathematician at MIT). Kan introduced the construction in his 1958 paper "Adjoint Functors," as part of the same project that gave us adjunctions. The original motivation: extending a functor $F : \mathcal{C} \to \mathcal{D}$ along a functor $p : \mathcal{C} \to \mathcal{E}$ to a functor $G : \mathcal{E} \to \mathcal{D}$ in the "best" way (left or right universal).

Kan extensions subsume limits, colimits, and adjunctions as special cases (Mac Lane's famous slogan: "Kan extensions are perhaps the most important concept of category theory"). Kan's other contributions to the foundations of homotopy theory and category theory are extensive: Kan complexes (Vol IV Ch. 39), the Kan model structure on simplicial sets (Vol IV Ch. 41), and the now-standard categorical conception of adjoint functors.

11.1 Extending Along a Functor

11.1.1 Informally

Suppose you have a small category $\mathcal{C}$, a functor $p : \mathcal{C} \to \mathcal{E}$, and a functor $F : \mathcal{C} \to \mathcal{D}$. You would like a functor $G : \mathcal{E} \to \mathcal{D}$ that "agrees with F on $\mathcal{C}$," meaning $G \circ p \cong F$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, "p"] \arrow[dr, "F"'] & \mathcal{E} \arrow[d, dashed, "G"] \\ & \mathcal{D} \end{tikzcd}$$

Most of the time there is no *unique* such G. Many functors could fill the dashed arrow. So we ask a softer question: among all functors $G : \mathcal{E} \to \mathcal{D}$ equipped with a natural transformation $\alpha : F \Rightarrow G \circ p$, is there a *universal* one — one that every other such pair factors through uniquely?

That universal pair is called the **left Kan extension** of F along p, written $\mathrm{Lan}_p F$. Its universal property says: any natural transformation $F \Rightarrow G \circ p$ factors uniquely through the canonical transformation $\eta : F \Rightarrow (\mathrm{Lan}_p F) \circ p$.

Dually, the **right Kan extension** $\mathrm{Ran}_p F$ is equipped with a natural transformation $\varepsilon : (\mathrm{Ran}_p F) \circ p \Rightarrow F$, and any transformation $G \circ p \Rightarrow F$ factors uniquely through ε.

Internal Dialog

"Is this just like an adjunction?"

Exactly. The Kan extension is defined by a universal property, and universal properties always define adjoints in disguise. We will make that precise in Section 11.3.

11.1.2 Expanding

Let us think about what "best approximation" means concretely. The functor p maps objects of $\mathcal{C}$ into $\mathcal{E}$. If $e \in \mathcal{E}$ happens to equal $p(c)$ for some c, then $G(e)$ had better relate to $F(c)$. But what if e is not in the image of p at all?

A natural answer is: look at all the ways e can be "approached" by objects in the image of p. Form a diagram indexed by those approaches, evaluate F on the objects of $\mathcal{C}$ involved, and take the colimit of that diagram. The colimit assembles all the relevant information about e from F's perspective.

This is the **colimit formula**:

$$(\mathrm{Lan}_p\, F)(e) \;=\; \mathrm{colim}_{(c,\,p(c)\to e)\,\in\,(p\downarrow e)}\, F(c).$$

The comma category $(p \downarrow e)$ has objects (c, ϕ) where $c \in \mathcal{C}$ and $\phi : p(c) \to e$ is a morphism in $\mathcal{E}$. It collects all the "windows into e" through the image of p.

Dually, the right Kan extension uses a limit:

$$(\mathrm{Ran}_p\, F)(e) \;=\; \lim_{(c,\,e\to p(c))\,\in\,(e\downarrow p)} F(c).$$

Internal Dialog

"Why colimit for left and limit for right?"
Left Kan extensions are left adjoints, and left adjoints preserve colimits — so the formula is built from colimits. Right Kan extensions are right adjoints, and right adjoints preserve limits — so the formula uses limits. The direction of the arrow in the comma category tracks which side you are on.

11.1.3 Formalizing

Definition 11.1. Let $p : \mathcal{C} \to \mathcal{E}$ and $F : \mathcal{C} \to \mathcal{D}$ be functors. A **left Kan extension** of F along p is a functor $\mathrm{Lan}_p\, F : \mathcal{E} \to \mathcal{D}$ together with a natural transformation

$$\eta : F \;\Rightarrow\; (\mathrm{Lan}_p\, F) \circ p,$$

called the **unit**, satisfying the following universal property: for every functor $G : \mathcal{E} \to \mathcal{D}$ and every natural transformation $\alpha : F \Rightarrow G \circ p$, there exists a unique natural transformation $\bar{\alpha} : \mathrm{Lan}_p\, F \Rightarrow G$ such that $(\bar{\alpha} \circ p) \cdot \eta = \alpha$.

How to Say This

$\mathrm{Lan}_p\, F$ is read "the left Kan extension of F along p." The subscript p names the functor you are extending along.

Definition 11.2. A **right Kan extension** of F along p is a functor $\mathrm{Ran}_p\, F : \mathcal{E} \to \mathcal{D}$ together with a natural transformation

$$\varepsilon : (\mathrm{Ran}_p\, F) \circ p \;\Rightarrow\; F,$$

called the **counit**, satisfying the dual universal property: for every functor $G : \mathcal{E} \to \mathcal{D}$ and every natural transformation $\beta : G \circ p \Rightarrow F$, there exists a unique natural transformation $\bar{\beta} : G \Rightarrow \mathrm{Ran}_p\, F$ such that $\varepsilon \cdot (\bar{\beta} \circ p) = \beta$.

11.2 The Colimit Formula

11.2.1 Formalizing

The existence of left Kan extensions, and an explicit formula for computing them, rests on the colimit formula stated informally above. We now make it precise.

Definition 11.3. The **comma category** $(p \downarrow e)$, for a functor $p : \mathcal{C} \to \mathcal{E}$ and an object $e \in \mathcal{E}$, has:

- **Objects:** pairs (c, ϕ) where $c \in \mathcal{C}$ and $\phi : p(c) \to e$ in $\mathcal{E}$.
- **Morphisms:** $(c, \phi) \to (c', \phi')$ is a morphism $f : c \to c'$ in $\mathcal{C}$ such that $\phi' \circ p(f) = \phi$.

There is a forgetful functor $\pi : (p \downarrow e) \to \mathcal{C}$ sending $(c, \phi) \mapsto c$.

Theorem 11.4 (Colimit Formula for Left Kan Extensions)**.** *If $\mathcal{D}$ has all small colimits and $\mathcal{C}$ is small, then $\mathrm{Lan}_p\, F$ exists and is given pointwise by*

$$(\mathrm{Lan}_p\, F)(e) \;\cong\; \mathrm{colim}\big(F \circ \pi_e\big),$$

where $\pi_e : (p \downarrow e) \to \mathcal{C}$ is the forgetful functor.

The colimit is over a diagram that assembles all $F(c)$ for all morphisms $p(c) \to e$. As e varies, these diagrams fit together naturally to make $\mathrm{Lan}_p\, F$ a functor.

Theorem 11.5 (Limit Formula for Right Kan Extensions)**.** *Dually, if $\mathcal{D}$ has all small limits, then $\mathrm{Ran}_p\, F$ exists and is given pointwise by*

$$(\mathrm{Ran}_p\, F)(e) \;\cong\; \lim\big(F \circ \pi^e\big),$$

where $\pi^e : (e \downarrow p) \to \mathcal{C}$ is the forgetful functor from the comma category $(e \downarrow p)$ whose objects are $(c, \phi : e \to p(c))$.

11.2.2 Reorganizing: Limits and Colimits Are Kan Extensions

The colimit formula makes a striking observation possible. Take $\mathcal{E} = \mathbf{1}$, the category with one object and one morphism, and let $p : \mathcal{C} \to \mathbf{1}$ be the unique functor. Then:

- The comma category $(p \downarrow \star)$, for the single object $\star \in \mathbf{1}$, is isomorphic to $\mathcal{C}$ itself (every morphism $p(c) \to \star$ is the identity).
- So $(\mathrm{Lan}_p\, F)(\star) = \mathrm{colim}_{\mathcal{C}}\, F$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, "p"] \arrow[dr, "F"'] & \mathbf{1} \arrow[d, dashed, "\mathrm{Lan}_p\, F"] \\ & \mathcal{D} \end{tikzcd}$$

In other words, *colimits are left Kan extensions along the unique functor to the terminal category*, and limits are right Kan extensions along the same functor. Kan extensions are more general: they push a functor along an arbitrary functor, not just along the collapse to a point.

11.3 Kan Extensions as Adjoints

11.3.1 Formalizing

Fix categories $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$ and a functor $p : \mathcal{C} \to \mathcal{E}$. There is a **restriction functor**

$$(-) \circ p \;:\; [\mathcal{E}, \mathcal{D}] \;\longrightarrow\; [\mathcal{C}, \mathcal{D}]$$

that sends a functor $G : \mathcal{E} \to \mathcal{D}$ to the composite $G \circ p : \mathcal{C} \to \mathcal{D}$.

Theorem 11.6. *When Kan extensions exist, they form adjoint pairs with the restriction functor:*

$$\mathrm{Lan}_p \dashv (-) \circ p,$$
$$(-) \circ p \dashv \mathrm{Ran}_p\,.$$

In explicit terms, there are natural bijections:

$$[\mathcal{E}, \mathcal{D}](\mathrm{Lan}_p\, F,\, G) \;\cong\; [\mathcal{C}, \mathcal{D}](F,\, G \circ p),$$
$$[\mathcal{C}, \mathcal{D}](G \circ p,\, F) \;\cong\; [\mathcal{E}, \mathcal{D}](G,\, \mathrm{Ran}_p\, F).$$

Proof sketch. The hom-set bijection for $\mathrm{Lan}_p \dashv (-) \circ p$ is precisely the universal property of the left Kan extension: natural transformations $\mathrm{Lan}_p\, F \Rightarrow G$ are in natural bijection with natural transformations $F \Rightarrow G \circ p$, mediated by pre- or post-composition with the unit η. ☐

This is the key insight: *forming a left Kan extension is left adjoint to restricting along p*. The construction that was built by universal property is, in the language of the previous chapters, exactly an adjunction between functor categories.

11.3.2 Simplifying

The four main constructions of the book now appear as Kan extensions:

Construction	**As a Kan extension**	**Along**
Colimit of F	$\mathrm{Lan}_p\, F$	$p : \mathcal{C} \to \mathbf{1}$
Limit of F	$\mathrm{Ran}_p\, F$	$p : \mathcal{C} \to \mathbf{1}$
Left adjoint of G	$\mathrm{Lan}_Y\, \mathrm{id}$	Yoneda embedding Y
Right adjoint of G	$\mathrm{Ran}_Y\, \mathrm{id}$	Yoneda embedding Y

11.4 Density and the Density Corollary

11.4.1 Formalizing

A functor $p : \mathcal{C} \to \mathcal{E}$ is called **dense** if every object of $\mathcal{E}$ is a colimit of objects in the image of p. The precise statement involves Kan extensions:

Definition 11.7. A functor $p : \mathcal{C} \to \mathcal{E}$ is **dense** if the left Kan extension of p along itself is naturally isomorphic to the identity:

$$\mathrm{Lan}_p\, p \;\cong\; \mathrm{Id}_{\mathcal{E}}\,.$$

The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{op}, \mathbf{Set}]$ is dense. This gives the **density corollary**: every presheaf is a canonical colimit of representable presheaves.

Theorem 11.8 (Density of the Yoneda Embedding)**.** *For any small category $\mathcal{C}$ and any presheaf $F \in [\mathcal{C}^{op}, \mathbf{Set}]$:*

$$F \;\cong\; \operatorname{colim}_{(c,\, Y(c) \to F) \,\in\, (Y \downarrow F)} Y(c).$$

In words: every presheaf F is the colimit of all representable presheaves mapping into it. The "shape" of a presheaf is entirely determined by how representables map into it — which is the content of the Yoneda lemma stated from a new angle.

Internal Dialog

"So the Yoneda lemma and Kan extensions are the same thing?"
Not quite, but they are deeply connected. The Yoneda lemma says that $[\mathcal{C}^{op}, \mathbf{Set}](Y(c), F) \cong F(c)$. The density formula says that F is the left Kan extension of itself along Y. Together, they say that presheaf categories are the canonical home where every object is freely built from representables.

11.4.2 Formally

Theorem 11.9 (Every Functor is a Left Kan Extension)**.** *For any functor $F : \mathcal{C} \to \mathcal{D}$:*

$$F \;\cong\; \operatorname{Lan}_{\mathrm{Id}_{\mathcal{C}}} F.$$

That is, F is its own left Kan extension along the identity. More substantively, if $Y : \mathcal{C} \to \hat{\mathcal{C}}$ is the Yoneda embedding, then:

$$\operatorname{Lan}_Y F \;\cong\; F^+,$$

where $F^+ : \hat{\mathcal{C}} \to \mathcal{D}$ is the unique (up to isomorphism) colimit-preserving extension of F to the presheaf category.

This is the universal property of the presheaf category: $\hat{\mathcal{C}}$ is the free cocompletion of $\mathcal{C}$.

11.5 Exercises

What is the left Kan extension $\operatorname{Lan}_p F$ of F along p?	A functor $\operatorname{Lan}_p F : \mathcal{E} \to \mathcal{D}$ with a unit $\eta : F \Rightarrow (\operatorname{Lan}_p F) \circ p$, universal among all such pairs.
What is the universal property of the unit η?	Any $\alpha : F \Rightarrow G \circ p$ factors uniquely through η via a natural transformation $\bar{\alpha} : \operatorname{Lan}_p F \Rightarrow G$.
How do you read $\operatorname{Lan}_p F$?	"The left Kan extension of F along p." The subscript names the functor you are extending along.
What is the colimit formula for $(\operatorname{Lan}_p F)(e)$?	$\operatorname{colim}\big(F \circ \pi_e\big)$, where $\pi_e : (p \downarrow e) \to \mathcal{C}$ is the forgetful functor from the comma category.
What are the objects of the comma category $(p \downarrow e)$?	Pairs (c, ϕ) where $c \in \mathcal{C}$ and $\phi : p(c) \to e$ in $\mathcal{E}$.
What are the morphisms in $(p \downarrow e)$?	A morphism $(c, \phi) \to (c', \phi')$ is $f : c \to c'$ in $\mathcal{C}$ such that $\phi' \circ p(f) = \phi$.
What is the limit formula for $(\operatorname{Ran}_p F)(e)$?	$\lim\big(F \circ \pi^e\big)$, where $\pi^e : (e \downarrow p) \to \mathcal{C}$ is the forgetful functor from the comma category whose objects have $\phi : e \to p(c)$.
How does a colimit of $F : \mathcal{C} \to \mathcal{D}$ arise as a Kan extension?	As $(\operatorname{Lan}_p F)(\star)$ where $p : \mathcal{C} \to \mathbf{1}$ is the unique functor.
What is the adjunction involving Lan_p?	$\operatorname{Lan}_p \dashv (-) \circ p$, as functors between functor categories $[\mathcal{C}, \mathcal{D}]$ and $[\mathcal{E}, \mathcal{D}]$.
What is the adjunction involving Ran_p?	$(-) \circ p \dashv \operatorname{Ran}_p$.
What does it mean for a functor $p : \mathcal{C} \to \mathcal{E}$ to be dense?	$\operatorname{Lan}_p p \cong \mathrm{Id}_{\mathcal{E}}$; every object of $\mathcal{E}$ is a canonical colimit of objects in the image of p.
Is the Yoneda embedding $Y : \mathcal{C} \to \hat{\mathcal{C}}$ dense?	Yes. Every presheaf is a colimit of representable presheaves.
What is the density corollary?	Every presheaf F is the colimit of all representables $Y(c)$ over the comma category $(Y \downarrow F)$.
What is the free cocompletion of $\mathcal{C}$?	The presheaf category $\hat{\mathcal{C}} = [\mathcal{C}^{op}, \mathbf{Set}]$: the universal category that receives $\mathcal{C}$ and has all colimits.

If $\mathcal{D}$ has all colimits, what is the unique colimit-preserving extension of $F : \mathcal{C} \to \mathcal{D}$ to $\hat{\mathcal{C}}$?	$\mathrm{Lan}_Y F$, the left Kan extension of F along the Yoneda embedding.
What does $\mathrm{Lan}_p F$ compute at e when every object of $\mathcal{E}$ is in the image of p?	It essentially evaluates F at a preimage of e; when p is an isomorphism the formula recovers $F \circ p^{-1}$.
Can $\mathrm{Lan}_p F$ fail to exist?	Yes, when $\mathcal{D}$ lacks the required colimits. The colimit formula requires $\mathcal{D}$ to have all small colimits.
What is the restriction functor?	$(-) \circ p : [\mathcal{E}, \mathcal{D}] \to [\mathcal{C}, \mathcal{D}]$, precomposing every functor $\mathcal{E} \to \mathcal{D}$ with p.
Which Kan extension is left adjoint to restriction, and which is right?	Lan_p is left adjoint; Ran_p is right adjoint.
What does it mean that "every concept is a Kan extension"?	Limits, colimits, adjoints, and many other universal constructions can each be expressed as a Kan extension of an appropriate functor along an appropriate functor.
How does the right Kan extension $\mathrm{Ran}_p F$ relate to adjoints?	If F has a right adjoint G, and $p = F$, then $\mathrm{Ran}_F \mathrm{Id} \cong G$; right adjoints can be computed as right Kan extensions along the left adjoint.
What structure does the comma category $(p \downarrow e)$ have as e varies?	It varies functorially: a morphism $e \to e'$ in $\mathcal{E}$ induces a functor $(p \downarrow e) \to (p \downarrow e')$ by composing with the morphism.
What is the unit η for a left Kan extension?	A natural transformation $\eta : F \Rightarrow (\mathrm{Lan}_p F) \circ p$; it is the canonical "comparison" map from F to its extension composed with p.
What is the counit ε for a right Kan extension?	A natural transformation $\varepsilon : (\mathrm{Ran}_p F) \circ p \Rightarrow F$; the canonical comparison in the other direction.
If p is an equivalence of categories, what can you say about $\mathrm{Lan}_p F$?	It is isomorphic to $F \circ p^{-1}$ (the composite with a quasi-inverse), and the unit η is a natural isomorphism.
What condition on p makes every functor F compute its Kan extension without taking genuine colimits?	If p is fully faithful, then $(\mathrm{Lan}_p F)(p(c)) \cong F(c)$: the extension recovers F exactly on the image of p.
What is a pointwise Kan extension?	A Kan extension computed by the colimit (or limit) formula at each object separately. Most Kan extensions in practice are pointwise.
Why are pointwise Kan extensions better-behaved?	They are preserved by representable functors; equivalently, they interact well with limits and colimits in $\mathcal{D}$.
How does sheafification relate to Kan extensions?	Sheafification is the left Kan extension of the inclusion of sheaves into presheaves, along the identity on presheaves — equivalently, the left adjoint to the forgetful functor from sheaves to presheaves.
What single adjunction encodes both Lan_p and Ran_p?	There is no single adjunction; they form two separate adjunctions: $\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p$, making $(-) \circ p$ a functor with a left and a right adjoint.

Chapter Summary

A **Kan extension** of $F : \mathcal{C} \to \mathcal{D}$ along $p : \mathcal{C} \to \mathcal{E}$ is the best approximation to extending F to $\mathcal{E}$. The **left Kan extension** $\mathrm{Lan}_p F$ is computed by a colimit over the comma category $(p \downarrow e)$ at each object $e \in \mathcal{E}$; the **right Kan extension** $\mathrm{Ran}_p F$ by a limit. Both are adjoints to the restriction functor $(-) \circ p$.
Limits and colimits are the special case where $\mathcal{E} = \mathbf{1}$. The Yoneda embedding is dense, giving the density corollary: every presheaf is canonically a colimit of representables. The presheaf category is the free cocompletion of $\mathcal{C}$.

11.6 Formal Restatement

Formal Restatement

Left Kan extension. Given $p : \mathcal{C} \to \mathcal{E}$ and $F : \mathcal{C} \to \mathcal{D}$, the pair $(\mathrm{Lan}_p F, \eta)$ where $\mathrm{Lan}_p F : \mathcal{E} \to \mathcal{D}$ and $\eta : F \Rightarrow (\mathrm{Lan}_p F) \circ p$ is terminal in the category of such pairs, yielding a natural bijection

$$[\mathcal{E}, \mathcal{D}](\mathrm{Lan}_p F,\, G) \;\cong\; [\mathcal{C}, \mathcal{D}](F,\, G \circ p).$$

Colimit formula. If $\mathcal{C}$ is small and $\mathcal{D}$ is cocomplete:

$$(\mathrm{Lan}_p F)(e) \;=\; \mathrm{colim}\Big((p \downarrow e) \xrightarrow{\pi_e} \mathcal{C} \xrightarrow{F} \mathcal{D}\Big).$$

Right Kan extension. Dually, $(\mathrm{Ran}_p F, \varepsilon)$ with $\varepsilon : (\mathrm{Ran}_p F) \circ p \Rightarrow F$, and:

$$[\mathcal{C}, \mathcal{D}](G \circ p,\, F) \;\cong\; [\mathcal{E}, \mathcal{D}](G,\, \mathrm{Ran}_p F).$$

Limit formula. If $\mathcal{D}$ is complete:

$$(\mathrm{Ran}_p F)(e) \;=\; \lim\Big((e \downarrow p) \xrightarrow{\pi^e} \mathcal{C} \xrightarrow{F} \mathcal{D}\Big).$$

Adjunction triple. $\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p$ as functors $[\mathcal{C}, \mathcal{D}] \rightleftarrows [\mathcal{E}, \mathcal{D}]$.
Density. p is dense iff $\mathrm{Lan}_p\, p \cong \mathrm{Id}_{\mathcal{E}}$. The Yoneda embedding $Y : \mathcal{C} \to \hat{\mathcal{C}}$ is dense; $\hat{\mathcal{C}}$ is the free cocompletion of $\mathcal{C}$.

Chapter 12

Monoidal and Enriched Categories

Chapter Overview

The categories you have encountered so far have *sets* of morphisms between any two objects. The hom-set $\mathcal{C}(a,b)$ is just a set: a collection of arrows from a to b, with no additional structure.

But in many important categories, those hom-sets carry extra structure. In the category of abelian groups, $\mathrm{Hom}(A,B)$ is itself an abelian group. In a category of chain complexes, morphism spaces are chain complexes. In a 2-category, the morphisms between two objects form a whole category.

The unifying concept is **enriched category theory**: a category enriched over a monoidal category $\mathcal{V}$ replaces its hom-sets with objects of $\mathcal{V}$. To make this work, $\mathcal{V}$ must itself be a **monoidal category** — a category equipped with a tensor product that plays the role of the cartesian product for sets.

This chapter builds monoidal categories from scratch, develops the coherence theorem that makes them well-behaved, introduces symmetric and closed monoidal categories, and then constructs the general theory of $\mathcal{V}$-enriched categories.

Where the Name Comes From

Monoidal category was introduced by *Jean Bénabou* in 1963. The term combines *monoid* (a set with an associative multiplication and unit) with the categorical extension: a monoidal category is a category whose "multiplication" is a tensor product $\otimes$ on objects (a bifunctor), with a coherent associator and unit isomorphisms playing the role of associativity and unit laws.

Enriched category (or $\mathcal{V}$-*category*) was developed by *Samuel Eilenberg* and *Max Kelly* (1966). The adjective *enriched* reflects that the hom-objects are richer than mere sets — they carry $\mathcal{V}$-structure. The terminology contrasts *enriched* categories with the *ordinary* (= **Set**-enriched) categories of Vol I Ch. 2. Major modern foundations (Vol III Ch. 20) develop enriched CT in depth.

Pentagon and **triangle** (the Mac Lane coherence axioms) are named for the geometric shapes of the commutative diagrams they require. Mac Lane's 1963 *Natural Associativity and Commutativity* proved the pentagon + triangle suffices to imply *all* higher coherences — the celebrated *Mac Lane coherence theorem.*

12.1 Monoidal Categories

12.1.1 Informally

A **monoidal category** is a category with a way to "multiply" objects and morphisms together. The archetype is $(\mathbf{Set}, \times, \{*\})$: sets with cartesian product, where the one-element set $\{*\}$ acts as the identity for $\times$. But many other examples exist:

- $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$: abelian groups with tensor product.
- $(\mathbf{Vect}_k, \otimes_k, k)$: vector spaces over a field k.
- $(\mathbf{Set}, +, \emptyset)$: sets with disjoint union.
- $([\mathcal{C},\mathcal{C}], \circ, \mathrm{Id})$: endofunctors on a category, with composition.

The defining data is: a category $\mathcal{V}$, a functor $\otimes : \mathcal{V}\times\mathcal{V}\to\mathcal{V}$ (the tensor product), and a unit object $I\in\mathcal{V}$, together with natural isomorphisms expressing associativity and left/right unitality.

12.1.2 Expanding

The tricky part is that associativity and unitality do not need to be equalities — they can be natural isomorphisms ("coherent up to isomorphism"):

$$\alpha_{A,B,C} : (A \otimes B) \otimes C \xrightarrow{\sim} A \otimes (B \otimes C) \qquad \text{(associator)},$$
$$\lambda_A : I \otimes A \xrightarrow{\sim} A \qquad \text{(left unitor)},$$
$$\rho_A : A \otimes I \xrightarrow{\sim} A \qquad \text{(right unitor)}.$$

Having these isomorphisms opens a question: if there are many ways to parenthesize an expression like $A \otimes B \otimes C \otimes D$, do all the paths between any two parenthesizations agree? **Mac Lane's coherence theorem** says yes: any two composites of associators and unitors between the same parenthesizations are equal. This means we can compute in monoidal categories as if the tensor product were strictly associative — writing $A \otimes B \otimes C$ without parentheses — as long as we are careful about what "equal" means.

12.1.3 Formalizing

Definition 12.1. A **monoidal category** $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ consists of:

- A category $\mathcal{V}$.
- A **tensor product** functor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$.
- A **unit object** $I \in \mathcal{V}$.
- Natural isomorphisms:

$$\alpha_{A,B,C} : (A \otimes B) \otimes C \xrightarrow{\sim} A \otimes (B \otimes C),$$
$$\lambda_A : I \otimes A \xrightarrow{\sim} A,$$
$$\rho_A : A \otimes I \xrightarrow{\sim} A.$$

These must satisfy the **pentagon axiom**:

$$\begin{tikzcd}
 & (A \otimes B) \otimes (C \otimes D) \arrow[dr, "\alpha_{A,B,C\otimes D}"] & \\
((A \otimes B) \otimes C) \otimes D \arrow[ur, "\alpha_{A\otimes B,C,D}"] \arrow[d, "\alpha_{A,B,C}\otimes \mathrm{id}_D"'] & & A \otimes (B \otimes (C \otimes D)) \\
(A \otimes (B \otimes C)) \otimes D \arrow[rr, "\alpha_{A,B\otimes C,D}"'] & & A \otimes ((B \otimes C) \otimes D) \arrow[u, "\mathrm{id}_A \otimes \alpha_{B,C,D}"']
\end{tikzcd}$$

and the **triangle axiom**:

$$\begin{tikzcd}
(A \otimes I) \otimes B \arrow[rr, "\alpha_{A,I,B}"] \arrow[dr, "\rho_A \otimes \mathrm{id}_B"'] & & A \otimes (I \otimes B) \arrow[dl, "\mathrm{id}_A \otimes \lambda_B"] \\
 & A \otimes B &
\end{tikzcd}$$

How to Say This

The tensor product $A \otimes B$ is read "A tensor B." The unit I may be called the *monoidal unit* or *unit object.* A monoidal category where α, λ, ρ are all identity morphisms is called **strict**.

Theorem 12.2 (Mac Lane's Coherence Theorem)**.** *Every monoidal category is monoidally equivalent to a strict monoidal category. Moreover, every diagram built from instances of α, λ, ρ and their inverses that connects the same two parenthesizations commutes.*

Coherence allows us to write $A \otimes B \otimes C$ without parentheses and to move freely between parenthesizations without tracking the specific isomorphisms.

12.2 Symmetric and Closed Monoidal Categories

12.2.1 Formalizing

Two additional structures enrich monoidal categories significantly.

Definition 12.3. A monoidal category $(\mathcal{V}, \otimes, I)$ is **symmetric** if it has a natural isomorphism

$$\sigma_{A,B} : A \otimes B \xrightarrow{\sim} B \otimes A$$

satisfying: $\sigma_{B,A} \circ \sigma_{A,B} = \mathrm{id}_{A\otimes B}$, and compatibility with α expressed by the hexagon axioms.

Definition 12.4. A monoidal category $(\mathcal{V}, \otimes, I)$ is **closed** if for each object B, the functor $(-) \otimes B : \mathcal{V} \to \mathcal{V}$ has a right adjoint, written $[B, -]$ or $B \multimap (-)$ *(read: B hom dash)*:

$$\mathcal{V}(A \otimes B,\ C) \ \cong\ \mathcal{V}(A,\ [B, C]).$$

The object $[B, C]$ is called the **internal hom** of B and C.

How to Say This

$[B, C]$ is read "B to C" or "the internal hom from B to C." In **Set**, $[B, C] = C^B$ is the set of functions from B to C; the adjunction becomes the familiar currying bijection $\mathrm{Hom}(A \times B, C) \cong \mathrm{Hom}(A, C^B)$.

The category **Set** with cartesian product is symmetric monoidal closed. The category **Ab** with $\otimes_{\mathbb{Z}}$ is symmetric monoidal closed, with internal hom $[A, B] = \mathrm{Hom}_{\mathbb{Z}}(A, B)$.

12.3 Enriched Categories

12.3.1 Informally

Fix a monoidal category $(\mathcal{V}, \otimes, I)$. A **$\mathcal{V}$-enriched category** $\mathcal{C}$, or **$\mathcal{V}$-category**, is like an ordinary category except that its hom-sets are replaced by objects of $\mathcal{V}$.

The idea is simple: instead of a *set* $\mathcal{C}(a, b)$ of morphisms from a to b, we have an *object* $\mathcal{C}(a, b) \in \mathcal{V}$. Composition becomes a morphism in $\mathcal{V}$:

$$\circ_{a,b,c} : \mathcal{C}(b, c) \otimes \mathcal{C}(a, b) \ \to\ \mathcal{C}(a, c).$$

Identity becomes a morphism from the unit:

$$j_a : I \ \to\ \mathcal{C}(a, a).$$

12.3.2 Expanding

Here are the key examples:

- **Set-enriched categories** are ordinary categories.
- **Ab-enriched categories** are **preadditive categories**: the hom-sets are abelian groups and composition is bilinear.
- **Vect$_k$-enriched categories** are **k-linear categories**.
- **Cat-enriched categories** are **2-categories**: hom-objects are themselves categories (categories of morphisms between two objects).
- **$([0, \infty], \geq, +, 0)$-enriched categories** are **Lawvere metric spaces**: objects are points, and $\mathcal{C}(a, b)$ is the "distance from a to b."

The last example is striking. A monoidal category need not be cartesian. The poset $([0, \infty], \geq)$ — real numbers ordered by $\geq$, with tensor product $a + b$ and unit 0 — is a monoidal category. An enriched category over it is a set with a distance function satisfying:

$$0 \geq d(a, a), \qquad d(b, c) + d(a, b) \geq d(a, c).$$

That is: identity has distance 0, and the triangle inequality. This is a (generalized) metric space.

12.3.3 Formalizing

Definition 12.5. Let $(\mathcal{V}, \otimes, I)$ be a monoidal category. A **$\mathcal{V}$-enriched category** (or **$\mathcal{V}$-category**) $\mathcal{C}$ consists of:

- A collection of **objects** $\mathrm{ob}(\mathcal{C})$.
- For each pair $a, b \in \mathrm{ob}(\mathcal{C})$, an object $\mathcal{C}(a, b) \in \mathcal{V}$, called the **hom-object**.
- For each triple a, b, c, a **composition morphism**

$$\circ_{a,b,c} : \mathcal{C}(b, c) \otimes \mathcal{C}(a, b) \ \to\ \mathcal{C}(a, c).$$

- For each object a, an **identity morphism**

$$j_a : I \ \to\ \mathcal{C}(a, a).$$

These must satisfy associativity and unit laws expressed as commutative diagrams in $\mathcal{V}$, analogous to the ordinary category axioms.

12.4 $\mathcal{V}$-Functors and $\mathcal{V}$-Natural Transformations

12.4.1 Formalizing

Definition 12.6. A **$\mathcal{V}$-functor** $F : \mathcal{C} \to \mathcal{D}$ between $\mathcal{V}$-categories consists of:

- A function $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$.
- For each pair $a, b \in \mathcal{C}$, a morphism in $\mathcal{V}$:

$$F_{a,b} : \mathcal{C}(a, b) \ \to \ \mathcal{D}(Fa, Fb).$$

These must be compatible with composition and identities via commutative diagrams in $\mathcal{V}$.

Definition 12.7. A **$\mathcal{V}$-natural transformation** $\alpha : F \Rightarrow G$ between $\mathcal{V}$-functors $F, G : \mathcal{C} \to \mathcal{D}$ assigns to each $a \in \mathcal{C}$ a morphism $\alpha_a : I \to \mathcal{D}(Fa, Ga)$ in $\mathcal{V}$, satisfying the naturality condition expressed as a commutative square in $\mathcal{V}$.

How to Say This

A $\mathcal{V}$-natural transformation has its "components" as morphisms out of the unit I into hom-objects, rather than as elements of a set. In the **Set**-enriched case, $I = \{*\}$ and a morphism $\{*\} \to \mathcal{D}(Fa, Ga)$ is the same as picking an element of $\mathcal{D}(Fa, Ga)$, recovering the ordinary definition.

12.4.2 Self-Enrichment

A monoidal category $\mathcal{V}$ that is also closed can be enriched over itself. The hom-object between two objects a, b is the internal hom $[a, b] \in \mathcal{V}$, and composition is given by the morphism:

$$[b, c] \otimes [a, b] \ \to \ [a, c]$$

obtained via the adjunction for closed structure. This self-enrichment is what makes closed monoidal categories the natural foundation for enriched category theory.

12.4.3 Reorganizing: Ordinary CT as Enriched CT

Every concept from Volumes I can be enriched:

Ordinary	**Enriched over $\mathcal{V}$**
Category	$\mathcal{V}$-category
Functor	$\mathcal{V}$-functor
Natural transformation	$\mathcal{V}$-natural transformation
Adjunction	$\mathcal{V}$-adjunction
Limit / Colimit	$\mathcal{V}$-limit / $\mathcal{V}$-colimit (weighted)
Yoneda lemma	$\mathcal{V}$-Yoneda lemma
Kan extension	$\mathcal{V}$-Kan extension

The passage from ordinary to enriched is not merely generalization for its own sake. Many important categories are naturally enriched (abelian groups, chain complexes, spectra in homotopy theory), and their enriched structure carries essential information that the underlying ordinary structure loses.

12.5 Exercises

What is a monoidal category?	A category $\mathcal{V}$ with a tensor product functor $\otimes$, a unit object I, and natural isomorphisms α (associator), λ (left unitor), ρ (right unitor), satisfying pentagon and triangle axioms.
What are the three natural isomorphisms in a monoidal category?	Associator $\alpha_{A,B,C} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$; left unitor $\lambda_A : I \otimes A \cong A$; right unitor $\rho_A : A \otimes I \cong A$.
What does the pentagon axiom say?	The five ways of reassociating $((A \otimes B) \otimes C) \otimes D$ into $A \otimes (B \otimes (C \otimes D))$ using α all agree.
What does the triangle axiom say?	$(\rho_A \otimes \mathrm{id}_B) = (\mathrm{id}_A \otimes \lambda_B) \circ \alpha_{A,I,B}$ as morphisms $(A \otimes I) \otimes B \to A \otimes B$.
What is a strict monoidal category?	One where α, λ, and ρ are all identity morphisms; the tensor product is strictly associative and unital.

What does Mac Lane's coherence theorem say?	Every monoidal category is monoidally equivalent to a strict one; any diagram of associators and unitors between the same two parenthesizations commutes.
What does coherence allow us to do in practice?	Write $A \otimes B \otimes C$ without parentheses and freely move between parenthesizations without tracking the specific isomorphisms.
What is a symmetric monoidal category?	A monoidal category with a natural isomorphism $\sigma_{A,B} : A \otimes B \cong B \otimes A$ satisfying $\sigma_{B,A} \circ \sigma_{A,B} = \mathrm{id}$ and hexagon axioms.
What is a closed monoidal category?	A monoidal category where $(-) \otimes B$ has a right adjoint $[B, -]$ (internal hom) for every B: $\mathcal{V}(A \otimes B, C) \cong \mathcal{V}(A, [B, C])$.
What is the internal hom in **Set**?	$[B, C] = C^B$, the set of functions $B \to C$. The adjunction is the currying bijection $\mathrm{Hom}(A \times B, C) \cong \mathrm{Hom}(A, C^B)$.
What is a $\mathcal{V}$-enriched category?	Like a category, but hom-sets $\mathcal{C}(a, b)$ are objects of $\mathcal{V}$; composition is a morphism $\mathcal{C}(b, c) \otimes \mathcal{C}(a, b) \to \mathcal{C}(a, c)$; identity is a morphism $I \to \mathcal{C}(a, a)$.
What is a **Set**-enriched category?	An ordinary category. Hom-sets are sets; composition is a function.
What is an **Ab**-enriched category?	A preadditive category: hom-sets are abelian groups and composition is bilinear (a $\mathbb{Z}$-bilinear map).
What is a **Cat**-enriched category?	A 2-category: hom-objects are categories; objects are 0-cells, morphisms are 1-cells, and morphisms between morphisms are 2-cells.
What is a $([0, \infty], \geq, +, 0)$-enriched category?	A (generalized) metric space: objects are points, $\mathcal{C}(a, b)$ is the distance from a to b, satisfying $d(a, a) = 0$ and the triangle inequality.
What is a $\mathcal{V}$-functor?	A function on objects plus, for each pair a, b, a morphism $F_{a,b} : \mathcal{C}(a, b) \to \mathcal{D}(Fa, Fb)$ in $\mathcal{V}$, compatible with composition and identities.
What is a $\mathcal{V}$-natural transformation?	Assigns to each a a morphism $\alpha_a : I \to \mathcal{D}(Fa, Ga)$ in $\mathcal{V}$, satisfying $\mathcal{V}$-naturality.
What does a $\mathcal{V}$-natural transformation reduce to when $\mathcal{V} = \mathbf{Set}$?	An ordinary natural transformation: a morphism $I = \{*\} \to \mathcal{D}(Fa, Ga)$ is just an element of the hom-set.
What is self-enrichment of a closed monoidal category?	A closed monoidal $\mathcal{V}$ enriches over itself with hom-objects $[a, b] \in \mathcal{V}$; composition given by the closed structure.
What monoidal category is used to enrich toward 2-categories?	**Cat** with product $\times$ and unit the terminal category **1**.
What is a monoidal functor?	A functor $F : \mathcal{V} \to \mathcal{W}$ between monoidal categories with natural morphisms $F(A) \otimes F(B) \to F(A \otimes B)$ and $I_{\mathcal{W}} \to F(I_{\mathcal{V}})$, compatible with associators and unitors.
What is the tensor product of abelian groups?	$A \otimes_{\mathbb{Z}} B$ is the abelian group generated by symbols $a \otimes b$ subject to bilinearity. It is the unit for the closed structure with $[A, B] = \mathrm{Hom}_{\mathbb{Z}}(A, B)$.
Why does the monoidal category need to be closed to allow self-enrichment?	The identity element of a $\mathcal{V}$-category requires a map $I \to \mathcal{C}(a, a)$; the composition requires a map $\mathcal{C}(b, c) \otimes \mathcal{C}(a, b) \to \mathcal{C}(a, c)$. In the self-enriched case, both are constructed from the closed structure.
What is the $\mathcal{V}$-Yoneda lemma?	For a $\mathcal{V}$-category $\mathcal{C}$ and $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{V}$: $\mathcal{V}$-$\mathbf{Nat}(\mathcal{C}(a, -), F) \cong F(a)$ as objects of $\mathcal{V}$, naturally in a and F.

What is a weighted limit in a $\mathcal{V}$-category?	A limit weighted by a $\mathcal{V}$-functor $W : \mathcal{J} \to \mathcal{V}$; generalizes ordinary limits (where $W = \Delta I$) and allows enriched universal properties.
How does enrichment help express that $\mathrm{Hom}(A, B)$ for abelian groups is itself an abelian group?	**Ab**-enrichment of **Ab** makes each hom-set $\mathrm{Hom}(A, B)$ an abelian group, with composition bilinear. This is the self-enriched structure.
What is a monoid in a monoidal category?	An object $M \in \mathcal{V}$ with morphisms $\mu : M \otimes M \to M$ (multiplication) and $\eta : I \to M$ (unit), satisfying associativity and unit laws. Monoids in **Set** are ordinary monoids; in **Ab** they are rings.
What is a comonoid in a monoidal category?	An object C with comultiplication $\delta : C \to C \otimes C$ and counit $\varepsilon : C \to I$, satisfying dual laws. Comonoids in **Set** with cartesian product are just sets (every set is a comonoid via the diagonal).
What is a braided monoidal category?	A monoidal category with a natural isomorphism $\gamma_{A,B} : A \otimes B \to B \otimes A$ satisfying hexagon axioms but *not* required to satisfy $\gamma_{B,A} \circ \gamma_{A,B} = \mathrm{id}$. Symmetric monoidal is the special case where it does.

Chapter Summary

A **monoidal category** $(\mathcal{V}, \otimes, I)$ is a category with a tensor product and unit object, with associativity and unitality natural isomorphisms satisfying the pentagon and triangle axioms. Mac Lane's **coherence theorem** says any monoidal category is equivalent to a strict one, so associativity can be ignored in practice.
A monoidal category is **symmetric** if $A \otimes B \cong B \otimes A$ naturally, and **closed** if $(-) \otimes B$ has a right adjoint $[B, -]$ (internal hom).
A **$\mathcal{V}$-enriched category** replaces hom-sets with objects of $\mathcal{V}$, with composition as a morphism in $\mathcal{V}$. **Set**-categories are ordinary categories; **Ab**-categories are preadditive; **Cat**-categories are 2-categories; $([0, \infty], +, 0)$-categories are metric spaces. Closed monoidal categories can enrich over themselves.

12.6 Formal Restatement

Formal Restatement

Monoidal category. $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ where $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, $I \in \mathcal{V}$, and natural isomorphisms $\alpha_{A,B,C} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$, $\lambda_A : I \otimes A \cong A$, $\rho_A : A \otimes I \cong A$ satisfying the pentagon and triangle coherence conditions.
Symmetric. Additional natural isomorphism $\sigma_{A,B} : A \otimes B \xrightarrow{\sim} B \otimes A$ with $\sigma_{B,A} \circ \sigma_{A,B} = \mathrm{id}$ and hexagon axioms.
Closed. $(-) \otimes B \dashv [B, -]$ for all B; internal hom $[B, C] \in \mathcal{V}$ with $\mathcal{V}(A \otimes B, C) \cong \mathcal{V}(A, [B, C])$.
$\mathcal{V}$-category. Objects $\mathrm{ob}(\mathcal{C})$; for each a, b: hom-object $\mathcal{C}(a, b) \in \mathcal{V}$; composition $\circ_{a,b,c} : \mathcal{C}(b, c) \otimes \mathcal{C}(a, b) \to \mathcal{C}(a, c)$; identity $j_a : I \to \mathcal{C}(a, a)$; satisfying associativity and unit laws as diagrams in $\mathcal{V}$.
$\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{D}$**.** Function on objects plus $F_{a,b} : \mathcal{C}(a, b) \to \mathcal{D}(Fa, Fb)$ in $\mathcal{V}$, compatible with $\circ$ and j.
$\mathcal{V}$-natural transformation $\alpha : F \Rightarrow G$**.** Components $\alpha_a : I \to \mathcal{D}(Fa, Ga)$ in $\mathcal{V}$ satisfying: $\circ(G_{a,b} \otimes \alpha_a) = \circ(\alpha_b \otimes F_{a,b})$ as morphisms $\mathcal{C}(a, b) \otimes I \to \mathcal{D}(Fa, Gb)$.
$\mathcal{V}$-Yoneda. $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \cong F(a)$ in $\mathcal{V}$, naturally in $a \in \mathcal{C}$ and $F \in [\mathcal{C}, \mathcal{V}]_{\mathcal{V}}$.

Volume II

The Architecture Revisited

Chapter 13

Categories as Algebraic Structures

The first volume introduced categories from the outside: objects, morphisms, composition, identity. This chapter looks at them from the inside, as algebraic structures in the sense of abstract algebra. The key insight is that a category is a *monoid with multiple objects* — and that monoids, groups, preorders, and groupoids are each a special type of category.

This algebraic viewpoint sharpens the intuition for what is "categorically natural" versus what is a coincidence of a specific example.

Where the Name Comes From

Monoid (Greek *monos* "single, one" + *-oid* "having the form of") is a set with an associative multiplication and a single (= "mon-") identity element. The term is 20th-century algebraic terminology, standard by the 1930s. **Groupoid** (Heinrich Brandt 1927) is the analogue with all elements invertible — and crucially, where the multiplication is *partial* (not every pair multiplies). Brandt's insight: a groupoid is to a group what a category is to a monoid. The slogan "a category is a many-object monoid" captures this exactly.

13.1 Categories as Generalized Monoids

A monoid is a set M with an associative binary operation and a unit element. A category has objects, morphisms, associative composition, and identity morphisms. The difference: in a monoid, any two elements can be multiplied. In a category, two morphisms $f : a \to b$ and $g : c \to d$ can only be composed when $b = c$. Composition is *partial*.

Internal Dialog

"A category is just a monoid where the multiplication might not be defined?"
Exactly. The technical term is **monoidoid** — rarely used — but the idea is precise: a category is to a monoid what a groupoid is to a group.

The correspondence is made exact by passing to one-object categories.

Theorem 13.1. *There is a bijection between monoids and categories with exactly one object.*

Let $\mathcal{C}$ be a category with one object, say $\star$.	*$\mathcal{C}$ has exactly the morphisms $\mathcal{C}(\star, \star)$.*
Composition $\circ : \mathcal{C}(\star, \star) \times \mathcal{C}(\star, \star) \to \mathcal{C}(\star, \star)$ is total, since all morphisms share domain and codomain $\star$.	*In a one-object category, every morphism has both domain and codomain equal to $\star$.*
Composition is associative (category axiom); $\mathrm{id}_\star$ is a two-sided unit.	*This is precisely the monoid axioms for $(\mathcal{C}(\star, \star), \circ, \mathrm{id}_\star)$.*
Conversely, given a monoid $(M, \cdot, e)$, define a category $\mathbf{B}M$ with one object $\star$ and morphisms $\mathbf{B}M(\star, \star) = M$.	*$\mathbf{B}$ stands for "delooping" — a standard construction.*
Composition in $\mathbf{B}M$ is multiplication in M; identity is e. Associativity and unit laws hold because M is a monoid.	*The two constructions are inverses: $\mathbf{B}M$ round-trips a monoid back to itself.*

How to Say This

$\mathbf{B}M$ is read "the delooping of M" or "M viewed as a one-object category." The "B" comes from the classifying space construction in topology.

The same pattern gives a bijection between groups and one-object groupoids (categories where every morphism is an isomorphism).

13.2 Preorders, Groupoids, and the Algebraic Taxonomy

A **preorder** $(P, \leq)$ gives a category: objects are elements, and there is exactly one morphism $a \to b$ if $a \leq b$, no morphism otherwise. The category axioms become: $a \leq a$ (reflexivity = identity) and $a \leq b$, $b \leq c$ implies $a \leq c$ (transitivity = composition). A preorder is a category where each hom-set has at most one element.

A **groupoid** is a category where every morphism is an isomorphism. Groups are one-object groupoids. Equivalence relations, viewed as preorders, are groupoids that are also preorders: since $a \sim b \Leftrightarrow b \sim a$, every morphism $a \to b$ has an inverse $b \to a$.

Algebraic structure	**As a category**	**Condition**
Monoid	1-object category	—
Group	1-object groupoid	every morphism invertible
Preorder	thin category	at most 1 morphism per hom-set
Partial order	thin category with no cycles	thin + no $a \neq b$ with $a \leq b \leq a$
Groupoid	category	every morphism invertible
Equivalence relation	thin groupoid	thin + invertible

These are not just analogies — they are literal special cases. The category **Mon** of monoids is a full subcategory of **Cat** via **B**. The category **Grp** embeds into **Gpd** (groupoids). The category **Pos** of partial orders embeds into **Cat**.

13.3 Ab-Enriched Categories and the Additive Setting

A category is **Ab-enriched** when each hom-set $\mathcal{C}(a, b)$ is an abelian group and composition is bilinear:

$$(f + f') \circ g = f \circ g + f' \circ g, \qquad f \circ (g + g') = f \circ g + f \circ g'.$$

Such categories are called **preadditive**. The zero morphism $0_{a,b} : a \to b$ is the zero element of the abelian group $\mathcal{C}(a, b)$.

When a preadditive category also has a zero object (0 with $\mathcal{C}(0, a)$ and $\mathcal{C}(a, 0)$ each having one element for all a), and all finite products and coproducts, it is called **additive**. In additive categories, products and coproducts coincide and are called **biproducts**, written $A \oplus B$.

Theorem 13.2. *In a preadditive category $\mathcal{C}$, if a product $A \times B$ and a coproduct $A + B$ both exist for objects A, B, then they are isomorphic.*

Write $\iota_1 : A \to A + B$, $\iota_2 : B \to A + B$ for the coproduct inclusions, and $\pi_1 : A \times B \to A$, $\pi_2 : A \times B \to B$ for the product projections.	*Standard notation for the universal morphisms.*
The universal property of $A \times B$ gives a unique morphism $\phi : A + B \to A \times B$ such that $\pi_1 \circ \phi \circ \iota_1 = \mathrm{id}_A$, $\pi_2 \circ \phi \circ \iota_2 = \mathrm{id}_B$, $\pi_1 \circ \phi \circ \iota_2 = 0$, $\pi_2 \circ \phi \circ \iota_1 = 0$.	*The cross-terms vanish because the only morphisms involving $0_{a,b}$ in the preadditive structure allow them to be zero.*
By the bilinearity of composition and the universal property of the coproduct, $\phi \circ (\iota_1 \circ \pi_1 + \iota_2 \circ \pi_2) = \mathrm{id}_{A \times B}$ and $(\iota_1 \circ \pi_1 + \iota_2 \circ \pi_2) \circ \phi = \mathrm{id}_{A+B}$.	*The sum is well-defined because hom-sets are abelian groups.*
So ϕ is an isomorphism $A + B \xrightarrow{\sim} A \times B$.	*This isomorphism is the biproduct structure.*

13.4 Functors Between Algebraic Categories

Under the algebraic interpretation, functors between one-object categories are precisely monoid homomorphisms. A functor $F : \mathbf{B}M \to \mathbf{B}N$ maps the single object $\star_M$ to $\star_N$ and acts on morphisms as $F : M \to N$; functoriality requires $F(m \cdot m') = F(m) \cdot F(m')$ and $F(e_M) = e_N$. That is precisely a monoid homomorphism.

Similarly, a functor $F : \mathcal{P} \to \mathcal{Q}$ between preorder categories is an order-preserving map: $a \leq b \Rightarrow F(a) \leq F(b)$.

Natural transformations between functors of one-object categories are elements of the target monoid satisfying a conjugation condition. A natural transformation $\alpha : F \Rightarrow G$ between functors $F, G : \mathbf{B}M \to \mathbf{B}N$ is a single element $n \in N$ such that for every $m \in M$:

$$G(m) \cdot n = n \cdot F(m).$$

This is a **bimodule map condition** or **intertwining condition**.

13.5 Exercises

What algebraic structure corresponds to a one-object category?	A monoid. The single hom-set $\mathcal{C}(\star, \star)$ with composition as multiplication and the identity morphism as unit.
What notation is used for the one-object category associated to a monoid M?	$\mathbf{B}M$, the "delooping" of M.
What algebraic structure corresponds to a one-object groupoid?	A group. Every morphism has an inverse, so the monoid is a group.
What is a thin category?	A category where each hom-set has at most one element. Preorders are the thin categories.
What category axioms become when a category is a preorder?	Reflexivity ($a \leq a$) = identity; transitivity = composition. No further axioms.
What is a groupoid?	A category in which every morphism is an isomorphism.
How does an equivalence relation embed into category theory?	As a thin groupoid: a thin category where every morphism (arrow $a \to b$ when $a \sim b$) is invertible.
What is a preadditive category?	An **Ab**-enriched category: hom-sets are abelian groups and composition is bilinear.
What is a zero morphism in a preadditive category?	The zero element $0_{a,b} \in \mathcal{C}(a, b)$; it factors through any zero object.
What is a biproduct?	An object $A \oplus B$ that is simultaneously a product and a coproduct of A and B; exists in additive categories.
In a preadditive category, when do products equal coproducts?	When both exist, they are isomorphic (the biproduct theorem). The isomorphism is constructed via the bilinear structure.
What is a functor between one-object categories?	A monoid homomorphism. Functoriality is exactly the homomorphism conditions.
What is a natural transformation between functors of one-object categories?	An element n of the target monoid satisfying $G(m) \cdot n = n \cdot F(m)$ for all m — an intertwining condition.
What is a functor between preorder categories?	An order-preserving map: $a \leq b \Rightarrow F(a) \leq F(b)$.
Why is the algebraic perspective useful?	It shows that classical algebraic structures (monoids, groups, orders) are not just analogous to categories but literally special cases, making their theorems instances of general categorical theorems.
What is the category **Gpd** of groupoids?	Objects are groupoids; morphisms are functors between them. It contains **Grp** as a full subcategory via **B**.
What does it mean for a category to be "skeletally small"?	It has only a set of isomorphism classes of objects; equivalently, it is equivalent to a small category.
What is a free category on a directed graph?	Objects are vertices; morphisms are finite paths (sequences of composable edges); composition is path concatenation; identity is the empty path at each vertex.
What universal property does the free category satisfy?	Any function from the graph to the underlying graph of a category $\mathcal{C}$ extends uniquely to a functor from the free category to $\mathcal{C}$.
How does the free-forgetful adjunction for categories go?	Free category functor $\mathbf{F} : \mathbf{Grph} \to \mathbf{Cat}$ is left adjoint to the underlying-graph functor $U : \mathbf{Cat} \to \mathbf{Grph}$.
What is a congruence on a category?	An equivalence relation on each hom-set, compatible with composition; used to form quotient categories.
What is the relationship between relations on a monoid and congruences?	A congruence on the one-object category $\mathbf{B}M$ is exactly a congruence on the monoid M in the algebraic sense.

Chapter Summary

A category is a **monoid with multiple objects**: composition is a partially defined associative operation. One-object categories are exactly monoids; one-object groupoids are groups; thin categories are preorders. Enriched over **Ab**, categories become preadditive; with biproducts and zero object, additive. Functors between algebraic categories recover algebraic homomorphisms; natural transformations recover intertwining conditions.

13.6 Formal Restatement

Formal Restatement

Algebraic taxonomy. A category $\mathcal{C}$ is:

- a **monoid** iff $|\operatorname{ob}\mathcal{C}| = 1$;
- a **group** iff it is a monoid and a groupoid;
- a **preorder** iff $|\mathcal{C}(a,b)| \leq 1$ for all a, b;
- a **groupoid** iff every morphism is an isomorphism.

Delooping. $\mathbf{B} : \mathbf{Mon} \hookrightarrow \mathbf{Cat}$, $\mathbf{B}M$ has one object $\star$ and $\mathbf{B}M(\star,\star) = M$.
Preadditive. $\mathcal{C}$ is preadditive iff it is **Ab**-enriched: each $\mathcal{C}(a,b) \in \mathbf{Ab}$; composition bilinear.
Biproduct theorem. In a preadditive category, if $A \times B$ and $A + B$ both exist, then $A + B \cong A \times B$.
Nat transf between BM-functors. $\alpha : F \Rightarrow G$ for $F, G : \mathbf{B}M \to \mathbf{B}N$ is an element $n \in N$ with $G(m) \cdot n = n \cdot F(m)$ for all $m \in M$.

Chapter 14

Functors as Structure Maps

A functor $F : \mathcal{C} \to \mathcal{D}$ carries objects to objects and morphisms to morphisms, preserving composition and identity. But what precisely is being preserved? This chapter examines the structural content of functors — what they must, may, and cannot preserve — and then extends the picture to **profunctors**, which generalize functors by replacing the target category with a category of relations.

14.1 What Functors Preserve and Reflect

A functor $F : \mathcal{C} \to \mathcal{D}$ *preserves* a property P of morphisms if: whenever f has property P in $\mathcal{C}$, $F(f)$ has property P in $\mathcal{D}$. It *reflects* P if: whenever $F(f)$ has P in $\mathcal{D}$, f has P in $\mathcal{C}$.

Property	**All functors preserve?**	**Notes**
Isomorphisms	Yes	$F(f)F(f^{-1}) = \mathrm{id}$
Monomorphisms	Not in general	Left adjoints need not preserve monos
Epimorphisms	Not in general	Right adjoints need not preserve epis
Limits	Right adjoints preserve limits	RAPL
Colimits	Left adjoints preserve colimits	LAPC

The classification matters: knowing whether a functor is a left or right adjoint immediately tells you what it preserves, before computing anything specific.

14.1.1 Full, Faithful, Essentially Surjective

Definition 14.1. A functor $F : \mathcal{C} \to \mathcal{D}$ is:

- **faithful** if each $F_{a,b} : \mathcal{C}(a, b) \to \mathcal{D}(Fa, Fb)$ is injective;
- **full** if each $F_{a,b}$ is surjective;
- **fully faithful** if each $F_{a,b}$ is bijective;
- **essentially surjective** if for every $d \in \mathcal{D}$, there exists $c \in \mathcal{C}$ with $F(c) \cong d$.

An equivalence of categories is a functor that is fully faithful and essentially surjective.

Fully faithful functors reflect isomorphisms:

Theorem 14.2. *If $F : \mathcal{C} \to \mathcal{D}$ is fully faithful and $F(f)$ is an isomorphism, then f is an isomorphism.*

Suppose $F(f) : F(a) \to F(b)$ is an isomorphism in $\mathcal{D}$, with inverse $g : F(b) \to F(a)$.	*We need to find $h : b \to a$ in $\mathcal{C}$ with $h \circ f = \mathrm{id}_a$ and $f \circ h = \mathrm{id}_b$.*
Since F is full, there exists $h : b \to a$ in $\mathcal{C}$ with $F(h) = g$.	*Fullness gives a preimage for every morphism in $\mathcal{D}(F(b), F(a))$.*
$F(h \circ f) = F(h) \circ F(f) = g \circ F(f) = \mathrm{id}_{F(a)} = F(\mathrm{id}_a)$.	*Functoriality of F and the fact that g is inverse to $F(f)$.*
Since F is faithful, $F(h \circ f) = F(\mathrm{id}_a)$ implies $h \circ f = \mathrm{id}_a$.	*Faithfulness: equal images under F imply equal morphisms.*
By the same argument, $f \circ h = \mathrm{id}_b$.	*Symmetric argument using $F(f \circ h) = F(f) \circ g = \mathrm{id}_{F(b)}$.*
Therefore f is an isomorphism with inverse h.	*Both composite conditions are verified.*

14.2 The Image and Subcategories

A functor $F : \mathcal{C} \to \mathcal{D}$ has an **essential image**: the full subcategory of $\mathcal{D}$ on objects isomorphic to $F(c)$ for some $c \in \mathcal{C}$. Full faithfulness means F is an equivalence onto its essential image.

Definition 14.3. A **subcategory** $\mathcal{S} \subseteq \mathcal{D}$ is full if for all $s, s' \in \mathcal{S}$, $\mathcal{S}(s, s') = \mathcal{D}(s, s')$. The inclusion functor $\iota : \mathcal{S} \hookrightarrow \mathcal{D}$ is then fully faithful.

Reflective subcategories arise when the inclusion functor has a left adjoint (the **reflector**):

Definition 14.4. A full subcategory $\iota : \mathcal{S} \hookrightarrow \mathcal{D}$ is **reflective** if ι has a left adjoint $R : \mathcal{D} \to \mathcal{S}$.

The reflector R sends each object of $\mathcal{D}$ to the "nearest" object in $\mathcal{S}$. Examples: sheafification (sheaves are a reflective subcategory of presheaves); abelianization (abelian groups as a reflective subcategory of groups).

Where the Name Comes From

Profunctor (also **distributor**, also **bimodule**) was introduced by *Jean Bénabou* (1973) in his work on bicategories. The name combines *pro-* (Latin "in favor of, in place of") with *functor*: a profunctor $P : \mathcal{C} \nrightarrow \mathcal{D}$ is "a functor in place of an ordinary functor" — generalizing functors $\mathcal{C} \to \mathcal{D}$ by allowing **Set**-valued data $P(c, d)$ for each pair $(c, d) \in \mathcal{C} \times \mathcal{D}$.
The alternative *distributor* (also Bénabou's term) emphasizes that a profunctor distributes morphisms across the two-sided structure. *Bimodule* emphasizes the analogy with bimodules over rings: a profunctor is a $\mathcal{D}$-$\mathcal{C}$-bimodule of sets. All three names denote the same thing.

14.3 Profunctors

A functor $F : \mathcal{C} \to \mathcal{D}$ gives a hom-set bijection $\mathcal{D}(F(c), d)$ for every c, d. This is a function of two variables — contravariant in c, covariant in d. This is the structure of a profunctor.

Definition 14.5. A **profunctor** (also called a **distributor** or **bimodule**) from $\mathcal{C}$ to $\mathcal{D}$, written $P : \mathcal{C} \nrightarrow \mathcal{D}$, is a functor

$$P : \mathcal{D}^{op} \times \mathcal{C} \;\to\; \mathbf{Set}.$$

That is, a presheaf on $\mathcal{D}^{op} \times \mathcal{C}$.

How to Say This

$P : \mathcal{C} \nrightarrow \mathcal{D}$ is read "P is a profunctor from $\mathcal{C}$ to $\mathcal{D}$." The slashed arrow $\nrightarrow$ distinguishes profunctors from ordinary functors.

Every functor $F : \mathcal{C} \to \mathcal{D}$ gives rise to two profunctors:

$$\begin{aligned} \mathcal{D}(-, F(-)) : \mathcal{C} \nrightarrow \mathcal{D}, &\qquad (d, c) \mapsto \mathcal{D}(d, F(c)), \\ \mathcal{D}(F(-), -) : \mathcal{D} \nrightarrow \mathcal{C}, &\qquad (c, d) \mapsto \mathcal{D}(F(c), d). \end{aligned}$$

14.3.1 Composition of Profunctors

Two profunctors $P : \mathcal{C} \nrightarrow \mathcal{D}$ and $Q : \mathcal{D} \nrightarrow \mathcal{E}$ can be composed. The composite $Q \circ P : \mathcal{C} \nrightarrow \mathcal{E}$ is the **coend**:

$$(Q \circ P)(e, c) = \int^{d \in \mathcal{D}} Q(e, d) \times P(d, c).$$

(Coends are introduced formally in Volume III; the intuition is a "tensor product" over $\mathcal{D}$: matching outputs of P with inputs of Q.)

This composition makes profunctors the morphisms in the **bicategory of profunctors Prof**. The identity profunctor on $\mathcal{C}$ is the hom-functor $\mathcal{C}(-, -) : \mathcal{C}^{op} \times \mathcal{C} \to \mathbf{Set}$.

14.3.2 Representable Profunctors

A profunctor $P : \mathcal{C} \nrightarrow \mathcal{D}$ is **representable** if it is isomorphic to $\mathcal{D}(-, F(-))$ for some functor $F : \mathcal{C} \to \mathcal{D}$. Profunctors generalize functors by dropping the representability requirement.

The profunctor perspective is powerful: adjunctions between $F \dashv G$ are exactly profunctor isomorphisms $\mathcal{D}(F(-), -) \cong \mathcal{C}(-, G(-))$.

14.4 Exercises

What does it mean for a functor to preserve a property P?	Whenever f has property P in the source, $F(f)$ has property P in the target.

What does it mean for a functor to reflect a property?	Whenever $F(f)$ has property P in the target, f has property P in the source.
Do all functors preserve isomorphisms?	Yes. If $f \circ g = \mathrm{id}$ and $g \circ f = \mathrm{id}$, then $F(f) \circ F(g) = \mathrm{id}$ and $F(g) \circ F(f) = \mathrm{id}$.
What is a faithful functor?	One where each function on hom-sets $F_{a,b}$ is injective.
What is a full functor?	One where each $F_{a,b}$ is surjective: every morphism between images lifts to a morphism in the source.
What is an equivalence of categories?	A functor that is fully faithful and essentially surjective.
Do fully faithful functors reflect isomorphisms?	Yes. If $F(f)$ is an isomorphism then f is an isomorphism.
What is a reflective subcategory?	A full subcategory whose inclusion has a left adjoint (the reflector).
What is a profunctor from $\mathcal{C}$ to $\mathcal{D}$?	A functor $P : \mathcal{D}^{op} \times \mathcal{C} \to \mathbf{Set}$.
What notation is used for a profunctor from $\mathcal{C}$ to $\mathcal{D}$?	$P : \mathcal{C} \nrightarrow \mathcal{D}$ with a slashed arrow.
What is the identity profunctor on $\mathcal{C}$?	The hom-functor $\mathcal{C}(-,-) : \mathcal{C}^{op} \times \mathcal{C} \to \mathbf{Set}$.
How does a functor $F : \mathcal{C} \to \mathcal{D}$ give a profunctor?	Two ways: $\mathcal{D}(-, F(-))$ (a profunctor $\mathcal{C} \nrightarrow \mathcal{D}$) or $\mathcal{D}(F(-), -)$ (a profunctor $\mathcal{D} \nrightarrow \mathcal{C}$).
When is a profunctor representable?	When it is isomorphic to $\mathcal{D}(-, F(-))$ for some functor F.
How are adjunctions expressed as profunctors?	$F \dashv G$ iff there is an isomorphism of profunctors $\mathcal{D}(F(-), -) \cong \mathcal{C}(-, G(-))$.
What is the composition of profunctors $P : \mathcal{C} \nrightarrow \mathcal{D}$ and $Q : \mathcal{D} \nrightarrow \mathcal{E}$?	$(Q \circ P)(e, c) = \int^d Q(e, d) \times P(d, c)$, a coend over $\mathcal{D}$.
What is the bicategory **Prof**?	Its objects are categories; morphisms are profunctors; 2-cells are natural transformations between profunctors; composition is via coend.
What is RAPL?	Right Adjoints Preserve Limits: if G is a right adjoint then G preserves all limits that exist in its source category.
What is LAPC?	Left Adjoints Preserve Colimits: if F is a left adjoint then F preserves all colimits.
Give an example of a functor that does not preserve monomorphisms.	The abelianization functor $\mathbf{Grp} \to \mathbf{Ab}$: injective group homomorphisms need not remain injective after abelianization.
What does "faithful" mean for a forgetful functor?	The underlying set function of a morphism determines the morphism; distinct structure-preserving maps have distinct underlying functions.
Is the forgetful functor $\mathbf{Ab} \to \mathbf{Set}$ faithful?	Yes: group homomorphisms with the same underlying function are equal.
Is the forgetful functor $\mathbf{Ab} \to \mathbf{Set}$ full?	No: not every function between abelian groups is a homomorphism.

Chapter Summary

Functors preserve all isomorphisms but need not preserve monomorphisms, epimorphisms, limits, or colimits unless they have adjoint structure (RAPL/LAPC). A functor is **faithful** if injective on hom-sets; **full** if surjective; an **equivalence** if fully faithful and essentially surjective. Fully faithful functors reflect isomorphisms. A **profunctor** $P : \mathcal{C} \nrightarrow \mathcal{D}$ is a functor $\mathcal{D}^{op} \times \mathcal{C} \to \mathbf{Set}$. Every functor gives two profunctors; adjunctions are profunctor isomorphisms. Profunctors form the bicategory **Prof**, with coend-composition.

14.5 Formal Restatement

Formal Restatement

Faithful/full/ff. $F : \mathcal{C} \to \mathcal{D}$ is faithful (resp. full, fully faithful) iff $F_{a,b} : \mathcal{C}(a, b) \to \mathcal{D}(Fa, Fb)$ is injective (resp. surjective, bijective) for all a, b.
Reflection of isos. F fully faithful $\Rightarrow$ F reflects isomorphisms.
Equivalence. F is an equivalence iff fully faithful and essentially surjective; iff it has a quasi-inverse G with $GF \cong \mathrm{Id}$ and $FG \cong \mathrm{Id}$.
Reflective subcategory. $\mathcal{S} \subseteq \mathcal{D}$ is reflective iff $\iota : \mathcal{S} \hookrightarrow \mathcal{D}$ has a left adjoint R.
Profunctor. $P : \mathcal{C} \nrightarrow \mathcal{D}$ is $P : \mathcal{D}^{op} \times \mathcal{C} \to \mathbf{Set}$. Identity: $\mathrm{Id}_{\mathcal{C}} = \mathcal{C}(-, -)$. Composition: $(Q \circ P)(e, c) = \int^{d} Q(e, d) \times P(d, c)$. $F \dashv G$ iff $\mathcal{D}(F-, -) \cong \mathcal{C}(-, G-)$ as profunctors.

Chapter 15

Representability and the Yoneda Philosophy

Volume I proved the Yoneda lemma and noted its importance. This chapter takes the Yoneda perspective seriously as the central organizing principle of all of category theory: *to know an object is to know how everything maps into and out of it.*

We prove the Yoneda lemma again, this time via the prooflog format that makes the structure of the argument visible. Then we unpack the philosophical content: universal properties are representability conditions, and the Yoneda embedding is an embedding precisely because knowing hom-sets determines objects up to isomorphism.

Where the Name Comes From

Representable (cross-ref Vol I Ch. 9 Yoneda) marks a functor as "representable by" a specific object — analogous to representable in classical algebra (a representation of a group acting on a space). The *Yoneda philosophy*: every important functor in mathematics is representable, or close to it. The terminology was popularized by Grothendieck's school in the 1960s; the modern emphasis on representability as a structural principle is largely due to Mac Lane, Lawvere, and Bénabou.

15.1 Representable Functors

Definition 15.1. A functor $F : \mathcal{C}^{op} \to \mathbf{Set}$ is **representable** if there exists an object $c \in \mathcal{C}$ and a natural isomorphism $F \cong \mathcal{C}(-, c)$.

Dually, a covariant functor $F : \mathcal{C} \to \mathbf{Set}$ is **corepresentable** if $F \cong \mathcal{C}(c, -)$ for some c.

The object c is called the **representing object**, and the natural isomorphism is the **representation**.

The Yoneda lemma says exactly how many representations an object has: natural transformations $\mathcal{C}(-, c) \Rightarrow F$ are in bijection with elements of $F(c)$. In particular, representations of F by c correspond to "universal elements" — special elements $u \in F(c)$ such that every element of $F(d)$ is in the image of some morphism $d \to c$.

15.2 The Yoneda Lemma: Full Proof

Theorem 15.2 (Yoneda Lemma)**.** *Let $\mathcal{C}$ be a locally small category, $c \in \mathcal{C}$, and $F : \mathcal{C}^{op} \to \mathbf{Set}$. There is a bijection*

$$\Phi : [\mathcal{C}^{op}, \mathbf{Set}](\mathcal{C}(-, c),\ F) \ \xrightarrow{\sim}\ F(c),$$

natural in both c and F, given by $\Phi(\alpha) = \alpha_c(\mathrm{id}_c)$.

Define $\Phi : \mathrm{Nat}(\mathcal{C}(-,c), F) \to F(c)$ by $\Phi(\alpha) = \alpha_c(\mathrm{id}_c)$.	*Evaluate the natural transformation α at c, then at the identity morphism $\mathrm{id}_c \in \mathcal{C}(c,c)$.*
Define $\Psi : F(c) \to \mathrm{Nat}(\mathcal{C}(-,c), F)$ as follows: for $x \in F(c)$, set $\Psi(x)_d(f) = F(f)(x)$ for $f : d \to c$.	*Given $x \in F(c)$ and a morphism $f : d \to c$, apply F to f to get $F(f) : F(c) \to F(d)$, then evaluate at x.*
Check that $\Psi(x)$ is natural: for $g : e \to d$, we need $\Psi(x)_e(f \circ g) = F(g)(\Psi(x)_d(f))$, i.e., $F(f \circ g)(x) = F(g)(F(f)(x))$.	*This holds because F is a functor: $F(f \circ g) = F(g) \circ F(f)$.*
$\Phi \circ \Psi = \mathrm{id}$: $\Phi(\Psi(x)) = \Psi(x)_c(\mathrm{id}_c) = F(\mathrm{id}_c)(x) = x$.	*$F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$ since F is a functor.*
$\Psi \circ \Phi = \mathrm{id}$: given α, $\Psi(\Phi(\alpha))_d(f) = F(f)(\alpha_c(\mathrm{id}_c))$.	*By definition of Ψ and Φ.*
Naturality of α at $f : d \to c$ says: $\alpha_d(f) = \alpha_d(\mathcal{C}(f,c)(\mathrm{id}_c)) = F(f)(\alpha_c(\mathrm{id}_c))$.	*The naturality square for α at f: $\alpha_d \circ \mathcal{C}(f,c) = F(f) \circ \alpha_c$.*
So $\Psi(\Phi(\alpha))_d(f) = F(f)(\alpha_c(\mathrm{id}_c)) = \alpha_d(f)$. Hence $\Psi(\Phi(\alpha)) = \alpha$.	*This holds for all d and all $f : d \to c$, confirming $\Psi \circ \Phi = \mathrm{id}$.*
The bijection Φ is natural in c: for $g : c \to c'$, $\Phi(\mathcal{C}(-,g) \circ \alpha) = \alpha_{c'}(g)$; and $F(g)(\Phi(\alpha)) = F(g)(\alpha_c(\mathrm{id}_c)) = \alpha_{c'}(g)$.	*Functoriality and the naturality of α at g.*

15.3 The Yoneda Embedding

Theorem 15.3 (Yoneda Embedding). *The functor $Y : \mathcal{C} \to [\mathcal{C}^{op}, \mathbf{Set}]$ defined by $Y(c) = \mathcal{C}(-,c)$ and $Y(f) = \mathcal{C}(-,f)$ is fully faithful.*

We need to show: for all $c, c' \in \mathcal{C}$, the function $Y_{c,c'} : \mathcal{C}(c,c') \to [\mathcal{C}^{op}, \mathbf{Set}](Y(c), Y(c'))$ is a bijection.	*Fully faithful = bijective on all hom-sets.*
Apply Yoneda with $F = Y(c') = \mathcal{C}(-,c')$: $[\mathcal{C}^{op}, \mathbf{Set}](\mathcal{C}(-,c), \mathcal{C}(-,c')) \cong \mathcal{C}(-,c')(c) = \mathcal{C}(c,c')$.	*The Yoneda bijection with $F = Y(c')$.*
Under this bijection, $f : c \to c'$ corresponds to the natural transformation $\mathcal{C}(-,f)$ given by post-composition with f.	*Tracing through Ψ: $\Psi(f)_d(g) = \mathcal{C}(d,f)(g) = f \circ g = \mathcal{C}(-,f)_d(g)$.*
So $Y_{c,c'}$ is the Yoneda bijection, hence a bijection. Y is fully faithful.	*The Yoneda lemma directly implies the Yoneda embedding is fully faithful.*

Internal Dialog

"So the Yoneda embedding says that $\mathcal{C}$ embeds into its presheaf category as the full subcategory of representable presheaves?"

Exactly. The Yoneda embedding is fully faithful, meaning $\mathcal{C}$ and the subcategory of representable presheaves in $[\mathcal{C}^{op}, \mathbf{Set}]$ are equivalent. Every category can be "linearized" into a presheaf category without losing any information.

15.4 Universal Properties as Representability

The Yoneda philosophy has a practical payoff: every universal property is a representability statement, and the Yoneda lemma converts between them.

A product $a \times b$ in $\mathcal{C}$ is an object representing the functor

$$T_{a,b}(c) = \mathcal{C}(c,a) \times \mathcal{C}(c,b) : \mathcal{C}^{op} \to \mathbf{Set}.$$

The universal pair of projections (π_1, π_2) is the universal element in $T_{a,b}(a \times b) = \mathcal{C}(a \times b, a) \times \mathcal{C}(a \times b, b)$.

A coequalizer of $f, g : a \to b$ is an object representing

$$E_{f,g}(c) = \{h : b \to c \mid h \circ f = h \circ g\} \subseteq \mathcal{C}(b,c).$$

An exponential b^a in a cartesian closed category represents $c \mapsto \mathcal{C}(c \times a, b)$, and the evaluation morphism is the universal element.

Theorem 15.4. *An object $r \in \mathcal{C}$ together with a natural isomorphism $F \cong \mathcal{C}(-,r)$ is precisely an initial object of the category of elements $\int F$ of F.*

The **category of elements** $\int F$ of $F : \mathcal{C}^{op} \to \mathbf{Set}$ has objects (c,x) with $x \in F(c)$ and morphisms $(c,x) \to (c',x')$ the morphisms $f : c' \to c$ in $\mathcal{C}$ (note the reversal) with $F(f)(x) = x'$. The universal element is initial in this category.

15.5 Exercises

What does it mean for $F : \mathcal{C}^{op} \to \mathbf{Set}$ to be representable?	There exists $c \in \mathcal{C}$ with $F \cong \mathcal{C}(-, c)$ naturally.
What is the representing object?	The object c such that $F \cong \mathcal{C}(-, c)$.
State the Yoneda lemma as a bijection.	$[\mathcal{C}^{op}, \mathbf{Set}](\mathcal{C}(-, c), F) \cong F(c)$, natural in c and F.
What is the bijection map Φ in the Yoneda lemma?	$\Phi(\alpha) = \alpha_c(\mathrm{id}_c)$: evaluate α at c then at the identity.
What is the inverse Ψ of Φ?	$\Psi(x)_d(f) = F(f)(x)$ for $f : d \to c$ and $x \in F(c)$.
What key step proves $\Psi \circ \Phi = \mathrm{id}$?	The naturality of α: $\alpha_d(f) = F(f)(\alpha_c(\mathrm{id}_c))$.
What does it mean for the Yoneda embedding to be fully faithful?	The function $\mathcal{C}(c, c') \to \mathrm{Nat}(\mathcal{C}(-, c), \mathcal{C}(-, c'))$ is a bijection for all c, c'.
Is the Yoneda embedding an equivalence of categories?	Not in general: it need not be essentially surjective (not every presheaf is representable).
What is the Yoneda philosophy in one sentence?	An object is completely determined by its hom-functors — knowing $\mathcal{C}(-, c)$ or $\mathcal{C}(c, -)$ determines c up to isomorphism.
What functor does a product $a \times b$ represent?	$T_{a,b}(c) = \mathcal{C}(c, a) \times \mathcal{C}(c, b)$: pairs of morphisms from c to both a and b.
What is a universal element of a representable functor F?	An element $u \in F(r)$ corresponding to id_r under the Yoneda bijection; equivalently, the image of id_r under the representation isomorphism.
What is the category of elements $\int F$ of $F : \mathcal{C}^{op} \to \mathbf{Set}$?	Objects: pairs (c, x) with $x \in F(c)$. Morphisms $(c, x) \to (c', x')$: morphisms $f : c' \to c$ in $\mathcal{C}$ with $F(f)(x) = x'$.
What does the universal element correspond to in $\int F$?	The initial object of $\int F$.
What does the Yoneda lemma say about natural transformations between representable functors?	$\mathrm{Nat}(\mathcal{C}(-, c), \mathcal{C}(-, c')) \cong \mathcal{C}(c, c')$: natural transformations between representables are exactly morphisms between the representing objects.
Does the Yoneda lemma hold for enriched categories?	Yes, with the appropriate enriched version: $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \cong F(a)$ in $\mathcal{V}$.
What does it mean that "the Yoneda embedding is the free cocompletion"?	$[\mathcal{C}^{op}, \mathbf{Set}]$ is the universal cocomplete category receiving $\mathcal{C}$ via a functor that preserves all colimits that exist in $\mathcal{C}$.
How does the Yoneda lemma imply that $\mathcal{C}$ determines $[\mathcal{C}^{op}, \mathbf{Set}]$?	The Yoneda embedding $Y : \mathcal{C} \hookrightarrow [\mathcal{C}^{op}, \mathbf{Set}]$ is fully faithful, so $\mathcal{C}$ embeds as a full subcategory, and $[\mathcal{C}^{op}, \mathbf{Set}]$ is the free cocompletion of this subcategory.
Why does $\mathcal{C}(-, c) \cong \mathcal{C}(-, c')$ imply $c \cong c'$?	The Yoneda embedding is fully faithful, so it reflects isomorphisms.
What is a contravariant representable presheaf?	A functor $\mathcal{C}^{op} \to \mathbf{Set}$ of the form $\mathcal{C}(-, c)$ for some c.
Give the Yoneda lemma as a statement about adjunctions.	The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{op}, \mathbf{Set}]$ gives a bijection: natural transformations $Y(c) \Rightarrow F$ are in natural bijection with elements of $F(c)$.

Chapter Summary

A functor $F : \mathcal{C}^{op} \to \mathbf{Set}$ is **representable** when $F \cong \mathcal{C}(-, c)$. The **Yoneda lemma** says $\mathrm{Nat}(\mathcal{C}(-, c), F) \cong F(c)$ naturally, with bijection $\alpha \mapsto \alpha_c(\mathrm{id}_c)$. The **Yoneda embedding** $Y : \mathcal{C} \hookrightarrow [\mathcal{C}^{op}, \mathbf{Set}]$ is fully faithful, embedding $\mathcal{C}$ into its presheaf category. Every universal property is a representability condition; the universal element is the initial object of the category of elements.

15.6 Formal Restatement

Formal Restatement

Representable. $F : \mathcal{C}^{op} \to \mathbf{Set}$ representable iff $\exists\, c,\ u \in F(c)$ with $\Psi(u) : \mathcal{C}(-, c) \xrightarrow{\sim} F$ a natural isomorphism.
Yoneda lemma. $\mathrm{Nat}(\mathcal{C}(-, c), F) \cong F(c)$, naturally in $c \in \mathcal{C}$ and $F \in [\mathcal{C}^{op}, \mathbf{Set}]$. Bijection: $\Phi(\alpha) = \alpha_c(\mathrm{id}_c)$; inverse: $\Psi(x)_d(f) = F(f)(x)$.
Yoneda embedding. $Y : \mathcal{C} \hookrightarrow [\mathcal{C}^{op}, \mathbf{Set}]$, $Y(c) = \mathcal{C}(-, c)$, fully faithful. Consequence: $c \cong c'$ iff $Y(c) \cong Y(c')$; $\mathcal{C}(-, c) \cong \mathcal{C}(-, c') \Rightarrow c \cong c'$.
Category of elements. $\int F$: objects (c, x) with $x \in F(c)$; morphisms $(c, x) \to (c', x')$ are $f : c' \to c$ with $F(f)(x) = x'$. Universal element = initial object of $\int F$.

Chapter 16

Limits as Representable Constructions

Volume I defined limits via cones and universal properties. This chapter reframes limits through the Yoneda lens: a limit of F is the object representing the functor of cones over F. This framing connects limits directly to the Yoneda lemma and makes the universal property automatic.

Where the Name Comes From

Cone (over a diagram) is from geometric intuition: visualize the apex of the cone as a single point connected by arrows down to all "base" objects of the diagram. The metaphor is geometric (a literal geometric cone has an apex and a base); the categorical generalization keeps the picture but replaces the geometric base with an arbitrary diagram. Mac Lane's terminology, standard from the 1960s.

16.1 Cones as a Functor

A cone over $F : \mathcal{J} \to \mathcal{C}$ with apex c is a natural transformation $\Delta c \Rightarrow F$, where $\Delta c : \mathcal{J} \to \mathcal{C}$ is the constant functor at c. Varying the apex, we get a functor:

$$\mathrm{Cone}(-, F) : \mathcal{C}^{op} \to \mathbf{Set}, \qquad c \mapsto \mathrm{Nat}(\Delta c,\, F).$$

This is a presheaf on $\mathcal{C}$.

Definition 16.1. A **limit** of $F : \mathcal{J} \to \mathcal{C}$ is a representing object for the presheaf $\mathrm{Cone}(-, F)$: an object $\lim F \in \mathcal{C}$ with a natural isomorphism

$$\mathcal{C}(c,\, \lim F) \;\cong\; \mathrm{Cone}(c,\, F) = \mathrm{Nat}(\Delta c,\, F),$$

natural in c.

Theorem 16.2. *Limits, when they exist, are unique up to unique isomorphism.*

Suppose (ℓ, λ) and (ℓ', λ') both represent $\mathrm{Cone}(-, F)$.	*Both are representing objects for the same functor.*
By the Yoneda lemma, any two representing objects for the same functor are uniquely isomorphic.	*The Yoneda embedding is fully faithful, so it reflects isomorphisms.*
Under the Yoneda embedding, this isomorphism corresponds to a unique isomorphism $\ell \cong \ell'$ in $\mathcal{C}$.	*Fully faithful functors reflect isomorphisms.*

16.2 Representable Functors Preserve Limits

Theorem 16.3. *For any $c \in \mathcal{C}$, the functor $\mathcal{C}(c, -) : \mathcal{C} \to \mathbf{Set}$ preserves all limits.*

We want: $\mathcal{C}(c, \lim F) \cong \lim_j \mathcal{C}(c, F(j))$.	*That is, $\mathcal{C}(c, -)$ commutes with limits.*
$\mathcal{C}(c, \lim F) \cong \mathrm{Nat}(\Delta c, F)$ by the representability of $\lim F$.	*Definition of limit.*
$\mathrm{Nat}(\Delta c, F)$ is the set of compatible families $(f_j : c \to F(j))_{j \in \mathcal{J}}$.	*A natural transformation from a constant functor is exactly a compatible family.*
This equals $\lim_j \mathcal{C}(c, F(j))$ in $\mathbf{Set}$.	*Limits in* **Set** *are sets of compatible tuples.*

16.3 Completeness from Products and Equalizers

Theorem 16.4. *$\mathcal{C}$ is complete if and only if it has all small products and all equalizers.*

($\Rightarrow$): Products and equalizers are limits.	*Special cases of the general definition.*
($\Leftarrow$): Given $F : \mathcal{J} \to \mathcal{C}$, form $P = \prod_j F(j)$ and $Q = \prod_{f:j\to j'} F(j')$.	*Products over all objects and all morphism targets.*
Define $e, e' : P \to Q$ encoding the naturality condition for cones.	*e uses projections; e' applies $F(f)$ then projects.*
The equalizer of e and e' is $\lim F$.	*An element of the equalizer is a compatible family – a cone.*

16.4 Pointwise Limits in Functor Categories

Theorem 16.5. *If $\mathcal{D}$ is complete, then limits in $[\mathcal{C}, \mathcal{D}]$ are computed pointwise:* $(\lim_i F_i)(c) \cong \lim_i(F_i(c))$. In particular, $[\mathcal{C}^{op}, \mathbf{Set}]$ is complete and cocomplete.

16.5 Exercises

What functor does a limit represent?	$\mathrm{Cone}(-, F) = \mathrm{Nat}(\Delta(-), F)$, the functor of cones over F.
What is the universal cone?	The cone corresponding to $\mathrm{id}_{\lim F}$ under the Yoneda bijection.
Why are limits unique up to unique isomorphism?	Representing objects for the same functor are uniquely isomorphic by Yoneda.
What does it mean for a functor to be continuous?	It preserves all small limits.
Do representable functors $\mathcal{C}(c, -)$ preserve limits?	Yes: $\mathcal{C}(c, \lim F) \cong \mathrm{Nat}(\Delta c, F) \cong \lim_j \mathcal{C}(c, F(j))$.
What two constructions suffice to build all limits?	Products and equalizers.
What two constructions suffice to build all colimits?	Coproducts and coequalizers.
How are limits computed in $[\mathcal{C}, \mathcal{D}]$?	Pointwise: $(\lim_i F_i)(c) \cong \lim_i(F_i(c))$ when $\mathcal{D}$ is complete.
Is $[\mathcal{C}^{op}, \mathbf{Set}]$ complete and cocomplete?	Yes; all limits and colimits are computed pointwise in **Set**.
What is the terminal object as a limit?	The limit of the empty diagram.
What is a pullback as a limit?	The limit of a cospan $A \to C \leftarrow B$.
What is an equalizer as a limit?	The limit of $A \rightrightarrows B$.
Does $\mathcal{C}(c, -)$ preserve colimits in general?	Not in general; only for compact/finitely-presented c.
How does the Yoneda embedding interact with limits?	Y preserves all limits: $Y(\lim F) \cong \lim(Y \circ F)$.
What is the limit formula as a Kan extension?	$\lim F = (\mathrm{Ran}_p F)(\star)$ where $p : \mathcal{J} \to \mathbf{1}$.
Why is **Set** both complete and cocomplete?	All products, equalizers, coproducts, and coequalizers exist as set-theoretic constructions.
What is the difference between a limit and colimit from the representability perspective?	Limit represents cones (maps in from variable apex); colimit corepresents cocones (maps out to variable apex).

Chapter Summary

A **limit** of F is the representing object for $\mathrm{Cone}(-, F)$; unique up to unique isomorphism by Yoneda. Representable functors are continuous. Every complete category is built from products and equalizers; every cocomplete from coproducts and coequalizers. Limits in functor categories are pointwise.

16.6 Formal Restatement

Formal Restatement

Limit as representable. $\mathcal{C}(c, \lim F) \cong \mathrm{Nat}(\Delta c, F)$, naturally in c.
Colimit as corepresentable. $\mathcal{C}(\operatorname{colim} F, c) \cong \mathrm{Nat}(F, \Delta c)$, naturally in c.
Continuity of representables. $\mathcal{C}(c, -) : \mathcal{C} \to \mathbf{Set}$ is continuous for all c.
Completeness. $\mathcal{C}$ complete $\Leftrightarrow$ has all products and equalizers. $\mathcal{C}$ cocomplete $\Leftrightarrow$ has all coproducts and coequalizers.
Pointwise. $(\lim_i F_i)(c) \cong \lim_i (F_i(c))$ in $[\mathcal{C}, \mathcal{D}]$ when $\mathcal{D}$ complete.

Chapter 17

Adjunctions: A Deeper Account

Chapter Overview

Chapter 7 introduced adjunctions through examples and established the triangle identities as a convenient alternative to the hom-set bijection. In this chapter we take adjunctions apart more carefully. The triangle identities are not merely *convenient* — they are *equivalent* to the hom-set bijection, and that equivalence has a genuine proof. We also show that the right adjoint of a functor is unique up to natural isomorphism (not merely asserted so), construct adjunctions from initial objects in comma categories, and ask what happens when the right adjoint itself has a right adjoint.

By the end you will understand adjunctions from multiple angles simultaneously: as bijections, as universal morphisms, and as structure in comma categories. That triangulation is the sign of deep understanding.

17.1 Triangle Identities Revisited

17.1.1 Informally: What Needs Proof

Chapter 7 stated that from the natural bijection $\varphi_{A,B} : \mathcal{D}(FA, B) \xrightarrow{\cong} \mathcal{C}(A, GB)$ one derives a unit $\eta : \mathrm{Id} \Rightarrow GF$ and a counit $\varepsilon : FG \Rightarrow \mathrm{Id}$ satisfying

$$(G\varepsilon)(\eta G) = \mathrm{id}_G,$$
$$(\varepsilon F)(F\eta) = \mathrm{id}_F.$$

But we also claimed the converse: given η and ε satisfying these equations, one can reconstruct the bijection φ. The full proof of this equivalence is the business of this section.

17.1.2 Expanding: The Two Directions

The key insight is that φ and its inverse can be written entirely in terms of η, ε, and the functors:

$$\varphi(f) = G(f) \circ \eta_A, \qquad f : FA \to B,$$
$$\varphi^{-1}(g) = \varepsilon_B \circ F(g), \qquad g : A \to GB.$$

That these formulas actually define a bijection, and that the bijection is natural, follow from the triangle identities. Let us verify this in a structured proof.

Theorem 17.1. *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. The following data are equivalent:*

1. *A natural bijection $\varphi_{A,B} : \mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$.*
2. *Natural transformations $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon : FG \Rightarrow \mathrm{Id}_{\mathcal{D}}$ satisfying the triangle identities.*

Proof. We prove that (1) implies (2), and that (2) implies (1), and that the constructions are mutually inverse.

(1) ⇒ (2). Define $\eta_A = \varphi_{A,FA}(\mathrm{id}_{FA})$ and $\varepsilon_B = \varphi^{-1}_{GB,B}(\mathrm{id}_{GB})$. Naturality of φ implies naturality of η and ε. The triangle identities follow from the naturality squares for φ evaluated at the identity morphisms.

(2) ⇒ (1). Define $\varphi(f) = G(f) \circ \eta_A$ and $\psi(g) = \varepsilon_B \circ F(g)$. The triangle identities ensure $\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$, as we verify in the proof log below. □

17.1.3 Formalizing: Triangle Identities ⇔ Bijection

Goal: $\psi(\varphi(f)) = f$ for $f : FA \to B$.	*We want to show the composite is the identity on each morphism.*
$\psi(\varphi(f)) = \varepsilon_B \circ F(\varphi(f))$	*By definition of ψ.*
$= \varepsilon_B \circ F(G(f) \circ \eta_A)$	*Substituting $\varphi(f) = G(f) \circ \eta_A$.*
$= \varepsilon_B \circ FG(f) \circ F(\eta_A)$	*F is a functor, so it preserves composition.*
$= f \circ \varepsilon_{FA} \circ F(\eta_A)$	*Naturality of ε: $\varepsilon_B \circ FG(f) = f \circ \varepsilon_{FA}$.*
$= f \circ \mathrm{id}_{FA} = f$	*The second triangle identity: $(\varepsilon F)(F\eta) = \mathrm{id}_F$, so $\varepsilon_{FA} \circ F(\eta_A) = \mathrm{id}_{FA}$.*
Goal: $\varphi(\psi(g)) = g$ for $g : A \to GB$.	*The dual computation.*
$\varphi(\psi(g)) = G(\psi(g)) \circ \eta_A$	*By definition of φ.*
$= G(\varepsilon_B \circ F(g)) \circ \eta_A$	*Substituting $\psi(g) = \varepsilon_B \circ F(g)$.*
$= G(\varepsilon_B) \circ GF(g) \circ \eta_A$	*G preserves composition.*
$= G(\varepsilon_B) \circ \eta_{GB} \circ g$	*Naturality of η: $GF(g) \circ \eta_A = \eta_{GB} \circ g$.*
$= \mathrm{id}_{GB} \circ g = g$	*First triangle identity: $(G\varepsilon)(\eta G) = \mathrm{id}_G$, so $G(\varepsilon_B) \circ \eta_{GB} = \mathrm{id}_{GB}$.*
Naturality of φ.	*For $h : A' \to A$ in $\mathcal{C}$ and $k : B \to B'$ in $\mathcal{D}$, the naturality squares for φ in both variables follow from the fact that η and ε are natural transformations and from functoriality of F and G.*

Internal Dialog

Why do we need naturality of ε in the fourth step above?
The naturality square for ε at $f : FA \to B$ says $\varepsilon_B \circ FG(f) = f \circ \varepsilon_{FA}$. This is exactly the equation that moves ε past $FG(f)$, trading the outer B-component for the inner FA-component. Without it, the proof would be stuck.

17.1.4 Reorganizing: Why Both Directions Matter

The theorem tells us that neither formulation is more fundamental than the other. In practice:

- The *hom-set bijection* form is easiest for computing and identifying adjunctions.
- The *unit/counit* form is easiest for working in 2-categories and proving things about composites of adjunctions.
- The formulas $\varphi(f) = Gf \circ \eta$ and $\varphi^{-1}(g) = \varepsilon \circ Fg$ let you move between the two whenever needed.

How to Say This

- The equations $(G\varepsilon)(\eta G) = \mathrm{id}_G$ and $(\varepsilon F)(F\eta) = \mathrm{id}_F$ are called the *triangle identities* because they correspond to two triangular diagrams (one for each functor).
- In the literature you will also see them written as $G\varepsilon_B \circ \eta_{GB} = \mathrm{id}_{GB}$ (the G-triangle) and $\varepsilon_{FA} \circ F\eta_A = \mathrm{id}_{FA}$ (the F-triangle).
- Some authors call these the "zig-zag identities," because the composites zig from $\mathcal{C}$ to $\mathcal{D}$ and zag back.

17.2 Uniqueness of Adjoints

17.2.1 Informally: Two Right Adjoints Must Be Isomorphic

Chapter 7 stated without full proof that adjoints are unique up to natural isomorphism. Now we can prove it rigorously. If $F \dashv G$ and $F \dashv G'$, then $G \cong G'$ via a canonical natural isomorphism. The proof is the cleanest illustration of the "Yoneda argument": two representable functors agreeing on all hom-sets must be isomorphic.

17.2.2 Expanding: The Idea

Suppose G and G' are both right adjoints to F. For each object $B \in \mathcal{D}$, we have natural bijections

$$\mathcal{C}(A, GB) \cong \mathcal{D}(FA, B),$$
$$\mathcal{C}(A, G'B) \cong \mathcal{D}(FA, B).$$

Composing, $\mathcal{C}(A, GB) \cong \mathcal{C}(A, G'B)$ naturally in A. By the Yoneda lemma, a natural isomorphism of representable functors $\mathcal{C}(-, GB) \cong \mathcal{C}(-, G'B)$ comes from a unique isomorphism $GB \cong G'B$. Since this works for every B, and the isomorphisms assemble naturally in B, we get a natural isomorphism $G \cong G'$.

Theorem 17.2. *If $F \dashv G$ and $F \dashv G'$, then there is a natural isomorphism $G \cong G'$. Dually, if $G \dashv F$ and $G' \dashv F$, then $G \cong G'$.*

Proof. We construct the isomorphism $\theta : G \Rightarrow G'$ explicitly. □

For each $B \in \mathcal{D}$, apply both bijections at $A = G'B$.	*We want a specific isomorphism $GB \to G'B$.*
The bijection for G': $\mathcal{D}(F(G'B), B) \cong \mathcal{C}(G'B, G'B)$.	*Set $A = G'B$.*
The counit $\varepsilon'_B : F(G'B) \to B$ corresponds to $\mathrm{id}_{G'B}$ under this bijection.	*Definition of counit as image of the identity.*
Apply the bijection for G with $A = G'B$, $B = B$: $\mathcal{D}(F(G'B), B) \cong \mathcal{C}(G'B, GB)$.	*Use the $F \dashv G$ bijection on ε'_B.*
Define $\theta_B := \varphi_G(\varepsilon'_B) = G(\varepsilon'_B) \circ \eta_{G'B} : G'B \to GB$.	*This is a specific morphism $G'B \to GB$ for each B.*
Symmetrically define $\theta_B^{-1} = G'(\varepsilon_B) \circ \eta'_{GB} : GB \to G'B$.	*Using the same construction with G and G' swapped.*
$\theta_B^{-1} \circ \theta_B = \mathrm{id}_{G'B}$: compute $G'(\varepsilon_B) \circ \eta'_{GB} \circ G(\varepsilon'_B) \circ \eta_{G'B}$.	*Composing the two morphisms.*
By naturality of η' and the triangle identity for G' this collapses to $\mathrm{id}_{G'B}$.	*Each triangle identity cancels one of the zigs.*
Similarly $\theta_B \circ \theta_B^{-1} = \mathrm{id}_{GB}$.	*Symmetric argument.*
Naturality: for $k : B \to B'$, both sides of $\theta_{B'} \circ G'(k) = G(k) \circ \theta_B$ follow from naturality of η, ε, η', ε'.	*Standard naturality check.*
Therefore $\theta : G \Rightarrow G'$ is a natural isomorphism.	*All components are isomorphisms and assembly is natural.*

17.2.3 Reorganizing: The Yoneda Shortcut

The proof above constructs θ_B by hand, but the Yoneda lemma offers a cleaner perspective. The composite

$$\mathcal{C}(-, GB) \cong \mathcal{D}(F(-), B) \cong \mathcal{C}(-, G'B)$$

is a natural isomorphism of representable presheaves on $\mathcal{C}$. The Yoneda lemma says that any natural isomorphism $\mathcal{C}(-, X) \cong \mathcal{C}(-, Y)$ arises from a unique isomorphism $X \cong Y$. Applied here, this gives $\theta_B : GB \cong G'B$ for each B, naturally in B.

The two approaches are equivalent, but the Yoneda version is shorter once you internalize the lemma.

17.2.4 Simplifying: The Slogan

$$\boxed{F \dashv G \text{ and } F \dashv G' \implies G \cong G' \text{ (naturally).}}$$

This means we may speak of *the* right adjoint to F without ambiguity, knowing it is determined up to unique natural isomorphism.

Where the Name Comes From

Comma category (notation $F \downarrow G$ or (F, G)) was introduced by *F. William Lawvere* in his 1963 PhD thesis. The name reflects the original notation: Lawvere wrote the objects of the category as triples (c, d, f) separated by commas. The construction unifies many ad-hoc categories of "maps with extra data" (slice categories, arrow categories, fibrations) under a single universal construction. Modern category theory uses $\downarrow$ or $/$ notation to indicate slice-like comma constructions.

17.3 Adjoints as Initial Objects in Comma Categories

17.3.1 Informally: The Comma Category Perspective

There is a third, very geometric way to understand adjunctions. The left adjoint $F(c)$ is characterized as the *initial object in the comma category* $(c \downarrow G)$. Let us unpack this.

The comma category $(c \downarrow G)$ has:

- Objects: pairs (d, f) where $d \in \mathcal{D}$ and $f : c \to G(d)$ is a morphism in $\mathcal{C}$.

- Morphisms from (d, f) to (d', f'): a morphism $h : d \to d'$ in $\mathcal{D}$ such that $G(h) \circ f = f'$.

An initial object in $(c \downarrow G)$ is an object (d_0, f_0) such that for every (d, f) there is a unique morphism $(d_0, f_0) \to (d, f)$.

Internal Dialog

What does an initial object in $(c \downarrow G)$ look like in the free-monoid example?
Take $c = \{a, b\}$ (a two-element set), $G =$ forgetful functor from monoids to sets, $\mathcal{D} = \mathbf{Mon}$. The comma category $(c \downarrow G)$ has objects: pairs (monoid M, function $c \to M$). The initial object is the free monoid on $\{a, b\}$: it has a canonical function $c \to F(c)$ (send each generator to itself), and any function $c \to M$ factors uniquely through $F(c)$ by the universal property.

17.3.2 Expanding: The Theorem

Theorem 17.3. *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$. The following are equivalent:*

1. *F is left adjoint to G.*
2. *For every $c \in \mathcal{C}$, there exists an initial object in the comma category $(c \downarrow G)$.*

When these hold, the initial object of $(c \downarrow G)$ is $(F(c), \eta_c)$, where $\eta_c : c \to GF(c)$ is the unit of the adjunction.

Proof. (1) $\Rightarrow$ (2): Given $F \dashv G$, we claim $(F(c), \eta_c)$ is initial in $(c \downarrow G)$.

Let $(d, f : c \to Gd)$ be any object of $(c \downarrow G)$. By the adjunction bijection, f corresponds uniquely to a morphism $\tilde{f} : F(c) \to d$ in $\mathcal{D}$ with $\varphi(\tilde{f}) = f$. That is, $G(\tilde{f}) \circ \eta_c = f$. This says $\tilde{f}$ is a morphism $(F(c), \eta_c) \to (d, f)$ in $(c \downarrow G)$. Uniqueness of $\tilde{f}$ from bijectivity of φ says it is the unique such morphism.

(2) $\Rightarrow$ (1): Given initial objects $(F(c), \eta_c)$ in each $(c \downarrow G)$, define $\varphi_{c,d}(h) = G(h) \circ \eta_c$ for $h : F(c) \to d$. Initiality gives a unique h for each $f : c \to Gd$, making φ a bijection. Naturality follows from functoriality of G and naturality of η.

The construction of F on morphisms: for $k : c \to c'$ in $\mathcal{C}$, the morphism $F(k) : F(c) \to F(c')$ is the unique morphism in $\mathcal{D}$ corresponding (under the adjunction at c) to $\eta_{c'} \circ k : c \to GF(c')$. ☐

17.3.3 Formalizing: The Universal Arrow

The pair $(F(c), \eta_c)$ is called a **universal arrow from c to G**. The universal property says: for every $d \in \mathcal{D}$ and morphism $f : c \to Gd$, there is a unique $\tilde{f} : F(c) \to d$ with $G(\tilde{f}) \circ \eta_c = f$.

$$\begin{tikzcd} c \arrow[r, "\eta_c"] \arrow[dr, "f"'] & GF(c) \arrow[d, "G(\tilde{f})"] & & \\ & Gd & \leftrightarrow & F(c) \arrow[r, dashed, "\exists!\,\tilde{f}"] & d \end{tikzcd}$$

This is the "initial universal arrow" form of the adjunction. Compare with the construction of free objects in algebra: the free group on S has a universal property of exactly this form, with G the forgetful functor.

17.3.4 Reorganizing: Three Forms Compared

Form	Data	Best for
Hom-set bijection	$\varphi_{A,B}$ natural	Computing examples
Unit/counit	η, ε + triangle ids	2-categorical arguments
Initial in $(c \downarrow G)$	Universal arrows	Proving existence

The Adjoint Functor Theorems (Chapter 10) use the third form: to construct a left adjoint to G, show that each comma category $(c \downarrow G)$ has an initial object.

17.4 Adjoint of an Adjoint

17.4.1 Informally: Can We Keep Going?

If $F \dashv G$, can G have a right adjoint too? Certainly. The question is whether it is related to F in any systematic way, and what the structure of an adjoint triple looks like.

An **adjoint triple** is a sequence $F \dashv G \dashv H$ meaning $F \dashv G$ and $G \dashv H$. In this situation F is a left adjoint of G and H is a right adjoint of G — so G is the middle piece with an adjoint on each side.

17.4.2 Expanding: Adjoint Triples in Nature

Example: Constant Diagram and Limit/Colimit

Let $\mathcal{J}$ be a small category and consider the functor categories. The **diagonal functor** $\Delta : \mathcal{C} \to \mathcal{C}^{\mathcal{J}}$ sends an object c to the constant diagram at c. We have:

$$\mathrm{colim} \dashv \Delta \dashv \lim,$$

an adjoint triple. The colimit functor is left adjoint to Δ (since $\mathrm{colim}\, D \to c$ corresponds to $D \to \Delta c$), and Δ is left adjoint to the limit functor.

Example: Restriction and Extensions of Sheaves

For an open inclusion $j : U \hookrightarrow X$ of topological spaces, the restriction functor $j^* : \mathbf{Sh}(X) \to \mathbf{Sh}(U)$ has both a left adjoint $j_!$ (extension by zero) and a right adjoint j_* (direct image):

$$j_! \dashv j^* \dashv j_*.$$

Example: Kan Extensions Along a Functor

For $p : \mathcal{C} \to \mathcal{E}$, the restriction functor $(-) \circ p : [\mathcal{E}, \mathcal{D}] \to [\mathcal{C}, \mathcal{D}]$ (which precomposes with p) sits in an adjoint triple

$$\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p\,.$$

The left Kan extension is the left adjoint, the right Kan extension is the right adjoint. This is the content of Chapter 18.

17.4.3 Formalizing: Structure of an Adjoint Triple

Definition 17.4. An **adjoint triple** (F, G, H) between categories $\mathcal{C}$ and $\mathcal{D}$ consists of functors $F, H : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ (or the transposed version) with $F \dashv G$ and $G \dashv H$.

An important structural consequence:

Proposition 17.5. *In an adjoint triple $F \dashv G \dashv H$, the functor F preserves colimits, G preserves both limits and colimits, and H preserves limits.*

Proof. F is a left adjoint, so it preserves colimits (LAPC). H is a right adjoint, so it preserves limits (RAPL). G is simultaneously a right adjoint (to F, so RAPL applies) and a left adjoint (to H, so LAPC applies). Thus G preserves both. □

17.4.4 Reorganizing: The Double Right Adjoint

Starting from $F \dashv G$, the "double right adjoint" question asks: is $G \dashv ?$. The answer depends entirely on the specific functors involved — there is no general construction of a right adjoint to G from F alone. The existence of a right adjoint to G is an independent piece of structure.

What *can* be said: by uniqueness of adjoints (Theorem 17.2), if G has any right adjoint, that right adjoint is unique up to natural isomorphism. So the set of adjoint triples containing a given $F \dashv G$ is either empty or essentially unique.

17.4.5 Simplifying: Higher Adjoints

One can ask about adjoint *quadruples*, quintuples, and so on. In good situations (locally presentable categories, Grothendieck toposes) these finite adjoint towers arise naturally. In the theory of ∞-categories there are even infinite adjoint chains. The pattern of limit/colimit preservation escalates accordingly.

Internal Dialog

Is there a way to tell in advance whether G has a right adjoint, without constructing one?
Yes: the Adjoint Functor Theorems (Chapter 10) give conditions. Roughly, G has a right adjoint iff it preserves colimits and a smallness/solution-set condition holds. The theorem tells you to check preservation rather than construct.

Exercises

State the triangle identities in terms of the unit and counit.	$(G\varepsilon)(\eta G) = \mathrm{id}_G$ and $(\varepsilon F)(F\eta) = \mathrm{id}_F$.

Write the adjunction bijection φ in terms of unit and counit.	$\varphi(f) = G(f) \circ \eta_A$ for $f : FA \to B$.
Write φ^{-1} in terms of unit and counit.	$\varphi^{-1}(g) = \varepsilon_B \circ F(g)$ for $g : A \to GB$.
Which naturality of ε is used in the proof that $\psi \circ \varphi = \mathrm{id}$?	The naturality $\varepsilon_B \circ FG(f) = f \circ \varepsilon_{FA}$.
What does it mean for $\psi \circ \varphi = \mathrm{id}$?	The formula $\varphi^{-1}(\varphi(f)) = f$ for all $f : FA \to B$.
Suppose $F \dashv G$ and $F \dashv G'$. What is the canonical isomorphism $\theta_B : G'B \to GB$?	$\theta_B = G(\varepsilon'_B) \circ \eta_{G'B}$, using the G'-counit applied via the G-unit.
What is the Yoneda argument for uniqueness of adjoints?	The bijection $\mathcal{C}(-, GB) \cong \mathcal{D}(F-, B) \cong \mathcal{C}(-, G'B)$ is a natural iso of representables, so the Yoneda lemma gives $GB \cong G'B$.
What is the comma category $(c \downarrow G)$?	Objects are pairs $(d, f : c \to Gd)$; morphisms are $h : d \to d'$ with $G(h) \circ f = f'$.
What is the initial object in $(c \downarrow G)$ when $F \dashv G$ exists?	$(F(c), \eta_c : c \to GF(c))$.
What is a universal arrow from c to G?	A pair $(F(c), \eta_c)$ such that every $f : c \to Gd$ factors uniquely as $G(\tilde{f}) \circ \eta_c$.
Reformulate the adjunction $F \dashv G$ purely using universal arrows.	For every $c \in \mathcal{C}$, there is a universal arrow from c to G.
Give an example of an adjoint triple involving the diagonal functor.	$\mathrm{colim} \dashv \Delta \dashv \lim$.
In $j_! \dashv j^* \dashv j_*$ for an open inclusion j, which functor preserves both limits and colimits?	The restriction j^*, being both a left and right adjoint.
In an adjoint triple $F \dashv G \dashv H$, which functor preserves limits?	Both G (as right adjoint to F) and H (as right adjoint to G).
Does $F \dashv G$ force G to have a right adjoint?	No; the existence of a right adjoint to G is independent structure.
If G has a right adjoint, how many essentially different ones does it have?	Exactly one, up to natural isomorphism (by uniqueness of adjoints).
What does RAPL say about the middle functor in an adjoint triple?	It preserves limits (being a right adjoint) and colimits (being a left adjoint).
What is the adjoint triple for Kan extensions along $p : \mathcal{C} \to \mathcal{E}$?	$\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p$.
What is the "solution-set condition" used for?	It is a smallness hypothesis in the Adjoint Functor Theorem that, combined with limit-preservation, guarantees the existence of a left adjoint.
Are there categories admitting infinite adjoint chains?	Yes: in sufficiently well-behaved categories (like presheaf toposes) one finds arbitrarily long adjoint chains.

Chapter Summary

Chapter Summary

Triangle identities ⇔ hom-set bijection. The formulas $\varphi(f) = G(f) \circ \eta_A$ and $\varphi^{-1}(g) = \varepsilon_B \circ F(g)$ convert between the two presentations of an adjunction. The triangle identities are exactly the condition that these formulas define inverse bijections.
Uniqueness. If $F \dashv G$ and $F \dashv G'$, then $G \cong G'$ canonically. The canonical isomorphism θ_B is constructed by applying one adjunction bijection to the counit of the other.
Comma categories. The left adjoint $F(c)$ is the initial object in the comma category $(c \downarrow G)$. Equivalently, $(F(c), \eta_c)$ is the universal arrow from c to G. This third formulation of adjunctions is the basis for the Adjoint Functor Theorems.

Adjoint triples. When G itself has a right adjoint H, we get $F \dashv G \dashv H$. The middle functor G then preserves both limits and colimits. Examples: $\mathrm{colim} \dashv \Delta \dashv \lim$, and Kan extensions $\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p$.

Formal Restatement

Formal Restatement

Triangle identities. Given $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$, $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$, $\varepsilon : FG \Rightarrow \mathrm{Id}_{\mathcal{D}}$, the following are equivalent: (i) $F \dashv G$ with unit η and counit ε; (ii) $(G\varepsilon)(\eta G) = \mathrm{id}_G$ and $(\varepsilon F)(F\eta) = \mathrm{id}_F$. Under these conditions, $\varphi_{A,B}(f) = G(f) \circ \eta_A$ and $\varphi^{-1}_{A,B}(g) = \varepsilon_B \circ F(g)$ are mutually inverse natural bijections.

Uniqueness. If $F \dashv G$ and $F \dashv G'$, then $G \cong G'$ via $\theta_B = G(\varepsilon'_B) \circ \eta_{G'B}$, a natural isomorphism.

Comma category characterization. $F \dashv G$ iff for each c, the comma category $(c \downarrow G)$ has an initial object, which is $(F(c), \eta_c : c \to GF(c))$.

Adjoint triple. $F \dashv G \dashv H$ means $F \dashv G$ and $G \dashv H$. The middle functor G preserves limits (RAPL applied to $F \dashv G$) and colimits (LAPC applied to $G \dashv H$).

Chapter 18 develops Kan extensions as adjoint triples and shows how every adjunction can be expressed using Kan extensions along the Yoneda embedding.

Chapter 18

Kan Extensions as Universal Adjoints

Chapter Overview

Chapter 11 introduced Kan extensions and their colimit/limit formulas. This chapter reveals the deeper organizational role of Kan extensions: they are themselves adjoint functors, and the restriction functor sits between a left and a right Kan extension as the middle of an adjoint triple. This triple

$$\mathrm{Lan}_p \;\dashv\; (-) \circ p \;\dashv\; \mathrm{Ran}_p$$

is one of the most important structural facts in category theory. It means every Kan extension is an instance of the universal adjoint phenomenon, and conversely, every adjunction can be *computed* as a Kan extension along the Yoneda embedding.

We also examine the distinction between pointwise and non-pointwise Kan extensions: only the former behave well with respect to representability and composition.

Where the Name Comes From

The **adjoint triple** $L \dashv M \dashv R$ describes three functors with two consecutive adjunctions: L is left adjoint to M, and M is left adjoint to R. Adjoint triples are a striking categorical phenomenon; they appear in geometry (base-change), logic (existential/ universal quantifiers as adjoints to substitution), and homotopy theory (suspension/loop). The Kan-extension adjoint triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$ is the foundational example. The terminology "adjoint string" is also used for the same or longer chains.

18.1 Recap: The Restriction Adjoint Triple

18.1.1 Informally: Restriction as the Middle

Fix a functor $p : \mathcal{C} \to \mathcal{E}$ and a target category $\mathcal{D}$. The **restriction functor** (or **precomposition functor**) along p is

$$p^* = (-) \circ p : [\mathcal{E}, \mathcal{D}] \to [\mathcal{C}, \mathcal{D}], \qquad p^*(H) = H \circ p.$$

It takes a functor defined on all of $\mathcal{E}$ and restricts it to $\mathcal{C}$.

The key fact: p^* has a left adjoint (Lan_p, the left Kan extension) and a right adjoint (Ran_p, the right Kan extension). These are not coincidental — they are exactly what it means for a functor to have best approximations from the left and from the right.

18.1.2 Expanding: The Triple in Full

Theorem 18.1. *When all relevant Kan extensions exist, there is an adjoint triple*

$$\mathrm{Lan}_p \;\dashv\; p^* \;\dashv\; \mathrm{Ran}_p$$

between $[\mathcal{C}, \mathcal{D}]$ *and* $[\mathcal{E}, \mathcal{D}]$.

The unit of $\mathrm{Lan}_p \dashv p^*$ is the natural transformation $\alpha : F \Rightarrow p^*(\mathrm{Lan}_p F) = (\mathrm{Lan}_p F) \circ p$, which is exactly the universal natural transformation into the Kan extension.

The counit of $p^* \dashv \mathrm{Ran}_p$ is the natural transformation $\varepsilon : p^*(\mathrm{Ran}_p F) \Rightarrow F$, i.e., the canonical map $(\mathrm{Ran}_p F) \circ p \Rightarrow F$.

18.1.3 Formalizing: What the Adjunctions Say

For $H \in [\mathcal{E}, \mathcal{D}]$ and $F \in [\mathcal{C}, \mathcal{D}]$, the bijection for $\mathrm{Lan}_p \dashv p^*$ is:

$$[\mathcal{E}, \mathcal{D}](\mathrm{Lan}_p F,\ H) \;\cong\; [\mathcal{C}, \mathcal{D}](F,\ H \circ p).$$

A natural transformation from the left Kan extension to H corresponds to a natural transformation from F to the restriction of H.

The bijection for $p^* \dashv \mathrm{Ran}_p$ is:

$$[\mathcal{C},\mathcal{D}](H \circ p,\ F) \;\cong\; [\mathcal{E},\mathcal{D}](H,\ \mathrm{Ran}_p F).$$

A natural transformation from the restriction of H to F corresponds to a natural transformation from H to the right Kan extension.

How to Say This

- $\mathrm{Lan}_p F$ is the *left Kan extension* of F along p. Read "Lan-sub-p of F."
- $\mathrm{Ran}_p F$ is the *right Kan extension* of F along p.
- The restriction p^* is sometimes written $- \circ p$ or p° in older literature.
- The triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$ is pronounced "Lan-p is left adjoint to restriction, which is left adjoint to Ran-p."

18.2 Proof That $\mathrm{Lan}_p \dashv p^*$

18.2.1 Informally: The Colimit Formula as the Key

When $\mathcal{D}$ has all small colimits, the left Kan extension can be computed by the **colimit formula**:

$$(\mathrm{Lan}_p F)(e) = \mathrm{colim}_{(c,f:p(c)\to e)} F(c),$$

where the colimit is taken over the **comma category** $(p \downarrow e)$. This formula is the explicit construction that witnesses the adjunction.

18.2.2 Expanding: The Adjunction Bijection via Colimits

To show $\mathrm{Lan}_p \dashv p^*$, we need a natural bijection

$$[\mathcal{E},\mathcal{D}](\mathrm{Lan}_p F, H) \;\cong\; [\mathcal{C},\mathcal{D}](F, H \circ p).$$

The key calculation uses the universal property of colimits on the left-hand side.

Let $\sigma : \mathrm{Lan}_p F \Rightarrow H$ be a natural transformation.	*This is an element of the left-hand side.*
At each $e \in \mathcal{E}$, $\sigma_e : (\mathrm{Lan}_p F)(e) \to H(e)$ is a morphism from the colimit over $(p \downarrow e)$ to $H(e)$.	*By the colimit formula.*
By the universal property of colimits, σ_e corresponds to a cocone: for each $(c, f : p(c) \to e)$ in $(p \downarrow e)$, a morphism $\sigma_e^{(c,f)} : F(c) \to H(e)$.	*Colimits are defined by their universal property: morphisms from the colimit biject with cocones.*
Taking $e = p(c')$ and $(c, f) = (c', \mathrm{id}_{p(c')})$, we obtain $\sigma_{p(c')}^{(c',\mathrm{id})} : F(c') \to H(p(c'))$.	*Choosing a specific object in $(p \downarrow p(c'))$.*
These morphisms assemble into a natural transformation $\tau : F \Rightarrow H \circ p$, with $\tau_{c'} = \sigma_{p(c')}^{(c',\mathrm{id})}$.	*Naturality follows from naturality of σ and functoriality of the colimit construction.*
Conversely, given $\tau : F \Rightarrow H \circ p$, define σ_e at each e using the composite $F(c) \xrightarrow{\tau_c} H(p(c)) \xrightarrow{H(f)} H(e)$ as the (c, f)-component of a cocone over $(p \downarrow e)$.	*$H(f) \circ \tau_c$ is the e-component of the candidate cocone.*
This cocone is compatible with morphisms in $(p \downarrow e)$: for $h : c \to c'$ with $f' \circ p(h) = f$, naturality of τ gives $H(f') \circ \tau_{c'} \circ F(h) = H(f') \circ H(p(h)) \circ \tau_c = H(f) \circ \tau_c$.	*The cocone condition follows from naturality of τ and functoriality of H.*
The universal property of the colimit then produces a unique $\sigma_e : (\mathrm{Lan}_p F)(e) \to H(e)$.	*Colimit universal property.*
The two constructions ($\sigma \mapsto \tau$ and $\tau \mapsto \sigma$) are inverse.	*Follows by unfolding the colimit universal property in each direction.*
Naturality of the bijection in F and H follows from the naturality of colimits in their parameters.	*Standard argument: colimits are functorial in F, and the bijection respects composition with natural transformations.*

18.2.3 Formalizing: The Coend Formula

When the comma category is expressed as a coend, the colimit formula becomes:

$$(\mathrm{Lan}_p\, F)(e) \;=\; \int^{c\in\mathcal{C}} \mathcal{E}(p(c), e)\cdot F(c),$$

where $S \cdot X$ denotes the copower (coproduct of $|S|$ copies of X). This coend formula is the "tensor product" of the hom-functor $\mathcal{E}(p(-), e)$ with F, and it makes precise the sense in which Kan extensions are "weighted colimits." The adjunction $\mathrm{Lan}_p \dashv p^*$ then becomes a special case of the general hom-tensor adjunction.

18.3 Pointwise Kan Extensions

18.3.1 Informally: When Kan Extensions Are "Really" Colimits

Not every Kan extension is computed by the colimit formula. When it is, we call it **pointwise**, and pointwise Kan extensions have much better properties than general ones.

A Kan extension $\mathrm{Lan}_p\, F$ is **pointwise** if for each $e \in \mathcal{E}$ and representable $\mathcal{E}(e, -)$, we have:

$$\mathrm{Lan}_p\, F(e) \cong \mathrm{colim}_{(p\downarrow e)}\, F.$$

Equivalently, the Kan extension is preserved by all representable functors $\mathcal{D}(d, -)$ for $d \in \mathcal{D}$.

18.3.2 Expanding: Why Pointwise Is Better

Example: Pointwise Kan Extensions Compose

If $\mathrm{Lan}_p\, F$ and $\mathrm{Lan}_q(\mathrm{Lan}_p\, F)$ are both pointwise, then they compose correctly: $\mathrm{Lan}_{q\circ p}\, F \cong \mathrm{Lan}_q(\mathrm{Lan}_p\, F)$. This fails for general (non-pointwise) Kan extensions.

Example: Pointwise Extensions Commute with Composition

If $\mathrm{Lan}_p\, F$ is pointwise and $G : \mathcal{D} \to \mathcal{D}'$ is any functor, then $G \circ \mathrm{Lan}_p\, F \cong \mathrm{Lan}_p(G \circ F)$ iff G preserves the relevant colimits. For a right adjoint G, this is automatic.

18.3.3 Formalizing: The Characterization

Theorem 18.2. *A left Kan extension $\mathrm{Lan}_p\, F$ is pointwise iff for every $e \in \mathcal{E}$, there is an isomorphism*

$$(\mathrm{Lan}_p\, F)(e) \;\cong\; \mathrm{colim}\Big((p\downarrow e) \to \mathcal{C} \xrightarrow{F} \mathcal{D}\Big)$$

natural in e, where the first arrow is the forgetful functor from the comma category to $\mathcal{C}$.

Corollary 18.3. *If $\mathcal{D}$ has all small colimits, then the colimit formula defines a pointwise Kan extension. In particular, for $\mathcal{D}$ cocomplete, all left Kan extensions along any p exist and are pointwise.*

18.3.4 Reorganizing: Absolute Kan Extensions

A further special case: an **absolute** Kan extension is one preserved by *every* functor $G : \mathcal{D} \to \mathcal{D}'$. Absolute Kan extensions exist even without colimits in $\mathcal{D}$ and play a role in formal category theory (the study of 2-categories via enrichment in **Cat**). Every adjunction gives rise to absolute Kan extensions: $\mathrm{Lan}_F(\mathrm{id}) \cong G$ whenever $F \dashv G$.

Internal Dialog

If pointwise Kan extensions are so much better, why bother with the general case?
General Kan extensions arise naturally when $\mathcal{D}$ does not have enough colimits, or when working in enriched or ∞-categorical settings where the colimit formula requires modification. The general notion is the correct one 2-categorically; pointwise-ness is a bonus when it holds.

18.4 Global Kan Extensions and Failure Modes

18.4.1 Informally: When Can Extensions Always Be Computed?

We say $\mathcal{D}$ **admits all left Kan extensions along** p if $\mathrm{Lan}_p\, F$ exists for every $F : \mathcal{C} \to \mathcal{D}$. When this holds for all p, we say $\mathcal{D}$ **admits global Kan extensions**.

For $\mathcal{D}$ cocomplete, global left Kan extensions exist (by the colimit formula). But the colimit formula requires colimits indexed by the comma categories $(p \downarrow e)$, which may be large even when $\mathcal{C}$ and $\mathcal{E}$ are small. This is a genuine issue in non-cocomplete $\mathcal{D}$.

18.4.2 Expanding: Failure Modes

Example: No Colimits, No Kan Extension

Let $\mathcal{D}$ be a category with no colimits (e.g., the category of fields and field maps, which has no pushouts in general). There will be functors F and p for which $\mathrm{Lan}_p F$ does not exist.

Example: Non-Pointwise from Small Size

In the enriched setting, there exist Kan extensions that satisfy the universal property (as adjunctions) but are not computed by the colimit formula. These "non-pointwise" extensions behave badly: they may not compose, may not be preserved by other functors, and do not have a clean colimit interpretation.

18.4.3 Formalizing: A Sufficient Condition

Theorem 18.4. *Let $p : \mathcal{C} \to \mathcal{E}$ with $\mathcal{C}$ small, and let $\mathcal{D}$ be cocomplete. Then for every $F : \mathcal{C} \to \mathcal{D}$, the left Kan extension $\mathrm{Lan}_p F$ exists and is pointwise. Moreover the functor $\mathrm{Lan}_p : [\mathcal{C}, \mathcal{D}] \to [\mathcal{E}, \mathcal{D}]$ is the left adjoint to p^*.*

When $\mathcal{C}$ is not small, one must work in a universe-relative setting or impose additional cocompleteness conditions on $\mathcal{D}$.

18.5 Kan Extensions as Formulas for Adjoints

18.5.1 Informally: Every Adjoint Is a Kan Extension

There is a profound theorem: every right adjoint $G : \mathcal{D} \to \mathcal{C}$ can be computed as a right Kan extension along the Yoneda embedding. Similarly for left adjoints. This means Kan extensions are not just a generalization of adjunctions — they *subsume* them.

Let $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ be the Yoneda embedding $Y(c) = \mathcal{C}(-, c)$.

Theorem 18.5 (Kan's Formula for Adjoints)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor with a right adjoint G. Then G is naturally isomorphic to the right Kan extension of F along Y, viewed appropriately:*

$$G \cong \mathrm{Ran}_Y F.$$

More precisely: if $\mathcal{D}$ is locally small and cocomplete, the left adjoint F to a given G can be computed *as*

$$F(c) \;=\; \mathrm{colim}_{(\mathcal{C}(c,-) \Rightarrow G)} \mathrm{Id}_{\mathcal{D}},$$

the colimit over all natural transformations from the representable $\mathcal{C}(c, -)$ to the composite $G \circ (-)$.

18.5.2 Expanding: The Nerve/Realization Paradigm

A particularly clean case: suppose $\mathcal{C}$ is small and we have a functor $i : \mathcal{C} \to \mathcal{D}$ (thinking of $\mathcal{C}$ as a category of "shapes"). Then:

- The **nerve functor** $N : \mathcal{D} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is $N(d) = \mathcal{D}(i(-), d)$.
- The **realization functor** $|-| : [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}] \to \mathcal{D}$ is $\mathrm{Lan}_Y i$ (when $\mathcal{D}$ is cocomplete).

The adjunction $|-| \dashv N$ is exactly $\mathrm{Lan}_Y i \dashv i^*$, where i^* is restriction along i. The left Kan extension along Yoneda always gives the realization, and its right adjoint is always the nerve.

Example: Geometric Realization

Take $\mathcal{C} = \Delta$ (the simplex category), $\mathcal{D} = \mathbf{Top}$, and $i : \Delta \to \mathbf{Top}$ sending $[n]$ to the standard n-simplex Δ^n. Then $\mathrm{Lan}_Y i$ is geometric realization, and the nerve N sends a space X to its singular simplicial set $\mathbf{Top}(\Delta^-, X)$. The adjunction $|-| \dashv N$ is the fundamental adjunction of homotopy theory.

Example: Free Colimit Completion

The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ exhibits the presheaf category as the free cocomplete category on $\mathcal{C}$. Any functor $F : \mathcal{C} \to \mathcal{D}$ with $\mathcal{D}$ cocomplete extends to $\mathrm{Lan}_Y F : [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}] \to \mathcal{D}$ preserving all colimits.

18.5.3 Formalizing: The Universal Property of Presheaves

The theorem captures the universal property of presheaf categories:

Theorem 18.6 (Free Cocompletion)**.** *For a small category $\mathcal{C}$ and a cocomplete category $\mathcal{D}$, restriction along the Yoneda embedding Y gives an equivalence*

$$\mathrm{Fun}_{\mathrm{colim}}([\mathcal{C}^{\mathrm{op}}, \mathbf{Set}], \mathcal{D}) \;\simeq\; \mathrm{Fun}(\mathcal{C}, \mathcal{D}),$$

where the left side is the category of colimit-preserving functors. The inverse sends F to $\mathrm{Lan}_Y F$.

This says: prescribing where objects of $\mathcal{C}$ go in $\mathcal{D}$ determines a unique colimit-preserving functor out of the presheaf category. The Kan extension is the unique colimit-preserving extension.

Exercises

Write the adjoint triple for Kan extensions along p.	$\mathrm{Lan}_p \dashv (-) \circ p \dashv \mathrm{Ran}_p$.
What is the restriction functor p^*?	Precomposition with p: $p^*(H) = H \circ p$.
What is the hom-set bijection for $\mathrm{Lan}_p \dashv p^*$?	$[\mathcal{E}, \mathcal{D}](\mathrm{Lan}_p F, H) \cong [\mathcal{C}, \mathcal{D}](F, H \circ p)$.
State the colimit formula for a left Kan extension.	$(\mathrm{Lan}_p F)(e) = \mathrm{colim}_{(c, f : p(c) \to e)} F(c)$.
What is the coend formula for the left Kan extension?	$(\mathrm{Lan}_p F)(e) = \int^c \mathcal{E}(p(c), e) \cdot F(c)$.
In the proof that $\mathrm{Lan}_p \dashv p^*$, what does the colimit universal property do?	It converts a natural transformation from the colimit to a cocone, bijecting with natural transformations $F \Rightarrow H \circ p$.
What does it mean for a Kan extension to be pointwise?	$(\mathrm{Lan}_p F)(e) \cong \mathrm{colim}_{(p \downarrow e)} F$ for each e.
Under what conditions do pointwise Kan extensions compose?	When both $\mathrm{Lan}_p F$ and $\mathrm{Lan}_q(\mathrm{Lan}_p F)$ are pointwise.
Does the composition law $\mathrm{Lan}_{q \circ p} \cong \mathrm{Lan}_q \mathrm{Lan}_p$ hold in general?	Only for pointwise Kan extensions; it can fail otherwise.
What is an absolute Kan extension?	One preserved by every functor out of $\mathcal{D}$.
Give an example of an absolute Kan extension.	If $F \dashv G$, then $\mathrm{Lan}_F(\mathrm{Id}) \cong G$ absolutely.
When does $\mathcal{D}$ admit all left Kan extensions along p?	When $\mathcal{D}$ is cocomplete, by the colimit formula.
What is the nerve functor for $i : \mathcal{C} \to \mathcal{D}$?	$N(d) = \mathcal{D}(i(-), d)$, a presheaf on $\mathcal{C}$.
What is the realization functor?	$\mathrm{Lan}_Y i$, the left Kan extension of i along Yoneda.
What adjunction does the nerve/realization pair form?	$\lvert - \rvert \dashv N$, i.e., $\mathrm{Lan}_Y i \dashv i^*$.
Give a geometric example of the nerve/realization adjunction.	Geometric realization $\lvert - \rvert$ left adjoint to singular simplicial set functor $\mathbf{Top}(\Delta^-, -)$.
What is the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ called in the context of Kan extensions?	The free cocompletion of $\mathcal{C}$.
State the free cocompletion theorem.	Colimit-preserving functors $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}] \to \mathcal{D}$ biject with all functors $\mathcal{C} \to \mathcal{D}$ when $\mathcal{D}$ is cocomplete.
Why does p^* preserve limits and colimits (as the middle of an adjoint triple)?	Because p^* is simultaneously a left adjoint ($p^* \dashv \mathrm{Ran}_p$) and a right adjoint ($\mathrm{Lan}_p \dashv p^*$).
What goes wrong with non-pointwise Kan extensions?	They may not compose, may not be preserved under composition with other functors, and lack a clean colimit interpretation.

Chapter Summary

Chapter Summary

Adjoint triple. The restriction functor $p^* = (-) \circ p$ sits in $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$. The hom-set bijection for $\mathrm{Lan}_p \dashv p^*$ follows directly from the universal property of colimits applied to the colimit formula.
Pointwise Kan extensions. A Kan extension is pointwise if it is computed at each e by $\mathrm{colim}_{(p \downarrow e)} F$. Pointwise extensions compose correctly, are preserved by representables, and exist whenever $\mathcal{D}$ is cocomplete.
Non-pointwise failure. Without cocompleteness, Kan extensions may exist abstractly (as adjoints) without being pointwise, and then lose the good compositional properties.

Kan formula for adjoints. Every adjunction can be computed as a Kan extension along Yoneda: $F(c) = \mathrm{Lan}_Y\, i(c)$ and the right adjoint is the nerve. The presheaf category is the free cocompletion: every functor $F : \mathcal{C} \to \mathcal{D}$ extends uniquely (up to iso) to a colimit-preserving $\mathrm{Lan}_Y\, F$ on the presheaf category.

Formal Restatement

Formal Restatement

Adjoint triple for Kan extensions. $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$ between $[\mathcal{C}, \mathcal{D}]$ and $[\mathcal{E}, \mathcal{D}]$, where $p : \mathcal{C} \to \mathcal{E}$ and $p^*(H) = H \circ p$.

Adjunction bijections.

$$[\mathcal{E}, \mathcal{D}](\mathrm{Lan}_p\, F, H) \cong [\mathcal{C}, \mathcal{D}](F, p^* H),$$
$$[\mathcal{C}, \mathcal{D}](p^* H, F) \cong [\mathcal{E}, \mathcal{D}](H, \mathrm{Ran}_p\, F).$$

Colimit formula. $(\mathrm{Lan}_p\, F)(e) = \mathrm{colim}_{(p\downarrow e)}\, F = \int^c \mathcal{E}(p(c), e) \cdot F(c)$. An extension is *pointwise* iff it is computed by this formula for each e.

Free cocompletion. $Y : \mathcal{C} \hookrightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the free cocompletion: any $F : \mathcal{C} \to \mathcal{D}$ ($\mathcal{D}$ cocomplete) extends uniquely to a colimit-preserving $\mathrm{Lan}_Y\, F$. The induced adjunction $\mathrm{Lan}_Y\, i \dashv i^*$ is the realization–nerve adjunction.

Chapter 19 turns to monoidal categories, which are what category theory says when we ask "what is a monoid in a category of categories?"

Chapter 19

Monoidal Categories as Categorified Monoids

Chapter Overview

Chapter 12 introduced monoidal categories and their coherence theorem. This chapter takes a step back and asks: what *is* a monoidal category from a higher categorical perspective? The answer is that a monoidal category is a **categorified monoid**: a monoid in which the underlying set has been replaced by a category, and the equations of a monoid have been replaced by natural isomorphisms.

This perspective makes the coherence theorem inevitable (the natural isos must be compatible), clarifies the distinction between strict and weak monoidal categories, and connects monoidal categories to the theory of monads through the concept of a "monoid in a monoidal category." We also develop monoidal functors in their three flavors (lax, oplax, strong) and conclude with the beautiful observation that monoids in various monoidal categories recover all the algebraic structures one cares about.

Where the Name Comes From

Categorification and its inverse **decategorification** are informal-but-precise terms popularized in the 1990s by *Louis Crane*, *Igor Frenkel*, and *John Baez*, in the context of constructing categorical analogues of algebraic structures (especially TQFTs and quantum groups). *Categorify* = "promote a set to a category, equations to natural isos, etc."; *decategorify* = "flatten back to underlying set / hom-set." The terminology has become standard for the philosophical operation of moving between n-categorical levels.

The **Eckmann-Hilton argument** (Eckmann-Hilton 1962) shows that two compatible monoid structures on a set must coincide and be commutative. Categorifying gives the $\mathbb{E}_2$/braided picture (Vol IV Ch. 56); the 1-categorical original is foundational to monoidal-category theory.

19.1 Decategorification

19.1.1 Informally: Going Down One Level

Decategorification means: take a categorical structure, forget about morphisms (or collapse them), and see what algebraic structure remains. It is a lossy map from the richer world to the poorer world, but it illuminates what the richer world is a categorification *of*.

For monoidal categories: if we take a monoidal category $(\mathcal{V}, \otimes, I)$ and decategorify by collapsing all morphisms (keeping only the objects and recording whether they are isomorphic), we obtain a *commutative monoid* when the monoidal category is symmetric — and a *monoid* in general.

19.1.2 Expanding: The Decategorification Map

Given a monoidal category $\mathcal{V}$, form the set $|\mathcal{V}|$ of isomorphism classes of objects. The tensor product $\otimes$ descends: if $A \cong A'$ and $B \cong B'$, then $A \otimes B \cong A' \otimes B'$. So $\otimes$ induces a binary operation on $|\mathcal{V}|$, with unit $[I]$.

The associator $\alpha_{A,B,C} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$ becomes an equality $([A][B])[C] = [A]([B][C])$ in $|\mathcal{V}|$. The unitors become $[I][A] = [A]$ and $[A][I] = [A]$.

Result: $(|\mathcal{V}|, \cdot, [I])$ is a monoid.

Internal Dialog

So a monoidal category is to a monoid as a category is to a set?

Exactly. A set is a category with only identity morphisms. A monoid is a set with a binary operation. A monoidal category is a category with a tensor product (binary operation on objects, functorial on morphisms). The equations of a monoid become isomorphisms in the monoidal category. This is categorification.

19.1.3 Formalizing: The Categorification Slogan

Monoid	Monoidal category
Set of elements	Category of objects and morphisms
Binary operation $m \cdot n$	Tensor functor $A \otimes B$
Unit element e	Unit object I
Associativity equation	Associator natural isomorphism α
Unit equations	Unitor natural isomorphisms λ, ρ
—	Coherence: pentagon and triangle commute

The coherence diagrams (pentagon for α, triangle for λ, ρ) have no monoid analogue because in a monoid, the corresponding equations are strict. In a monoidal category, the isomorphisms need to be "glued together" consistently — that is what coherence demands.

19.2 Strict vs. Weak Monoidal Categories

19.2.1 Informally: When Coherence Is Trivial

A **strict monoidal category** is one where the associator and unitors are *identities*, not merely natural isomorphisms:

$$(A \otimes B) \otimes C = A \otimes (B \otimes C), \qquad I \otimes A = A = A \otimes I.$$

These are on-the-nose equalities of objects. In a strict monoidal category, parenthesization is irrelevant by definition, not merely up to isomorphism.

The prototypical example is $(\mathbf{Set}, \times, \{*\})$ *if* we choose a specific product, but most constructions in nature produce only weak monoidal categories with non-identity coherence isomorphisms.

19.2.2 Expanding: Why Strict Is Rare

Consider $(\mathbf{Vect}_k, \otimes_k, k)$: the tensor product of vector spaces. We have $(U \otimes V) \otimes W \cong U \otimes (V \otimes W)$, but this is a canonical isomorphism, not an equality. The two sides have the same dimension but are *different sets* of tensors. Similarly, $k \otimes V \cong V$ via the scalar multiplication isomorphism, but $k \otimes V$ and V are not literally equal.

Strict monoidal categories do arise: the endofunctor category $[\mathcal{C}, \mathcal{C}]$ with composition is strict (functor composition is associative and has strict units). The category of strict 2-functors between 2-categories is also strict.

19.2.3 Formalizing: The Definitions

Definition 19.1. A **monoidal category** $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ is called **strict** if α, λ, ρ are all identity natural transformations (equivalently, identity morphisms at every object).

Definition 19.2. A **weak monoidal category** (or simply monoidal category) has α, λ, ρ as natural isomorphisms satisfying the pentagon and triangle axioms.

The pentagon axiom says the two ways to reassociate $(A \otimes B) \otimes (C \otimes D)$ to $A \otimes (B \otimes (C \otimes D))$ agree. The triangle axiom says $\rho_A \otimes \mathrm{id}_B = (\mathrm{id}_A \otimes \lambda_B) \circ \alpha_{A,I,B}$.

19.3 Mac Lane's Coherence Theorem

19.3.1 Informally: Why Every Diagram Commutes

When we write $A \otimes B \otimes C \otimes D$ in a monoidal category, there are many ways to parenthesize this expression, and between any two parenthesizations there are multiple composite natural isomorphisms built from α, λ, ρ. Mac Lane's coherence theorem guarantees they all agree.

Theorem 19.3 (Mac Lane's Coherence Theorem)**.** *In any monoidal category, every diagram built from instances of α, λ, ρ and their inverses, using tensor product and composition, commutes.*

This is why we write $A \otimes B \otimes C$ without parentheses: any parenthesization gives an isomorphic result, and the isomorphisms between different parenthesizations are all equal.

19.3.2 Expanding: The Normal Form Strategy

The proof strategy is to reduce every parenthesization to a **normal form** (fully right-associated, with no I factors) using the coherence isomorphisms, and then show that any two paths to the normal form give the same result.

The key lemma: any two composites of coherence isomorphisms between the same source and target are equal. This is proved by induction on the complexity of the composites, using the pentagon and triangle axioms as the base cases.

19.3.3 Formalizing: Key Step via Prooflog

We prove the key step: the pentagon axiom implies that any two ways to reassociate a four-fold tensor product agree.

Let P be the pentagon: $(\mathrm{id}_A \otimes \alpha_{B,C,D}) \circ \alpha_{A,B\otimes C,D} \circ (\alpha_{A,B,C} \otimes \mathrm{id}_D) = \alpha_{A,B,C\otimes D} \circ \alpha_{A\otimes B,C,D}$.	*Pentagon axiom for (A, B, C, D).*
There are five distinct parenthesizations of $A \otimes B \otimes C \otimes D$, corresponding to the five vertices of the Mac Lane pentagon.	*Vertices: $((AB)C)D$, $(A(BC))D$, $A((BC)D)$, $A(B(CD))$, $(AB)(CD)$.*
The five edges of the pentagon are instances of α. The pentagon axiom says the diagram of these five edges commutes.	*Each edge applies α at one level.*
For a general diagram of coherence morphisms, we define the *normal form functor* N that reduces any parenthesization to a unique right-associated form $A_1 \otimes (A_2 \otimes (\cdots \otimes A_n) \cdots)$ using α.	*Definition of N by structural recursion.*
Any coherence morphism $\theta : P \to Q$ factors as $N(Q)^{-1} \circ n_\theta \circ N(P)$ where n_θ is a morphism in normal form.	*By induction on the structure of θ.*
The pentagon axiom implies that n_θ is uniquely determined by its source and target normal forms.	*The pentagon says there is only one way to pass between adjacent parenthesizations in normal form.*
Therefore any two coherence morphisms with the same source and target have the same normal form component, hence are equal.	*Two paths through the same source/target reduce to the same normal-form morphism.*

19.3.4 Reorganizing: What Coherence Buys

The coherence theorem has a slick reformulation:

Corollary 19.4. *Every monoidal category is monoidally equivalent to a strict monoidal category.*

Sketch. Build the "strictification" $\mathcal{V}^{\mathrm{st}}$ whose objects are *finite lists* of objects of $\mathcal{V}$ and whose tensor product is list concatenation (strict!). The functor $\mathcal{V}^{\mathrm{st}} \to \mathcal{V}$ sends a list to its tensor product. The coherence theorem implies this is an equivalence. □

This means we may always *assume without loss of generality* that a monoidal category is strict when doing abstract arguments. In concrete situations (like **Vect**) the non-strict form is necessary, but for proving abstract theorems, strictification reduces to the easier case.

19.4 Monoidal Functors

19.4.1 Informally: Preserving the Tensor Structure

A functor between monoidal categories should "preserve the tensor product," but there are three flavors of how well it does so.

A **lax monoidal functor** $F : \mathcal{V} \to \mathcal{W}$ comes with a *comparison morphism*

$$\phi_{A,B} : F(A) \otimes F(B) \to F(A \otimes B)$$

and a unit comparison $\phi_0 : I_{\mathcal{W}} \to F(I_{\mathcal{V}})$, satisfying coherence conditions. The morphism ϕ goes in the direction "product of images maps to image of product," but is not required to be an isomorphism.

An **oplax monoidal functor** has ϕ going the other direction: $\phi_{A,B} : F(A \otimes B) \to F(A) \otimes F(B)$.

A **strong monoidal functor** (sometimes just "monoidal functor") has ϕ an isomorphism in both cases.

19.4.2 Expanding: Examples

Example: Free Monoid Is Lax

The free monoid functor L : **Set** → **Mon** is lax monoidal (with **Set** and **Mon** both having cartesian product as their tensor). The comparison $L(A) \times L(B) \to L(A \times B)$ sends a pair of words (w, v) to the sequence of pairs. This is not an isomorphism (a word in $L(A \times B)$ contains more interleaved information than a pair of words).

Example: Forgetful Functor Is Lax

The forgetful functor from monoids to sets is lax monoidal in the reverse direction: it is *oplax*, with $|M \times N| \to |M| \times |N|$ the identity (here it happens to be strict).

Example: Symmetric Monoidal Functors in Algebra

A strong monoidal functor $F : \mathbf{Ab} \to \mathbf{Ab}$ (with tensor product) that is additionally symmetric carries the structure of a Hopf algebra or similar in some categorical sense.

19.4.3 Formalizing: The Definitions

Definition 19.5. A **lax monoidal functor** $(F, \phi, \phi_0) : (\mathcal{V}, \otimes, I) \to (\mathcal{W}, \otimes, J)$ consists of:

- A functor $F : \mathcal{V} \to \mathcal{W}$.
- A natural transformation $\phi_{A,B} : FA \otimes FB \to F(A \otimes B)$.
- A morphism $\phi_0 : J \to FI$.

These must satisfy the coherence conditions: two diagrams expressing compatibility of ϕ with α and with λ, ρ.

Definition 19.6. A **strong monoidal functor** is a lax monoidal functor where ϕ and ϕ_0 are isomorphisms.

How to Say This

- "Lax monoidal" means the comparison goes *toward* the image: $FA \otimes FB \to F(A \otimes B)$.
- "Oplax monoidal" means the comparison goes *away* from the image: $F(A \otimes B) \to FA \otimes FB$.
- "Strong monoidal" means both are isomorphisms (so it doesn't matter which direction you specify).
- In the literature, a "monoidal functor" with no adjective often means *strong* monoidal. Check the convention.

19.4.4 Reorganizing: Functors and Algebras

A key reason to study lax monoidal functors: a lax monoidal functor sends monoids to monoids. If M is a monoid in $\mathcal{V}$ (see Section 19.5) and F is lax monoidal, then FM is a monoid in $\mathcal{W}$. The multiplication of FM is $FM \otimes FM \xrightarrow{\phi} F(M \otimes M) \xrightarrow{F\mu} FM$.

This is why lax monoidal functors are more useful than mere functors between monoidal categories: they respect the algebraic structure.

19.5 Monoids in Monoidal Categories

19.5.1 Informally: Algebra Inside a Category

Once we have a monoidal category $(\mathcal{V}, \otimes, I)$, we can ask: what does a "monoid" look like inside $\mathcal{V}$? The answer is an object M with a multiplication $\mu : M \otimes M \to M$ and a unit $\eta : I \to M$, satisfying associativity and unit laws in the form of commutative diagrams.

This is the correct abstraction because it recovers all the classical notions of "algebra" for different choices of $\mathcal{V}$:

$\mathcal{V}$	Monoidal structure	Monoid in $\mathcal{V}$
Set	Cartesian $\times$	Ordinary monoid
Ab	Tensor $\otimes_{\mathbb{Z}}$	Ring
$\mathbf{Vect}_k$	Tensor $\otimes_k$	k-algebra
Grp	Direct product	Group (with extra structure)
$[\mathcal{C}, \mathcal{C}]$	Composition	Monad

19.5.2 Expanding: The Monad Connection

The last row deserves emphasis: a *monad* on $\mathcal{C}$ is precisely a monoid in the monoidal category $[\mathcal{C}, \mathcal{C}]$ of endofunctors with composition.

- The object M is an endofunctor $T : \mathcal{C} \to \mathcal{C}$.
- The multiplication $\mu : T \circ T \to T$ is the monad multiplication.
- The unit $\eta : \mathrm{Id} \to T$ is the monad unit.
- Associativity: $\mu \circ (T\mu) = \mu \circ (\mu T)$.
- Unit laws: $\mu \circ (T\eta) = \mathrm{id}_T = \mu \circ (\eta T)$.

These are exactly the monad axioms from Chapter 8. The monoidal structure of $[\mathcal{C}, \mathcal{C}]$ is the reason monads are "monoids in endofunctors."

19.5.3 Formalizing: Monoids Precisely

Definition 19.7. Let $(\mathcal{V}, \otimes, I)$ be a monoidal category. A **monoid** in $\mathcal{V}$ is a triple (M, μ, η) where $M \in \mathcal{V}$, $\mu : M \otimes M \to M$, and $\eta : I \to M$, such that:

$$\mu \circ (\mu \otimes \mathrm{id}_M) = \mu \circ (\mathrm{id}_M \otimes \mu) \circ \alpha_{M,M,M}, \quad \text{(associativity)}$$
$$\mu \circ (\eta \otimes \mathrm{id}_M) = \lambda_M, \quad \text{(left unit)}$$
$$\mu \circ (\mathrm{id}_M \otimes \eta) = \rho_M. \quad \text{(right unit)}$$

A **monoid homomorphism** $(M, \mu, \eta) \to (M', \mu', \eta')$ is a morphism $f : M \to M'$ with $f \circ \mu = \mu' \circ (f \otimes f)$ and $f \circ \eta = \eta'$.

Definition 19.8. A **comonoid** in $\mathcal{V}$ is a monoid in $\mathcal{V}^{\mathrm{op}}$: an object C with comultiplication $\delta : C \to C \otimes C$ and counit $\varepsilon : C \to I$ satisfying co-associativity and co-unit laws.

19.5.4 Reorganizing: The Eckmann–Hilton Argument

In a braided monoidal category (one with a coherent swap $\sigma_{A,B} : A \otimes B \cong B \otimes A$), monoids with compatible braidings are *commutative*. The Eckmann–Hilton argument shows that a monoid in a category of monoids must be commutative:

Proposition 19.9 (Eckmann–Hilton)**.** *If M is simultaneously a monoid (M, μ, η) and (M, ν, ϵ) in* **Set** *and the two operations are compatible (μ is a monoid homomorphism for ν), then $\mu = \nu$ and both are commutative.*

Categorically, this says that a monoid in $(\mathbf{Mon}, \times, \{e\})$ is the same as a commutative monoid. This is why double loop spaces are abelian group objects: they carry two compatible multiplication structures that force commutativity.

19.5.5 Simplifying: The Internal Algebra Slogan

$$\boxed{\text{Monoid in } (\mathcal{V}, \otimes, I) = \text{algebra internal to } \mathcal{V} \text{ with respect to } \otimes}$$

This single observation subsumes rings, algebras, and monads as special cases. The monoidal structure of $\mathcal{V}$ determines what kind of algebra is possible; different monoidal structures give different theories.

Exercises

What is decategorification of a monoidal category?	Forget morphisms, keep isomorphism classes; the tensor product descends to a binary operation making $\lvert\mathcal{V}\rvert$ a monoid.
What monoid does $(\mathbf{Vect}_k, \otimes_k, k)$ decategorify to?	The monoid of isomorphism classes of vector spaces under $\otimes$, isomorphic to $(\mathbb{N}, +, 0)$ via dimension.
What is a strict monoidal category?	One where the associator and unitors are identities (not merely natural isomorphisms).
Give an example of a strict monoidal category.	The endofunctor category $[\mathcal{C}, \mathcal{C}]$ with composition.
Why is $(\mathbf{Vect}_k, \otimes_k, k)$ not strict?	Because $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ are isomorphic but not literally equal as sets.
State Mac Lane's coherence theorem.	Every diagram built from α, λ, ρ and their inverses commutes.
What is the key proof strategy for coherence?	Reduce each parenthesization to a normal form (right-associated, I-free) and show any two composites agree by the pentagon and triangle axioms.
What does the coherence theorem imply about strictification?	Every monoidal category is monoidally equivalent to a strict one.
What are the five vertices of the Mac Lane pentagon for (A, B, C, D)?	$((AB)C)D$, $(A(BC))D$, $A((BC)D)$, $A(B(CD))$, $(AB)(CD)$.
What is a lax monoidal functor?	A functor with comparison morphisms $\phi_{A,B} : FA \otimes FB \to F(A \otimes B)$ and $\phi_0 : J \to FI$ satisfying coherence, but not required to be isomorphisms.
What is an oplax monoidal functor?	Same as lax but with comparisons going the other direction: $F(A \otimes B) \to FA \otimes FB$.

What is a strong monoidal functor?	A lax monoidal functor where all comparison morphisms are isomorphisms.
Why do lax monoidal functors send monoids to monoids?	Given $\mu : M \otimes M \to M$ and lax comparison ϕ, compose $FM \otimes FM \xrightarrow{\phi} F(M \otimes M) \xrightarrow{F\mu} FM$.
What is a monoid in a monoidal category $(\mathcal{V}, \otimes, I)$?	An object M with $\mu : M \otimes M \to M$ and $\eta : I \to M$ satisfying associativity and unit diagrams.
What kind of structure is a monoid in $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$?	A ring (associative and unital).
What kind of structure is a monoid in $([\mathcal{C}, \mathcal{C}], \circ, \mathrm{Id})$?	A monad on $\mathcal{C}$.
State the Eckmann–Hilton argument.	A set with two compatible monoid structures must have them equal and commutative.
What does Eckmann–Hilton say about a monoid in $(\mathbf{Mon}, \times)$?	It is a commutative monoid.
What coherence data does a monoidal category have beyond the tensor functor?	Associator α, left unitor λ, right unitor ρ satisfying pentagon and triangle axioms.
What is a comonoid in $\mathcal{V}$?	An object with comultiplication $C \to C \otimes C$ and counit $C \to I$ satisfying co-associativity and co-unit laws.

Chapter Summary

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Categorification. A monoidal category is a categorified monoid: equations become natural isomorphisms (coherence data). Decategorifying a monoidal category recovers a monoid on isomorphism classes of objects.
Strict vs. weak. Strict monoidal categories have identity coherence morphisms. Most natural examples are only weakly monoidal. By Mac Lane's coherence theorem, every weak monoidal category is equivalent to a strict one, so we can always strictify for abstract arguments.
Monoidal functors. Lax monoidal functors carry comparison morphisms $FA \otimes FB \to F(A \otimes B)$ that need not be isomorphisms. They send monoids to monoids. Oplax and strong are the directional variants.
Monoids in monoidal categories. An object M with multiplication and unit morphisms satisfying categorical associativity/unit laws. This subsumes: ordinary monoids (in **Set**), rings (in **Ab**), k-algebras (in $\mathbf{Vect}_k$), and monads (in $[\mathcal{C}, \mathcal{C}]$). The Eckmann–Hilton argument shows that double-monoid structures force commutativity.

Formal Restatement

Formal Restatement

Monoidal category. $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ with $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, natural isos $\alpha_{A,B,C} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$, $\lambda_A : I \otimes A \cong A$, $\rho_A : A \otimes I \cong A$, satisfying the pentagon and triangle axioms.

Coherence theorem. Every diagram of coherence morphisms commutes. Every monoidal category is monoidally equivalent to a strict one.

Lax monoidal functor. (F, ϕ, ϕ_0) with $\phi_{A,B} : FA \otimes FB \to F(A \otimes B)$ and $\phi_0 : J \to FI$ natural and coherent. Strong if ϕ, ϕ_0 invertible; oplax if reversed.

Monoid in $\mathcal{V}$. $(M, \mu : M \otimes M \to M, \eta : I \to M)$ with $\mu(\mu \otimes \mathrm{id}) = \mu(\mathrm{id} \otimes \mu)\alpha$ and $\mu(\eta \otimes \mathrm{id}) = \lambda$, $\mu(\mathrm{id} \otimes \eta) = \rho$. Instances: monoids, rings, algebras, monads.

Chapter 20 enriches this picture by replacing the hom-sets of a category with objects of a monoidal category $\mathcal{V}$, yielding the full theory of $\mathcal{V}$-enriched categories.

Chapter 20

Enriched Category Theory

Chapter Overview

Chapter 12 introduced $\mathcal{V}$-enriched categories, where hom-sets are replaced by objects of a monoidal category $\mathcal{V}$. This chapter develops the deep theory: the enriched Yoneda lemma, enriched adjunctions, weighted limits and colimits, base change, and self-enrichment of closed categories.

The enriched Yoneda lemma is the heart of the chapter. It states that the "set of natural transformations from a representable $\mathcal{V}$-functor to F" is (up to the right notion of $\mathcal{V}$-natural transformation) just $F(a)$ in $\mathcal{V}$ — the same statement as ordinary Yoneda, but now the conclusion lands in $\mathcal{V}$, not **Set**.

The payoff: enriched limits, enriched adjunctions, and the general theory of weighted limits that subsume ordinary limits as a special case. Weighted limits are the correct notion of limit in the enriched world, and understanding them is essential for working with chain complexes, spectra, and higher-categorical structures.

Where the Name Comes From

Enriched category theory as a systematic discipline was developed by *Max Kelly* (1930–2007, Australian mathematician) in his 1982 book *Basic Concepts of Enriched Category Theory*. The foundational papers by *Eilenberg-Kelly* (1966) established the theory. The slogan: *ordinary category theory is the* **Set**-*enriched case of enriched category theory*, and many constructions generalize uniformly across enrichments.

Weighted limit (sometimes **indexed limit**) was developed to extend ordinary limits to the enriched setting, where the index category is replaced by a $\mathcal{V}$-functor (the *weight*). The weighted limit $\{W, F\}$ generalizes ordinary limits ($W = \text{const } \mathbf{1}$) and many other constructions (cotensors, ends). Kelly's book is the standard reference.

20.1 The Enriched Yoneda Lemma

20.1.1 Informally: Yoneda in $\mathcal{V}$

The ordinary Yoneda lemma says: for a locally small category $\mathcal{C}$ and a functor $F : \mathcal{C} \to \mathbf{Set}$, natural transformations from the representable $\mathcal{C}(a, -)$ to F are in bijection with elements of $F(a)$:

$$[\mathcal{C}, \mathbf{Set}](\mathcal{C}(a, -), F) \;\cong\; F(a).$$

When $\mathcal{C}$ is enriched over $\mathcal{V}$, the hom-objects $\mathcal{C}(a, b)$ are objects of $\mathcal{V}$, not sets. The Yoneda lemma must be rewritten to have its conclusion in $\mathcal{V}$ rather than **Set**, and "natural transformations" must be replaced by the $\mathcal{V}$*-natural transformations* that live in $\mathcal{V}$.

Theorem 20.1 ($\mathcal{V}$-Yoneda Lemma)**.** *Let $\mathcal{C}$ be a $\mathcal{V}$-enriched category and $F : \mathcal{C} \to \mathcal{V}$ a $\mathcal{V}$-functor. For each object $a \in \mathcal{C}$, there is a $\mathcal{V}$-natural isomorphism*

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -),\; F) \;\cong\; F(a),$$

where the left-hand side is the $\mathcal{V}$-object of $\mathcal{V}$-natural transformations.

20.1.2 Expanding: What the Statement Means

The $\mathcal{V}$-object of $\mathcal{V}$-natural transformations between two $\mathcal{V}$-functors $G, H : \mathcal{C} \to \mathcal{V}$ is defined as the *end*:

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(G, H) \;=\; \int_{c \in \mathcal{C}} \mathcal{V}(G(c), H(c)).$$

This is an object of $\mathcal{V}$ built from all the individual hom-objects $\mathcal{V}(G(c), H(c))$, glued together by the end construction (Chapter 26).

For the Yoneda case, $G = \mathcal{C}(a, -)$ and $H = F$:

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \;=\; \int_c \mathcal{V}(\mathcal{C}(a, c), F(c)).$$

The Yoneda lemma says this end is isomorphic to $F(a)$ in $\mathcal{V}$.

20.1.3 Formalizing: Full Proof via Prooflog

We construct a morphism $\phi : \int_c \mathcal{V}(\mathcal{C}(a, c), F(c)) \to F(a)$.	*We want the forward direction of the isomorphism.*
An element of the end $\int_c \mathcal{V}(\mathcal{C}(a, c), F(c))$ is a compatible family of morphisms $(\theta_c : \mathcal{C}(a, c) \to F(c))_c$ in $\mathcal{V}$.	*By the universal property of the end.*
Apply $\theta_a : \mathcal{C}(a, a) \to F(a)$ to the identity $j_a : I \to \mathcal{C}(a, a)$ (the enriched identity map).	*The identity morphism gives a specific element of $\mathcal{C}(a, a)$ over I.*
Define $\phi(\theta) = \theta_a \circ j_a : I \to F(a)$.	*This is a morphism in $\mathcal{V}$.*
We construct the inverse $\psi : F(a) \to \int_c \mathcal{V}(\mathcal{C}(a, c), F(c))$.	*For this we need to produce a compatible family $(\psi(x)_c)_c$ from $x : I \to F(a)$.*
At each c, define $\psi(x)_c$ as the composite $\mathcal{C}(a, c) \xrightarrow{\cong} I \otimes \mathcal{C}(a, c) \xrightarrow{x \otimes \mathrm{id}} F(a) \otimes \mathcal{C}(a, c) \xrightarrow{F_{a,c}} F(c)$.	*The action of F on hom-objects is the $\mathcal{V}$-functor structure map $F_{a,c} : F(a) \otimes \mathcal{C}(a, c) \to F(c)$.*
The family $(\psi(x)_c)_c$ is compatible (i.e., defines an element of the end) because F satisfies the $\mathcal{V}$-functoriality axioms.	*The end condition is: for $f : c \to c'$ in $\mathcal{C}$, the square $\mathcal{V}(\mathcal{C}(a, c'), F(c')) \to \mathcal{V}(\mathcal{C}(a, c), F(c))$ commutes. This follows from functoriality of F.*
$\phi \circ \psi = \mathrm{id}_{F(a)}$: compute $\phi(\psi(x)) = \psi(x)_a \circ j_a$.	*Substituting the definitions.*
$\psi(x)_a \circ j_a = F_{a,a}(x \otimes j_a) \circ \lambda^{-1}_{\mathcal{C}(a,a)} = F_{a,a}(x \otimes j_a) \circ \lambda^{-1} = x$.	*The $\mathcal{V}$-functor axiom says $F_{a,a}(x \otimes j_a) = x$ when applied with the enriched unit j_a (unit axiom for $\mathcal{V}$-functors).*
$\psi \circ \phi = \mathrm{id}$: For a compatible family (θ_c), we need $\psi(\phi(\theta))_c = \theta_c$ for all c.	*The converse direction.*
$\psi(\phi(\theta))_c = F_{a,c}((\theta_a \circ j_a) \otimes \mathrm{id}_{\mathcal{C}(a,c)})$.	*Substituting $\phi(\theta) = \theta_a \circ j_a$.*
$= F_{a,c}(\theta_a \otimes \mathrm{id}) \circ (j_a \otimes \mathrm{id}) = \theta_c$	*By the naturality (end condition) for (θ_c): the compatibility of θ with morphisms in $\mathcal{C}$ translates exactly to this equality.*
Both composites are identities; ϕ and ψ are inverse isomorphisms in $\mathcal{V}$.	*The enriched Yoneda isomorphism.*
$\mathcal{V}$-naturality in a: the isomorphism is natural in a because all constructions (end, functor action) are natural.	*Standard argument: the maps ϕ_a and ψ_a assemble into $\mathcal{V}$-natural transformations.*

20.1.4 Reorganizing: Corollaries

Corollary 20.2 (Enriched Representability)**.** *A $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{V}$ is representable iff there exists $a \in \mathcal{C}$ with $F \cong \mathcal{C}(a, -)$ as $\mathcal{V}$-functors.*

Corollary 20.3 (Enriched Yoneda Embedding)**.** *The enriched Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ is fully faithful as a $\mathcal{V}$-functor.*

Internal Dialog

Is the enriched Yoneda lemma really the same as the ordinary one?
Yes, in a precise sense. If $\mathcal{V} = \mathbf{Set}$ (with cartesian product), then $\mathcal{V}$-enriched categories are ordinary locally small categories, ends are ordinary natural transformation sets, and the enriched Yoneda lemma reduces to the ordinary one. The enriched version is a strict generalization.

20.2 $\mathcal{V}$-Adjunctions

20.2.1 Informally: Hom-Object Bijections

In ordinary category theory, an adjunction is a bijection of hom-sets: $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$. In the enriched world, hom-sets are replaced by hom-objects in $\mathcal{V}$, and the bijection is replaced by an isomorphism in $\mathcal{V}$:

$$\mathcal{D}(FA, B) \;\cong\; \mathcal{C}(A, GB) \quad \text{(isomorphism in } \mathcal{V})$$

$\mathcal{V}$-natural in A and B. This is a **$\mathcal{V}$-adjunction** (or **enriched adjunction**).

20.2.2 Expanding: Unit and Counit in the Enriched Case

The unit of a $\mathcal{V}$-adjunction is a $\mathcal{V}$-natural transformation $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$. But the naturality of η is now $\mathcal{V}$-naturality: the naturality square is a commutative diagram in $\mathcal{V}$, not merely in **Set**.

Explicitly, $\mathcal{V}$-naturality of η at A, A' is the commutativity of the diagram in $\mathcal{V}$:

$$\begin{tikzcd}
\mathcal{C}(A, A') \arrow[r, "G_{FA,FA'}F_{A,A'}"] \arrow[d, "\eta_{A'*}"'] & \mathcal{C}(GFA, GFA') \arrow[d, "\eta_A^*"] \\
\mathcal{C}(A, GFA') \arrow[r, equal] & \mathcal{C}(A, GFA')
\end{tikzcd}$$

where $\eta_{A'*}$ and η_A^* denote post- and pre-composition with η.

20.2.3 Formalizing: The Definition

Definition 20.4. Let $\mathcal{C}$ and $\mathcal{D}$ be $\mathcal{V}$-categories. A **$\mathcal{V}$-adjunction** $F \dashv_{\mathcal{V}} G$ (with $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ $\mathcal{V}$-functors) consists of a $\mathcal{V}$-natural isomorphism

$$\varphi_{A,B} : \mathcal{D}(FA, B) \xrightarrow{\cong} \mathcal{C}(A, GB)$$

for all $A \in \mathcal{C}$, $B \in \mathcal{D}$, natural in both A and B in the $\mathcal{V}$-sense.

Remark 20.5. Every $\mathcal{V}$-adjunction has an underlying ordinary adjunction: applying $\mathcal{V}(I, -)$ to the hom-object isomorphism gives an ordinary bijection of hom-sets (since $\mathcal{V}(I, \mathcal{C}(A, B))$ is the set of morphisms $A \to B$ in the underlying ordinary category of $\mathcal{C}$).

20.2.4 Reorganizing: Enriched vs. Ordinary Adjunctions

The gap between ordinary and enriched adjunctions is subtle: an ordinary adjunction between $\mathcal{V}$-categories (functors that happen to be $\mathcal{V}$-functors, but with only a set bijection on hom-sets) need not be a $\mathcal{V}$-adjunction. The stronger requirement of isomorphism in $\mathcal{V}$ is genuinely richer.

However, in many practical cases (when $\mathcal{V}$ is the base of an enrichment and the relevant categories are "nice"), ordinary adjunctions between $\mathcal{V}$-functors automatically lift to $\mathcal{V}$-adjunctions. This is guaranteed, for example, when $\mathcal{C}$ and $\mathcal{D}$ are $\mathcal{V}$-tensored and cotensored.

20.3 Weighted Limits and Colimits

20.3.1 Informally: Why Ordinary Limits Are Not Enough

In ordinary category theory, a limit of $F : \mathcal{J} \to \mathcal{C}$ is characterized by $\mathcal{C}(c, \lim F) \cong [\mathcal{J}, \mathbf{Set}](\Delta 1, \mathcal{C}(c, F-))$: cones from c to F.

In enriched category theory, the relevant functor category is a $\mathcal{V}$-functor category, and the "constant diagram" functor Δc should be a $\mathcal{V}$-functor. But more general shapes arise naturally: we may want to weight the limit by a functor $W : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$ that assigns different "multiplicities" or "shapes" to each object of $\mathcal{J}$.

A **W-weighted limit** of $F : \mathcal{J} \to \mathcal{C}$ is an object $\{W, F\} \in \mathcal{C}$ characterized by the enriched universal property:

$$\mathcal{C}(c, \{W, F\}) \;\cong\; [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$$

as an isomorphism in $\mathcal{V}$, $\mathcal{V}$-natural in c.

20.3.2 Expanding: Ordinary Limits as Special Cases

Setting $\mathcal{V} = \mathbf{Set}$ and $W = \Delta 1$ (the constant weight sending every object to the one-element set): the weighted limit $\{W, F\}$ recovers the ordinary limit $\lim F$, since $[\mathcal{J}, \mathbf{Set}](\Delta 1, \mathcal{C}(c, F-))$ is the set of natural transformations from the constant functor to $\mathcal{C}(c, F-)$, i.e., the set of cones.

More generally, taking $W = \mathcal{J}(j, -)$ (the representable weight at j) and applying the Yoneda lemma gives $\{W, F\} \cong F(j)$: the limit weighted by a representable recovers evaluation at that object.

Example: Cotensor as Weighted Limit

For $\mathcal{V} = \mathbf{Ab}$ and $\mathcal{J} = \{*\}$ (one object), a weight $W : \{*\}^{\mathrm{op}} \to \mathbf{Ab}$ is just an abelian group A, and a functor $F : \{*\} \to \mathcal{C}$ is just an object $M \in \mathcal{C}$. The weighted limit $\{A, M\}$ is the *cotensor* of M by A: the object representing the functor $\mathcal{C}(-, M)^A$ in an appropriate sense. For $\mathcal{C} = \mathbf{Ab}$, this is $\mathrm{Hom}(A, M)$.

Example: The Homotopy Limit

In the $(\infty, 1)$-categorical setting, homotopy limits are weighted limits where the weight encodes simplicial resolution data. The classical Bousfield–Kan formula for homotopy limits is a weighted colimit formula with the weight coming from the nerve of the indexing category.

20.3.3 Formalizing: The Definition and its Dual

Definition 20.6. Let $\mathcal{C}$ be a $\mathcal{V}$-category, $\mathcal{J}$ a small $\mathcal{V}$-category, $W : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$ a $\mathcal{V}$-functor (the **weight**), and $F : \mathcal{J} \to \mathcal{C}$ a $\mathcal{V}$-functor. The **W-weighted limit** $\{W, F\}$, when it exists, is an object of $\mathcal{C}$ with a $\mathcal{V}$-natural isomorphism

$$\mathcal{C}(c, \{W, F\}) \;\cong\; [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W,\ \mathcal{C}(c, F(-))).$$

Dually, the **W-weighted colimit** $W \star F$, when it exists, is an object with

$$\mathcal{C}(W \star F,\ c) \;\cong\; [\mathcal{J}, \mathcal{V}]_{\mathcal{V}}(W,\ \mathcal{C}(F(-), c)).$$

20.3.4 Reorganizing: The Yoneda Reduction

The reason ordinary limits are a special case is the Yoneda lemma: for $W = \Delta I$ (constant weight at the unit object $I \in \mathcal{V}$),

$$[\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\Delta I,\ \mathcal{C}(c, F(-))) \;\cong\; \int_j \mathcal{V}(I, \mathcal{C}(c, F(j))) \;\cong\; \int_j \mathcal{C}(c, F(j)),$$

the last step using $\mathcal{V}(I, -) \cong (-)$ (since I is the unit). For $\mathcal{V} = \mathbf{Set}$ this is the set of natural cones, i.e., ordinary limit cones.

How to Say This

- $\{W, F\}$ is pronounced "W hom F" or "the W-weighted limit of F." Some authors write $\lim^W F$ or $[W, F]$.
- $W \star F$ is the *coend* or weighted colimit; also written $\mathrm{colim}^W F$ or $W \otimes F$ in older texts.
- The ΔI-weighted limit is the ordinary conical limit; the ΔI-weighted colimit is the ordinary conical colimit.
- Weighted limits subsume: ordinary limits, ends, cotensors, homotopy limits, and mapping spaces.

20.4 The Change of Base

20.4.1 Informally: Moving Between Enrichments

If $\Phi : \mathcal{V} \to \mathcal{W}$ is a lax monoidal functor (Chapter 19), we can use Φ to turn any $\mathcal{V}$-category into a $\mathcal{W}$-category by applying Φ to all hom-objects. This is the **change of base**.

20.4.2 Expanding: The Construction

Let $\mathcal{C}$ be a $\mathcal{V}$-category. The **base change** along $\Phi : \mathcal{V} \to \mathcal{W}$ is the $\mathcal{W}$-category $\mathcal{C}_\Phi$ with:

- The same objects as $\mathcal{C}$.
- Hom-objects $\mathcal{C}_\Phi(a, b) = \Phi(\mathcal{C}(a, b))$ in $\mathcal{W}$.
- Composition $\mathcal{C}_\Phi(b, c) \otimes_{\mathcal{W}} \mathcal{C}_\Phi(a, b) \to \mathcal{C}_\Phi(a, c)$ defined as

  $$\Phi(\mathcal{C}(b, c)) \otimes_{\mathcal{W}} \Phi(\mathcal{C}(a, b)) \xrightarrow{\phi_{-,-}} \Phi(\mathcal{C}(b, c) \otimes_{\mathcal{V}} \mathcal{C}(a, b)) \xrightarrow{\Phi(\circ_{\mathcal{C}})} \Phi(\mathcal{C}(a, c)),$$

  using the lax monoidal comparison ϕ of Φ.
- Identities $j_a^{\mathcal{W}} : I_{\mathcal{W}} \xrightarrow{\phi_0} \Phi(I_{\mathcal{V}}) \xrightarrow{\Phi(j_a^{\mathcal{C}})} \Phi(\mathcal{C}(a, a))$.

The coherence conditions for $\mathcal{V}$-categories (associativity and unit laws for $\circ_{\mathcal{C}}$) together with the coherence of the lax monoidal functor Φ guarantee that $\mathcal{C}_\Phi$ is a valid $\mathcal{W}$-category.

20.4.3 Formalizing: Functoriality of Base Change

Base change is functorial: if $F : \mathcal{C} \to \mathcal{D}$ is a $\mathcal{V}$-functor, then applying Φ to the hom-object maps $F_{a,b} : \mathcal{C}(a, b) \to \mathcal{D}(Fa, Fb)$ gives $\mathcal{W}$-functor maps $\Phi(F_{a,b}) : \mathcal{C}_\Phi(a, b) \to \mathcal{D}_\Phi(Fa, Fb)$, yielding a $\mathcal{W}$-functor $F_\Phi : \mathcal{C}_\Phi \to \mathcal{D}_\Phi$.

Proposition 20.7. *The change-of-base construction defines a 2-functor $(-)_\Phi : \mathcal{V}\text{-}\mathbf{Cat} \to \mathcal{W}\text{-}\mathbf{Cat}$.*

20.4.4 Reorganizing: Examples of Base Change

Example: Forgetting the Enrichment

Let $\Phi = \mathcal{V}(I, -) : \mathcal{V} \to \mathbf{Set}$. This is lax monoidal (it sends $\otimes$ to $\times$ via the composition map). For any $\mathcal{V}$-category $\mathcal{C}$, the base change $\mathcal{C}_\Phi$ is the **underlying ordinary category** of $\mathcal{C}$: its hom-sets are $\mathcal{V}(I, \mathcal{C}(a, b))$.

Example: Complexes to Graded

The cohomology functor $H : \mathbf{Ch}(\mathbf{Ab}) \to \mathbf{GrAb}$ (from chain complexes to graded abelian groups) is lax monoidal with respect to tensor products, and induces a base change from $\mathbf{Ch}(\mathbf{Ab})$-enriched to $\mathbf{GrAb}$-enriched categories.

20.5 Self-Enrichment of Closed Symmetric Monoidal Categories

20.5.1 Informally: A Category Enriching Over Itself

The most important source of enriched categories: a **closed symmetric monoidal category** $(\mathcal{V}, \otimes, I, [-, -])$ enriches over itself. Here $[A, B]$ is the *internal hom* object, characterized by:

$$\mathcal{V}(C \otimes A, B) \;\cong\; \mathcal{V}(C, [A, B]).$$

This is the **closed structure**: $(- \otimes A)$ has a right adjoint $[A, -]$.

Internal Dialog

Why does closedness give enrichment?
To enrich $\mathcal{V}$ over itself, we need hom-objects in $\mathcal{V}$. The internal hom $[A, B]$ is exactly the right object to use. The composition law $[B, C] \otimes [A, B] \to [A, C]$ ("evaluation followed by application") is the standard map in a closed category, and the identity $I \to [A, A]$ comes from the unit isomorphism.

20.5.2 Expanding: The Self-Enrichment

Let $(\mathcal{V}, \otimes, I)$ be a closed symmetric monoidal category with internal hom $[-, -]$. Define a $\mathcal{V}$-category (also called $\mathcal{V}$):

- Objects: the objects of $\mathcal{V}$.
- $\mathcal{V}$-hom-objects: $\mathcal{V}(A, B) = [A, B] \in \mathcal{V}$.
- Composition: the standard morphism $[B, C] \otimes [A, B] \to [A, C]$ obtained by currying the evaluation composite $[B, C] \otimes [A, B] \otimes A \to [B, C] \otimes B \to C$.
- Identity: the morphism $I \to [A, A]$ corresponding (under the adjunction) to the identity $A \otimes I \cong A$.

These satisfy the $\mathcal{V}$-category axioms by the coherence of the closed structure.

20.5.3 Formalizing: Examples

Example: Ab-enrichment

$(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ is closed symmetric monoidal with internal hom $[A, B] = \mathrm{Hom}(A, B)$ (the group of homomorphisms). The self-enrichment says: $\mathbf{Ab}$ is enriched over itself, with hom-objects $\mathrm{Hom}(A, B)$ being abelian groups. This is the standard $\mathbf{Ab}$-enrichment of the category of abelian groups.

Example: Vect_k-enrichment

$(\mathbf{Vect}_k, \otimes_k, k)$ is closed with $[V, W] = \mathrm{Hom}_k(V, W)$. The self-enrichment gives $\mathbf{Vect}_k$ as a $\mathbf{Vect}_k$-enriched category, with hom-objects being vector spaces of linear maps.

Example: sSet-enrichment

The category of simplicial sets $\mathbf{sSet} = [\Delta^{\mathrm{op}}, \mathbf{Set}]$ is closed monoidal with the cartesian product. The internal hom is the simplicial mapping space $\mathbf{Map}(X, Y)_n = \mathbf{sSet}(X \times \Delta^n, Y)$. Self-enrichment gives $\mathbf{sSet}$ as an $\mathbf{sSet}$-enriched category, which is the starting point for simplicial homotopy theory.

20.5.4 Reorganizing: Tensors, Cotensors, and Completeness

A $\mathcal{V}$-category $\mathcal{C}$ is **$\mathcal{V}$-complete** if all weighted limits $\{W, F\}$ exist. It is **$\mathcal{V}$-tensored** if for each $V \in \mathcal{V}$ and $c \in \mathcal{C}$, the tensor $V \otimes c$ exists, characterized by $\mathcal{C}(V \otimes c, d) \cong \mathcal{V}(V, \mathcal{C}(c, d))$. Dually for cotensors $[V, c]$.

When $\mathcal{C}$ is both tensored and cotensored over $\mathcal{V}$, all weighted limits and colimits reduce to ordinary conical limits plus the (co)tensor action. This is a powerful computational tool.

Theorem 20.8. *A $\mathcal{V}$-category $\mathcal{C}$ that is $\mathcal{V}$-tensored and $\mathcal{V}$-cotensored has all weighted limits iff it has all conical limits:*

$$\{W, F\} \;\cong\; \lim_{j \in \mathcal{J}} [W(j), F(j)],$$

where $[W(j), F(j)]$ denotes the cotensor of $F(j)$ by $W(j)$.

Exercises

State the enriched Yoneda lemma.	$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \cong F(a)$ in $\mathcal{V}$, $\mathcal{V}$-naturally in a.
What is the $\mathcal{V}$-object of $\mathcal{V}$-natural transformations?	$\int_c \mathcal{V}(G(c), H(c))$, an end in $\mathcal{V}$.
In the enriched Yoneda proof, what is $\phi(\theta)$?	$\phi(\theta) = \theta_a \circ j_a : I \to F(a)$, where $j_a : I \to \mathcal{C}(a, a)$ is the enriched identity.
What is a $\mathcal{V}$-adjunction?	A $\mathcal{V}$-natural isomorphism $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ in $\mathcal{V}$.
Does a $\mathcal{V}$-adjunction imply an ordinary adjunction?	Yes: apply $\mathcal{V}(I, -)$ to get a bijection of underlying hom-sets.
Can an ordinary adjunction between $\mathcal{V}$-functors fail to be a $\mathcal{V}$-adjunction?	Yes, in general. But in tensored/cotensored $\mathcal{V}$-categories, ordinary adjoints between $\mathcal{V}$-functors lift to $\mathcal{V}$-adjunctions.
What is a W-weighted limit?	An object $\{W, F\} \in \mathcal{C}$ with $\mathcal{C}(c, \{W, F\}) \cong [\mathcal{J}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$ $\mathcal{V}$-naturally in c.
What ordinary limit corresponds to $W = \Delta I$?	The ordinary conical limit $\lim F$.
What weighted limit corresponds to $W = \mathcal{J}(j, -)$?	Evaluation $F(j)$, by the enriched Yoneda lemma.
What is the cotensor $[A, M]$ as a weighted limit?	The A-weighted limit of the single-object diagram at M, for $A \in \mathcal{V}$ and $M \in \mathcal{C}$.
What is a closed symmetric monoidal category?	One where $(- \otimes A)$ has a right adjoint $[A, -]$ (the internal hom), and the tensor is symmetric.
What is self-enrichment?	Using $[A, B]$ as the hom-object of A, B in the $\mathcal{V}$-category structure on $\mathcal{V}$ itself.
What is the internal hom in **Ab**?	$[A, B] = \text{Hom}(A, B)$, the abelian group of homomorphisms.
What is the change of base for enriched categories?	Given a lax monoidal $\Phi : \mathcal{V} \to \mathcal{W}$, a $\mathcal{V}$-category $\mathcal{C}$ becomes a $\mathcal{W}$-category $\mathcal{C}_\Phi$ with hom-objects $\Phi(\mathcal{C}(a, b))$.
What lax monoidal functor gives the underlying ordinary category?	$\mathcal{V}(I, -) : \mathcal{V} \to \mathbf{Set}$, sending each hom-object to its underlying set of "generalized elements."
What is a $\mathcal{V}$-tensored category?	One where $V \otimes c$ exists with $\mathcal{C}(V \otimes c, d) \cong \mathcal{V}(V, \mathcal{C}(c, d))$.
In a tensored and cotensored $\mathcal{V}$-category, how do weighted limits reduce?	$\{W, F\} \cong \lim_j [W(j), F(j)]$, a conical limit of cotensors.
What is the enriched Yoneda embedding?	$Y : \mathcal{C} \to [\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}$, $Y(a) = \mathcal{C}(-, a)$; it is fully faithful as a $\mathcal{V}$-functor.
What does $\mathcal{V}$-naturality require beyond ordinary naturality?	The naturality squares must commute in $\mathcal{V}$ (as morphisms in $\mathcal{V}$), not just in **Set**.
What is the simplicial mapping space in **sSet**?	$\mathbf{Map}(X, Y)_n = \mathbf{sSet}(X \times \Delta^n, Y)$, the internal hom in the closed monoidal category **sSet**.

Chapter Summary

Chapter Summary

Enriched Yoneda. For a $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{V}$, $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \cong F(a)$ in $\mathcal{V}$. The proof uses the enriched identity morphism and the $\mathcal{V}$-functor action, with mutual inverse $\psi(x)_c = F_{a,c} \circ (x \otimes \mathrm{id})$.
$\mathcal{V}$-adjunctions. The ordinary hom-set bijection is replaced by a $\mathcal{V}$-natural isomorphism $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ in $\mathcal{V}$. Every $\mathcal{V}$-adjunction has an underlying ordinary adjunction.
Weighted limits. The W-weighted limit $\{W, F\}$ satisfies $\mathcal{C}(c, \{W, F\}) \cong [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$. Ordinary limits are ΔI-weighted. In tensored and cotensored categories, $\{W, F\} \cong \lim_j [W(j), F(j)]$.
Change of base. A lax monoidal functor $\Phi : \mathcal{V} \to \mathcal{W}$ induces $(-)_\Phi : \mathcal{V}\text{-}\mathbf{Cat} \to \mathcal{W}\text{-}\mathbf{Cat}$ by applying Φ to all hom-objects. The special case $\mathcal{V}(I, -)$ gives the underlying ordinary category.
Self-enrichment. A closed symmetric monoidal category $(\mathcal{V}, \otimes, I, [-, -])$ enriches over itself using $[A, B]$ as the hom-object. Examples: **Ab**, $\mathbf{Vect}_k$, **sSet**.

Formal Restatement

Formal Restatement

Enriched Yoneda. For $\mathcal{C}$ a $\mathcal{V}$-category and $F : \mathcal{C} \to \mathcal{V}$ a $\mathcal{V}$-functor:

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \;=\; \int_c \mathcal{V}(\mathcal{C}(a, c), F(c)) \;\cong\; F(a)$$

$\mathcal{V}$-naturally in a.

$\mathcal{V}$-adjunction. $F \dashv_{\mathcal{V}} G$ iff there is a $\mathcal{V}$-natural isomorphism $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ in $\mathcal{V}$. Underlying ordinary adjunction obtained by applying $\mathcal{V}(I, -)$.

Weighted limit.

$$\mathcal{C}(c, \{W, F\}) \;\cong\; [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-)).$$

Ordinary limit: $W = \Delta I$. In tensored/cotensored $\mathcal{C}$: $\{W, F\} \cong \lim_j [W(j), F(j)]$.

Change of base. Lax monoidal $\Phi : \mathcal{V} \to \mathcal{W}$ induces $\mathcal{C}_\Phi$ with $\mathcal{C}_\Phi(a, b) = \Phi(\mathcal{C}(a, b))$, composites via ϕ-comparisons.

Self-enrichment. Closed symmetric monoidal $(\mathcal{V}, \otimes, I, [-, -])$ is a $\mathcal{V}$-category with $\mathcal{V}(A, B) = [A, B]$, composition via the evaluation map $[B, C] \otimes [A, B] \otimes A \to C$.

Chapter 21 returns to monads via Beck's monadicity theorem, which characterizes when a functor is equivalent to a category of algebras for a monad.

Chapter 21

Monads: Algebraic Theory and Beck's Theorem

Chapter Overview

Chapter 8 introduced monads and showed how every adjunction $F \dashv G$ gives rise to a monad $T = GF$ on $\mathcal{C}$. But it left a gap: different adjunctions can produce the *same* monad. Which adjunction is the "right" one? And conversely, given a monad (T, η, μ), can we always recover an adjunction that produces it?

The answer is yes — and in fact there are two canonical adjunctions, one minimal (the **Kleisli category** $\mathcal{C}_T$) and one maximal (the **Eilenberg–Moore category** $\mathcal{C}^T$). All adjunctions that resolve T factor through these two extremes via a **comparison functor**.

The climax of this chapter is **Beck's monadicity theorem**: a precise criterion for when a functor $F : \mathcal{C} \to \mathcal{D}$ is *monadic* — meaning $\mathcal{D}$ is equivalent, over $\mathcal{C}$, to the Eilenberg–Moore category of the induced monad. Beck's theorem transforms an abstract question about equivalence of categories into a concrete checklist involving coequalizers, and is one of the most useful recognition theorems in all of category theory.

Along the way we meet **F-split coequalizers**, a remarkable class of coequalizers that "know how to split themselves" when lifted along F, and we see why they are the right notion of coequalizer for algebraic categories.

21.1 Recap: Monads from Adjunctions

21.1.1 The Induced Monad

Let $F : \mathcal{C} \rightleftharpoons \mathcal{D} : G$ be an adjunction $F \dashv G$ with unit $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$ and counit $\varepsilon : FG \Rightarrow \mathrm{Id}_{\mathcal{D}}$.

Set $T = GF : \mathcal{C} \to \mathcal{C}$. The monad structure is:

- **Unit:** $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow T = GF$. This is already the adjunction unit.
- **Multiplication:** $\mu = G\varepsilon F : GFGF \Rightarrow GF$. At component c: $\mu_c = G(\varepsilon_{Fc}) : GFGFc \to GFc$.

The monad axioms — associativity $\mu \circ \mu T = \mu \circ T\mu$ and the unit laws $\mu \circ \eta T = \mathrm{id}_T = \mu \circ T\eta$ — follow directly from the triangle identities for $F \dashv G$.

Internal Dialog

So the monad carries only the "shadow" of the adjunction that lands back on $\mathcal{C}$. The functor F disappears; only the composite GF survives. This information loss is precisely what Beck's theorem measures.

21.1.2 The Eilenberg–Moore Category

Given a monad (T, η, μ) on $\mathcal{C}$, the **Eilenberg–Moore category** $\mathcal{C}^T$ has:

- **Objects:** **T-algebras**: pairs (a, h) where $a \in \mathcal{C}$ and $h : Ta \to a$ (the *structure map*), satisfying:

$$h \circ \eta_a = \mathrm{id}_a \qquad \text{(unit law)}$$
$$h \circ \mu_a = h \circ Th \qquad \text{(associativity law)}$$

- **Morphisms:** $(a, h) \to (b, k)$: morphisms $f : a \to b$ in $\mathcal{C}$ such that $k \circ Tf = f \circ h$.

The forgetful functor $G^T : \mathcal{C}^T \to \mathcal{C}$ sends $(a, h) \mapsto a$ and has a left adjoint $F^T : \mathcal{C} \to \mathcal{C}^T$ sending $a \mapsto (Ta, \mu_a)$. The induced monad of $F^T \dashv G^T$ is exactly (T, η, μ).

21.1.3 The Kleisli Category

The **Kleisli category** $\mathcal{C}_T$ has:

- **Objects:** the same objects as $\mathcal{C}$.

- **Morphisms:** $a \to b$ in $\mathcal{C}_T$ are morphisms $a \to Tb$ in $\mathcal{C}$ ("T-decorated" or effectful maps).
- **Composition:** if $f : a \to Tb$ and $g : b \to Tc$ in $\mathcal{C}$, then $g \star f = \mu_c \circ Tg \circ f : a \to Tc$.
- **Identity on** a**:** the unit $\eta_a : a \to Ta$.

There is a free functor $F_T : \mathcal{C} \to \mathcal{C}_T$ (identity on objects, $f \mapsto \eta_b \circ f$ on morphisms) and a forgetful functor $G_T : \mathcal{C}_T \to \mathcal{C}$ (objects to Ta, morphisms f to $\mu_b \circ Tf$). These form an adjunction $F_T \dashv G_T$ inducing (T, η, μ).

How to Say This

- A morphism $f : a \to Tb$ in $\mathcal{C}$ is called a **Kleisli morphism** or an **effectful computation** from a to b.
- Kleisli composition is the categorical form of "bind" (>>= in Haskell).
- $\mathcal{C}_T$ is the minimal resolution; $\mathcal{C}^T$ is the maximal resolution.

21.2 The Comparison Functor

21.2.1 Every Resolution Factors Between the Two Extremes

Suppose $F \dashv G$ resolves the monad $T = GF$. We define a functor

$$K : \mathcal{D} \to \mathcal{C}^T$$

called the **comparison functor**, by:

- On objects: $d \mapsto (Gd, G\varepsilon_d)$.
- On morphisms: $f \mapsto Gf$.

Lemma 21.1. *The pair $(Gd, G\varepsilon_d)$ is indeed a T-algebra: the map $G\varepsilon_d : GFGd \to Gd$ satisfies the T-algebra axioms.*

Proof. Unit law: $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ by the triangle identity $G\varepsilon \circ \eta G = \mathrm{id}_G$.

Associativity: $G\varepsilon_d \circ \mu_{Gd} = G\varepsilon_d \circ GFG\varepsilon_d$ is the equation $G\varepsilon_d \circ G(\varepsilon_{FGd}) = G\varepsilon_d \circ GF(G\varepsilon_d)$, which is the naturality of ε applied to ε_d. □

Furthermore, K commutes with the forgetful functors: $G^T \circ K = G$. The comparison functor measures "how much of the structure of $\mathcal{D}$ is captured by T-algebras."

21.2.2 When is K an Equivalence?

The functor $F \dashv G$ is called **monadic** if the comparison functor $K : \mathcal{D} \to \mathcal{C}^T$ is an equivalence of categories. In this case, $\mathcal{D}$ *is* (up to equivalence) the category of algebras for the monad T on $\mathcal{C}$.

Examples:

- The free/forgetful adjunction for groups is monadic: $\mathbf{Grp} \simeq \mathbf{Set}^T$ for the free-group monad.
- The free/forgetful adjunction for rings is monadic.
- The global-sections/constant-sheaf adjunction for sheaves is *not* monadic in general.

Beck's theorem gives the exact criterion.

Where the Name Comes From

Beck's monadicity theorem is named for *Jonathan Mock Beck* (1935–2014, American mathematician), who proved it in his 1967 PhD thesis at Columbia (advisor: Eilenberg). The theorem gives a precise criterion for when an adjunction is "monadic" — i.e., when the adjunction is equivalent to the canonical Eilenberg-Moore adjunction for the induced monad. Beck's theorem is one of the most useful recognition theorems in category theory; it appears in modern form across higher algebra (Vol IV Ch. 51 for the ∞-version, Lurie HA Ch. 4). Beck also discovered distributive laws between monads (Vol III Ch. 35).

21.3 Beck's Monadicity Theorem

21.3.1 The Statement

Theorem 21.2 (Beck's Monadicity Theorem)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor with right adjoint $G : \mathcal{D} \to \mathcal{C}$, and let $T = GF$ be the induced monad. The comparison functor $K : \mathcal{D} \to \mathcal{C}^T$ is an equivalence of categories if and only if:*

*(B1) F **reflects isomorphisms**: if Ff is an iso in $\mathcal{C}$, then f is an iso in $\mathcal{D}$.*

*(B2) $\mathcal{D}$ has coequalizers of all **F-split pairs**, and F preserves these coequalizers.*

21.3.2 Proof Sketch via the Prooflog

We prove that under (B1) and (B2), the comparison functor K is full, faithful, and essentially surjective.

Proof. We trace the essential surjectivity argument, as it is the most illuminating. Let $(a, h : Ta \to a)$ be any T-algebra. We need to find $d \in \mathcal{D}$ with $Kd \cong (a, h)$.

Consider the pair $Fa \overset{Fh}{\underset{\eta_{Fa}}{\rightrightarrows}} FTa = FGFa$	*$h : Ta \to a$ in $\mathcal{C}$; apply F to get $Fh : FTa \to Fa$ and also have $\varepsilon_{Fa} : FGFa \to Fa$*
Actually the relevant pair is $FTa \overset{Fh}{\underset{\varepsilon_{Fa}}{\rightrightarrows}} Fa$	*Both Fh and ε_{Fa} are morphisms $FTa = FGFa \to Fa$; the pair (Fh, ε_{Fa}) is F-split*
This pair is F-split via: $h : Ta \to a$, $\eta_a : a \to Ta$	*Check: $G(Fh) \circ \eta_a = Th \circ \eta_a$; $G(\varepsilon_{Fa}) \circ \eta_a = \mathrm{id}_a$ (triangle identity)*
By (B2), the coequalizer $d = \mathrm{coeq}(Fh, \varepsilon_{Fa})$ exists in $\mathcal{D}$	*Let $q : Fa \to d$ be the coequalizer morphism*
Apply G: since F preserves this coequalizer and G is right adjoint, Gq is a split epi in $\mathcal{C}$	*The F-split condition propagates through the adjunction*
One shows $Kd = (Gd, G\varepsilon_d) \cong (a, h)$ as T-algebras	*The algebra map $Gd \to a$ is constructed from q via the universal property of the coequalizer; the T-algebra laws follow from the F-split splitting data*
Faithfulness of K follows from F reflecting isomorphisms (B1): K-faithful + F-iso-reflecting forces K-faithful	*If $Kf = Kg$ then $GFf = GFg$; since the coequalizer structure forces uniqueness, $f = g$*
Fullness follows from the coequalizer presentation of free algebras	*Every T-algebra morphism lifts uniquely to a morphism in $\mathcal{D}$ via the universal property*

Therefore K is an equivalence. The converse direction checks that if K is an equivalence, then conditions (B1) and (B2) must hold by abstract properties of equivalences and the structure of $\mathcal{C}^T$. ☐

Internal Dialog

Why are F-split pairs the right notion? Because split coequalizers are *absolute*: they are preserved by every functor. So requiring the coequalizer to exist only after applying F is the minimal assumption needed to do the construction. If you demand more — say, all reflexive coequalizers — you get a stronger ("crude") form of monadicity that is easier to apply but less sharp.

21.4 F-Split Coequalizers

21.4.1 Definition

Definition 21.3 (F-split pair and split coequalizer)**.** A **split coequalizer** is a diagram

$$a \overset{f}{\underset{g}{\rightrightarrows}} b \xrightarrow{q} c$$

together with morphisms $s : c \to b$ and $t : b \to a$ satisfying:

$$q \circ s = \mathrm{id}_c$$
$$f \circ t = \mathrm{id}_b$$
$$g \circ t = s \circ q$$

Such a diagram is a coequalizer of (f, g), and it is preserved by every functor (the splitting data witnesses the coequalizer purely equationally).

A pair $(f, g) : a \rightrightarrows b$ in $\mathcal{D}$ is called an **F-split pair** (for a functor $F : \mathcal{D} \to \mathcal{C}$) if the pair (Ff, Fg) has a split coequalizer in $\mathcal{C}$.

21.4.2 The Key Example: Free Algebra Pairs

Given a T-algebra $(a, h : Ta \to a)$, the pair

$$T^2a \overset{\mu_a}{\underset{Th}{\rightrightarrows}} Ta$$

is always split in $\mathcal{C}$ (via η_{Ta} and η_a), and its coequalizer is a with coequalizer map h. This is the prototype of every F-split pair that arises in Beck's theorem.

Example: F-split pairs for the free-module monad

Let R be a ring, $F = R \otimes_{\mathbb{Z}} - : \mathbf{Ab} \to R\text{-}\mathbf{Mod}$ the free R-module functor (left adjoint to the forgetful functor G). A pair $(f, g) : M \rightrightarrows N$ in R-**Mod** is F-split iff the underlying abelian group pair $(Gf, Gg) : GM \rightrightarrows GN$ has a split coequalizer in **Ab**.
In practice: this pair is split iff the coequalizer in **Ab** happens to be the underlying abelian group of an R-module that R can act on in a compatible way.

21.4.3 Absolute Coequalizers

The splitting data (s, t) is remarkable: it makes the coequalizer *absolute*, meaning it is preserved by every functor without any assumption on that functor. Contrast:

- Ordinary coequalizers: preserved by left adjoints.
- Reflexive coequalizers: preserved by categories satisfying additional exactness properties.
- **Split coequalizers: preserved by every functor.**

This is why split coequalizers are the right setting for Beck's theorem: the condition "F preserves the coequalizer" is automatically satisfied once the coequalizer is split.

21.5 Applications

21.5.1 Rings and Modules

Let R be a commutative ring. The free R-module functor $F = R[-] : \mathbf{Set} \to R\text{-}\mathbf{Mod}$, sending a set X to the free R-module $R[X] = \bigoplus_{x \in X} R$, is left adjoint to the forgetful functor G.

Theorem 21.4. *The adjunction $F \dashv G$ is monadic: $R\text{-}\mathbf{Mod} \simeq \mathbf{Set}^T$ where $T = GF$ is the free R-module monad.*

Proof. We verify Beck's conditions.

G reflects isomorphisms	*A map of R-modules is an iso iff its map is a bijection; G is the forgetfu*
G creates coequalizers of G-split pairs	*Given a G-split pair $f, g : M \rightrightarrows N$ i set-level split coequalizer has a uniq structure making it the coequalizer i*
G preserves these coequalizers	*Since the coequalizer is split at the s preserved by every functor, includin*
Both Beck conditions satisfied; apply the theorem	*$K : R\text{-}\mathbf{Mod} \to \mathbf{Set}^T$ is an equivalen*

∎

This result generalizes: for any algebraic theory (a set of operations and equations), the category of models is equivalent to $\mathbf{Set}^T$ for the corresponding monad.

21.5.2 Algebraic Theories as Monads

A **Lawvere theory** or **algebraic theory** is a category $\mathbf{T}$ with finite products, whose objects are finite powers $n = 1^n$ of a distinguished object 1. A model of $\mathbf{T}$ in **Set** is a product-preserving functor $M : \mathbf{T} \to \mathbf{Set}$.

Main fact: the category of models $\mathrm{Mod}(\mathbf{T}, \mathbf{Set})$ is equivalent to the Eilenberg–Moore category $\mathbf{Set}^T$ for a naturally constructed monad T on **Set**. The monad $T(X)$ is the set of $\mathbf{T}$-terms in variables X.

This connection — Beck's theorem applied to Lawvere theories — is one of the deepest results in categorical algebra: *monads on* **Set** *are exactly algebraic theories* (in a precise sense, with appropriate finiteness conditions).

21.5.3 Crude Monadicity

In practice, one often uses a simpler sufficient criterion:

Theorem 21.5 (Crude Monadicity / Barr–Beck). *$F \dashv G$ is monadic if F reflects isomorphisms and $\mathcal{D}$ has and G preserves coequalizers of all* reflexive pairs *(pairs (f, g) that admit a common right inverse).*

This handles many algebraic cases without having to identify F-split pairs explicitly.

21.6 Exercises

What is the multiplication μ of the monad $T = GF$ arising from an adjunction $F \dashv G$?	$\mu = G\varepsilon F : GFGF \Rightarrow GF$, where ε is the counit of the adjunction. At component c: $\mu_c = G(\varepsilon_{Fc})$.

What are the two canonical adjunctions that resolve a monad T?	The Kleisli adjunction $F_T \dashv G_T$ with $\mathcal{C}_T$ (minimal, initial among all resolutions), and the Eilenberg–Moore adjunction $F^T \dashv G^T$ with $\mathcal{C}^T$ (maximal, terminal among resolutions).
What is a T-algebra for a monad (T, η, μ)?	A pair $(a, h : Ta \to a)$ where $h \circ \eta_a = \mathrm{id}_a$ (unit law) and $h \circ \mu_a = h \circ Th$ (associativity law). The map h is the "structure map" or "action."
What is the comparison functor K?	Given an adjunction $F \dashv G$ inducing $T = GF$, the functor $K : \mathcal{D} \to \mathcal{C}^T$ sending $d \mapsto (Gd, G\varepsilon_d)$. It commutes with the forgetful functors over $\mathcal{C}$.
State Beck's monadicity theorem.	K is an equivalence iff: (B1) F reflects isomorphisms, and (B2) $\mathcal{D}$ has coequalizers of F-split pairs and F preserves them.
What is a split coequalizer?	A diagram $a \rightrightarrows b \to c$ with extra maps $s : c \to b$ and $t : b \to a$ satisfying $qs = \mathrm{id}$, $ft = \mathrm{id}$, $gt = sq$. These equations make c the coequalizer of (f, g) without requiring any universal property argument.
Why are split coequalizers preserved by every functor?	Because the splitting is expressed purely by equations between morphisms, and every functor preserves equations. No colimit universal property needs to be checked.
What does "F-split pair" mean?	A pair $(f, g) : a \rightrightarrows b$ in $\mathcal{D}$ such that (Ff, Fg) has a split coequalizer in $\mathcal{C}$. The split structure may not lift to $\mathcal{D}$; it only needs to exist downstairs.
Why is the free-module adjunction monadic?	The forgetful functor $G : R\text{-}\mathbf{Mod} \to \mathbf{Set}$ reflects isomorphisms (module isos are exactly underlying-set bijections) and creates coequalizers of G-split pairs (the coequalizer set has a unique R-module structure).
What is a morphism in the Kleisli category $\mathcal{C}_T$?	A morphism $a \to b$ in $\mathcal{C}_T$ is a morphism $a \to Tb$ in $\mathcal{C}$. Kleisli composition of $f : a \to Tb$ and $g : b \to Tc$ is $\mu_c \circ Tg \circ f : a \to Tc$.
What is the identity morphism on a in $\mathcal{C}_T$?	The unit $\eta_a : a \to Ta$.
What does it mean for a functor to reflect isomorphisms?	F reflects isomorphisms if: whenever Ff is an isomorphism in the codomain, f was already an isomorphism in the domain.
Give an example of an adjunction that is *not* monadic.	The inclusion $i : \mathbf{Ab} \hookrightarrow \mathbf{Grp}$ with left adjoint the abelianization functor $(-)/[G, G]$ is not monadic: the comparison functor is not full.
What is the Crude Monadicity Theorem?	$F \dashv G$ is monadic if F reflects isomorphisms and the domain has and G preserves coequalizers of reflexive pairs.
What is a reflexive pair?	A pair of morphisms $f, g : a \rightrightarrows b$ that share a common right inverse: there exists $s : b \to a$ with $fs = \mathrm{id}_b = gs$.
Why does the Eilenberg–Moore category earn the name "maximal resolution"?	Every adjunction $F' \dashv G'$ resolving T factors through $\mathcal{C}^T$ via the comparison functor K: $\mathcal{D} \xrightarrow{K} \mathcal{C}^T$. So $\mathcal{C}^T$ is final among all such factorizations.
Why does the Kleisli category earn the name "minimal resolution"?	Every adjunction resolving T receives a functor from $\mathcal{C}_T$: there is a canonical $J : \mathcal{C}_T \to \mathcal{D}$ for any resolution, making $\mathcal{C}_T$ initial.
What is the free T-algebra on a?	The pair (Ta, μ_a), where the structure map is $\mu_a : T^2a \to Ta$. The associativity and unit laws for (Ta, μ_a) are exactly the monad axioms for (T, η, μ).
What pair of morphisms does every T-algebra (a, h) present as a coequalizer?	The pair $\mu_a, Th : T^2a \rightrightarrows Ta$. The coequalizer is a with map $h : Ta \to a$. This pair is always split by η_{Ta} and η_a.

In functional programming, what corresponds to the Kleisli composition?	The "bind" operation (>>= in Haskell). If $f : a \to M\,b$ and $g : b \to M\,c$, then $g \star f =$ 'a »= f »= g'. The monad laws for >>= are exactly the Kleisli category axioms.

Chapter Summary

The two canonical resolutions. Every monad (T, η, μ) on $\mathcal{C}$ arises from an adjunction. The minimal such adjunction lives in the Kleisli category $\mathcal{C}_T$ (objects of $\mathcal{C}$, morphisms are T-decorated maps); the maximal one lives in the Eilenberg–Moore category $\mathcal{C}^T$ (T-algebras). Every other resolution factors between these two extremes.
The comparison functor. Any adjunction $F \dashv G$ inducing T gives a comparison functor $K : \mathcal{D} \to \mathcal{C}^T$, $d \mapsto (Gd, G\varepsilon_d)$. The question "is this adjunction monadic?" is the question "is K an equivalence?"
Beck's theorem. K is an equivalence iff F reflects isomorphisms and $\mathcal{D}$ has, with F preserving, coequalizers of F-split pairs. The key notion is an F-split pair: one whose image under F has a *split* (absolute) coequalizer.
Algebraic significance. The Eilenberg–Moore category of the free-algebra monad over **Set** is exactly the category of algebras for the corresponding algebraic theory. This makes Beck's theorem the precise categorical statement that monads on **Set** are the same as equational algebraic theories.

21.7 Formal Restatement

Formal Restatement

Monad. A monad on $\mathcal{C}$ is a monoid object in the strict monoidal category $(\mathrm{End}(\mathcal{C}), \circ, \mathrm{Id})$ of endofunctors: a triple $(T, \eta : \mathrm{Id} \Rightarrow T, \mu : T^2 \Rightarrow T)$ with $\mu \circ \mu T = \mu \circ T\mu$, $\mu \circ \eta T = \mathrm{id}_T = \mu \circ T\eta$.
Eilenberg–Moore category. $\mathcal{C}^T$ is the category of T-algebras $(a, h : Ta \to a)$ satisfying the two coherence laws, with morphisms the T-equivariant maps. The forgetful $G^T : \mathcal{C}^T \to \mathcal{C}$ and free $F^T : \mathcal{C} \to \mathcal{C}^T$ $(a \mapsto (Ta, \mu_a))$ form an adjunction inducing T.
Kleisli category. $\mathcal{C}_T$ has the same objects as $\mathcal{C}$ and $\mathrm{Hom}_{\mathcal{C}_T}(a, b) = \mathrm{Hom}_{\mathcal{C}}(a, Tb)$, with Kleisli composition $g \star f = \mu \circ Tg \circ f$ and identity η_a. The inclusion $F_T : \mathcal{C} \to \mathcal{C}_T$ and resolution $G_T : \mathcal{C}_T \to \mathcal{C}$ form an adjunction inducing T.
Comparison functor. For any adjunction $F \dashv G$ with induced monad T, the comparison functor $K : \mathcal{D} \to \mathcal{C}^T$ is $Kd = (Gd, G\varepsilon_d)$, $Kf = Gf$. It satisfies $G^T \circ K = G$.
Split coequalizer. A diagram $a \overset{f}{\underset{g}{\rightrightarrows}} b \xrightarrow{q} c$ with $s : c \to b$ and $t : b \to a$ satisfying $qs = \mathrm{id}_c$, $ft = \mathrm{id}_b$, $gt = sq$. Every functor preserves split coequalizers.
F-split pair. A pair $(f, g) : a \rightrightarrows b$ in $\mathcal{D}$ such that (Ff, Fg) admits a split coequalizer in $\mathcal{C}$.
Beck's Monadicity Theorem. K is an equivalence of categories iff (B1) F reflects isomorphisms; and (B2) $\mathcal{D}$ has coequalizers of all F-split pairs, and F preserves them.
Crude monadicity. Sufficient: (B1) holds and $\mathcal{D}$ has, with G preserving, coequalizers of all reflexive pairs in $\mathcal{D}$.

Chapter 22

Natural Transformations as 2-Morphisms

Chapter Overview

Chapter 4 introduced natural transformations and showed that they organize functors into a more structured whole. But we only used one operation on natural transformations: vertical composition. There are actually three canonical ways to compose natural transformations — **vertical**, **horizontal**, and **whiskering** — and they interact through a single remarkable law: the **interchange law**.

The structure formed by categories, functors, and natural transformations, with all three compositions, is called the **2-category Cat**. Understanding **Cat** as a 2-category is not just organizational: it reveals that the three compositions are not ad hoc, but arise from the fact that **Cat** has *two levels of morphism*, each with its own composition. Natural transformations are the **2-morphisms**.

The interchange law is the key coherence axiom: it says that composing in the horizontal direction and then vertically gives the same result as composing vertically and then horizontally. Without this, the two compositions would conflict; with it, they mesh into a unified structure.

This chapter prepares the ground for enriched category theory (Volume III) by making explicit the 2-categorical structure that has been implicit throughout the book.

Where the Name Comes From

2-category was introduced by *Jean Bénabou* (1963) and developed by Kelly, Street, and many others. The terminology comes from the dimensional count: a 2-category has objects (0-cells), 1-morphisms (1-cells), and 2-morphisms (2-cells), with each "cell" a higher-dimensional generalization of an arrow. The generalization continues: n-categories have k-cells for $0 \leq k \leq n$; ∞-categories (Vol IV) have cells at every level.

Whiskering (composing a natural transformation with a functor on one side) is named for the visual picture: you imagine the natural transformation as a "hair" or "whisker" attached to the side, then "brushing it" along the functor's input or output. The whisker metaphor is Mac Lane's; the term is universal in 2-categorical literature.

Interchange law expresses the compatibility of vertical and horizontal composition of 2-cells; it is the central coherence axiom of a 2-category. The terminology is from algebra (interchange of operations) and dates to the early 1960s.

22.1 The 2-Category Cat

22.1.1 Objects, 1-Morphisms, 2-Morphisms

The structure **Cat** has three levels:

- **Objects (0-cells):** small categories $\mathcal{A}, \mathcal{B}, \mathcal{C}, \ldots$
- **1-morphisms (1-cells):** functors $F, G : \mathcal{A} \to \mathcal{B}$.
- **2-morphisms (2-cells):** natural transformations $\alpha : F \Rightarrow G$ between parallel functors.

A **2-category** requires that between any two 0-cells $\mathcal{A}$ and $\mathcal{B}$, the collection of 1-cells forms a *category* — a category $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ whose objects are functors $\mathcal{A} \to \mathcal{B}$ and whose morphisms are natural transformations. The composition in this hom-category is called **vertical composition**.

Additionally, the 1-cell composition (functor composition) must interact with the 2-cells (natural transformations) coherently. This interaction is captured by **horizontal composition** and **whiskering**.

How to Say This

- "$\alpha : F \Rightarrow G$" reads "α is a 2-cell from F to G," or "α goes between F and G."
- We draw 2-cells as double arrows: $\Rightarrow$.

- The hom-category $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ is also written $[\mathcal{A}, \mathcal{B}]$ or $\mathcal{B}^{\mathcal{A}}$.

22.1.2 Globular Diagrams

It helps to draw the three cell levels as nested diagrams:

- 0-cells (categories) are points: $\bullet$.
- 1-cells (functors) are arrows between points: $\bullet \to \bullet$.
- 2-cells (natural transformations) are regions bounded by two parallel arrows:

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \end{tikzcd}$$

This picture makes it visual that natural transformations "fill in" the region between two parallel functors.

22.2 Vertical Composition

22.2.1 Definition

Given natural transformations $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$ (both $\mathcal{A} \to \mathcal{B}$), their **vertical composite** $\beta \cdot \alpha : F \Rightarrow H$ is defined componentwise:

$$(\beta \cdot \alpha)_c = \beta_c \circ \alpha_c : Fc \to Gc \to Hc \qquad \text{for each } c \in \mathcal{A}.$$

Internal Dialog

Vertical composition stacks natural transformations on top of each other: two consecutive transformations between the same pair of categories collapse into one. The name "vertical" comes from the globular picture: you are composing within a single column (between $\mathcal{A}$ and $\mathcal{B}$), stacking transformation regions vertically.

The naturality of $\beta \cdot \alpha$ is immediate: for any $f : c \to c'$,

$$H(f) \circ (\beta \cdot \alpha)_c = H(f) \circ \beta_c \circ \alpha_c = \beta_{c'} \circ G(f) \circ \alpha_c = \beta_{c'} \circ \alpha_{c'} \circ F(f) = (\beta \cdot \alpha)_{c'} \circ F(f).$$

The identity for vertical composition is the **identity natural transformation** $\mathrm{id}_F : F \Rightarrow F$, with $(\mathrm{id}_F)_c = \mathrm{id}_{Fc}$.

Vertical composition is associative (since composition in $\mathcal{B}$ is), so $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ is indeed a category.

22.2.2 The Vertical Composition Diagram

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left=60, "F"] \arrow[r, "G" description] \arrow[r, bend right=60, "H"'] \arrow[r, phantom, shift left=2, "\Downarrow\alpha"] \arrow[r, phantom, shift right=2, "\Downarrow\beta"] & \mathcal{B} \end{tikzcd} \quad = \quad \begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\beta\cdot\alpha"] & \mathcal{B} \end{tikzcd}$$

22.3 Horizontal Composition

22.3.1 The Setup

Vertical composition works within a single hom-category. Horizontal composition works *across* hom-categories. Suppose we have:

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C} \end{tikzcd}$$

Here $\alpha : F \Rightarrow G : \mathcal{A} \to \mathcal{B}$ and $\beta : H \Rightarrow K : \mathcal{B} \to \mathcal{C}$ are "side by side." Their horizontal composite is a natural transformation $\beta * \alpha : H \circ F \Rightarrow K \circ G : \mathcal{A} \to \mathcal{C}$.

22.3.2 The Formula (Two Versions)

At a component $a \in \mathcal{A}$, the horizontal composite $(\beta * \alpha)_a : HFa \to KGa$ is defined by either of two equal formulas:

$$\begin{aligned} (\beta * \alpha)_a &= K(\alpha_a) \circ \beta_{Fa} && \text{(apply } \beta \text{ first, then } K \text{ on } \alpha) \\ &= \beta_{Ga} \circ H(\alpha_a) && \text{(apply } H \text{ on } \alpha \text{ first, then } \beta). \end{aligned}$$

Theorem 22.1. *The two formulas agree:* $K(\alpha_a) \circ \beta_{Fa} = \beta_{Ga} \circ H(\alpha_a)$.

Proof.

Consider the morphism $\alpha_a : Fa \to Ga$ in $\mathcal{B}$	*This is the a-component of* $\alpha : F \Rightarrow G$
$\beta : H \Rightarrow K$ is a natural transformation, so it is natural at the morphism α_a	*Naturality of* β *at* $\alpha_a : Fa \to Ga$ *gives:* $K(\alpha_a) \circ \beta_{Fa} = \beta_{Ga} \circ H(\alpha_a)$
This is exactly the required equation	*The two formulas for* $(\beta * \alpha)_a$ *are equal by the naturality square of* β *at* α_a

□

The naturality of $\beta * \alpha$ (as a transformation $HF \Rightarrow KG$) follows from the naturalities of α and β and functoriality of H, K.

22.3.3 Horizontal Composition is Functorial

The assignment $(\alpha, \beta) \mapsto \beta * \alpha$ is the action on morphisms of a functor

$$* : \mathbf{Cat}(\mathcal{A}, \mathcal{B}) \times \mathbf{Cat}(\mathcal{B}, \mathcal{C}) \;\to\; \mathbf{Cat}(\mathcal{A}, \mathcal{C}).$$

This is the *composition functor* for the 2-category **Cat**. Preservation of identity transformations: $\mathrm{id}_H * \mathrm{id}_F = \mathrm{id}_{HF}$, since $(\mathrm{id}_H * \mathrm{id}_F)_a = \mathrm{id}_{HFa}$.

22.4 Whiskering

22.4.1 Whiskering as a Special Case of Horizontal Composition

Whiskering is the horizontal composition of a natural transformation with an identity natural transformation, or equivalently, applying a functor to a natural transformation.

Right-whiskering. Given $\alpha : F \Rightarrow G$ (functors $\mathcal{A} \to \mathcal{B}$) and $H : \mathcal{C} \to \mathcal{A}$, define

$$\alpha H : F \circ H \Rightarrow G \circ H \qquad (\alpha H)_c = \alpha_{Hc} \text{ for } c \in \mathcal{C}.$$

Right-whiskering by H precomposes with H.

Left-whiskering. Given $\alpha : F \Rightarrow G$ (functors $\mathcal{A} \to \mathcal{B}$) and $K : \mathcal{B} \to \mathcal{C}$, define

$$K\alpha : K \circ F \Rightarrow K \circ G \qquad (K\alpha)_a = K(\alpha_a) \text{ for } a \in \mathcal{A}.$$

Left-whiskering by K postcomposes with K.

How to Say This

- "αH" reads "α right-whiskered by H" or "α at H" (the subscript H reminds us it acts as a right argument).
- "$K\alpha$" reads "α left-whiskered by K" or "K applied to α."
- In index notation: $(K\alpha)_a = K(\alpha_a)$, which looks just like applying the functor K to each component.

22.4.2 The Monad Axioms via Whiskering

Whiskering gives an elegant way to state the monad axioms. For a monad (T, η, μ) on $\mathcal{C}$:

$$\mu \circ \mu T = \mu \circ T\mu \qquad \text{(associativity: whiskering } \mu \text{ on both sides of } T)$$
$$\mu \circ \eta T = \mathrm{id}_T = \mu \circ T\eta \qquad \text{(unit laws: whiskering } \eta \text{ on both sides)}$$

Here $\mu T = \mu \cdot T$ means right-whiskering μ by T, and $T\mu$ means left-whiskering μ by T.

22.4.3 Adjunction Unit and Counit via Whiskering

The monad multiplication from an adjunction is $\mu = G\varepsilon F$: left-whisker the counit $\varepsilon : FG \Rightarrow \mathrm{Id}$ by G to get $G\varepsilon : GFG \Rightarrow G$, then right-whisker by F to get $G\varepsilon F : GFGF \Rightarrow GF$. The two triangle identities for an adjunction are:

$$\varepsilon F \cdot F\eta = \mathrm{id}_F \qquad \text{and} \qquad G\varepsilon \cdot \eta G = \mathrm{id}_G$$

— both expressed using whiskering.

22.5 The Interchange Law

22.5.1 Statement

The **interchange law** is the core compatibility axiom between vertical and horizontal composition. Suppose we have four natural transformations arranged in a 2×2 grid:

$$\begin{tikzcd}
\mathcal{A} \arrow[r, bend left=60, "F"] \arrow[r, "G" description] \arrow[r, bend right=60, "H"'] \arrow[r, phantom, shift left=2, "\Downarrow\alpha"] \arrow[r, phantom, shift right=2, "\Downarrow\beta"] & \mathcal{B} \arrow[r, bend left=60, "J"] \arrow[r, "K" description] \arrow[r, bend right=60, "L"'] \arrow[r, phantom, shift left=2, "\Downarrow\gamma"] \arrow[r, phantom, shift right=2, "\Downarrow\delta"] & \mathcal{C}
\end{tikzcd}$$

where $\alpha : F \Rightarrow G$, $\beta : G \Rightarrow H$, $\gamma : J \Rightarrow K$, $\delta : K \Rightarrow L$.

Theorem 22.2 (Interchange Law)**.**

$$(\delta \cdot \gamma) * (\beta \cdot \alpha) \;=\; (\delta * \beta) \cdot (\gamma * \alpha).$$

Reading this: "horizontal composite of vertical composites equals vertical composite of horizontal composites."

Proof.

Evaluate both sides at a component $a \in \mathcal{A}$	*We need:* $[(\delta \cdot \gamma) * (\beta \cdot \alpha)]_a = [(\delta * \beta) \cdot (\gamma * \alpha)]_a$
Left side using horizontal formula	$((\delta \cdot \gamma) * (\beta \cdot \alpha))_a = L(\beta \cdot \alpha)_a \circ (\delta \cdot \gamma)_{Fa} = L(\beta_a \circ \alpha_a) \circ \delta_{Fa} \circ \gamma_{Fa}$
Right side: $(\gamma * \alpha)_a = K\alpha_a \circ \gamma_{Fa}$ and $(\delta * \beta)_a = L\beta_a \circ \delta_{Ga}$	*So* $[(\delta * \beta) \cdot (\gamma * \alpha)]_a = (L\beta_a \circ \delta_{Ga}) \circ (K\alpha_a \circ \gamma_{Fa})$
Rewrite right side: $L\beta_a \circ \delta_{Ga} \circ K\alpha_a \circ \gamma_{Fa}$	*Insert* $\delta_{Ga} \circ K\alpha_a$*; naturality of* δ *at* $\alpha_a : Fa \to Ga$ *gives* $\delta_{Ga} \circ K\alpha_a = L\alpha_a \circ \delta_{Fa}$
Right side becomes $L\beta_a \circ L\alpha_a \circ \delta_{Fa} \circ \gamma_{Fa}$	*Using* $L\beta_a \circ L\alpha_a = L(\beta_a \circ \alpha_a)$ *(functoriality of* L*)*
Right side is $L(\beta_a \circ \alpha_a) \circ \delta_{Fa} \circ \gamma_{Fa}$	*This equals the left side; done*

□

22.5.2 Why the Interchange Law Matters

The interchange law is not just a technical identity — it is the *reason* that the three compositions are compatible. Without it, one could not freely rewrite a 2-categorical expression by changing the order of vertical and horizontal operations. The interchange law says:

> *It does not matter whether you compose vertically first and then horizontally, or horizontally first and then vertically.*

This is precisely the 2-dimensional analogue of the axiom that in a monoid, the binary product is well-defined.

Internal Dialog

Think of the 2×2 grid of natural transformations. You can "collapse" it into a single transformation either by:

1. First composing each row horizontally (getting two transformations), then composing those vertically.
2. First composing each column vertically (getting two transformations), then composing those horizontally.

The interchange law says you get the same answer either way.

22.6 Strict 2-Categories and Bicategories

22.6.1 Strict 2-Categories

A **strict 2-category** is a category enriched over the category **Cat** of small categories: it has a set of 0-cells, a hom-category between each pair of 0-cells, and composition functors satisfying associativity and unit laws *strictly* (as equations, not just up to isomorphism).

Cat itself (with small categories as objects) is a strict 2-category. The hom-category $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ is the functor category $[\mathcal{A}, \mathcal{B}]$; functor composition is strictly associative; the interchange law holds as a strict equality.

Other examples of strict 2-categories:

- **2Cat**: strict 2-categories, strict 2-functors, strict natural transformations.
- **MonCat**: monoidal categories, strong monoidal functors, monoidal natural transformations.
- R-**Bimod**: rings as objects, bimodules as 1-morphisms, bimodule maps as 2-morphisms.

22.6.2 Bicategories: Weakening the Axioms

In many naturally occurring examples, 1-cell composition is associative only up to coherent isomorphism, not strictly. A **bicategory** (or **weak 2-category**) relaxes this:

- Composition of 1-cells: $c \circ (b \circ a) \cong (c \circ b) \circ a$ via an **associator** 2-cell $\alpha_{a,b,c}$.

- Unit laws: $a \circ \mathrm{id} \cong a$ and $\mathrm{id} \circ a \cong a$ via **unitor** 2-cells.
- These 2-cell isomorphisms must satisfy coherence conditions (the pentagon axiom for associators, the triangle axiom for unitors).

Example: Bimodules as a Bicategory

Objects: rings. 1-morphisms: (R, S)-bimodules (R-S-bimodules as morphisms from R to S). 2-morphisms: bimodule maps. Composition: tensor product $M \otimes_S N$. This is a bicategory: the tensor product is associative only up to a canonical isomorphism, not strictly.

Example: Spans as a Bicategory

Objects: sets. 1-morphisms $A \to B$: spans, i.e., diagrams $A \leftarrow X \to B$. 2-morphisms: maps of spans. Composition: pullback. This is a bicategory because pullback is only associative up to isomorphism.

A key theorem (Mac Lane's coherence theorem, generalized) says: every bicategory is biequivalent to a strict 2-category. So, up to biequivalence, the weak and strict notions coincide. But in practice, the weak version is more natural for many examples.

22.7 Exercises

What are the three levels of **Cat** as a 2-category?	0-cells (objects): small categories; 1-cells: functors; 2-cells: natural transformations.
Define vertical composition of natural transformations $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$.	$(\beta \cdot \alpha)_c = \beta_c \circ \alpha_c$ for each c. It is the composition in the hom-category $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$.
Define horizontal composition of $\alpha : F \Rightarrow G : \mathcal{A} \to \mathcal{B}$ and $\beta : H \Rightarrow K : \mathcal{B} \to \mathcal{C}$.	$(\beta * \alpha)_a = K(\alpha_a) \circ \beta_{Fa} = \beta_{Ga} \circ H(\alpha_a)$. Both formulas are equal by naturality of β.
Why do the two formulas for horizontal composition agree?	Because β is natural at $\alpha_a : Fa \to Ga$: the naturality square for β at this morphism gives exactly $K(\alpha_a) \circ \beta_{Fa} = \beta_{Ga} \circ H(\alpha_a)$.
State the interchange law.	$(\delta \cdot \gamma) * (\beta \cdot \alpha) = (\delta * \beta) \cdot (\gamma * \alpha)$: for a 2×2 grid of composable natural transformations.
What is right-whiskering αH for $\alpha : F \Rightarrow G$ and $H : \mathcal{C} \to \mathcal{A}$?	$\alpha H : FH \Rightarrow GH$ with $(\alpha H)_c = \alpha_{Hc}$.
What is left-whiskering $K\alpha$ for $\alpha : F \Rightarrow G$ and $K : \mathcal{B} \to \mathcal{C}$?	$K\alpha : KF \Rightarrow KG$ with $(K\alpha)_a = K(\alpha_a)$.
How are whiskering and horizontal composition related?	Whiskering is horizontal composition with an identity transformation: $K\alpha = \mathrm{id}_K * \alpha$ and $\alpha H = \alpha * \mathrm{id}_H$.
Express the monad associativity axiom $\mu \circ \mu T = \mu \circ T\mu$ using whiskering notation.	$\mu \cdot (\mu \cdot T) = \mu \cdot (T \cdot \mu)$ where μT is μ right-whiskered by T (i.e., $(\mu T)_c = \mu_{Tc}$) and $T\mu$ is T left-whiskered with μ (i.e., $(T\mu)_c = T(\mu_c)$).
What are the triangle identities for $F \dashv G$ written using whiskering?	$\varepsilon F \cdot F\eta = \mathrm{id}_F$ and $G\varepsilon \cdot \eta G = \mathrm{id}_G$.
What is a strict 2-category?	A category enriched over **Cat**: 0-cells, hom-categories between pairs of 0-cells, and strictly associative and unital composition functors satisfying interchange.
What is a bicategory?	A weakened 2-category where 1-cell composition is associative only up to coherent associator 2-cells satisfying the pentagon axiom, and unital only up to coherent unitor 2-cells satisfying the triangle axiom.
Give an example of a bicategory that is not a strict 2-category.	Rings and bimodules: composition (tensor product) is only associative up to canonical isomorphism, not strictly.
What does the identity natural transformation at F look like?	$(\mathrm{id}_F)_c = \mathrm{id}_{Fc}$: the identity morphism at Fc for each c. It is the identity for vertical composition.

What is the horizontal composite of two identity transformations $\mathrm{id}_H * \mathrm{id}_F$?	id_{HF}: the identity transformation at the composite functor HF.
In a strict 2-category, how many independent axioms govern the interactions of 0-, 1-, and 2-cells?	The 1-cells form a category (associativity, units). The 2-cells between each pair of 1-cells form a category (associativity, units for vertical composition). Horizontal composition is functorial. And the interchange law holds. These account for all the axioms.
What does "**Cat** enriched over **Cat**" mean?	Each hom-set $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ is not just a set but a category (the functor category). The composition map is a functor between these categories, not just a function of sets.
State Mac Lane's coherence theorem for bicategories.	Every bicategory is biequivalent to a strict 2-category.
What is the vertical identity for the hom-category $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$?	The identity natural transformation id_F at each functor F.
For a 2-category, what axiom ensures that composing 2-cells horizontally and then vertically gives the same result as composing vertically and then horizontally?	The interchange law: $(\delta \cdot \gamma) * (\beta \cdot \alpha) = (\delta * \beta) \cdot (\gamma * \alpha)$.

Chapter Summary

Three compositions. Natural transformations support three compositions: *vertical* ($\beta \cdot \alpha$: stacking within one hom-category), *horizontal* ($\beta * \alpha$: placing side by side across composable hom-categories), and *whiskering* (a special case of horizontal composition involving an identity).
Horizontal composition formula. $(\beta * \alpha)_a = K(\alpha_a) \circ \beta_{Fa} = \beta_{Ga} \circ H(\alpha_a)$: the two formulas agree by naturality of β. This is a micro-proof of the interchange law in disguise.
The interchange law. $(\delta \cdot \gamma) * (\beta \cdot \alpha) = (\delta * \beta) \cdot (\gamma * \alpha)$: the law that reconciles horizontal and vertical composition. It is proved using naturality at a component and functoriality.
2-category Cat. The triple (categories, functors, natural transformations) with these compositions forms a strict 2-category. This is the prototype for all 2-categorical thinking.
Bicategories. The weak version of a 2-category, where 1-cell composition is only associative up to coherent isomorphism. Examples: bimodules (tensor product), spans (pullback). Every bicategory is biequivalent to a strict 2-category.

22.8 Formal Restatement

Formal Restatement

2-Category. A strict 2-category $\mathcal{K}$ consists of: a set of 0-cells; for each pair (A, B) of 0-cells, a hom-category $\mathcal{K}(A, B)$ (whose objects are 1-cells, morphisms are 2-cells, and composition is vertical); for each triple (A, B, C), a composition functor $\circ : \mathcal{K}(B, C) \times \mathcal{K}(A, B) \to \mathcal{K}(A, C)$ that is strictly associative and unital.
Vertical composition. The composition in the hom-category: if $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$ in $\mathcal{K}(A, B)$, then $\beta \cdot \alpha : f \Rightarrow h$.
Horizontal composition. The action of the composition functor on 2-cells: if $\alpha : f \Rightarrow g$ in $\mathcal{K}(A, B)$ and $\beta : h \Rightarrow k$ in $\mathcal{K}(B, C)$, then $\beta * \alpha : hf \Rightarrow kg$ in $\mathcal{K}(A, C)$.
Interchange law. Since $\circ$ is a functor: $(\delta \cdot \gamma) * (\beta \cdot \alpha) = (\delta * \beta) \cdot (\gamma * \alpha)$. This is functoriality of $\circ$ applied to the pair of composable pairs $((\gamma, \alpha), (\delta, \beta))$.
Whiskering. Special cases of horizontal composition: right-whiskering by a 1-cell $f : A \to B$ gives $(\alpha f)_x = \alpha_{fx}$; left-whiskering by $g : B \to C$ gives $(g\alpha)_x = g(\alpha_x)$.
Cat as a 2-category. Objects: small categories. Hom-categories $\mathbf{Cat}(\mathcal{A}, \mathcal{B}) = [\mathcal{A}, \mathcal{B}]$: functor categories. Composition: functor composition (on objects) and horizontal composition of natural transformations (on morphisms in the hom-categories). Strictly associative.
Bicategory. As above but with associator isomorphisms $a_{f,g,h} : (fg)h \overset{\cong}{\Longrightarrow} f(gh)$ and unitor isomorphisms, satisfying the pentagon and triangle coherence conditions.

Chapter 23

Adjoint Functor Theorems: Complete Treatment

Chapter Overview

Chapter 10 presented the Adjoint Functor Theorems as culmination of Volume I. This chapter revisits them from the perspective of Volume II, with complete proofs and deeper applications. The question — when does a functor $F : \mathcal{C} \to \mathcal{D}$ have a left adjoint? — is one of the most important existence questions in category theory.

The two main answers are:

- **General Adjoint Functor Theorem (GAFT)**: F has a left adjoint iff it preserves all small limits (is **continuous**) and every comma category $(d \downarrow F)$ has a **weakly initial set** (the solution set condition).
- **Special Adjoint Functor Theorem (SAFT)**: if additionally $\mathcal{C}$ is complete and has a small **cogenerating set**, then continuity of F alone suffices.

Both theorems reduce the existence question to two ingredients: a *limit* argument (build a candidate from the solution set by taking a limit) and a *smallness* argument (ensure the limit is over a small diagram). The SAFT uses the cogenerating set to automatically produce a small solution set for any continuous functor.

Applications include free algebras, sheafification, and the Stone–Čech compactification: each is a left adjoint whose existence is guaranteed by the SAFT.

23.1 The Adjoint Functor Question

23.1.1 Recap: Adjunctions via Universal Arrows

Recall from Chapter 7: a functor $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint $L : \mathcal{D} \to \mathcal{C}$ if and only if, for every $d \in \mathcal{D}$, there exists a **universal arrow** from d to F — a pair $(Ld, \eta_d : d \to F(Ld))$ such that every $f : d \to F(c)$ factors uniquely as $f = F(\bar{f}) \circ \eta_d$ for a unique $\bar{f} : Ld \to c$.

This universal arrow is exactly the initial object of the **comma category** $(d \downarrow F)$:

- Objects: pairs $(c \in \mathcal{C},\ f : d \to Fc)$.
- Morphisms $(c, f) \to (c', f')$: maps $g : c \to c'$ with $Fg \circ f = f'$.

So the existence of a left adjoint reduces to: does every comma category $(d \downarrow F)$ have an initial object?

23.1.2 Two Ingredients for an Initial Object

An initial object in a category can be constructed as a limit over a *weakly initial set*:

Definition 23.1 (Weakly initial set)**.** A set $\mathcal{S} \subseteq \mathrm{Ob}(\mathcal{K})$ is called **weakly initial** if every object $k \in \mathcal{K}$ admits a morphism from some $s \in \mathcal{S}$.

Lemma 23.2 (Weakly initial set $\Rightarrow$ initial object)**.** *If $\mathcal{K}$ is a locally small category that has small limits and $\mathcal{S}$ is a weakly initial set, then $\mathcal{K}$ has an initial object.*

Proof.

Let $i = \prod_{s\in\mathcal{S}} s$ be the product of all objects in $\mathcal{S}$ (a small product)	*Exists because $\mathcal{S}$ is a set and $\mathcal{K}$ has small limits*
For each k, there is some $s \in \mathcal{S}$ and a map $s \to k$; composing with the projection $i \to s$ gives a map $i \to k$	*So i is a weakly initial object: every k has at least one map from i*
Not all maps $i \to k$ need be equal; take the equalizer of all maps $i \to k$ for all k	*For each k, let $e_k : \mathrm{eq}(i \rightrightarrows k) \to i$ be the equalizer of the (possibly large) family of all maps $i \to k$*
Since $\mathcal{K}$ is locally small, the hom-set $\mathcal{K}(i, k)$ is a set, so its equalizer is a small limit	*e_k equalizes all $f : i \to k$ at once*
The limit of the family $\{e_k\}_{k\in\mathcal{K}}$: let $\ell = \lim_k \mathrm{eq}(i \rightrightarrows k)$	*Another small (in fact set-indexed) limit*
ℓ is initial: it maps to i, hence to every k; and the unique map $\ell \to k$ is forced by the equalizer at k	*One verifies that any two maps $\ell \to k$ must be equal by unpacking the equalizer and limit construction*

□

23.2 The General Adjoint Functor Theorem

23.2.1 The Solution Set Condition

Definition 23.3 (Solution set condition)**.** A functor $F : \mathcal{C} \to \mathcal{D}$ satisfies the **solution set condition** if for every $d \in \mathcal{D}$, there exists a *set* $\{(c_i, f_i : d \to Fc_i)\}_{i\in I}$ of objects of $(d \downarrow F)$ such that every $(c, f : d \to Fc)$ in $(d \downarrow F)$ admits a morphism from some (c_i, f_i). That is, $\{(c_i, f_i)\}_{i\in I}$ is a weakly initial set in $(d \downarrow F)$.

Informally: there is a *set* of "generators" for all maps from d into F's image. The construction Ld will be a limit over this set.

23.2.2 The Theorem

Theorem 23.4 (General Adjoint Functor Theorem)**.** *Let $\mathcal{C}$ be locally small and complete (has all small limits), and let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Then F has a left adjoint if and only if:*

(G1) F is ***continuous****: it preserves all small limits.*

(G2) F satisfies the solution set condition.

Proof. We prove the direction (G1)+(G2) $\Rightarrow \exists L$. The converse — that any left adjoint is continuous and satisfies the solution set condition — is standard.

Fix $d \in \mathcal{D}$; we need an initial object of $(d \downarrow F)$	*$(d \downarrow F)$ is locally small (since $\mathcal{C}$ is) and has small limits by (G1): F continuous implies $(d \downarrow F)$ has small limits*
Why $(d \downarrow F)$ has small limits when F is continuous	*A limit $\lim_j (c_j, f_j)$ in $(d \downarrow F)$ has underlying object $\lim_j c_j$ in $\mathcal{C}$. The map $d \to F(\lim_j c_j)$ is given by $d \xrightarrow{f_j} Fc_j$ factoring through $F(\lim_j c_j) \cong \lim_j Fc_j$ (using F continuous)*
By (G2), $(d \downarrow F)$ has a weakly initial set $\{(c_i, f_i)\}_{i\in I}$	*This is exactly what the solution set condition provides*
Apply Lemma 23.2 to $(d \downarrow F)$: it has small limits and a weakly initial set	*Therefore $(d \downarrow F)$ has an initial object $(Ld, \eta_d : d \to FLd)$*
Do this for every d; the assignment $d \mapsto Ld$ extends to a functor $L : \mathcal{D} \to \mathcal{C}$	*The universal property of the initial object gives the action of L on morphisms*
$\eta : \mathrm{Id}_{\mathcal{D}} \Rightarrow FL$ is the unit; the adjunction $L \dashv F$ holds by construction	*Each $\eta_d : d \to FLd$ is the initial object; universality gives the adjunction bijection $\mathcal{C}(Ld, c) \cong \mathcal{D}(d, Fc)$*

□

Internal Dialog

The proof has a clean two-step structure. Step one: use F-continuity to give $(d \downarrow F)$ limits. Step two: use the solution set to give it a weakly initial set. Then the lemma combines these to produce an initial object, which is the adjoint value Ld. The solution set condition is the *smallness* valve — without it, the product $\prod_{s\in\mathcal{S}} s$ might be over a proper class, which is not permissible.

23.2.3 The Converse

If F has a left adjoint L, then:

- F is continuous: right adjoints preserve all limits.
- Solution set condition: for each d, take the singleton set $\{(Ld, \eta_d)\}$. Every $(c, f : d \to Fc)$ factors through (Ld, η_d) by the universal property: this is the definition of the adjunction bijection.

So (G1) and (G2) are not just sufficient but necessary.

Where the Name Comes From

The **Special Adjoint Functor Theorem** (SAFT) is the version applicable when the source category is locally small, complete, and admits a **cogenerating set**. **Cogenerator**: an object (or set of objects) S such that any two distinct morphisms $f \neq g : A \to B$ are distinguished by a morphism into S — dually to a generating set. The terminology is from algebra (generators of a module/group); the "co-" marks the dual direction (Vol I Ch. 2 opposite categories).
Stone–Čech compactification is named for *Marshall Stone* (1903–1989) and *Eduard Čech* (1893–1960), who independently constructed the universal compactification of a Tychonoff space in the 1930s. It is a left adjoint to the forgetful **CompHaus** $\hookrightarrow$ **Tych** and one of the classical SAFT applications.

23.3 The Special Adjoint Functor Theorem

23.3.1 Cogenerating Sets

Definition 23.5 (Cogenerating set)**.** A set $\mathcal{G} \subseteq \mathrm{Ob}(\mathcal{C})$ is a **cogenerating set** (or **coseparating set**) for $\mathcal{C}$ if: for any two distinct morphisms $f \neq g : a \rightrightarrows b$, there exists $s \in \mathcal{G}$ and $h : b \to s$ such that $hf \neq hg$.

Equivalently: the collection of functors $\{\mathrm{Hom}(-, s) : s \in \mathcal{G}\}$ is jointly faithful.

Example: Cogenerating sets in Set

The single set $\{0, 1\}$ cogenerates **Set**: two functions $f, g : A \to B$ are equal iff for every $b \in B$, the preimages $f^{-1}(b)$ and $g^{-1}(b)$ agree, which is detected by maps $B \to \{0, 1\}$.

Example: Cogenerating sets in Ab

The group $\mathbb{Q}/\mathbb{Z}$ cogenerates **Ab**: this is the content of Baer's criterion for injectivity and the fact that $\mathbb{Q}/\mathbb{Z}$ is an injective cogenerator.

23.3.2 The Theorem

Theorem 23.6 (Special Adjoint Functor Theorem)**.** *Let $\mathcal{C}$ be locally small, complete, and have a small cogenerating set $\mathcal{G}$. Then every continuous functor $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint.*

Proof sketch. By GAFT, it suffices to produce a solution set for each d.

Fix $d \in \mathcal{D}$; we must find a set of objects $(c_i, f_i : d \to Fc_i)$ weakly generating $(d \downarrow F)$	*Strategy: bound the "size" of c using the cogenerating set $\mathcal{G}$*
For any $(c, f : d \to Fc)$ in $(d \downarrow F)$, consider the maps from c into elements of $\mathcal{G}$	*Since $\mathcal{G}$ cogenerates, c embeds into the product $\prod_{s \in \mathcal{G},\, h : c \to s} s$*
The cardinality of the indexing set for this product is bounded by $\lvert\mathcal{G}\rvert \times \lvert\mathcal{D}(d, Fs)\rvert$ for $s \in \mathcal{G}$	*All these hom-sets are sets (locally small) and $\mathcal{G}$ is a set*
Let S be the set of all isomorphism classes of objects c that embed into products of elements of $\mathcal{G}$ with index set of this bounded cardinality	*S is a set, as it is bounded in cardinality*
Claim: the set $\{(c, f : d \to Fc) : c \in S\}$ is a weakly initial set in $(d \downarrow F)$	*For any $(c', f' : d \to Fc')$, the object c' embeds into some product from $\mathcal{G}$; the image of f' under this embedding gives an object in S that maps to c'*
We have a set-indexed weakly initial set; apply GAFT	*GAFT gives the initial object Ld and the left adjoint L*

□

Internal Dialog

The cogenerating set acts as a "measuring stick" for objects of $\mathcal{C}$. It turns a potentially proper-class-sized collection of objects into a set-sized one: every object, up to its maps into $\mathcal{G}$, is represented by a bounded-size product. This is the key move that eliminates the solution set condition from the hypotheses.

23.4 Applications

23.4.1 Free Algebras

Example: The Free Group Functor

The forgetful functor $G : \mathbf{Grp} \to \mathbf{Set}$ has a left adjoint, the **free group functor** $F : \mathbf{Set} \to \mathbf{Grp}$.
Via GAFT: G is continuous (it creates limits: the underlying set of a limit of groups is the limit of the underlying sets). The solution set for a set X is: all groups with cardinality at most $\aleph_0 \cdot |X|$ (a set, bounded by the number of words in the generators). So GAFT applies and F exists.
Via SAFT: $\mathbf{Grp}$ is complete, locally small, and the single group $\mathbb{Z}/2\mathbb{Z}$ together with $\mathbb{Z}$ forms a cogenerating set (or one can use the family of all finite groups plus $\mathbb{Z}$). So SAFT gives the free group functor directly.

Example: Free Algebras for Any Algebraic Theory

Any algebraic theory $\mathbf{T}$ (in the sense of a Lawvere theory or a signature with equations) gives a forgetful functor $G : \mathrm{Alg}(\mathbf{T}) \to \mathbf{Set}$ that is continuous and whose domain is complete, locally small, and has a cogenerating set. By SAFT, the free algebra functor $F : \mathbf{Set} \to \mathrm{Alg}(\mathbf{T})$ always exists.

23.4.2 Sheafification

Let X be a topological space and $\mathbf{PSh}(X)$ the category of presheaves on X. The full subcategory $\mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ of sheaves has an inclusion functor ι.

Theorem 23.7. *The inclusion $\iota : \mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ has a left adjoint, the* ***sheafification functor*** $\mathrm{sh} : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$.

Sketch via GAFT.

ι preserves limits: limits of sheaves are sheaves, and ι reflects limits by definition	*Verified by checking the sheaf condition passes to limits*
Solution set for a presheaf P: let S be the set of all sheaves $\mathcal{F}$ together with a map $P \to \iota\mathcal{F}$, up to isomorphism, with $\lvert\mathcal{F}(U)\rvert$ bounded by $\lvert P(U)\rvert$ for all U	*This bounds are achievable since sheafification doesn't increase sections beyond the presheaf*
By GAFT, the initial object $\mathrm{sh}(P)$ exists; it is the sheafification	*The sheaf $\mathrm{sh}(P)$ with the unit map $P \to \iota\,\mathrm{sh}(P)$ is universal among all sheaf maps from P*

□

The sheafification functor is one of the most important examples of a left adjoint in algebraic topology and algebraic geometry. It is the reason why the category of sheaves is reflective in the category of presheaves.

23.4.3 Stone–Čech Compactification

Let **CompHaus** be the category of compact Hausdorff spaces and continuous maps, and $i : \mathbf{CompHaus} \hookrightarrow \mathbf{Top}$ the inclusion into all topological spaces.

Theorem 23.8. *The functor i has a left adjoint $\beta : \mathbf{Top} \to \mathbf{CompHaus}$, the* ***Stone–Čech compactification****.*

Sketch via SAFT.

CompHaus is complete, locally small	*Limits of compact Hausdorff spaces are compact Hausdorff*
CompHaus has a cogenerating set: $\{[0,1]\}$ alone cogenerates	*By Urysohn's theorem, continuous functions $X \to [0,1]$ separate points of X; Gelfand duality confirms $[0,1]$ cogenerates*
The inclusion i is continuous: limits of compact Hausdorff spaces computed in **Top** remain compact Hausdorff	*Standard compactness arguments*
SAFT applies: continuous functor out of a complete, locally small category with a cogenerating set has a left adjoint	*Therefore $\beta = Li$ exists*

□

The left adjoint βX is the Stone–Čech compactification: the "largest" compact Hausdorff space through which all maps $X \to K$ (for compact Hausdorff K) factor. It is a genuinely analytic construction, but category theory reveals its existence as a formal consequence of SAFT.

23.4.4 Geometric Morphisms and Toposes

In topos theory, the right adjoint ("inverse image") of a geometric morphism $f : \mathcal{E} \to \mathcal{F}$ always has a left adjoint ("direct image"). This follows from SAFT applied to the topos $\mathcal{E}$, which is complete, locally small, and has a cogenerating set (given by the subobject classifier Ω and the terminal object). The AFTs are thus fundamental to the theory of toposes and sheaves.

23.5 Exercises

State the General Adjoint Functor Theorem.	A functor $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint iff $\mathcal{C}$ is locally small and complete, F is continuous (preserves all small limits), and F satisfies the solution set condition (each $(d \downarrow F)$ has a weakly initial set).
What is the comma category $(d \downarrow F)$?	Objects: pairs $(c, f : d \to Fc)$ with $c \in \mathcal{C}$. Morphisms $(c, f) \to (c', f')$: maps $g : c \to c'$ with $Fg \circ f = f'$. The initial object of $(d \downarrow F)$, if it exists, is the value Ld of the left adjoint.
What is a weakly initial set?	A set $\mathcal{S}$ of objects in a category $\mathcal{K}$ such that every $k \in \mathcal{K}$ admits at least one morphism from some $s \in \mathcal{S}$. Not every such morphism needs to be unique.
How does a weakly initial set yield an initial object?	If $\mathcal{K}$ is locally small with small limits, take the product $i = \prod_{s \in \mathcal{S}} s$, then the limit over all maps from i (equalizing all parallel pairs $i \rightrightarrows k$) produces an initial object.
What is the solution set condition for $F : \mathcal{C} \to \mathcal{D}$?	For every $d \in \mathcal{D}$, there is a set $\{(c_i, f_i : d \to Fc_i)\}$ such that every $(c, f : d \to Fc)$ in $(d \downarrow F)$ admits a map from some (c_i, f_i).
If F has a left adjoint L, what is the solution set for d?	The singleton $\{(Ld, \eta_d : d \to FLd)\}$. Every map $f : d \to Fc$ factors uniquely through η_d, so this single pair is already weakly initial (in fact initial).
State the Special Adjoint Functor Theorem.	If $\mathcal{C}$ is locally small, complete, and has a small cogenerating set, then every continuous functor $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint.
What is a cogenerating set for $\mathcal{C}$?	A set $\mathcal{G} \subseteq \mathrm{Ob}(\mathcal{C})$ such that for any $f \neq g : a \rightrightarrows b$, some $h : b \to s$ with $s \in \mathcal{G}$ satisfies $hf \neq hg$. Equivalently, the functors $\mathrm{Hom}(-, s)$ for $s \in \mathcal{G}$ are jointly faithful.
How does the SAFT avoid the explicit solution set condition?	The cogenerating set provides a size bound: every object c "fits into" a product of elements of $\mathcal{G}$, and the cardinality of this product is bounded by the size of $\mathcal{G}$ and the hom-sets. This automatically constructs a solution set from the cogenerating set.
What cogenerates **Set**?	The two-element set $\{0, 1\}$ alone cogenerates **Set**: for any $f \neq g : A \to B$, some characteristic function $B \to \{0, 1\}$ distinguishes f and g.
What cogenerates **Ab**?	$\mathbb{Q}/\mathbb{Z}$ cogenerates **Ab**: it is an injective cogenerator. The Pontryagin duality $A \mapsto \mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z})$ is faithful.
Why does every continuous functor out of **Set** automatically satisfy the solution set condition?	Because **Set** has a cogenerating set ($\{0, 1\}$), so SAFT applies. But also: for a continuous functor $F : \mathbf{Set} \to \mathcal{D}$, one can take the solution set for d to be all (X, f) with $\lvert X \rvert$ bounded by $\lvert \mathcal{D}(d, F\{*\}) \rvert$ and similar cardinality bounds.
What makes sheafification a left adjoint?	The inclusion $\iota : \mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ is continuous (limits of sheaves are sheaves). The GAFT/SAFT then guarantees a left adjoint, which is sheafification.

What is the Stone–Čech compactification categorically?	The left adjoint to the inclusion $i : \mathbf{CompHaus} \hookrightarrow \mathbf{Top}$. Its existence follows from SAFT: **CompHaus** is complete, locally small, and cogerated by $[0, 1]$.
Why does $[0, 1]$ cogenerate **CompHaus**?	By Urysohn's lemma, for any compact Hausdorff space X and distinct points $x \neq y$, there is a continuous $f : X \to [0, 1]$ with $f(x) \neq f(y)$. So maps into $[0, 1]$ separate points, hence separate morphisms.
What is a reflective subcategory?	A full subcategory $\mathcal{D} \hookrightarrow \mathcal{C}$ is reflective if the inclusion has a left adjoint, called the **reflector**. Examples: sheaves in presheaves, compact Hausdorff spaces in topological spaces, abelian groups in groups.
Give the GAFT condition in the form "F has a left adjoint iff . . . "	$F : \mathcal{C} \to \mathcal{D}$ (with $\mathcal{C}$ locally small and complete) has a left adjoint iff F preserves all small limits and for each $d \in \mathcal{D}$, the category $(d \downarrow F)$ has a weakly initial set.
Does the free abelian group functor $\mathbf{Set} \to \mathbf{Ab}$ exist, and why?	Yes. The forgetful $G : \mathbf{Ab} \to \mathbf{Set}$ is continuous and **Ab** has a cogenerating set $(\mathbb{Q}/\mathbb{Z})$. By SAFT, G has a left adjoint — the free abelian group functor.
What is the key step in the proof of GAFT that uses local smallness?	Local smallness is needed to ensure that the hom-set $\mathcal{K}(i, k)$ is a set (not a proper class) for each k, so that the equalizer of all maps $i \to k$ is a small limit.
Why does continuity of F imply $(d \downarrow F)$ has small limits?	A limit $\lim_j (c_j , f_j)$ in $(d \downarrow F)$ has underlying object $\lim_j c_j$ in $\mathcal{C}$. The map $d \to F(\lim_j c_j)$ is given by the limit cone factoring through $F(\lim_j c_j) \cong \lim_j Fc_j$ (by F preserving limits); this makes $(\lim_j c_j ,$ this map) the limit in $(d \downarrow F)$.

Chapter Summary

Reducing to initial objects. F has a left adjoint iff each comma category $(d \downarrow F)$ has an initial object. This reduces the existence question to one about small categories.
GAFT. A complete, locally small $\mathcal{C}$ and a continuous F with the solution set condition: then $(d \downarrow F)$ has small limits (by continuity) and a weakly initial set (by the solution set), so an initial object exists by the limit-over-weakly-initial-set construction.
SAFT. If additionally $\mathcal{C}$ has a small cogenerating set, the solution set condition is automatic: the cogenerating set provides a cardinality bound that turns any potential solution set into a genuine set.
Applications. Free algebras (GAFT or SAFT applied to the forgetful functor), sheafification (GAFT applied to the sheaf inclusion), Stone–Čech compactification (SAFT applied to the compact Hausdorff inclusion). Each is a left adjoint whose existence is categorical, even if the construction is analytic.
The pattern. Every AFT application has the same shape: identify a continuous functor out of a complete, well-powered category; check the solution set or cogenerator hypothesis; conclude the left adjoint exists; identify what that left adjoint "is" concretely.

23.6 Formal Restatement

Formal Restatement

Continuous functor. $F : \mathcal{C} \to \mathcal{D}$ preserves all small limits: for any small diagram $D : \mathcal{J} \to \mathcal{C}$, the canonical map $F(\lim D) \to \lim(F \circ D)$ is an isomorphism.
Solution set condition. For $F : \mathcal{C} \to \mathcal{D}$ and each $d \in \mathcal{D}$: there is a set I and a family $(c_i, f_i : d \to Fc_i)_{i\in I}$ such that for every $f : d \to Fc$, there exist $i \in I$ and $g : c_i \to c$ with $Fg \circ f_i = f$.
General Adjoint Functor Theorem (GAFT). Let $\mathcal{C}$ be locally small and complete. A functor $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint iff F is continuous and satisfies the solution set condition.
Cogenerating set. A set $\mathcal{G} \subseteq \mathrm{Ob}(\mathcal{C})$ is cogenerating if the family of functors $(\mathrm{Hom}_{\mathcal{C}}(-, s))_{s\in\mathcal{G}}$ is jointly faithful: $f \neq g$ implies $hs \circ f \neq hs \circ g$ for some $s \in \mathcal{G}$ and $h : \mathrm{cod}(f) \to s$.
Special Adjoint Functor Theorem (SAFT). Let $\mathcal{C}$ be locally small, complete, and have a small cogenerating set. Then every continuous $F : \mathcal{C} \to \mathcal{D}$ has a left adjoint.

Key lemma. If $\mathcal{K}$ is locally small with small limits, and has a weakly initial set $\mathcal{S}$, then $\mathcal{K}$ has an initial object. *Proof*: the limit $\lim(\prod_{s\in\mathcal{S}} s \rightrightarrows k)$ over all k is initial.
Reflective subcategory. A full subcategory $i : \mathcal{D} \hookrightarrow \mathcal{C}$ is reflective if i has a left adjoint (the reflector). By SAFT: if $\mathcal{D}$ is complete, locally small, with a cogenerating set, and i is continuous, then $\mathcal{D}$ is reflective.

Chapter 24

The Representable Universe

Chapter Overview

We have arrived at the synthesis chapter for Volume II. Over these two volumes we have built the entire toolkit of basic category theory: categories, functors, natural transformations, limits and colimits, adjunctions, monads, the Yoneda lemma, and the adjoint functor theorems.

Looking back, every construction we have studied shares a single organizing principle: **representability**. A terminal object is a representable functor. A product is a representable functor. A limit is a representable functor. An adjunction is a natural isomorphism of hom-functors. A monad algebra is a representable structure on a category. The Yoneda lemma is the foundational fact about representability itself.

This chapter makes that thread explicit. We trace representability through every major construction, state it as a uniform principle, and use the resulting picture to understand the **presheaf category** $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ as the "free completion" of $\mathcal{C}$ — the universe in which all universal constructions live.

We close with a preview of Volume III, where enriched categories, weighted limits, ends/coends, and the full 2-categorical language will allow us to state everything in maximum generality.

24.1 The Representability Thread

24.1.1 The Unifying Principle

The **representability principle** says: to understand a construction in $\mathcal{C}$, express it as the object that *represents* a certain functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.

Concretely: a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is **representable** if there exists $c \in \mathcal{C}$ and an isomorphism $F \cong \mathrm{Hom}(-, c)$.

When F is representable, the representing object c is the *universal solution*: every element of $F(a)$, for any a, corresponds to a unique morphism $a \to c$. The Yoneda lemma then says this correspondence is natural, and the representing object is unique up to unique isomorphism.

24.1.2 Terminal Objects

Let $\mathcal{C}$ be a category. Define $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ by $F(a) = \{*\}$ (the single-element set, constantly).

F is representable iff there exists $1 \in \mathcal{C}$ with $\mathrm{Hom}(-, 1) \cong F = \{*\}$, i.e., iff for every a there is a unique map $a \to 1$. This is exactly the definition of a **terminal object**.

> *A terminal object is the representing object for the constant functor* $a \mapsto \{*\}$.

24.1.3 Products

Fix $b, c \in \mathcal{C}$. Define $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ by $F(a) = \mathrm{Hom}(a, b) \times \mathrm{Hom}(a, c)$ (pairs of maps from a to b and c).

F is representable iff there exists $b \times c$ with $\mathrm{Hom}(-, b \times c) \cong \mathrm{Hom}(-, b) \times \mathrm{Hom}(-, c)$. By Yoneda, this means: for every a, a map $a \to b \times c$ is the same data as a pair of maps $a \to b$ and $a \to c$. This is the universal property of the **binary product**.

> *A binary product* $b \times c$ *represents the functor* $a \mapsto \mathrm{Hom}(a, b) \times \mathrm{Hom}(a, c)$.

24.1.4 Limits

Let $D : \mathcal{J} \to \mathcal{C}$ be a diagram. Define $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ by

$$F(a) = \mathrm{Cone}(a, D) = \text{set of cones from } a \text{ to } D.$$

Explicitly, $F(a)$ is the set of families $(f_j : a \to Dj)_{j \in \mathcal{J}}$ compatible with the diagram.

F is representable iff there exists $\lim D$ with $\mathrm{Hom}(-, \lim D) \cong F$: a map $a \to \lim D$ corresponds to a compatible cone from a. This is the definition of the **limit**.

The limit $\lim D$ *represents the functor of cones:* $\mathrm{Hom}(-, \lim D) \cong \mathrm{Cone}(-, D)$.

24.1.5 Adjunctions

An adjunction $L \dashv R$ is precisely a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{D}}(L-, -) \;\cong\; \mathrm{Hom}_{\mathcal{C}}(-, R-) \qquad : \mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathbf{Set}.$$

Reading this: the functor $d \mapsto \mathrm{Hom}_{\mathcal{D}}(Lc, d)$ is naturally isomorphic to $d \mapsto \mathrm{Hom}_{\mathcal{C}}(c, Rd)$.

For fixed $c \in \mathcal{C}$, the functor $\mathrm{Hom}_{\mathcal{D}}(Lc, -)$ is representable (represented by Lc). The adjunction says that Lc also represents $\mathrm{Hom}_{\mathcal{C}}(c, R-)$: the left adjoint value Lc is the representing object for the "maps from c that go through R."

An adjunction is a statement that two families of representable functors are naturally isomorphic.

24.1.6 Monads and Kan Extensions

A monad algebra $(a, h : Ta \to a)$ is the structure that makes a a *model* of the monad: the structure map h is the data that represents the monad's action on a. In the language of Kan extensions (Volume III), a monad is itself a Kan extension — the left Kan extension of the identity functor along itself — and its algebras represent certain functors out of the Kleisli category.

The pattern is: every piece of structure in category theory is captured by which functor it represents, or by what it is the left/right Kan extension of.

24.2 The Yoneda Philosophy

24.2.1 The Yoneda Embedding

The **Yoneda embedding** is the functor

$$\mathcal{Y} : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}], \qquad c \mapsto \mathrm{Hom}(-, c).$$

The Yoneda lemma says this functor is *fully faithful*: the natural bijection

$$\mathrm{Nat}(\mathrm{Hom}(-, c), \mathrm{Hom}(-, c')) \;\cong\; \mathrm{Hom}(c, c')$$

means that natural transformations between representable functors are exactly the morphisms between the representing objects. No information is lost; $\mathcal{C}$ embeds into the presheaf category without distortion.

24.2.2 The Presheaf Category as the Free World

The presheaf category $\hat{\mathcal{C}} = [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is, in a precise sense, the "free cocompletion" of $\mathcal{C}$:

Theorem 24.1 (Presheaf category as free cocompletion)**.** *Every presheaf $P \in \hat{\mathcal{C}}$ is a colimit of representables:*

$$P \;\cong\; \mathrm{colim}_{(c,x) \in \int P} \mathrm{Hom}(-, c)$$

*where $\int P$ is the **category of elements** of P (objects: pairs $(c \in \mathcal{C}, x \in Pc)$; morphisms: maps $f : c \to c'$ with $Pf(x') = x$).*

This is the **co-Yoneda lemma**. It says: in the presheaf category, every object is "built" from representables by colimits. The representables $\mathrm{Hom}(-, c)$ are the building blocks; colimits are the glue.

Internal Dialog

Think of $\hat{\mathcal{C}}$ as the "free world" where any diagram in $\mathcal{C}$ can be formally completed. If $\mathcal{C}$ is a category of geometric shapes (simplices, cubes, globes), then $\hat{\mathcal{C}}$ is the category of spaces (simplicial sets, cubical sets, globular sets) built by gluing those shapes. Every space is a colimit of elementary shapes, and the elementary shapes are exactly the representables.

24.2.3 The Universal Property of $\hat{\mathcal{C}}$

The Yoneda embedding $\mathcal{Y} : \mathcal{C} \to \hat{\mathcal{C}}$ is universal among functors from $\mathcal{C}$ to cocomplete categories that preserve colimits:

Theorem 24.2 (Universal property of the presheaf category)**.** *For any cocomplete category $\mathcal{E}$ and functor $F : \mathcal{C} \to \mathcal{E}$, there is an essentially unique colimit-preserving functor $\bar{F} : \hat{\mathcal{C}} \to \mathcal{E}$ with $\bar{F} \circ \mathcal{Y} \cong F$. The functor $\bar{F}$ is given by left Kan extension: $\bar{F}(P) = \int^{c} P(c) \cdot Fc$ (the coend formula).*

In this sense, $\hat{\mathcal{C}}$ is the **free cocompletion** of $\mathcal{C}$: it freely adds all colimits, and any functor from $\mathcal{C}$ extends uniquely to a colimit-preserving functor from $\hat{\mathcal{C}}$.

Where the Name Comes From

The **presheaf category** $\widehat{\mathcal{C}} = [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the "free cocompletion" of $\mathcal{C}$ — the universal cocomplete category containing $\mathcal{C}$. The "hat" notation $\widehat{\mathcal{C}}$ is standard in the Grothendieck-school literature, evoking the analogy with $\widehat{R}$ (the completion of a ring): both add "ideal" or "universal" missing constructions. *Cocompletion* = closing under colimits, freely. Cross-ref Vol IV Ch. 48 for the ∞-version.
Density theorem: every presheaf is canonically a colimit of representables. The terminology echoes topology (a dense subset of a space). Mac Lane's standard form; the slogan is that *representables are dense in the presheaf category.*

24.3 The Universe of Constructions

24.3.1 Everything as Representability or Adjointness

The following table summarizes the universal constructions of Volumes I and II as representability or adjointness statements. The pattern is uniform: every construction is either (a) the representing object for a canonically defined functor, or (b) one side of an adjunction that expresses a natural isomorphism of hom-functors.

Construction	Functor Represented	Adjointness Statement
Terminal object 1	$a \mapsto \{*\}$ (constant singleton)	$\mathrm{Hom}(-, 1) \cong !$
Product $b \times c$	$a \mapsto \mathrm{Hom}(a, b) \times \mathrm{Hom}(a, c)$	$\Delta \dashv \times$ (diagonal left adjoint to product)
Equalizer	$a \mapsto \{f : a \to b \mid uf = vf\}$	Limit of the equalizer diagram
Limit $\lim D$	$a \mapsto \mathrm{Cone}(a, D)$	$\Delta \dashv \lim$ (constant diagram left adjoint to limit)
Left adjoint Lc	$d \mapsto \mathrm{Hom}(c, Rd)$	$\mathrm{Hom}(Lc, d) \cong \mathrm{Hom}(c, Rd)$
Free algebra TX	$A \mapsto \mathrm{Hom}(X, GA)$ in $\mathrm{Alg}(T)$	Free/forgetful adjunction
Monad algebra (a, h)	Action of T on a via $h : Ta \to a$	EM category $\mathcal{C}^T$ as maximal resolution
Sheafification $\mathrm{sh}(P)$	$\mathcal{F} \mapsto \mathrm{Hom}(P, i\mathcal{F})$ in **Sh**	$\mathrm{sh} \dashv i$ (sheafification left adjoint to inclusion)
Kan extension $\mathrm{Lan}_K F$	$G \mapsto \mathrm{Nat}(F, G \circ K)$	$\mathrm{Lan}_K \dashv (- \circ K)$ (precomposition)

Internal Dialog

The table makes a philosophical point: category theory does not invent new constructions — it observes that constructions already present in mathematics share the same logical form. Each entry in the table is a well-known mathematical object; what category theory contributes is the observation that they all live in the same conceptual house.

24.3.2 The Two Dual Perspectives

There are two dual ways to see this uniformity:

Limits / representability: Every object representing a "right-hand" hom-functor (cone functor, mapping-in functor) is a limit. Limits live in $\mathcal{C}$ and represent *contravariant* functors $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.

Colimits / corepresentability: Every object representing a "left-hand" hom-functor (cocone functor, mapping-out functor) is a colimit. Colimits live in $\mathcal{C}$ and represent *covariant* functors $\mathcal{C} \to \mathbf{Set}$.

The two perspectives are related by Yoneda: every covariant representable functor on $\mathcal{C}$ is a contravariant representable functor on $\mathcal{C}^{\mathrm{op}}$. Duality in category theory is literally *reversing the variance of the hom-functor.*

24.4 Preview of Volume III

24.4.1 What Remains

Volumes I and II have laid the foundation. But every construction we have studied still has a more general form:

- **Enriched categories**: Categories where the hom-sets are not sets but objects of some monoidal category $\mathcal{V}$ — abelian groups, chain complexes, topological spaces, simplicial sets. The ordinary category theory of Volumes I–II is the case $\mathcal{V} = \mathbf{Set}$.
- **Weighted limits**: The "cone" definition of a limit uses the terminal weight. Weighted limits allow arbitrary $\mathcal{V}$-valued weight functors. This covers homotopy limits, ends, coends, and much more.

- **Ends and coends**: Generalizations of limit and colimit to "mixed-variance" functors $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$. The coend $\int^c T(c,c)$ and end $\int_c T(c,c)$ are fundamental in enriched category theory, the Yoneda extension formula, and the calculus of natural transformations.
- **2-categorical structure**: The interchange law and bicategories (Chapter 22) are the beginning. Volume III develops the full theory of 2-adjunctions, 2-limits, and pseudofunctors.

24.4.2 Enriched Category Theory

An **enriched category** $\mathcal{C}$ over a monoidal category $(\mathcal{V}, \otimes, I)$ has:

- Objects: as usual.
- Hom-objects: $\mathcal{C}(a,b) \in \mathcal{V}$ (not a set, but an object of $\mathcal{V}$).
- Composition: a morphism $\mathcal{C}(b,c) \otimes \mathcal{C}(a,b) \to \mathcal{C}(a,c)$ in $\mathcal{V}$.
- Identity: a morphism $I \to \mathcal{C}(a,a)$ for each a.

When $\mathcal{V} = \mathbf{Ab}$, enriched categories are $\mathbf{Ab}$-categories (additive categories). When $\mathcal{V} = \mathbf{Ch}$ (chain complexes), they are dg-categories (differential graded categories), the setting for homological algebra. When $\mathcal{V} = \mathbf{sSet}$ (simplicial sets), they are simplicial categories, the setting for homotopy theory.

24.4.3 Weighted Limits and Coends

An ordinary limit of $D : \mathcal{J} \to \mathcal{C}$ is the limit *weighted* by the constant functor $W : \mathcal{J} \to \mathbf{Set}$ sending everything to $\{*\}$. A **weighted limit** allows $W : \mathcal{J} \to \mathbf{Set}$ to be any functor; the weighted limit $\{W, D\}$ (if it exists) represents the functor $c \mapsto \mathrm{Nat}(W, \mathcal{C}(c, D-))$.

The **coend** $\int^c T(c,c)$ of $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$ is a colimit that "contracts" the two copies of c into one. The Yoneda lemma has a coend form:

$$\int^c \mathrm{Hom}(a,c) \cdot Fc \;\cong\; Fa$$

(the "ninja Yoneda lemma"), which is the coend expression for the co-Yoneda formula. Ends and coends are the universal tools for computing with natural transformations and enriched functors.

24.4.4 The Same Constructions, More General

Volume III will revisit every construction of Volumes I and II:

- Limits $\rightsquigarrow$ weighted limits in enriched categories.
- Adjunctions $\rightsquigarrow$ enriched adjunctions.
- The Yoneda lemma $\rightsquigarrow$ the enriched Yoneda lemma.
- Kan extensions $\rightsquigarrow$ enriched Kan extensions (which, via coend formulas, become the foundational operation).
- Beck's theorem $\rightsquigarrow$ enriched monadicity.

The philosophical punchline: the *form* of every theorem stays the same. Enrichment generalizes the coefficients from **Set** to an arbitrary $\mathcal{V}$. The representability principle, the Yoneda philosophy, the AFTs — all survive the generalization, often becoming *cleaner* in the enriched setting because more structure is available.

24.5 Exercises

State the representability principle informally.	Every universal construction in $\mathcal{C}$ is the representing object for a canonically defined functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.
Which functor does a terminal object represent?	The constant functor $a \mapsto \{*\}$.
Which functor does a product $b \times c$ represent?	The functor $a \mapsto \mathrm{Hom}(a,b) \times \mathrm{Hom}(a,c)$.
State the co-Yoneda lemma (also called the density theorem).	Every presheaf $P \in [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is a colimit of representables: $P \cong \mathrm{colim}_{(c,x) \in \int P} \mathrm{Hom}(-,c)$ over the category of elements of P.
What is the category of elements $\int P$ of a presheaf $P : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$?	Objects: pairs $(c \in \mathcal{C}, x \in Pc)$. Morphisms $(c,x) \to (c',x')$: maps $f : c \to c'$ in $\mathcal{C}$ with $Pf(x') = x$.
What is the universal property of the presheaf category $\hat{\mathcal{C}}$?	For any cocomplete $\mathcal{E}$ and $F : \mathcal{C} \to \mathcal{E}$, there is an essentially unique colimit-preserving $\bar{F} : \hat{\mathcal{C}} \to \mathcal{E}$ with $\bar{F} \circ \mathcal{Y} \cong F$. So $\hat{\mathcal{C}}$ is the free cocompletion of $\mathcal{C}$.

What is an enriched category over $(\mathcal{V}, \otimes, I)$?	A category where hom-sets are replaced by hom-objects in $\mathcal{V}$, with composition $\mathcal{C}(b,c) \otimes \mathcal{C}(a,b) \to \mathcal{C}(a,c)$ and identity $I \to \mathcal{C}(a,a)$ satisfying associativity and unit axioms in $\mathcal{V}$.
What is a coend $\int^c T(c,c)$?	A colimit that "contracts" a functor $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{V}$ along the diagonal: it satisfies the universal property for "dinatural transformations," natural in a mixed covariant/contravariant sense.
Write the Yoneda lemma as a coend.	$Fa \cong \int^c \mathrm{Hom}(a,c) \cdot Fc$. This "ninja Yoneda" formula says every functor value is a coend of representables weighted by the hom-sets.
What is a weighted limit?	A limit $\{W, D\}$ of $D : \mathcal{J} \to \mathcal{C}$ weighted by $W : \mathcal{J} \to \mathbf{Set}$, representing the functor $c \mapsto \mathrm{Nat}(W, \mathcal{C}(c, D-))$. Ordinary limits are the case $W = \Delta\{*\}$.
What monoidal category does ordinary category theory correspond to in the enriched setting?	$\mathcal{V} = (\mathbf{Set}, \times, \{*\})$: ordinary category theory is the $\mathbf{Set}$-enriched case.
What are dg-categories?	Categories enriched over chain complexes $\mathbf{Ch}$. Their hom-objects are chain complexes; composition respects the differential. They are the natural home for homological algebra and derived categories.
Express the adjunction bijection $\mathrm{Hom}(Lc, d) \cong \mathrm{Hom}(c, Rd)$ in terms of representability.	For fixed c: the functor $d \mapsto \mathrm{Hom}(c, Rd)$ is represented by Lc. For fixed d: the functor $c \mapsto \mathrm{Hom}(Lc, d)$ is represented by Rd.
What is the left Kan extension formula involving coends?	$(\mathrm{Lan}_K F)(d) \cong \int^c \mathrm{Hom}(Kc, d) \cdot Fc$ — the left Kan extension of $F : \mathcal{C} \to \mathcal{E}$ along $K : \mathcal{C} \to \mathcal{D}$ is a coend formula.
What does the "free cocompletion" statement say concretely for $\mathcal{C} = \Delta$ (the simplex category)?	Presheaves on Δ are simplicial sets. Every simplicial set is a colimit of standard simplices $\Delta[n] = \mathrm{Hom}(-, [n])$. The simplex category Δ is the "generating shapes," and simplicial sets are their free colimit completion.

Chapter Summary

The representability thread. Every universal construction — terminal objects, products, equalizers, limits, adjoints, free algebras, sheafification — is the representing object for a canonically defined functor. The Yoneda lemma is the foundation: it says representing objects are unique up to unique isomorphism and are determined by their universal properties.

The presheaf category as free world. Every presheaf is a colimit of representables (co-Yoneda / density theorem). The category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the free cocompletion of $\mathcal{C}$: any functor from $\mathcal{C}$ to a cocomplete category extends uniquely (as a colimit-preserving functor) to the presheaf category.

The table of constructions. Terminal object, product, equalizer, limit, left adjoint, free algebra, monad algebra, sheafification, Kan extension: each is a representability or adjointness statement. They form a coherent family organized by the single principle that every construction is a universal solution to a mapping problem.

Volume III preview. Enriched category theory generalizes hom-sets to hom-objects in a monoidal category $\mathcal{V}$. Weighted limits and coends are the enriched generalizations of limits. Ends and coends are the computational tools for working with natural transformations and enriched functors. All theorems of Volumes I–II survive in enriched form, often in cleaner statements.

The philosophical punchline. Category theory does not construct new mathematics from nothing. It observes that all of mathematics is organized by representability — by the pattern of which objects can play the role of "universal recipient" for which kinds of morphisms. Everything we have seen is a variation on this single theme.

Volume III

The Formal Spine

Chapter 25

2-Categories and Bicategories

Chapter Overview

Chapter 22 introduced the 2-category **Cat** — objects are categories, 1-cells are functors, 2-cells are natural transformations — and proved the interchange law for its vertical and horizontal compositions. This chapter develops the full theory of 2-categorical structure: strict 2-categories, *bicategories* (where associativity and unitality hold only up to coherent 2-isomorphism), the three flavours of functor between them (*strict*, *pseudo*, *lax*), and the higher cells (*pseudonatural transformations* and *modifications*) that organize them into a 3-categorical hierarchy. Two payoffs. First, examples beyond **Cat** — **Span**, **Prof**, **Rel**, **Bimod** — none of which are strictly associative; bicategory language is mandatory to handle them. Second, the strictification theorem: every bicategory is biequivalent to a strict 2-category. This is the higher-categorical analogue of Mac Lane's coherence (Chapter 19) and explains why we may often pretend strictness is no loss of generality.

25.1 Strict 2-Categories Revisited

25.1.1 Formalizing

Definition 25.1 (Strict 2-category)**.** A **strict 2-category** $\mathcal{K}$ consists of:

- A class of *objects* (0-cells) $a, b, \ldots$
- For each pair a, b, a category $\mathcal{K}(a, b)$ — the *hom-category*. Its objects are 1-cells $f : a \to b$; its morphisms are 2-cells $\alpha : f \Rightarrow g$.
- For each triple a, b, c, a *composition functor* $\circ : \mathcal{K}(b, c) \times \mathcal{K}(a, b) \to \mathcal{K}(a, c)$.
- For each a, an identity 1-cell $1_a \in \mathcal{K}(a, a)$.

These satisfy *strict* associativity and unit laws: $h \circ (g \circ f) = (h \circ g) \circ f$ and $1_b \circ f = f = f \circ 1_a$.

The composition functor being a *functor* (not merely a function) is what packages the interchange law: applied to 2-cells it gives horizontal composition, applied to identity 2-cells it gives whiskering, and functoriality in each variable says exactly that vertical and horizontal composites commute.

Remark 25.2 (2-categories as **Cat**-enriched categories)**.** A strict 2-category is precisely a category *enriched over* **Cat** with the cartesian product. Hom-categories are the $\mathcal{V}$-objects; composition functors are the $\mathcal{V}$-composition. All structure descends from Chapter 12's enriched-category machinery.

25.1.2 Formally

Example 25.3. **Cat** itself, with hom-categories $\mathbf{Cat}(\mathcal{A}, \mathcal{B}) = [\mathcal{A}, \mathcal{B}]$ the functor category.

Example 25.4. For each ordinary category $\mathcal{C}$, the *locally discrete* 2-category $\mathcal{C}_{\text{disc}}$: same objects and 1-cells; only identity 2-cells. This realises every category as a 2-category.

Example 25.5. For each ordinary category $\mathcal{C}$ with all pullbacks, the 2-category $\mathbf{Cat}/\mathcal{C}$ of *categories sliced over* $\mathcal{C}$: objects are functors into $\mathcal{C}$, 1-cells are triangles, 2-cells are natural transformations between the triangle's tops compatible with the projection.

Example 25.6 (Monoidal categories as one-object 2-categories)**.** A monoidal category $(\mathcal{V}, \otimes, I)$ is the same data as a strict 2-category $B\mathcal{V}$ with a single object $\star$: 1-cells are objects of $\mathcal{V}$, composition is $\otimes$, 2-cells are morphisms of $\mathcal{V}$, the identity 1-cell is I. This works when $\otimes$ is strictly associative; otherwise the identification is up to a coherent isomorphism, which is the bicategorical case (§22.6).

Where the Name Comes From

Bicategory was introduced by *Jean Bénabou* in 1967. The prefix *bi-* (Latin "two") reflects that there are two levels of morphisms (1-cells and 2-cells), but with associativity and unit laws holding only "up to coherent 2-isomorphism" rather than strictly. This is in contrast to the *strict* 2-category, where the laws hold on the nose. The strict-vs-weak distinction is one of the central themes of higher category theory.
Pseudofunctor (a functor between bicategories that preserves composition only up to coherent isomorphism) is also Bénabou's term; the prefix *pseudo-* (Greek "false, almost") indicates that the preservation is up to coherent iso rather than strict equality. **Lax functor**: even weaker, with only a (not necessarily invertible) 2-cell as the comparison; "lax" from Latin *laxus* ("loose, slack").

25.2 Bicategories

25.2.1 Informally

A bicategory is a 2-category in which the strict equality $h \circ (g \circ f) = (h \circ g) \circ f$ is replaced by an *associator*: a specified invertible 2-cell $\alpha_{h,g,f} : (h \circ g) \circ f \xRightarrow{\cong} h \circ (g \circ f)$. Likewise the strict unit equations are replaced by invertible *unitors*. Coherence axioms — the pentagon for associators, the triangle for unitors — guarantee that all reassociation diagrams commute, just as in the monoidal case of Chapter 19.

25.2.2 Formalizing

Definition 25.7 (Bicategory)**.** A **bicategory** $\mathcal{B}$ consists of:

- Objects (0-cells) $a, b, \ldots$
- For each pair a, b, a hom-category $\mathcal{B}(a, b)$.
- Composition functors $\circ_{a,b,c} : \mathcal{B}(b, c) \times \mathcal{B}(a, b) \to \mathcal{B}(a, c)$.
- Identity 1-cells $1_a \in \mathcal{B}(a, a)$.
- For each composable triple (h, g, f), an invertible **associator** $\alpha_{h,g,f} : (h \circ g) \circ f \xRightarrow{\cong} h \circ (g \circ f)$, natural in h, g, f.
- For each $f : a \to b$, invertible **unitors** $\lambda_f : 1_b \circ f \xRightarrow{\cong} f$ and $\rho_f : f \circ 1_a \xRightarrow{\cong} f$, natural in f.

These are required to satisfy the **pentagon** and **triangle** axioms:

- Pentagon: for composable k, h, g, f, the two ways to reassociate $((k \circ h) \circ g) \circ f \to k \circ (h \circ (g \circ f))$ via α's agree.
- Triangle: $\rho_g \circ_h \mathrm{id}_h = \mathrm{id}_g \circ_h \lambda_h$ composed with $\alpha_{g,1_b,h}$ for $g : b \to c$, $h : a \to b$.

How to Say This

- $\alpha_{h,g,f}$ *(read: "alpha sub h, g, f")* or *(read: "the associator at h, g, f")*.
- $f \circ_h g$ *(read: "f horizontal g")* — horizontal composition of 2-cells, the action of the composition *functor* on morphisms.

A strict 2-category is the special case where all associators and unitors are identity 2-cells. Conversely:

Proposition 25.8 (Coherence for bicategories — preview)**.** *In any bicategory, every diagram built from associators and unitors commutes.*

This is the analogue of Mac Lane's coherence theorem (Chapter 19) for the monoidal case. The proof, via strictification, appears in §25.6.

25.2.3 Reorganizing: Monoidal Categories as One-Object Bicategories

Proposition 25.9. *A monoidal category $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ is the same data as a bicategory with a single object.*

Proof. A one-object bicategory $\mathcal{B}$ with object $\star$ has a single hom-category $\mathcal{B}(\star, \star)$ whose objects (1-cells) and morphisms (2-cells) are precisely the objects and morphisms of a category $\mathcal{V}$. Composition $\circ : \mathcal{B}(\star, \star) \times \mathcal{B}(\star, \star) \to \mathcal{B}(\star, \star)$ is a functor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$. The identity 1-cell is the unit I. The associator and unitors are precisely α, λ, ρ. The pentagon and triangle axioms transfer identically. The correspondence is exact. □

Remark 25.10. This is the same kind of one-object delooping as "monoid = one-object category" and "group = one-object groupoid". Each level of categorification adds one dimension to the diagram and one axis of coherence.

25.3 Pseudofunctors, Lax Functors, and Oplax Functors

25.3.1 Formalizing

For ordinary categories, functoriality is an equation: $F(g \circ f) = F(g) \circ F(f)$ and $F(1_a) = 1_{Fa}$. For bicategories, equality is too rigid: when $\mathcal{B}$ has non-strict associators, we cannot expect a map of bicategories to preserve composition on the nose. The right notion is preservation *up to a coherent 2-isomorphism.*

Definition 25.11 (Pseudofunctor)**.** A **pseudofunctor** (also: **weak 2-functor**) $F : \mathcal{B} \to \mathcal{C}$ between bicategories consists of:

- A function $F : \mathrm{ob}\,\mathcal{B} \to \mathrm{ob}\,\mathcal{C}$ on objects.
- For each pair a, b, a functor $F_{a,b} : \mathcal{B}(a, b) \to \mathcal{C}(Fa, Fb)$.
- For each composable pair (g, f), an invertible 2-cell $\phi_{g,f} : F(g) \circ F(f) \xRightarrow{\cong} F(g \circ f)$ — the **compositor**.
- For each a, an invertible 2-cell $\phi_a : 1_{Fa} \xRightarrow{\cong} F(1_a)$ — the **unitor**.

The compositor and unitor are required to be *natural* in their arguments and to satisfy three coherence diagrams: one matching $F(\alpha)$ to the associator of $\mathcal{C}$ via ϕ, and two matching $F(\lambda)$, $F(\rho)$ to the unitors via ϕ.

Definition 25.12 (Lax and oplax functors)**.** A **lax functor** $F : \mathcal{B} \to \mathcal{C}$ has the same data as a pseudofunctor except $\phi_{g,f}$ and ϕ_a are *not required to be invertible* — they are merely 2-cells in the displayed direction. An **oplax functor** reverses the direction: $\phi^{\mathrm{op}}_{g,f} : F(g \circ f) \Rightarrow F(g) \circ F(f)$ and $\phi^{\mathrm{op}}_a : F(1_a) \Rightarrow 1_{Fa}$.

Remark 25.13. Lax functors with values in **Cat** have a striking interpretation that returns in Chapter 32: a lax functor $\mathbf{1} \to \mathbf{Cat}$ is precisely a *monad.* The compositor ϕ is the multiplication μ; the unitor ϕ_a is η; the coherence diagrams are exactly the monad associativity and unit axioms.

25.3.2 Reorganizing: When Pseudofunctors Are Strict

A **strict 2-functor** is a pseudofunctor whose $\phi_{g,f}$ and ϕ_a are all identity 2-cells. Between strict 2-categories these are the "honest" 2-functors. Between bicategories, asking for strictness is too much; even the identity functor of a non-strict bicategory cannot be strict, because the associator is non-identity. *Pseudo* is the natural notion.

25.3.3 Formally

Example 25.14 (Identity pseudofunctor)**.** $\mathrm{Id}_{\mathcal{B}} : \mathcal{B} \to \mathcal{B}$ with $\phi_{g,f} = \mathrm{id}$ and $\phi_a = \mathrm{id}$. This is strict — but only because the source equals the target.

Example 25.15 (Underlying ordinary category)**.** The forgetful pseudofunctor $U : \mathcal{B} \to \mathcal{B}_0$ to the underlying ordinary category collapses 2-cells. It is *lax*, with $\phi_{g,f} : g \circ_{\mathcal{B}} f \to g \circ_{\mathcal{B}_0} f$ given by quotienting; not invertible in general.

Example 25.16 (Free monoidal functor)**.** The free symmetric monoidal completion $\mathcal{V} \to \mathrm{Sym}(\mathcal{V})$ is a pseudofunctor of monoidal-as-bicategories. Compositors arise from the canonical isomorphisms in the symmetric monoidal closure.

25.4 Pseudonatural Transformations and Modifications

25.4.1 Formalizing

For ordinary functors $F, G : \mathcal{A} \to \mathcal{B}$, a natural transformation $\eta : F \Rightarrow G$ is a family $\eta_a : Fa \to Ga$ satisfying a square equality. For pseudofunctors $F, G : \mathcal{B} \to \mathcal{C}$ between bicategories, the square is replaced by an invertible 2-cell.

Definition 25.17 (Pseudonatural transformation)**.** A **pseudonatural transformation** $\eta : F \Rightarrow G$ between pseudofunctors $F, G : \mathcal{B} \to \mathcal{C}$ consists of:

- For each $a \in \mathcal{B}$, a 1-cell $\eta_a : Fa \to Ga$ in $\mathcal{C}$.
- For each 1-cell $f : a \to b$ in $\mathcal{B}$, an invertible 2-cell

$$\eta_f : Gf \circ \eta_a \xRightarrow{\cong} \eta_b \circ Ff.$$

These satisfy two coherence axioms: *respect for composition* ($\eta_{g\circ f}$ assembles from η_g and η_f via the compositors of F, G) and *respect for identities* (η_{1_a} matches the unitor data).

Definition 25.18 (Lax/oplax transformations)**.** Drop the invertibility of η_f to get a **lax** transformation; reverse its direction to get an **oplax** transformation.

25.4.2 Reorganizing

Definition 25.19 (Modification)**.** For two pseudonatural transformations $\eta, \eta' : F \Rightarrow G$, a **modification** $m : \eta \Rrightarrow \eta'$ is a family of 2-cells $m_a : \eta_a \Rightarrow \eta'_a$ in $\mathcal{C}$ — one for each object $a \in \mathcal{B}$ — satisfying compatibility with the structure 2-cells η_f, η'_f: for each $f : a \to b$, the equation

$$(\mathrm{id}_{\eta'_b} \circ_h Ff) \cdot m_a^{Gf} \;=\; m_b^{Ff} \cdot (\mathrm{id}_{Gf} \circ_h \eta_a)$$

in $\mathcal{C}(Fa, Gb)$, with $m^{(-)}$ denoting whiskering of m by the indicated 1-cell.

Proposition 25.20 (The tricategorical hierarchy)**.** *For any pair of bicategories $\mathcal{B}, \mathcal{C}$, pseudofunctors, pseudonatural transformations, and modifications form a (strict) 2-category* $\mathrm{Bicat}(\mathcal{B}, \mathcal{C})$*.*

Sketch. 1-cell composition is pasting of 2-cells; 2-cell vertical and horizontal compositions are inherited from $\mathcal{C}$. The interchange law and unit laws of the strict 2-category structure follow from the corresponding axioms in $\mathcal{C}$ together with the coherence axioms of pseudonaturality. □

Remark 25.21. Globally, bicategories, pseudofunctors, pseudonatural transformations, and modifications form a **tricategory** **Bicat**. The tricategorical structure is needed because composition of pseudofunctors is associative only up to a coherent pseudonatural transformation, not strictly.

25.5 Examples Beyond Cat

The strength of bicategory theory shows in examples where strict associativity fails. We sketch four.

25.5.1 Span($\mathcal{C}$)

For a category $\mathcal{C}$ with finite limits, the bicategory **Span**($\mathcal{C}$) has:

- Objects: objects of $\mathcal{C}$.
- 1-cells $a \to b$: spans $a \xleftarrow{p} x \xrightarrow{q} b$.
- 2-cells: morphisms of spans (a single map of apexes commuting with both legs).
- Composition: pullback. Given $a \leftarrow x \to b$ and $b \leftarrow y \to c$, the composite is $a \leftarrow x \times_b y \to c$.

Remark 25.22. Composition is associative only up to the canonical isomorphism between iterated pullbacks $(x \times_b y) \times_c z \cong x \times_b (y \times_c z)$ — a genuine non-trivial associator. This is why **Span** cannot be made strict (its strict version would require choosing pullbacks coherently, which is rarely possible).

25.5.2 Prof

The bicategory of **profunctors**:

- Objects: small categories.
- 1-cells $\mathcal{A} \to \mathcal{B}$: profunctors $P : \mathcal{B}^{\mathrm{op}} \times \mathcal{A} \to \mathbf{Set}$.
- 2-cells: natural transformations of profunctors.
- Composition: the coend formula $(Q \circ P)(c, a) = \int^b Q(c, b) \times P(b, a)$.

The associator for **Prof** encodes the Fubini theorem for coends (Chapter 26). Composition is not strict because the iterated coends $\int^b \int^c$ and $\int^c \int^b$ are isomorphic but distinct constructions.

25.5.3 Rel

The bicategory of relations:

- Objects: sets.
- 1-cells $A \to B$: relations $R \subseteq A \times B$.
- 2-cells: containment $R \subseteq R'$ (a poset structure).
- Composition: relational composition.

Rel is in fact *locally posetal* — every hom-category is a poset — and the composition is strictly associative on relations, so it can be presented as a strict 2-category. We keep it under "bicategories" for uniformity with **Span**.

25.5.4 Bimod

For commutative rings, the bicategory **Bimod**:

- Objects: commutative rings.
- 1-cells $R \to S$: (S, R)-bimodules.
- 2-cells: bimodule homomorphisms.
- Composition: tensor product $M \otimes_S N$.

The associator is the canonical isomorphism $(M \otimes_S N) \otimes_T P \cong M \otimes_S (N \otimes_T P)$. Like **Span**, this fails strict associativity because the tensor product is only associative up to a chosen isomorphism.

25.6 Biequivalences and Strictification

25.6.1 Formalizing

Definition 25.23 (Biequivalence)**.** A pseudofunctor $F : \mathcal{B} \to \mathcal{C}$ is a **biequivalence** if it is:

- *Essentially surjective on objects*: every $c \in \mathcal{C}$ is equivalent (via 1-cells with invertible 2-cells) to some Fb;
- *Locally an equivalence*: each $F_{a,b} : \mathcal{B}(a, b) \to \mathcal{C}(Fa, Fb)$ is an equivalence of categories.

Remark 25.24. This is the analogue of "fully faithful and essentially surjective ⇒ equivalence" for ordinary functors, but at one categorical level higher. The witness data is a pair $(F, G, \eta, \varepsilon)$ where η, ε are pseudonatural *equivalences* (themselves invertible up to invertible modifications).

25.6.2 Formally

Theorem 25.25 (Strictification)**.** *Every bicategory $\mathcal{B}$ is biequivalent to a strict 2-category $\mathcal{B}^{\mathrm{st}}$.*

Define $\mathcal{B}^{\mathrm{st}}$ with the same objects as $\mathcal{B}$; 1-cells $a \to b$ are finite *strings* of composable 1-cells of $\mathcal{B}$ from a to b (including the empty string at a, acting as identity).	*We replace composites by formal strings to gain strict associativity.*
2-cells from one string to another are 2-cells in $\mathcal{B}$ between their respective composites $\widehat{(-)}$: strings $\to$ 1-cells.	*The map $\widehat{(-)}$ takes a string $(f_1, \ldots, f_n)$ to $f_n \circ \cdots \circ f_1$, left-associated.*
Composition of strings is concatenation; identity 1-cell at a is the empty string.	*Both operations are strictly associative and unital* at the level of strings.
The pseudofunctor $\widehat{(-)} : \mathcal{B}^{\mathrm{st}} \to \mathcal{B}$ sends a string to its left-associated composite.	*Provides the candidate biequivalence.*
Compositor of $\widehat{(-)}$: for strings s, t, the canonical $\widehat{t \cdot s} = \widehat{t} \circ \widehat{s}$ is built from associators in $\mathcal{B}$. Invertible by definition.	*The associator chain is well-defined by Mac Lane coherence (Proposition 25.8).*
$\widehat{(-)}$ is essentially surjective: each 1-cell $f \in \mathcal{B}$ is the composite of the singleton string (f).	*Trivially.*
$\widehat{(-)}$ is locally an equivalence: 2-cells between strings, by construction, are 2-cells between their composites in $\mathcal{B}$; so $\widehat{(-)}_{a,b}$ is fully faithful, and surjective on objects by the singleton-string argument.	*Both fullness/faithfulness and essential surjectivity at the hom-category level.*
Conclude: $\widehat{(-)} : \mathcal{B}^{\mathrm{st}} \to \mathcal{B}$ is a biequivalence.	*By definition of biequivalence.*

Corollary 25.26 (Mac Lane coherence for bicategories)**.** *Every diagram of associators and unitors in $\mathcal{B}$ commutes.*

Proof. Such a diagram, transported via the strictification $\widehat{(-)}$, becomes a diagram in the strict 2-category $\mathcal{B}^{\mathrm{st}}$ where associators and unitors are identities. There, every diagram of identity 2-cells commutes trivially. Pulling back via $\widehat{(-)}$ (which is faithful on 2-cells) gives commutativity in $\mathcal{B}$. □

Remark 25.27 (Why strictification is not the whole story)**.** Strictification does not say every bicategory *equals* a strict 2-category, only that it is biequivalent. The distinction matters when we demand strict preservation of structure (as in strict Ω-spectra of algebraic topology). Mostly, however, the strictification theorem licenses the working mathematician to write as if all bicategories were strict — the "up to coherent isomorphism" caveats can be left implicit.

25.7 Exercises

Define a strict 2-category in one line.	A category enriched over **Cat** (with cartesian product).
What is the hom-category of a strict 2-category $\mathcal{K}$?	For each pair a, b, a category $\mathcal{K}(a, b)$ whose objects are 1-cells $a \to b$ and morphisms are 2-cells between them.
What additional data does a bicategory have beyond a strict 2-category?	Invertible 2-cells: an associator $\alpha_{h,g,f}$ for each composable triple, and left/right unitors λ_f, ρ_f for each 1-cell.
What axioms must the associator and unitors satisfy?	The pentagon (for four-fold composites) and the triangle (for unitors).
What is the bicategorical analogue of Mac Lane's coherence theorem?	Every diagram of associators and unitors in a bicategory commutes (Corollary 25.26).

What does a monoidal category correspond to as a bicategory?	A one-object bicategory. The single hom-category is the underlying category of the monoidal category.
State the strictification theorem.	Every bicategory is biequivalent to a strict 2-category.
What is a pseudofunctor?	A map of bicategories preserving composition and identities up to invertible 2-cells (compositor $\phi_{g,f}$ and unitor ϕ_a) satisfying coherence axioms.
What is a lax functor?	Same data as a pseudofunctor, but $\phi_{g,f}$ and ϕ_a are not required to be invertible.
What is the difference between lax and oplax functors?	Lax: $\phi_{g,f} : F(g) \circ F(f) \Rightarrow F(g \circ f)$. Oplax: arrows reversed.
A lax functor $\mathbf{1} \to \mathbf{Cat}$ from the terminal bicategory is what?	A monad. The compositor is μ, the unitor is η, and the coherence axioms are the monad laws.
What is a pseudonatural transformation?	Between pseudofunctors F, G: a family of 1-cells $\eta_a : Fa \to Ga$ and an invertible 2-cell $\eta_f : Gf \circ \eta_a \cong \eta_b \circ Ff$ for each 1-cell f, satisfying coherence with composition and identities.
What is a modification?	Between pseudonatural transformations $\eta, \eta' : F \Rightarrow G$: a family of 2-cells $m_a : \eta_a \Rightarrow \eta'_a$ compatible with the structure 2-cells of η and η'.
Why are bicategories, pseudofunctors, pseudonatural transformations, and modifications a tricategory rather than a strict 3-category?	Composition of pseudofunctors is associative only up to coherent pseudonatural transformation, not strictly.
What is $\mathbf{Span}(\mathcal{C})$?	The bicategory whose objects are objects of $\mathcal{C}$, 1-cells are spans $a \leftarrow x \to b$, 2-cells are span morphisms, and composition is pullback.
Why is $\mathbf{Span}(\mathcal{C})$ not strictly associative?	Pullbacks are only associative up to canonical isomorphism: $(x \times_b y) \times_c z \cong x \times_b (y \times_c z)$.
What is $\mathbf{Prof}$?	The bicategory of profunctors: small categories, profunctors as 1-cells (functors $\mathcal{B}^{\mathrm{op}} \times \mathcal{A} \to \mathbf{Set}$), natural transformations as 2-cells, composition via coend.
What is the composition rule in $\mathbf{Prof}$?	$(Q \circ P)(c, a) = \int^b Q(c, b) \times P(b, a)$, a coend.
What is $\mathbf{Rel}$?	The locally posetal bicategory of sets with relations as 1-cells, containments as 2-cells, and relational composition.
What is $\mathbf{Bimod}$?	The bicategory of commutative rings, (S, R)-bimodules as 1-cells, bimodule maps as 2-cells, and tensor product as composition.
Define biequivalence.	A pseudofunctor that is essentially surjective on objects (up to equivalence) and locally an equivalence (each hom-functor is an equivalence of categories).
What is the analogue of "fully faithful and essentially surjective" for biequivalences?	Locally an equivalence and essentially surjective on objects up to equivalence — the bicategorical version.
What is a strict 2-functor?	A pseudofunctor whose compositors and unitors are identities.
Why can't the identity of a non-strict bicategory be strict?	Because the associator is non-identity; strict preservation would require $F(\alpha) = \mathrm{id}$, contradicting non-strictness.
In $\mathbf{Bimod}$, what is the identity 1-cell at a ring R?	The bimodule R itself (the regular (R, R)-bimodule).
In $\mathbf{Span}(\mathbf{Set})$, what is the identity 1-cell at a set A?	The span $A \leftarrow A \to A$ with both legs the identity.

What 2-categorical structure on **Cat** is implied by the interchange law?	Strict 2-category. The interchange law is exactly the functoriality of horizontal composition, which is what makes **Cat** strictly **Cat**-enriched.
Is a monoidal functor a pseudofunctor of one-object bicategories?	Yes: a strong (= pseudo) monoidal functor is precisely a pseudofunctor of one-object bicategories; lax monoidal is lax; oplax is oplax.
What is the relationship between **Bicat** as a tricategory and **Cat** as a 2-category?	**Cat** is a sub-2-category of **Bicat**: strict 2-categories with strict 2-functors and strict 2-natural transformations.
In what sense does strictification leave us in no loss of generality?	Up to biequivalence: every bicategorical statement can be checked in the strictified 2-category. The "up to coherent iso" caveats can be left implicit.

Chapter Summary

Chapter Summary

Strict 2-categories. Categories enriched over **Cat**: objects, hom-categories, composition functors, with strict associativity and unitality. **Cat** itself, locally discrete 2-categories, and $\mathbf{Cat}/\mathcal{C}$ are examples.
Bicategories. Generalization where associativity and unitality hold only up to coherent invertible 2-cells (associators α, unitors λ, ρ) satisfying pentagon and triangle axioms. Monoidal categories are exactly one-object bicategories.
Pseudofunctors, lax/oplax functors. Maps of bicategories preserving composition up to invertible 2-cells (pseudo); up to (not necessarily invertible) 2-cells in one direction (lax) or the other (oplax). A lax functor $\mathbf{1} \to \mathbf{Cat}$ is a monad.
Pseudonatural transformations and modifications. Natural transformations between pseudofunctors with invertible 2-cell components η_f instead of square equalities; modifications between pseudonatural transformations. Together, bicategories form a tricategory **Bicat**.
Examples beyond Cat. $\mathbf{Span}(\mathcal{C})$ (pullback composition), **Prof** (coend composition), **Rel** (locally posetal), **Bimod** (tensor over a ring).
Strictification. Every bicategory is biequivalent to a strict 2-category, via the string-replacement construction. As a corollary, every diagram of associators and unitors commutes (Mac Lane coherence for bicategories). The strictification licenses the convention of treating bicategories as strict in informal arguments.

Formal Restatement

Formal Restatement

Bicategory data. Objects; hom-categories $\mathcal{B}(a, b)$; composition functors $\circ_{a,b,c}$; identity 1-cells 1_a; invertible 2-cells $\alpha_{h,g,f}, \lambda_f, \rho_f$. Pentagon and triangle axioms.

Pseudofunctor. Object map; hom-functors $F_{a,b}$; invertible 2-cells $\phi_{g,f} : F(g) \circ F(f) \cong F(g \circ f)$ and $\phi_a : 1_{Fa} \cong F(1_a)$; coherence with α, λ, ρ. Lax: drop invertibility. Oplax: reverse direction.

Pseudonatural transformation $\eta : F \Rightarrow G$. 1-cells $\eta_a : Fa \to Ga$ and invertible 2-cells $\eta_f : Gf \circ \eta_a \cong \eta_b \circ Ff$ satisfying compatibility with ϕ^F, ϕ^G.

Modification $m : \eta \Rrightarrow \eta'$. 2-cells $m_a : \eta_a \Rightarrow \eta'_a$ compatible with η_f, η'_f.

Biequivalence. Pseudofunctor that is essentially surjective on objects (up to equivalence) and locally an equivalence.

Strictification (Theorem 25.25). For every bicategory $\mathcal{B}$ there is a strict 2-category $\mathcal{B}^{\mathrm{st}}$ and a biequivalence $\widehat{(-)} : \mathcal{B}^{\mathrm{st}} \to \mathcal{B}$.

Mac Lane coherence (Corollary 25.26). In any bicategory, every diagram of associators and unitors commutes.

Chapter 26 introduces ends and coends, *the foundational universal constructions for working with profunctors, weighted limits, and enriched Yoneda. Coends will be the calculus that powers most of the remaining chapters of Volume III.*

Chapter 26

Ends and Coends

Chapter Overview

Volume II made repeated unsupported promises about *ends* and *coends*: profunctor composition (Chapter 14) used the formula $(Q \circ P)(c,a) = \int^b Q(c,b) \times P(b,a)$ without explaining the integral sign; the Kan extension colimit formula $(\mathrm{Lan}_p\, F)(e) = \int^c \mathcal{E}(p(c),e) \cdot F(c)$ likewise sidestepped what $\int^c$ means; the enriched Yoneda lemma (Chapter 20) even *stated* its conclusion as an end $\int_c \mathcal{V}(G(c), H(c))$.

This chapter delivers on those promises. Ends and coends are the universal constructions for "mixed-variance" functors $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ — limits or colimits that, on the diagonal, "contract" the two copies of $\mathcal{C}$ into one. They are the foundational vocabulary of formal category theory: weighted limits, enriched Kan extensions, Day convolution, and the formal Yoneda lemma are all expressed in coend calculus. After this chapter, the rest of Volume III flows.

Where the Name Comes From

Dinatural transformation was introduced by *Eilenberg* and *Kelly* (1966) for transformations between mixed-variance functors. The prefix *di-* (Greek "two, twice") reflects the two-sided naturality condition: a dinatural transformation is natural in both variables of a bifunctor $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ *simultaneously*, via a hexagonal coherence condition rather than the square of ordinary naturality.

End ($\int_c$) and **coend** ($\int^c$) are due to *Yoneda*, popularized by *Day-Kelly* (1969). The integral notation is metaphorical (categorical-accumulation parallels classical integration) and surprisingly powerful: Fubini's theorem, Yoneda reduction, and many other coend-calculus tools mirror their classical analytic counterparts.

26.1 Dinatural Transformations

26.1.1 Formalizing

Let $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ be a functor — a *mixed-variance* or *bifunctor* in the contravariant- then-covariant sense. The objects of interest are the diagonal values $T(c,c)$ and how they fit together across $\mathcal{C}$.

Definition 26.1 (Dinatural transformation)**.** A **dinatural transformation** $\alpha : T \mathrel{\ddot{\Longrightarrow}} S$ between two such functors $T, S : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ is a family of morphisms $\alpha_c : T(c,c) \to S(c,c)$, one for each $c \in \mathcal{C}$, such that for every $f : c \to c'$ the hexagon

$$\begin{tikzcd}
& T(c,c) \arrow[r, "\alpha_c"] & S(c,c) \arrow[dr, "S(\mathrm{id},f)"] & \\
T(c',c) \arrow[ur, "T(f,\mathrm{id})"] \arrow[dr, "T(\mathrm{id},f)"'] & & & S(c,c') \\
& T(c',c') \arrow[r, "\alpha_{c'}"'] & S(c',c') \arrow[ur, "S(f,\mathrm{id})"'] &
\end{tikzcd}$$

commutes. We call this the **hexagonal** or **dinaturality** condition.

How to Say This

$\alpha : T \mathrel{\ddot{\Longrightarrow}} S$ *(read: "dinatural alpha from T to S")*, with the dotted arrow signaling that the naturality holds in the hexagonal (dinatural) sense rather than the square (natural) sense.

Remark 26.2 (When the hexagon collapses to a square)**.** If T and S are constant in their first argument — i.e., they factor through the projection $\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$ — the hexagon collapses to an ordinary naturality square. Thus ordinary natural transformations are the special case of dinatural transformations between covariant functors. The dinatural notion is strictly more general.

26.1.2 Reorganizing: Wedges as Dinatural Cones

Definition 26.3 (Wedge)**.** A **wedge** from an object $d \in \mathcal{D}$ to a functor $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ is a dinatural transformation $\omega : \Delta d \overset{..}{\Longrightarrow} T$ from the constant functor at d. Concretely: a family $\omega_c : d \to T(c,c)$ such that for every $f : c \to c'$ the square

$$\begin{tikzcd} d \arrow[r, "\omega_{c'}"] \arrow[d, "\omega_c"'] & T(c',c') \arrow[d, "T(f,\mathrm{id})"] \\ T(c,c) \arrow[r, "T(\mathrm{id},f)"'] & T(c,c') \end{tikzcd}$$

commutes. (The hexagon for Δd collapses because Δd has no non-trivial action.) Dually, a **cowedge** from T to d is a dinatural $\nu : T \overset{..}{\Longrightarrow} \Delta d$, i.e., a family $\nu_c : T(c,c) \to d$ satisfying the analogous square.

A wedge replaces "a cone from d" (in the limit picture) and a cowedge replaces "a cone over d" (in the colimit picture), now for the mixed-variance functor T.

26.2 The End

26.2.1 Formalizing

Definition 26.4 (End)**.** Let $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$. An **end** of T is an object $\int_c T(c,c) \in \mathcal{D}$ together with a universal wedge $\pi : \Delta \int_c T(c,c) \overset{..}{\Longrightarrow} T$: for every wedge $\omega : \Delta d \overset{..}{\Longrightarrow} T$ there exists a unique morphism $\hat{\omega} : d \to \int_c T(c,c)$ with $\pi_c \circ \hat{\omega} = \omega_c$ for all c.

How to Say This

$\int_c T(c,c)$ *(read: "the end over c of T(c, c)")* or just *(read: "end of T").* The subscript c on the integral sign marks the bound variable; it does not denote a particular value of c.

26.2.2 Reorganizing: Ends as Equalizers

Theorem 26.5 (End as an equalizer)**.** *The end $\int_c T(c,c)$, when $\mathcal{D}$ has products and equalizers and $\mathcal{C}$ is small, exists and is computed as the equalizer*

$$\int_c T(c,c) \xrightarrow{e} \prod_c T(c,c) \rightrightarrows \prod_{f:c\to c'} T(c,c')$$

where the two parallel arrows are induced respectively by $T(\mathrm{id},f)$ and $T(f,\mathrm{id})$, indexed over all morphisms f of $\mathcal{C}$.

An element of $\prod_c T(c,c)$ is a family $(\omega_c)_c$ with $\omega_c \in T(c,c)$.	*Universal property of the product.*
The two parallel arrows send (ω_c) to the families $(T(\mathrm{id},f)\omega_c)_f$ and $(T(f,\mathrm{id})\omega_{c'})_f$ in $\prod_f T(c,c')$.	*By definition of the induced maps into the product over morphisms.*
The equalizer condition $T(\mathrm{id},f)\omega_c = T(f,\mathrm{id})\omega_{c'}$ for every $f : c \to c'$ is precisely the wedge condition of Definition 26.3.	*Direct comparison of the equations.*
So the equalizer is the universal object equipped with a wedge $\pi_c : E \to T(c,c)$ for each c, satisfying exactly the wedge condition.	*Equalizer universal property.*
Therefore the equalizer is the end.	*By Definition 26.4.*

Corollary 26.6. *Ends exist whenever $\mathcal{D}$ is complete (has small limits) and $\mathcal{C}$ is small. They are limits — limits of a specific diagram of shape determined by T.*

26.2.3 Formally: Examples

Example 26.7 (Natural transformations as an end)**.** For functors $F, G : \mathcal{C} \to \mathcal{D}$, define $T(c,c') = \mathcal{D}(Fc, Gc')$. Then

$$\int_c \mathcal{D}(Fc, Gc) \;\cong\; \mathrm{Nat}(F,G).$$

The end's wedge $\pi_c : \mathrm{Nat}(F,G) \to \mathcal{D}(Fc,Gc)$ is evaluation at c; the wedge condition is exactly the naturality square.

Example 26.8 (Enriched natural transformations)**.** For $\mathcal{V}$-functors $G, H : \mathcal{C} \to \mathcal{V}$ between $\mathcal{V}$-categories, the object of $\mathcal{V}$-natural transformations is the end

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(G,H) \;=\; \int_c \mathcal{V}(Gc, Hc)$$

in $\mathcal{V}$ (cf. Chapter 20). This is the enriched generalization of Example 26.7.

Example 26.9 (Center of a category)**.** For a category $\mathcal{C}$ viewed as a one-object 2-category, the end $\int_c \mathcal{C}(c,c)$ over $\mathcal{C}$ itself (with $T(c,c') = \mathcal{C}(c,c')$) is the monoid of **natural endomorphisms** of the identity functor — the *center* of $\mathcal{C}$.

26.3 The Coend

26.3.1 Formalizing

Definition 26.10 (Coend)**.** The **coend** $\int^c T(c,c) \in \mathcal{D}$ of a functor $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ is an object equipped with a universal cowedge $\iota : T \stackrel{..}{\Longrightarrow} \Delta \int^c T(c,c)$: for every cowedge $\nu : T \stackrel{..}{\Longrightarrow} \Delta d$ there is a unique morphism $\hat{\nu} : \int^c T(c,c) \to d$ with $\hat{\nu} \circ \iota_c = \nu_c$ for all c.

Theorem 26.11 (Coend as a coequalizer)**.** *When $\mathcal{D}$ has coproducts and coequalizers and $\mathcal{C}$ is small, the coend exists and is the coequalizer*

$$\coprod_{f:c\to c'} T(c',c) \rightrightarrows \coprod_c T(c,c) \xrightarrow{\iota} \int^c T(c,c).$$

The two parallel arrows are induced by $T(f,\mathrm{id}) : T(c',c) \to T(c,c)$ and $T(\mathrm{id},f) : T(c',c) \to T(c',c')$.

Proof. Dual to Theorem 26.5, with products replaced by coproducts, equalizers by coequalizers, and the universal property reversed. A cowedge from T to d is precisely a family (ν_c) satisfying $\nu_c \circ T(f,\mathrm{id}) = \nu_{c'} \circ T(\mathrm{id},f)$ for each f, which is the coequalizer condition. □

26.3.2 Reorganizing: The Tensor-Product Picture

The coend $\int^c T(c,c)$ admits a useful informal reading as a *tensor product*. For $\mathcal{D} = \mathbf{Set}$ and $T(c,c') = P(c) \times Q(c')$ with P contravariant and Q covariant,

$$\int^c P(c) \times Q(c) \;\cong\; \frac{\coprod_c P(c) \times Q(c)}{\sim}$$

where $\sim$ identifies $(P(f)(p),q) \sim (p,Q(f)(q))$ for $f : c \to c'$. This is the standard quotient defining a tensor product. In general, "$\int^c$" is the tensor product over $\mathcal{C}$ in the same sense that tensor products of modules are coequalizers over a ring.

26.3.3 Formally: Examples

Example 26.12 (Profunctor composition)**.** For profunctors $P : \mathcal{A} \to \mathcal{B}$ (i.e., $P : \mathcal{B}^{\mathrm{op}} \times \mathcal{A} \to \mathbf{Set}$) and $Q : \mathcal{B} \to \mathcal{C}$, their composition is

$$(Q \circ P)(c,a) \;=\; \int^{b\in\mathcal{B}} Q(c,b) \times P(b,a).$$

This is the formula promised in Chapter 14. The coend identifies pairs $(P(f)(p),q) \sim (p,Q(f)(q))$ along the morphism $f : b \to b'$, the correct "mixing" of P's output with Q's input.

Example 26.13 (Left Kan extension)**.** For $F : \mathcal{C} \to \mathcal{E}$ and $K : \mathcal{C} \to \mathcal{D}$,

$$(\mathrm{Lan}_K F)(d) \;\cong\; \int^c \mathcal{D}(Kc,d) \cdot F(c)$$

where $S \cdot X$ for $S \in \mathbf{Set}$ and $X \in \mathcal{E}$ denotes the copower $\coprod_S X$. This is the formula from Chapter 18, now with the integral sign defined.

Example 26.14 (Geometric realization)**.** The geometric realization of a simplicial set $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$ is

$$|X| \;=\; \int^{[n]\in\Delta} X_n \cdot \Delta^n$$

where Δ^n is the topological n-simplex and $\cdot$ is the topological copower $S \cdot Y = \coprod_S Y$. The coend identifies face and degeneracy maps with their topological counterparts.

26.4 The Coend Calculus

The reason ends and coends earn the integral notation is that they obey calculus-like rules: a Fubini theorem for swapping the order of integration, a substitution rule, and a Yoneda reduction. These rules turn coend manipulations into mechanical computations.

26.4.1 Fubini

Theorem 26.15 (Fubini for ends)**.** *For a functor $T : (\mathcal{C} \times \mathcal{D})^{\mathrm{op}} \times (\mathcal{C} \times \mathcal{D}) \to \mathcal{E}$,*

$$\int_{(c,d)} T((c,d),(c,d)) \;\cong\; \int_c \int_d T((c,d),(c,d)) \;\cong\; \int_d \int_c T((c,d),(c,d))$$

whenever the relevant limits exist.

Theorem 26.16 (Fubini for coends)**.** *Dually,*

$$\int^{(c,d)} T((c,d),(c,d)) \;\cong\; \int^{c}\int^{d} T((c,d),(c,d)) \;\cong\; \int^{d}\int^{c} T((c,d),(c,d)).$$

Sketch. Both statements follow from the fact that ends are limits and coends are colimits, and limits/colimits over a product category factor through limits/ colimits over each factor (Fubini for ordinary limits). The end of a mixed-variance functor uses the same Fubini at the level of equalizer diagrams. □

26.4.2 Yoneda Reduction

Theorem 26.17 (Yoneda reduction — coend form)**.** *For a (covariant) functor $F : \mathcal{C} \to \mathbf{Set}$,*

$$\int^{c} \mathcal{C}(c,c') \times F(c) \;\cong\; F(c').$$

Dually for $G : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$,

$$\int^{c} G(c) \times \mathcal{C}(c',c) \;\cong\; G(c').$$

How to Say This

This is called the **ninja Yoneda lemma**: the variable c is "integrated out" against the representable $\mathcal{C}(c,c')$, leaving only the value at c'. The name comes from how silently and decisively the coend disappears.

Compute the coend explicitly as the coequalizer $\coprod_{f:c\to d} \mathcal{C}(d,c') \times F(c) \rightrightarrows \coprod_c \mathcal{C}(c,c') \times F(c)$.	*By Theorem 26.11.*
The two parallel arrows identify $(\mathcal{C}(d,c') \ni g, F(c) \ni x) \sim (g \circ f, x)$ vs. $(g, F(f)(x))$ along $f : c \to d$.	*Direct substitution.*
Define $\phi : \coprod_c \mathcal{C}(c,c') \times F(c) \to F(c')$ by $(g,x) \mapsto F(g)(x)$.	*The candidate map to $F(c')$.*
ϕ respects the equivalence relation: $\phi(g \circ f, x) = F(g \circ f)(x) = F(g)(F(f)(x)) = \phi(g, F(f)(x))$.	*By functoriality of F.*
Therefore ϕ factors through the coend to give $\hat{\phi} : \int^c \mathcal{C}(c,c') \times F(c) \to F(c')$.	*By the universal property of the coequalizer.*
Construct the inverse $\psi : F(c') \to \int^c \mathcal{C}(c,c') \times F(c)$ by $x \mapsto [\mathrm{id}_{c'}, x]$ (the equivalence class).	*The standard "identity-and-itself" element.*
$\hat{\phi} \circ \psi = \mathrm{id}_{F(c')}$: $\hat{\phi}([\mathrm{id}, x]) = F(\mathrm{id})(x) = x$.	*Identity action.*
$\psi \circ \hat{\phi} = \mathrm{id}$ on classes: for (g,x) with $g : c \to c'$, $\psi(\phi(g,x)) = [\mathrm{id}_{c'}, F(g)(x)] \sim [g,x]$ via the relation at $f = g$.	*The relation collapses every class to a singleton with identity.*
Both composites are the identity; isomorphism established.	*Yoneda reduction.*

Remark 26.18 (Why this is Yoneda)**.** Theorem 26.17 is the Yoneda lemma in disguise. The ordinary Yoneda lemma says elements of $F(c')$ correspond to natural transformations $\mathcal{C}(c',-) \Rightarrow F$, which are an end: $F(c') \cong \int_c \mathbf{Set}(\mathcal{C}(c',c), F(c))$. Taking the hom-set adjunction backwards (using that the coend is the colimit and the end is the limit) gives the coend formulation. The two forms are duals: ends pair with hom-functors covariantly; coends pair with hom-functors contravariantly.

26.4.3 The Co-Yoneda Lemma

Theorem 26.19 (Co-Yoneda / density)**.** *For any functor $F : \mathcal{C} \to \mathbf{Set}$,*

$$F \;\cong\; \int^{c} \mathcal{C}(c,-) \cdot F(c)$$

as functors $\mathcal{C} \to \mathbf{Set}$. Every functor is a colimit of representables.

Proof. At each $c' \in \mathcal{C}$, applying the formula gives $\int^c \mathcal{C}(c,c') \times F(c) \cong F(c')$ by Theorem 26.17. Naturality in c' is automatic because the coend is functorial in its non-integrated arguments. □

This is the precise statement that "the presheaf category is the free cocompletion of $\mathcal{C}$" (Chapter 18, Chapter 24): every presheaf is canonically a colimit of representables, witnessed by the coend formula.

26.5 Ends and Coends in Enriched Settings

26.5.1 Formalizing

The development so far has assumed $\mathcal{D}$ is an ordinary category. In the enriched setting, $\mathcal{D} = \mathcal{V}$ for a complete or cocomplete monoidal category $\mathcal{V}$, and the functor T is a $\mathcal{V}$-functor. The definitions transpose verbatim:

Definition 26.20 (Enriched end and coend)**.** For a $\mathcal{V}$-functor $T : \mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$ (where $\mathcal{C}^{\mathrm{op}} \otimes \mathcal{C}$ is the tensor product of $\mathcal{V}$-categories), the **enriched end** $\int_c T(c,c)$ is the equalizer in $\mathcal{V}$ of two maps $\prod_c T(c,c) \to \prod_{c,c'} \mathcal{V}(\mathcal{C}(c,c'), T(c,c'))$ encoding the wedge condition in $\mathcal{V}$. Dually for coend.

Proposition 26.21 (Enriched natural transformations as enriched end)**.** *For $\mathcal{V}$-functors $G, H : \mathcal{C} \to \mathcal{V}$,*

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(G, H) \;=\; \int_c \mathcal{V}(Gc, Hc).$$

This is the statement that closed (and complete enough) Chapter 20 made without proof; the foundation is now in place.

26.5.2 Reorganizing

In an enriched setting, the coend calculus rules — Fubini and Yoneda reduction — continue to hold with the same proofs, replacing “set products” by “$\mathcal{V}$-tensors” and “set sums” by “$\mathcal{V}$-cotensors”. The ninja Yoneda lemma in enriched form reads

$$\int^c \mathcal{C}(c, c') \otimes F(c) \;\cong\; F(c')$$

in $\mathcal{V}$, where $\otimes$ is the $\mathcal{V}$-tensor.

26.6 Exercises

What is a dinatural transformation $\alpha : T \stackrel{..}{\Longrightarrow} S$?	A family $\alpha_c : T(c,c) \to S(c,c)$ satisfying a hexagonal coherence condition involving both $T(f, \mathrm{id})$ and $T(\mathrm{id}, f)$ for each $f : c \to c'$.
What is a wedge from d to T?	A dinatural transformation $\Delta d \stackrel{..}{\Longrightarrow} T$, equivalently a family $\omega_c : d \to T(c,c)$ satisfying $T(\mathrm{id}, f)\omega_c = T(f, \mathrm{id})\omega_{c'}$ for every $f : c \to c'$.
Define the end of $T : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$.	An object $\int_c T(c,c) \in \mathcal{D}$ together with a universal wedge $\pi : \Delta \int_c T(c,c) \stackrel{..}{\Longrightarrow} T$.
Express the end as an equalizer.	$\int_c T(c,c) = \mathrm{eq}(\prod_c T(c,c) \rightrightarrows \prod_{f : c \to c'} T(c, c'))$, where the two arrows are induced by $T(\mathrm{id}, f)$ and $T(f, \mathrm{id})$.
When does an end exist?	When $\mathcal{D}$ is complete (has small limits) and $\mathcal{C}$ is small. Ends are limits.
Show that natural transformations form an end.	$\mathrm{Nat}(F, G) = \int_c \mathcal{D}(Fc, Gc)$: the wedge condition is the naturality square.
What is the coend of T?	An object $\int^c T(c,c)$ with universal cowedge $\iota : T \stackrel{..}{\Longrightarrow} \Delta \int^c T(c,c)$ — dual to the end.
Express the coend as a coequalizer.	$\int^c T(c,c) = \mathrm{coeq}(\coprod_{f : c \to c'} T(c', c) \rightrightarrows \coprod_c T(c,c))$.
What is the tensor-product picture of a coend in **Set**?	$\int^c P(c) \times Q(c) \cong (\coprod_c P(c) \times Q(c)) / \sim$, where $(P(f)(p), q) \sim (p, Q(f)(q))$ for $f : c \to c'$.
Write the profunctor composition formula.	$(Q \circ P)(c, a) = \int^b Q(c, b) \times P(b, a)$ for $P : \mathcal{B}^{\mathrm{op}} \times \mathcal{A} \to \mathbf{Set}$, $Q : \mathcal{C}^{\mathrm{op}} \times \mathcal{B} \to \mathbf{Set}$.
Write the left Kan extension formula as a coend.	$(\mathrm{Lan}_K F)(d) = \int^c \mathcal{D}(Kc, d) \cdot F(c)$, where $S \cdot X$ is the copower.
State the Fubini theorem for coends.	$\int^{(c,d)} T = \int^c \int^d T = \int^d \int^c T$, when the coends exist.
State the Yoneda reduction (ninja Yoneda).	$\int^c \mathcal{C}(c, c') \times F(c) \cong F(c')$ for $F : \mathcal{C} \to \mathbf{Set}$.

Why is the ninja Yoneda lemma equivalent to ordinary Yoneda?	Both express the same fact that F is determined by its values at all c via the representables $\mathcal{C}(c,-)$; one as an end (Hom-out of representables), the other as a coend (tensor with representables).
State the co-Yoneda / density theorem in coend form.	For $F : \mathcal{C} \to \mathbf{Set}$, $F(-) \cong \int^c \mathcal{C}(c,-) \cdot F(c)$. Every functor is a colimit of representables.
What is the contravariant Yoneda reduction?	$\int^c G(c) \times \mathcal{C}(c', c) \cong G(c')$ for $G : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.
Why does the hexagonal condition collapse to a square for wedges?	Because the constant functor Δd has trivial action, so $\Delta d(f, \mathrm{id}) = \Delta d(\mathrm{id}, f) = \mathrm{id}_d$; the hexagon's two "$\Delta d$ sides" become the identity.
What is the center of a category as an end?	$Z(\mathcal{C}) = \int_c \mathcal{C}(c, c)$, the monoid of natural endomorphisms of $\mathrm{Id}_{\mathcal{C}}$.
Geometric realization as a coend?	$\lvert X\rvert = \int^{[n]\in\Delta} X_n \cdot \Delta^n$, where $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$, Δ^n is the topological n-simplex, and $\cdot$ is the copower.
Define the enriched end.	For $T : \mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$, the equalizer in $\mathcal{V}$ of $\prod_c T(c, c) \rightrightarrows \prod_{c,c'} \mathcal{V}(\mathcal{C}(c, c'), T(c, c'))$.
What is the enriched object of $\mathcal{V}$-natural transformations?	$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(G, H) = \int_c \mathcal{V}(Gc, Hc)$ in $\mathcal{V}$.
When are ordinary natural transformations the special case of dinatural ones?	When T and S factor through the projection to $\mathcal{C}$ (i.e., are covariant), the hexagonal condition collapses to the naturality square.
What is a cowedge?	A dinatural transformation $T \overset{..}{\Longrightarrow} \Delta d$: a family $\nu_c : T(c, c) \to d$ with $\nu_c \circ T(f, \mathrm{id}) = \nu_{c'} \circ T(\mathrm{id}, f)$.
What is the relationship between ends/coends and ordinary limits/colimits?	Ends are limits and coends are colimits of specific diagrams. The integral notation captures the "contraction along the diagonal" aspect that ordinary limit notation hides.
Why does the Yoneda reduction fail without an integration variable?	Because the reduction $\int^c \mathcal{C}(c, c') \cdot F(c) \cong F(c')$ is exactly the substitution of $c = c'$ that the integral notation indicates; without integration, the formula collapses to a single class and loses naturality.
Coend version of currying for **Set**-valued profunctors?	$[\mathcal{A}^{\mathrm{op}}, [\mathcal{B}, \mathbf{Set}]] \simeq [\mathcal{A}^{\mathrm{op}} \times \mathcal{B}, \mathbf{Set}]$ — both expressible via coends/ends in symmetric ways.
Why is Fubini for coends "the same" as Fubini for double colimits?	Because a coend is the colimit of a specific double diagram, and Fubini for colimits says colimits over $\mathcal{C} \times \mathcal{D}$ may be computed iteratively.
Write the coend formula for "functor evaluation at a" using representables.	$F(a) \cong \int^c \mathcal{C}(c, a) \cdot F(c)$ — special case of Yoneda reduction with $c' = a$.
If $\mathcal{C}$ is discrete, what is $\int^c F(c) \times G(c)$?	$\coprod_c F(c) \times G(c)$ — no morphisms means no identifications, so the coend reduces to the coproduct.
Why is the coend the "right" way to write profunctor composition?	Because both projections from $T(c, c) = Q(c, b) \times P(b, a)$ — the contravariant action of b on Q and the covariant action of b on P — must be identified, exactly the coend's quotient relation.

Chapter Summary

Chapter Summary

Dinatural transformations. For mixed-variance functors $T, S : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$, a family $\alpha_c : T(c,c) \to S(c,c)$ satisfying a hexagonal coherence. The constant case collapses to wedges and cowedges — the dinatural analogues of cones.
End. $\int_c T(c,c)$ is the universal wedge to T; equivalently, the equalizer of two maps from $\prod_c T(c,c)$ to $\prod_{c \to c'} T(c,c')$. Natural transformations are ends: $\mathrm{Nat}(F,G) = \int_c \mathcal{D}(Fc, Gc)$.
Coend. $\int^c T(c,c)$ is the universal cowedge from T; equivalently, the coequalizer of two maps to $\coprod_c T(c,c)$. Profunctor composition and left Kan extensions are coends.
The coend calculus. Fubini for ends and coends (swap order of integration); Yoneda reduction (the "ninja Yoneda lemma": $\int^c \mathcal{C}(c,c') \times F(c) \cong F(c')$); the co-Yoneda / density theorem (every functor is a colimit of representables).
Enriched ends and coends. The same definitions transpose to $\mathcal{V}$-functors, with $\mathcal{V}$-products and $\mathcal{V}$-coproducts replacing set products and sums. The enriched natural transformations object of Chapter 20 is the enriched end $\int_c \mathcal{V}(Gc, Hc)$.

Formal Restatement

Formal Restatement

Dinatural transformation $T \overset{\cdot\cdot}{\Longrightarrow} S$**.** Family $\alpha_c : T(c,c) \to S(c,c)$ with hexagonal coherence: $S(\mathrm{id}, f)\alpha_c T(f, \mathrm{id}) = S(f, \mathrm{id})\alpha_{c'} T(\mathrm{id}, f)$ for each $f : c \to c'$.

End.

$$\int_c T(c,c) \;=\; \mathrm{eq}\Big(\prod_c T(c,c) \rightrightarrows \prod_{f : c \to c'} T(c,c')\Big).$$

Coend.

$$\int^c T(c,c) \;=\; \mathrm{coeq}\Big(\coprod_{f : c \to c'} T(c',c) \rightrightarrows \coprod_c T(c,c)\Big).$$

Natural transformations as end. $\mathrm{Nat}(F,G) = \int_c \mathcal{D}(Fc, Gc)$.

Fubini. $\int_{(c,d)} T = \int_c \int_d T = \int_d \int_c T$; coend version dual.

Yoneda reduction (ninja Yoneda). $\int^c \mathcal{C}(c,c') \times F(c) \cong F(c')$ for $F : \mathcal{C} \to \mathbf{Set}$; dually, $\int^c G(c) \times \mathcal{C}(c',c) \cong G(c')$ for $G : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$.

Co-Yoneda / density. $F \cong \int^c \mathcal{C}(c, -) \cdot F(c)$.

Profunctor composition. $(Q \circ P)(c,a) = \int^b Q(c,b) \times P(b,a)$.

Left Kan extension formula. $(\mathrm{Lan}_K F)(d) = \int^c \mathcal{D}(Kc, d) \cdot F(c)$.

Enriched end. For $T : \mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$, $\int_c T(c,c) \in \mathcal{V}$ defined via equalizer of $\mathcal{V}$-products encoding the $\mathcal{V}$-wedge condition.

Chapter 27 uses the coend calculus to develop weighted limits and colimits *in full generality. The Yoneda reduction is the calculation tool that powers most of the enriched-categorical theory to follow.*

Chapter 27

Weighted Limits and Colimits

Chapter Overview

Chapter 20 introduced weighted limits in the enriched setting: a W-weighted limit $\{W, F\}$ of $F : \mathcal{J} \to \mathcal{C}$ weighted by $W : \mathcal{J} \to \mathcal{V}$ represents the $\mathcal{V}$-functor $c \mapsto [\mathcal{J}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$. Ordinary limits were the case $W = \Delta I$. This chapter uses the coend calculus of Chapter 26 to develop the full theory. We re-derive weighted limits and colimits via ends and coends, prove the central representability theorem, work out the principal examples (cotensor, conical, lax limit, comma object), and show how all of ordinary limits-and-colimits theory transports to the weighted setting. The chapter closes with a preview of weighted Kan extensions (Chapter 31): in the enriched world, *every* (co)limit is a Kan extension and *every* Kan extension is a weighted (co)limit.

Where the Name Comes From

Weighted limit (also *indexed limit*) generalizes ordinary limits to the enriched setting by allowing the "shape" of the limit to be modulated by a weighting functor $W : \mathcal{J} \to \mathcal{V}$. The terminology is Kelly's (Vol II Ch. 20). The intuition: ordinary limits compute by taking the "equal-weight average" of a diagram (weight $W = \text{const } \mathbf{1}$); weighted limits allow nontrivial weight distributions. Modern enriched category theory works almost entirely in the weighted setting, since ordinary limits are too restrictive in $\mathcal{V}$-categories.

27.1 Weighted Limits via Ends

27.1.1 Formalizing

Fix a complete symmetric monoidal closed category $\mathcal{V}$ and a small $\mathcal{V}$-category $\mathcal{J}$. Let $F : \mathcal{J} \to \mathcal{C}$ be a $\mathcal{V}$-functor (a *diagram*) and $W : \mathcal{J} \to \mathcal{V}$ another $\mathcal{V}$-functor (a *weight*).

Definition 27.1 (Weighted limit)**.** The **weighted limit** of F by W, written $\{W, F\}$ or $\lim^W F$, is an object of $\mathcal{C}$ — when it exists — together with a $\mathcal{V}$-natural isomorphism

$$\mathcal{C}(c, \{W, F\}) \;\cong\; [\mathcal{J}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$$

in $\mathcal{V}$, naturally in $c \in \mathcal{C}$.

How to Say This

$\{W, F\}$ *(read: "the W-weighted limit of F")* or *(read: "the limit of F weighted by W").*

Theorem 27.2 (End formula for weighted limits)**.** *The right-hand side of Definition 27.1 is an end:*

$$[\mathcal{J}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-)) \;=\; \int_{j \in \mathcal{J}} \mathcal{V}(W(j), \mathcal{C}(c, Fj)).$$

Proof. This is the end formula for $\mathcal{V}$-natural transformations (Chapter 26, Proposition on enriched ends). Apply it with $G = W$ and $H = \mathcal{C}(c, F-)$. ☐

Combining:

$$\mathcal{C}(c, \{W, F\}) \;\cong\; \int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj)).$$

27.1.2 Reorganizing: Conical Limits as ΔI-Weighted

Proposition 27.3 (Ordinary limits as weighted limits)**.** *For $\mathcal{V} = \mathbf{Set}$ and $W = \Delta\{*\}$ the terminal weight, $\{\Delta\{*\}, F\} = \lim F$, the ordinary (**conical**) limit.*

Proof. The end formula gives

$$\mathcal{C}(c, \{\Delta\{*\}, F\}) = \int_j \mathbf{Set}(\{*\}, \mathcal{C}(c, Fj)) = \int_j \mathcal{C}(c, Fj) = \mathrm{Nat}(\Delta c, F) = \mathrm{cones}(c, F),$$

the set of cones from c to F. By the universal property of the limit, this is $\mathcal{C}(c, \lim F)$, so $\{\Delta\{*\}, F\} \cong \lim F$. □

27.1.3 Formally: Other Weighted Limits Worth Naming

Example 27.4 (Cotensor)**.** For a single-object indexing category $\mathcal{J} = \mathbf{1}$ (terminal $\mathcal{V}$-category), F is a single object $X \in \mathcal{C}$, W is a single object $V \in \mathcal{V}$, and the weighted limit is the **cotensor** $V \pitchfork X$ (also written X^V or $[V, X]_{\mathcal{C}}$):

$$\mathcal{C}(c, V \pitchfork X) \;\cong\; \mathcal{V}(V, \mathcal{C}(c, X)).$$

When $\mathcal{C} = \mathcal{V}$ self-enriched, this is the internal hom $[V, X]$.

Example 27.5 (Lax limit)**.** For $\mathcal{J} = \mathbf{2}$ (the walking arrow) and $W : \mathbf{2} \to \mathbf{Cat}$ sending the arrow to a specific 1-cell, the weighted limit is the **lax limit** or *comma object*. In **Cat**, this is the comma category $(F \downarrow G)$ when $F = W(0)$ and $G = W(1)$.

Example 27.6 (Inserter, equifier, descent)**.** More elaborate weights generate *inserters* (a 1-cell making two given 2-cells equal up to a new mediating 2-cell), *equifiers* (objects that equify two 2-cells), and *descent objects*. These are the standard *flexible limits* of 2-category theory and arise as weighted limits with explicit small **Cat**-valued weights.

27.2 Weighted Colimits via Coends

27.2.1 Formalizing

Definition 27.7 (Weighted colimit)**.** For a diagram $F : \mathcal{J} \to \mathcal{C}$ and weight $W : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$ (note the variance), the **weighted colimit** is an object $W \star F \in \mathcal{C}$ — when it exists — with a $\mathcal{V}$-natural isomorphism

$$\mathcal{C}(W \star F, c) \;\cong\; [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(F-, c)).$$

How to Say This

$W \star F$ *(read: "W-weighted colimit of F").* The star is the standard shorthand for the weighted colimit construction.

Theorem 27.8 (Coend formula for weighted colimits)**.** *In a tensored $\mathcal{V}$-category $\mathcal{C}$ (where $V \otimes c$ exists for $V \in \mathcal{V}$ and $c \in \mathcal{C}$ with the adjunction $\mathcal{C}(V \otimes c, d) \cong \mathcal{V}(V, \mathcal{C}(c, d))$), the weighted colimit is given by the coend*

$$W \star F \;\cong\; \int^j W(j) \otimes F(j).$$

Compute $\mathcal{C}(W \star F, c)$ using the proposed coend formula.	*Substitute the candidate into the hom-functor.*
$\mathcal{C}\left(\int^j W(j) \otimes F(j), c\right) \cong \int_j \mathcal{C}(W(j) \otimes F(j), c)$.	*Hom-functor turns coends into ends in the first variable (continuity).*
$\int_j \mathcal{C}(W(j) \otimes F(j), c) \cong \int_j \mathcal{V}(W(j), \mathcal{C}(F(j), c))$.	*By the tensor-cotensor adjunction of Definition assumption.*
$\int_j \mathcal{V}(W(j), \mathcal{C}(F(j), c)) = [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(F-, c))$.	*End formula for enriched natural transformations (Theorem 27.2).*
Therefore $\mathcal{C}\left(\int^j W(j) \otimes F(j), c\right) \cong \mathcal{C}(W \star F, c)$ $\mathcal{V}$-naturally in c.	*Composing the chain of isomorphisms.*
By the Yoneda lemma, $\int^j W(j) \otimes F(j) \cong W \star F$.	*Two objects with $\mathcal{V}$-naturally isomorphic hom-out functors are isomorphic.*

Corollary 27.9 (Conical colimit as weighted)**.** $\Delta I \star F \cong \operatorname{colim} F$ *(the ordinary colimit), where $\Delta I : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$ is the constant unit.*

27.2.2 Reorganizing: Tensor as Weighted Colimit

Example 27.10 (Tensor)**.** For $\mathcal{J} = \mathbf{1}$, F a single object X, W a single object V, the weighted colimit is the **tensor** $V \otimes X$ (also written $V \cdot X$ in **Set**-enriched settings):

$$\mathcal{C}(V \otimes X, c) \;\cong\; \mathcal{V}(V, \mathcal{C}(X, c)).$$

This is the dual of cotensor (Example 27.4); together they characterize $\mathcal{V}$-tensored-and-cotensored categories.

Example 27.11 (Coend as weighted colimit of a profunctor)**.** For $T : \mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$, treating T as a diagram $\mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$ and the hom-profunctor $\mathcal{C}(-,-)$ as the weight, the coend $\int^c T(c, c)$ is the weighted colimit of T along the hom-weight.

27.3 Functoriality and Yoneda Reduction

27.3.1 Formalizing

Proposition 27.12 (Functoriality of weighted limits)**.** *$\{-,-\} : [\mathcal{J},\mathcal{V}]^{\mathrm{op}} \otimes [\mathcal{J},\mathcal{C}] \to \mathcal{C}$ is a $\mathcal{V}$-functor: weighted limits are functorial in both weight and diagram, contravariantly in the weight.*

Proof. The defining isomorphism $\mathcal{C}(c,\{W,F\}) \cong \int_j \mathcal{V}(Wj,\mathcal{C}(c,Fj))$ is $\mathcal{V}$-natural in c, but also in W (contravariantly) and F (covariantly) by the universal property of the end. Apply Yoneda in c to lift naturality from the hom-functors to $\{W,F\}$ itself. □

Theorem 27.13 (Yoneda reduction for weighted (co)limits)**.** *For $F : \mathcal{J} \to \mathcal{C}$ and $j_0 \in \mathcal{J}$:*

$$\begin{aligned} \{\mathcal{J}(j_0,-),F\} &\cong F(j_0) \quad \textit{(weighted limit)} \\ \mathcal{J}(-,j_0) \star F &\cong F(j_0) \quad \textit{(weighted colimit)} \end{aligned}$$

For the weighted limit: $\mathcal{C}(c,\{\mathcal{J}(j_0,-),F\}) \cong \int_j \mathcal{V}(\mathcal{J}(j_0,j),\mathcal{C}(c,Fj))$.	*Apply the defining formula.*
$\int_j \mathcal{V}(\mathcal{J}(j_0,j),\mathcal{C}(c,Fj))$ is the $\mathcal{V}$-natural transformations from $\mathcal{J}(j_0,-)$ to $\mathcal{C}(c,F-)$.	*By the end formula for $\mathcal{V}$-naturals.*
By the (enriched) Yoneda lemma, this equals $\mathcal{C}(c,F(j_0))$.	*Yoneda applied to the $\mathcal{V}$-functor $\mathcal{C}(c,F-) : \mathcal{J} \to \mathcal{V}$.*
So $\mathcal{C}(c,\{\mathcal{J}(j_0,-),F\}) \cong \mathcal{C}(c,F(j_0))$ $\mathcal{V}$-naturally in c.	*Combining the above.*
By Yoneda in c, $\{\mathcal{J}(j_0,-),F\} \cong F(j_0)$.	*Yoneda for objects.*
For the weighted colimit: $\mathcal{C}(\mathcal{J}(-,j_0) \star F, c) \cong \int_j \mathcal{V}(\mathcal{J}(j,j_0),\mathcal{C}(Fj,c))$.	*Apply the defining formula.*
This is the $\mathcal{V}$-naturals $\mathcal{J}(-,j_0) \Rightarrow \mathcal{C}(F-,c)$.	*End formula again.*
By contravariant Yoneda, equals $\mathcal{C}(F(j_0),c)$.	*Yoneda applied to $\mathcal{C}(F-,c) : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$.*
Hence $\mathcal{J}(-,j_0) \star F \cong F(j_0)$.	*Yoneda for objects.*

Remark 27.14 (Representable weights pick out values)**.** Theorem 27.13 says: *representable weights evaluate the diagram.* This is the weighted analogue of the fact that "the limit over a singleton diagram is the object itself."

27.3.2 Reorganizing: Bilinearity and Distributivity

Proposition 27.15 (Colimits-into-limits commutation)**.** *For $W : \mathcal{J} \to \mathcal{V}$, $F : \mathcal{J} \to \mathcal{C}$, and $G : \mathcal{K} \to \mathcal{C}$:*

$$\{W, \overset{V}{\lim}\, G \circ F^*\} \;\cong\; \overset{V}{\lim}\{W, G \circ F^*\}$$

whenever the relevant weighted limits exist. Symbolically: limits commute with limits (here weighted limits with weighted limits in the diagram).

Proof. Apply the defining isomorphism, push the hom-functor through the inner limit (right adjoints preserve limits), and recognize the end-of-end as the end-over-the-product (Fubini for ends, Chapter 26). □

Remark 27.16. The dual statement — weighted colimits commute with weighted colimits — also holds, by Fubini for coends. These two statements together make the *calculus* of weighted (co)limits as smooth as the calculus of ordinary (co)limits, with one extra dimension of flexibility (the weight).

27.4 Existence and Completeness

27.4.1 Formalizing

Definition 27.17 (Weighted completeness)**.** A $\mathcal{V}$-category $\mathcal{C}$ is **$\mathcal{V}$-complete** if $\{W,F\}$ exists for every small diagram $F : \mathcal{J} \to \mathcal{C}$ and weight $W : \mathcal{J} \to \mathcal{V}$. Dually, **$\mathcal{V}$-cocomplete** for weighted colimits.

Theorem 27.18 (Weighted completeness from cotensors and conical limits)**.** *A tensored and cotensored $\mathcal{V}$-category $\mathcal{C}$ is $\mathcal{V}$-complete iff its underlying ordinary category is complete (has all small conical limits). In that case,*

$$\{W,F\} \;\cong\; \lim_{j \in \mathcal{J}} W(j) \pitchfork F(j),$$

where the outer lim *is the ordinary conical limit and $W(j) \pitchfork F(j)$ is the cotensor.*

Compute $\mathcal{C}(c, \{W, F\}) \cong \int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj))$.	*End formula.*
$\int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj)) \cong \int_j \mathcal{C}(c, W(j) \pitchfork F(j))$.	*By the cotensor adjunction* $\mathcal{V}(V, \mathcal{C}(c, X)) \cong \mathcal{C}(c, V \pitchfork X)$.
$\int_j \mathcal{C}(c, W(j) \pitchfork F(j)) \cong \mathcal{C}(c, \lim_j W(j) \pitchfork F(j))$.	*Hom-functor is continuous; takes ends (limits) in to limits.*
Therefore $\{W, F\} \cong \lim_j W(j) \pitchfork F(j)$ by Yoneda.	*Two objects with isomorphic hom-out are isomorphic.*
Existence: if conical limits and cotensors exist, the right-hand side does, so does the weighted limit.	*Reduction.*
Conversely, taking $W = \Delta I$ recovers conical limits; taking $\mathcal{J}$ discrete recovers cotensors.	*The two pieces are special weighted limits.*

Corollary 27.19. *A symmetric monoidal closed complete category $\mathcal{V}$ is $\mathcal{V}$-complete when self-enriched. (Cotensors are internal homs.)*

27.4.2 Reorganizing: Cocompleteness Dually

Theorem 27.20 (Weighted cocompleteness from tensors and conical colimits)**.** *A tensored $\mathcal{V}$-category $\mathcal{C}$ with all conical colimits is $\mathcal{V}$-cocomplete, and*

$$W \star F \;\cong\; \operatorname{colim}_j W(j) \otimes F(j).$$

Proof. Dual to Theorem 27.18. Replace ends by coends, products by coproducts, $\pitchfork$ by $\otimes$, lim by colim. □

27.5 Weighted Kan Extensions: A Preview

The weighted-limit framework gives an immediate definition of Kan extensions in the enriched setting. We state the formula here; Chapter 31 develops the general theory.

Definition 27.21 (Weighted Kan extension)**.** For a $\mathcal{V}$-functor $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$, the **(pointwise) left Kan extension** $\operatorname{Lan}_K F : \mathcal{D} \to \mathcal{E}$ is defined by the weighted colimit

$$(\operatorname{Lan}_K F)(d) \;=\; \mathcal{D}(K-, d) \star F.$$

Dually, the right Kan extension is the weighted limit

$$(\operatorname{Ran}_K F)(d) \;=\; \{\mathcal{D}(d, K-), F\}.$$

Remark 27.22 (Coend form)**.** Using the coend formula:

$$(\operatorname{Lan}_K F)(d) \;=\; \int^c \mathcal{D}(Kc, d) \otimes F(c), \qquad (\operatorname{Ran}_K F)(d) \;=\; \int_c [\mathcal{D}(d, Kc), F(c)],$$

the latter using cotensor $[-,-]$. These recover the formulas of Chapter 18 (in the ordinary case) and Chapter 24's promised enriched versions.

Remark 27.23 (Every weighted (co)limit is a Kan extension)**.** A weighted limit $\{W, F\}$ for $F : \mathcal{J} \to \mathcal{C}$ and $W : \mathcal{J} \to \mathcal{V}$ is the right Kan extension of F along $W : \mathcal{J} \to \mathcal{V}$ evaluated at $I \in \mathcal{V}$:

$$\{W, F\} \;\cong\; (\operatorname{Ran}_W F)(I).$$

Dually for colimits. This identifies (weighted) limit theory with Kan extension theory — they are the same thing presented from different viewpoints.

27.6 Exercises

Define the weighted limit $\{W, F\}$ of $F : \mathcal{J} \to \mathcal{C}$ by $W : \mathcal{J} \to \mathcal{V}$.	The object with $\mathcal{C}(c, \{W, F\}) \cong [\mathcal{J}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(c, F-))$ in $\mathcal{V}$, naturally in c.
Write the end formula for the weighted limit.	$\mathcal{C}(c, \{W, F\}) \cong \int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj))$.
What is the conical limit as a weighted limit?	$\lim F = \{\Delta I, F\}$ (or $\{\Delta\{*\}, F\}$ in the **Set**-case): weighted by the constant unit.
What is the cotensor as a weighted limit?	For $\mathcal{J} = \mathbf{1}$, F a single object X, W a single object V: $V \pitchfork X = \{V, X\}$, satisfying $\mathcal{C}(c, V \pitchfork X) \cong \mathcal{V}(V, \mathcal{C}(c, X))$.

Define the weighted colimit $W \star F$.	For $W : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$, the object with $\mathcal{C}(W \star F, c) \cong [\mathcal{J}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(W, \mathcal{C}(F-, c))$ in $\mathcal{V}$.
Coend formula for the weighted colimit?	$W \star F \cong \int^j W(j) \otimes F(j)$, in a tensored category.
Conical colimit as a weighted colimit?	$\operatorname{colim} F = \Delta I \star F$.
What is the tensor $V \otimes X$ as a weighted colimit?	For $\mathcal{J} = \mathbf{1}$, the colimit weighted by V of the diagram at X. Satisfies $\mathcal{C}(V \otimes X, c) \cong \mathcal{V}(V, \mathcal{C}(X, c))$.
State the Yoneda reduction for weighted limits.	$\{\mathcal{J}(j_0, -), F\} \cong F(j_0)$: representable weights pick out values.
Coend Yoneda reduction for weighted colimits?	$\mathcal{J}(-, j_0) \star F \cong F(j_0)$.
What is a comma object as a weighted limit?	The lax limit of a 2-cell, with weight given by a specific **Cat**-valued functor $\mathbf{2} \to \mathbf{Cat}$.
When is a $\mathcal{V}$-category $\mathcal{V}$-complete?	When $\{W, F\}$ exists for every small diagram F and weight W.
How is weighted completeness reduced to cotensors plus conical limits?	$\{W, F\} \cong \lim_j W(j) \pitchfork F(j)$ in a tensored/cotensored $\mathcal{V}$-category.
Dually for weighted cocompleteness?	$W \star F \cong \operatorname{colim}_j W(j) \otimes F(j)$.
Why do weighted limits commute with weighted limits (limits commute with limits)?	Because they are both ends, and ends commute by Fubini.
Why does the coend formula for weighted colimits require $\mathcal{C}$ to be tensored?	Because the formula uses $W(j) \otimes F(j)$; without tensors, this is not a well-defined object of $\mathcal{C}$.
Define the (pointwise) left Kan extension in the weighted setting.	$(\mathrm{Lan}_K\, F)(d) = \mathcal{D}(K-, d) \star F = \int^c \mathcal{D}(Kc, d) \otimes F(c)$.
Right Kan extension in the weighted setting?	$(\mathrm{Ran}_K\, F)(d) = \{\mathcal{D}(d, K-), F\} = \int_c [\mathcal{D}(d, Kc), F(c)]$.
Is every weighted (co)limit a Kan extension?	Yes: $\{W, F\} \cong (\mathrm{Ran}_W\, F)(I)$ and $W \star F \cong (\mathrm{Lan}_W\, F)(I)$, where W is treated as a functor $\mathcal{J} \to \mathcal{V}$ and the Kan extensions are along it.
Why do representable weights "evaluate" the diagram?	Because $\mathcal{J}(j_0, -)$ acts as the "point at j_0" under the Yoneda lemma; the weighted limit reduces by Yoneda to $F(j_0)$.
What is a homotopy limit in this framework?	A weighted limit where the weight encodes higher coherence (e.g., simplicial structure). When $\mathcal{V} = \mathbf{sSet}$ and W is a fibrant replacement of the constant simplicial set, $\{W, F\}$ is the homotopy limit.
What is the relationship between the end of a $\mathcal{V}$-functor and a weighted limit?	An end is the weighted limit by the hom-profunctor: $\int_c T(c, c) = \{\mathcal{C}(-, -), T\}$ where $T : \mathcal{C}^{\mathrm{op}} \otimes \mathcal{C} \to \mathcal{V}$.
In **Set**-enriched setting, what does "tensored" mean?	**Set**-tensored means having copowers: $S \cdot X = \coprod_S X$ exists. Every cocomplete category is **Set**-tensored.
Why is the dual variance condition on W for colimits ($\mathcal{J}^{\mathrm{op}}$ vs. $\mathcal{J}$) needed?	Because the weighted colimit's defining isomorphism contains $\mathcal{C}(F-, c)$ which is contravariant in the diagram source; the weight must match this contravariance.
What is the underlying ordinary functor of $\{-, -\} : [\mathcal{J}, \mathcal{V}]^{\mathrm{op}} \otimes [\mathcal{J}, \mathcal{C}] \to \mathcal{C}$?	The ordinary bifunctor taking (W, F) to the weighted limit $\{W, F\}$.
If $\mathcal{V}$ is cartesian closed, what is the cotensor in $[\mathcal{C}, \mathcal{V}]$?	$(V \pitchfork F)(c) = V \pitchfork F(c) = [V, F(c)]_{\mathcal{V}}$, computed pointwise.
What is a "descent object" as a weighted limit?	A weighted limit of a coherence diagram (a cosimplicial object in $\mathcal{C}$, truncated). It encodes "descent data" from a cover to a base.

Why does flexible weighted limits theory work for 2-categories?	Because **Cat** is closed under (small) products and cotensors with categories; tensors are categories indexed by a category.
What is the relationship between weighted (co)limits and ordinary natural transformations?	Weighted limits represent "natural transformations from the weight to the hom-functor"; the weight becomes a probe for the diagram.
Why does the chapter's title pair "weighted" with "limits"?	Because ordinary limits are the case $W = \Delta I$ (a single "constant" weight); weighted limits add the flexibility of an arbitrary weight, generalizing the limit notion to all of enriched/2-categorical settings.

Chapter Summary

Chapter Summary

Weighted limits via ends. $\{W, F\}$ is defined by $\mathcal{C}(c, \{W, F\}) \cong \int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj))$. Conical limits are the case $W = \Delta I$.
Weighted colimits via coends. $W \star F = \int^j W(j) \otimes F(j)$ in a tensored $\mathcal{V}$-category. Conical colimits are the case $W = \Delta I$. Tensors are the case $\mathcal{J} = \mathbf{1}$.
Yoneda reduction. Representable weights evaluate the diagram: $\{\mathcal{J}(j_0, -), F\} \cong F(j_0)$ and $\mathcal{J}(-, j_0) \star F \cong F(j_0)$.
Weighted completeness from pieces. In a tensored/cotensored $\mathcal{V}$-category, weighted completeness is equivalent to conical completeness; $\{W, F\} \cong \lim_j W(j) \pitchfork F(j)$.
Functoriality and Fubini. Weighted (co)limits are functorial in both arguments. Limits commute with limits (Fubini for ends); colimits with colimits (Fubini for coends). This makes weighted-limit calculus mechanical.
Kan extension preview. The (pointwise) left Kan extension is $(\mathrm{Lan}_K F)(d) = \int^c \mathcal{D}(Kc, d) \otimes F(c)$ and the right is $\int_c [\mathcal{D}(d, Kc), F(c)]$. Every weighted (co)limit is a Kan extension; every Kan extension is a weighted (co)limit. Full development in Chapter 31.

Formal Restatement

Formal Restatement

Weighted limit.

$$\mathcal{C}(c, \{W, F\}) \;\cong\; \int_j \mathcal{V}(Wj, \mathcal{C}(c, Fj)) \quad \text{in } \mathcal{V}, \text{ naturally in } c.$$

Weighted colimit.

$$W \star F \;\cong\; \int^j W(j) \otimes F(j), \qquad \mathcal{C}(W \star F, c) \;\cong\; \int_j \mathcal{V}(Wj, \mathcal{C}(Fj, c)).$$

Conical (co)limits. $\lim F = \{\Delta I, F\}$; $\operatorname{colim} F = \Delta I \star F$.

Cotensor, tensor. $V \pitchfork X = \{V, X\}$ for $\mathcal{J} = \mathbf{1}$; $V \otimes X = V \star X$.

Yoneda reduction. $\{\mathcal{J}(j_0, -), F\} \cong F(j_0)$ and $\mathcal{J}(-, j_0) \star F \cong F(j_0)$.

Completeness reduction. In a tensored/cotensored $\mathcal{V}$-category, $\{W, F\} \cong \lim_j W(j) \pitchfork F(j)$ and $W \star F \cong \operatorname{colim}_j W(j) \otimes F(j)$. Weighted (co)completeness reduces to conical (co)completeness plus (co)tensors.

Kan extension formulas.

$$(\mathrm{Lan}_K\, F)(d) = \mathcal{D}(K-, d) \star F = \int^c \mathcal{D}(Kc, d) \otimes F(c),$$

$$(\mathrm{Ran}_K\, F)(d) = \{\mathcal{D}(d, K-), F\} = \int_c [\mathcal{D}(d, Kc), F(c)].$$

Universal identification. Every weighted (co)limit is a Kan extension; every Kan extension is a weighted (co)limit.

Chapter 28 makes the universality of Yoneda formal in this weighted/enriched setting: the Yoneda embedding is the free cocompletion *in the precise sense of weighted colimit theory.*

Chapter 28

The Formal Yoneda Lemma

Chapter Overview

Yoneda has appeared three times: as a hom-set bijection (Chapter 9), as a representability principle (Chapter 15), and in enriched form (Chapter 20). These three statements are facets of one theorem. With ends and coends in hand (Chapter 26) and weighted (co)limits available (Chapter 27), we can unify them.

The *formal* Yoneda lemma states a single isomorphism in any sufficiently complete enriched setting:

$$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \;=\; \int_c \mathcal{V}(\mathcal{C}(a, c), Fc) \;\cong\; F(a).$$

The proof is now mechanical: apply the ninja Yoneda lemma. From this we derive: the **density theorem** (every presheaf is a coend of representables); the **free cocompletion** property of the presheaf category; the **coend Yoneda** and **end Yoneda** forms; and the *ninja duality* between left and right Kan extensions through the Yoneda embedding. The chapter closes with the formal Yoneda lemma in the 2-categorical language of **Prof**.

Where the Name Comes From

The **ninja Yoneda lemma** is a popular informal name for the coend form of Yoneda: $\int^c \mathcal{C}(c, c') \cdot F(c) \cong F(c')$. The name "ninja" echoes the way the lemma is invoked in proofs: suddenly, without warning, a coend collapses to a single value — like a ninja striking. The terminology is jocular but universal in category-theoretic blog posts, lectures, and even some textbooks. The formal name is **co-Yoneda** or *Yoneda reduction*.

Density theorem (cross-ref Vol II Ch. 24, Vol IV Ch. 48) is the coend / colimit statement: every presheaf is a canonical colimit of representables. The terminology echoes topology (a "dense" subset has closure equal to the whole space); categorically, representables are dense in $\mathcal{P}(\mathcal{C})$.

28.1 The Unified Statement

28.1.1 Formalizing

Theorem 28.1 (Yoneda lemma, unified form)**.** *Let $\mathcal{V}$ be a complete symmetric monoidal closed category and $\mathcal{C}$ a small $\mathcal{V}$-category. For any $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{V}$ and object $a \in \mathcal{C}$, there is a $\mathcal{V}$-natural isomorphism*

$$\mathrm{Yo}^a_F \;:\; \int_c \mathcal{V}(\mathcal{C}(a, c), Fc) \;\xrightarrow{\cong}\; F(a)$$

in $\mathcal{V}$, natural in a and in F.

The end on the left is precisely the $\mathcal{V}$-object of $\mathcal{V}$-natural transformations: $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F)$.	*End formula for enriched naturals (Chapter 26).*
We construct a candidate map $\mathrm{Yo}^a_F : [\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(a, -), F) \to F(a)$ by evaluation at a: precompose with the identity $\mathcal{V}$-morphism $j_a : I \to \mathcal{C}(a, a)$.	*This is the forward direction; explicit on representables.*
Construct the inverse via the $\mathcal{V}$-functoriality of F: given $x : I \to F(a)$ in $\mathcal{V}$, define $\widetilde{x}_c : \mathcal{C}(a, c) \to F(c)$ as the composite $\mathcal{C}(a, c) \cong I \otimes \mathcal{C}(a, c) \xrightarrow{x \otimes \mathrm{id}} F(a) \otimes \mathcal{C}(a, c) \xrightarrow{F_{a,c}} F(c)$.	*Uses the $\mathcal{V}$-functor structure $F_{a,c} : F(a) \otimes \mathcal{C}(a, c) \to F(c)$.*
The family $(\widetilde{x}_c)_c$ is a wedge to $T(c, c') = \mathcal{V}(\mathcal{C}(a, c), Fc')$ — i.e., it satisfies the dinatural condition.	*By the $\mathcal{V}$-functor axioms of F applied to morphisms $c \to c'$ in $\mathcal{C}$.*
So $(\widetilde{x}_c)$ factors through the end to give a morphism $\widetilde{x} : I \to \int_c \mathcal{V}(\mathcal{C}(a, c), Fc)$.	*Universal property of the end.*
$\mathrm{Yo}^a_F \circ \widetilde{(-)} = \mathrm{id}_{F(a)}$: evaluating $\widetilde{x}$ at a gives $F_{a,a}(x \otimes j_a) = x$ by the $\mathcal{V}$-functor unit axiom.	*Unit-of-F.*
$\widetilde{(-)} \circ \mathrm{Yo}^a_F = \mathrm{id}$: starting from a $\mathcal{V}$-natural transformation, evaluating at a then transporting via $F_{a,-}$ reproduces the original transformation, by the $\mathcal{V}$-functor composition axiom (the naturality of θ matches F's action exactly).	*Composition axiom of F.*
Both composites are identities, so Yo^a_F is an iso in $\mathcal{V}$.	*Yoneda established at a.*
Naturality in a and F is automatic: all constructions (end, evaluation, F's action) are $\mathcal{V}$-natural in both arguments.	*Standard.*

28.1.2 Reorganizing: The Three Forms Recovered

Corollary 28.2 (Three forms of Yoneda). *Theorem 28.1 specializes to:*

- ***Set-Yoneda*** *(Chapter 9): $\mathcal{V} = \mathbf{Set}$, $\mathcal{C}$ ordinary locally small. Then $\mathrm{Nat}(\mathcal{C}(a, -), F) \cong F(a)$.*
- ***Enriched Yoneda*** *(Chapter 20): $\mathcal{V}$ symmetric monoidal closed, $\mathcal{C}$ a $\mathcal{V}$-category. The statement above.*
- ***Coend Yoneda*** *(the ninja Yoneda): the dual form, $\int^c \mathcal{C}(c, a) \otimes F(c) \cong F(a)$, obtained by the end-coend duality for hom-objects.*

Remark 28.3. The "three forms" are not three different theorems. They are one theorem expressed in different language: ends (the explicit hom-set / hom-object form), set-natural transformations (the original statement), and coends (the "tensor-out" dualization). Each is the others' shadow.

28.2 The Yoneda Embedding

28.2.1 Formalizing

Definition 28.4 (Enriched Yoneda embedding). For a small $\mathcal{V}$-category $\mathcal{C}$, the **Yoneda embedding** is the $\mathcal{V}$-functor

$$Y \;:\; \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}, \qquad a \mapsto \mathcal{C}(-, a).$$

Theorem 28.5 (Yoneda embedding is fully faithful). *The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ is a fully faithful $\mathcal{V}$-functor: it induces $\mathcal{V}$-isomorphisms*

$$Y_{a,b} \;:\; \mathcal{C}(a, b) \xrightarrow{\cong} [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(-, a), \mathcal{C}(-, b)).$$

Proof. By Theorem 28.1 applied to $F = \mathcal{C}(-, b) : \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$ at the object a:

$$[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(-, a), \mathcal{C}(-, b)) \;\cong\; \mathcal{C}(a, b).$$

That this isomorphism is exactly the hom-functor of Y (i.e., $Y_{a,b}$) is the content of the construction. □

28.2.2 Reorganizing: The Universal Property

Theorem 28.6 (Free cocompletion). *For any cocomplete $\mathcal{V}$-category $\mathcal{E}$ and $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{E}$, there is an essentially unique colimit-preserving (i.e., $\mathcal{V}$-cocontinuous) $\mathcal{V}$-functor*

$$\widetilde{F} \;:\; [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{E}$$

with $\widetilde{F} \circ Y \cong F$. Explicitly, on a presheaf P:

$$\widetilde{F}(P) \;=\; P \star F \;=\; \int^{c} P(c) \otimes F(c).$$

Define $\widetilde{F}(P) = \int^c P(c) \otimes F(c)$ (a weighted colimit with P as weight).	*Coend formula.*
$\widetilde{F}$ is a $\mathcal{V}$-functor by functoriality of weighted colimits in the weight (Chapter 27).	*Functoriality.*
$\widetilde{F} \circ Y(a) = \widetilde{F}(\mathcal{C}(-,a)) = \int^c \mathcal{C}(c,a) \otimes F(c) \cong F(a)$.	*By the coend Yoneda lemma (Theorem 28.1 dual / ninja Yoneda).*
$\widetilde{F}$ preserves all small colimits because coends commute with colimits (in their non-integration arguments).	*Coend = colimit, and colimits commute with colimits.*
For uniqueness, suppose $G : [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{E}$ is colimit-preserving with $G \circ Y \cong F$.	*Compare to any candidate.*
Every presheaf is a coend of representables: $P \cong \int^c P(c) \otimes Y(c)$ (co-Yoneda).	*Density theorem in coend form.*
Applying G: $G(P) \cong G\big(\int^c P(c) \otimes Y(c)\big) \cong \int^c P(c) \otimes G(Y(c)) \cong \int^c P(c) \otimes F(c) = \widetilde{F}(P)$.	*G preserves colimits; $G \circ Y = F$.*
Therefore $G \cong \widetilde{F}$, essentially uniquely.	*Uniqueness.*

Corollary 28.7 (Universal property of $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$)**.** *The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ exhibits the presheaf category as the free $\mathcal{V}$-cocompletion of $\mathcal{C}$: it is initial among $\mathcal{V}$-functors from $\mathcal{C}$ into cocomplete $\mathcal{V}$-categories.*

28.3 The Density Theorem

28.3.1 Formalizing

Theorem 28.8 (Density / co-Yoneda)**.** *For any $\mathcal{V}$-functor $P : \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$,*

$$P \;\cong\; \int^{c} P(c) \otimes \mathcal{C}(-,c)$$

as $\mathcal{V}$-functors. Every presheaf is canonically a coend of representables.

Proof. At each $a \in \mathcal{C}$, the right-hand side evaluates to $\int^c P(c) \otimes \mathcal{C}(a,c) \cong P(a)$ by ninja Yoneda (Chapter 26, Theorem 26.17). Naturality in a is automatic. □

28.3.2 Reorganizing: The Yoneda Embedding is Dense

Definition 28.9 (Dense functor)**.** A $\mathcal{V}$-functor $K : \mathcal{C} \to \mathcal{D}$ is **dense** if the canonical map

$$\int^{c} \mathcal{D}(Kc, -) \otimes Kc \;\longrightarrow\; \mathrm{Id}_{\mathcal{D}}$$

is an isomorphism — i.e., the identity functor of $\mathcal{D}$ is the left Kan extension of K along itself.

Corollary 28.10. *The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ is dense.*

Proof. The density condition at a presheaf P becomes

$$\int^{c} [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(-,c), P) \otimes \mathcal{C}(-,c) \;\cong\; P.$$

By the Yoneda lemma applied to the inner $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(-,c), P) \cong P(c)$, this reduces to $\int^c P(c) \otimes \mathcal{C}(-,c) \cong P$ — which is the density theorem. □

28.4 Adjoint Functors via Yoneda

28.4.1 Formalizing

The Yoneda lemma gives a clean formulation of when a functor has an adjoint.

Theorem 28.11 (Adjoint as represented hom-functor)**.** *A $\mathcal{V}$-functor $G : \mathcal{D} \to \mathcal{C}$ has a left adjoint iff for every $c \in \mathcal{C}$, the $\mathcal{V}$-functor*

$$\mathcal{C}(c, G-) \ : \ \mathcal{D} \to \mathcal{V}$$

is representable. In that case, the representing object is $L(c)$, where $L \dashv G$.

Proof. *($\Rightarrow$):* If $L \dashv G$, then $\mathcal{C}(c, G-) \cong \mathcal{D}(Lc, -)$, manifestly representable by Lc.
($\Leftarrow$): If $\mathcal{C}(c, G-)$ is represented by Lc, the correspondence $c \mapsto Lc$ is functorial (by uniqueness of representing objects), and the isomorphism is exactly the adjoint hom-bijection. □

Corollary 28.12 (Right Kan extension formula for adjoints)**.** *A right adjoint $G : \mathcal{D} \to \mathcal{C}$ is the right Kan extension of itself along the Yoneda embedding:*

$$G \ \cong \ \mathrm{Ran}_Y(G \circ Y^{-1}) \quad \textit{informally,}$$

or, more precisely: every right adjoint is, up to natural isomorphism, $\mathrm{Ran}_Y F$ *for some $F : \mathcal{C} \to \mathcal{C}$, where Y is the Yoneda embedding of an appropriate source.*

28.4.2 Reorganizing: Universal Constructions as Kan Extensions

The combination of Theorem 28.11, the free cocompletion Theorem 48.10, and the weighted Kan extension formulas (Chapter 27) yields the slogan:

> *Every universal construction is the left Kan extension of an evident functor along the Yoneda embedding.*

Example 28.13 (Free abelian group)**.** The free abelian group functor **Set** $\to$ **Ab** is $\mathrm{Lan}_Y(\mathbb{Z}[-])$ where $\mathbb{Z}[-]$ sends a finite set to the free abelian group on it and Y is the Yoneda embedding of finite sets into **Set**. Computing the coend formula recovers the standard construction.

Example 28.14 (Geometric realization)**.** $|-| :$ **sSet** $\to$ **Top** is $\mathrm{Lan}_Y(\Delta^\bullet)$ where $\Delta^\bullet : \Delta \to$ **Top** sends $[n]$ to the topological n-simplex, and $Y : \Delta \to$ **sSet** is the Yoneda embedding $[n] \mapsto \Delta[n]$. This recovers the coend formula $|X| = \int^{[n]} X_n \cdot \Delta^n$ from Chapter 26.

28.5 Yoneda in Prof

28.5.1 Formalizing

The bicategory **Prof** (Chapter 25, § 5.2) houses profunctors as 1-cells. Its unit at $\mathcal{C}$ is the hom-profunctor $\mathcal{C}(-,-) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to$ **Set**.

Theorem 28.15 (Yoneda lemma in **Prof**)**.** *For a small category $\mathcal{C}$, the hom-profunctor $\mathcal{C}(-,-) : \mathcal{C} \to \mathcal{C}$ is the* identity 1-cell *of* **Prof** *at $\mathcal{C}$, in the sense that for any profunctor $P : \mathcal{C} \to \mathcal{D}$:*

$$P \circ \mathcal{C}(-,-) \ \cong \ P, \qquad \mathcal{D}(-,-) \circ P \ \cong \ P.$$

Proof. The left identity: $(P \circ \mathcal{C}(-,-))(d,c) = \int^{c'} P(d,c') \times \mathcal{C}(c',c) \cong P(d,c)$ by the ninja Yoneda lemma applied to $P(d,-)$. The right identity is the dual computation. □

Remark 28.16 (Yoneda is unitality)**.** This is the most distilled form of Yoneda's content: *the hom-functor is the identity of profunctor composition.* All the other forms of Yoneda follow from this single fact, because profunctor composition is the coend and the unit is the hom-profunctor.

28.5.2 Reorganizing: Cauchy Completeness

Definition 28.17 (Cauchy completeness)**.** A small $\mathcal{V}$-category $\mathcal{C}$ is **Cauchy complete** (or *Karoubian* in the **Set**-case) if every $\mathcal{V}$-profunctor $P : \mathcal{C} \to \mathbf{1}$ representable in **Prof** is representable by an object of $\mathcal{C}$ — equivalently, every *absolute* weighted colimit exists in $\mathcal{C}$.

Remark 28.18. Cauchy completion is the universal Cauchy-complete enlargement of $\mathcal{C}$. It is the most refined version of the "presheaf category as free cocompletion" theme: not all colimits, but exactly the colimits that are forced by profunctor representability.

28.6 Exercises

State the unified Yoneda lemma.	For $F : \mathcal{C} \to \mathcal{V}$ a $\mathcal{V}$-functor and $a \in \mathcal{C}$: $\int_c \mathcal{V}(\mathcal{C}(a,c), Fc) \cong F(a)$ in $\mathcal{V}$, $\mathcal{V}$-naturally in a and F.
What is the "end form" of natural transformations?	$[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}(G,H) = \int_c \mathcal{V}(Gc, Hc)$.
What is the "coend form" of Yoneda (ninja Yoneda)?	$\int^c \mathcal{C}(c,a) \otimes F(c) \cong F(a)$.

How does ordinary Set-Yoneda emerge from the unified statement?	Take $\mathcal{V} = \mathbf{Set}$, $\mathcal{C}$ locally small. The end becomes $\mathrm{Nat}(\mathcal{C}(a,-), F)$ and the conclusion is $\mathrm{Nat}(\mathcal{C}(a,-), F) \cong F(a)$.
Define the (enriched) Yoneda embedding.	$Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$, $a \mapsto \mathcal{C}(-,a)$.
Why is the Yoneda embedding fully faithful?	Because $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(\mathcal{C}(-,a), \mathcal{C}(-,b)) \cong \mathcal{C}(a,b)$ by the unified Yoneda lemma applied to $F = \mathcal{C}(-,b)$.
State the free cocompletion theorem.	For any cocomplete $\mathcal{V}$-category $\mathcal{E}$ and $F : \mathcal{C} \to \mathcal{E}$, there is an essentially unique colimit-preserving $\widetilde{F} : [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{E}$ with $\widetilde{F} \circ Y \cong F$.
What is the explicit formula for $\widetilde{F}$ on a presheaf P?	$\widetilde{F}(P) = \int^{c} P(c) \otimes F(c) = P \star F$.
State the density theorem in coend form.	$P \cong \int^{c} P(c) \otimes \mathcal{C}(-,c)$. Every presheaf is canonically a coend of representables.
What does "dense functor" mean?	A $\mathcal{V}$-functor $K : \mathcal{C} \to \mathcal{D}$ such that the canonical map $\int^{c} \mathcal{D}(Kc,-) \otimes Kc \to \mathrm{Id}_{\mathcal{D}}$ is an iso — equivalently, $\mathrm{Id}_{\mathcal{D}} = \mathrm{Lan}_K K$.
Why is the Yoneda embedding dense?	Because the density condition reduces, via the Yoneda lemma, to the density theorem itself.
When does $G : \mathcal{D} \to \mathcal{C}$ have a left adjoint?	Iff $\mathcal{C}(c, G-) : \mathcal{D} \to \mathcal{V}$ is representable for every $c \in \mathcal{C}$.
What is the relationship between universal constructions and Kan extensions?	Every universal construction is a left Kan extension along the Yoneda embedding of an appropriate sub-category.
Express the free abelian group functor as a Kan extension.	$F_{\mathrm{ab}} = \mathrm{Lan}_Y(\mathbb{Z}[-])$, where $Y : \mathbf{FinSet} \to \mathbf{Set}$ and $\mathbb{Z}[-] : \mathbf{FinSet} \to \mathbf{Ab}$.
Express geometric realization as a Kan extension.	$\lvert - \rvert = \mathrm{Lan}_Y(\Delta^\bullet)$, $Y : \Delta \to \mathbf{sSet}$, $\Delta^\bullet : \Delta \to \mathbf{Top}$.
What is the identity 1-cell of **Prof** at $\mathcal{C}$?	The hom-profunctor $\mathcal{C}(-,-)$.
State Yoneda as the unitality of **Prof**.	For any profunctor $P : \mathcal{C} \to \mathcal{D}$: $P \circ \mathcal{C}(-,-) \cong P$ and $\mathcal{D}(-,-) \circ P \cong P$.
Why is "Yoneda is unitality" the most distilled form?	Because all other Yoneda statements (representable functor, hom-bijection, density) follow from this single profunctor-composition fact.
What is a Cauchy complete category?	A category in which every absolute weighted colimit exists; equivalently, every profunctor to the terminal category that is representable in **Prof** is representable by an object.
What is the Karoubian (idempotent splitting) view of Cauchy completion?	In the **Set**-enriched case, Cauchy completion is the splitting of all idempotents.
Why is Cauchy completion less than full cocompletion?	Because only *absolute* colimits are forced; the presheaf category (full cocompletion) typically contains many more objects than the Cauchy completion.
What is the relationship between $\widetilde{F}$ and $\mathrm{Lan}_Y F$?	They are equal: $\widetilde{F} = \mathrm{Lan}_Y F$. The free cocompletion is the left Kan extension along Yoneda.
Why does the colimit-preserving extension $\widetilde{F}$ exist and is unique?	Existence: the coend formula constructs it. Uniqueness: every presheaf is a coend of representables (density), so any colimit-preserving extension is determined on representables and is then forced by coend preservation.
What does "natural in F" mean in the unified Yoneda statement?	The isomorphism $\int_c \mathcal{V}(\mathcal{C}(a,c), Fc) \cong F(a)$ is a natural transformation between two $\mathcal{V}$-functors $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{V}$: the end of the hom and evaluation at a.
What does the Yoneda embedding "embed" a category into?	The presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$, by viewing each object a as the representable presheaf $\mathcal{C}(-,a)$.

Why is the Yoneda embedding a $\mathcal{V}$-functor, not just an ordinary functor?	Because the action on hom-objects is the $\mathcal{V}$-natural enrichment of the action on objects; the $\mathcal{V}$-structure is preserved.
What is the dual of the free cocompletion theorem?	The free completion theorem: for any complete $\mathcal{V}$-category $\mathcal{E}$ and $F : \mathcal{C}^{\text{op}} \to \mathcal{E}$, the right Kan extension $\text{Ran}_{Y^{\text{op}}} F : [\mathcal{C}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{E}$ is the essentially unique limit-preserving extension.
In what sense is Yoneda "trivial"?	Once expressed as the unitality of **Prof** or as the ninja Yoneda formula via coend, the proof is mechanical: substitute the identity, apply functoriality, conclude.
What is the "coherence" the unified statement encodes?	The compatibility of three pictures: hom-natural transformations as ends; representability as Yoneda; density as the co-Yoneda formula. All are facets of a single coend/end identity.
Why is this chapter called "Formal Yoneda" rather than just "Yoneda Revisited"?	Because the statement is now formal in the sense of formal category theory: it works in any enriched / 2-categorical setting where ends, coends, and weighted (co)limits make sense.

Chapter Summary

Chapter Summary

The unified Yoneda lemma. For a $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{V}$ and $a \in \mathcal{C}$:

$$\int_c \mathcal{V}(\mathcal{C}(a, c), Fc) \;\cong\; F(a)$$

in $\mathcal{V}$. This unifies the set-, enriched-, and coend-forms of Yoneda into one statement.
The Yoneda embedding. $Y : \mathcal{C} \to [\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}$ is fully faithful: the Yoneda lemma identifies $\mathcal{C}(a, b)$ with $\mathcal{V}$-natural transformations $\mathcal{C}(-, a) \Rightarrow \mathcal{C}(-, b)$.
Free cocompletion. $[\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}$ is the free $\mathcal{V}$-cocompletion of $\mathcal{C}$: every $\mathcal{V}$-functor $F : \mathcal{C} \to \mathcal{E}$ to a cocomplete $\mathcal{V}$-category extends uniquely (up to iso) to a colimit-preserving $\widetilde{F} = \text{Lan}_Y F$, given by the coend formula $\widetilde{F}(P) = \int^c P(c) \otimes F(c)$.
Density. Every presheaf is a coend of representables: $P \cong \int^c P(c) \otimes \mathcal{C}(-, c)$. The Yoneda embedding is dense: the identity functor of the presheaf category is the left Kan extension of Y along itself.
Adjoints via representability. A $\mathcal{V}$-functor G has a left adjoint iff $\mathcal{C}(c, G-)$ is representable for every c. Every adjoint construction is a Kan extension along Yoneda.
Yoneda in Prof. The hom-profunctor is the identity 1-cell of **Prof** at $\mathcal{C}$. This is the most distilled form of Yoneda's content; all other forms follow.
Cauchy completeness. The category where exactly the absolute weighted colimits exist; in **Set**-enriched, the splitting of idempotents.

Formal Restatement

Formal Restatement

Unified Yoneda. For $F : \mathcal{C} \to \mathcal{V}$ a $\mathcal{V}$-functor:

$$\int_c \mathcal{V}(\mathcal{C}(a, c), Fc) \;\cong\; F(a) \qquad \int^c \mathcal{C}(c, a) \otimes F(c) \;\cong\; F(a)$$

$\mathcal{V}$-naturally in a and F.

Yoneda embedding. $Y : \mathcal{C} \to [\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}$, $a \mapsto \mathcal{C}(-, a)$. Fully faithful as a $\mathcal{V}$-functor.

Free cocompletion. For cocomplete $\mathcal{E}$ and $F : \mathcal{C} \to \mathcal{E}$, $\widetilde{F} = \text{Lan}_Y F$ is the essentially unique colimit-preserving extension with $\widetilde{F} \circ Y \cong F$. Formula: $\widetilde{F}(P) = \int^c P(c) \otimes F(c)$.

Density. $P \cong \int^c P(c) \otimes \mathcal{C}(-, c)$ for any $\mathcal{V}$-functor $P : \mathcal{C}^{\text{op}} \to \mathcal{V}$. Y is dense: $\text{Id}_{[\mathcal{C}^{\text{op}}, \mathcal{V}]} = \text{Lan}_Y Y$.

Adjoint as represented hom. $G : \mathcal{D} \to \mathcal{C}$ has $L \dashv G$ iff $\mathcal{C}(c, G-) \cong \mathcal{D}(Lc, -)$ for each c.

Yoneda as unitality in Prof. $\mathcal{C}(-,-)$ is the identity 1-cell at $\mathcal{C}$ in **Prof**: $P \circ \mathcal{C}(-,-) \cong P$ for all profunctors out of $\mathcal{C}$ (and dually).

Chapter 29 returns to monoidal categories with the strictification theorem in full generality (Mac Lane coherence), now in the 2-categorical / bicategorical language of Chapter 25.

Chapter 29

Monoidal Categories: Coherence

Chapter Overview

Chapter 19 proved Mac Lane's coherence theorem for monoidal categories via the normal-form strategy: every formal diagram of associators and unitors in a free monoidal category collapses to a canonical bracketing, hence commutes. Chapter 25 proved its bicategorical generalization, that every bicategory is biequivalent to a strict 2-category.

This chapter is the synthesis. We re-derive monoidal coherence as a *special case* of bicategorical strictification — a monoidal category is a one-object bicategory, so the strictification of Chapter 25 specializes to the strictification of monoidal categories. We then push further into material untouched by Chapter 19: *coherence for monoidal functors* (when does a monoidal functor's structure 2-cells satisfy "all" the coherence diagrams automatically?), *free monoidal categories* and their universal property, and the symmetric/braided variants. The chapter closes with a glimpse of operads — the algebraic objects that make "monoidal coherence" a workable computational tool.

Where the Name Comes From

Coherence (as in "Mac Lane's coherence theorem") is from *Saunders Mac Lane*'s 1963 paper *Natural Associativity and Commutativity.* The word *coherent* (Latin *cohaerere*, "to stick together") captures the meaning: a coherence theorem says that "all the diagrams that ought to commute do commute," once a small fundamental set (pentagon, triangle) is required to commute. Mac Lane's foundational coherence theorem was the prototype for the entire genre.

Strictification means converting a weak (= coherent up to iso) structure into a strict (= equality on the nose) one. The terminology is standard across n-category theory: bicategory $\to$ strict 2-category (Mac Lane / Bénabou), monoidal $\to$ strict monoidal (Mac Lane 1963), A_∞ $\to$ associative (Markl, in some settings). Always available; not always desirable (some constructions are more natural in their weak form).

29.1 Monoidal Strictification via Bicategorical Strictification

29.1.1 Formalizing

Recall (Chapter 25, Proposition on monoidal $\Leftrightarrow$ one-object bicategory): a monoidal category $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ is the same data as a one-object bicategory $B\mathcal{V}$.

Theorem 29.1 (Strictification for monoidal categories)**.** *Every monoidal category $\mathcal{V}$ is monoidally equivalent to a* strict *monoidal category $\mathcal{V}^{\mathrm{st}}$.*

Form the one-object bicategory $B\mathcal{V}$ (Chapter 25, § 25.2).	*Monoidal $\leftrightarrow$ one-object bicategory.*
By Theorem 25.25 (Chapter 25, bicategorical strictification), $B\mathcal{V}$ is biequivalent to a strict 2-category $(B\mathcal{V})^{\mathrm{st}}$.	*Apply the strictification construction.*
The string-replacement construction of Chapter 25 preserves the one-object property: $(B\mathcal{V})^{\mathrm{st}}$ is again one-object.	*Strings are constructed pointwise; the single 0-cell of $B\mathcal{V}$ remains the single 0-cell of $(B\mathcal{V})^{\mathrm{st}}$.*
Therefore $(B\mathcal{V})^{\mathrm{st}}$ corresponds to a strict one-object 2-category, i.e., a strict monoidal category $\mathcal{V}^{\mathrm{st}}$.	*Monoidal $\leftrightarrow$ one-object reverse direction.*
The biequivalence of bicategories $\widehat{(-)} : (B\mathcal{V})^{\mathrm{st}} \to B\mathcal{V}$ corresponds to a monoidal equivalence $\widehat{(-)} : \mathcal{V}^{\mathrm{st}} \to \mathcal{V}$.	*A pseudofunctor between one-object bicategories is the same data as a strong monoidal functor between the corresponding monoidal categories.*
Conclude: every monoidal category is monoidally equivalent to a strict one.	*Strictification.*

Corollary 29.2 (Mac Lane's coherence, repackaged)**.** *In any monoidal category, every diagram of associators and unitors commutes.*

Proof. By Theorem 29.1, $\mathcal{V}$ is monoidally equivalent to $\mathcal{V}^{\mathrm{st}}$, in which all α, λ, ρ are identities and hence every formal diagram trivially commutes. The equivalence reflects commutativity (it is fully faithful). □

Remark 29.3. Chapter 19's normal-form proof of coherence is constructive in $\mathcal{V}$ itself, while the strictification-based proof above factors through the larger category $\mathcal{V}^{\mathrm{st}}$. Both yield the same theorem; the strictification proof has the conceptual virtue of making coherence a special case of the general bicategorical phenomenon.

29.1.2 Reorganizing: The Strictification Construction Explicitly

The string-replacement construction of Chapter 25, specialized:

Definition 29.4 (Explicit strictification $\mathcal{V}^{\mathrm{st}}$)**.** For a monoidal category $\mathcal{V}$:

- Objects of $\mathcal{V}^{\mathrm{st}}$: finite strings (lists) of objects of $\mathcal{V}$, including the empty string (which serves as the strict unit).
- Morphisms from $(A_1, \ldots, A_m)$ to $(B_1, \ldots, B_n)$: morphisms $A_1 \otimes \cdots \otimes A_m \to B_1 \otimes \cdots \otimes B_n$ in $\mathcal{V}$ (left-associated), where the empty string composes to I.
- Tensor: concatenation of strings.
- Unit: empty string.
- Associator, unitors: identities.

Proposition 29.5. *$\mathcal{V}^{\mathrm{st}}$ is a strict monoidal category. The functor $\widehat{(-)} : \mathcal{V}^{\mathrm{st}} \to \mathcal{V}$, $(A_1, \ldots, A_n) \mapsto A_1 \otimes \cdots \otimes A_n$ (left-bracketed), is a monoidal equivalence.*

Proof. Concatenation is strictly associative; the empty string is a strict identity. $\widehat{(-)}$ is essentially surjective (every $A \in \mathcal{V}$ is $\widehat{(A)}$) and fully faithful (by definition of morphisms in $\mathcal{V}^{\mathrm{st}}$). □

29.2 Coherence for Monoidal Functors

A monoidal functor $F : \mathcal{V} \to \mathcal{W}$ between monoidal categories carries comparison 2-cells $\phi_{A,B} : FA \otimes FB \to F(A \otimes B)$ and $\phi_I : I_{\mathcal{W}} \to FI_{\mathcal{V}}$. The coherence axioms in Chapter 19 ensure ϕ is compatible with the associators and unitors of $\mathcal{V}, \mathcal{W}$.

29.2.1 Formalizing

Theorem 29.6 (Coherence for monoidal functors)**.** *In a strong (resp. lax, oplax) monoidal functor $F : \mathcal{V} \to \mathcal{W}$, every diagram built from $\phi_{A,B}$, ϕ_I, F applied to the structure morphisms of $\mathcal{V}$, and the structure morphisms of $\mathcal{W}$ — where the diagram has a single fixed source and target — commutes.*

Remark 29.7. The proof again reduces to strictification: replace $\mathcal{V}$ and $\mathcal{W}$ by their strictifications, transport F to a functor between strict monoidal categories. In the strict case the coherence diagrams trivially commute, and equivalence reflects.

29.2.2 Reorganizing: The Two Hexagons

Definition 29.8 (Hexagonal coherence axioms)**.** A lax monoidal functor $F : (\mathcal{V}, \otimes, I) \to (\mathcal{W}, \otimes, I)$ satisfies the **associativity hexagon**: for $A, B, C \in \mathcal{V}$,

$$\begin{tikzcd}
(FA \otimes FB) \otimes FC \arrow[r, "\alpha"] \arrow[d, "\phi_{A,B} \otimes 1"'] & FA \otimes (FB \otimes FC) \arrow[d, "1 \otimes \phi_{B,C}"] \\
F(A \otimes B) \otimes FC \arrow[d, "\phi_{A \otimes B, C}"'] & FA \otimes F(B \otimes C) \arrow[d, "\phi_{A, B \otimes C}"] \\
F((A \otimes B) \otimes C) \arrow[r, "F(\alpha)"] & F(A \otimes (B \otimes C))
\end{tikzcd}$$

and the **unitality squares**:

$$\begin{tikzcd}
I_{\mathcal{W}} \otimes FA \arrow[r, "\lambda_{\mathcal{W}}"] \arrow[d, "\phi_I \otimes 1"'] & FA \\
FI_{\mathcal{V}} \otimes FA \arrow[r, "\phi_{I,A}"] & F(I \otimes A) \arrow[u, "F(\lambda)"']
\end{tikzcd}
\qquad
\begin{tikzcd}
FA \otimes I_{\mathcal{W}} \arrow[r, "\rho_{\mathcal{W}}"] \arrow[d, "1 \otimes \phi_I"'] & FA \\
FA \otimes FI_{\mathcal{V}} \arrow[r, "\phi_{A,I}"] & F(A \otimes I) \arrow[u, "F(\rho)"']
\end{tikzcd}$$

Proposition 29.9 (Coherence axioms imply all coherence diagrams)**.** *The associativity hexagon and the unitality squares of Definition 29.8 imply Theorem 29.6: all diagrams of formal coherence built from ϕ, F, and the structure 2-cells of $\mathcal{V}, \mathcal{W}$ commute.*

Sketch. The strategy is the same as the monoidal-category case. Construct the strictifications $\mathcal{V}^{\mathrm{st}}, \mathcal{W}^{\mathrm{st}}$; transport F to $F^{\mathrm{st}} : \mathcal{V}^{\mathrm{st}} \to \mathcal{W}^{\mathrm{st}}$. In $\mathcal{W}^{\mathrm{st}}$, all coherence diagrams have identity sides; the strictified ϕ^{st} is forced (by the two hexagons and squares) to be an identity. Pulling back, all original diagrams commute. □

29.3 Free Monoidal Categories

29.3.1 Formalizing

The universal monoidal category on a given category $\mathcal{C}$ is the "one with no relations beyond what monoidal structure requires." Formally:

Definition 29.10 (Free monoidal category)**.** For an ordinary category $\mathcal{C}$, the **free monoidal category** on $\mathcal{C}$, written $F^{\otimes}(\mathcal{C})$, is the monoidal category characterized by the universal property: for every monoidal category $\mathcal{V}$ and every ordinary functor $F : \mathcal{C} \to \mathcal{V}$, there is an essentially unique strong monoidal functor $\widetilde{F} : F^{\otimes}(\mathcal{C}) \to \mathcal{V}$ extending F.

Theorem 29.11 (Construction)**.** *$F^{\otimes}(\mathcal{C})$ has:*

- *Objects: finite lists (with bracketings, possibly nested) of objects of $\mathcal{C}$.*
- *Morphisms: generated by morphisms of $\mathcal{C}$ on singletons, identity for tensor combinations, with associators and unitors formally adjoined as isomorphisms.*

The universal property follows from a normal-form argument akin to strictification.

Sketch. Existence: the explicit construction. Universality: given $F : \mathcal{C} \to \mathcal{V}$, define $\widetilde{F}$ on singletons by F; extend to tensor combinations by $F(A_1) \otimes \cdots \otimes F(A_n)$ (with appropriate bracketing) and on morphisms by transport along the $\mathcal{V}$-associators and unitors. Coherence (Theorem 29.6) guarantees the extension is well-defined and a strong monoidal functor. Uniqueness up to monoidal natural isomorphism is standard. □

Remark 29.12 (Strict variant)**.** The *strict* free monoidal category $F^{\otimes,\mathrm{st}}(\mathcal{C})$ has objects flat lists (no nested brackets) and morphisms generated by $\mathcal{C}$ without formal associators. It is the strict 2-category one-object specialization of the strictification construction.

29.3.2 Reorganizing: The Yoneda Description

Proposition 29.13 (Free monoidal as Day convolution)**.** *For a category $\mathcal{C}$ (with no monoidal structure), $F^{\otimes}(\mathcal{C})$ is the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ equipped with* Day convolution *— see Chapter 30. The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ becomes the unit of $F^{\otimes}$.*

This identification — free cocompletion = free monoidal category, in the appropriate sense — is the link between Yoneda theory (Chapter 28) and monoidal theory developed in Chapter 30.

29.4 Symmetric and Braided Coherence

Symmetric and braided monoidal categories add an extra structure 2-cell $\sigma_{A,B} : A \otimes B \cong B \otimes A$ (the **braiding** or **symmetry**), satisfying additional hexagons.

29.4.1 Formalizing

Definition 29.14 (Braided monoidal category)**.** A **braided monoidal category** adds to a monoidal category a natural isomorphism $\sigma_{A,B} : A \otimes B \to B \otimes A$ — the **braiding** — satisfying the two **braiding hexagons** (compatibility with the associator).

Definition 29.15 (Symmetric monoidal category)**.** A **symmetric monoidal category** is a braided monoidal category whose braiding is **self-inverse**: $\sigma_{B,A} \circ \sigma_{A,B} = \mathrm{id}_{A \otimes B}$.

29.4.2 Formally

Theorem 29.16 (Coherence for symmetric monoidal categories)**.** *In a symmetric monoidal category, every diagram built from $\alpha, \lambda, \rho, \sigma$ between two parenthesizations of the same word —* where both words use the same permutation of the input letters *— commutes.*

Remark 29.17 (Note the constraint)**.** Theorem 29.16 is *not* "every formal diagram commutes." Two different ways of permuting $A \otimes B \otimes C$ via braidings can give two different morphisms $A \otimes B \otimes C \to B \otimes A \otimes C$. The theorem says: if both ends of the diagram correspond to the *same* permutation, then any path connecting them commutes. Braids are real morphisms, not coherence noise.

Theorem 29.18 (Coherence for braided monoidal categories)**.** *In a braided monoidal category, the coherence "free model" is the braid category* **Br***: objects are natural numbers, morphisms from m to m are braids on m strands. Two formal diagrams commute iff the corresponding braids are equal as braids.*

Remark 29.19. The contrast: symmetric coherence quotients out the over-and-under information in braids, treating each as a permutation; braided coherence keeps the full braid. This is why braided monoidal categories model topological information (knots, links, ribbon categories) that symmetric ones miss.

29.5 Operads: A Glimpse

The combinatorial structure underlying monoidal coherence is captured by **operads** (and their relatives — props, properads, multicategories).

29.5.1 Formalizing

Definition 29.20 (Operad)**.** A **(symmetric, $\mathcal{V}$-enriched) operad** $\mathcal{O}$ consists of:

- For each $n \geq 0$, an object $\mathcal{O}(n) \in \mathcal{V}$ equipped with a right action of the symmetric group $\mathfrak{S}_n$.
- Composition maps $\gamma : \mathcal{O}(n) \otimes \mathcal{O}(k_1) \otimes \cdots \otimes \mathcal{O}(k_n) \to \mathcal{O}(k_1 + \cdots + k_n)$ for all $n, k_1, \ldots, k_n$.
- Unit $\eta : I \to \mathcal{O}(1)$.

Satisfying associativity, unit, and equivariance axioms.

Remark 29.21 (Algebras over an operad)**.** For an operad $\mathcal{O}$ in $\mathcal{V}$, an **$\mathcal{O}$-algebra** is an object $A \in \mathcal{V}$ equipped with structure morphisms $\mathcal{O}(n) \otimes A^{\otimes n} \to A$ satisfying compatibility with γ and η.

Examples:

- The *associative operad* Assoc in **Set**: $\mathrm{Assoc}(n) = \mathfrak{S}_n$. Algebras are non-commutative monoids.
- The *commutative operad* Comm: $\mathrm{Comm}(n) = \{*\}$. Algebras are commutative monoids.
- The *little disks operad* (in **Top**): E_2-algebras are braided monoidal structures up to homotopy.

29.5.2 Reorganizing: Operads as Monads

Proposition 29.22 (Operad-monad correspondence)**.** *A symmetric $\mathcal{V}$-operad $\mathcal{O}$ in a cocomplete monoidal $\mathcal{V}$ gives a monad $T_{\mathcal{O}}$ on $\mathcal{V}$:*

$$T_{\mathcal{O}}(X) \;=\; \coprod_{n\geq 0} \mathcal{O}(n) \otimes_{\mathfrak{S}_n} X^{\otimes n}.$$

$\mathcal{O}$-algebras and $T_{\mathcal{O}}$-algebras coincide.

Remark 29.23. This is the link between operad theory and the formal theory of monads (Chapter 32). An operad presents an algebraic theory with n-ary operations for every n; the monad presents the same theory as a coalgebraic gadget.

29.6 Exercises

State the monoidal strictification theorem.	Every monoidal category is monoidally equivalent to a strict monoidal category.
How does monoidal strictification follow from bicategorical strictification?	A monoidal category is a one-object bicategory. The bicategorical strictification of Chapter 25 preserves the one-object property; the resulting strict 2-category corresponds to a strict monoidal category.
Describe $\mathcal{V}^{\mathrm{st}}$ explicitly.	Objects: finite lists of objects of $\mathcal{V}$. Morphisms: morphisms of $\mathcal{V}$ between the left-associated tensor of the lists. Tensor: concatenation. Unit: empty list. All structure 2-cells are identities.
What is the equivalence $\widehat{(-)} : \mathcal{V}^{\mathrm{st}} \to \mathcal{V}$?	$(A_1, \ldots, A_n) \mapsto A_1 \otimes \cdots \otimes A_n$ (left-bracketed). It is essentially surjective and fully faithful.
State Mac Lane's coherence theorem (in this language).	In any monoidal category, every diagram of associators and unitors with the same source and target commutes.
Why does Mac Lane's coherence follow from strictification?	In the strict category, all diagrams of identity 2-cells trivially commute. The equivalence with $\mathcal{V}$ reflects this.

State coherence for monoidal functors.	In a (lax, oplax, or strong) monoidal functor $F : \mathcal{V} \to \mathcal{W}$, every diagram of formal F-coherence morphisms (involving $\phi_{A,B}$, ϕ_I, F applied to $\mathcal{V}$'s structure morphisms, and $\mathcal{W}$'s structure morphisms) commutes.
What axioms guarantee monoidal-functor coherence?	The associativity hexagon and the unitality squares of Definition 29.8.
Define the free monoidal category $F^{\otimes}(\mathcal{C})$.	The monoidal category with the universal property: every functor $\mathcal{C} \to \mathcal{V}$ extends uniquely (up to monoidal natural iso) to a strong monoidal functor $F^{\otimes}(\mathcal{C}) \to \mathcal{V}$.
Describe $F^{\otimes}(\mathcal{C})$ explicitly.	Objects: finite bracketed lists of objects of $\mathcal{C}$. Morphisms: generated by morphisms of $\mathcal{C}$ on singletons and the associators/unitors as formal isomorphisms.
What is the relationship between $F^{\otimes}(\mathcal{C})$ and the presheaf category?	$F^{\otimes}(\mathcal{C})$ is the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ with Day convolution as the tensor (Chapter 30).
Define a braided monoidal category.	A monoidal category with a natural isomorphism $\sigma_{A,B} : A \otimes B \to B \otimes A$ — the braiding — satisfying two compatibility hexagons with the associator.
Define a symmetric monoidal category.	A braided monoidal category whose braiding is self-inverse: $\sigma_{B,A} \circ \sigma_{A,B} = \mathrm{id}$.
State coherence for symmetric monoidal categories.	Every diagram of $\alpha, \lambda, \rho, \sigma$ between two parenthesizations of the same word — using the same permutation — commutes.
Why is symmetric coherence not "every diagram commutes"?	Because different choices of braiding can give genuinely different permutations; coherence quotients by permutation equivalence, not all rearrangements.
State coherence for braided monoidal categories.	The coherence-free model is the braid category $\mathbf{Br}$. Two formal diagrams commute iff their corresponding braids are equal as braids.
What is the difference between symmetric and braided coherence?	Symmetric coherence forgets the "over-vs. under" information in a braid (keeps only the permutation). Braided coherence retains the full braid. Hence braided monoidal categories model knots/links/ribbon structures while symmetric ones don't.
Define an operad.	A family $\{\mathcal{O}(n)\}_{n \geq 0}$ of $\mathfrak{S}_n$-equipped objects with composition maps $\gamma : \mathcal{O}(n) \otimes \prod_i \mathcal{O}(k_i) \to \mathcal{O}(\sum k_i)$ and unit, satisfying associativity, unit, and equivariance.
What is an algebra over an operad $\mathcal{O}$?	An object A with structure maps $\mathcal{O}(n) \otimes A^{\otimes n} \to A$ compatible with γ and η.
Give three example operads.	Assoc (associative operad, algebras = monoids); Comm (commutative operad, algebras = commutative monoids); E_2 (little disks operad in $\mathbf{Top}$, algebras = braided monoidal structures up to homotopy).
State the operad-monad correspondence.	Each operad $\mathcal{O}$ gives a monad $T_{\mathcal{O}}(X) = \coprod_n \mathcal{O}(n) \otimes_{\mathfrak{S}_n} X^{\otimes n}$; algebras coincide.
Why do operads matter for coherence?	They make the combinatorial structure of monoidal coherence (and braided/symmetric variants) into a workable algebraic gadget. Operad theory governs which coherence diagrams must commute.
What is the strict free monoidal category $F^{\otimes,\mathrm{st}}(\mathcal{C})$?	Objects: flat lists. Morphisms: generated by $\mathcal{C}$ alone, no formal associators. It is the strict 2-category specialization of strictification.

How does the strictification of monoidal functors work?	Given $F : \mathcal{V} \to \mathcal{W}$, transport to $F^{\text{st}} : \mathcal{V}^{\text{st}} \to \mathcal{W}^{\text{st}}$. Since $\mathcal{W}^{\text{st}}$ is strict, F^{st}'s coherence comparators must reduce; the axioms force them to be identities.
In what sense is coherence "conceptual" rather than "combinatorial"?	Once strictification is in hand, coherence reduces to the trivial fact that diagrams in strict categories have identity sides. The strictification theorem absorbs all combinatorial intricacy.
Why is the bicategorical strictification proof of Chapter 25 "deeper" than the monoidal case of Chapter 19?	Because Chapter 19 used a normal-form combinatorial argument specialized to monoidal coherence. Chapter 25's strictification works for arbitrary bicategories — and monoidal is just one example. The bicategorical proof unifies many coherence theorems.
What is the "free strict monoidal category on a strictly monoidal generating data"?	The free strict monoidal category $F^{\otimes,\text{st}}(\mathcal{C})$ is its own strictification.
What is the monoidal natural transformation between two strong monoidal functors?	A natural transformation $\eta : F \Rightarrow G$ such that the diagrams $\eta_{A\otimes B} \circ \phi^F_{A,B} = \phi^G_{A,B} \circ (\eta_A \otimes \eta_B)$ and $\eta_I \circ \phi^F_I = \phi^G_I$ commute.
What does monoidal equivalence buy beyond monoidal isomorphism?	Monoidal equivalence is weaker: it requires only essentially surjective fully faithful strong monoidal functors with monoidal natural isomorphisms as units/counits. Strictification gives *equivalence*, not isomorphism, between $\mathcal{V}$ and $\mathcal{V}^{\text{st}}$.
Summarize the chapter's contribution beyond Chapter 19.	Coherence is now a bicategorical phenomenon (one-object special case of strictification); monoidal-functor coherence is established formally; free monoidal categories and operads give the algebraic backbone.

Chapter Summary

Chapter Summary

Strictification, derived twice. Every monoidal category is monoidally equivalent to a strict one — once proved combinatorially in Chapter 19, here re-proved as the one-object special case of bicategorical strictification (Chapter 25). Mac Lane's coherence becomes the trivial observation that in the strict category, all diagrams have identity sides.

Coherence for monoidal functors. The associativity hexagon and unitality squares of $\phi_{A,B}, \phi_I$ imply that every formal coherence diagram involving F, its comparators, and the structure of source/target commutes. Same strictification proof.

Free monoidal categories. $F^{\otimes}(\mathcal{C})$ — objects = bracketed lists, morphisms generated by $\mathcal{C}$ and formal coherence isos — is universal among monoidal categories receiving a functor from $\mathcal{C}$. It equals the presheaf category with Day convolution (Chapter 30).

Symmetric/braided coherence. Symmetric coherence: every diagram between two parenthesizations of the same word using the same permutation commutes (different permutations are genuinely different morphisms). Braided coherence: governed by the braid category **Br**. The distinction explains why braided monoidal categories model knots and ribbons.

Operads. The algebraic gadget capturing n-ary structure under monoidal coherence. Every operad yields a monad; algebras coincide. Examples: Assoc, Comm, E_2. Operads are the combinatorial spine of higher coherence theory.

Formal Restatement

Formal Restatement

Strictification. For every monoidal category $\mathcal{V}$, there exist a strict monoidal category $\mathcal{V}^{\text{st}}$ and a strong monoidal equivalence $\widehat{(-)} : \mathcal{V}^{\text{st}} \to \mathcal{V}$. Construction: objects are finite lists of objects of $\mathcal{V}$; morphisms are $\mathcal{V}$-morphisms between left-bracketed tensors; tensor is concatenation.

Mac Lane coherence. Every diagram of associators and unitors in a monoidal category commutes (special case of strictification).

Monoidal-functor coherence. If $F : \mathcal{V} \to \mathcal{W}$ is monoidal (lax/oplax/strong) satisfying the associativity hexagon and unitality squares, every diagram of formal coherence commutes.

Free monoidal category. For any category $\mathcal{C}$, $F^{\otimes}(\mathcal{C})$ exists and satisfies the universal property: every functor $\mathcal{C} \to \mathcal{V}$ extends essentially uniquely to a strong monoidal functor $F^{\otimes}(\mathcal{C}) \to \mathcal{V}$.

Symmetric coherence. In a symmetric monoidal category, every diagram of $\alpha, \lambda, \rho, \sigma$ between parenthesizations representing the *same permutation* of the input letters commutes.

Braided coherence. The coherence-free model is the braid category $\mathbf{Br}$; formal diagrams commute iff their braids are equal.

Operad to monad. For a symmetric $\mathcal{V}$-operad $\mathcal{O}$ in cocomplete $\mathcal{V}$:

$$T_{\mathcal{O}}(X) \;=\; \coprod_{n \geq 0} \mathcal{O}(n) \otimes_{\mathfrak{S}_n} X^{\otimes n},$$

with $\mathcal{O}$-algebras $= T_{\mathcal{O}}$-algebras.

Chapter 30 introduces Day convolution *— the monoidal structure on enriched functor categories that completes the identification of* $F^{\otimes}(\mathcal{C})$ *with* $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.

Chapter 30

Enriched Categories: Day Convolution

Chapter Overview

Chapter 20 developed enriched category theory: $\mathcal{V}$-categories, enriched Yoneda, weighted limits, base change, self-enrichment. Chapter 26 delivered the coend calculus that those constructions depend on. Chapter 29 identified free monoidal categories with presheaf categories "with the right tensor." This chapter delivers that tensor: **Day convolution**, the canonical monoidal structure on $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ when $\mathcal{C}$ is itself a $\mathcal{V}$-monoidal category.
Day convolution is a single coend formula that turns the Yoneda embedding into a strong monoidal functor and exhibits the presheaf category as the free $\mathcal{V}$-cocompletion-with-monoidal-structure of $\mathcal{C}$. We develop its construction, prove its universal property, derive the closed structure, and apply it to enriched Kan extensions and to the formal theory of monads (preview, Chapter 32). The chapter closes with Cauchy completion in the Day-convolution-aware sense and a sketch of where "monoidal Yoneda" takes us next.

Where the Name Comes From

Day convolution is named for *Brian J. Day*, who introduced the construction in his 1970 paper *On Closed Categories of Functors*. The terminology echoes *convolution* from analysis (the convolution of two functions $f * g$); Day's categorical convolution is the coend formula $(F * G)(c) = \int^{c_1, c_2} \mathcal{C}(c_1 \otimes c_2, c) \otimes F(c_1) \otimes G(c_2)$, which combines two functors using the monoidal structure of the source category. Day convolution is the canonical way to lift monoidal structure from $\mathcal{C}$ to the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]$.

30.1 The Day Convolution Construction

30.1.1 Formalizing

Fix a complete and cocomplete symmetric monoidal closed category $\mathcal{V}$ and a small $\mathcal{V}$-monoidal category $\mathcal{C}$. That is, $\mathcal{C}$ is a $\mathcal{V}$-category equipped with a $\mathcal{V}$-functor $\otimes_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$ and a unit object $I_{\mathcal{C}} \in \mathcal{C}$, satisfying the $\mathcal{V}$-enriched monoidal axioms.

Definition 30.1 (Day convolution)**.** For $\mathcal{V}$-functors $F, G : \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$, the **Day convolution** $F \star_{\mathrm{Day}} G : \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$ is defined by the coend

$$(F \star_{\mathrm{Day}} G)(c) \;=\; \int^{a, b \in \mathcal{C}} \mathcal{C}(c, a \otimes_{\mathcal{C}} b) \otimes F(a) \otimes G(b).$$

The Day-convolution unit is the representable $\mathcal{C}(-, I_{\mathcal{C}}) = Y(I_{\mathcal{C}})$.

How to Say This

$F \star_{\mathrm{Day}} G$ *(read: "F Day-convolution G")* or simply *(read: "F star G")* when the context is clear. The integral signs are over the pair (a, b) — i.e., a double coend over $\mathcal{C}$ in both arguments.

30.1.2 Reorganizing: Why This Formula?

Remark 30.2 (Conceptual reading)**.** The Day convolution is the "left Kan extension along the monoidal product" of the pair (F, G). Explicitly: $\mathcal{C} \otimes \mathcal{C} \xrightarrow{F \otimes G} \mathcal{V} \otimes \mathcal{V} \xrightarrow{\otimes_{\mathcal{V}}} \mathcal{V}$, then take the left Kan extension along $\otimes_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$. The coend formula (Chapter 26, Example 26.13) gives the displayed integral.

Theorem 30.3 (Day convolution is a $\mathcal{V}$-functor monoidal structure)**.** *$\star_{\mathrm{Day}}$ makes $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ into a $\mathcal{V}$-enriched monoidal category. Its associator and unitors are induced from those of $\mathcal{C}$ and $\mathcal{V}$.*

Show $F \star_{\text{Day}} (G \star_{\text{Day}} H) \cong (F \star_{\text{Day}} G) \star_{\text{Day}} H$ via the associator of $\mathcal{C}$.	*Compute both sides by Fubini and the coend manipulations.*
$(F \star_{\text{Day}} (G \star_{\text{Day}} H))(c) = \int^{a,b} \mathcal{C}(c, a \otimes b) \otimes F(a) \otimes \int^{x,y} \mathcal{C}(b, x \otimes y) \otimes G(x) \otimes H(y)$.	*Substitute the definition.*
Apply Fubini (Theorem 26.16) and the ninja Yoneda lemma to integrate over b against $\mathcal{C}(b, x \otimes y)$.	*Yoneda reduction reduces* $\int^b \mathcal{C}(c, a \otimes b) \otimes \mathcal{C}(b, x \otimes y)$ *to* $\mathcal{C}(c, a \otimes (x \otimes y))$.
Result: $\int^{a,x,y} \mathcal{C}(c, a \otimes (x \otimes y)) \otimes F(a) \otimes G(x) \otimes H(y)$.	*After Yoneda reduction.*
Symmetrically, $((F \star_{\text{Day}} G) \star_{\text{Day}} H)(c) \cong \int^{a,x,y} \mathcal{C}(c, (a \otimes x) \otimes y) \otimes F(a) \otimes G(x) \otimes H(y)$.	*By the same calculation with the inner coend in the first argument.*
The associator of $\mathcal{C}$ gives an isomorphism $\mathcal{C}(c, a \otimes (x \otimes y)) \cong \mathcal{C}(c, (a \otimes x) \otimes y)$, natural in a, x, y, c.	*Functoriality of* $\mathcal{C}$ *along* $\alpha_{\mathcal{C}}$.
This induces $F \star_{\text{Day}} (G \star_{\text{Day}} H) \cong (F \star_{\text{Day}} G) \star_{\text{Day}} H$.	*Coend functoriality.*
Unit: $Y(I_{\mathcal{C}}) \star_{\text{Day}} F \cong F$ via the unit-law isomorphism of $\mathcal{C}$. Symmetrically for the right unit.	*Same Yoneda-reduction argument.*
Pentagon and triangle axioms for $\star_{\text{Day}}$ follow from those for $\otimes_{\mathcal{C}}$.	*Functoriality transfers commutativity.*

30.1.3 Formally

Theorem 30.4 (Day convolution is closed)**.** *The Day-monoidal category* $([\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\text{Day}}, Y(I_{\mathcal{C}}))$ *is symmetric monoidal closed when* $\mathcal{C}$ *is symmetric monoidal. The internal hom is*

$$[G, H]_{\text{Day}}(c) \;=\; \int_b \mathcal{V}(G(b), H(c \otimes b)).$$

Sketch. Verify the adjunction $[\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}(F \star_{\text{Day}} G, H) \cong [\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}(F, [G, H]_{\text{Day}})$ by direct calculation: substitute the coend formula for the convolution, use Yoneda reduction on the $\mathcal{C}(c, a \otimes b)$ factor, and identify the resulting double end with $[G, H]_{\text{Day}}$. Symmetry comes from that of $\mathcal{C}$. □

30.2 The Universal Property

30.2.1 Formalizing

Day convolution exhibits the presheaf category as the *free* $\mathcal{V}$*-cocompletion that is also a* $\mathcal{V}$*-monoidal extension* of $\mathcal{C}$.

Theorem 30.5 (Universal property of Day convolution)**.** *Let* $\mathcal{C}$ *be a small* $\mathcal{V}$*-monoidal category and let* $(\mathcal{E}, \otimes_{\mathcal{E}}, I_{\mathcal{E}})$ *be a cocomplete* $\mathcal{V}$*-monoidal category whose tensor product preserves colimits in each variable. Then every* $\mathcal{V}$*-monoidal functor* $F : \mathcal{C} \to \mathcal{E}$ *extends, up to essentially unique* $\mathcal{V}$*-monoidal natural isomorphism, to a* $\mathcal{V}$*-cocontinuous strong monoidal functor*

$$\widetilde{F} \;:\; ([\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\text{Day}}) \to \mathcal{E}$$

with $\widetilde{F} \circ Y \cong F$ *as* $\mathcal{V}$*-monoidal functors.*

Define $\widetilde{F}(P) = \int^c P(c) \otimes_{\mathcal{E}} F(c)$ (weighted colimit; Chapter 28, Theorem 48.10).	*This is the $\mathcal{V}$-cocontinuous extension.*
Show $\widetilde{F}$ is strong monoidal: $\widetilde{F}(F_1 \star_{\mathrm{Day}} F_2) \cong \widetilde{F}(F_1) \otimes_{\mathcal{E}} \widetilde{F}(F_2)$.	*The key computation.*
$\widetilde{F}(F_1 \star_{\mathrm{Day}} F_2) = \int^c (F_1 \star F_2)(c) \otimes_{\mathcal{E}} F(c) = \int^{c,a,b} \mathcal{C}(c, a \otimes b) \otimes_{\mathcal{V}} F_1(a) \otimes_{\mathcal{V}} F_2(b) \otimes_{\mathcal{E}} F(c)$.	*Substituting definitions.*
Apply Yoneda reduction to the integral over c against $\mathcal{C}(c, a \otimes b)$: $\int^c \mathcal{C}(c, a \otimes b) \otimes_{\mathcal{E}} F(c) \cong F(a \otimes b) \cong F(a) \otimes_{\mathcal{E}} F(b)$.	*F is monoidal.*
So $\widetilde{F}(F_1 \star_{\mathrm{Day}} F_2) \cong \int^{a,b} F_1(a) \otimes_{\mathcal{V}} F_2(b) \otimes_{\mathcal{E}} F(a) \otimes_{\mathcal{E}} F(b)$.	*Substituting and rearranging.*
Right-hand side: $\widetilde{F}(F_1) \otimes_{\mathcal{E}} \widetilde{F}(F_2) = \left(\int^a F_1(a) \otimes_{\mathcal{E}} F(a)\right) \otimes_{\mathcal{E}} \left(\int^b F_2(b) \otimes_{\mathcal{E}} F(b)\right)$.	*Substituting the definition of $\widetilde{F}$.*
Since $\otimes_{\mathcal{E}}$ preserves colimits, we can move the $\otimes_{\mathcal{E}}$ inside the coends; the two expressions agree.	*Cocontinuity of $\otimes_{\mathcal{E}}$ plus Fubini.*
Unit: $\widetilde{F}(Y(I_{\mathcal{C}})) = \int^c \mathcal{C}(c, I_{\mathcal{C}}) \otimes_{\mathcal{E}} F(c) \cong F(I_{\mathcal{C}}) \cong I_{\mathcal{E}}$.	*Yoneda reduction then F preserves the unit.*
Uniqueness: any cocontinuous strong monoidal G extending F must agree with $\widetilde{F}$ on representables (by $G \circ Y \cong F$), hence on all presheaves (by cocontinuity and density).	*Standard uniqueness argument.*

30.2.2 Reorganizing: The Slogan

Corollary 30.6 (Day convolution = free monoidal cocompletion). *For a small $\mathcal{V}$-monoidal category $\mathcal{C}$, the Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ exhibits the Day-monoidal presheaf category as the free $\mathcal{V}$-cocomplete monoidal extension of $\mathcal{C}$.*

Remark 30.7 (The full picture)**.** Combining Chapter 28 (Yoneda as free cocompletion) and the present theorem:

- Cocompletion alone: $\mathcal{C} \xrightarrow{Y} [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$. Universal among cocomplete extensions.
- Monoidal cocompletion: same target with Day convolution. Universal among cocomplete monoidal extensions.

Day convolution is the canonical lift of the universal property from "cocompletion" to "monoidal cocompletion."

30.3 Enriched Kan Extensions

The coend calculus and Day convolution combine to give the cleanest formulation of enriched Kan extensions.

30.3.1 Formalizing

Theorem 30.8 (Enriched Kan extension formulas)**.** *For $\mathcal{V}$-functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$ (with $\mathcal{E}$ tensored and cocomplete), the* ***enriched left Kan extension*** $\mathrm{Lan}_K F : \mathcal{D} \to \mathcal{E}$ *exists and is given by*

$$(\mathrm{Lan}_K F)(d) \;=\; \int^{c} \mathcal{D}(Kc, d) \otimes_{\mathcal{E}} F(c).$$

Dually, the ***enriched right Kan extension*** $\mathrm{Ran}_K F$ *(with $\mathcal{E}$ cotensored and complete) is*

$$(\mathrm{Ran}_K F)(d) \;=\; \int_{c} [\mathcal{D}(d, Kc), F(c)]_{\mathcal{E}}.$$

Proof. The formulas are the weighted colimit and weighted limit, respectively (Chapter 27, § 27.5). Existence follows from cocompleteness (resp. completeness) of $\mathcal{E}$ via the reduction of weighted (co)limits to conical (co)limits + (co)tensors (Chapter 27, Theorem 27.18). □

Theorem 30.9 (Kan extensions form an adjoint triple along Yoneda)**.** *For the Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$, there is an adjoint triple*

$$\mathrm{Lan}_Y \dashv Y^* \dashv \mathrm{Ran}_Y$$

between $[\mathcal{C}, \mathcal{E}]_{\mathcal{V}}$ (functor $\mathcal{V}$-category) and $[[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}, \mathcal{E}]_{\mathcal{V}}$ (presheaf functor category). Here Y^* is precomposition with Y.

Sketch. This is the enriched analogue of Chapter 18's adjoint triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$ specialized to $p = Y$. The proof uses the coend formulas of Theorem 30.8 and the universal property of the presheaf category. □

30.3.2 Reorganizing: Day Convolution as Kan Extension

Proposition 30.10 (Day convolution as Lan)**.** *The Day convolution $F \star_{\mathrm{Day}} G$ is the left Kan extension of the composite*

$$\mathcal{C} \otimes \mathcal{C} \xrightarrow{Y \otimes Y} [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}} \otimes [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}} \xrightarrow{\star_{\mathrm{Day}}} [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$$

along $\otimes_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$, evaluated at the pair (F, G).

Proof. Direct from Definition 30.1 and the left Kan coend formula of Theorem 30.8. □

30.4 Monoids in the Day-Monoidal Category

The most striking applications of Day convolution come from looking at *monoids* in the resulting monoidal category $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$.

30.4.1 Formalizing

Theorem 30.11 (Monoids in Day convolution)**.** *For a small $\mathcal{V}$-monoidal category $\mathcal{C}$:*

$$\mathrm{Mon}([\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\mathrm{Day}}) \;\simeq\; [\mathcal{C}, \mathcal{V}]_{\mathcal{V}}^{\text{lax monoidal}},$$

the category of ***lax monoidal $\mathcal{V}$-functors*** *$\mathcal{C} \to \mathcal{V}$.*

Remark 30.12 (Interpretation)**.** A monoid in $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ with Day tensor is precisely a lax monoidal functor $\mathcal{C} \to \mathcal{V}$. This is the *universal monoid construction*: Day convolution turns "lax monoidal functor" into "monoid in a single category."

30.4.2 Reorganizing: Examples and Special Cases

Example 30.13 (Algebras over a $\mathcal{V}$-operad)**.** For an operad $\mathcal{O}$ in $\mathcal{V}$, monoids in the Day-monoidal $[\mathbb{F}^{\mathrm{op}}, \mathcal{V}]$ (with $\mathbb{F}$ the category of finite sets and bijections, monoidal under disjoint union) recover $\mathcal{O}$-algebras. Operads-as-monoids becomes a special case of Day-convolution monoid theory.

Example 30.14 (The Hochschild cohomology setup)**.** For an associative algebra A, the Hochschild cohomology of A arises as the cohomology of the Day-monoidal presheaf category $[\Delta^{\mathrm{op}}, \mathcal{V}]$ at the specific monoid A. Day convolution captures the "cosimplicial homotopical structure" uniformly.

30.5 Cauchy Completion Revisited

Chapter 28 introduced Cauchy completeness in the (unenriched) profunctor setting. In the enriched setting, Cauchy completeness interacts richly with Day convolution.

30.5.1 Formalizing

Definition 30.15 (Enriched Cauchy completion)**.** A $\mathcal{V}$-category $\mathcal{C}$ is **Cauchy complete** if for every $\mathcal{V}$-profunctor $P : \mathbf{1} \to \mathcal{C}$ admitting a right adjoint as a 1-cell in $\mathcal{V}$-**Prof**, P is representable by an object of $\mathcal{C}$.

The **Cauchy completion** $\bar{\mathcal{C}}$ is the universal Cauchy-complete enlargement; it is precisely the full sub-$\mathcal{V}$-category of $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ consisting of the **absolute presheaves**.

Remark 30.16 (Concrete cases)**.**

- In the **Set**-enriched case, Cauchy completion is the idempotent (Karoubian) completion.
- In the **Ab**-enriched case, Cauchy completion adds direct summands of free objects.
- In the **Cat**-enriched case (i.e., 2-categories), Cauchy completion adds objects represented by 1-cells with a right adjoint.

30.6 Exercises

Define Day convolution.	For $F, G : \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$, $(F \star_{\mathrm{Day}} G)(c) = \int^{a,b} \mathcal{C}(c, a \otimes_{\mathcal{C}} b) \otimes F(a) \otimes G(b)$.
What is the Day-convolution unit?	The Yoneda embedding of the monoidal unit: $Y(I_{\mathcal{C}}) = \mathcal{C}(-, I_{\mathcal{C}})$.
Why is Day convolution monoidal?	Because the associator and unitors of $\mathcal{C}$ transfer (via Yoneda reduction and functoriality of coends) to associator and unitors of $\star_{\mathrm{Day}}$.

Show that $Y(I_{\mathcal{C}}) \star_{\text{Day}} F \cong F$.	$(Y(I_{\mathcal{C}}) \star F)(c) = \int^{a,b} \mathcal{C}(c, a \otimes b) \otimes \mathcal{C}(a, I) \otimes F(b)$; Yoneda-reduce a to get $\int^{b} \mathcal{C}(c, I \otimes b) \otimes F(b) \cong \int^{b} \mathcal{C}(c, b) \otimes F(b) \cong F(c)$.
What is the internal hom of Day convolution?	$[G, H]_{\text{Day}}(c) = \int_{b} \mathcal{V}(G(b), H(c \otimes b))$.
When is Day convolution closed?	When $\mathcal{C}$ is symmetric monoidal and $\mathcal{V}$ is symmetric monoidal closed and complete.
State the universal property of Day convolution.	For every cocomplete $\mathcal{V}$-monoidal $\mathcal{E}$ and $\mathcal{V}$-monoidal $F : \mathcal{C} \to \mathcal{E}$, there is an essentially unique cocontinuous strong monoidal $\widetilde{F} : [\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}} \to \mathcal{E}$ extending F via Yoneda.
Why is Day convolution the "free monoidal cocompletion"?	Because $[\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}$ with Day tensor is initial among cocomplete monoidal extensions of $\mathcal{C}$; equivalently, the universal property uniquely characterizes the construction.
Write the enriched left Kan extension formula.	$(\text{Lan}_K F)(d) = \int^{c} \mathcal{D}(Kc, d) \otimes_{\mathcal{E}} F(c)$.
Right Kan extension formula?	$(\text{Ran}_K F)(d) = \int_{c} [\mathcal{D}(d, Kc), F(c)]_{\mathcal{E}}$.
Why is $\text{Lan}_Y \dashv Y^* \dashv \text{Ran}_Y$ an adjoint triple?	Because Yoneda is the universal embedding into the free cocompletion; the restriction Y^* has left and right adjoints by the colimit/limit formulas (special case of Chapter 18's adjoint triple).
Why is Day convolution the left Kan extension of $\star_{\mathcal{V}} \circ (Y \otimes Y)$ along $\otimes_{\mathcal{C}}$?	Because Yoneda extends pointwise to the presheaf category, and the Day formula is exactly the left Kan extension of the composite "embed-then-tensor" along $\otimes_{\mathcal{C}}$.
State the monoids-in-Day-convolution theorem.	Monoids in $([\mathcal{C}^{\text{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\text{Day}})$ are equivalent to lax monoidal $\mathcal{V}$-functors $\mathcal{C} \to \mathcal{V}$.
Interpret this for operads.	For $\mathcal{C} = \mathbb{F}$ (finite sets and bijections, disjoint union), monoids in Day convolution = lax monoidal $\mathbb{F} \to \mathcal{V}$ = operads.
Why is Day convolution natural in the base category?	Because the formula's coend integrates over all morphisms; functoriality in $\mathcal{C}$ is preserved by the coend construction.
What is an enriched profunctor $\mathcal{C} \to \mathcal{D}$?	A $\mathcal{V}$-functor $P : \mathcal{D}^{\text{op}} \otimes \mathcal{C} \to \mathcal{V}$.
What is enriched Cauchy completion?	The universal Cauchy-complete enlargement: full sub-$\mathcal{V}$-category of presheaves consisting of absolute (representable-up-to-equivalence) ones.
What is **Ab**-Cauchy completion?	Adding direct summands of free objects (i.e., the idempotent splitting for direct sum decompositions).
What does "absolute" mean for a presheaf in this context?	A presheaf P is absolute if its Yoneda extension $\widetilde{Y}^{-1}(P)$ exists in $\mathcal{C}$ — i.e., P is represented by an object of the Cauchy completion.
Why is the Day-convolution monoidal structure "the right one"?	Because it is uniquely determined by the universal property: it is the unique monoidal structure on the free cocompletion that makes Yoneda strongly monoidal.
What is the relationship between Day convolution and the convolution of L^1-functions on a group?	Same formula: $(f * g)(x) = \int f(a)g(x - a)\, da$ is the special case for $\mathcal{C}$ = a group and $\mathcal{V} = \mathbf{Set}$; the coend becomes an ordinary integral over the group.
State the Day convolution formula for a discrete (groupoid) $\mathcal{C}$.	$(F \star_{\text{Day}} G)(c) = \coprod_{a \otimes b \cong c} F(a) \times G(b)/\sim$, where the coend reduces to a coproduct over a single representative; in a group, this is the convolution sum.

Why does $\otimes_{\mathcal{E}}$ need to preserve colimits for the universal property to hold?	Because $\widetilde{F}$ is a colimit-preserving functor; for it to be *monoidal*, the target tensor must distribute over colimits. Otherwise the convolution formula doesn't carry the monoidal structure cleanly.
What's the simplest example of Day convolution?	$\mathcal{C}$ = a discrete monoidal category (a monoid). Then $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the category of $\mathcal{C}$-graded sets, and Day convolution is exactly the convolution on graded sets.
What is the dual of the Day convolution theorem (using right Kan extension)?	The cofree comonoidal cocompletion: $[\mathcal{C}, \mathcal{V}]_{\mathcal{V}}$ with the dual "Day cotensor" has the universal property for comonoids and oplax monoidal functors.
How does Day convolution relate to Chapter 29's free monoidal category?	$F^{\otimes}(\mathcal{C}) \cong ([\mathcal{C}^{\mathrm{op}}, \mathbf{Set}], \star_{\mathrm{Day}})$ when $\mathcal{C}$ is itself monoidal; Day convolution provides the missing tensor on the presheaf category.
What is the explicit formula for an $\mathcal{O}$-algebra in $\mathcal{V}$ via Day convolution?	An $\mathcal{O}$-algebra A corresponds to a lax monoidal functor $\mathbb{F} \to \mathcal{V}$ sending n to $A^{\otimes n}$, with the operad's composition giving the structure 2-cells.
State Day convolution closure as an adjunction.	$[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(F \star_{\mathrm{Day}} G, H) \cong [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}(F, [G, H]_{\mathrm{Day}})$.
Why is the proof of associativity for Day convolution a coend calculation?	Because the formula is itself a coend; associativity reduces to Fubini-plus-Yoneda-reduction, the two pillars of the coend calculus.
What chapter develops the formal theory of monads that uses Day convolution?	Chapter 32 (The Formal Theory of Monads); monoids in Day-monoidal categories provide the foundation for monad transformers and distributive laws (Chapter 35).

Chapter Summary

Chapter Summary

Day convolution. For a $\mathcal{V}$-monoidal $\mathcal{C}$, the convolution $(F \star_{\mathrm{Day}} G)(c) = \int^{a,b} \mathcal{C}(c, a \otimes b) \otimes F(a) \otimes G(b)$ makes $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ into a $\mathcal{V}$-monoidal category. The unit is $Y(I_{\mathcal{C}})$.
Universal property. The Yoneda embedding $Y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ exhibits the Day-monoidal presheaf category as the free $\mathcal{V}$-cocomplete monoidal extension of $\mathcal{C}$: any $\mathcal{V}$-monoidal $F : \mathcal{C} \to \mathcal{E}$ extends uniquely to a cocontinuous strong monoidal $\widetilde{F}$.
Closed structure. For symmetric monoidal $\mathcal{C}$, Day convolution is symmetric monoidal closed, with internal hom $[G, H]_{\mathrm{Day}}(c) = \int_b \mathcal{V}(G(b), H(c \otimes b))$.
Enriched Kan extensions. $\mathrm{Lan}_K F(d) = \int^c \mathcal{D}(Kc, d) \otimes F(c)$ and $\mathrm{Ran}_K F(d) = \int_c [\mathcal{D}(d, Kc), F(c)]$. These give the adjoint triple $\mathrm{Lan}_Y \dashv Y^* \dashv \mathrm{Ran}_Y$ along the Yoneda embedding.
Day-monoid theorem. Monoids in $([\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\mathrm{Day}})$ are equivalent to lax monoidal $\mathcal{V}$-functors $\mathcal{C} \to \mathcal{V}$. Operads and algebraic structures are special cases.
Cauchy completion. The full sub-presheaf category of *absolute* presheaves. In **Set**-enriched: idempotent splitting; in **Ab**-enriched: direct summand completion; in **Cat**-enriched: left-adjoint completion.

Formal Restatement

Formal Restatement

Day convolution.

$$(F \star_{\mathrm{Day}} G)(c) = \int^{a,b \in \mathcal{C}} \mathcal{C}(c, a \otimes_{\mathcal{C}} b) \otimes F(a) \otimes G(b), \qquad \mathbf{1}_{\mathrm{Day}} = Y(I_{\mathcal{C}}).$$

Internal hom (Day-closed).

$$[G, H]_{\mathrm{Day}}(c) = \int_b \mathcal{V}(G(b), H(c \otimes b)).$$

Universal property. The Day-monoidal $[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ is the free $\mathcal{V}$-cocomplete monoidal extension of $\mathcal{C}$. $\widetilde{F}(P) = \int^c P(c) \otimes_{\mathcal{E}} F(c)$.

Enriched Kan extensions.

$$(\mathrm{Lan}_K F)(d) \;=\; \int^c \mathcal{D}(Kc, d) \otimes_{\mathcal{E}} F(c), \qquad (\mathrm{Ran}_K F)(d) \;=\; \int_c [\mathcal{D}(d, Kc), F(c)]_{\mathcal{E}}.$$

Adjoint triple along Y: $\mathrm{Lan}_Y \dashv Y^* \dashv \mathrm{Ran}_Y$.

Day-monoid theorem.

$$\mathrm{Mon}([\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}, \star_{\mathrm{Day}}) \;\simeq\; [\mathcal{C}, \mathcal{V}]_{\mathcal{V}}^{\mathrm{lax\,mon}}.$$

Cauchy completion. $\bar{\mathcal{C}} \subset [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ is the full sub-$\mathcal{V}$-category of absolute presheaves. Examples: idempotent splitting (**Set**), direct-summand closure (**Ab**), left-adjoint splitting (**Cat**).

Chapter 31 develops the general theory of Kan extensions in the enriched setting, building on the formulas of this chapter and the adjoint triple along Yoneda. Pointwise extensions, absolute extensions, and applications to coherence and to the formal theory of monads (Chapter 32) follow.

Chapter 31

Kan Extensions: The General Theory

Chapter Overview

Chapter 18 developed Kan extensions in the **Set**-enriched setting: the adjoint triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$, the colimit formula, pointwise and absolute extensions, the nerve/realization paradigm. Chapters 27 and 30 lifted the formulas to the enriched setting. This chapter completes the theory: pointwise and absolute Kan extensions in $\mathcal{V}$-**Cat**, the *codensity monad* (every functor $G : \mathcal{B} \to \mathcal{C}$ defines a monad $\mathrm{Ran}_G\, G$ on $\mathcal{C}$), algebraic theories as Kan extensions of representables, and the unifying slogan: *every adjoint is a Kan extension; every monad is a Kan extension; every limit is a Kan extension.*

Where the Name Comes From

Pointwise Kan extension (also *absolute* when stronger conditions hold) is the variant of Kan extension that is computable "at each point" via a (co)limit formula. The terminology distinguishes from *non-pointwise* extensions, which may exist but are not computable from local data. Most natural Kan extensions are pointwise.
Codensity monad (Kock 1966): for a functor $G : \mathcal{B} \to \mathcal{C}$, the right Kan extension $\mathrm{Ran}_G G$ is a canonical monad on $\mathcal{C}$. The terminology marks the dual of "density": if G is dense (every $\mathcal{C}$-object is a colimit of G-images), the codensity monad is trivial; obstruction to density *is* the codensity monad. The classical Manes example: the ultrafilter monad on **Set** is the codensity of **FinSet** $\hookrightarrow$ **Set**.

31.1 Pointwise Kan Extensions in $\mathcal{V}$-Cat

31.1.1 Formalizing

Definition 31.1 (Pointwise Kan extension)**.** Let $\mathcal{V}$ be a complete, cocomplete symmetric monoidal closed category. For $\mathcal{V}$-functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$, the **pointwise left Kan extension** $\mathrm{Lan}_K\, F : \mathcal{D} \to \mathcal{E}$ is the $\mathcal{V}$-functor defined object-wise by the weighted colimit

$$(\mathrm{Lan}_K\, F)(d) \;=\; \mathcal{D}(K-, d) \star F \;=\; \int^{c} \mathcal{D}(Kc, d) \otimes_{\mathcal{E}} F(c).$$

Dually, the **pointwise right Kan extension** is

$$(\mathrm{Ran}_K\, F)(d) \;=\; \{\mathcal{D}(d, K-), F\} \;=\; \int_{c} [\mathcal{D}(d, Kc), F(c)]_{\mathcal{E}}.$$

Theorem 31.2 (Pointwise Kan extensions are Kan extensions)**.** *The pointwise extensions are Kan extensions in the classical sense: they satisfy the universal property of left/right Kan extension as initial/terminal objects in the appropriate comma category of functors.*

$(\mathrm{Lan}_K F)$ comes with a $\mathcal{V}$-natural unit $\eta : F \Rightarrow (\mathrm{Lan}_K F) \circ K$.	*Defined at c by the universal cowedge into the coend* $\int^{c'} \mathcal{D}(Kc', Kc) \otimes F(c')$ *at the identity* $\mathcal{D}(Kc, Kc)$.
For any $G : \mathcal{D} \to \mathcal{E}$ and $\sigma : F \Rightarrow G \circ K$, define $\widetilde{\sigma} : \mathrm{Lan}_K F \Rightarrow G$ by $\widetilde{\sigma}_d : \int^c \mathcal{D}(Kc, d) \otimes F(c) \to G(d)$ sending $(f : Kc \to d, x \in F(c))$ to $G(f) \circ \sigma_c(x)$.	*Universal property of the coend.*
$\widetilde{\sigma}$ is well-defined (compatible with the coend quotient) by naturality of σ in c.	*Cowedge condition.*
$\widetilde{\sigma}$ satisfies $\widetilde{\sigma}_K \circ \eta = \sigma$ and is unique with that property.	*Direct calculation; standard.*
Conclude: $\mathrm{Lan}_K F$ is the universal $\mathcal{V}$-functor with a $\mathcal{V}$-natural transformation $F \Rightarrow (\mathrm{Lan}_K F) \circ K$.	*Initial in the appropriate comma category.*
Dually for $\mathrm{Ran}_K F$.	*Reverse all arrows.*

31.1.2 Reorganizing: Pointwise vs. "Global"

A Kan extension as defined in Chapter 17 (Kan extension along a functor, defined by the universal property in the comma category of functors) need not be pointwise — i.e., it may not satisfy the weighted-colimit formula at each object. Pointwise Kan extensions are the well-behaved kind; in the enriched setting they are essentially the only kind worth considering.

Theorem 31.3 (Pointwise extensions exist under cocompleteness)**.** *If $\mathcal{E}$ is $\mathcal{V}$-cocomplete and $\mathcal{C}$ is small, then $\mathrm{Lan}_K F$ exists and is pointwise for every K, F. Dually, if $\mathcal{E}$ is $\mathcal{V}$-complete, $\mathrm{Ran}_K F$ exists and is pointwise.*

Proof. Existence of the weighted colimits/limits in the formulas of Definition 31.1 follows from cocompleteness/completeness (Chapter 27, Theorem 27.18). That they satisfy the universal property of Kan extension is Theorem 31.2. □

31.1.3 Formally

Example 31.4 (Left Kan extension along an inclusion)**.** For $K : \mathcal{C} \hookrightarrow \mathcal{D}$ a fully faithful inclusion and $F : \mathcal{C} \to \mathcal{E}$,

$$(\mathrm{Lan}_K F)(Kc) \;\cong\; F(c)$$

(by Yoneda reduction applied to $\mathcal{D}(Kc', Kc) \cong \mathcal{C}(c', c)$). The extension fixes the embedded subcategory; only the new objects in $\mathcal{D}$ get genuinely new values.

31.2 Absolute Kan Extensions

31.2.1 Formalizing

Definition 31.5 (Absolute Kan extension)**.** A Kan extension $\mathrm{Lan}_K F$ is **absolute** if it is preserved by every $\mathcal{V}$-functor $G : \mathcal{E} \to \mathcal{F}$:

$$G \circ \mathrm{Lan}_K F \;\cong\; \mathrm{Lan}_K(G \circ F).$$

Theorem 31.6 (Characterization of absolute Kan extensions)**.** *$\mathrm{Lan}_K F$ is absolute iff every constituent weighted colimit $\mathcal{D}(K-, d) \star F$ is an absolute colimit (preserved by every functor).*

Remark 31.7. Absolute colimits exist in only special cases. In **Set**-enriched: filtered colimits of split idempotents, and not much more. In other enrichments: e.g., **Ab**-absolute colimits include direct summand splittings. The absoluteness condition is restrictive.

31.2.2 Reorganizing

Proposition 31.8 (Cauchy completeness via absoluteness)**.** *A $\mathcal{V}$-category $\mathcal{C}$ is Cauchy complete iff every absolute $\mathcal{V}$-colimit-diagram has a colimit in $\mathcal{C}$. The Cauchy completion $\bar{\mathcal{C}}$ is the smallest enlargement with this property.*

31.3 The Codensity Monad

31.3.1 Formalizing

Definition 31.9 (Codensity monad)**.** For a $\mathcal{V}$-functor $G : \mathcal{B} \to \mathcal{C}$, the **codensity monad** $\mathbb{T}_G$ is the monad on $\mathcal{C}$ defined by

$$T_G(c) \;=\; (\mathrm{Ran}_G G)(c) \;=\; \int_b [\mathcal{C}(c, Gb), Gb]_{\mathcal{C}}.$$

Its unit is the unit of the Kan extension; its multiplication comes from the natural composition action of $\mathrm{Ran}_G G$ on itself.

Theorem 31.10 (Codensity is a monad)**.** *$\mathbb{T}_G = (T_G, \eta, \mu)$ is a $\mathcal{V}$-monad on $\mathcal{C}$, called the* ***codensity monad of*** *G.*

$T_G = \mathrm{Ran}_G\, G$ comes with a $\mathcal{V}$-natural unit $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow T_G$.	*The unit of the right Kan extension.*
The composite $T_G \circ T_G : \mathcal{C} \to \mathcal{C}$ factors through the colim that defines T_G.	*T_G preserves the relevant limits because it is defined as a limit.*
Construct $\mu : T_G \circ T_G \Rightarrow T_G$: using the universal property of $\mathrm{Ran}_G\, G$, $T_G \circ T_G$ also admits a natural transformation from G, and the universality factors it through T_G.	*Detailed argument; uses the fact that Kan extensions form an adjunction.*
Associativity: $\mu \circ T_G\mu = \mu \circ \mu T_G$.	*Both factor through the same universal arrow from G.*
Unit axiom: $\mu \circ \eta T_G = \mu \circ T_G \eta = \mathrm{id}_{T_G}$.	*Unit of the Kan extension.*
Conclude: (T_G, η, μ) is a monad on $\mathcal{C}$.	*Standard.*

31.3.2 Reorganizing: Adjunctions Give Codensity

Theorem 31.11 (Adjoint gives its left's codensity)**.** *If $F \dashv G$ with $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$, then $\mathbb{T}_G = GF$ is the standard adjunction monad.*

Proof. $\mathrm{Ran}_G\, G = GF$ when $F \dashv G$: the right Kan extension of a right adjoint along itself is the adjunction composite. Hence $\mathbb{T}_G = GF$, the standard monad from the adjunction (Chapter 8). □

Remark 31.12 (The codensity is a generalization)**.** For functors G that are *not* right adjoints, the codensity monad exists but does not arise from an adjunction. This is the situation Beck sought to characterize with monadicity (Chapter 21): the codensity monad provides "the closest monad to G," adjunction or not.

31.3.3 Formally: Examples

Example 31.13 (Ultrafilter monad as codensity)**.** The **ultrafilter monad** on **Set** is the codensity monad of the inclusion **FinSet** $\hookrightarrow$ **Set**. Explicitly, $T_G(X) = \int_F [\mathbf{Set}(X, F), F]$ where F ranges over finite sets; this end equals the set of ultrafilters on X. *(Kennison–Gildenhuys 1971.)*

Example 31.14 (Double dualization monad)**.** For a field k, the codensity monad of $\mathbf{FinDim}_k \hookrightarrow \mathbf{Vect}_k$ is the **double dualization monad** $V \mapsto V^{**}$. Its algebras are precisely the linearly compact vector spaces.

Example 31.15 (Topological codensity)**.** The codensity monad of **CompactHaus** $\hookrightarrow$ **Top** is the **Stone–Čech monad** $\beta : \mathbf{Top} \to \mathbf{Top}$: its algebras are the compact Hausdorff spaces. Stone–Čech compactification is monadic over **Top** via the codensity construction.

31.4 Algebraic Theories as Kan Extensions

The codensity monad reveals an organizing principle: many "algebraic theories" over a base category arise as Kan extensions of representable functors.

31.4.1 Formalizing

Theorem 31.16 (Algebraic theories via Kan)**.** *For a small category $\mathcal{T}$ of "operations" and a functor $J : \mathcal{T} \to$ **Set** presenting the theory as Lawvere did (each $J(t)$ is the underlying set of the free algebra on $|t|$ generators), the free-algebra monad on **Set** is*

$$T \;=\; \mathrm{Lan}_J\, J.$$

Algebras for this monad are precisely $\mathcal{T}$-models.

Sketch. Lawvere's classical correspondence between Lawvere theories and finitary monads on **Set** (Chapter 21, application section) is recast as "the monad is the left Kan extension of J along itself." The construction's universal property — every algebra for the monad satisfies the operations encoded by $\mathcal{T}$ — follows from the universal property of the Kan extension. □

31.4.2 Reorganizing

Remark 31.17 (The two faces of monads)**.** Theorem 31.16 and Theorem 31.11 together show: every monad on $\mathcal{C}$ is *either* the codensity of an inclusion (the "categorical face") *or* the left Kan extension of a representing functor (the "algebraic face"). Many monads are both simultaneously. Vol III Chapter 32 makes the formal theory explicit.

31.5 The General Slogan

31.5.1 Formalizing

The work of Chapters 26–31 unifies under a single slogan:

> **The slogan.** *Every universal construction in category theory is a Kan extension.*

Theorem 31.18 (Kan extensions subsume universal constructions)**.** *The following are all Kan extensions:*

- *Limits and colimits (Chapter 27, §27.5): every weighted (co)limit is a Kan extension.*
- *Adjoint functors (Chapter 28, Corollary 28.12): every right adjoint is* Ran_Y *for some Yoneda embedding; every left adjoint is* Lan_Y*.*
- *Monads (this chapter, §31.3, §31.4): every monad is either a codensity* $\mathrm{Ran}_G\, G$ *or an algebra-theoretic* $\mathrm{Lan}_J\, J$*.*
- *Free constructions (Chapter 28, §28.4, Example 26.12): every free algebraic construction is the left Kan extension along the Yoneda embedding of a generating sub-category.*

Remark 31.19 (The unifying principle)**.** This is not a coincidence. Kan extensions are exactly the universal constructions that solve the universal mapping problem "find an object equipped with a universal natural transformation to/from a given functor." Every universal construction reduces to this same template; the only variable is which functor and along which arrow.

31.6 Exercises

Define the pointwise left Kan extension.	$(\mathrm{Lan}_K\, F)(d) = \mathcal{D}(K-, d) \star F = \int^c \mathcal{D}(Kc, d) \otimes F(c)$, a weighted colimit.
Define the pointwise right Kan extension.	$(\mathrm{Ran}_K\, F)(d) = \{\mathcal{D}(d, K-), F\} = \int_c [\mathcal{D}(d, Kc), F(c)]$, a weighted limit.
When do pointwise Kan extensions exist?	When the target is $\mathcal{V}$-cocomplete (for Lan) or $\mathcal{V}$-complete (for Ran) and the source is small.
State the universal property of $\mathrm{Lan}_K\, F$.	Initial: there is a unit $\eta : F \Rightarrow (\mathrm{Lan}_K\, F) \circ K$, and for any $\sigma : F \Rightarrow GK$ there is a unique $\widetilde{\sigma} : \mathrm{Lan}_K\, F \Rightarrow G$ with $\widetilde{\sigma}K \circ \eta = \sigma$.
Why is the pointwise extension also a (classical) Kan extension?	Because it satisfies the universal property of Theorem 31.2; the coend formula explicitly constructs the universal arrow.
What is a non-pointwise Kan extension?	One satisfying the universal property in the functor category but not given by the weighted-colimit formula at each object. In enriched settings these are pathological; pointwise is the standard.
Define an absolute Kan extension.	One preserved by every functor: $G \circ \mathrm{Lan}_K\, F \cong \mathrm{Lan}_K(G \circ F)$ for all G.
When is a weighted colimit absolute?	When every functor preserves it. In **Set**: split idempotents and filtered colimits (with finite limits). Rarer in non-**Set** enrichments.
Define the codensity monad of G.	$\mathbb{T}_G = \mathrm{Ran}_G\, G$: a monad on the target of G, with $T_G(c) = \int_b [\mathcal{C}(c, Gb), Gb]$.
Why is the codensity monad a monad?	Because the right Kan extension's universal property gives natural unit and multiplication structures satisfying the monad axioms.
State the adjoint-codensity theorem.	If $F \dashv G$ then $\mathbb{T}_G = GF$, the standard adjunction monad.
What is the codensity monad of **FinSet** $\hookrightarrow$ **Set**?	The ultrafilter monad: $T(X)$ = set of ultrafilters on X.
What is the codensity monad of $\mathbf{FinDim}_k \hookrightarrow \mathbf{Vect}_k$?	The double dualization monad: $V \mapsto V^{**}$.
Codensity of **CompactHaus** $\hookrightarrow$ **Top**?	The Stone–Čech monad: algebras are compact Hausdorff spaces.

Why is the codensity construction a generalization of adjunctions?	Because it exists for any functor G, not just right adjoints. When G has a left adjoint, the codensity recovers the standard monad; otherwise it provides the "best monad" arising from G.
State the algebraic-theory-as-Kan theorem.	For a Lawvere theory $\mathcal{T}$ with $J : \mathcal{T} \to \mathbf{Set}$ presenting it, the free-algebra monad is $T = \mathrm{Lan}_J\, J$, and algebras for this monad are $\mathcal{T}$-models.
What are the "two faces" of monads via Kan extensions?	Codensity ($\mathrm{Ran}_G\, G$, for G from a subcategory) and algebraic ($\mathrm{Lan}_J\, J$, for J presenting a theory). Many monads are both.
State the unifying Kan extension slogan.	Every universal construction in category theory is a Kan extension: limits, colimits, adjoints, monads, free constructions all fit.
Why is this slogan not a coincidence?	Because Kan extensions are exactly the universal solutions to "find an object with a universal natural transformation along a given functor." Every universal construction reduces to this template.
Express the limit of $F : \mathcal{J} \to \mathcal{C}$ as a Kan extension.	$\lim F = (\mathrm{Ran}_{!_{\mathcal{J}}}\, F)(\star)$ where $!_{\mathcal{J}} : \mathcal{J} \to \mathbf{1}$ is the unique functor to the terminal category and $\star$ is the unique object of $\mathbf{1}$.
Express the free algebra functor over a monad T as a Kan extension.	The free T-algebra functor $\mathcal{C} \to \mathcal{C}^T$ is $\mathrm{Lan}_{!^T_{\mathcal{C}}}\, \mathrm{Id}_{\mathcal{C}^T}$ (in suitable form).
Why is Cauchy completion characterized via absolute Kan extensions?	Because Cauchy complete categories are exactly those where every absolute weighted colimit diagram has a colimit; absolute Kan extensions are the "forced" ones.
What is the relationship between codensity monads and the formal theory of monads (Chapter 32)?	Codensity provides examples of monads that are not adjunction-induced; the formal theory of monads provides the 2-categorical setting in which codensity monads naturally live.
Why do pointwise Kan extensions exist in enriched settings?	Because they are weighted (co)limits, and weighted (co)completeness reduces to conical (co)completeness plus (co)tensors (Chapter 27).
Why is the unit $\eta : F \Rightarrow (\mathrm{Lan}_K\, F) \circ K$ the universal arrow?	Because every natural transformation $F \Rightarrow GK$ factors uniquely through $\mathrm{Lan}_K\, F$, by definition of the Kan extension.
What is the relationship between Lan_Y for the Yoneda embedding Y and free cocompletion?	$\mathrm{Lan}_Y\, F$ is the free $\mathcal{V}$-cocompletion of F along Y (Chapter 28, Theorem 48.10).
Why is the Yoneda embedding dense?	Because $\mathrm{Lan}_Y\, Y = \mathrm{Id}_{[\mathcal{C}^{\mathrm{op}}, \mathcal{V}]}$ (Chapter 28, Corollary 28.10); the identity is the Kan extension of Y along itself.
What is the Kan extension along the Yoneda embedding in **Prof**?	The hom-profunctor is the identity 1-cell; Kan extensions along the embedding into **Prof** realize various universal constructions in profunctor language.
In what sense is the codensity monad "the right one" even for non-adjoint G?	Because it provides the universal monad whose algebras are "things satisfying all the relations G does"; the codensity captures the equational content of G as a structure-preserving functor.
How does Theorem 31.18 summarize Volume III's contribution?	It establishes that the entire universal-constructions toolkit of category theory reduces to a single primitive (Kan extension) plus the coend calculus that computes it; this is the formal-spine perspective Volume III sought.

Chapter Summary

Chapter Summary

Pointwise Kan extensions. In the enriched setting, the pointwise extensions are the weighted-colimit and weighted-limit formulas: $(\mathrm{Lan}_K F)(d) = \int^c \mathcal{D}(Kc, d) \otimes F(c)$ and $(\mathrm{Ran}_K F)(d) = \int_c [\mathcal{D}(d, Kc), F(c)]$. They exist whenever the target is cocomplete (resp. complete).
Pointwise satisfies the universal property. These extensions are universal: they are initial/terminal in the relevant functor comma categories. Non-pointwise Kan extensions are pathological in enriched settings.
Absolute Kan extensions. Preserved by every functor. Rare: in **Set**-enriched, mainly split idempotents and filtered colimits. Cauchy completion: the universal closure under absolute colimits.
The codensity monad. $\mathbb{T}_G = \mathrm{Ran}_G G$ on the codomain of G. A monad even when G is not a right adjoint. Examples: ultrafilter monad (codensity of **FinSet** $\hookrightarrow$ **Set**); double dualization (codensity of $\mathbf{FinDim}_k \hookrightarrow \mathbf{Vect}_k$); Stone–Čech monad (codensity of **CompactHaus** $\hookrightarrow$ **Top**).
Algebraic theories as Kan extensions. The free-algebra monad for a Lawvere theory $J : \mathcal{T} \to \mathbf{Set}$ is $\mathrm{Lan}_J J$. This gives the second face of monads (the first being codensity).
The slogan. Every universal construction in category theory is a Kan extension: limits, adjoints, monads, free algebras, geometric realization. The coend calculus (Chapter 26) and weighted (co)limits (Chapter 27) provide the computational tools.

Formal Restatement

Formal Restatement

Pointwise Kan extensions.

$$(\mathrm{Lan}_K F)(d) = \int^c \mathcal{D}(Kc, d) \otimes_{\mathcal{E}} F(c),$$
$$(\mathrm{Ran}_K F)(d) = \int_c [\mathcal{D}(d, Kc), F(c)]_{\mathcal{E}}.$$

Existence. Pointwise Kan extensions exist when $\mathcal{C}$ is small and $\mathcal{E}$ is $\mathcal{V}$-cocomplete (Lan) or complete (Ran).

Universal property. $\mathrm{Lan}_K F$ is initial in the comma category of natural transformations $F \Rightarrow GK$; dually for Ran.

Codensity monad. $\mathbb{T}_G = \mathrm{Ran}_G G$ is a monad on $\mathcal{C}$ for every functor $G : \mathcal{B} \to \mathcal{C}$. For $F \dashv G$: $\mathbb{T}_G = GF$.

Algebraic-theory monads. For $J : \mathcal{T} \to \mathbf{Set}$ presenting a Lawvere theory, the free-algebra monad is $T = \mathrm{Lan}_J J$; algebras $= \mathcal{T}$-models.

Absolute Kan extension. $\mathrm{Lan}_K F$ is absolute iff every constituent weighted colimit is preserved by every functor.

The slogan. Every universal construction in category theory is a Kan extension. Limits and colimits, adjoints, monads, free algebras, nerve/realization, geometric realization, sheafification — all are specific instances of $\mathrm{Lan}_K F$ or $\mathrm{Ran}_K F$ for appropriate K, F.

Chapter 32 develops the formal theory of monads*: monads in a 2-category, Eilenberg–Moore objects, the algebra of monads, and the structures that codify the "two faces" (codensity and algebraic) of monad theory.*

Chapter 32

The Formal Theory of Monads

Chapter Overview

Chapter 8 introduced monads as triples (T, η, μ) of endofunctor + unit + multiplication. Chapter 21 developed their Eilenberg–Moore and Kleisli categories and proved Beck's monadicity theorem. Chapter 31 described the codensity monad and the algebraic-theory monad as Kan extensions. This chapter is the synthesis: the *formal theory of monads* (Street 1972) sees monads as 0-dimensional algebraic gadgets in *any* 2-category $\mathcal{K}$. The Eilenberg–Moore construction becomes a 2-categorical right adjoint, the Kleisli construction a left adjoint, and Beck's theorem the assertion that a 2-functor admits a "monad presentation".

By treating monads formally, we make several constructions of Chapter 21 into special cases of single 2-categorical formalisms, prepare the way for *distributive laws* (Chapter 35) as monad maps in a 2-category of 2-categories, and exhibit the deep parallel between monads in different 2-categories: monads in **Cat** (ordinary), monads in **Prof** (profunctors), monads in **Span** (categories enriched in sets/spans).

Where the Name Comes From

The **formal theory of monads** is due to *Ross Street* (1972), who developed the theory in his paper *The Formal Theory of Monads*. The adjective "formal" marks the contrast with the concrete 1-categorical monads of Vol I Ch. 8: Street's monads live in any 2-category $\mathcal{K}$, with **Cat** as one example among many. The formal viewpoint reveals that the Eilenberg-Moore and Kleisli constructions are themselves universal 2-categorical adjoints, and that many monadic phenomena are special cases of 2-categorical patterns. Cross-ref Vol IV Ch. 51 for the ∞-categorical generalization.

32.1 Monads in a 2-Category

32.1.1 Formalizing

Definition 32.1 (Monad in a 2-category)**.** Let $\mathcal{K}$ be a 2-category. A **monad** in $\mathcal{K}$ on an object $a \in \mathcal{K}$ consists of:

- A 1-cell $t : a \to a$.
- A 2-cell $\eta : 1_a \Rightarrow t$ (the **unit**).
- A 2-cell $\mu : tt \Rightarrow t$ (the **multiplication**).

satisfying associativity ($\mu \cdot \mu t = \mu \cdot t\mu$) and unit laws ($\mu \cdot \eta t = \mu \cdot t\eta = 1_t$), where all compositions are in $\mathcal{K}$.

Example 32.2 (Ordinary monads)**.** A monad in **Cat** is precisely an ordinary monad on a category (Chapter 8). The object is the category; the 1-cell is the endofunctor; the 2-cells are the natural transformations η, μ.

Example 32.3 (Monads in **Span**)**.** A monad in **Span**(**Set**) on an object A is a relation $R \subseteq A \times A$ with: a unit (reflexivity — the identity span $A \hookrightarrow R$); a multiplication ($R \circ R \subseteq R$, i.e., transitivity). Hence monads in **Span** are *preorders*.

Example 32.4 (Monads in **Prof**)**.** A monad in **Prof** on a category $\mathcal{A}$ is a profunctor $P : \mathcal{A} \to \mathcal{A}$ with unit $\eta : \mathcal{A}(-,-) \Rightarrow P$ and multiplication $\mu : P \circ P \Rightarrow P$ satisfying the monad axioms. Such structures are equivalent to **category extensions** of $\mathcal{A}$: small categories $\mathcal{B}$ with an identity-on-objects functor $\mathcal{A} \to \mathcal{B}$.

32.2 Eilenberg–Moore and Kleisli, Formally

32.2.1 Formalizing

The construction of Eilenberg–Moore and Kleisli categories from a monad generalizes to any 2-category — when those 2-categorical objects exist.

Definition 32.5 (Eilenberg–Moore object). For a monad $t = (t, \eta, \mu)$ on a in $\mathcal{K}$, the **Eilenberg–Moore object** a^t — when it exists — is an object of $\mathcal{K}$ equipped with a 1-cell $u^t : a^t \to a$ and a 2-cell $\alpha^t : tu^t \Rightarrow u^t$ satisfying:

- $\alpha^t \cdot \eta u^t = 1_{u^t}$.
- $\alpha^t \cdot \mu u^t = \alpha^t \cdot t\alpha^t$.

The universal property: for any $(b, k : b \to a, \beta : tk \Rightarrow k)$ satisfying the same axioms, there is a unique 1-cell $\hat{k} : b \to a^t$ with $u^t\hat{k} = k$ and $\alpha^t\hat{k} = \beta$, 2-cells equivariant.

Theorem 32.6 (Eilenberg–Moore as 2-categorical right adjoint). *In any 2-category $\mathcal{K}$ with Eilenberg–Moore objects, the construction $t \mapsto a^t$ is the right adjoint of an inclusion*

$$\mathcal{K} \xhookrightarrow{(-)^{\text{free}}} \mathrm{Mnd}(\mathcal{K})$$

where $\mathrm{Mnd}(\mathcal{K})$ is the 2-category of monads in $\mathcal{K}$ and "free monads" on each object.

Definition 32.7 (Kleisli object). The **Kleisli object** a_t — when it exists — is dual to Eilenberg–Moore: an object with a 1-cell $f_t : a \to a_t$ and a 2-cell $\rho_t : f_t t \Rightarrow f_t$, universally so. In any 2-category, a_t is the *left* adjoint to $(-)^{\text{free}}$ — that is, the free Kleisli construction.

Remark 32.8 (The two extremes). For any monad t on a in $\mathcal{K}$, the Kleisli and Eilenberg–Moore objects are the *initial* and *terminal* "resolutions" of t (adjunctions $L \dashv R$ with $RL = t$ as monad). This recovers in 2-categorical terms the well-known fact (Chapter 21) that every monad's adjunctions form a poset with Kleisli at the bottom and Eilenberg–Moore at the top.

32.2.2 Reorganizing: When Do EM/Kleisli Objects Exist?

Theorem 32.9 (Existence of EM objects in **Cat**). **Cat** *has all Eilenberg–Moore and Kleisli objects. They coincide with the classical constructions of Chapter 21.*

Theorem 32.10 (Existence in **Prof**). **Prof** *has all Eilenberg–Moore objects. The Eilenberg–Moore object of a monad-profunctor is the corresponding "Eilenberg–Moore profunctor"; explicitly, it is the category of P-algebras for the monad-profunctor P.*

32.3 Monad Maps and the 2-Category of Monads

32.3.1 Formalizing

Definition 32.11 (Monad map). For monads (a, t, η^t, μ^t) and (b, s, η^s, μ^s) in $\mathcal{K}$, a **monad map** is a pair (f, ϕ):

- A 1-cell $f : a \to b$.
- A 2-cell $\phi : sf \Rightarrow ft$.

satisfying:

- $\phi \cdot \eta^s f = f\eta^t$ (compatibility with units).
- $\phi \cdot \mu^s f = f\mu^t \cdot \phi t \cdot s\phi$ (compatibility with multiplications).

Definition 32.12 ($\mathrm{Mnd}(\mathcal{K})$). The **2-category of monads in** $\mathcal{K}$, $\mathrm{Mnd}(\mathcal{K})$, has:

- 0-cells: monads in $\mathcal{K}$.
- 1-cells: monad maps.
- 2-cells: 2-cells $\beta : f \Rightarrow f'$ in $\mathcal{K}$ compatible with the monad map structure.

Theorem 32.13 (Free monad and Eilenberg–Moore as adjoints). *In a 2-category $\mathcal{K}$ with EM objects,*

$$(-)^{\text{free}} \;:\; \mathcal{K} \rightleftarrows \mathrm{Mnd}(\mathcal{K}) \;:\; \mathrm{EM}$$

is a 2-adjunction (in the appropriate sense).

$(-)^{\text{free}}$ sends $a \in \mathcal{K}$ to the identity monad $(1_a, 1, 1)$ on a.	*The trivial monad.*
EM sends (a, t) to a^t (the EM object).	*Functorial in monad maps by the universal property of EM.*
Adjunction unit at a: the trivial monad on a becomes $a^{1_a} = a$ (the EM object of the identity monad is the original object).	*Trivial monad has trivial EM object.*
Adjunction counit at (a, t): the natural arrow $a^t \to a$ forgetting algebra structure.	*u^t from the EM universal property.*
Triangle identities: by the universal properties of EM and $(-)^{\text{free}}$.	*Direct calculation.*

32.3.2 Reorganizing

Remark 32.14 (Beck's monadicity in formal terms)**.** Beck's theorem (Chapter 21) characterizes when an adjunction $L \dashv R$ is *monadic*, i.e., when the comparison functor to the EM category is an equivalence. In formal 2-categorical terms: when does an adjunction in $\mathcal{K}$ factor through an EM object? The answer is what Theorem 32.15 below provides.

Theorem 32.15 (Formal Beck's monadicity)**.** *In a 2-category $\mathcal{K}$ with EM objects, an adjunction $f \dashv g$ between a and b is monadic — i.e., $b \cong a^{gf}$ over a — iff g* creates EM algebras*: every algebra structure on g uniquely lifts to an EM-object structure on b.*

32.4 Distributive Laws — Preview

The 2-categorical theory of monads sets up the natural language for *distributive laws*: ways of composing two monads on the same object.

32.4.1 Formalizing

Definition 32.16 (Distributive law)**.** For monads $s = (s, \eta^s, \mu^s)$ and $t = (t, \eta^t, \mu^t)$ on an object a in $\mathcal{K}$, a **distributive law of s over t** is a 2-cell $\lambda : ts \Rightarrow st$ satisfying four coherence axioms involving $\eta^s, \eta^t, \mu^s, \mu^t$ and λ.

Theorem 32.17 (Distributive law $\Leftrightarrow$ composite monad)**.** *A distributive law $\lambda : ts \Rightarrow st$ is equivalent to the data of a monad structure on the composite st extending those of s and t.*

Remark 32.18. Chapter 35 develops this fully. The 2-categorical formulation of monads makes distributive laws into "2-cells with structure," immediately suggesting their interaction with the rest of formal monad theory.

32.5 Bicategorical Monads and Higher Structure

32.5.1 Formalizing

The formal theory extends to bicategories. In a bicategory $\mathcal{B}$, a monad on an object a is a 1-cell $t : a \to a$ with 2-cell unit and multiplication satisfying the monad axioms *up to coherent isomorphism* in $\mathcal{B}$.

Example 32.19 (Monoidal monads as bicategorical monads)**.** A monoidal monad on a monoidal category $(\mathcal{V}, \otimes)$ — i.e., a monad T that is also a lax monoidal functor with the unit/multiplication being monoidal natural transformations — is precisely a monad in the bicategory $B\mathcal{V}^{\text{lax}}$ of $\mathcal{V}$-enriched lax monoidal functors and lax monoidal natural transformations.

Remark 32.20 (Higher monads)**.** Higher categories give higher monads. A monad in a 3-category is a 1-cell together with 2-cell unit/multiplication and 3-cell coherence (associator and unitor) for the multiplication. The cycle continues; this is the beginning of *higher monad theory* and its connection to higher algebra.

32.6 Exercises

Define a monad in a 2-category.	An object a, a 1-cell $t : a \to a$, 2-cells $\eta : 1_a \Rightarrow t$ and $\mu : tt \Rightarrow t$, satisfying associativity and unit axioms.
What is a monad in **Cat**?	An ordinary monad on a category.
What is a monad in **Span**(**Set**)?	A preorder (reflexive transitive relation).
What is a monad in **Prof** on $\mathcal{A}$?	A profunctor $P : \mathcal{A} \to \mathcal{A}$ with unit and multiplication; equivalent to an identity-on-objects extension of $\mathcal{A}$.
Define an Eilenberg–Moore object in a 2-category.	Object a^t with universal 1-cell $u^t : a^t \to a$ and 2-cell $\alpha^t : tu^t \Rightarrow u^t$ satisfying the EM axioms.
Define a Kleisli object in a 2-category.	Dual to EM: object a_t with universal 1-cell $f_t : a \to a_t$ and 2-cell $\rho_t : f_t t \Rightarrow f_t$.
In what sense are EM and Kleisli the "extremes"?	Among all adjunctions $L \dashv R$ in $\mathcal{K}$ with $RL = t$ (as monad), Kleisli is initial and EM terminal.
Does **Cat** have all EM objects?	Yes, the classical Eilenberg–Moore construction.
Does **Prof** have EM objects?	Yes; the EM object of a monad-profunctor is the category of its algebras (a profunctor itself).

Define a monad map between monads in $\mathcal{K}$.	A pair (f, ϕ) where $f : a \to b$ is a 1-cell and $\phi : sf \Rightarrow ft$ is a 2-cell satisfying compatibility with units and multiplications.
What is $\mathrm{Mnd}(\mathcal{K})$?	The 2-category whose 0-cells are monads in $\mathcal{K}$, 1-cells are monad maps, 2-cells are compatible 2-cells.
State the EM-adjoint theorem.	$(-)^{\mathrm{free}} \dashv \mathrm{EM}$ is a 2-adjunction between $\mathcal{K}$ and $\mathrm{Mnd}(\mathcal{K})$ (in any 2-category with EM objects).
State formal Beck's monadicity.	An adjunction $f \dashv g : a \to b$ is monadic iff g creates EM algebras: every algebra structure on g uniquely lifts to an EM-object structure on b.
Define a distributive law of s over t.	A 2-cell $\lambda : ts \Rightarrow st$ satisfying four coherence axioms involving $\eta^s, \eta^t, \mu^s, \mu^t$.
State the distributive-law-composite-monad theorem.	A distributive law $\lambda : ts \Rightarrow st$ is equivalent to a monad structure on the composite st extending the monad structures of s and t.
Why is the formal theory of monads "the right one"?	Because it makes the constructions of Chapter 21 (EM, Kleisli, comparison) into 2-categorical universal constructions valid in any 2-category, recovering them as adjoints of natural functors.
What does "creates EM algebras" mean?	g creates the EM object: any algebra-structure 2-cell on g uniquely lifts to make b into an EM object of gf over a.
How does codensity (Chapter 31) fit into this picture?	The codensity monad $\mathrm{Ran}_G\, G$ is a monad in **Cat** canonically associated to any functor; it provides the formal counterpart of the algebraic theory presented by G.
What's the simplest nontrivial example of a monad in **Span**?	Equality on a set: a preorder, hence a monad. More elaborate examples come from arbitrary preorders.
What 2-categorical structure does a monoidal monad live in?	The bicategory $B\mathcal{V}^{\mathrm{lax}}$ of $\mathcal{V}$-enriched lax monoidal functors.
Why is the formal monad theory the foundation for distributive laws?	Because composition of monads — which is what distributive laws regulate — is naturally a 2-categorical operation; the formal framework makes the composition meaningful.
What is the relationship between $\mathrm{Mnd}(\mathcal{K})$ and the codensity construction?	The codensity monad provides an embedding $G^*\mathcal{K} \to \mathrm{Mnd}(\mathcal{K})$ sending each functor G to its codensity monad.
Why are bicategorical monads more general than 2-categorical ones?	Because the bicategorical version allows the associativity and unit axioms to hold only up to coherent isomorphism, accommodating more examples.
Sketch why a profunctor monad $P : \mathcal{A} \to \mathcal{A}$ corresponds to an extension $\mathcal{A} \to \mathcal{B}$.	The EM object of P in **Prof** is the category $\mathcal{B}$; the unit $\mathcal{A}(-,-) \to P$ becomes the identity-on-objects functor $\mathcal{A} \to \mathcal{B}$.
What's the relationship between formal monads and bicategorical strictification (Chapter 25)?	Both involve replacing weak structure with strict structure up to equivalence; for monads, strictification yields strict 2-monads from bicategorical monads.
Why is the 2-category $\mathrm{Mnd}(\mathcal{K})$ "2-categorical"?	Because it carries the structure of a 2-category: not just monads (objects) and monad maps (1-cells), but also coherent 2-cells between monad maps.
What is the comparison functor from Chapter 21 in this formal language?	For an adjunction $F \dashv G$ in $\mathcal{K}$, the comparison $K : b \to a^{GF}$ is the unique 1-cell from the codomain of F to the EM object of GF over a.

Why is Beck's monadicity stated for "creates EM algebras"?	Because the EM creation condition is the 2-categorical universal property; Beck's classical conditions (reflecting isos, F-split coequalizers) are equivalent reformulations in **Cat**.
What is the relationship between the formal monad theory and homotopical algebra?	In homotopical algebra, one works with ∞-monads in ∞-categories; the formal monad theory provides the strict-categorical foundation that the ∞-categorical version generalizes.
Why does "formal" here mean "2-categorical"?	Because the constructions are defined uniformly in any 2-category, with the same proofs/diagrams working everywhere; the "formal" theory is the one that's independent of the specific 2-category $\mathcal{K}$.

Chapter Summary

Chapter Summary

Monads in a 2-category. A 1-cell $t : a \to a$ with 2-cell unit η and multiplication μ satisfying associativity and unit axioms. Examples: monads in **Cat** (ordinary), in **Span** (preorders), in **Prof** (category extensions).
Eilenberg–Moore and Kleisli objects. 2-categorical universal constructions extending the classical Chapter 21 constructions. EM is the right adjoint of $(-)^{\text{free}} : \mathcal{K} \to \text{Mnd}(\mathcal{K})$; Kleisli is the left adjoint. They are the terminal/initial extremes among adjunctions inducing t.
$\text{Mnd}(\mathcal{K})$**.** The 2-category whose 0-cells are monads in $\mathcal{K}$, 1-cells are monad maps (f, ϕ) with coherence axioms, 2-cells are compatible 2-cells from $\mathcal{K}$.
Formal Beck. An adjunction $f \dashv g$ is monadic iff g creates EM algebras: algebra-structure 2-cells on g uniquely lift to EM-object structures.
Distributive laws. 2-cells $\lambda : ts \Rightarrow st$ satisfying four coherence axioms; equivalent to a monad structure on the composite st. Chapter 35 develops fully.
Bicategorical and higher monads. Bicategorical generalizations where the monad axioms hold up to coherent isomorphism. Foundation for higher monad theory.

Formal Restatement

Formal Restatement

Monad in $\mathcal{K}$. (t, η, μ) on an object a with $\eta : 1_a \Rightarrow t$, $\mu : tt \Rightarrow t$, satisfying $\mu \cdot t\mu = \mu \cdot \mu t$ and $\mu \cdot \eta t = \mu \cdot t\eta = 1_t$.

Eilenberg–Moore object. a^t with universal $u^t : a^t \to a$, $\alpha^t : tu^t \Rightarrow u^t$, satisfying the EM axioms. Universally factors all adjunctions with monad t.

Kleisli object. Dual: a_t with $f_t : a \to a_t$, $\rho_t : f_t t \Rightarrow f_t$. The initial resolution of t.

Monad map. $(f, \phi) : (a, t) \to (b, s)$ with $\phi : sf \Rightarrow ft$ compatible with η, μ.

2-adjunction. $(-)^{\text{free}} \dashv \text{EM}$ between $\mathcal{K}$ and $\text{Mnd}(\mathcal{K})$, when EM objects exist.

Formal Beck. $f \dashv g$ monadic $\iff$ g creates EM algebras.

Distributive law. $\lambda : ts \Rightarrow st$ with four coherence axioms; bijective with monad structures on st extending those of s, t.

Chapter 33 develops adjunctions formally in 2-categories and bicategories, providing the dual framework to this chapter's monad theory.

Chapter 33

Adjunctions: The Full Formal Account

Chapter Overview

Chapter 7 introduced adjunctions via hom-set bijections. Chapter 17 gave the deeper account: triangle identities, uniqueness via Yoneda, the comma-category characterization. Chapter 32 placed monads in any 2-category. This chapter does the same for adjunctions and harvests the consequences.

An **adjunction in a 2-category** $\mathcal{K}$ is a pair of 1-cells $f \dashv g$ with unit and counit 2-cells satisfying the triangle identities, all formulated in $\mathcal{K}$. Such adjunctions automatically give monads (by composition) and play the role of universal-property witnesses. The chapter develops adjunctions in $\mathcal{K}$, in bicategories (where the triangle identities hold only up to coherent iso), and in the enriched setting $\mathcal{V}$-**Cat**. The unifying observation: adjunctions form the *free Eilenberg–Moore resolution* of monads, while monads form the *free decomposition* of adjunctions — they are formally dual constructions in the 2-category of 2-categories.

Where the Name Comes From

Adjunction in a 2-category extends the **Cat**-specific adjunctions of Vol I Ch. 7 to any 2-category. The formalization is due to *Ross Street*, *Max Kelly*, and others working in the formal-theory-of-monads tradition (early 1970s). The terminology ("adjunction in $\mathcal{K}$," "2-categorical adjunction") is standard; some authors use "adjoint pair" for the same thing. Cross-ref Vol III Ch. 32 (formal theory of monads) and Vol IV Ch. 50 (∞-categorical adjunctions).

33.1 Adjunctions in a 2-Category

33.1.1 Formalizing

Definition 33.1 (Adjunction in a 2-category)**.** An **adjunction** $f \dashv g$ in a 2-category $\mathcal{K}$ between objects a, b consists of:

- 1-cells $f : a \to b$ and $g : b \to a$;
- 2-cells $\eta : 1_a \Rightarrow gf$ (**unit**) and $\varepsilon : fg \Rightarrow 1_b$ (**counit**);

satisfying the **triangle identities**: $\varepsilon f \cdot f\eta = 1_f$ and $g\varepsilon \cdot \eta g = 1_g$.

Theorem 33.2 (Each adjunction gives a monad)**.** *For an adjunction $f \dashv g$ in $\mathcal{K}$, the composite $gf : a \to a$ with unit η and multiplication $g\varepsilon f : gfgf \Rightarrow gf$ is a monad on a in $\mathcal{K}$.*

Proof. Direct verification: associativity from the interchange law plus naturality of ε; unit laws from the triangle identities applied at the appropriate variables. □

33.1.2 Reorganizing: Examples

Example 33.3 (Adjunctions in **Cat**)**.** Adjunctions in **Cat** are precisely ordinary adjunctions $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ with $F \dashv G$.

Example 33.4 (Adjunctions in **Span**)**.** An adjunction in **Span**(**Set**) between A, B consists of a span $f : A \leftarrow X \to B$ left-adjoint to a span $g : B \leftarrow Y \to A$. Such adjunctions correspond to *functions* $f : A \to B$ — the left adjoint span is the graph of f; the right adjoint is its converse.

Example 33.5 (Adjunctions in **Prof**)**.** An adjunction $f \dashv g$ in **Prof** between $\mathcal{A}, \mathcal{B}$ consists of profunctors $f : \mathcal{A} \to \mathcal{B}$ and $g : \mathcal{B} \to \mathcal{A}$ with $fg \cong \mathcal{B}(-,-)$ (counit is an iso) and unit a natural transformation $\mathcal{A}(-,-) \to gf$. By the Yoneda lemma in **Prof** (Chapter 28), adjunctions in **Prof** correspond to ordinary functors $\mathcal{A} \to \mathcal{B}$: f is the representable profunctor of the functor, g is the corepresentable.

33.2 Adjunctions and Monads, Formally

33.2.1 Formalizing

Theorem 33.6 (Free adjunctions from monads)**.** *In a 2-category $\mathcal{K}$ with Eilenberg–Moore objects, every monad $t = (t, \eta, \mu)$ on a gives rise to an adjunction*

$$f_t \;:\; a \rightleftarrows a^t \;:\; u^t$$

with $u^t f_t = t$ as monads. This is the free *adjunction realizing t.*

Theorem 33.7 (Formal monadicity)**.** *An adjunction $f \dashv g$ in $\mathcal{K}$ is* monadic *iff the canonical comparison 1-cell $b \to a^{gf}$ is an equivalence in $\mathcal{K}$.*

Remark 33.8 (The duality)**.** Chapter 32 said EM objects are the *right adjoint* of $(-)^{\mathrm{free}}$. Here Theorem 33.6 says EM objects yield adjunctions with prescribed monad. These two facts together exhibit the precise duality: monads $\leftrightarrow$ adjunctions through the EM construction. Adjunctions are the "free things from which monads arise"; monads are the "shadows of adjunctions".

33.2.2 Reorganizing: The Adjoint-Triple Yoga

Theorem 33.9 (Adjoint triple from adjunction)**.** *For an adjunction $f \dashv g$ in $\mathcal{K}$, suppose $\mathcal{K}$ has Kan extensions along f. Then there is an adjoint triple*

$$\mathrm{Lan}_f\, g \;\dashv\; (- \circ f) \;\dashv\; \mathrm{Ran}_f\, g$$

between $[a, ?]$ and $[b, ?]$ (functor 2-categories). In particular, when $\mathcal{K} = \mathbf{Cat}$, this recovers the classical result that $\mathrm{Lan}_F \dashv F^* \dashv \mathrm{Ran}_F$ along any functor F.

33.3 Adjunctions in Bicategories

33.3.1 Formalizing

In a bicategory $\mathcal{B}$, the strict triangle identities of Definition 33.1 must be replaced by coherent isomorphisms.

Definition 33.10 (Adjunction in a bicategory)**.** An **adjunction in a bicategory** $\mathcal{B}$ consists of:

- 1-cells $f : a \to b$, $g : b \to a$;
- 2-cells $\eta : 1_a \Rightarrow gf$ and $\varepsilon : fg \Rightarrow 1_b$;

satisfying the *coherent* triangle identities, i.e., the standard triangle equations modified by the coherence isomorphisms of $\mathcal{B}$.

Theorem 33.11 (Adjunctions strictify with the bicategory)**.** *The strictification $\mathcal{B}^{\mathrm{st}}$ of a bicategory $\mathcal{B}$ (Chapter 25, Theorem 25.25) transports adjunctions: every adjunction in $\mathcal{B}$ corresponds to a strict adjunction in $\mathcal{B}^{\mathrm{st}}$, and conversely.*

This is the analogue for adjunctions of the bicategorical coherence theorem.

33.3.2 Reorganizing

Example 33.12 (Adjoint pairs of bimodules)**.** In **Bimod**, an adjunction between commutative rings R, S consists of:

- An (S, R)-bimodule M;
- An (R, S)-bimodule N;

with $M \otimes_S N \cong R$ as (R, R)-bimodules and $N \otimes_R M \cong S$ as (S, S)-bimodules. Such bimodules are precisely *Morita equivalences* between R and S. Adjunctions in **Bimod** thus recover the classical Morita theory of rings.

33.4 Enriched Adjunctions

33.4.1 Formalizing

Definition 33.13 ($\mathcal{V}$-adjunction)**.** A **$\mathcal{V}$-adjunction** $F \dashv_{\mathcal{V}} G$ between $\mathcal{V}$-categories $\mathcal{C}, \mathcal{D}$ consists of $\mathcal{V}$-functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ with a $\mathcal{V}$-natural isomorphism $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ in $\mathcal{V}$.

Remark 33.14. This is exactly Chapter 20's $\mathcal{V}$-adjunction. Here we revisit it formally: $\mathcal{V}$-adjunctions are adjunctions in the 2-category $\mathcal{V}$-**Cat** (whose 1-cells are $\mathcal{V}$-functors and 2-cells are $\mathcal{V}$-natural transformations).

Theorem 33.15 ($\mathcal{V}$-adjunctions and $\mathcal{V}$-monads)**.** *Every $\mathcal{V}$-adjunction $F \dashv_{\mathcal{V}} G$ gives a $\mathcal{V}$-monad GF on $\mathcal{C}$. Conversely, every $\mathcal{V}$-monad arises this way (via the EM and Kleisli constructions in $\mathcal{V}$-**Cat**).*

33.4.2 Reorganizing: When Ordinary $\Leftrightarrow$ Enriched

Theorem 33.16 (Lifting ordinary to enriched)**.** *For tensored and cotensored $\mathcal{V}$-categories $\mathcal{C}, \mathcal{D}$, an ordinary adjunction $F \dashv G$ between $\mathcal{V}$-functors (that need not be $\mathcal{V}$-natural) lifts to a $\mathcal{V}$-adjunction iff G preserves cotensors.*

Sketch. The bijection $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ between **Set**s lifts to an isomorphism in $\mathcal{V}$ iff the bijection respects the $\mathcal{V}$-action, which is exactly the cotensor-preservation condition for G. □

33.5 Adjunctions and the Yoneda Embedding

33.5.1 Formalizing

The formal Yoneda lemma (Chapter 28) characterizes adjoints via representability. We restate it here in 2-categorical language.

Theorem 33.17 (Adjunctions as Yoneda-representable)**.** *For a $\mathcal{V}$-functor $G : \mathcal{D} \to \mathcal{C}$, the following are equivalent:*

1. *G has a left adjoint L.*
2. *For each $c \in \mathcal{C}$, the functor $\mathcal{C}(c, G-) : \mathcal{D} \to \mathcal{V}$ is representable.*
3. *For each c, the precomposition $G \circ Y : \mathcal{D} \to [\mathcal{C}^{\mathrm{op}}, \mathcal{V}]_{\mathcal{V}}$ has $\mathcal{C}(c, -)$ as a value somewhere.*

The witnessing left adjoint L is the functor whose component at c is the representing object of $\mathcal{C}(c, G-)$.

33.5.2 Reorganizing

Remark 33.18 (Kan extension perspective)**.** The right adjoint G is the right Kan extension of itself along the Yoneda embedding (Chapter 28, Corollary 28.12). Adjunctions become Kan extensions in this language; the unifying slogan of Chapter 31 holds at the adjunction level.

33.6 Exercises

Define an adjunction in a 2-category.	1-cells $f \dashv g$, 2-cells $\eta : 1 \Rightarrow gf$ and $\varepsilon : fg \Rightarrow 1$, satisfying the triangle identities $\varepsilon f \cdot f\eta = 1_f$ and $g\varepsilon \cdot \eta g = 1_g$.
State the adjunction-monad theorem.	Every adjunction $f \dashv g$ gives a monad gf with unit η and multiplication $g\varepsilon f$.
What are adjunctions in **Cat**?	Ordinary adjunctions $F \dashv G$ between categories.
What are adjunctions in **Span**?	Functions $f : A \to B$: the graph of f is left adjoint to its converse.
What are adjunctions in **Prof**?	Ordinary functors $\mathcal{A} \to \mathcal{B}$, presented via the representable and corepresentable profunctors.
State free adjunction from monads.	Each monad t on a in a 2-category with EM objects yields an adjunction $f_t : a \rightleftarrows a^t : u^t$ with $u^t f_t = t$.
State formal monadicity.	An adjunction $f \dashv g$ is monadic iff the canonical $b \to a^{gf}$ is an equivalence in $\mathcal{K}$.
Define an adjunction in a bicategory.	Same as 2-categorical, but the triangle identities are required to hold up to coherent isomorphisms (the structure 2-cells of the bicategory).
State the strictification theorem for adjunctions.	Every adjunction in a bicategory corresponds to a strict adjunction in the strictified 2-category; the correspondence is one-to-one.
What are adjunctions in **Bimod**?	Morita equivalences between commutative rings: (S, R)-bimodule M and (R, S)-bimodule N with $M \otimes_S N \cong R$ and $N \otimes_R M \cong S$.
Define a $\mathcal{V}$-adjunction.	$\mathcal{V}$-functors $F \dashv_{\mathcal{V}} G$ with $\mathcal{V}$-natural iso $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$ in $\mathcal{V}$.
When does an ordinary adjunction between $\mathcal{V}$-functors lift to a $\mathcal{V}$-adjunction?	When G preserves cotensors (and $\mathcal{C}, \mathcal{D}$ are tensored/cotensored).
State Theorem 33.17.	G has left adjoint iff $\mathcal{C}(c, G-)$ is representable for every c, with Lc being the representing object.

Why is every right adjoint a right Kan extension along Yoneda?	Because $\mathcal{C}(c, G-) \cong \mathcal{D}(Lc, -) = Y(Lc)$, so $G \cong \mathrm{Ran}_Y(G \circ Y^{-1})$ informally; precisely, G is determined by its action through the Yoneda embedding.
What's the adjoint triple from an adjunction?	For $f \dashv g$: $\mathrm{Lan}_f\, g \dashv (- \circ f) \dashv \mathrm{Ran}_f\, g$ between functor 2-categories, the dual of Chapter 18's adjoint triple of restrictions.
Why is the duality monads $\leftrightarrow$ adjunctions encoded in the EM construction?	Because EM is the right adjoint of $(-)^{\mathrm{free}}$ from $\mathcal{K}$ to $\mathrm{Mnd}(\mathcal{K})$. Going monad $\to$ adjunction is the EM functor; going adjunction $\to$ monad is its inverse on the relevant subcategory.
How does Theorem 33.7 subsume Beck's classical theorem?	In **Cat**, a^{gf} is the classical EM category; the comparison being an equivalence is exactly Beck's conclusion.
State why all adjunctions in **Prof** come from ordinary functors.	Because the Yoneda lemma in **Prof** (Theorem 28.15) says hom-profunctors are the identity 1-cells; a left adjoint profunctor must therefore be representable, hence corresponds to a functor.
What is a Morita equivalence?	An adjunction in **Bimod** (or in the corresponding category of derived/triangulated categories of modules). It is an isomorphism of categories at the bimodule-derived level.
Why does the triangle identity hold up to coherent iso in bicategorical adjunctions?	Because in a bicategory the composition of 1-cells (and the action on 2-cells) is associative and unital only up to the bicategory's structure 2-cells; the strict triangle identity is the special case where these are identities.
Sketch how cotensor preservation gives a $\mathcal{V}$-adjunction.	$\mathcal{V}(V, \mathcal{D}(FA, B)) \cong \mathcal{V}(V, \mathcal{C}(A, GB))$ by ordinary adjunction; if G preserves cotensors, this isomorphism is $\mathcal{V}$-natural, lifting to an iso in $\mathcal{V}$.
What 2-categorical adjunctions do not arise from ordinary adjunctions?	Adjunctions in **Prof** where the profunctor is not representable — but by Theorem 28.15 this case is empty. Adjunctions in 2-categories with no **Cat**-structure (e.g., **Span**) can be "functional" without coming from **Cat**.
Why is the free-monad/EM duality stated in $\mathrm{Mnd}(\mathcal{K})$?	Because the duality is internal to that 2-category: $\mathcal{K}$ sits inside $\mathrm{Mnd}(\mathcal{K})$ as the trivial monads, and EM provides the inverse direction.
What's the relationship between adjoint Kan extensions and adjoint pairs?	Each adjoint pair $\mathrm{Lan}_f \dashv f^*$ gives an adjunction in the 2-category $[\mathcal{C}, ?]$; the Kan extension language is the 2-categorical version of adjunctions between functor categories.
Why is Beck's monadicity "creating EM algebras" equivalent to "comparison is an equivalence"?	Because creating EM algebras gives the inverse functor of the comparison; together they make $b \to a^{gf}$ an equivalence (by direct construction of the inverse).
What's the formal-monadicity version of "F-split coequalizers exist"?	Equivalent to "g preserves the absolute coequalizers of EM resolutions" — a 2-categorical statement that reduces to Beck's condition in **Cat**.
In what sense are adjunctions "primitive" relative to monads?	Because every monad arises from an adjunction (its EM resolution), but not every adjunction is reconstructable from its monad (the comparison may not be an equivalence). Adjunctions carry strictly more information.
What's the dual of Theorem 33.6?	For each comonad c on a in a 2-category with co-Eilenberg–Moore objects, there's an adjunction $f^c : a^c \rightleftarrows a : u_c$ with $f^c u_c = c$ as comonad.

How does the formal theory of adjunctions inform ∞-categorical theory?	In an ∞-category $\mathcal{K}$, an adjunction is data $f \dashv g$ with unit/counit and coherence going all the way up. The formal monadicity theorem extends, but the proofs become more elaborate.
Summarize the unifying perspective.	Adjunctions and monads are formally dual 2-categorical constructions. EM provides the equivalence between adjunctions inducing t and the universal a^t. The Kan extension formula makes the duality computational.

Chapter Summary

Chapter Summary

Adjunctions in a 2-category. $f \dashv g$ with η, ε satisfying the triangle identities (strict in a 2-category; up to coherent iso in a bicategory). Adjunctions in **Cat** are ordinary; in **Span** are functions; in **Prof** are functors; in **Bimod** are Morita equivalences.
Adjunction $\to$ monad. Each adjunction $f \dashv g$ gives a monad gf on the source. Conversely, each monad in $\mathcal{K}$ (with EM objects) yields an adjunction (free realization). The two constructions are formally dual via the EM functor.
Enriched adjunctions. $\mathcal{V}$-adjunctions are adjunctions in $\mathcal{V}$-**Cat**. They lift from ordinary adjunctions of $\mathcal{V}$-functors when the right adjoint preserves cotensors.
Yoneda-representability. G has a left adjoint iff $\mathcal{C}(c, G-)$ is representable for every c; the representing object is the left adjoint's value at c. Adjoints are right Kan extensions along Yoneda (Chapter 28).
Strictification. Adjunctions in a bicategory strictify with the bicategory: every adjunction in $\mathcal{B}$ corresponds to a strict adjunction in $\mathcal{B}^{\mathrm{st}}$.
Formal Beck monadicity. An adjunction is monadic iff the comparison $b \to a^{gf}$ is an equivalence. In **Cat** this is classical Beck; in general 2-categories it gives the right formulation of "algebras for a monad".

Formal Restatement

Formal Restatement

Adjunction in $\mathcal{K}$. $f \dashv g$ with η, ε, triangle identities $\varepsilon f \cdot f\eta = 1_f$ and $g\varepsilon \cdot \eta g = 1_g$.

Adjunction $\to$ monad. $(gf, \eta, g\varepsilon f)$ is a monad on a.

Formal monadicity. $f \dashv g$ monadic $\iff$ $b \to a^{gf}$ is an equivalence in $\mathcal{K}$.

$\mathcal{V}$-adjunction. $F \dashv_{\mathcal{V}} G$ with $\mathcal{V}$-natural iso $\mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$. Ordinary $\to$ $\mathcal{V}$-: condition is cotensor preservation.

Adjunction $\to$ Kan extension. For $f \dashv g$ and Kan extensions: $\mathrm{Lan}_f\, g \dashv (- \circ f) \dashv \mathrm{Ran}_f\, g$.

Yoneda-representable adjoints. G has $L \dashv G \iff \mathcal{C}(c, G-)$ representable $\forall c$; Lc is the representing object.

Strictification. Adjunctions in a bicategory $\leftrightarrow$ strict adjunctions in $\mathcal{B}^{\mathrm{st}}$.

Examples by 2-category. **Cat**: ordinary. **Span**: functions. **Prof**: functors. **Bimod**: Morita equivalences.

Chapter 34 unifies the adjoint functor theorems of Chapter 10/23 in light of the formal adjunction theory and the enriched weighted-limits perspective. Special focus: AFTs in enriched and locally presentable settings.

Chapter 34

Adjoint Functor Theorems: Unified

Chapter Overview

Chapter 10 introduced GAFT and SAFT in the **Set**-enriched setting. Chapter 23 gave their complete treatments with prooflog-style proofs. This chapter unifies them under a single framework based on the **locally presentable / accessible category** machinery and the enriched language of weighted limits. We:

- Recast GAFT and SAFT via Yoneda and representability;
- Prove the AFTs in the enriched setting;
- Connect them to the codensity-monad picture of Chapter 31;
- Treat locally presentable and accessible categories as the natural home for the AFTs;
- Apply: free monoids, sheafification, Stone–Čech, accessible equalizers.

The chapter concludes with the *small object argument* (Quillen's construction), a separate but parallel "existence of adjoints" principle used in homotopical algebra.

34.1 AFT as Representability

34.1.1 Formalizing

Chapter 28 established: G has a left adjoint iff $\mathcal{C}(c, G-)$ is representable for every c. The adjoint functor theorems are precisely *existence theorems for representing objects.*

Theorem 34.1 (Yoneda-representability form of GAFT)**.** *A functor $G : \mathcal{D} \to \mathcal{C}$ with $\mathcal{D}$ locally small and complete has a left adjoint iff:*

1. *G preserves all small limits.*
2. *For each $c \in \mathcal{C}$, the comma category $(c \downarrow G)$ has a small weakly initial set.*

Remark 34.2. Compared to Chapter 10/23 phrasing: "preserves small limits" and "solution set condition" jointly enable Freyd's construction of an initial object in $(c \downarrow G)$, which by representability is the value Lc of the left adjoint.

34.1.2 Reorganizing: SAFT via Cogeneration

Theorem 34.3 (SAFT, Yoneda form)**.** *A functor $G : \mathcal{D} \to \mathcal{C}$ with $\mathcal{D}$ locally small, complete, well-powered, and admitting a cogenerating set has a left adjoint iff G preserves small limits.*

Remark 34.4 (Why SAFT is "easier" than GAFT)**.** SAFT replaces the solution-set condition by structural conditions on $\mathcal{D}$ (well-powered, cogenerating set). These conditions automatically yield a solution set for every c, so SAFT is GAFT applied to "especially nice" categories.

34.2 Enriched AFT

34.2.1 Formalizing

Theorem 34.5 (Enriched GAFT)**.** *Let $\mathcal{V}$ be a complete symmetric monoidal closed category. A $\mathcal{V}$-functor $G : \mathcal{D} \to \mathcal{C}$ between $\mathcal{V}$-categories, with $\mathcal{D}$ $\mathcal{V}$-complete, has a $\mathcal{V}$-left adjoint iff:*

1. *G preserves all (small) weighted limits.*
2. *For each $c \in \mathcal{C}$, the $\mathcal{V}$-comma category $(c \downarrow G)$ has a small weakly initial set (in the enriched sense).*

Remark 34.6. The proof transposes the classical Freyd argument to the $\mathcal{V}$-enriched setting using the weighted-limit machinery of Chapter 27. The "weighted-completeness $\Rightarrow$ conical-completeness" reduction makes the $\mathcal{V}$-enriched proof manage with the same constructive ingredients as the **Set**-enriched one.

34.2.2 Reorganizing

Theorem 34.7 (Enriched SAFT)**.** *A $\mathcal{V}$-functor $G : \mathcal{D} \to \mathcal{C}$ with $\mathcal{D}$ $\mathcal{V}$-complete, $\mathcal{V}$-well-powered, and admitting a small cogenerating set, has a $\mathcal{V}$-left adjoint iff G preserves all $\mathcal{V}$-weighted limits.*

Example 34.8 (**Ab**-enriched AFT)**.** For $\mathcal{V} = \mathbf{Ab}$: a functor between additive (Ab-enriched) categories has a left adjoint iff it preserves all finite limits and satisfies the additive solution-set condition. Application: the existence of **flat replacements** in the derived category of a Grothendieck abelian category.

Example 34.9 (**sSet**-enriched AFT)**.** In simplicial set-enriched categories, the AFT recovers (with care) **Quillen's small object argument**-style results: existence of fibrant replacements, cofibrant replacements, derived functors.

> **Where the Name Comes From**
>
> **Locally presentable category** (also *locally finitely presentable*, when $\kappa = \omega$) was introduced by *Pierre Gabriel* and *Friedrich Ulmer* in 1971 (*Lokal präsentierbare Kategorien*). The term "locally presentable" means "presentable at each object via a small presentation" — every object is a κ-filtered colimit of κ-compact objects. The terminology was extended to ∞-categories by Lurie (HTT 2009; cross-ref Vol IV Ch. 52).

34.3 Locally Presentable Categories

34.3.1 Formalizing

Definition 34.10 (Locally presentable)**.** For a regular cardinal κ, a category $\mathcal{C}$ is **locally κ-presentable** if:

- $\mathcal{C}$ is cocomplete;
- $\mathcal{C}$ has a small set $\mathcal{G}$ of **κ-presentable objects** (those whose hom-functor preserves κ-filtered colimits);
- Every object of $\mathcal{C}$ is a κ-filtered colimit of objects from $\mathcal{G}$.

Theorem 34.11 (AFT in locally presentable categories)**.** *For locally presentable $\mathcal{C}, \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$:*

- *G has a left adjoint iff G preserves all small limits.*
- *G has a right adjoint iff G preserves all small colimits.*

Remark 34.12. This is the cleanest form of the AFT: in locally presentable categories, "preserves limits" suffices for adjoint existence; no solution-set condition is needed. Locally presentable categories are precisely the "nice" categories for which SAFT-like results hold without restrictive hypotheses.

34.3.2 Reorganizing

Example 34.13 (Locally presentable examples)**.**

- **Set** is locally finitely presentable.
- $\mathbf{Ab}, \mathbf{Mod}_R, \mathbf{Top}, \mathbf{sSet}$ are locally presentable.
- Categories of algebras for finitary monads on **Set** are locally finitely presentable.
- Categories of sheaves on a small site are locally presentable.

Theorem 34.14 (Accessible categories)**.** *A category is* ***accessible*** *if it has κ-filtered colimits and a small generating set of κ-presentable objects for some κ. Locally presentable categories are accessible and cocomplete; many "non-presentable but accessible" categories arise in homotopical and ∞-categorical settings.*

34.4 The Codensity-Monad Approach to AFT

34.4.1 Formalizing

Chapter 31 introduced the codensity monad $\mathbb{T}_G = \mathrm{Ran}_G\, G$. The AFT condition can be reformulated via codensity.

Theorem 34.15 (Adjoint via codensity)**.** *A functor $G : \mathcal{D} \to \mathcal{C}$ has a left adjoint iff the codensity monad $\mathbb{T}_G$ is the* identity monad *on $\mathcal{C}$, i.e.,* $\mathrm{Ran}_G\, G \cong \mathrm{Id}_{\mathcal{C}}$.

G has a left adjoint L iff $\mathcal{C}(c, G-)$ is representable for every c (Chapter 28).	*Yoneda-representability of adjoints.*
Representability means $\mathrm{Ran}_G\, G(c) \cong c$ naturally.	*The right Kan extension formula evaluated at c.*
Naturally in c means $\mathrm{Ran}_G\, G \cong \mathrm{Id}_{\mathcal{C}}$ as functors.	*Pointwise iso $\to$ functorial iso.*
Conversely, if $\mathrm{Ran}_G\, G \cong \mathrm{Id}$, then each $\mathrm{Ran}_G\, G(c) \cong c$, so $\mathcal{C}(c, G-)$ is represented by $G^{-1}(c)$... wait, this needs more care.	*Sketch only.*
Detailed proof: when $\mathrm{Ran}_G\, G \cong \mathrm{Id}$, the codensity monad is trivial, hence G is essentially the identity on its image, and a left adjoint exists if it is essentially a (fully faithful) right adjoint.	*The technical fine print.*

34.4.2 Reorganizing

Remark 34.16 (The codensity perspective)**.** The codensity monad measures "how far" G is from being a right adjoint: if $\mathbb{T}_G \cong \mathrm{Id}$, G is monadic-trivial and has a left adjoint. The general AFT becomes a question about computing or estimating the codensity monad.

34.5 The Small Object Argument

34.5.1 Formalizing

For categories carrying a notion of "cofibration" or "relative cell complex," Quillen's **small object argument** provides an alternative existence principle.

Theorem 34.17 (Small object argument)**.** *Let $\mathcal{C}$ be a cocomplete category and I a set of morphisms in $\mathcal{C}$. Assume each domain of a morphism in I is κ-small for some cardinal κ. Then every morphism in $\mathcal{C}$ factors as I-cell $\circ$ (I-injective).*

Remark 34.18 (How it relates to AFT)**.** The small object argument constructs "free I-cell complexes" as transfinite colimits, producing left adjoints to the right-orthogonal functors. It is an existence theorem for adjoints in the same spirit as SAFT, but uses transfinite induction rather than the comma-category condition.

34.5.2 Reorganizing: Applications in Model Categories

Example 34.19 (Fibrant replacement)**.** In a model category, fibrant replacement is the right adjoint of an inclusion-like functor; the small object argument provides its existence when the set of generating trivial cofibrations satisfies the smallness condition.

Example 34.20 (Free monad on an accessible endofunctor)**.** For an accessible endofunctor $F : \mathcal{C} \to \mathcal{C}$ on a locally presentable $\mathcal{C}$, the free monad on F exists via the small object argument: it is the left adjoint of the forgetful functor from F-algebras to $\mathcal{C}$.

34.6 Exercises

State GAFT in Yoneda-representability form.	$G : \mathcal{D} \to \mathcal{C}$ (with $\mathcal{D}$ locally small and complete) has a left adjoint iff G preserves small limits and each $(c \downarrow G)$ has a small weakly initial set.
State SAFT in Yoneda-representability form.	G with $\mathcal{D}$ complete, well-powered, cogenerating set has a left adjoint iff G preserves small limits.
Why is SAFT "cleaner" than GAFT?	Because structural conditions on $\mathcal{D}$ (well-powered + cogenerating set) replace the case-by-case solution-set condition, so the hypotheses are uniform.
State enriched GAFT.	For $\mathcal{V}$-functors with $\mathcal{D}$ $\mathcal{V}$-complete, left adjoint exists iff weighted limits are preserved and the enriched comma categories admit weakly initial sets.
State enriched SAFT.	For $\mathcal{V}$-functors with $\mathcal{D}$ $\mathcal{V}$-complete, $\mathcal{V}$-well-powered, with cogenerating set: left adjoint iff weighted limits preserved.
What is a locally κ-presentable category?	A cocomplete category $\mathcal{C}$ with a small set of κ-presentable objects whose κ-filtered colimits give every object.
What is a κ-presentable object?	An object c whose hom-functor $\mathcal{C}(c, -)$ preserves κ-filtered colimits.
State the AFT for locally presentable categories.	Between locally presentable $\mathcal{C}, \mathcal{D}$: G has a left adjoint iff G preserves small limits; G has a right adjoint iff G preserves small colimits.
Why does locally presentable give the cleanest AFT?	Because the structural conditions (small generating set of presentable objects) automatically supply solution sets, so the limit-preservation hypothesis alone suffices.
Give three examples of locally presentable categories.	**Set** (locally finitely presentable), **Ab** (locally finitely presentable), **Top** (locally presentable but only for κ exceeding $\aleph_0$).

What's the difference between accessible and locally presentable?	Locally presentable = accessible + cocomplete. Accessible categories may lack colimits.
State the adjoint-via-codensity theorem.	G has a left adjoint iff $\mathrm{Ran}_G\, G \cong \mathrm{Id}_{\mathcal{C}}$.
What does the codensity monad measure?	How far G is from being a right adjoint. Trivial codensity (identity monad) = G has a left adjoint.
State the small object argument.	Every morphism in a cocomplete category $\mathcal{C}$ factors as (transfinite I-cell) $\circ$ (I-injective) when the domains of I are κ-small.
How does the small object argument differ from SAFT?	SAFT uses comma-category solution sets; the small object argument uses transfinite induction over a set of generators I. Both construct left adjoints.
Where do small object arguments appear?	In model category theory (fibrant/cofibrant replacement), Quillen's path object construction, and the construction of free monads on accessible endofunctors.
Why is "preserves limits" the universal sufficient condition for adjoint existence (with appropriate niceness)?	Because by Yoneda, G has a left adjoint iff G is the right Kan extension along Yoneda; limit preservation makes G continuous, and the niceness ensures the Kan extension exists.
What is the relationship between AFT and the formal theory of monads (Chapter 32)?	An adjoint functor pair $f \dashv g$ gives a monad gf; the AFT is the existence theorem for the adjunction. The formal theory says when this adjunction is determined by an EM construction.
Why does the AFT framework also produce *right* adjoints in locally presentable settings?	Because the dual situation — G preserves colimits — works in locally presentable categories with the same generators-and-presentability machinery.
What is a fibrant replacement in model category theory?	The right adjoint of an inclusion of fibrant objects; constructed via the small object argument applied to a generating set of trivial cofibrations.
What is a free monad on an endofunctor?	The left adjoint of the forgetful functor from algebras to the underlying category; exists for accessible endofunctors via the small object argument.
What's the relationship between the AFT and the codensity-monad of Chapter 31?	The AFT says "G has a left adjoint when codensity is trivial." This unifies adjoint-existence with the formal-theory observation that monads are codensity monads.
What's the "enriched solution set condition"?	For each $c \in \mathcal{C}$, there should be a small set of $\mathcal{V}$-morphisms $c \to Gb_i$ such that any $c \to Gb$ factors through one of them, in the $\mathcal{V}$-enriched sense.
Why does enriched AFT need "$\mathcal{V}$-well-powered"?	Because the SAFT uses the well-powered hypothesis to bound the size of monomorphisms entering each object; the enriched version requires the same bound for $\mathcal{V}$-subobjects.
State a typical application of SAFT in a locally presentable category.	Sheafification: the inclusion of sheaves into presheaves preserves limits; SAFT applies (using cogenerating sets) to give the left-adjoint sheafification functor.
What's the locally presentable analogue of compactness?	κ-presentability: objects that look "small" relative to κ-filtered colimits, generalizing finite presentability.
Why is the codensity perspective on AFT "conceptual"?	Because it reduces existence of adjoints to a structural property of the right Kan extension, eliminating the need for hand-construction of the left adjoint.
State the AFT in terms of preserving weighted limits in the enriched case.	$\mathcal{V}$-functor G has $\mathcal{V}$-left adjoint iff G preserves all $\mathcal{V}$-weighted limits (subject to size/niceness conditions on the categories).

What's the relationship between AFT and Yoneda?	Both say: a functor is determined by its restriction to a small generating subcategory, and adjoint existence is a representability condition on this restriction.
Summarize this chapter's contribution beyond Chapter 23.	Three things: enriched and weighted-limit forms of AFT; locally presentable categories as the clean home for AFT; the codensity-monad reformulation; and the small object argument as a parallel existence principle.

Chapter Summary

Chapter Summary

Yoneda-representability form. GAFT: complete $\mathcal{D}$, G preserves limits, and comma categories have weakly initial sets $\Rightarrow L \dashv G$. SAFT: replace SSC by well-powered + cogenerating set on $\mathcal{D}$.
Enriched AFTs. Same statements transposed to $\mathcal{V}$-**Cat**. Limit preservation becomes *weighted*-limit preservation; the enriched solution set condition applies to the $\mathcal{V}$-comma categories.
Locally presentable categories. Cocomplete categories with a small set of κ-presentable generators. Between locally presentable categories, G has a left adjoint iff G preserves small limits (no SSC needed). Examples: **Set**, **Ab**, **Top**.
Codensity reformulation. G has a left adjoint iff $\mathrm{Ran}_G\, G \cong \mathrm{Id}_{\mathcal{C}}$. The codensity monad measures the obstruction.
Small object argument. Quillen's transfinite-induction construction of "I-cell complex factorizations." Used to construct fibrant replacements, free monads on accessible endofunctors, derived functors. A parallel existence theorem for adjoints in model-categorical settings.

Formal Restatement

Formal Restatement

GAFT (Yoneda form). For $G : \mathcal{D} \to \mathcal{C}$ with $\mathcal{D}$ locally small and complete: $L \dashv G$ iff G preserves small limits and each $(c \downarrow G)$ has a small weakly initial set.

SAFT. As above with: $\mathcal{D}$ well-powered, has a small cogenerating set; SSC replaced by these.

Enriched AFTs. Same with "small limits" $\to$ "weighted limits"; "well-powered" $\to$ "$\mathcal{V}$-well-powered."

Locally presentable AFT. Between locally presentable categories: $L \dashv G \iff G$ preserves limits; $G \dashv R \iff G$ preserves colimits.

Codensity reformulation. $L \dashv G \iff \mathrm{Ran}_G\, G \cong \mathrm{Id}_{\mathcal{C}}$.

Small object argument. In a cocomplete $\mathcal{C}$ with set I of morphisms whose domains are κ-small: every morphism factors as I-cell composite then I-injective.

Applications. Sheafification (SAFT in $[\mathcal{O}^{\mathrm{op}}, \mathbf{Set}] \to \mathrm{Sh}(\mathcal{O})$). Stone–Čech compactification (SAFT in **Top** $\to$ **CHaus**, or codensity of inclusion). Fibrant replacement (small object argument in model categories).

Chapter 35 develops distributive laws and composed monads*: the central new topic of Volume III, not covered in Volume II.*

Chapter 35

Distributive Laws and Composed Monads

Chapter Overview

This chapter develops a topic untouched by Volumes I and II: when can two monads s, t on the same object be *composed* to give a new monad st? Beck (1969) answered: precisely when there is a **distributive law** $\lambda : ts \Rightarrow st$ satisfying four coherence axioms. The composed monad st then has unit $\eta^{st} = \eta^s\eta^t$ and multiplication built from μ^s, μ^t, λ.

Distributive laws are the algebraic vocabulary for "two monad structures that play well together." Their applications: *monad transformers* in programming languages (Reader-of-State, State-of-List, etc.); the construction of *rings as compositions of abelian groups and monoids*; *double dualization* as a monoidal composite; *traced monads*; the *distributive law approach to operads*. The chapter closes with mixed distributive laws (monad over comonad) and the connection to bimonads and Hopf monads.

Where the Name Comes From

Distributive law was introduced by *Jon Beck* (1969, "Distributive Laws") — the same Beck of Beck's monadicity theorem (Vol II Ch. 21). The terminology comes from the analogy with classical distributivity in algebra ($a \cdot (b + c) = ab + ac$): just as multiplication "distributes over" addition in a ring, a monad s "distributes over" a monad t via the 2-cell $\lambda : ts \Rightarrow st$. The four axioms (D1–D4) ensure the distribution respects the unit and multiplication of both monads.

Distributive laws are the algebraic vocabulary of *monad composition*: st is a monad iff s distributes over t. The same pattern recurs in ∞-category theory (Vol IV Ch. 51) and in operad theory.

35.1 The Distributive Law Axioms

35.1.1 Formalizing

Definition 35.1 (Distributive law of s over t)**.** For monads s, t on a in a 2-category $\mathcal{K}$, a **distributive law of s over t** is a 2-cell $\lambda : ts \Rightarrow st$ such that the four diagrams below commute:

$$\begin{aligned}
\text{(D1) Unit-on-left:}\quad & \lambda \cdot t\eta^s = \eta^s t\\
\text{(D2) Unit-on-right:}\quad & \lambda \cdot \eta^t s = s\eta^t\\
\text{(D3) Mult-on-left:}\quad & \lambda \cdot t\mu^s = \mu^s t \cdot s\lambda \cdot \lambda s\\
\text{(D4) Mult-on-right:}\quad & \lambda \cdot \mu^t s = s\mu^t \cdot \lambda t \cdot t\lambda
\end{aligned}$$

How to Say This

$\lambda : ts \Rightarrow st$ *(read: "lambda from t-s to s-t")*: the distributive law moves t past s, swapping their order. (D1), (D2) are the **unit** laws (compatibility of λ with the monad units), and (D3), (D4) are the **multiplication** laws.

35.1.2 Reorganizing

Remark 35.2 (Symmetry)**.** Distributive laws are inherently directional: $\lambda : ts \Rightarrow st$ is not the same as $\lambda' : st \Rightarrow ts$. The composed monad on the result depends on which direction is chosen. For symmetric monoidal contexts, one can sometimes find a self-dual law.

Example 35.3 (Trivial distributive law)**.** For any monad s on a, the identity $ss \Rightarrow ss$ is the trivial distributive law of s over itself. The composed monad ss may or may not agree with s; it does iff μ^s acts "as expected."

35.2 Beck's Theorem on Distributive Laws

35.2.1 Formalizing

Theorem 35.4 (Beck, 1969)**.** *For monads (s, η^s, μ^s) and (t, η^t, μ^t) on a in a 2-category $\mathcal{K}$, the following data are equivalent:*

1. *A distributive law $\lambda : ts \Rightarrow st$.*
2. *A monad structure $(st, \eta^{st}, \mu^{st})$ on the composite st* extending *those of s and t (i.e., the inclusions $s \to st$ and $t \to st$ via η are monad maps).*
3. *A lift of t to a monad on the EM object a^s of s.*

(1) ⇒ (2): given λ, define $\eta^{st} = \eta^s \cdot \eta^t$ and $\mu^{st} = (\mu^s \cdot s\lambda) \cdot (t\mu^t)$ (reading right-to-left).	*Composing the basic data through λ.*
Verify μ^{st} is associative: chain the four distributive-law axioms (D1–D4) to reduce both $\mu^{st} \cdot st\mu^{st}$ and $\mu^{st} \cdot \mu^{st}st$ to the same expression.	*Mechanical but lengthy: this is the "coherence pyramid" for λ.*
Verify μ^{st} satisfies unit laws: use (D1) and (D2) to check $\mu^{st} \cdot \eta^{st}st = 1_{st} = \mu^{st} \cdot st\eta^{st}$.	*Unit-law verification.*
(2) ⇒ (1): given a monad structure on st, define λ as the composite $ts \xrightarrow{\eta^s ts\eta^t} stst \xrightarrow{\mu^{st}} st$.	*Inverse construction.*
(1) ⇔ (3): a distributive law lifts t to a $\mathcal{K}$-monad on a^s, because (D3) and (D4) provide the necessary naturality and compatibility with the algebra structure on a^s.	*Algebra lifting.*
All three forms thus carry equivalent data.	*Standard.*

35.2.2 Reorganizing: Why the Three Forms Are Useful

Remark 35.5. The three forms of Beck's theorem support different applications:

- Form (1) — distributive law — is the most explicit, easiest to check, used in computations.
- Form (2) — composite monad — emphasizes the algebraic conclusion: an algebra for the composite is an algebra for both monads, compatibly.
- Form (3) — lift of t to a^s — is the conceptual heart: λ encodes how t becomes a monad on s-algebras.

35.3 Classical Examples

35.3.1 Formalizing

Example 35.6 (Rings = (Abelian groups) ⊗ (Monoids))**.** Let Ab be the free abelian group monad on **Set**, and let Mon be the free monoid monad. A distributive law $\lambda : \mathrm{Ab}\,\mathrm{Mon} \Rightarrow \mathrm{Mon}\,\mathrm{Ab}$ sends a formal $\mathbb{Z}$-linear combination of words to a sum of products, expanding the linear combination through the monoid structure.

The composite monad Mon Ab on **Set** produces the **free ring** on a set: an algebra is a ring.

Example 35.7 (Modules = (Abelian groups) ⊗ (Action of a monoid))**.** For a fixed ring R, the free R-module monad on **Set** factors as a composite of: (i) the free abelian group monad on **Set**, (ii) the "action of R" monad on **Ab**. The distributive law expresses that scalar multiplication distributes over addition.

Example 35.8 (Monad transformers in programming)**.** In functional programming, **monad transformers** like `ReaderT s m`, `StateT s m`, `ListT m` construct composite monads from basic ones plus a distributive law. The λ axioms verify that the operations of one monad commute correctly with the operations of the other.

35.3.2 Reorganizing

Example 35.9 (Self-distributive monoidal monads)**.** On **Set**, the powerset monad P is a monad. Asking when P distributes over P — i.e., when sets-of-sets can be flattened consistently — amounts to choosing a coherent "element-wise" versus "set-wise" ordering. Such laws give rise to "$\mathbb{Z}_2$-graded monads."

35.4 Mixed Distributive Laws

35.4.1 Formalizing

A monad and a comonad can interact via a *mixed* distributive law.

Definition 35.10 (Mixed distributive law)**.** For a monad $t = (t, \eta^t, \mu^t)$ and a comonad $g = (g, \varepsilon^g, \delta^g)$ on the same object a in $\mathcal{K}$, a **mixed distributive law** is a 2-cell $\lambda : gt \Rightarrow tg$ satisfying four coherence diagrams involving $\eta^t, \mu^t, \varepsilon^g, \delta^g$

and λ.

Theorem 35.11 (Mixed Beck)**.** *Mixed distributive laws are in bijection with:*

- *Monad structures on the composite tg compatible with both t and g.*
- *Lifts of t to a monad on the (co-)EM object a^g.*

35.4.2 Reorganizing: Bimonads and Hopf Monads

Definition 35.12 (Bimonad)**.** A **bimonad** (sometimes "Hopf monad" in some authors' terminology) is a monad H on a monoidal category $\mathcal{V}$ together with a comonoidal structure (mediating its own multiplication and the tensor product of $\mathcal{V}$) such that the two structures are compatible in a coherent way. Equivalently, H is a comonad in the bicategory of monoidal categories and lax monoidal functors.

Example 35.13 (Group rings as bimonads)**.** For a group G, the group ring monad $-[G]$ on $\mathbf{Vect}_k$ has a natural bimonad structure: it is at once a free-module monad and a "representation" bimonad. Its algebras are **G-modules**, and its bimonad structure is what gives them the monoidal structure of G-modules under tensor product.

35.5 Distributive Laws in ∞-Categorical Settings

35.5.1 Formalizing

In an ∞-category, monads and distributive laws carry higher coherence data: not just 2-cells but also higher morphisms making the "distributive law axioms" coherent.

Remark 35.14 (Higher distributive laws)**.** For 2-monads (monads in a 3-category), a distributive law is a 2-cell with 3-cell coherence. The pattern continues; ∞-monad theory exhibits a *tower of distributive laws*, where each level's coherence becomes the next level's basic structure.

35.5.2 Reorganizing

Remark 35.15 (Why distributive laws matter for Vol III's synthesis)**.** Distributive laws are the formal mechanism by which compound monadic structures (rings, modules, groups acting on rings, etc.) decompose into "layers." They are the categorification of distributivity in arithmetic. Just as distributivity $(a + b)c = ac + bc$ in ring theory expresses how multiplication interacts with addition, distributive laws express how *any* two monadic structures interact. Vol III sees this as a foundational pattern: the entire "algebra of structures" refines through distributive laws.

35.6 Exercises

Define a distributive law of s over t.	A 2-cell $\lambda : ts \Rightarrow st$ satisfying four coherence axioms: two unit laws (D1, D2) and two multiplication laws (D3, D4).
State the four axioms.	(D1) $\lambda \cdot t\eta^s = \eta^s t$. (D2) $\lambda \cdot \eta^t s = s\eta^t$. (D3) $\lambda \cdot t\mu^s = \mu^s t \cdot s\lambda \cdot \lambda s$. (D4) $\lambda \cdot \mu^t s = s\mu^t \cdot \lambda t \cdot t\lambda$.
Is a distributive law symmetric in s and t?	No. $\lambda : ts \Rightarrow st$ moves t past s in one direction. The reverse direction $\lambda' : st \Rightarrow ts$ is a different (and not generally existing) structure.
State Beck's distributive-law theorem.	Distributive laws $\lambda : ts \Rightarrow st$ are in bijection with monad structures on st extending those of s, t, and with lifts of t to a monad on a^s.
What is the formula for the composite monad's multiplication?	$\mu^{st} = (\mu^s \cdot s\lambda) \cdot (t\mu^t)$: apply t's multiplication, swap via λ, then apply s's multiplication.
What is the formula for the composite monad's unit?	$\eta^{st} = \eta^s \cdot \eta^t$: the composition of both monad units.
Give the ring-as-composite-monad example.	The free ring monad Ring on **Set** factors as Mon $\circ$ Ab via a distributive law sending formal linear combinations of words to expanded sums of products.
Why does this give rings?	Because an algebra over Mon $\circ$ Ab is a set with both an abelian group structure and a monoid structure such that multiplication distributes over addition — i.e., a ring.

State the module-monad decomposition for a ring R.	Free R-module = (free abelian group) composed with (action of R), with the distributive law expressing that scalar multiplication distributes over addition.
What are monad transformers in functional programming?	Type constructors that combine multiple monad effects: e.g., `StateT` $s\,m$ combines State and m, requiring a distributive law of s and m.
Define a mixed distributive law.	For a monad t and comonad g: $\lambda : gt \Rightarrow tg$ satisfying four coherence axioms involving both η, μ of t and ε, δ of g.
State mixed Beck's theorem.	Mixed distributive laws are in bijection with monad structures on tg compatible with both, or equivalently with lifts of t to a monad on the co-EM object a^g.
Define a bimonad.	A monad on a monoidal category equipped with a comonoidal structure such that the two are coherent. Algebras for a bimonad have additional monoidal structure.
Give an example of a bimonad.	The group ring monad $-[G]$ on $\mathbf{Vect}_k$ for a group G.
What is a Hopf monad?	A bimonad whose comonoidal structure has an antipode-like operation; algebras for a Hopf monad have natural left/right module/comodule structures.
Why are distributive laws inherently 2-categorical?	Because they are 2-cells (natural transformations between functors). They live in the 2-categorical structure of $\mathcal{K}$ and depend on it.
What is the dual statement: "law of s over t" vs. "law of t over s"?	$\lambda : ts \Rightarrow st$ is a distributive law of s over t; the dual $\lambda' : st \Rightarrow ts$ would be a law of t over s. The two are different structures (generally).
What is the relationship between distributive laws and monad maps?	A distributive law λ provides a way to "transport" algebras of one monad through the other, i.e., the natural lifting of one monad to algebras for another.
Can two monads have no distributive law between them?	Yes: e.g., the powerset monad and the set-of-bags monad have no compatible distributive law. The composite cannot be made into a monad.
Why is the composite monad's existence equivalent to the distributive law?	Because (a) given λ, you construct μ^{st} from μ^s, μ^t, λ; and (b) given a compatible μ^{st}, you recover λ as a specific composite. The two structures carry the same information.
State the categorical interpretation of (D1).	λ extends η^s from the unit of s to a unit for the composite st: applying λ after $t\eta^s$ should give the same as $\eta^s t$ (transporting through).
Give an example of a distributive law in **Cat**.	The free abelian group functor distributes over the free monoid functor; the law assigns to each "element of (free monoid of (free abelian group on X))" a corresponding element of the swapped composite.
Why does a distributive law lift one monad to algebras for the other?	Because the four axioms encode exactly the compatibility needed for t to define operations on s-algebras: the unit and multiplication of t become natural with respect to the s-algebra structure.
What does the (D3) axiom say informally?	It says "the law λ respects the multiplication of s": transporting through λ commutes with the way s's multiplication acts.
Why are distributive laws sometimes called "compatibility 2-cells"?	Because they encode the precise way two monad structures must agree to be combined; they are the algebraic certificate of compatibility.

Where do distributive laws appear in topology?	In topological algebra: distributive laws relate operations such as group multiplication and topological structure, producing topological groups, rings, etc.
Why is the Hopf monad framework valuable?	Because it encodes Hopf-algebraic structure (group rings, quantum groups, Hopf algebras) in monadic terms, making representations into the natural "algebras of a bimonad."
What's the ∞-categorical version of a distributive law?	An ∞-cell with higher coherence data, providing distributive structure in higher categories; appears in derived algebraic geometry and homotopy theory.
Summarize Beck's theorem's three faces.	(1) Distributive law $\lambda : ts \Rightarrow st$. (2) Composite monad structure on st. (3) Lift of t to a monad on a^s. All three carry the same data.
What's the strategic role of distributive laws in Vol III?	They are the mechanism by which compound monadic structures decompose into "layers," generalizing arithmetic distributivity to a structural primitive of category theory.

Chapter Summary

Chapter Summary

Distributive law $\lambda : ts \Rightarrow st$. A 2-cell satisfying four coherence axioms (D1, D2 unit; D3, D4 multiplication). Encodes the compatibility allowing two monads on the same object to be composed.
Beck's theorem. Distributive laws are in bijection with: (a) monad structures on the composite st; (b) lifts of t to a monad on the Eilenberg–Moore object a^s.
Classical examples. Rings = (Abelian groups) $\otimes$ (Monoids). Modules over R = (Abelian groups) $\otimes$ (action of R). Monad transformers in functional programming.
Mixed distributive laws. Between a monad and a comonad on the same object: lift the monad to a structure on the co-Eilenberg–Moore object of the comonad.
Bimonads and Hopf monads. Monads on monoidal categories with compatible comonoidal structures. Algebras of a bimonad on $\mathbf{Vect}_k$ have monoidal structure; algebras of a Hopf monad have additional module/comodule duality.
∞-categorical aspects. Distributive laws extend to ∞-monads in ∞-categories with higher coherence data; basic for derived algebraic geometry and homotopy theory.

Formal Restatement

Formal Restatement

Distributive law axioms (D1–D4).

$$\lambda \cdot t\eta^s = \eta^s t$$
$$\lambda \cdot \eta^t s = s\eta^t$$
$$\lambda \cdot t\mu^s = \mu^s t \cdot s\lambda \cdot \lambda s$$
$$\lambda \cdot \mu^t s = s\mu^t \cdot \lambda t \cdot t\lambda$$

Composite monad. $\eta^{st} = \eta^s \cdot \eta^t$; $\mu^{st} = (\mu^s \cdot s\lambda) \cdot (t\mu^t)$.

Beck's bijection. $\{\lambda : ts \Rightarrow st$ satisfying D1–D4$\} \leftrightarrow \{$monad structures on $st\} \leftrightarrow \{$lifts of t to monad on $a^s\}$.

Examples. Ring = Monoid $\circ$ AbGroup; R-Module = (action of R) $\circ$ AbGroup; monad transformers in programming.

Mixed distributive law. $\lambda : gt \Rightarrow tg$ between a monad t and a comonad g, with four axioms involving $\eta^t, \mu^t, \varepsilon^g, \delta^g$.

Bimonad. Monad on monoidal $\mathcal{V}$ + compatible comonoidal structure. Hopf monad: bimonad + antipode-like operation.

Chapter 36 synthesizes Volume III as a whole and points forward to topics beyond Volume III's scope: ∞-categories, higher algebra, derived algebraic geometry, homotopy type theory.

Chapter 36

The Formal Landscape

Chapter Overview

We have arrived. Volume I built category theory from scratch through the six-stage cycle (Informal $\to \ldots \to$ Formally) over twelve chapters on the central concepts: objects, morphisms, functors, natural transformations, universal properties, limits, adjoints, monads, Yoneda, adjoint functor theorems, Kan extensions, monoidal and enriched categories.

Volume II revisited the same architecture at half familiarity. Each chapter rotated its viewpoint: *categories as algebra*, *functors as structure maps*, *representability as the unifying principle*, *limits as representable constructions*, and so on. The recapitulation was synthetic: it organized what Volume I built around the structural themes that emerge once the parts are understood individually.

Volume III completes the foundational arc. With the coend calculus (Chapter 26), weighted (co)limits (Chapter 27), the formal Yoneda lemma (Chapter 28), and the 2-categorical formal theories of monads and adjunctions (Chapters 32, 33), every universal construction of Volumes I–II has acquired its enriched, 2-categorical, formal counterpart. Day convolution (Chapter 30), the codensity monad (Chapter 31), the unified AFT (Chapter 34), and distributive laws (Chapter 35) added genuinely new material that exhibits the deep coherence of the resulting picture.

This final chapter synthesizes Volume III, restates the unifying theses of the entire book, and points forward to the topics beyond our scope: ∞-categories, higher algebra, derived algebraic geometry, and the applications of category theory to other branches of mathematics.

Reader's Annotation

This chapter is synthesis, not new material: every term used has been introduced in its home chapter with etymology and historical context. The retrospective "one picture" framing assembles those threads into the formal-spine landscape that Volume III delivers. Volume IV ("Toward Higher Category Theory") begins immediately after.

36.1 Volume III in One Picture

36.1.1 The Layers

Volume III's eleven preceding chapters arranged themselves in three layers:

Layer 1: 2-Categorical Foundations (Chapters 25, 32, 33). Bicategories with examples beyond **Cat**; the formal theory of monads (monads in any 2-category, with EM and Kleisli as adjoints of $(-)^{\text{free}}$); adjunctions in 2-categories, bicategories, and $\mathcal{V}$-**Cat**. The unifying view: every concept in Volumes I–II generalizes to "a 2-categorical version," and the strict **Cat**-case is one example.

Layer 2: Coend Calculus (Chapters 26, 27, 28, 30, 31). Ends and coends as the computational tool for "mixed-variance" functors; weighted (co)limits as the enriched generalization of limits; the formal Yoneda lemma in unified form; Day convolution as the canonical monoidal structure on presheaves; Kan extensions in full generality with codensity monads. The unifying view: every universal construction is a weighted (co)limit and hence a Kan extension and hence a coend.

Layer 3: Synthesis (Chapters 29, 34, 35, this chapter). Monoidal coherence as bicategorical strictification; AFTs unified across enrichments and locally presentable categories; distributive laws giving composed monads. The unifying view: the algebraic structures of category theory decompose, layer by layer, through these systematic constructions.

36.1.2 The Picture

Remark 36.1. Imagine a tower. The ground floor is Volume I (objects, morphisms, basic universal properties). The second floor is Volume II (the same concepts reorganized around the structural principles of representability and adjointness). The third floor — Volume III — has three rooms: a 2-categorical room (bicategories, formal monads, formal adjunctions), a coend room (ends, coends, weighted limits, Kan extensions, Day convolution), and a synthesis room (coherence, AFT, distributive laws).

From the synthesis room, several doors open: to ∞-category theory, to higher algebra, to derived algebraic geometry, to homotopical algebra, to applications in algebra and topology. These rooms — the entire next floor — are Volume IV, never written here.

36.2 The Unifying Theses, Restated

The book's overarching theses, now fully justified:

36.2.1 Thesis 1: Representability

> **Every universal construction in category theory is the representing object for a canonically defined functor.**

Stated informally in Chapter 5; sharpened in Chapter 9 (Yoneda) and Chapter 15 (representability philosophy); generalized to enriched and weighted settings in Chapters 20 and 28; rigorously concluded in Chapter 31: *every universal construction is a Kan extension along a Yoneda embedding*, computable as a coend.

36.2.2 Thesis 2: Adjointness

> **Every adjoint pair encodes a universal-property pair, and conversely every universal property is captured by an adjunction.**

The fundamental link of Volumes I/II, formalized in Chapter 17 (adjunctions deepened) and Chapter 28 (adjoints via representability). The 2-categorical version of Chapter 33 closes the circle: monads and adjunctions are formally dual in any 2-category.

36.2.3 Thesis 3: Kan Extensions Are Primitive

> **Every universal construction is, equivalently, a Kan extension.**

The strongest unification of Volume III: Kan extensions subsume limits, colimits, adjunctions, monads (via codensity and algebraic-theory forms), free constructions, geometric realizations, sheafifications. The coend calculus (Chapter 26) gives the computational tools.

36.2.4 Thesis 4: 2-Categorical Coherence

> **Strictification reduces "weak" to "strict."**

Mac Lane's coherence theorem for monoidal categories (Chapter 19); bicategorical strictification (Chapter 25); monoidal-functor coherence (Chapter 29); strict adjunctions from bicategorical ones (Chapter 33). The recurring observation: in many practical contexts, weak structures may be replaced by strict ones up to equivalence.

36.2.5 Thesis 5: Enrichment as Generalization

> **Categorical structures generalize from Set-enrichment to $\mathcal{V}$-enrichment with no loss of theory.**

The enriched yoga of Chapter 20 and Chapter 30: every theorem of Volumes I–II survives the generalization, often becoming cleaner. Weighted limits, enriched Kan extensions, $\mathcal{V}$-adjunctions, $\mathcal{V}$-monads are all natural lifts.

36.2.6 Thesis 6: Algebraic Structures Decompose

> **Compound algebraic structures decompose, through distributive laws, into compositions of simpler structures.**

The newest thesis (Chapter 35): rings = abelian groups $\otimes$ monoids, modules = abelian groups $\otimes$ action, etc., all through distributive laws. The categorification of arithmetic distributivity.

36.3 What Remains: Toward Volume IV

Several major themes are beyond the scope of this book but flow naturally from Volume III's foundations.

36.3.1 ∞-Categories

The natural next step. An ∞-category has objects, 1-morphisms, 2-morphisms, 3-morphisms, ... ad infinitum, with all the coherence laws of category theory holding up to higher morphisms. Quasi-categories, simplicial categories, and complete Segal spaces are common models. Volume IV would develop ∞-categorical analogues of every Volume III theorem: ∞-Kan extensions, ∞-Yoneda, ∞-monads, ∞-adjunctions.

36.3.2 Higher Algebra

The natural home for "algebraic structures up to homotopy." $\mathbb{E}_n$-algebras, A_∞-algebras, L_∞-algebras — all generalizations of associative or commutative algebra structures to homotopy-coherent versions. The starting point is Volume III: operads (Chapter 29), monads (Chapter 32), distributive laws (Chapter 35).

36.3.3 Derived Algebraic Geometry

Applies ∞-categories and higher algebra to algebraic geometry. Stacks, derived schemes, formal moduli problems. The basic objects are "things up to coherent isomorphism," which is precisely what ∞-categories enable.

36.3.4 Homotopy Type Theory

The intersection of category theory and dependent type theory. The univalence principle says "isomorphic things are equal"; this is, philosophically, the same as the Yoneda philosophy.

36.3.5 Applications to Other Branches

The original promise of the book: *connections to other branches of mathematics*. These appear after Volume IV's foundations are laid:

- **Algebra**: derived functors, derived categories, triangulated/stable categories.
- **Topology**: simplicial sets, spectra, stable homotopy theory.
- **Logic**: classifying topoi, geometric morphisms, internal languages.
- **Computer science**: categorical semantics, models of dependent type theory, functional programming.

36.4 Reading the Book Backward

36.4.1 Formalizing

A reader who has completed Volume III may profitably reread the earlier material with the formal-spine perspective:

Yoneda's lemma (Chapters 9, 15, 20, 28). The most read theorem of category theory. Each iteration adds an angle: in Chapter 9, a hom-set bijection; in Chapter 15, the representability principle; in Chapter 20, the enriched form; in Chapter 28, the unified formal statement encompassing ends, coends, density, free cocompletion, and "Yoneda is unitality in **Prof**."

Kan extensions (Chapters 11, 18, 31). Beginning as a coding trick for representing functors as colimits, Kan extensions become the *primitive* of category theory by Chapter 31: every universal construction is one.

Monads (Chapters 8, 21, 32). From a 4-axiom diagram-chase in Chapter 8 to a 2-categorical universal construction in Chapter 32, the notion is the same; the perspective deepens. Monads become *algebraic theories presented in compact form*.

Adjunctions (Chapters 7, 17, 33). From a hom-set bijection to a formal 2-categorical construction. Adjunctions and monads, in the formal view, are dual constructions that together generate every universal property of category theory.

36.4.2 Reorganizing

Remark 36.2 (The reader's progression)**.** This book builds capacity slowly. The Volume I reader manipulates objects and morphisms in concrete examples. The Volume II reader recognizes patterns: representability, adjointness, the structural reasoning behind universal properties. The Volume III reader uses the formal-spine tools — coend calculus, weighted limits, 2-categorical thinking — to navigate abstract problems with the same fluency. The progression is logarithmic: each volume halves the friction of the next-higher abstraction.

36.5 Volume III's Final Theorem: Coherence of the Edifice

If Volume III has a single overarching theorem, it is:

The universal constructions of category theory cohere, through the language of (weighted) limits and (co)ends, into a single computational calculus.

Every construction we encountered — products, coproducts, equalizers, pullbacks, free monoids, sheafifications, geometric realizations, nerves, fundamental groupoids, Cauchy completions, Stone–Čech compactifications, ultrafilter monads, double dualizations — all fit into the same coend formula:

$$\int^{c} (\text{weight on } c) \otimes (\text{diagram value at } c).$$

The choices of weight and diagram parameterize the construction. The coend identifies the universal solution.

Remark 36.3 (The conceptual achievement)**.** The reader who finishes Volume III has acquired not just facts about category theory but a unified mental representation: every universal construction is a single computational pattern. The pattern can be applied mechanically — once you know the weight and diagram, the construction is forced.

36.6 Exercises

What are Volume III's three layers?	2-categorical foundations (Chapters 25, 32, 33); coend calculus (Chapters 26, 27, 28, 30, 31); synthesis (Chapters 29, 34, 35, 36).
State the representability thesis.	Every universal construction in category theory is the representing object for a canonically defined functor.
State the Kan extension thesis.	Every universal construction is, equivalently, a Kan extension.
State the adjointness thesis.	Every adjoint pair encodes a universal-property pair, and conversely every universal property is captured by an adjunction.
State the strictification thesis.	Weak structures reduce to strict ones up to equivalence: bicategories to 2-categories, monoidal to strict monoidal, etc.
State the enrichment-as-generalization thesis.	Categorical structures generalize from **Set**-enrichment to $\mathcal{V}$-enrichment with no loss of theory.
State the algebraic-decomposition thesis.	Compound algebraic structures decompose, through distributive laws, into compositions of simpler structures.
What is the single coend formula that unifies universal constructions?	$\int^c W(c) \otimes F(c)$: weight W times diagram F, integrated over the indexing category c.
What does "Yoneda is unitality in **Prof**" mean?	The hom-profunctor $\mathcal{C}(-,-)$ is the identity 1-cell of **Prof** at $\mathcal{C}$: $P \circ \mathcal{C}(-,-) \cong P$ and $\mathcal{D}(-,-) \circ P \cong P$.
Why is Yoneda revisited four times across the book?	Each iteration adds an angle: hom-set bijection (Ch 9), representability (Ch 15), enriched (Ch 20), unified formal (Ch 28).
Why is Kan extension "primitive" in Vol III's view?	Because every universal construction (limits, colimits, adjoints, monads, free constructions) is a Kan extension, and the coend calculus computes them all.
What is ∞-category theory's relationship to Vol III?	It's the natural successor: ∞-categorical analogues of every Vol III theorem (Kan extensions, Yoneda, monads, adjunctions) make up Volume IV.
Name three topics beyond Volume III's scope.	∞-category theory, higher algebra, derived algebraic geometry. (Also: homotopy type theory; applications to other branches.)
What is the original goal of the book?	Full formal understanding of category theory, its complete suite of theorems, and connections to other branches of mathematics — in that order.

Why is the book structured in three volumes?	Vol I: pedagogically slow, full cycle structure, no connections. Vol II: same content reorganized around structural principles. Vol III: formal spine with enriched, 2-categorical, and synthesis content.
Why is Vol III written densely?	Because the reader has been incrementally trained for two volumes. Vol III rewards that training with formal/terse coverage at maximum proof density.
What does “the reader’s progression is logarithmic” mean?	Each volume halves the friction of the next abstraction: Vol I = manipulating objects; Vol II = recognizing patterns; Vol III = fluent navigation of the formal spine.
What is the role of representability across all three volumes?	The unifying organizing principle. Vol I introduces it (Ch 5). Vol II makes it the thesis. Vol III shows it’s also Yoneda, Kan extensions, codensity, and the coend calculus.
Why is “every adjunction is a Kan extension” a deep theorem?	Because it identifies two perspectives on universal constructions: the adjoint-functor view (Vol I) and the colimit-formula view (Vol III). They are the same.
Why do distributive laws (Ch 35) deserve a chapter on their own?	Because they are the formal mechanism by which compound monadic structures (rings, modules, monad transformers) decompose into layers — a foundational pattern not covered in Vols I–II.
What’s the Vol III pedagogical model?	High familiarity assumed; minimal prose expansion; maximum proof density; the prooflog environment for two-column step/justification proofs.
Why is coherence (bicategorical, monoidal) a recurring theme?	Because the gap between strict and weak structures is unavoidable: associativity rarely holds on the nose. Coherence is the bridge.
What is the relationship between Yoneda and Kan extensions?	The Yoneda embedding makes every functor a Kan extension along itself (density). Conversely, every Kan extension is a weighted (co)limit along the Yoneda embedding.
Where does the formal monad theory live?	In any 2-category. The theory is specialized to **Cat** for the classical case; to **Prof** for category extensions; to **Span** for preorders; etc.
What is the dual of monad theory in Vol III?	Adjunction theory. Monads arise from adjunctions; adjunctions are universally factored through Eilenberg–Moore objects. The duality is internal to $\mathrm{Mnd}(\mathcal{K})$.
Why is the “two faces” of monads in Ch 31 important?	Because it shows monads = codensity (categorical face) AND monads = algebraic theory presentation (algebraic face). Many monads are both; the formal theory unifies them.
How does Day convolution fit into the synthesis?	It’s the canonical monoidal structure on presheaves, exhibiting the presheaf category as a free *monoidal* cocompletion (Ch 30). It makes the free-cocompletion theorem “monoidal-aware.”
What’s the role of weighted limits in Vol III’s calculus?	They are the universal computational tool: every limit, every colimit, every (co)end is a weighted (co)limit. The Yoneda reduction makes them mechanical.
Summarize Vol III’s final theorem.	The universal constructions of category theory cohere, through the language of weighted (co)limits and (co)ends, into a single computational calculus. The coend formula $\int^c W(c) \otimes F(c)$ parameterizes all of them.

What should the reader who finishes the book do next?	Open Lurie's "Higher Topos Theory" or Riehl–Verity's "Elements of ∞-Category Theory" to begin Volume IV.

Chapter Summary

Chapter Summary

Volume III in three layers. 2-categorical foundations (Chapters 25, 32, 33); coend calculus (26, 27, 28, 30, 31); synthesis (29, 34, 35, 36).

The unifying theses. Representability, adjointness, Kan extensions as primitive, 2-categorical strictification, enrichment as generalization, algebraic decomposition through distributive laws.

The coend formula. $\int^c W(c) \otimes F(c)$ — a single template parameterizing every universal construction by weight and diagram.

Reading the book backward. Yoneda's lemma in four iterations (Ch 9, 15, 20, 28); Kan extensions in three (Ch 11, 18, 31); monads in three (Ch 8, 21, 32); adjunctions in three (Ch 7, 17, 33). Each iteration deepens the perspective.

Beyond Volume III. ∞-categories, higher algebra, derived algebraic geometry, homotopy type theory. Applications to algebra, topology, logic, computer science (the original promise) await Volume IV.

Vol III's final theorem. The universal constructions of category theory cohere, through the language of weighted (co)limits and (co)ends, into a single computational calculus.

Formal Restatement

Formal Restatement

The coend formula. For weight $W : \mathcal{J}^{\mathrm{op}} \to \mathcal{V}$ and diagram $F : \mathcal{J} \to \mathcal{C}$ in a $\mathcal{V}$-tensored cocomplete $\mathcal{C}$:

$$W \star F \;=\; \int^c W(c) \otimes F(c) \;\in\; \mathcal{C}.$$

Every universal construction in category theory specializes this formula.

Specializations.

$$\begin{array}{rlr}
\text{Limit (conical):} & \Delta I \star F & = \operatorname{colim} F. \\
\text{Cotensor:} & \{V, X\} & = V \pitchfork X. \\
\text{Tensor:} & V \star X & = V \otimes X. \\
\text{Left Kan extension:} & \mathcal{D}(K-, d) \star F & = (\operatorname{Lan}_K F)(d). \\
\text{Right Kan extension:} & \{\mathcal{D}(d, K-), F\} & = (\operatorname{Ran}_K F)(d). \\
\text{Free abelian group:} & |X| \star \mathbb{Z}[-] & = F_{\mathrm{ab}}(X). \\
\text{Geometric realization:} & X \star \Delta^\bullet & = |X|. \\
\text{Codensity monad:} & \operatorname{Ran}_G G & = \mathbb{T}_G.
\end{array}$$

The slogan. Every universal construction is a Kan extension is a weighted (co)limit is a coend.

The categorical picture. Category theory is the study of universal constructions, organized into a single computational calculus. The calculus is the coend calculus. The pictures are: Yoneda (every representable is a free object); Kan extensions (every universal construction is one); distributive laws (compound structures decompose); strictification (weak reduces to strict).

Volume III is complete. The path forward — ∞-categories, higher algebra, derived algebraic geometry, and applications — is the territory of Volume IV.

Volume IV

Toward Higher Category Theory

Chapter 37

The Simplex Category Δ

Chapter Overview

Volume IV opens on the combinatorial substrate that every model of ∞-category theory rests upon. Before we can speak of quasi-categories (Chapter 43), simplicial categories, or complete Segal spaces, we need the indexing category Δ — the category of finite non-empty totally ordered sets and order-preserving maps between them.

We approach Δ from three angles, walking the cycle at a gentle pace. *Geometrically*, each object $[n] = \{0 < 1 < \cdots < n\}$ is the vertex set of an n-dimensional simplex (point, line segment, triangle, tetrahedron, ...), and the morphisms are the combinatorial shadows of affine maps between simplices that preserve vertex orderings. *Combinatorially*, the morphisms are generated by elementary *face maps* δ^i (one-point omissions) and *degeneracy maps* σ^i (one-point duplications) subject to a small list of relations, the cosimplicial identities. *Universally*, the augmented simplex category Δ_+ is the free strict monoidal category on a single monoid object — a remarkable theorem of André Joyal that explains why Δ appears *everywhere a monoid acts.*

The next two chapters use Δ to build first $\mathbf{sSet} = \mathbf{Set}^{\Delta^{\mathrm{op}}}$ (the category of *simplicial sets*, Chapter 38) and then the geometric realization $|-| : \mathbf{sSet} \to \mathbf{Top}$ (Chapter 39). The chapter after that turns to model-categorical machinery (Chapters 40–42). The journey to quasi-categories begins here, with monotone maps between ordered sets.

37.1 The Category Δ

37.1.1 Informally

Before any definitions, fix a mental picture. Imagine a sequence of objects, one for each dimension:

- A *point* (dimension 0): one vertex.
- A *line segment* (dimension 1): two vertices, joined by an edge.
- A *triangle* (dimension 2): three vertices, three edges, one 2-dimensional interior.
- A *tetrahedron* (dimension 3): four vertices, six edges, four triangular faces, one 3-dimensional interior.
- And so on: an *n-simplex* has $n + 1$ vertices and is the "simplest possible" shape of dimension n.

Each of these shapes has a *combinatorial skeleton*: the vertex set, totally ordered (we always agree on which vertex is "first," which is "second," and so on). For an n-simplex with vertices labeled $0, 1, \ldots, n$, the skeleton is simply the ordered set $\{0 < 1 < \cdots < n\}$.

Now imagine ways to move between simplices that respect both the shape and the ordering. An edge $\{0 < 1\}$ has three natural maps into a triangle $\{0 < 1 < 2\}$: send both endpoints into a single vertex ("collapse the edge to a point"), or include them as one of the three edges of the triangle. A triangle has many natural maps into a tetrahedron: include as a face, or collapse two vertices, or send everything to a single vertex. All of these — collapses, inclusions, and combinations — are exactly the *order-preserving* (or *monotone*) maps between the underlying ordered vertex sets.

The category Δ collects every such map. Its objects are the combinatorial skeletons; its morphisms are the order-preserving maps; composition is the obvious composition of functions. The category captures, in pure combinatorial form, every way that simplices fit together.

Where the Name Comes From

The symbol Δ is the Greek capital letter *delta*, chosen because the standard 2-simplex — a triangle — has shape Δ. The convention dates to Eilenberg and Steenrod's *Foundations of Algebraic Topology* (1952), where the simplex category was first isolated as the indexing object for singular homology. The word **simplex** is from Latin *simplex* ("simple, single-fold"), since an n-simplex is the simplest convex shape of dimension n.

Reader's Annotation

The single most useful thing to remember about Δ is that its morphisms are abundant: between any two objects $[m]$ and $[n]$ there are *many* monotone maps (we will count them in Example 37.5). This contrasts with categories like posets where you might expect maps to be rare. The abundance is what makes Δ useful as an *indexing* category: many maps means many ways to "compare" simplicial data of different dimensions.

37.1.2 Expanding

Three lenses through which the same category appears.

The combinatorial lens. Δ is the category of finite non-empty totally ordered sets. Choosing one representative of each isomorphism class (the standard shape $\{0 < 1 < \cdots < n\}$) gives the objects $[0], [1], [2], \ldots$. The morphisms are exactly the monotone functions.

The geometric lens. Δ is the category of *combinatorial simplices*. The object $[n]$ corresponds to the standard n-simplex $|\Delta^n| \subset \mathbb{R}^{n+1}$ (the convex hull of the standard basis vectors). Each monotone map $f : [m] \to [n]$ extends linearly to an affine map of simplices $|f| : |\Delta^m| \to |\Delta^n|$ that sends the i-th vertex of the source to the $f(i)$-th vertex of the target.

The algebraic lens. Δ is generated by two families of elementary maps — the face maps δ^i (inclusions of $(n-1)$-faces) and degeneracy maps σ^i (collapses of adjacent vertex pairs) — subject to a finite set of relations called the *cosimplicial identities* (§37.2). This is the generators-and-relations presentation that makes Δ usable in software and explicit computations.

Internal Dialog

Reader: "Why ordered sets, and not just sets?"
Author: The orderings give the simplices a notion of "orientation." An ordered triangle $\{0 < 1 < 2\}$ knows that "vertex 0 comes before vertex 1, which comes before vertex 2" — and this matters as soon as we want to define face maps without ambiguity ("the face opposite vertex 2") or degeneracy maps ("collapse vertex 1 onto vertex 2"). Unordered simplices would require us to keep track of all possible relabellings; the ordering fixes a canonical form.
Reader: "So Δ is rigid in a way ordinary set theory isn't."
Author: Exactly. Δ is rigid *combinatorially*: it has no non-identity automorphisms (Exercise: prove this). Every isomorphism $[n] \to [n]$ is the identity. This is the structural payoff of the ordering.

Example: Three monotone maps in pictures

Consider three monotone maps $[1] \to [2]$:

- $0 \mapsto 0$, $1 \mapsto 1$: includes the edge as "front" of the triangle.
- $0 \mapsto 0$, $1 \mapsto 2$: includes the edge as the long "base."
- $0 \mapsto 0$, $1 \mapsto 0$: collapses the edge to a single vertex.

The first two are injections; the third is the constant. Each is a distinct map. (See Example 37.5 for the full count.)

37.1.3 Formalizing

We now collect the picture into a precise definition.

Definition 37.1 (The simplex category Δ)**.** The **simplex category** Δ has:

- **Objects.** For each integer $n \geq 0$, an object $[n] = \{0, 1, \ldots, n\}$ regarded as a totally ordered set under the usual $<$.
- **Morphisms.** A morphism $f : [m] \to [n]$ is a function $f : \{0, \ldots, m\} \to \{0, \ldots, n\}$ that is **monotone**: $i \leq j$ in $[m]$ implies $f(i) \leq f(j)$ in $[n]$.
- **Composition.** Function composition; the composite of two monotone maps is monotone.
- **Identities.** The identity function $\mathrm{id}_{[n]}$.

How to Say This

Δ *(read: "the simplex category" or "capital delta")*; $[n]$ *(read: "n-bracket" or "the standard n-simplex (combinatorial)")*. A monotone map $f : [m] \to [n]$ may be read aloud as *(read: "f from [m] to [n]" or "a monotone map m-to-n.")* Saying "$[n]$ has $n + 1$ elements" is a useful catch-phrase to keep the off-by-one convention front-of-mind.

Where the Name Comes From

Monotone comes from Greek *mónos* (single) + *tónos* (tone, tension, stretch). A monotone function "moves in a single direction" — never reversing. The mathematical usage dates to the 19th century in real analysis (*monotonically increasing* sequences of real numbers). In Δ we use it for functions of ordered finite sets; the meaning is the same.

Reader's Annotation

The order-preservation condition $i \le j \Rightarrow f(i) \le f(j)$ is *weak* (it uses $\le$, not $<$), so a monotone map may send two distinct elements to the same image. This is what allows the "collapse" morphisms (the degeneracies) to exist in Δ.

Remark 37.2 (Cardinality vs. index)**.** Beware: $[n]$ has $n+1$ *elements*, not n. The off-by-one is geometric: an n-simplex (an n-dimensional shape) has $n+1$ vertices. Authors who write $\boldsymbol{n} = \{0, 1, \ldots, n-1\}$ shift the indexing by one but pay for it in formulas; we follow the geometric convention throughout.

Remark 37.3 (Where Δ appears)**.** Δ is the indexing category for *simplicial objects*: contravariant functors $\Delta^{\mathrm{op}} \to \mathcal{C}$ for any category $\mathcal{C}$. When $\mathcal{C} = \mathbf{Set}$ we get *simplicial sets* (Chapter 38); when $\mathcal{C} = \mathbf{Grp}$ we get *simplicial groups* (classifying-space machinery); when $\mathcal{C} = \mathbf{CommRing}$ we get *simplicial commutative rings* (one foundation of derived algebraic geometry, Chapter 58). In each case, the combinatorial geometry lives in Δ and the values live in the chosen target.

37.1.4 Formally: Small cases

Example 37.4 (The objects $[0]$ and $[1]$)**.** $[0] = \{0\}$ is the one-element ordered set. It is a *terminal* object of Δ: for every $[n]$ there is exactly one monotone map $[n] \to [0]$ (the constant function 0). Geometrically, $[0]$ is the 0-simplex — a single point.

$[1] = \{0 < 1\}$ has three monotone endomorphisms (the identity and the two constants), and exactly two non-constant monotone maps $[0] \to [1]$ (picking out the element 0 or the element 1). $[1]$ is *not* initial in Δ: there is not a unique map $[0] \to [1]$, but two — and an initial object must have a *unique* map to every other object. Geometrically, $[1]$ is the 1-simplex — a line segment.

Example 37.5 (Counting monotone maps)**.** The number of monotone maps $[m] \to [n]$ equals $\binom{m+n+1}{m+1}$. Equivalently: choose a non-decreasing sequence

$$0 \le f(0) \le f(1) \le \cdots \le f(m) \le n.$$

This is the same as choosing a multiset of size $m+1$ from $\{0, 1, \ldots, n\}$, equivalently $\binom{(n+1)+(m+1)-1}{m+1} = \binom{m+n+1}{m+1}$ stars-and-bars combinations.

Example: Spot-checking the count

Monotone maps $[1] \to [2]$: $\binom{1+2+1}{1+1} = \binom{4}{2} = 6$. Listing them: $(0,0), (0,1), (0,2), (1,1), (1,2), (2,2)$ — six pairs of non-decreasing values. Of these, three are injections (rows $1, 2, 4$), two are surjections-on-image (rows $1, 2, 4$ again — these inject), and the remaining three are constants. The count checks out.

Reader's Annotation

$[0]$ is terminal but not initial in Δ. The asymmetry is characteristic: Δ has a terminal object (the one-point set) but *no initial object*, because the empty set is not present (it would need cardinality zero, but $[n]$ has $n+1 \ge 1$ elements). The augmented variant Δ_+ in §37.3 adjoins the empty set $[-1] = \emptyset$ as an initial object — exactly what is needed to make Δ_+ strict-monoidal with $[-1]$ as unit.

Remark 37.6 (Δ is small and locally finite)**.** Δ has countably many objects and, for each pair $[m], [n]$, *finitely* many morphisms (by Example 37.5). It is therefore a small category, with finite hom-sets — making Δ a useful indexing category for both set-valued diagrams and topological diagrams.

37.2 Face and Degeneracy Maps

37.2.1 Informally

A monotone map $f : [m] \to [n]$ does one of three things:

- Possibly *omits* some elements of $[n]$ from its image.
- Possibly *repeats* some images (sending two adjacent elements of $[m]$ to the same place in $[n]$).
- Otherwise: passes elements through in order.

It turns out that every monotone map factors uniquely as a sequence of *elementary omissions* (called face maps) followed by a sequence of *elementary repeats* (called degeneracy maps) — or vice versa, depending on the factorization convention. The face and degeneracy maps are the atomic building blocks of Δ.

Where the Name Comes From

Face map is geometric: in an n-simplex, the face opposite vertex i is the $(n-1)$-simplex obtained by deleting i from the vertex set. The face map $\delta^i : [n-1] \to [n]$ *embeds* the $(n-1)$-simplex into the n-simplex as that face. **Degeneracy map** comes from Latin *degenerare* (to depart from one's lineage, to deteriorate). In mathematics "degenerate" generally means "collapsed to a simpler form" (a degenerate triangle is one where two vertices coincide). The degeneracy map $\sigma^i : [n+1] \to [n]$ collapses two adjacent vertices, making the source $(n+1)$-simplex "degenerate" onto a non-degenerate face of the target n-simplex.

37.2.2 Expanding

The face map $\delta^i : [n-1] \to [n]$ "inserts i into $[n-1]$'s image without using it." Concretely: it sends $0, 1, \ldots, i-1$ to themselves, and $i, i+1, \ldots, n-1$ all shifted up by one to become $i+1, i+2, \ldots, n$. The vertex $i \in [n]$ is the one omitted.

The degeneracy map $\sigma^i : [n+1] \to [n]$ "duplicates the image at i." Concretely: it sends $0, 1, \ldots, i$ to themselves, and $i+1, i+2, \ldots, n+1$ all shifted down by one to become $i, i+1, \ldots, n$. The fibre over $i \in [n]$ has two elements (namely i and $i+1$ in $[n+1]$); the fibres over every other element are singletons.

Example: Faces and degeneracies in low dimension

The maps $\delta^0, \delta^1 : [0] \to [1]$: δ^0 skips $0 \in [1]$, so $0 \in [0]$ maps to $1 \in [1]$; δ^1 skips $1 \in [1]$, so $0 \in [0]$ maps to $0 \in [1]$. Geometrically, $[0]$ is a point and $[1]$ is an edge: δ^0 and δ^1 pick out the two endpoints.
The map $\sigma^0 : [1] \to [0]$: The unique map $[1] \to [0]$ — both elements collapse onto the single target. Geometrically, the edge collapses to a point.
The maps $\delta^0, \delta^1, \delta^2 : [1] \to [2]$: The three edges of a triangle. δ^0 skips 0: gives the edge $\{1, 2\}$ (opposite vertex 0). δ^1 skips 1: gives the "long" edge $\{0, 2\}$. δ^2 skips 2: gives the edge $\{0, 1\}$.
The maps $\sigma^0, \sigma^1 : [2] \to [1]$: Two ways to collapse a triangle onto an edge by identifying two adjacent vertices. σ^0 identifies $\{0, 1\}$ in $[2]$ to $0 \in [1]$; σ^1 identifies $\{1, 2\}$ in $[2]$ to $1 \in [1]$.

Reader's Annotation

Why the superscripts? The convention δ^i and σ^i (superscripts, not subscripts) emphasizes the *covariant* direction: the maps go $[n-1] \to [n]$ and $[n+1] \to [n]$, both in Δ itself. When we later pass to the opposite category Δ^{op} to define simplicial sets, the induced face and degeneracy operators on the simplicial set are written d_i and s_i (subscripts) — Chapter 38. The convention helps signal which direction you are working in.

37.2.3 Formalizing

Definition 37.7 (Face map)**.** For $0 \le i \le n$, the ***i*-th face map**

$$\delta^i : [n-1] \longrightarrow [n]$$

is the unique *injective* monotone map whose image omits exactly the element $i \in [n]$:

$$\delta^i(j) \;=\; \begin{cases} j & \text{if } j < i, \\ j+1 & \text{if } j \ge i. \end{cases}$$

Definition 37.8 (Degeneracy map)**.** For $0 \le i \le n$, the ***i*-th degeneracy map**

$$\sigma^i : [n+1] \longrightarrow [n]$$

is the unique *surjective* monotone map whose fibre over i has two elements (namely i and $i+1$):

$$\sigma^i(j) \;=\; \begin{cases} j & \text{if } j \le i, \\ j-1 & \text{if } j > i. \end{cases}$$

How to Say This

δ^i *(read: "face delta-i")* "skips i"; σ^i *(read: "degeneracy sigma-i")* "doubles i." Some texts write d_i and s_i instead; we follow the more common "cosimplicial" (covariant) convention with δ, σ throughout this chapter and reserve d_i, s_i for the contravariant simplicial-set operators of Chapter 38.

37.2.4 Reorganizing: the cosimplicial identities

Face and degeneracy maps interact in a small, finite number of ways. The package of relations is called the **cosimplicial identities**.

Where the Name Comes From

Cosimplicial is the dual ("co-") of **simplicial**: where a *simplicial* object is a contravariant functor on Δ (i.e., a functor $\Delta^{\mathrm{op}} \to \mathcal{C}$), a *cosimplicial* object is a covariant functor $\Delta \to \mathcal{C}$. The face and degeneracy maps live in Δ itself; their relations are the *cosimplicial* identities. The dual relations among d_i, s_i (face and degeneracy *operators* on a simplicial object) are called the **simplicial identities** (Chapter 38).

Theorem 37.9 (Cosimplicial identities)**.** *The face and degeneracy maps in Δ satisfy:*

$$\begin{aligned}
\delta^j\delta^i &= \delta^i\delta^{j-1} && \text{if } i < j,\\
\sigma^j\sigma^i &= \sigma^i\sigma^{j+1} && \text{if } i \le j,\\
\sigma^j\delta^i &= \delta^i\sigma^{j-1} && \text{if } i < j,\\
\sigma^j\delta^i &= \mathrm{id} && \text{if } i = j \text{ or } i = j+1,\\
\sigma^j\delta^i &= \delta^{i-1}\sigma^j && \text{if } i > j+1.
\end{aligned}$$

Every identity among composites of δ's and σ's in Δ is a consequence of these five.

Proof sketch. Each identity is checked by tracking what each side does to a generic element. For example, the first: $\delta^j\delta^i : [n-2] \to [n]$ skips i and then j (now shifted); under the assumption $i < j$, the "j after i has shifted" becomes $j-1$ in the codomain $[n-1]$, and the order $\delta^i\delta^{j-1}$ — skip $j-1$, then i — produces the same omission set $\{i,j\}$ with the same target indexing.

The completeness claim (every relation follows) is a consequence of the epi-mono factorization in Δ (Section 37.3). □ □

Reader's Annotation

The cosimplicial identities are best remembered as: *faces commute past faces with a shift; degeneracies commute past degeneracies with a shift; faces and degeneracies either commute or annihilate.* The annihilation case $\sigma^j\delta^i = \mathrm{id}$ for $i \in \{j, j+1\}$ is the key geometric content: *doubling a vertex and then deleting one copy of the doubled pair recovers the original vertex.*
A useful first practice: pick small n and write out a few cases by hand. The patterns are easy to see in dimension $n = 2$ or 3, where all maps are visualizable.

37.2.5 Formally: generators

Theorem 37.10 (Generators of Δ)**.** *Every morphism in Δ is a composite of face and degeneracy maps. Specifically, any non-identity morphism $f : [m] \to [n]$ admits a unique factorization*

$$f = \delta^{i_1}\delta^{i_2}\cdots\delta^{i_p}\,\sigma^{j_1}\sigma^{j_2}\cdots\sigma^{j_q}$$

with $n \ge i_1 > i_2 > \cdots > i_p \ge 0$ and $0 \le j_1 < j_2 < \cdots < j_q < m$, where p is the number of elements of $[n]$ not in the image of f and q is the number of repetitions in the image (counting multiplicity).

Proof sketch. Every monotone map $f : [m] \to [n]$ factors uniquely as a surjection followed by an injection (the standard epi-mono factorization, which exists in Δ because monotone maps are determined by their images): write $f = e \circ s$ where $s : [m] \twoheadrightarrow [k]$ is the surjective part (collapsing repeated values) and $e : [k] \hookrightarrow [n]$ is the injective part (the image inclusion).

Surjective monotone maps $[m] \twoheadrightarrow [k]$ are uniquely composites of degeneracies in increasing index order; injective monotone maps $[k] \hookrightarrow [n]$ are uniquely composites of faces in decreasing index order. The unique factorization claim follows. □ □

Remark 37.11 (Why the index order matters)**.** The "decreasing i's, increasing j's" convention is the unique *normal form*: any other ordering can be rewritten into this one using the cosimplicial identities. The convention is one of the cleanest in combinatorial topology — once you internalize it, you can read off the structure of any monotone map at a glance.

Remark 37.12 (Cosimplicial vs. simplicial)**.** The dual category Δ^{op} has the same objects but reverses arrows. A **simplicial object** in a category $\mathcal{C}$ is a functor $X : \Delta^{\mathrm{op}} \to \mathcal{C}$ — equivalently, a sequence of objects $X_0, X_1, \ldots$ together with maps $d_i = X(\delta^i) : X_n \to X_{n-1}$ (*face* operators on the simplicial side) and $s_i = X(\sigma^i) : X_n \to X_{n+1}$ (*degeneracy* operators), satisfying the **simplicial identities** — the relations of Theorem 37.9 with arrows reversed and indices renamed. Chapter 38 develops this.

37.3 The Universal Property of Δ

37.3.1 Informally

We have seen that Δ has a clean generators-and-relations presentation. There is also a much deeper structural reason for Δ to be "the right" category for indexing simplicial data: Δ is the *free* object of a particular categorical type.

More precisely: its augmented variant Δ_+ is the free strict monoidal category on a single monoid object. This means: *specifying a monoid in* any *strict monoidal category is the same as specifying a strict monoidal functor out of* Δ_+.

The consequence: anywhere a monoid appears (and monoids are everywhere in mathematics), the data is automatically organized by Δ. This is the structural origin of the omnipresence of simplicial objects.

Reader's Annotation

The universal property in this section is Joyal's theorem (Theorem 37.17). It is one of the most important theorems of combinatorial category theory, but it requires the augmented variant Δ_+ — the version of Δ with the empty ordered set $[-1] = \emptyset$ added as an initial object. The empty ordinal provides the *unit* for ordinal sum. Without it, Δ would not be strict-monoidal.

37.3.2 Formalizing

Definition 37.13 (Augmented simplex category Δ_+)**.** The **augmented simplex category** Δ_+ has the same data as Δ together with an additional object

$$[-1] \;=\; \emptyset,$$

the empty ordered set. There is exactly one morphism $[-1] \to [n]$ for each $[n]$ (the empty function), and no morphisms $[n] \to [-1]$ for $n \geq 0$ (since $[n]$ is non-empty and $[-1]$ is empty).

Where the Name Comes From

Augmented from Latin *augere* ("to increase, to enlarge"). Δ_+ is the "augmented" simplex category because we have *augmented* Δ by adding an initial object below it. The notation Δ_+ is older than current conventions for general augmentations; some references write Δ_a or $\Delta_{\leq}$.

How to Say This

Δ_+ *(read: "*Δ*-plus" or "augmented Delta")*; $[-1]$ *(read: "the empty ordinal" or "negative-one bracket")*. The phrase "augmented simplex category" is standard in modern higher category theory; older topology texts sometimes call it the "simplicial category with augmentation."

Definition 37.14 (Ordinal sum)**.** The **ordinal sum** on Δ_+ is the functor $\oplus : \Delta_+ \times \Delta_+ \to \Delta_+$ given on objects by

$$[m] \oplus [n] \;=\; [m+n+1],$$

with the convention that $[-1] \oplus [n] = [n] = [n] \oplus [-1]$. On morphisms $f : [m] \to [m']$ and $g : [n] \to [n']$, the morphism $f \oplus g : [m+n+1] \to [m'+n'+1]$ acts as f on the first $m+1$ elements (sending $i \mapsto f(i)$) and as a shifted g on the remaining $n+1$ elements (sending $m+1+j \mapsto m'+1+g(j)$).

Where the Name Comes From

The **ordinal sum** is the standard set-theoretic operation on well-ordered sets: place $[m]$ first, then $[n]$ after — with every element of $[m]$ less than every element of $[n]$. The result is the ordered set of cardinality $(m+1)+(n+1) = m+n+2$, which is $[m+n+1]$ in our indexing. Note that $\oplus$ is *not* symmetric: $[m] \oplus [n]$ has $[m]$'s elements coming first, but as ordered sets is canonically isomorphic to $[n] \oplus [m]$.

Theorem 37.15 (Strict monoidal structure on Δ_+)**.** *The functor* $\oplus$ *makes* Δ_+ *a* strict monoidal category *with unit* $[-1]$. *Associativity is strict:*

$$([m] \oplus [n]) \oplus [p] \;=\; [m+n+p+2] \;=\; [m] \oplus ([n] \oplus [p]).$$

Remark 37.16 (A monoid object in Δ_+)**.** The object $[0]$ carries a canonical monoid structure in the strict monoidal category $(\Delta_+, \oplus, [-1])$. The structure maps are:

- Multiplication: $\mu = \sigma^0 : [0] \oplus [0] = [1] \to [0]$ (the unique monotone surjection).
- Unit: $\eta : [-1] \to [0]$ (the unique empty-function).

The unit and associativity laws hold strictly by the uniqueness of monotone maps with prescribed targets in Δ_+.

37.3.3 Reorganizing: Joyal's theorem

The remarkable fact is that this single monoid object *generates* Δ_+ in a precise sense.

Where the Name Comes From

Joyal's universal property of Δ_+ is attributed to *André Joyal* (born 1943, Montréal), a foundational figure in modern category theory. Joyal's contributions include species (combinatorial structures), the Joyal model structure on simplicial sets (making **sSet** a model for ∞-categories — Chapter 44), and many results in higher categorical algebra. The universal property theorem here is one of the cleanest applications of the "free-monoid-on-an-object" viewpoint in classical category theory.

Theorem 37.17 (Joyal's universal property of Δ_+)**.** *The pair* $(\Delta_+, [0])$ *is the* ***free strict monoidal category on a monoid object****: for any strict monoidal category* $(\mathcal{V}, \otimes, I)$ *equipped with a monoid object* $(M, \mu : M \otimes M \to M, \eta : I \to M)$*, there is a unique strict monoidal functor*

$$F : \Delta_+ \to \mathcal{V} \quad \textit{with} \quad F([0]) = M,\ F(\sigma^0_{[1]}) = \mu,\ F(\eta_{[0]}) = \eta.$$

Proof sketch. On objects, F is forced: $F([n]) = M^{\otimes(n+1)}$ and $F([-1]) = I$. On morphisms, the strict monoidal extension is unique because every morphism in Δ_+ factors as a composite of ⊕-tensorings of the elementary morphisms $\sigma^0 : [1] \to [0]$ (the multiplication on [0]), $[-1] \to [0]$ (the unit), and their identities — and these are forced to map to μ, η, and identities. The cosimplicial identities translate exactly into the monoid axioms, so F is well-defined. □ □

Reader's Annotation

Joyal's theorem is the *universal explanation* for why Δ appears in so many contexts:

- Wherever a monoid acts (and monoids are everywhere — groups, rings, monads, A_∞-algebras, etc.), there is automatically a strict monoidal functor out of Δ_+.
- Restricting that functor to $\Delta \subset \Delta_+$ produces a simplicial object — the canonical *bar construction* (Corollary 37.18).
- This is a single sentence that explains the bar resolution in homological algebra, the nerve of a category, the simplicial resolution of a group, the bar-cobar duality of operads (Chapter 57), and the ∞-monadic bar (Chapter 51).

Internalize this and you will see Δ wherever you look.

37.3.4 Formally: Consequences

Corollary 37.18 (Bar construction)**.** *For any monoid* M *in a strict monoidal category* $\mathcal{V}$*, the corresponding strict monoidal functor* $F : \Delta_+ \to \mathcal{V}$ *restricted to* $\Delta \subset \Delta_+$ *defines a simplicial object*

$$B_\bullet M : \Delta^{\mathrm{op}} \to \mathcal{V}, \quad B_n M \ = \ M^{\otimes(n+1)},$$

known as the ***bar construction*** *on* M*. Its face maps come from multiplication; its degeneracy maps come from the unit.*

Where the Name Comes From

The **bar construction** gets its name from the vertical-bar notation used by Eilenberg and Mac Lane (1953) for its elements: they wrote $[a_1|a_2|\cdots|a_n]$ for an element of $M^{\otimes n}$, with vertical bars separating the tensor factors. The construction itself predates the notation by years, but the bars stuck.

Remark 37.19 (Bar construction in Volume IV's arc)**.** The bar construction appears repeatedly in Volume IV. In Chapter 42 it provides resolutions for derived functors; in Chapter 51 it computes the universal ∞-monad; in Chapter 56 it relates A_∞- to $\mathbb{E}_1$-algebras; in Chapter 57 it is the keystone of A_∞ / L_∞ Koszul duality. Each appearance is a special case of the construction here, applied in a different monoidal category.

Corollary 37.20 (The epi-mono factorization revisited)**.** *Theorem 37.10's unique factorization is the* orthogonal factorization system *on* Δ *generated by* $\mathcal{E}$ *= surjective monotone maps (composites of degeneracies) and* $\mathcal{M}$ *= injective monotone maps (composites of faces). Every monotone map factors uniquely (up to canonical isomorphism) as* $\mathcal{E}$ *followed by* $\mathcal{M}$*.*

37.4 Geometric Realization (Preview)

37.4.1 Informally

We have been carrying a geometric mental picture all along — $[n]$ as the vertex set of an n-simplex. The picture is more than a metaphor: each object $[n] \in \Delta$ has a literal topological avatar, and the morphisms of Δ encode literal continuous maps between simplices.

Where the Name Comes From

Geometric realization is so named because it "realizes" the combinatorial simplex $[n]$ as a literal geometric object in $\mathbb{R}^{n+1}$. The verb *realize* in mathematics typically means "construct a concrete instance of"; the noun *realization* (or *realisation*, British) names that instance. The terminology dates to early algebraic topology and was standardized by Milnor's *Geometric realization of a semi-simplicial complex* (1957).

37.4.2 Formally

Definition 37.21 (Standard topological n-simplex)**.** The **standard topological n-simplex** is

$$|\Delta^n| \;=\; \left\{(t_0,\dots,t_n) \in \mathbb{R}^{n+1} \;\middle|\; t_i \geq 0,\; \sum_i t_i = 1\right\}.$$

The i-th vertex is the standard basis vector $e_i \in \mathbb{R}^{n+1}$; the simplex itself is the convex hull of $e_0, e_1, \dots, e_n$.

How to Say This

$|\Delta^n|$ *(read: "the geometric n-simplex" or "the realization of Δ^n").* The vertical bars $|-|$ are the standard notation for realization throughout algebraic topology; we use them consistently for the realization functor of Chapter 39.

Remark 37.22 (The cosimplicial space $|\Delta^\bullet|$)**.** The assignment $[n] \mapsto |\Delta^n|$ defines a functor $|\Delta^\bullet| : \Delta \to \mathbf{Top}$, the **cosimplicial space of standard simplices**. Face maps δ^i correspond to inclusions of $(n-1)$-faces into the n-simplex (literally: send $e_j \mapsto e_{\delta^i(j)}$ and extend affinely). Degeneracy maps σ^i correspond to collapses of n-simplices onto their lower-dimensional faces (send $e_j \mapsto e_{\sigma^i(j)}$ and extend affinely; this collapses an edge of the n-simplex onto a vertex of the $(n-1)$-simplex).

Theorem 37.23 ($|\Delta^\bullet|$ is fully faithful)**.** *The functor $|\Delta^\bullet| : \Delta \to \mathbf{Top}$ is faithful, and its image consists of the standard simplices and the affine simplicial maps between them.*

The functor $|\Delta^\bullet|$ is the engine of **geometric realization**, which Chapter 39 extends along the Yoneda embedding $\Delta \hookrightarrow \mathbf{sSet}$ to a left adjoint $|-| : \mathbf{sSet} \to \mathbf{Top}$. The full development is the content of Chapter 39; here we only note the existence as motivation for the indexing combinatorics.

Reader's Annotation

The functor $|\Delta^\bullet|$ is not full — there are continuous maps $|\Delta^m| \to |\Delta^n|$ that are *not* affine (e.g., any non-affine continuous map). Its faithfulness (in the sense of injection on morphisms) is what matters for combinatorial applications: distinct monotone maps yield distinct affine simplicial maps.

37.5 Exercises

What are the objects of Δ?	Finite non-empty totally ordered sets, presented as $[n] = \{0 < 1 < \cdots < n\}$ for $n \geq 0$. Up to canonical isomorphism, one per positive cardinality.
What are the morphisms?	Monotone (order-preserving) functions: $f : [m] \to [n]$ with $i \leq j \Rightarrow f(i) \leq f(j)$.
Why does $[n]$ have $n+1$ elements, not n?	Geometric convention: $[n]$ is the combinatorial n-simplex, which is an n-dimensional shape with $n+1$ vertices.
Why ordered sets and not just sets?	The ordering gives simplices orientation: a canonical labeling of vertices, so face maps and degeneracies can be defined unambiguously. Δ has no non-identity automorphisms — every $[n]$ is rigid.
Where does the name "simplex" come from?	Latin *simplex*, "simple" — an n-simplex is the simplest convex shape of dimension n.
Is $[0]$ initial, terminal, both, or neither in Δ?	Terminal: from any $[n]$ there is exactly one monotone map to $[0]$ (the constant). Not initial: e.g., two distinct monotone maps $[0] \to [1]$ exist.
How many monotone maps $[m] \to [n]$ are there?	$\binom{m+n+1}{m+1}$ — equivalently, the number of multisets of size $m+1$ from an $(n+1)$-element set.
What is the face map $\delta^i : [n-1] \to [n]$?	The unique injective monotone map whose image omits exactly $i \in [n]$: $\delta^i(j) = j$ if $j < i$, else $j+1$.
What is the degeneracy map $\sigma^i : [n+1] \to [n]$?	The unique surjective monotone map whose fibre over i has two elements: $\sigma^i(j) = j$ if $j \leq i$, else $j-1$.
Why are the maps named "face" and "degeneracy"?	"Face" is geometric: δ^i embeds the $(n-1)$-simplex as the face opposite vertex i. "Degeneracy" from Latin *degenerare*: σ^i degenerates an $(n+1)$-simplex by collapsing two adjacent vertices.

State one cosimplicial identity.	$\sigma^j\delta^i = \mathrm{id}$ if $i \in \{j, j+1\}$ — doubling a point and then deleting one copy is the identity.
What does $\delta^j\delta^i = \delta^i\delta^{j-1}$ for $i < j$ say geometrically?	Two non-adjacent face inclusions of an $(n-2)$-face into an n-face commute, with the second-applied face index shifted to account for the absence of the first.
State the generation theorem.	Every morphism in Δ factors uniquely as $\delta^{i_1} \cdots \delta^{i_p} \sigma^{j_1} \cdots \sigma^{j_q}$ with $i_1 > \cdots > i_p$ and $j_1 < \cdots < j_q$.
What is the epi-mono factorization in Δ?	Every monotone map factors as a surjection (degeneracies) followed by an injection (faces). The factorization is unique.
What is the augmented simplex category Δ_+?	Δ with the empty ordered set $[-1] = \emptyset$ added as an object; the empty function $[-1] \to [n]$ is the unique morphism from $[-1]$.
Why is $[-1]$ included in Δ_+?	So that ordinal sum has a unit, making Δ_+ a strict monoidal category. In Δ alone, no object plays that role.
What does "augmented" mean here?	From Latin *augere* ("to increase"); Δ_+ is Δ augmented by an initial object.
Define ordinal sum $[m] \oplus [n]$.	$[m] \oplus [n] = [m+n+1]$, with $[-1]$ as the unit. The first $m+1$ elements come from $[m]$; the last $n+1$ come from $[n]$ shifted.
What monoid object lives in Δ_+?	$[0]$, with multiplication $[0] \oplus [0] = [1] \to [0]$ (the unique surjection σ^0) and unit $[-1] \to [0]$ (the empty function).
State Joyal's universal property.	$(\Delta_+, [0])$ is the free strict monoidal category on a monoid object: strict monoidal functors out of Δ_+ correspond bijectively to monoids in the target monoidal category.
Who is André Joyal?	Born 1943, Montréal. Foundational figure in modern category theory; introduced species (combinatorial structures), the Joyal model structure on **sSet** (Chapter 44), and many results in higher categorical algebra. Theorem 37.17 is one of his cleanest results.
What is the bar construction $B_\bullet M$?	The simplicial object $\Delta^{\mathrm{op}} \to \mathcal{V}$ with $B_n M = M^{\otimes(n+1)}$ obtained by restricting Joyal's monoid-classifying functor to $\Delta \subset \Delta_+$.
Where does the "bar" in "bar construction" come from?	The vertical bars in the original notation $[a_1 \vert a_2 \vert \cdots \vert a_n]$ for elements; Eilenberg–MacLane introduced the notation in 1953. The bars separated tensor factors.
Where does the bar construction appear in Volume IV?	In resolutions for derived functors (Ch. 42); in the universal ∞-monad (Ch. 51); in relating A_∞- to $\mathbb{E}_1$-algebras (Ch. 56); in Koszul duality (Ch. 57).
What is the standard topological n-simplex?	$\lvert\Delta^n\rvert = \{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum t_i = 1\}$: the convex hull of the standard basis vectors.
What does the functor $\lvert\Delta^\bullet\rvert : \Delta \to \mathbf{Top}$ do on morphisms?	It extends a monotone $f : [m] \to [n]$ affinely: the i-th vertex of $\lvert\Delta^m\rvert$ goes to the $f(i)$-th vertex of $\lvert\Delta^n\rvert$, interpolated linearly.
Why does Δ appear "everywhere a monoid acts"?	Because Joyal's theorem identifies Δ_+ as the universal monoid-classifying monoidal category: every monoid in any strict monoidal category factors through Δ_+.
Is Δ small? Locally finite?	Small (countably many objects) and locally finite (finite hom-sets, by the binomial count).

What is the relationship between Δ and Δ^{op}?	Opposite categories: Δ^{op} has the same objects but reverses morphisms. A presheaf $\Delta^{\mathrm{op}} \to \mathbf{Set}$ is a simplicial set (Chapter 38).
Why are the operators on a simplicial set written d_i, s_i (subscripts) instead of δ^i, σ^i (superscripts)?	To signal the direction: δ^i, σ^i live in Δ (covariant); $d_i = X(\delta^i)$ and $s_i = X(\sigma^i)$ are induced by a contravariant functor $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$, hence go "the other way."

Chapter Summary

Chapter Summary

The category Δ. Objects are the finite non-empty totally ordered sets $[n] = \{0 < 1 < \cdots < n\}$. Morphisms are monotone (order-preserving) functions. $[0]$ is terminal; the hom-sets are finite ($\binom{m+n+1}{m+1}$ monotone maps from $[m]$ to $[n]$).
Faces and degeneracies. The face map $\delta^i : [n-1] \to [n]$ is the injection skipping i; the degeneracy $\sigma^i : [n+1] \to [n]$ is the surjection doubling i. Together they generate every morphism: every $f : [m] \to [n]$ factors uniquely as $\delta^{i_1} \cdots \delta^{i_p} \sigma^{j_1} \cdots \sigma^{j_q}$ in normal form.
Cosimplicial identities. Five relations governing how δ's and σ's commute past each other or annihilate. They encode exactly the relations among monotone maps.
The augmented simplex category Δ_+. Add the empty ordered set $[-1]$ to make ordinal sum $[m] \oplus [n] = [m+n+1]$ into a strict monoidal product with unit $[-1]$.
Joyal's theorem. Δ_+ is the free strict monoidal category on a monoid object: strict monoidal functors $\Delta_+ \to \mathcal{V}$ correspond bijectively to monoid objects in $\mathcal{V}$. The bar construction $B_\bullet M$ is the resulting simplicial object.
Geometric realization (preview). The functor $\Delta \to \mathbf{Top}$ sending $[n]$ to the standard n-simplex $|\Delta^n|$ is faithful and extends along $\Delta \hookrightarrow \mathbf{sSet}$ to the geometric realization of Chapter 39.

Formal Restatement

Formal Restatement

Simplex category. $\Delta = \{[n] : n \geq 0\}$, objects $[n] = \{0 < 1 < \cdots < n\}$; morphisms are monotone functions. Δ is small, locally finite, with $[0]$ terminal.

Face map. $\delta^i : [n-1] \to [n]$, $\delta^i(j) = j$ if $j < i$, else $j + 1$. Injective; image omits i.

Degeneracy map. $\sigma^i : [n+1] \to [n]$, $\sigma^i(j) = j$ if $j \leq i$, else $j - 1$. Surjective; fibre over i has two elements.

Cosimplicial identities.

$$\begin{aligned}
\delta^j \delta^i &= \delta^i \delta^{j-1}, && i < j;\\
\sigma^j \sigma^i &= \sigma^i \sigma^{j+1}, && i \leq j;\\
\sigma^j \delta^i &= \delta^i \sigma^{j-1}, && i < j;\\
\sigma^j \delta^i &= \mathrm{id}, && i \in \{j, j+1\};\\
\sigma^j \delta^i &= \delta^{i-1} \sigma^j, && i > j+1.
\end{aligned}$$

Generators. Every $f \in \Delta([m], [n])$ has a unique normal form $f = \delta^{i_1} \cdots \delta^{i_p} \sigma^{j_1} \cdots \sigma^{j_q}$ with $i_1 > \cdots > i_p$ and $j_1 < \cdots < j_q$. Equivalently: Δ admits a unique orthogonal factorization system (surjections, injections).

Augmented simplex category. $\Delta_+ = \Delta \cup \{[-1]\}$ with $[-1] = \emptyset$ and unique empty maps $[-1] \to [n]$.

Ordinal sum. $\oplus : \Delta_+ \times \Delta_+ \to \Delta_+$, $[m] \oplus [n] = [m+n+1]$, with $[-1]$ as strict unit. Strictly associative. $(\Delta_+, \oplus, [-1])$ is a strict monoidal category.

Joyal's universal property. For any strict monoidal $(\mathcal{V}, \otimes, I)$ and monoid object (M, μ, η) in $\mathcal{V}$, there is a unique strict monoidal functor $F : \Delta_+ \to \mathcal{V}$ with $F([0]) = M$, $F(\sigma^0_{[1]}) = \mu$, $F(\eta_{[0]}) = \eta$.

Bar construction. For $M \in \mathrm{Mon}(\mathcal{V})$, the restriction of Joyal's F to Δ^{op} gives the simplicial object $B_\bullet M$, $B_n M = M^{\otimes(n+1)}$, with face/degeneracy operators induced by μ/η.

Geometric realization (preview). $|\Delta^n| = \{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum t_i = 1\}$; $|\Delta^\bullet| : \Delta \to \mathbf{Top}$ faithful. Extends along Yoneda to $|-| : \mathbf{sSet} \to \mathbf{Top}$ (Chapter 39).

Chapter 38

Simplicial Sets

Chapter Overview

With Δ in hand we can now name the combinatorial structures it indexes. A **simplicial set** is a contravariant functor $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$ — equivalently, a sequence of sets $X_0, X_1, X_2, \ldots$ together with face operators $d_i : X_n \to X_{n-1}$ and degeneracy operators $s_i : X_n \to X_{n+1}$ satisfying the *simplicial identities*, which are the cosimplicial identities of Chapter 37 with arrows reversed.

This chapter develops the category $\mathbf{sSet} = \mathbf{Set}^{\Delta^{\mathrm{op}}}$ and its core combinatorial objects: the **standard n-simplices** Δ^n, the **boundaries** $\partial\Delta^n$, and the **horns** Λ^n_k. The distinction between *inner* horns ($0 < k < n$) and *outer* horns ($k = 0$ or $k = n$) is the central technical fact organizing the next chapter and all of ∞-category theory.

Examples drive the chapter. We build the simplicial circle S^1 by gluing two endpoints of an edge, exhibit the nerve $\mathrm{N}(\mathcal{C})$ of an ordinary category (the first taste of Chapter 39), and preview the singular simplicial set $\mathrm{Sing}(X)$ of a topological space. Skeleta and coskeleta provide the truncation tools — every simplicial set is determined by its n-simplices for n large enough.

38.1 The Category of Simplicial Sets

38.1.1 Informally

Let us begin with a non-categorical description, before reorganizing into a clean functor definition.

A *simplicial set* is a recipe for a combinatorial shape built out of simplices of every dimension. To specify one, we say:

- How many 0-simplices there are (vertices).
- How many 1-simplices there are (edges), each with two endpoints (specified by face operators).
- How many 2-simplices there are (triangles), each with three edges and three vertices (again specified by face operators).
- And so on for each dimension n.

On top of the counts, we record *how* the simplices fit together:

- **Face operators** $d_i : X_n \to X_{n-1}$: tell us, for each n-simplex, what its $(n-1)$-dimensional faces are.
- **Degeneracy operators** $s_i : X_n \to X_{n+1}$: produce higher-dimensional "degenerate" simplices by repeating vertices.

The face and degeneracy operators must satisfy a tower of compatibility conditions — the *simplicial identities* — that ensure the combinatorial geometry hangs together coherently. The miracle of Chapter 37 is that all those compatibility conditions are subsumed by a single sentence: "X is a contravariant functor from Δ to $\mathbf{Set}$."

Where the Name Comes From

Simplicial is the adjectival form of *simplex* (Latin *simplex*, "simple"). A simplicial set is "a set indexed by simplices," just as a *topological* set is one indexed by topology or a *measurable* set one indexed by measurable structure. The terminology dates to Eilenberg–Zilber (1950), who isolated the category of *semisimplicial* complexes (face maps only); the modern "simplicial sets" (with both face and degeneracy operators) was standardized by Kan in the late 1950s.

Reader's Annotation

The set of all simplicial-set data is enormous: even small simplicial sets have infinitely many degeneracies (every simplex of every dimension generates an infinite tower of higher-dimensional degenerate simplices). The bookkeeping looks daunting at first. The trick is to focus on the *non-degenerate* simplices — these are finitely many for many simplicial sets of interest, and they determine everything else.

38.1.2 Expanding

A simplicial set X has *three* layers of structure, building on one another:

Layer 1: the sets X_n. For each $n \geq 0$, a set X_n — the "set of n-simplices."

Layer 2: the face operators. For each i, n with $0 \leq i \leq n$, a function $d_i : X_n \to X_{n-1}$. Reading: "the i-th face operator on X_n." Applied to an n-simplex $x \in X_n$, $d_i x \in X_{n-1}$ is the face of x opposite vertex i.

Layer 3: the degeneracy operators. For each i, n with $0 \leq i \leq n$, a function $s_i : X_n \to X_{n+1}$. Applied to $x \in X_n$, $s_i x$ is the "degenerate" $(n+1)$-simplex obtained by repeating vertex i.

The simplicial identities. The face and degeneracy operators must satisfy a small finite list of compatibility conditions — the **simplicial identities** (Theorem 38.2). These are the duals of the cosimplicial identities of Chapter 37, obtained by reversing arrow order.

Why a functor? Specifying the three layers above and verifying the identities is exactly equivalent to specifying a contravariant functor $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$, where:

- $X([n]) = X_n$.
- $X(\delta^i) = d_i$.
- $X(\sigma^i) = s_i$.

The simplicial identities become automatic consequences of X being a functor on Δ^{op} (functoriality = all relations among generators preserved). This functorial repackaging is the modern style.

Internal Dialog

Reader: Why a *contravariant* functor? Why not just go in the same direction as Δ?
Author: The contravariance lines up the geometric expectation with the algebra. In Δ, the face map $\delta^i : [n-1] \to [n]$ goes *up* in dimension. A simplicial set X assigns sets to ordinals, and a face operator $d_i : X_n \to X_{n-1}$ goes *down* in dimension (because a face has *lower* dimension than its ambient simplex). To get the directions to match — face maps go up, face operators go down — we use the opposite category.
Reader: So *cosimplicial* objects, which use Δ directly, have the dimensions going up?
Author: Yes. Cosimplicial = covariant on Δ = "dimensions go up." Simplicial = contravariant on Δ = "dimensions go down." We will see both kinds: cosimplicial spaces (Chapter 37); simplicial spaces (this chapter).

Where the Name Comes From

A **presheaf** on a category $\mathcal{C}$ is a contravariant functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$. The name *presheaf* predates the modern definition: Jean Leray (1946) introduced *faisceau* (French "sheaf"), and *préfaisceau* (presheaf) was coined for the underlying data of a sheaf *before* imposing the gluing axiom. "Pre-" from Latin *prae-* ("before, in front of") — the data exists *prior to* the sheaf condition. Simplicial sets are presheaves on Δ.

38.1.3 Formalizing

Definition 38.1 (Simplicial set)**.** A **simplicial set** is a functor

$$X : \Delta^{\mathrm{op}} \to \mathbf{Set}.$$

We write $X_n = X([n])$ and call its elements **n-simplices** of X. For each $\delta^i : [n-1] \to [n]$ in Δ, the contravariant $X(\delta^i)$ is a function $X_n \to X_{n-1}$ written d_i; for each $\sigma^i : [n+1] \to [n]$, the contravariant $X(\sigma^i)$ is a function $X_n \to X_{n+1}$ written s_i.

A **morphism** of simplicial sets is a natural transformation $X \to Y$. The category of simplicial sets is denoted $\mathbf{sSet} := [\Delta^{\mathrm{op}}, \mathbf{Set}]$.

How to Say This

sSet *(read: "simplicial sets")*; X_n *(read: "the n-simplices of X")* or *(read: "X at level n")*; d_i *(read: "i-th face of")* or *(read: "d-sub-i")*; s_i *(read: "i-th degeneracy of")* or *(read: "s-sub-i")*. The boldface **sSet** distinguishes the category from individual simplicial sets; some texts write Set_Δ or $\widehat{\Delta}$ (the latter using the standard "hat" notation for presheaf categories).

Reader's Annotation

The notation d_i and s_i uses subscripts where Chapter 37 used superscripts (δ^i, σ^i). This is intentional: the superscript variants live in Δ itself (covariant); the subscript variants are induced by the contravariant functor X, so they "go the other way." Keep the subscript/superscript convention in mind whenever you see a face or degeneracy.

Theorem 38.2 (Equivalent combinatorial description)**.** *The data of a simplicial set* $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$ *is equivalent to the data of:*

- *A sequence of sets* $\{X_n\}_{n\geq 0}$.
- *For each* $0 \leq i \leq n$, *a face operator* $d_i : X_n \to X_{n-1}$.
- *For each* $0 \leq i \leq n$, *a degeneracy operator* $s_i : X_n \to X_{n+1}$.

subject to the ***simplicial identities****:*

$$\begin{aligned}
d_i d_j &= d_{j-1} d_i && \text{if } i < j,\\
s_i s_j &= s_{j+1} s_i && \text{if } i \leq j,\\
d_i s_j &= s_{j-1} d_i && \text{if } i < j,\\
d_i s_j &= \mathrm{id} && \text{if } i = j \text{ or } i = j+1,\\
d_i s_j &= s_j d_{i-1} && \text{if } i > j+1.
\end{aligned}$$

Proof sketch. The face/degeneracy maps δ^i, σ^i in Δ generate all morphisms (Theorem 37.37.10). Hence specifying a functor X on Δ^{op} is the same as specifying its values on δ^i and σ^i, namely the d_i and s_i. The relations that hold between δ^i and σ^i (the cosimplicial identities) become, under the contravariant X, the simplicial identities above. □ □

Remark 38.3 (Dualizing the indexing)**.** The simplicial identities differ from the cosimplicial identities of Chapter 37 only by reversed arrow order: $\delta^j \delta^i$ in Δ becomes $d_i d_j$ in Δ^{op} (the order of composition flips under the duality). The shift constants ± 1 also flip sides; this is the source of the off-by-one variations between cosimplicial and simplicial identities that one sees in different references.

38.2 Standard Simplices, Boundaries, and Horns

38.2.1 Informally

There are three foundational simplicial sets that every other will be built from:

- The **standard n-simplex** Δ^n: the "free" simplicial set with one distinguished non-degenerate n-simplex. This is the Yoneda image of $[n] \in \Delta$.
- The **boundary** $\partial\Delta^n$: the standard n-simplex with its top-dimensional simplex removed, leaving only the "shell."
- The **horn** Λ^n_k: the boundary with one more face (δ^k) also removed, leaving an open "U-shaped" or "L-shaped" partial simplex.

These three sit in a sequence $\Lambda^n_k \hookrightarrow \partial\Delta^n \hookrightarrow \Delta^n$, removing one n-simplex and one face at a time. The combinatorics of filling horns and boundaries — extending a partial map back up to a full Δ^n-map — is the central technical content of ∞-category theory.

38.2.2 Formalizing

Definition 38.4 (Standard n-simplex)**.** The **standard n-simplex** is the simplicial set Δ^n representable by $[n] \in \Delta$:

$$\Delta^n \;=\; \Delta(-, [n]) : \Delta^{\mathrm{op}} \to \mathbf{Set}.$$

Its k-simplices are the monotone maps $[k] \to [n]$. The face operators act by precomposition with $\delta^i : [k-1] \to [k]$, and the degeneracy operators by precomposition with $\sigma^i : [k+1] \to [k]$.

How to Say This

Δ^n *(read: "Delta-n" or "the standard n-simplex")*; the "standard" modifier distinguishes the simplicial set Δ^n from the topological simplex $|\Delta^n|$ (§37.37.4). Context usually makes which is meant clear; when ambiguous, "combinatorial" vs. "topological" disambiguates.

Where the Name Comes From

The notation Δ^n (with superscript n, not the bracket $[n]$ of Δ itself) names the *simplicial set represented by* $[n]$ under the Yoneda embedding $\Delta \hookrightarrow \mathbf{sSet}$. The convention comes from algebraic topology, where the "standard n-simplex" was the canonical name for the geometric object $|\Delta^n|$; when the combinatorial avatar emerged, the notation Δ^n (without bars) was kept for the simplicial set.

Example 38.5 (Low-dimensional Δ^n)**.** Δ^0 is the constant simplicial set with one k-simplex in every dimension (the unique monotone $[k] \to [0]$). It is the terminal object of **sSet**. Δ^1 has two 0-simplices (the maps $[0] \to [1]$ named 0 and 1), three 1-simplices (the identity and the two constants), and a growing number of degenerate k-simplices for $k \geq 2$.

Lemma 38.6 (Yoneda for simplicial sets)**.** *For any simplicial set X, the set of morphisms $\Delta^n \to X$ in* **sSet** *is naturally bijective with X_n:*

$$\mathbf{sSet}(\Delta^n, X) \;\cong\; X_n.$$

Proof. This is the Yoneda lemma (Chapter 9 / Chapter 15) applied to the representable $\Delta^n = \Delta(-, [n])$ in $\mathbf{sSet} = [\Delta^{\mathrm{op}}, \mathbf{Set}]$. The bijection sends $f : \Delta^n \to X$ to $f_{[n]}(\mathrm{id}_{[n]}) \in X_n$. □ □

Reader's Annotation

Yoneda for **sSet** is the single most-used computational fact in the chapter. It says: *"an n-simplex of X is the same as a map $\Delta^n \to X$"*. We will switch fluidly between the two descriptions. A given n-simplex $x \in X_n$ corresponds to its "picking-out map" $\widehat{x} : \Delta^n \to X$, and the universal n-simplex of Δ^n itself is $\mathrm{id}_{[n]} \in \Delta^n_n$.

Definition 38.7 (Boundary of Δ^n)**.** The **boundary** $\partial\Delta^n \subset \Delta^n$ is the simplicial subset whose k-simplices are the *non-surjective* monotone maps $[k] \to [n]$. Equivalently, $\partial\Delta^n$ is the union of the images of the $n+1$ face maps $\delta^i : \Delta^{n-1} \to \Delta^n$,

$$\partial\Delta^n \;=\; \bigcup_{i=0}^{n} \delta^i(\Delta^{n-1}).$$

For $n = 0$, we set $\partial\Delta^0 = \emptyset$ (the empty simplicial set).

Where the Name Comes From

The symbol ∂ for "boundary" is from classical topology, where it denoted the boundary operator on chain complexes (Poincaré's work in the late 1800s). The combinatorial boundary $\partial\Delta^n$ is the simplicial set version: it removes the one top-dimensional non-degenerate simplex, leaving the union of the $n+1$ codimension-1 faces. Geometrically, $|\partial\Delta^n|$ is homeomorphic to the $(n-1)$-sphere S^{n-1}.

Definition 38.8 (Horn Λ^n_k)**.** For $0 \leq k \leq n$ with $n \geq 1$, the **k-th horn** Λ^n_k is the simplicial subset of $\partial\Delta^n$ obtained by omitting the k-th face:

$$\Lambda^n_k \;=\; \bigcup_{i \neq k} \delta^i(\Delta^{n-1}).$$

We call Λ^n_k an **inner horn** if $0 < k < n$, and an **outer horn** if $k = 0$ or $k = n$.

How to Say This

Λ^n_k *(read: "Lambda n-k" or "the k-th horn of Δ^n").* The geometric picture: Δ^n with the interior *and* the face opposite vertex k removed, leaving an open "horn" that bends toward the missing k-th face.

Where the Name Comes From

The letter Λ (Greek capital lambda, "L") was chosen by Joyal in the late 1990s for the horn simplicial set because:

- The shape Λ visually resembles the case $n = 2$: Λ^2_1 is two edges joined at the middle vertex, forming a wide "Λ" shape (or an inverted "V").
- More mundanely, Λ is one of the few unused Greek capitals close to Δ, making the typographic pairing natural.

The terminology **horn** reflects the geometric intuition: Λ^n_k is the boundary $\partial\Delta^n$ with one face removed, leaving an "open horn" that bends or curves around the missing face.

Example 38.9 (Horns of low dimension)**.** Λ^2_1 is the inner horn of the 2-simplex (triangle): two edges $0 \to 1$ and $1 \to 2$ joined at the middle vertex, missing the long edge $0 \to 2$ and the 2-cell. Λ^2_0 and Λ^2_2 are the two outer horns, each consisting of two edges joined at an endpoint (forming an "L"-shape).

Λ^3_1 and Λ^3_2 are the two inner horns of the 3-simplex (tetrahedron): three of the four triangular faces, missing either the face opposite vertex 1 or the face opposite vertex 2.

38.2.3 Reorganizing: Inner vs. outer horns

The inner/outer horn distinction is the central technical fact of ∞-category theory.

Where the Name Comes From

The distinction between **inner** and **outer** horns refers to *which vertex* is at the bend:

- *Inner* ($0 < k < n$): the missing face is opposite an "interior" vertex (not the first or last). The horn closes around an inner vertex.
- *Outer* ($k = 0$ or $k = n$): the missing face is opposite the first or last vertex. The horn is at the "outer" edge

of the index range.
The terminology dates to Boardman–Vogt (1973), who originally called the inner-horn-filling condition the "restricted Kan condition." Joyal's preferred "inner/outer" terminology stuck.

Remark 38.10 (Foreshadowing the Kan condition)**.** A simplicial set in which every inner horn $\Lambda^n_k \to X$ (with $0 < k < n$) extends to $\Delta^n \to X$ is called a **quasi-category** (Chapter 43). If every horn (inner or outer) extends, it is called a **Kan complex** (Chapter 39).

The composition law of a quasi-category arises from extending inner 2-horns: a $\Lambda^2_1 \to X$ is a pair of composable 1-simplices, and the existence of a Δ^2 filling provides their composite as the long edge $0 \to 2$. The inner-horn condition makes composition existential but non-unique; the choice of filler is the witness that an ∞-category has *coherent* rather than strict associativity.

Reader's Annotation

The mnemonic: *inner horns let you compose; outer horns let you invert.* Filling Λ^2_1 (inner) takes two composable arrows and produces their composite. Filling Λ^2_0 (outer) takes $f, g : a \to b$ with f given, and asks for an arrow $h : a \to a$ making the triangle commute — effectively asking f to have a "left inverse." Quasi-categories impose only the composition condition; Kan complexes (= ∞-groupoids) impose both.

38.2.4 Formally

Lemma 38.11 (k-simplices of $\partial\Delta^n$ and Λ^n_k)**.** *The k-simplices of $\partial\Delta^n$ are the monotone maps $[k] \to [n]$ that miss at least one element; the k-simplices of Λ^n_j are those missing at least one element other than j.*

38.3 Skeleta and Coskeleta

38.3.1 Informally

The category **sSet** has objects of *every* dimension, so it is reasonable to ask: *how much information is in the first n dimensions of a simplicial set?* Two limit-style answers:

- Take the simplicial subset generated by the $\leq n$-simplices, with higher dimensions added back only as degeneracies of those. This is the **n-skeleton** $\mathrm{sk}_n X$.
- Take the largest simplicial set whose $\leq n$-simplices agree with X's, freely adding back simplices of higher dimension subject only to boundary constraints. This is the **n-coskeleton** $\mathrm{cosk}_n X$.

Both come from the same kind of construction (Kan extension along a truncation functor), but they go in opposite directions: sk_n adds as little as possible, cosk_n adds as much as possible.

Where the Name Comes From

The terms **skeleton** and **coskeleton** are loans from algebraic topology and the theory of CW complexes. A CW complex's *n-skeleton* is the union of its cells of dimension $\leq n$ — the skeletal structure beneath the higher cells. In simplicial sets, the n-skeleton plays the same role: the part of X in dimensions $\leq n$, with everything higher derived from it.
The prefix *co-* in "coskeleton" signals *dual* (the right adjoint instead of the left), matching the long category-theoretic tradition: cofibration, colimit, cohomology — all duals of the corresponding "un-prefixed" notions.

38.3.2 Formalizing

For each $n \geq 0$ there is a full subcategory $\Delta_{\leq n} \subset \Delta$ on objects $[0], [1], \ldots, [n]$. The corresponding restriction functor on presheaves yields a chain of adjoints that decomposes **sSet** by dimension.

Definition 38.12 (n-skeleton, n-coskeleton)**.** Let $\iota_n : \Delta_{\leq n} \hookrightarrow \Delta$ be the inclusion. The restriction functor

$$\iota_n^* : \mathbf{sSet} \longrightarrow \mathbf{Set}^{\Delta^{\mathrm{op}}_{\leq n}}$$

has both a left adjoint sk_n (left Kan extension, the **n-skeleton**) and a right adjoint cosk_n (right Kan extension, the **n-coskeleton**):

$$\mathrm{sk}_n \dashv \iota_n^* \dashv \mathrm{cosk}_n.$$

A simplicial set X is **n-skeletal** if the counit $\mathrm{sk}_n \iota_n^* X \to X$ is an isomorphism; it is **n-coskeletal** if the unit $X \to \mathrm{cosk}_n \iota_n^* X$ is.

How to Say This

$\mathrm{sk}_n X$ *(read: "the n-skeleton of X")*; $\mathrm{cosk}_n X$ *(read: "the n-coskeleton of X")*. "Skeletal" and "coskeletal" are adjectives applied to simplicial sets: X is *n-skeletal* if it agrees with its own n-skeleton; X is *n-coskeletal* if it agrees with its own n-coskeleton.

Remark 38.13 (The skeleta in pictures)**.** $\mathrm{sk}_n X$ is the smallest simplicial subset of X containing all k-simplices of X for $k \le n$, with all simplices of higher dimension being degeneracies of those. $\mathrm{cosk}_n X$ is the largest simplicial set agreeing with X in dimensions $\le n$: it freely adds back k-simplices for $k > n$ subject only to the boundary constraints.

In particular: Δ^n is n-skeletal (it is generated by its unique non-degenerate n-simplex $\mathrm{id}_{[n]}$); $\partial\Delta^n$ is $(n-1)$-skeletal (it has no non-degenerate n-simplex); Λ^n_k is also $(n-1)$-skeletal.

Lemma 38.14 (Skeleton filtration)**.** *For any $X \in \mathbf{sSet}$, the canonical maps assemble into a filtration*

$$\emptyset \;=\; \mathrm{sk}_{-1}X \;\hookrightarrow\; \mathrm{sk}_0 X \;\hookrightarrow\; \mathrm{sk}_1 X \;\hookrightarrow\; \cdots \;\hookrightarrow\; X,$$

with $X = \mathrm{colim}_n \mathrm{sk}_n X$.

Reader's Annotation

The skeleton filtration is the simplicial-set version of the cell-complex filtration in topology. Inductive arguments on simplicial sets routinely proceed up this filtration: prove the statement for $\mathrm{sk}_n X$, assuming it for $\mathrm{sk}_{n-1}X$. The step usually amounts to filling in non-degenerate n-simplices one at a time, via boundary inclusions $\partial\Delta^n \hookrightarrow \Delta^n$. We will see this argument pattern repeatedly in Chapters 41 (model categories) and 43 (quasi-categories).

38.4 Examples

38.4.1 The simplicial circle

Example 38.15 (The simplicial circle S^1)**.** Let $S^1 = \Delta^1/\partial\Delta^1$ — the simplicial set obtained from Δ^1 by identifying the two endpoints. In S^1 there is one non-degenerate 0-simplex $*$ (the common image of the endpoints) and one non-degenerate 1-simplex ℓ (the image of the edge of Δ^1), both face operators of which return $*$. All higher-dimensional non-degenerate simplices are absent; the higher k-simplices are exclusively degeneracies $s_{i_1}\cdots s_{i_p}\ell$ or $s_{i_1}\cdots s_{i_p}*$.

Geometrically, S^1's realization (Chapter 39) is the topological circle.

38.4.2 The nerve of a category (preview)

Where the Name Comes From

The **nerve** of a category was introduced by Alexander Grothendieck in correspondence around 1960 and developed in detail by Graeme Segal (1968) in his work on classifying spaces of categories and topological operads. The word "nerve" is a biological metaphor: just as the nervous system of an organism is the structural backbone that transmits information throughout the body, the nerve of a category is the simplicial structure that "transmits" the categorical information into the world of simplicial sets. The metaphor predates Grothendieck — Hilbert used "nerve of a covering" (Nerv einer Bedeckung) in topology around 1932 — but Grothendieck's categorical nerve is the modern usage.

Example 38.16 (Nerve of a category)**.** For an ordinary category $\mathcal{C}$, the **nerve** $\mathrm{N}(\mathcal{C}) \in \mathbf{sSet}$ has:

- $\mathrm{N}(\mathcal{C})_0 = \mathrm{Ob}(\mathcal{C})$;
- $\mathrm{N}(\mathcal{C})_1 = \mathrm{Mor}(\mathcal{C})$;
- $\mathrm{N}(\mathcal{C})_n = \{\, c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} c_n : n\text{-fold composable chain}\,\}$.

Face operators compose or drop end morphisms; degeneracy operators insert identities.

Remark 38.17 (A preview of the next chapter)**.** The nerve construction defines a functor $\mathrm{N} : \mathbf{Cat} \to \mathbf{sSet}$ that is fully faithful (Chapter 39) and whose image consists of exactly those simplicial sets in which every inner horn extends *uniquely*. That uniqueness — versus the merely-existence-of-a-filler of quasi-categories — is what distinguishes ordinary 1-categories from ∞-categories within **sSet**.

38.4.3 The singular complex of a space (preview)

Where the Name Comes From

The **singular** simplicial set, $\mathrm{Sing}(X)$, was introduced by Samuel Eilenberg (1944) as the foundation of singular homology. The adjective "singular" contrasts with "smooth" or "polyhedral": a singular simplex in X is merely a continuous map $|\Delta^n| \to X$, which can land on or pass through "singular points" (corners, edges, non-smooth locations) of X. Earlier theories required smoother conditions; Eilenberg's insight was that allowing arbitrary continuous maps gave a homology theory defined for *all* topological spaces.

Example 38.18 (Singular complex $\mathrm{Sing}(X)$)**.** For a topological space X, the **singular simplicial set** is

$$\mathrm{Sing}(X)_n \;=\; \mathbf{Top}\big(|\Delta^n|, X\big),$$

the set of continuous maps from the standard topological n-simplex to X. Face and degeneracy operators come from precomposition with the cosimplicial space $|\Delta^\bullet|$ of Chapter 37.

Remark 38.19 (Sing is a Kan complex)**.** For any space X, every horn $\Lambda^n_k \to \mathrm{Sing}(X)$ extends to $\Delta^n \to \mathrm{Sing}(X)$ — both inner and outer horns. This is the classical **Kan condition**, and the proof uses only the contractibility of $|\Lambda^n_k|$ inside $|\Delta^n|$. Kan complexes will be our model for "spaces up to homotopy" in Chapter 39.

Reader's Annotation

The pair (nerve, singular complex) gives two contrasting examples of simplicial sets:

- Nerve of a category: *rigid*; inner horns extend uniquely; outer horns might not extend at all.
- Singular complex of a space: *flexible*; *all* horns extend; the extensions are non-unique (every continuous deformation gives one).

The two examples bookend the spectrum: Chapter 39 develops both, and Chapter 43 introduces quasi-categories as the in-between case (inner horns extend, but possibly not uniquely).

38.5 Exercises

What is a simplicial set?	A functor $X : \Delta^{\mathrm{op}} \to \mathbf{Set}$; equivalently, a sequence of sets X_n with face and degeneracy operators satisfying the simplicial identities.
What is X_n?	The set of n-simplices of X, i.e., $X([n])$.
Why is the indexing Δ^{op} and not Δ?	So that face operators decrease dimension: $d_i : X_n \to X_{n-1}$. A covariant functor $\Delta \to \mathbf{Set}$ is a *cosimplicial* set, where "face" goes the other way.
Where does "presheaf" come from?	Leray (1946) coined *préfaisceau* as the data of a sheaf *before* (Latin *prae-*) imposing the gluing axiom. A simplicial set is a presheaf on Δ.
What are the face operators d_i?	For each $0 \le i \le n$, $d_i = X(\delta^i) : X_n \to X_{n-1}$. They project an n-simplex onto its i-th $(n-1)$-face.
Why subscripts d_i, s_i rather than superscripts?	To signal direction: δ^i, σ^i live in Δ (covariant); d_i, s_i are induced by the contravariant functor X and thus "go the other way."
State one simplicial identity.	$d_i s_i = \mathrm{id}$ (and $d_{i+1} s_i = \mathrm{id}$): degenerate then face-extract returns the original.
What is the standard n-simplex Δ^n?	The representable simplicial set $\Delta(-, [n])$. Its k-simplices are the monotone maps $[k] \to [n]$.
State Yoneda for simplicial sets.	$\mathbf{sSet}(\Delta^n, X) \cong X_n$, natural in both variables. An n-simplex of X is a map $\Delta^n \to X$.
What is the boundary $\partial\Delta^n$?	The simplicial subset of Δ^n consisting of non-surjective monotone maps into $[n]$. Equivalently, the union of the $n+1$ face inclusions $\Delta^{n-1} \hookrightarrow \Delta^n$.
Why the symbol ∂?	From topology: ∂ is the classical boundary operator on chains (Poincaré, late 1800s). The simplicial $\partial\Delta^n$ is the combinatorial version of the topological boundary sphere S^{n-1}.
What is the horn Λ^n_k?	The simplicial subset of $\partial\Delta^n$ obtained by removing the k-th face: $\Lambda^n_k = \bigcup_{i \ne k} \delta^i(\Delta^{n-1})$.
Why the letter Λ for horns?	Joyal's choice. The Greek capital lambda visually resembles Λ^2_1 — two edges joined at a middle vertex form an inverted "V" shape, like the letter Λ. The term *horn* reflects the geometric "open" shape with one face removed.

Inner vs. outer horns?	Inner: $0 < k < n$ (the missing face is opposite an interior vertex). Outer: $k = 0$ or $k = n$ (opposite the first or last vertex). The distinction is central to the Kan/quasi-category dichotomy.
Picture Λ^2_1 and Λ^2_0.	Λ^2_1: two edges $0 \to 1$ and $1 \to 2$ joined at the middle (inner). Λ^2_0: two edges sharing endpoint 0 (outer, "L"-shape).
Inner horn vs. outer horn mnemonic?	*Inner horns let you compose; outer horns let you invert.* Filling Λ^2_1 produces a composite; filling Λ^2_0 produces a left inverse.
When does a horn fill?	Inner horn fills always: quasi-categories. All horns fill: Kan complexes.
What is the n-skeleton $\mathrm{sk}_n X$?	The smallest simplicial subset of X generated by simplices of dimension $\leq n$. Equivalently: the left Kan extension along $\Delta_{\leq n} \hookrightarrow \Delta$ applied to $X\|_{\Delta_{\leq n}}$.
What is the n-coskeleton $\mathrm{cosk}_n X$?	The right Kan extension: the largest simplicial set agreeing with X in dimensions $\leq n$. Higher simplices are freely filled subject to boundary constraints.
Where does "skeleton" come from?	From CW-complex topology: the n-skeleton of a CW complex is the union of its cells of dimension $\leq n$ — the structural backbone beneath higher cells. The simplicial-set version plays the same role.
Why are $\Delta^n, \partial\Delta^n, \Lambda^n_k$ all n- or $(n-1)$-skeletal?	Because they are built from finitely many low-dimensional simplices and their degenerations: Δ^n is generated by one non-degenerate n-simplex (so n-skeletal); $\partial\Delta^n$ and Λ^n_k have no non-degenerate n-simplex (so $(n-1)$-skeletal).
Describe the simplicial circle S^1.	$S^1 = \Delta^1/\partial\Delta^1$. One non-degenerate 0-simplex and one non-degenerate 1-simplex, both face operators of the latter returning the former.
What are the n-simplices of $\mathrm{N}(\mathcal{C})$?	n-fold composable chains $c_0 \to c_1 \to \cdots \to c_n$ of morphisms in $\mathcal{C}$.
Where does "nerve" come from?	Biological metaphor (Grothendieck $\sim$ 1960; Segal 1968). The nervous system transmits information through an organism; the nerve of a category transmits its categorical information into **sSet**.
Faces of an n-simplex in the nerve?	d_0 drops the leftmost morphism (and object); d_n drops the rightmost; for $0 < i < n$, d_i composes adjacent morphisms $f_{i+1} \circ f_i$ to shorten the chain.
What is the singular simplicial set $\mathrm{Sing}(X)$?	$\mathrm{Sing}(X)_n = \mathbf{Top}(\|\Delta^n\|, X)$: continuous maps from the topological n-simplex into X.
Where does "singular" come from?	Eilenberg (1944): allowed arbitrary continuous maps from $\|\Delta^n\|$ into X, not just smooth or polyhedral ones. "Singular" contrasts with "smooth" — the simplex may pass through singular points of X.
Is $\mathrm{Sing}(X)$ a Kan complex?	Yes — every horn $\Lambda^n_k \to \mathrm{Sing}(X)$ extends to $\Delta^n \to \mathrm{Sing}(X)$, because $\|\Lambda^n_k\|$ is a deformation retract of $\|\Delta^n\|$.
Why does Yoneda for **sSet** pay off practically?	It identifies maps out of Δ^n with n-simplices of the target — converting universal mapping properties into concrete simplex-counting statements.
What is $\mathrm{sk}_0 X$?	The discrete simplicial set on X_0: just the 0-simplices of X, with all higher simplices degenerate.

Is **sSet** complete and cocomplete?	Yes — as a presheaf category $[\Delta^{\mathrm{op}}, \mathbf{Set}]$, it inherits all (co)limits from **Set**, computed pointwise (Chapter 6).

Chapter Summary

Chapter Summary

Simplicial sets. $\mathbf{sSet} = [\Delta^{\mathrm{op}}, \mathbf{Set}]$. A simplicial set is a sequence of sets X_n together with face operators d_i and degeneracy operators s_i satisfying the simplicial identities (the dual of the cosimplicial identities of Chapter 37).
Standard simplices, boundaries, horns. Δ^n is the representable simplicial set; an n-simplex of X is a map $\Delta^n \to X$ (Yoneda). $\partial\Delta^n$ is the boundary (non-surjective monotone maps into $[n]$); Λ^n_k is the horn omitting the k-th face. *Inner* horns have $0 < k < n$; *outer* horns have $k = 0$ or $k = n$. Mnemonic: *inner horns let you compose; outer horns let you invert.*
Skeleta and coskeleta. $\mathrm{sk}_n \dashv \iota_n^* \dashv \mathrm{cosk}_n$ gives the canonical filtration $\emptyset \subset \mathrm{sk}_0 X \subset \mathrm{sk}_1 X \subset \cdots$, with X as the colimit. Δ^n is n-skeletal; $\partial\Delta^n$ and Λ^n_k are $(n-1)$-skeletal. Borrowed from CW-complex topology.
Key examples. The simplicial circle $S^1 = \Delta^1/\partial\Delta^1$; the nerve $\mathrm{N}(\mathcal{C})$ of a category (Grothendieck/Segal, $\sim$ 1968); the singular simplicial set $\mathrm{Sing}(X)$ of a space (Eilenberg, 1944). The latter is a Kan complex — all horns fill, not merely the inner ones. The distinction foreshadows quasi-categories (Chapter 43).

Formal Restatement

Formal Restatement

Category. $\mathbf{sSet} = [\Delta^{\mathrm{op}}, \mathbf{Set}]$; complete, cocomplete, locally small.

Simplicial identities. For $d_i = X(\delta^i)$, $s_i = X(\sigma^i)$:

$$\begin{aligned}
d_i d_j &= d_{j-1} d_i, && i < j;\\
s_i s_j &= s_{j+1} s_i, && i \le j;\\
d_i s_j &= s_{j-1} d_i, && i < j;\\
d_i s_j &= \mathrm{id}, && i \in \{j, j+1\};\\
d_i s_j &= s_j d_{i-1}, && i > j+1.
\end{aligned}$$

Yoneda for sSet. $\mathbf{sSet}(\Delta^n, X) \cong X_n$; $\Delta^n_k = \Delta([k],[n])$.

Boundary and horn.

$$\partial\Delta^n = \bigcup_{i=0}^{n} \delta^i(\Delta^{n-1}), \quad \Lambda^n_k = \bigcup_{i \ne k} \delta^i(\Delta^{n-1}).$$

Inner: $0 < k < n$; outer: $k \in \{0, n\}$.

Skeleta. $\mathrm{sk}_n \dashv \iota_n^* \dashv \mathrm{cosk}_n$ with $\iota_n : \Delta_{\le n} \hookrightarrow \Delta$. Canonical filtration $\mathrm{sk}_n X \hookrightarrow X$, $X = \mathrm{colim}_n \mathrm{sk}_n X$.

Key simplicial sets.
- Δ^n — representable, n-skeletal; terminal of **sSet** is Δ^0.
- $\partial\Delta^n$ — boundary, $(n-1)$-skeletal.
- Λ^n_k — horn, $(n-1)$-skeletal.
- $\mathrm{N}(\mathcal{C})$ — nerve; $\mathrm{N}(\mathcal{C})_n = n$-fold composable chains.
- $\mathrm{Sing}(X)$ — singular; $\mathrm{Sing}(X)_n = \mathbf{Top}(|\Delta^n|, X)$; always a Kan complex.

Chapter 39

Geometric Realization, the Nerve, and Kan Complexes

Chapter Overview

Two adjunctions package the relationship between simplicial sets and the two "natural" categories they look like: topological spaces and ordinary categories.

The first is **geometric realization** $|-| : \mathbf{sSet} \to \mathbf{Top}$, left adjoint to the singular complex functor Sing of Chapter 38. It is the left Kan extension of the cosimplicial space $|\Delta^\bullet|$ along the Yoneda embedding — concretely, the coend $|X| = \int^{[n]} X_n \cdot |\Delta^n|$. Under this adjunction, simplicial sets serve as a combinatorial model for topological spaces up to weak equivalence.

The second is the **nerve** $\mathrm{N} : \mathbf{Cat} \to \mathbf{sSet}$, right adjoint to the fundamental category functor $\tau_1 : \mathbf{sSet} \to \mathbf{Cat}$. The nerve is fully faithful, and a theorem of Grothendieck identifies its image: the simplicial sets in which every *inner* horn has a *unique* filler. Relaxing "unique" to "exists" gives quasi-categories (Chapter 43); insisting on fillers for *all* horns gives Kan complexes — the topic of the final section.

Kan complexes are the simplicial-set avatars of ∞-groupoids: spaces whose homotopical content is fully captured by simplicial combinatorics. By the end of the chapter we have the picture

$$\text{ordinary categories} \hookleftarrow \text{simplicial sets} \hookrightarrow \text{Kan complexes} \rightsquigarrow \text{topological spaces},$$

which Chapters 40–42 refine via model-categorical machinery.

39.1 Geometric Realization

39.1.1 Informally

Every simplicial set X has a topological realization $|X|$ built by gluing one copy of the standard topological n-simplex $|\Delta^n|$ for each non-degenerate n-simplex $x \in X_n$, along the gluing prescribed by X's face operators. Degenerate simplices contribute nothing new: they are already present as faces of lower non-degenerate simplices.

In the other direction (Chapter 38), every topological space T has a singular simplicial set $\mathrm{Sing}(T)$ whose n-simplices are continuous maps $|\Delta^n| \to T$. The two operations are adjoint — and not by accident: realization is the left Kan extension along the Yoneda embedding $\Delta \hookrightarrow \mathbf{sSet}$ of the cosimplicial space of standard simplices.

Reader's Annotation

A useful way to picture realization: imagine the simplicial set as a "set of instructions" for assembling a topological space.

- "Take this many vertices."
- "Glue these vertices into edges according to the face data."
- "Glue these edges into triangles, these triangles into tetrahedra, etc."

The realization is the topological space the instructions describe. Crucially, only *non-degenerate* simplices add new content; degeneracies are bookkeeping.

39.1.2 Expanding

The coend formula $|X| = \int^{[n]} X_n \cdot |\Delta^n|$ hides a familiar construction. Unrolling:

- For each n, take a disjoint union of $|X_n|$-many copies of $|\Delta^n|$: this is $X_n \cdot |\Delta^n|$ (the *copower*).
- Glue along face and degeneracy maps: identify a face of an n-simplex copy with the corresponding $(n-1)$-simplex copy, and collapse degenerate copies onto their underlying non-degenerate ones.

The coend is the universal way to perform all these identifications at once.

Where the Name Comes From

The **coend** (notation $\int^{[n]} \cdots$) is a categorical construction introduced by Yoneda and developed by Day–Kelly (1969). The *co-* prefix marks it as the dual of the *end* ($\int_{[n]}$, a limit-style construction). The integral notation is a metaphor: just as $\int f$ accumulates values of f across the domain, $\int^{[n]} F$ accumulates the values of $F([n],[n])$ in a categorically-correct way. The notation has stuck because the analogies with classical integration (Fubini's theorem, etc.) carry over surprisingly well to the categorical setting.

39.1.3 Formalizing

Definition 39.1 (Geometric realization)**.** The **geometric realization** of a simplicial set X is the topological space

$$|X| \;=\; \int^{[n]\in\Delta} X_n \cdot |\Delta^n|,$$

where $S \cdot T = S \times T$ for a set S and space T (the copower in **Top**). This defines a functor $|{-}| : \mathbf{sSet} \to \mathbf{Top}$.

How to Say This

$|X|$ *(read: "the realization of X")*; the coend formula *(read: "integrate over Δ, copowering X_n by the topological n-simplex")*. The integral notation reflects that the construction identifies $X_m \cdot |\Delta^m|$ and $X_n \cdot |\Delta^n|$ along all face/degeneracy maps (Chapter 26 coend calculus).

Theorem 39.2 (Realization–singular adjunction)**.** *The realization functor is left adjoint to the singular simplicial set functor:*

$$|{-}| \;\dashv\; \mathrm{Sing} : \mathbf{Top} \to \mathbf{sSet}, \qquad \mathbf{Top}(|X|, T) \;\cong\; \mathbf{sSet}(X, \mathrm{Sing}\, T).$$

Proof sketch. Both sides are continuous in X. By the coend formula and the hom-coend adjunction of Chapter 26,

$$\mathbf{Top}\Big(\int^{[n]} X_n \cdot |\Delta^n|, T\Big) \;\cong\; \int_{[n]} \mathbf{Top}\big(X_n \cdot |\Delta^n|, T\big) \;\cong\; \int_{[n]} \mathbf{Set}\Big(X_n, \mathbf{Top}(|\Delta^n|, T)\Big).$$

The rightmost end is $\mathrm{Nat}(X_\bullet, \mathrm{Sing}(T)_\bullet) = \mathbf{sSet}(X, \mathrm{Sing}\, T)$. □ □

Remark 39.3 (Realization is a left Kan extension)**.** Equivalently: $|{-}|$ is the left Kan extension of $|\Delta^\bullet| : \Delta \to \mathbf{Top}$ along the Yoneda embedding $y : \Delta \to \mathbf{sSet}$,

$$|{-}| \;=\; \mathrm{Lan}_y |\Delta^\bullet|.$$

This is the general construction: a cosimplicial object in any cocomplete category $\mathcal{D}$ extends along Yoneda to a colimit-preserving functor $\mathbf{sSet} \to \mathcal{D}$ (Chapter 11/18 Kan extension theory).

Reader's Annotation

The adjunction $|{-}| \dashv \mathrm{Sing}$ is the precise bridge between combinatorial and topological category theory. The unit and counit are not isomorphisms in general — there are simplicial sets that don't recover from their realization, and topological spaces that don't recover from their singular complex on the nose — but they *are* weak equivalences after suitable replacement (Chapter 42's homotopy hypothesis).

39.1.4 Formally

Example 39.4 (Realization in low dimension)**.** $|\Delta^n|$ in the abstract is the topological n-simplex already encountered (§37.37.4); $|\partial\Delta^n|$ is the topological boundary sphere S^{n-1} (up to homeomorphism); $|\Lambda^n_k|$ is a contractible piece of the boundary obtained by removing one $(n-1)$-face.

The simplicial circle $S^1 = \Delta^1/\partial\Delta^1$ (Example 38.38.15) realizes to the topological circle.

Remark 39.5 (A common topological subtlety)**.** The codomain **Top** in Theorem 39.2 should strictly be a category of *nice* spaces — compactly generated weak Hausdorff is standard — to ensure colimits behave well. In Chapter 41 we discuss the model-category framework that makes this precise; for our purposes any cartesian-closed category of spaces with the standard simplices will do.

39.2 The Nerve of a Category

39.2.1 Informally

The second adjunction lives between simplicial sets and ordinary categories. Every ordinary category $\mathcal{C}$ has a simplicial-set shadow — its nerve $\mathrm{N}(\mathcal{C})$ — whose n-simplices are the length-n composable chains in $\mathcal{C}$:

$$c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} c_n.$$

The face operators perform either dropping (at the ends) or composing (in the middle); the degeneracy operators insert identity arrows.

The construction is fully faithful: $\mathbf{Cat} \hookrightarrow \mathbf{sSet}$ via the nerve. Grothendieck's theorem identifies the image *exactly*: a simplicial set is in the image of the nerve iff every inner horn extends *uniquely*. Relax "uniquely" to "at least one" and the resulting class is the quasi-categories.

39.2.2 Formalizing

A finite ordinal $[n] = \{0 < 1 < \cdots < n\}$ is itself an ordinary category: objects are the elements; there is a unique morphism $i \to j$ whenever $i \leq j$, and composition is forced. This gives a functor

$$[-] : \Delta \longrightarrow \mathbf{Cat}, \qquad [n] \longmapsto [n] \text{ as a poset}.$$

Definition 39.6 (Nerve of a category)**.** For a small category $\mathcal{C}$, the **nerve** $\mathrm{N}(\mathcal{C})$ is the simplicial set

$$\mathrm{N}(\mathcal{C})_n \;=\; \mathbf{Cat}\big([n], \mathcal{C}\big).$$

An n-simplex is a functor $[n] \to \mathcal{C}$, equivalently an n-fold composable chain $c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} c_n$ of morphisms in $\mathcal{C}$.

How to Say This

$\mathrm{N}(\mathcal{C})$ *(read: "the nerve of $\mathcal{C}$")*; the chain description *(read: "length-n string of composable arrows in $\mathcal{C}$")*. Some texts write $\mathrm{N}_\bullet(\mathcal{C})$ or $\mathrm{Ner}(\mathcal{C})$; the bullet $\bullet$ emphasizes simpliciality but is omitted when context is clear.

Where the Name Comes From

Nerve (recall from Chapter 38): biological metaphor introduced by Grothendieck around 1960 and developed by Graeme Segal (1968). Just as a nervous system transmits information through an organism, the nerve of a category transmits its categorical content into the world of simplicial sets — letting the tools of simplicial homotopy theory operate on categorical structure.

Remark 39.7 (Face and degeneracy on the nerve)**.** For a chain $c_0 \to c_1 \to \cdots \to c_n$:

- d_0 drops c_0 (and the morphism f_1): gives $c_1 \to \cdots \to c_n$.
- d_n drops c_n (and f_n): gives $c_0 \to \cdots \to c_{n-1}$.
- For $0 < i < n$, d_i composes the two adjacent morphisms $f_{i+1} \circ f_i$ to give a length-$(n-1)$ chain.
- s_i inserts an identity morphism after position i, lengthening the chain.

Reader's Annotation

The asymmetry between d_0 / d_n (which drop) and d_i for $0 < i < n$ (which compose) is the geometric reason that *inner horns* are special. Filling an inner horn Λ^n_k $(0 < k < n)$ asks: "given all the faces except d_k, can we recover d_k and the whole n-simplex?" For nerves, d_k is forced — it is the composite of two adjacent arrows. For an outer horn ($k = 0$ or $k = n$), d_k is a *drop*, which is not forced by the other faces (you would need an inverse of a morphism, which doesn't generally exist).

Theorem 39.8 (Fundamental category–nerve adjunction)**.** *The nerve functor* $\mathrm{N} : \mathbf{Cat} \to \mathbf{sSet}$ *has a left adjoint* $\tau_1 : \mathbf{sSet} \to \mathbf{Cat}$*, the* ***fundamental category*** *(or* homotopy category*) functor:*

$$\tau_1 \dashv \mathrm{N}, \qquad \mathbf{Cat}(\tau_1 X, \mathcal{C}) \;\cong\; \mathbf{sSet}(X, \mathrm{N}(\mathcal{C})).$$

$\tau_1 X$ has objects X_0 and is freely generated by the 1-simplices of X subject to the relations imposed by the 2-simplices (each 2-simplex says "$f_2 \circ f_1 = f_3$" for its three boundary 1-simplices).

Where the Name Comes From

The notation τ_1 comes from *truncation*: τ_1 takes a simplicial set (which has data in every dimension) and produces a 1-categorical *truncation* of it — discarding the higher homotopical information and keeping only the categorical part. The subscript 1 marks the target dimension (an ordinary 1-category). Generalizations τ_n produce n-categorical truncations (Chapter 44).

Proof sketch. $\tau_1 = \mathrm{Lan}_y[-]$, the left Kan extension along Yoneda of the cosimplicial category $[-] : \Delta \to \mathbf{Cat}$ — exactly parallel to the realization construction of Remark 39.3. Adjointness follows from the Yoneda adjunction for Kan extensions. □ □

39.2.3 Reorganizing: characterizing the nerve

Where the Name Comes From

The **characterization of nerves via unique inner-horn filling** is attributed to *Alexander Grothendieck* (1928–2014), one of the most influential mathematicians of the 20th century. The result appears implicitly in Grothendieck's work on classifying spaces and explicitly in subsequent literature on Segal's nerve construction. Grothendieck's broader project of reorganizing algebraic geometry through category theory pervades Volume IV: the Grothendieck construction (Chapter 46 fibrations), Grothendieck universes, Grothendieck topologies (Chapter 58 DAG), and the homotopy hypothesis (§39.3.3 below) are all his.

Theorem 39.9 (Grothendieck's characterization of the nerve)**.** *The nerve functor* $\mathrm{N} : \mathbf{Cat} \to \mathbf{sSet}$ *is fully faithful. Its essential image consists of the simplicial sets X in which every inner horn admits a* unique *filler: for each* $0 < k < n$ *and each map* $\Lambda^n_k \to X$*, there is exactly one* $\Delta^n \to X$ *extending it.*

Proof sketch. Fully faithful. A natural transformation $\mathrm{N}(F) \to \mathrm{N}(G)$ is determined by its action on 0- and 1-simplices — i.e., by an assignment to objects and morphisms compatible with identities and composition — and that is the same as a functor $F \to G$.

Image. If $X = \mathrm{N}(\mathcal{C})$, an inner horn $\Lambda^n_k \to X$ for $0 < k < n$ supplies a chain of $n-1$ morphisms with a "gap" at position k; the unique filler is forced because composition in $\mathcal{C}$ is uniquely defined. Conversely, if all inner horns fill uniquely, the unique-composite $\Lambda^2_1 \to X$ extension defines a composition law on X_1 that is associative (forced by Λ^3_1 and Λ^3_2 fillings) and unital (forced by degeneracies), making X the nerve of a category. □ □

Remark 39.10 (The quasi-category relaxation)**.** Theorem 39.9 is the gateway to ∞-category theory. *Drop "unique"* from the characterization — require inner horns to admit at least one filler, not necessarily unique — and the resulting class of simplicial sets is the **quasi-categories** (Chapter 43). Composition becomes existential (a composite exists) rather than constructive (a composite is given); associativity holds up to higher coherent homotopies rather than on the nose; and the entire structure of ordinary category theory generalizes to a "coherent" version.

39.3 Kan Complexes

39.3.1 Informally

A Kan complex is a simplicial set in which *every* horn — inner or outer — admits a filler. The condition is symmetric: morphisms can be composed both forward (inner-horn filling, as for the nerve) and "inverted" or "transported backward" (outer-horn filling).

In an ordinary category nerve, outer horns do *not* fill in general: $\Lambda^2_0 \to \mathrm{N}(\mathcal{C})$ is two arrows out of a common source, and a Δ^2 filler would force the existence of an arrow relating the two targets — which exists only if the category has limits/colimits or all morphisms are isomorphisms. Outer horns force invertibility; that is why Kan complexes are sometimes called **∞-groupoids**.

Where the Name Comes From

Kan complex is named for *Daniel M. Kan* (1927–2013), Israeli-American mathematician primarily at MIT. Kan introduced the horn-filling condition in his 1958 paper "On c.s.s. complexes" (*c.s.s.* = "complete semi-simplicial," an older term that has since been superseded by "simplicial set"). Kan's other major contributions include the theory of Kan extensions (Chapters 11, 18, 31, 51), adjoint functors (Chapter 7), and the Kan model structure on **sSet** (Chapter 41).

39.3.2 Formalizing

Definition 39.11 (Kan complex)**.** A simplicial set X is a **Kan complex** if every horn $\Lambda^n_k \to X$ (for $n \geq 1$ and $0 \leq k \leq n$) extends to $\Delta^n \to X$.

How to Say This

"Kan complex" *(read: "Kan complex" or "Kan simplicial set")*, after Daniel Kan. The term "fibrant simplicial set" is also standard once Chapter 41's model structure is in place — Kan complexes are exactly the fibrant objects of the standard (Kan–Quillen) model structure on **sSet**.

Where the Name Comes From

The term **∞-groupoid** dates to Grothendieck's *Pursuing Stacks* manuscript (1983, unpublished but widely circulated). The word **groupoid** itself was coined by Heinrich Brandt (1927) for a generalization of groups in which composition is only partially defined. A groupoid has all morphisms invertible; an ∞-groupoid has "all morphisms invertible at every level" — including the homotopies between morphisms, the homotopies between those homotopies, and so on ad infinitum. Kan complexes are the simplicial-set realization of this "invertibility-at-every-level" picture.

Reader's Annotation

The slogan "Kan complex = ∞-groupoid" is one of the most important slogans of higher category theory. It says: the simplicial combinatorial structure where "all horns fill" is exactly the homotopy-theoretic structure where "everything is invertible." The outer-horn condition encodes the invertibility; the inner-horn condition encodes the composition. Removing only the outer-horn requirement gives quasi-categories — ∞-categories where invertibility is not forced.

Theorem 39.12 (Sing(T) is a Kan complex)**.** *For any topological space T, the singular simplicial set* Sing(T) *is a Kan complex.*

Proof sketch. A horn $\Lambda^n_k \to \mathrm{Sing}(T)$ corresponds, by adjunction (Theorem 39.2) and the unit on representables, to a continuous map $|\Lambda^n_k| \to T$. The space $|\Lambda^n_k|$ is a deformation retract of $|\Delta^n|$ — collapsing the missing face inward gives the retraction — and any retract extends maps along itself, so $|\Lambda^n_k| \to T$ extends to $|\Delta^n| \to T$. Apply the adjunction in reverse. □ □

39.3.3 The homotopy hypothesis

Where the Name Comes From

The **homotopy hypothesis** was formulated by *Alexander Grothendieck* in his 1983 manuscript *Pursuing Stacks* (*À la poursuite des champs*). It is the thesis that the homotopy theory of topological spaces and the homotopy theory of ∞-groupoids are *the same thing* — i.e., ∞-groupoids are a complete model for homotopy types of spaces. The hypothesis ramifies through all of modern algebraic topology and provides the philosophical foundation for homotopy type theory (Chapter 59).

Remark 39.13 (The homotopy hypothesis: a preview)**.** The functors $|-|$ and Sing are not equivalences of categories: they restrict to a *Quillen equivalence* between the model category of simplicial sets (Kan model structure, Chapter 41) and the model category of topological spaces (Quillen model structure). Localizing both sides at weak equivalences produces equivalent homotopy categories.

Within **sSet**, the full subcategory of Kan complexes models the homotopy category of spaces. This statement — that "∞-groupoids = homotopy types" — is Grothendieck's **homotopy hypothesis**, formalized within the Kan-complex framework.

39.3.4 Formally: nerve vs. Kan vs. quasi-category

Remark 39.14 (The three horn conditions)**.** The three horn conditions partition the simplicial-set landscape:

- **Nerve of a category**: every inner horn fills *uniquely*.
- **Quasi-category**: every inner horn fills (existence only).
- **Kan complex**: every horn — inner *or outer* — fills.

The nerve condition is sharpest; the quasi-category condition relaxes uniqueness; the Kan condition adds outer-horn filling. Every nerve of a groupoid is a Kan complex; every Kan complex is a quasi-category; and both ordinary categories and topological spaces embed into the simplicial landscape under these descriptions.

Reader's Annotation

A useful diagram of the simplicial-set landscape:
- *Nerves of categories* (inner horns fill *uniquely*) — the rigid case.
- *Quasi-categories* (inner horns fill, possibly non-uniquely) — the ∞-category case (Ch. 43).
- *Kan complexes* (all horns fill, possibly non-uniquely) — the ∞-groupoid case.
- Intersection: *nerves of groupoids* — both rigid and invertible.

The whole of ∞-category theory lives in the space between these conditions, and the rest of Volume IV explores it.

Example 39.15 (Nerves and groupoids)**.** If $\mathcal{G}$ is a groupoid (every morphism invertible), then N($\mathcal{G}$) is a Kan complex: outer horns fill because the required morphism exists and is unique. Conversely, a Kan complex in which every inner horn fills *uniquely* is the nerve of a groupoid. The fully faithful embedding **Grpd** $\hookrightarrow$ **sSet** lands in the intersection "nerve $\cap$ Kan."

39.4 Exercises

Define the geometric realization $\lvert X\rvert$.	$\lvert X\rvert = \int^{[n]} X_n \cdot \lvert\Delta^n\rvert$ — the coend that glues X_n copies of the topological n-simplex along all face and degeneracy identifications.
What adjunction does realization fit into?	$\lvert-\rvert \dashv \mathrm{Sing}$: $\mathbf{Top}(\lvert X\rvert, T) \cong \mathbf{sSet}(X, \mathrm{Sing}\, T)$.
Realization as a Kan extension?	$\lvert-\rvert = \mathrm{Lan}_y \lvert\Delta^\bullet\rvert$, the left Kan extension along $y : \Delta \hookrightarrow \mathbf{sSet}$ of the cosimplicial space of standard simplices.
Where does the coend notation $\int^{[n]}$ come from?	Yoneda and Day–Kelly (1969). The *co-* prefix marks it as the dual of the end $\int_{[n]}$; the integral metaphor captures categorical-accumulation analogous to classical integration.
What is $\lvert\partial\Delta^n\rvert$?	Homeomorphic to the $(n-1)$-sphere S^{n-1}.
What is $\lvert\Lambda^n_k\rvert$?	A deformation retract of $\lvert\Delta^n\rvert$ — geometrically, the n-simplex with one $(n-1)$-face removed; contractible.
Define the nerve $\mathrm{N}(\mathcal{C})$.	$\mathrm{N}(\mathcal{C})_n = \mathbf{Cat}([n], \mathcal{C})$, where $[n]$ is the ordinal $0 \to 1 \to \cdots \to n$ regarded as a category. Equivalently, n-fold composable chains in $\mathcal{C}$.
Why "nerve"?	Biological metaphor: as a nervous system transmits information through an organism, the nerve of a category transmits its categorical content into **sSet**, letting simplicial-homotopy tools operate on it.
What does d_i do on a chain $c_0 \to \cdots \to c_n$ in the nerve?	d_0 drops the leftmost arrow; d_n drops the rightmost; d_i for $0 < i < n$ composes the i-th and $(i+1)$-th arrows.
What does s_i do?	Inserts an identity morphism on c_i, lengthening the chain by one.
What is the fundamental category $\tau_1 X$?	The category with objects X_0, freely generated by 1-simplices subject to relations from 2-simplices. $\tau_1 \dashv \mathrm{N}$.
Where does the notation τ_1 come from?	From *truncation*: τ_1 truncates a simplicial set to its 1-categorical content (discarding higher coherences). The subscript 1 marks the target dimension.
Why is N fully faithful?	Natural transformations $\mathrm{N}(F) \to \mathrm{N}(G)$ are determined by their 1-simplex action, which is exactly a functor $F \to G$.
State Grothendieck's characterization of the nerve's image.	X is in the essential image of N iff every inner horn $\Lambda^n_k \to X$ $(0 < k < n)$ extends *uniquely* to $\Delta^n \to X$.
Who is Grothendieck?	Alexander Grothendieck (1928–2014), one of the most influential mathematicians of the 20th century. Reorganized algebraic geometry through category theory; introduced ∞-groupoids and the homotopy hypothesis in *Pursuing Stacks* (1983).
What changes when you drop "uniquely"?	You get quasi-categories (existence-only inner-horn filling). Composition becomes existential; associativity holds up to coherent homotopy. This is the foundation of ∞-category theory.
Define a Kan complex.	A simplicial set in which every horn $\Lambda^n_k \to X$ — inner or outer — extends to $\Delta^n \to X$.
Who is Kan?	Daniel M. Kan (1927–2013), Israeli-American mathematician at MIT. Introduced the horn-filling condition in 1958; also developed Kan extensions, adjoint functors, and the Kan model structure.
What is an ∞-groupoid?	A categorical structure in which all morphisms are invertible at every level (1-morphisms, 2-morphisms between them, and so on). Realized by Kan complexes.

Where does "groupoid" come from?	Heinrich Brandt (1927) coined the term for a generalization of groups in which composition is only partially defined.
Is $\mathrm{Sing}(T)$ a Kan complex?	Yes — because $\lvert\Lambda^n_k\rvert$ is a deformation retract of $\lvert\Delta^n\rvert$, every map from $\lvert\Lambda^n_k\rvert$ extends to $\lvert\Delta^n\rvert$, and the adjunction transports this to $\mathrm{Sing}(T)$.
Is the nerve of an ordinary category always a Kan complex?	No — outer horns fail in general. $\Lambda^2_0 \to \mathrm{N}(\mathcal{C})$ is two arrows out of a common source; filling forces an arrow relating their targets, which exists only if morphisms are invertible.
When IS $\mathrm{N}(\mathcal{C})$ a Kan complex?	When $\mathcal{C}$ is a groupoid (every morphism invertible). The intersection "nerve ∩ Kan complex" is exactly the image of $\mathbf{Grpd} \hookrightarrow \mathbf{sSet}$.
How do nerve, quasi-category, and Kan complex relate?	Nerve ⊂ quasi-category (drop "unique"); Kan complex ⊂ quasi-category (specialize to all horns). Nerve of a groupoid = intersection of the two.
What is the "homotopy hypothesis"?	Grothendieck's thesis (∼ 1983) that ∞-groupoids are equivalent to homotopy types of spaces — i.e., the homotopy theory of Kan complexes is equivalent to the homotopy theory of topological spaces, by the realization–singular adjunction.
Why is composition in a quasi-category not strict?	Because the inner-horn filler gives *a* composite, not *the* composite — different choices of Δ^2 filling a Λ^2_1 give different witnesses, all linked by higher-dimensional simplices.
What category is $\tau_1\mathrm{Sing}(T)$?	The fundamental groupoid $\Pi_1(T)$: objects are points, morphisms are homotopy classes of paths, all invertible.
Why do face operators on a nerve include composition?	Because the i-th face for $0 < i < n$ drops the object at position i from the chain, and the only way to restore composability is to compose the two adjacent morphisms — exactly what $f_{i+1} \circ f_i$ does.
Is the realization functor a left adjoint? Why?	Yes, it preserves colimits by construction (a coend is a colimit). The right adjoint is Sing.
What is the unit $X \to \mathrm{Sing}\,\lvert X\rvert$?	The map sending an n-simplex $x \in X_n$ to the topological n-simplex it generates in $\lvert X\rvert$, regarded as a continuous map $\lvert\Delta^n\rvert \to \lvert X\rvert$.

Chapter Summary

Chapter Summary

Geometric realization. $\lvert X\rvert = \int^{[n]} X_n \cdot \lvert\Delta^n\rvert$, the coend that glues topological n-simplices along simplicial face/degeneracy data. Left adjoint to Sing; left Kan extension of $\lvert\Delta^\bullet\rvert : \Delta \to \mathbf{Top}$ along Yoneda.
The nerve. $\mathrm{N}(\mathcal{C})_n = \mathbf{Cat}([n], \mathcal{C})$, the simplicial set of composable chains in $\mathcal{C}$. Fully faithful $\mathbf{Cat} \hookrightarrow \mathbf{sSet}$, with left adjoint τ_1 (fundamental category; "τ" from *truncation*).
Grothendieck's characterization. A simplicial set is a nerve iff every inner horn fills *uniquely*. Relaxing "uniquely" to "exists" gives quasi-categories.
Kan complexes. Simplicial sets in which every horn — inner or outer — fills. Named for Daniel Kan (1958). $\mathrm{Sing}(T)$ is always a Kan complex. Kan complexes are ∞-groupoids: outer-horn filling forces invertibility.
The homotopy hypothesis. Grothendieck's thesis (1983): ∞-groupoids and homotopy types of spaces are equivalent. Realized by the realization–singular Quillen equivalence (Chapter 42).

Formal Restatement

Formal Restatement

Realization. $|-| : \mathbf{sSet} \to \mathbf{Top}$, $|X| = \int^{[n]} X_n \cdot |\Delta^n| = \mathrm{Lan}_y |\Delta^\bullet|$.

Adjunction. $|-| \dashv \mathrm{Sing}$. $\mathrm{Sing}(T)_n = \mathbf{Top}(|\Delta^n|, T)$.

Nerve. $\mathrm{N} : \mathbf{Cat} \to \mathbf{sSet}$, $\mathrm{N}(\mathcal{C})_n = \mathbf{Cat}([n], \mathcal{C})$, with left adjoint $\tau_1 = \mathrm{Lan}_y[-]$. N is fully faithful.

Characterization of the nerve's image.

$$X \in \mathrm{ess.im}(\mathrm{N}) \iff \forall n \geq 2,\ 0 < k < n,\ \forall \alpha : \Lambda^n_k \to X,\ \exists!\, \tilde{\alpha} : \Delta^n \to X \text{ extending } \alpha.$$

Three horn conditions. For $X \in \mathbf{sSet}$:
- *Nerve*: inner horns fill uniquely.
- *Quasi-category*: inner horns fill (Chapter 43).
- *Kan complex*: all horns ($0 \leq k \leq n$, $n \geq 1$) fill.

Theorem. $\mathrm{Sing}(T)$ is a Kan complex for any $T \in \mathbf{Top}$.

Inclusions. $\{\mathrm{N}(\mathbf{Grpd})\} = \{\mathrm{N}(\mathbf{Cat})\} \cap \{\mathrm{Kan}\} \subset \{\mathrm{Kan}\} \subset \{\text{quasi-cat}\} \subset \mathbf{sSet}$.

The homotopy hypothesis (preview). $\mathrm{Kan} \simeq_{\mathrm{Q}} \mathbf{Top}$: the homotopy theory of Kan complexes coincides with the homotopy theory of topological spaces. Made precise by the Quillen equivalence of Chapter 42.

Chapter 40

Localization at Weak Equivalences

Chapter Overview

Many constructions of modern category theory follow a common recipe: take a category $\mathcal{C}$ of "concrete" objects, pick out a class $\mathcal{W}$ of morphisms that "ought to be isomorphisms even though they aren't," and form a new category $\mathcal{C}[\mathcal{W}^{-1}]$ in which every arrow of $\mathcal{W}$ has been turned into a genuine isomorphism. The result is the **localization** of $\mathcal{C}$ at $\mathcal{W}$.

You have met three motivating special cases. The *derived category* $\mathcal{D}(\mathcal{A})$ of an abelian category (Chapter 54), where $\mathcal{W}$ = quasi-isomorphisms; the *homotopy category* $\mathrm{Ho}(\mathbf{Top})$ of topological spaces, where $\mathcal{W}$ = weak homotopy equivalences; and Gabriel–Zisman's original construction of ring localizations, where $\mathcal{W}$ is a multiplicatively-closed subset.

This chapter develops the general theory in three layers. First, the **universal property**: $\mathcal{C}[\mathcal{W}^{-1}]$ is universal among categories receiving a $\mathcal{W}$-inverting functor from $\mathcal{C}$. Second, the **Gabriel–Zisman construction**: morphisms are *zigzags* of arrows in $\mathcal{C}$ with formal inverses inserted. Third, the **calculus of fractions**: conditions under which every zigzag reduces to a single span or cospan. The chapter closes with the three motivating examples, in preparation for the model-categorical machinery of Chapter 41 that makes localizations tractable.

40.1 Categories with Weak Equivalences

40.1.1 Informally

A *weak equivalence* is a morphism that ought to be considered an isomorphism, even though it isn't one in $\mathcal{C}$. Common patterns:

- In **Top**, a continuous map $f : X \to Y$ that induces an isomorphism on π_n for all n — without being a homeomorphism.
- In $\mathbf{Ch}(\mathcal{A})$ (chain complexes), a chain map that induces an isomorphism on homology — without being invertible at the chain level.
- In **sSet**, a map of simplicial sets that becomes a homotopy equivalence after geometric realization — without having an inverse in **sSet**.

In each case the original category retains too much rigidity. Localizing forces the weak equivalences to be invertible, recovering the homotopical content of the situation.

Where the Name Comes From

The term **localization** is borrowed from commutative algebra and algebraic geometry. To *localize* a ring R at a multiplicative set S is to invert the elements of S — making them units — producing $S^{-1}R$. "Local" because the result captures the behavior of R near (= local to) the prime ideal $\{r : r \notin S\}$. The categorical *localization* $\mathcal{C}[\mathcal{W}^{-1}]$ is exactly the same operation lifted up a level: invert a class of arrows by formally adjoining inverses. The terminology was carried into homotopy theory by Bousfield, Quillen, and others in the 1960s–70s.

Where the Name Comes From

Weak equivalence contrasts with the stricter notions of *isomorphism* or *equivalence of categories*. "Weak" here means "up to homotopy or some other higher equivalence relation," versus "on the nose." The terminology is from algebraic topology: a *weak homotopy equivalence* of spaces is one inducing isos on all π_n — weaker than an honest homotopy equivalence (Whitehead's theorem makes the two coincide for CW complexes, but not in general).

40.1.2 Expanding

Why bother with weak equivalences at all, when isomorphism already captures the strictest sense of "same"? Two reasons.

Pragmatic. In many natural categories, isomorphism is too strict. Two topological spaces of "the same homotopy type" are rarely homeomorphic; two chain complexes with "the same homology" are rarely chain-isomorphic. The natural notion of sameness is *homotopy* or *quasi-iso*, not honest isomorphism. Localizing puts that natural notion on equal footing with strict isomorphism by forcing the formal collapse.

Structural. The localization functor $Q : \mathcal{C} \to \mathcal{C}[\mathcal{W}^{-1}]$ has a universal property: $\mathcal{W}$-inverting functors out of $\mathcal{C}$ correspond to ordinary functors out of $\mathcal{C}[\mathcal{W}^{-1}]$. This makes $\mathcal{C}[\mathcal{W}^{-1}]$ the "universal home for $\mathcal{W}$-invariant functors," and the construction itself a left adjoint at the 2-categorical level.

Internal Dialog

Reader: Why not just quotient $\mathcal{C}$ by the equivalence relation "connected by a zigzag of weak equivalences"?
Author: You can, and that's exactly what the localization does at the level of objects. But the localization also has to manage *morphisms*: a morphism $X \to Y$ in $\mathcal{C}[\mathcal{W}^{-1}]$ should include not only morphisms in $\mathcal{C}$ but also "formal inverses" of $\mathcal{W}$-arrows. The zigzag description handles both layers at once.
Reader: And the universal property pins it down uniquely?
Author: Up to canonical isomorphism, yes. Two categories with the same universal property are canonically isomorphic; this is the abstract-nonsense way of saying the construction is well-defined.

40.1.3 Formalizing

Definition 40.1 (Category with weak equivalences)**.** A **category with weak equivalences** is a pair $(\mathcal{C}, \mathcal{W})$ where $\mathcal{C}$ is a category and $\mathcal{W} \subseteq \mathrm{Mor}(\mathcal{C})$ is a subclass closed under composition and containing all identities.

We typically also assume the **2-out-of-3 property**: if any two of f, g, gf are in $\mathcal{W}$, then so is the third.

How to Say This

$\mathcal{W}$ *(read: "the weak equivalences")*; the 2-out-of-3 property *(read: "two-out-of-three")* reflects that knowing two of the three pieces of a triangle suffices to deduce the third. Sometimes the stronger **2-out-of-6 property** (closure under sub-composition: if hg, gf are weak equivalences, so are f, g, h, hgf) is assumed.

Where the Name Comes From

The **2-out-of-3 property** is named for the combinatorial fact it expresses: among three morphisms in a composable triple f, g, gf, knowing two of them are in $\mathcal{W}$ forces the third. The stronger **2-out-of-6 property** expresses a similar pattern about six morphisms (two long composites and four atoms); it is necessary for $\mathcal{W}$ to be *saturated* — exactly equal to the class of morphisms made invertible by localization.

Example 40.2 (Three running examples)**. (a) Topological spaces.** $\mathcal{W}_{\mathrm{Top}}$ = weak homotopy equivalences: maps inducing isomorphisms on π_n for all $n \geq 0$. The category **Top** together with $\mathcal{W}_{\mathrm{Top}}$ is a category with weak equivalences satisfying 2-out-of-3.

(b) Chain complexes. $\mathcal{W}_{\mathrm{Ch}(\mathcal{A})}$ = **quasi-isomorphisms**: chain maps inducing isomorphisms on all homology groups. Forms the entry point to homological algebra.

(c) Simplicial sets. $\mathcal{W}_{\mathrm{sSet}}$ = weak equivalences: maps whose realization is a weak homotopy equivalence of spaces. Chapter 41 will give a combinatorial characterization not involving topological spaces.

Where the Name Comes From

A **quasi-isomorphism** is, literally, an "almost-isomorphism": a chain map inducing isomorphisms on homology, but not necessarily invertible at the chain level. "Quasi-" is from Latin *quasi* ("as if, almost"). The term was standardized in the 1950s–60s homological algebra literature (Cartan–Eilenberg, Grothendieck).

Reader's Annotation

The three examples are sometimes called the "trinity" of homotopical algebra: spaces, chain complexes, simplicial sets. Each has its own notion of weak equivalence, but all three are connected by adjunctions that turn out to be Quillen equivalences (Chapter 42), making their homotopy categories "the same." The localization construction below handles all three uniformly.

40.2 The Gabriel–Zisman Localization

40.2.1 Formalizing

Where the Name Comes From

The **Gabriel–Zisman localization** is named for *Pierre Gabriel* (b. 1933, German–French mathematician) and *Michel Zisman* (b. 1932, French mathematician), whose 1967 book *Calculus of Fractions and Homotopy Theory* systematized the construction below in the simplicial setting. The construction itself is older — versions appear in ring theory, in Grothendieck's work on derived categories, and in much earlier formal-inverse arguments — but Gabriel–Zisman gave the definitive abstract treatment.

Definition 40.3 (Localization)**.** The **localization** of $(\mathcal{C}, \mathcal{W})$ is a category $\mathcal{C}[\mathcal{W}^{-1}]$ together with a functor

$$Q : \mathcal{C} \to \mathcal{C}[\mathcal{W}^{-1}]$$

such that:

1. $Q(w)$ is an isomorphism for every $w \in \mathcal{W}$.
2. For any functor $F : \mathcal{C} \to \mathcal{D}$ that sends every $w \in \mathcal{W}$ to an isomorphism in $\mathcal{D}$, there is a unique $\bar{F} : \mathcal{C}[\mathcal{W}^{-1}] \to \mathcal{D}$ with $\bar{F} \circ Q = F$.

The universal property determines $(\mathcal{C}[\mathcal{W}^{-1}], Q)$ uniquely up to canonical isomorphism.

How to Say This

Q *(read: "the localization functor" or "the quotient")*; the universal property *(read: "functors out of $\mathcal{C}[\mathcal{W}^{-1}]$ correspond bijectively to functors out of $\mathcal{C}$ that already invert $\mathcal{W}$")*. Some texts write $L_{\mathcal{W}}$ or just γ for the localization functor.

Theorem 40.4 (Gabriel–Zisman construction)**.** *For any $(\mathcal{C}, \mathcal{W})$, the localization $\mathcal{C}[\mathcal{W}^{-1}]$ exists. It can be constructed as the quotient of the free category on the graph $\mathcal{C} \sqcup \mathcal{W}^{-1}$ — where $\mathcal{W}^{-1}$ adds a formal inverse edge $w^{-1} : Y \to X$ for each $w : X \to Y$ in $\mathcal{W}$ — by the relations:*

- *Composition relations of $\mathcal{C}$;*
- $w^{-1} \circ w = \mathrm{id}$ *and* $w \circ w^{-1} = \mathrm{id}$ *for each* $w \in \mathcal{W}$.

Where the Name Comes From

The morphisms of $\mathcal{C}[\mathcal{W}^{-1}]$ are called **zigzags** because they look like zigzag paths: arrows alternately pointing forward (in $\mathcal{C}$) and backward (a formal w^{-1}). The geometric description is descriptive and goes back to the 1950s combinatorial literature on free groups.

Proof sketch. Construct $\mathcal{C}[\mathcal{W}^{-1}]$ as described. Morphisms are **zigzags** of arrows

$$X \xrightarrow{f_1} Y_1 \xleftarrow{w_1} Z_1 \xrightarrow{f_2} \cdots \xleftarrow{w_n} Z_n \xrightarrow{f_{n+1}} Y,$$

with each backward arrow in $\mathcal{W}$, modulo the relations of the free quotient. The functor $Q : \mathcal{C} \to \mathcal{C}[\mathcal{W}^{-1}]$ sends each morphism to itself (a length-one zigzag). The universal property holds by direct verification: any F inverting $\mathcal{W}$ extends uniquely to zigzags by sending each w^{-1} to $F(w)^{-1}$. ☐ ☐

Remark 40.5 (Size issues)**.** The free quotient construction is set-theoretically problematic when $\mathcal{C}$ is large: hom-classes of $\mathcal{C}[\mathcal{W}^{-1}]$ may not be small even when $\mathcal{C}$'s are. The model-categorical machinery of Chapter 41 will resolve this for the examples we care about by producing *explicit small-hom-set models* for $\mathcal{C}[\mathcal{W}^{-1}]$ (the homotopy category as morphisms between bifibrant objects).

Reader's Annotation

The Gabriel–Zisman construction proves existence but is rarely useful for *computation*: a zigzag of length 17 may represent the same morphism as a zigzag of length 3, but checking equivalence requires following a long chain of relations. The calculus of fractions (§40.3) is the standard remedy: under suitable conditions, every zigzag reduces to length 2, giving a concrete description of $\mathcal{C}[\mathcal{W}^{-1}]$'s morphisms.

40.3 Calculus of Fractions

40.3.1 Informally

The Gabriel–Zisman construction describes a morphism of $\mathcal{C}[\mathcal{W}^{-1}]$ as a zigzag of arbitrary length. Under additional hypotheses on $\mathcal{W}$ — the **calculus of fractions** — every zigzag reduces to a single span (or cospan), yielding a much more manageable description.

Where the Name Comes From

The **calculus of fractions** is named by analogy with the calculus of fractions in (non-commutative) ring theory: *Øystein Ore* (1899–1968, Norwegian mathematician) gave conditions in 1931 under which a non-commutative ring R has a ring of fractions $S^{-1}R$. The *Ore condition* below is the same combinatorial condition lifted to categories. "Fractions" because the morphisms of the localization look like $w^{-1}f$ (formal "f over w"), like ratios in algebra.

40.3.2 Formalizing

Definition 40.6 (Left calculus of fractions)**.** A class $\mathcal{W}$ admits a **left calculus of fractions** if:

1. *Identities and composition.* $\mathcal{W}$ contains all identities and is closed under composition.
2. *Ore condition.* For every span $X \xleftarrow{w} A \xrightarrow{f} B$ with $w \in \mathcal{W}$, there exists a commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "w"'] & B \arrow[d, "w'"] \\ X \arrow[r, "f'"'] & Y \end{tikzcd}$$

   with $w' \in \mathcal{W}$.
3. *Cancellation.* If $fw = gw$ with $w \in \mathcal{W}$, there exists $w' \in \mathcal{W}$ such that $w'f = w'g$.

How to Say This

"Calculus of fractions" *(read: "calculus of fractions")*; the *Ore condition* *(read: "the Ore (or "or-AY") condition")*, sometimes *(read: "the left Ore condition")* to distinguish from its dual. Sometimes also called **Gabriel's axiom** or **Hartshorne's localization theorem** depending on context.

Remark 40.7 (Right calculus)**.** The dual notion of **right calculus of fractions** reverses arrows: the Ore condition completes *cospans* $X \xrightarrow{f} B \xleftarrow{w} A$ to commutative squares. Each side of the calculus gives a different reduction (left fractions are spans $X \xleftarrow{w} \cdot \xrightarrow{f} Y$; right fractions are cospans).

Theorem 40.8 (Reduction to fractions)**.** *If $\mathcal{W}$ admits a left calculus of fractions, every morphism of $\mathcal{C}[\mathcal{W}^{-1}]$ can be represented by a single* ***left fraction***

$$X \xleftarrow{w} A \xrightarrow{f} Y, \qquad w \in \mathcal{W},$$

and two left fractions (w, f) and (w', f') represent the same morphism iff there is a common refinement

$$\begin{tikzcd} & A \arrow[ddl, bend right, "w"'] \arrow[ddr, bend left, "f"] & \\ & B \arrow[u, "v"'] \arrow[d, "u"'] & \\ X & A' \arrow[l, "w'"] \arrow[r, "f'"'] & Y \end{tikzcd}$$

with $u, v \in \mathcal{W}$ commuting both squares.

Proof sketch. Every zigzag $f_1, w_1^{-1}, f_2, w_2^{-1}, \ldots$ is reduced by repeatedly applying the Ore condition (interchange a backward w^{-1} with the following forward f by completing the span). Zigzag length strictly decreases at each step; the procedure terminates with a single left fraction. Equivalence of two fractions is the equivalence relation generated by common refinement; the cancellation axiom guarantees that the relation is transitive. $\square$ $\square$

Remark 40.9 (Composition of fractions)**.** With the calculus in hand, composition is concrete: given fractions $X \xleftarrow{w} A \xrightarrow{f} Y$ and $Y \xleftarrow{v} B \xrightarrow{g} Z$, complete the span $A \xrightarrow{f} Y \xleftarrow{v} B$ to a commutative square by Ore, obtaining $A \xleftarrow{v'} C \xrightarrow{f'} B$ with $v' \in \mathcal{W}$; the composite is then $X \xleftarrow{wv'} C \xrightarrow{gf'} Z$.

Reader's Annotation

The Ore condition is best remembered geometrically: it says *spans with one $\mathcal{W}$-leg can be completed to commutative squares with $\mathcal{W}$-legs on both top and bottom.* This is exactly the "rewriting rule" used to push backwards arrows to the left of forwards arrows, simplifying zigzags. Without the Ore condition, the rewriting gets stuck and zigzags remain arbitrarily long.

40.4 Examples

40.4.1 The derived category of an abelian category

Example 40.10 (Derived category $\mathcal{D}(\mathcal{A})$)**.** Let $\mathcal{A}$ be an abelian category and $\mathbf{Ch}(\mathcal{A})$ its category of chain complexes. The **derived category**

$$\mathcal{D}(\mathcal{A}) = \mathbf{Ch}(\mathcal{A})\left[\mathcal{W}^{-1}\right]$$

is the localization at $\mathcal{W}$ = quasi-isomorphisms.

The intermediate *homotopy category* $\mathbf{K}(\mathcal{A}) = \mathbf{Ch}(\mathcal{A})/\sim$ (chain maps modulo chain homotopy) admits a calculus of fractions with respect to quasi-isomorphisms — both left and right. Therefore $\mathcal{D}(\mathcal{A}) = \mathbf{K}(\mathcal{A})[\mathcal{W}^{-1}]$ has the concrete description: morphisms are spans (or cospans) of chain maps with quasi-isomorphism legs, modulo refinement.

Where the Name Comes From

The **derived category** $\mathcal{D}(\mathcal{A})$ was introduced by Grothendieck's student *Jean-Louis Verdier* (1935–1989) in his 1967 thesis. "Derived" because morphisms in $\mathcal{D}(\mathcal{A})$ are equivalent to *derived functor* information (Tor, Ext, etc.): in $\mathcal{D}(\mathcal{A})$, the natural map $\mathrm{Hom}(M,N) \to \mathrm{Hom}(M^\bullet, N^\bullet)$ encodes Tor/Ext via the chain representatives.

40.4.2 The homotopy category of spaces

Example 40.11 ($\mathrm{Ho}(\mathbf{Top})$)**.** The homotopy category $\mathrm{Ho}(\mathbf{Top}) = \mathbf{Top}[\mathcal{W}^{-1}]$ inverts weak homotopy equivalences. The calculus of fractions does *not* apply directly in **Top**; constructing the localization requires explicit replacement of spaces by CW complexes (an essentially model-categorical operation). After replacement, every morphism of $\mathrm{Ho}(\mathbf{Top})$ between CW complexes is a homotopy class of continuous maps, by Whitehead's theorem (Chapter 41).

Where the Name Comes From

Whitehead's theorem is named for *J.H.C. Whitehead* (1904–1960, British algebraic topologist; not to be confused with Alfred North Whitehead). Whitehead's theorem says that a weak homotopy equivalence between CW complexes is an honest homotopy equivalence — a non-trivial fact tying the weak (homotopy-group-iso) and strong (homotopy-inverse) notions together for sufficiently nice spaces.

40.4.3 Quillen's localization

Remark 40.12 (Model-categorical localization)**.** A **model category** (Chapter 41) is a category with three classes of maps — weak equivalences, cofibrations, fibrations — satisfying axioms that together provide an explicit construction of $\mathcal{C}[\mathcal{W}^{-1}]$: the homotopy category $\mathrm{Ho}(\mathcal{C})$ has the same objects as $\mathcal{C}$, and its morphisms between bifibrant objects X (cofibrant and fibrant) are homotopy classes $[X,Y]/\sim$. The model-categorical localization is small-hom-class by construction, sidestepping the size issue of Remark 40.5.

Reader's Annotation

The pattern across all three examples — chain complexes, topological spaces, simplicial sets — is the same: a raw category with weak equivalences is too rigid to localize directly, but admits a *replacement* (chain homotopy classes, CW complexes, bifibrant simplicial sets) where the calculus of fractions does work, or equivalently, where homotopy classes give an honest description of localized morphisms. Chapter 41 axiomatizes this pattern.

40.5 Exercises

Define a category with weak equivalences.	A pair $(\mathcal{C},\mathcal{W})$ where $\mathcal{W} \subseteq \mathrm{Mor}(\mathcal{C})$ contains identities and is closed under composition.
What is the 2-out-of-3 property?	If two of f, g, gf are in $\mathcal{W}$, the third is too.
Where does "localization" come from?	Algebra: localizing a ring R at a multiplicative set S inverts the elements of S. "Local" because the result captures behavior near a prime ideal. Lifted to categories in the 1960s–70s.
Where does "weak equivalence" come from?	Contrast with strict isomorphism: a weak equivalence is iso "up to homotopy or some other higher equivalence," not on the nose. Standard in algebraic topology.
Define the localization $\mathcal{C}[\mathcal{W}^{-1}]$ by its universal property.	A category with a functor $Q : \mathcal{C} \to \mathcal{C}[\mathcal{W}^{-1}]$ inverting $\mathcal{W}$, universal: any functor F inverting $\mathcal{W}$ factors uniquely through Q.
Why does the localization always exist?	Gabriel–Zisman: build the free category on $\mathcal{C} \sqcup \mathcal{W}^{-1}$ and quotient by the appropriate relations.

Who are Gabriel and Zisman?	Pierre Gabriel (b. 1933) and Michel Zisman (b. 1932), authors of the 1967 book *Calculus of Fractions and Homotopy Theory*, which systematized the localization construction.
What is a zigzag in $\mathcal{C}[\mathcal{W}^{-1}]$?	A sequence of forward arrows in $\mathcal{C}$ and backward arrows in $\mathcal{W}^{-1}$ from X to Y — the generic morphism description from the Gabriel–Zisman construction.
Why is the localization set-theoretically subtle?	Hom-classes in $\mathcal{C}[\mathcal{W}^{-1}]$ may not be small (proper classes of zigzag equivalence classes) even when those of $\mathcal{C}$ are.
Define a left calculus of fractions.	$\mathcal{W}$ contains identities and closes under composition; the Ore condition completes spans with $\mathcal{W}$-leg to commutative squares with $\mathcal{W}$-leg; cancellation: $fw = gw$ implies $w'f = w'g$ for some $w' \in \mathcal{W}$.
Who is Ore?	Øystein Ore (1899–1968), Norwegian mathematician. Gave conditions in 1931 for non-commutative ring localization; the Ore condition above is the categorical version.
What does the Ore condition replace zigzags with?	Single left fractions $X \xleftarrow{w} A \xrightarrow{f} Y$ with $w \in \mathcal{W}$. Every zigzag reduces to a span by repeated application of Ore.
When are two left fractions equivalent?	When they admit a common refinement: a span $A' \to A$ and $A' \to B$ both in $\mathcal{W}$ that fits between the two.
How do you compose left fractions?	Given $X \xleftarrow{w} A \xrightarrow{f} Y$ and $Y \xleftarrow{v} B \xrightarrow{g} Z$, Ore-complete f, v to a square with v', f', obtaining $X \xleftarrow{wv'} C \xrightarrow{gf'} Z$.
What is the derived category of an abelian category?	$\mathcal{D}(\mathcal{A}) = \mathbf{Ch}(\mathcal{A})[\mathcal{W}^{-1}]$ where $\mathcal{W}$ = quasi-isomorphisms (maps inducing isomorphism on homology).
Who introduced derived categories?	Jean-Louis Verdier (Grothendieck's student) in his 1967 thesis. "Derived" because morphisms encode derived-functor information (Tor, Ext).
What is a quasi-isomorphism?	A chain map inducing isomorphisms on homology. "Quasi-" (Latin *quasi*, "almost") because it's almost an isomorphism — invertible at the level of homology, not necessarily at the chain level.
Why is the chain-homotopy category $\mathbf{K}(\mathcal{A})$ a useful stepping stone?	Because quasi-isomorphisms admit a calculus of fractions in $\mathbf{K}(\mathcal{A})$ (but not in $\mathbf{Ch}(\mathcal{A})$). So $\mathcal{D}(\mathcal{A}) = \mathbf{K}(\mathcal{A})[\mathcal{W}^{-1}]$ has the concrete left/right-fractions description.
What is Ho(**Top**)?	$\mathbf{Top}[\mathcal{W}^{-1}]$ with $\mathcal{W}$ = weak homotopy equivalences. Concretely, the homotopy category of CW complexes (by Whitehead).
Who is Whitehead?	J.H.C. Whitehead (1904–1960), British algebraic topologist. His theorem says weak homotopy equivalences between CW complexes are honest homotopy equivalences — tying weak and strong notions.
Why isn't the calculus of fractions enough for **Top**?	The Ore condition fails: completing a span involving a weak homotopy equivalence may require a CW replacement that lives outside **Top** in any natural sense. Model categories handle this systematically.
What's the model-category alternative to Gabriel–Zisman?	Replace objects by bifibrant ones and define $\mathrm{Ho}(\mathcal{C})(X,Y) = [X,Y]/\sim$ (homotopy classes). Gives the same localization but with small hom-classes by construction.
State the universal property of Q in equation form.	$\mathrm{Fun}(\mathcal{C}[\mathcal{W}^{-1}], \mathcal{D}) \xrightarrow{-\circ Q} \mathrm{Fun}_{\mathcal{W}\text{-inv}}(\mathcal{C}, \mathcal{D})$ is an isomorphism of categories.

What does inverting $\mathcal{W}$ feel like categorically?	The functor Q is essentially surjective but not faithful; it identifies parallel morphisms that differ by a chain of $\mathcal{W}$-zigzags.
Does $\mathcal{C}[\mathcal{W}^{-1}]$ have the same objects as $\mathcal{C}$?	Yes, by construction. The localization changes only morphisms.
What's an example where the calculus of fractions fails?	**Top** with weak homotopy equivalences: the Ore square cannot in general be filled within **Top** without CW replacement.
Why is the 2-out-of-6 property sometimes assumed?	It makes a class of weak equivalences "saturated": closed under recovering weak equivalences from longer composites. Implies that $\mathcal{W}$ contains exactly the morphisms made invertible by Q.
Is every isomorphism in $\mathcal{C}$ in $\mathcal{W}$?	Not required by the bare definition, but yes if $\mathcal{W}$ is *saturated* (closed under invertible-becoming-after-localization). Most natural classes of weak equivalences contain all isomorphisms.
What's the relationship between $\mathcal{C}[\mathcal{W}^{-1}]$ and Chapter 43's quasi-categories?	$\mathcal{C}[\mathcal{W}^{-1}]$ is the homotopy 1-category of an underlying ∞-category — the latter retains higher coherence data that the localization discards. Volume IV's ∞-categorical Yoneda (Chapter 48) lives at this richer level.
Why is localization a left adjoint?	$Q : \mathcal{C} \to \mathcal{C}[\mathcal{W}^{-1}]$ has the universal property of an initial object in the comma category of $\mathcal{W}$-inverting functors out of $\mathcal{C}$; this is left-adjoint-style universality on the level of the 2-category **Cat**.

Chapter Summary

Chapter Summary

Category with weak equivalences. $(\mathcal{C}, \mathcal{W})$ with $\mathcal{W}$ closed under composition and containing identities, typically with the 2-out-of-3 property. Origin: "localization" from algebra/geometry; "weak equivalence" from algebraic topology.
Localization (Gabriel–Zisman, 1967). $\mathcal{C}[\mathcal{W}^{-1}]$ is universal among categories receiving a $\mathcal{W}$-inverting functor from $\mathcal{C}$. Always exists, with morphisms presented as zigzags modulo relations.
Calculus of fractions (Ore, 1931 + Gabriel–Zisman). An Ore-type condition under which zigzags reduce to single left (or right) fractions. Composition becomes concrete; equivalence becomes common refinement.
Three examples. The derived category $\mathcal{D}(\mathcal{A})$ (Verdier 1967); the homotopy category Ho(**Top**) (Whitehead 1949); Quillen's model categories (1967). The last unifies the previous by giving an explicit small-hom-class construction via bifibrant replacement (Chapter 41).

Formal Restatement

Formal Restatement

Localization. For $(\mathcal{C}, \mathcal{W})$, the localization $(Q, \mathcal{C}[\mathcal{W}^{-1}])$ satisfies:

$$\mathrm{Fun}_{\mathcal{W}\text{-inv}}(\mathcal{C}, \mathcal{D}) \;\cong\; \mathrm{Fun}(\mathcal{C}[\mathcal{W}^{-1}], \mathcal{D}), \qquad F \longmapsto F \circ Q.$$

Left calculus of fractions. $\mathcal{W}$ admits left fractions if:

- $\mathcal{W}$ has identities and closes under composition.
- (Ore) Every span $X \xleftarrow{w} A \xrightarrow{f} B$ with $w \in \mathcal{W}$ completes to a commutative square with the right leg in $\mathcal{W}$.
- (Cancellation) $fw = gw$ with $w \in \mathcal{W}$ implies $w'f = w'g$ for some $w' \in \mathcal{W}$.

Fractions theorem. If $\mathcal{W}$ admits left fractions, $\mathcal{C}[\mathcal{W}^{-1}](X, Y)$ is the set of equivalence classes of left fractions $X \xleftarrow{w} A \xrightarrow{f} Y$ under common refinement.

Three running examples.

- **Ch**$(\mathcal{A})$ with quasi-isomorphisms: $\mathcal{D}(\mathcal{A})$.
- **Top** with weak homotopy equivalences: Ho(**Top**).

- **sSet** with realization-weak-equivalences: $\mathrm{Ho}(\mathbf{sSet})$.

Chapter 41

Model Categories: Axioms and Examples

Chapter Overview

A **model category** is a category equipped with three distinguished classes of morphisms — *weak equivalences*, *cofibrations*, and *fibrations* — satisfying axioms that together provide an explicit, small-hom-class construction of the localization $\mathcal{C}[\mathcal{W}^{-1}]$ of Chapter 40. The framework is due to Quillen (1967), and it transformed homotopy theory from an art form into a systematic apparatus.

The five axioms are: 2-out-of-3 for weak equivalences; retract-closure for all three classes; *factorization* of every map as cofibration followed by trivial fibration, or as trivial cofibration followed by fibration; *lifting* of cofibrations against trivial fibrations (and dually, trivial cofibrations against fibrations); and completeness of the underlying category.

This chapter develops the axioms, introduces lifting properties as the combinatorial heart of the theory, and concludes with the two foundational examples: the Quillen model structure on simplicial sets (fibrations = Kan fibrations, cofibrations = monomorphisms) and the Quillen model structure on topological spaces (fibrations = Serre fibrations, cofibrations generated by sphere inclusions). The **Quillen equivalence** $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : \mathrm{Sing}$ is stated; its proof — the homotopy hypothesis at the level of model categories — is the climax of Chapter 42.

Joyal's model structure on **sSet**, in which fibrant objects are exactly the quasi-categories, appears at the end of the chapter as a preview of Chapter 43.

41.1 Quillen's Axioms

41.1.1 Informally

Before the formal axioms, here is the picture they capture. A *model category* is a category $\mathcal{C}$ together with three distinguished classes of morphisms — *weak equivalences*, *cofibrations*, and *fibrations* — that interact in a specific way:

- Every morphism factors as *cofibration followed by trivial fibration* (or as *trivial cofibration followed by fibration*). Think: every map can be "set up" as a nice cofibration before settling into a homotopy equivalence, or as a homotopy equivalence before a nice fibration.
- Cofibrations *lift* on the left against trivial fibrations: given a commutative square with a cofibration on the left and a trivial fibration on the right, a diagonal lift exists. Symmetrically for trivial cofibrations and fibrations.

Out of this small package of axioms emerges an entire homotopical calculus: cofibrant and fibrant replacements, derived functors, the homotopy category as morphisms-between-bifibrant-objects-modulo-homotopy, and (Chapter 42) Quillen adjunctions and equivalences.

Where the Name Comes From

Model category was introduced by *Daniel Quillen* in his 1967 monograph *Homotopical Algebra*. Quillen's project was to axiomatize the structure that algebraic topology had been using implicitly for decades — the interaction of cofibrations, fibrations, and weak equivalences in CW complexes, chain complexes, and simplicial sets — into a single framework. The name reflects that the axioms specify a "model" for doing homotopy theory: a working language with formal rules for the moves you can make.

Where the Name Comes From

Cofibration and **fibration** are from algebraic topology. A *fibration* is the homotopy-theoretic generalization of a fibre bundle: a map $p : E \to B$ for which paths in B lift to paths in E (the **homotopy lifting property**). The term goes back to Serre (1951). A *cofibration* is the dual notion: a map $i : A \to B$ for which homotopies extending over A extend over B (**homotopy extension property**); the term is due to Hurewicz (1955). Quillen's axiomatization promotes the homotopy-lifting and homotopy-extension properties from special cases to defining axioms.

41.1.2 Formalizing

Definition 41.1 (Model category)**.** A **model category** is a complete and cocomplete category $\mathcal{C}$ together with three classes of morphisms — **weak equivalences** $\mathcal{W}$, **cofibrations** $\mathcal{C}\mathrm{of}$, and **fibrations** $\mathcal{F}\mathrm{ib}$ — satisfying the following five axioms.

(M1) Two-out-of-three. If two of f, g, gf are weak equivalences, so is the third.

(M2) Retracts. If f is a retract of g (in the arrow category of $\mathcal{C}$) and g belongs to $\mathcal{W}$, $\mathcal{C}\mathrm{of}$, or $\mathcal{F}\mathrm{ib}$, then so does f.

(M3) Factorization. Every morphism f admits factorizations

$$f \;=\; q \circ i \quad \text{with } i \in \mathcal{C}\mathrm{of},\, q \in \mathcal{W} \cap \mathcal{F}\mathrm{ib}$$

and

$$f \;=\; p \circ j \quad \text{with } j \in \mathcal{W} \cap \mathcal{C}\mathrm{of},\, p \in \mathcal{F}\mathrm{ib}.$$

(M4) Lifting. Cofibrations have the left lifting property against trivial fibrations; trivial cofibrations have the left lifting property against fibrations. That is, for every commutative square

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r] \arrow[ur, dashed, "h"] & Y \end{tikzcd}$$

with $i \in \mathcal{C}\mathrm{of}$ and $p \in \mathcal{W} \cap \mathcal{F}\mathrm{ib}$, a lift h exists making both triangles commute. The same holds if $i \in \mathcal{W} \cap \mathcal{C}\mathrm{of}$ and $p \in \mathcal{F}\mathrm{ib}$.

(M5) Completeness. $\mathcal{C}$ has all small limits and colimits.

How to Say This

Three classes pronounced "weak equivalences," "cofibrations," and "fibrations." **Trivial cofibration** = cofibration $\cap$ weak equivalence (also called **acyclic cofibration**); **trivial fibration** = fibration $\cap$ weak equivalence. Common diagrammatic shorthand: $\xrightarrow{\sim}$ for a weak equivalence, $\hookrightarrow$ for a cofibration, $\twoheadrightarrow$ for a fibration.

Reader's Annotation

The five axioms M1–M5 are easy to remember once you have a mnemonic:

- **M1** 2-out-of-3 (weak equivalences are a 2-out-of-3 class).
- **M2** Retracts (all three classes are closed under retracts).
- **M3** Factorizations exist (two of them).
- **M4** Lifting (cofibrations lift against trivial fibrations).
- **M5** (Co)completeness.

M3 and M4 together are the technical heart: factorization plus lifting give you the ability to construct homotopies and replacements systematically.

Remark 41.2 (Determination of two classes by the third)**.** The five axioms imply that any two of $\mathcal{W}, \mathcal{C}\mathrm{of}, \mathcal{F}\mathrm{ib}$ determine the third:

- Cofibrations = maps with LLP against trivial fibrations.
- Fibrations = maps with RLP against trivial cofibrations.
- Weak equivalences = maps factoring as trivial cofibration $\circ$ trivial fibration (and equivalent characterizations).

Hence model structures are often specified by giving only the classes $\mathcal{W}$ and one other class — the third being determined.

41.2 Lifting Properties and Small Objects

41.2.1 Formalizing

Definition 41.3 (Left and right lifting properties)**.** A morphism $i : A \to B$ has the **left lifting property** (LLP) with respect to $p : X \to Y$ if every commutative square

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

admits a diagonal lift. Equivalently, p has the **right lifting property** (RLP) with respect to i. For a class $\mathcal{S}$ of morphisms, $\mathrm{LLP}(\mathcal{S})$ is the class of morphisms with LLP against every $p \in \mathcal{S}$; dually $\mathrm{RLP}(\mathcal{S})$.

Lemma 41.4 (Closure properties of LLP/RLP)**.** $\mathrm{LLP}(\mathcal{S})$ *is closed under:*

- *Composition.*
- *Pushouts (in $\mathcal{C}$).*
- *Transfinite composition (sequential colimits along ordinals).*
- *Retracts.*

RLP($\mathcal{S}$) *is closed under composition, pullbacks, transfinite limits, and retracts.*

Proof sketch. For composition: a lift in $A \xrightarrow{i} B \xrightarrow{j} C$ against p is constructed in two stages, lifting i against p then j against p. For pushouts and transfinite composition: standard diagram-chase, constructing the lift component-by-component. For retracts: any commutative square involving the retract is filled in by lifting against the retracting square. □ □

Where the Name Comes From

The **lifting property** terminology is broadly used in homotopy theory and algebraic topology. The dichotomy "left lifting property" (LLP) vs. "right lifting property" (RLP) refers to *which side of the square* the cofibration / fibration is on. The first systematic use as an axiomatic device is in Quillen's 1967 monograph; earlier versions appear in the work of Kan and Serre.

Theorem 41.5 (Small object argument)**.** *Let $\mathcal{I}$ be a set of morphisms in a cocomplete category $\mathcal{C}$ whose domains are* ***small*** *(every map from such a domain into a transfinite colimit factors through a finite stage). Then every morphism $f : X \to Y$ in $\mathcal{C}$ admits a functorial factorization $f = p \circ i$ with $i \in \mathrm{LLP}(\mathrm{RLP}(\mathcal{I}))$ (i.e., i a cell-complex-like map built from $\mathcal{I}$-cells) and $p \in \mathrm{RLP}(\mathcal{I})$.*

Where the Name Comes From

The **small object argument** is named for the size hypothesis on the domains: they must be "small" relative to the transfinite sequential colimit, meaning that maps out of them factor through a stage. The argument itself is a transfinite iteration that builds cell-complex-like structure. Quillen attributes the technique to John Frank Adams and broader homotopy-theoretic folklore; the modern formulation is due to Quillen (1967) and Garner (2008, "algebraic" small object argument).

Remark 41.6 (Why the small object argument matters)**.** The small object argument is the engine producing factorizations (M3) in practice. **Cofibrantly generated model categories** are those whose cofibrations and trivial cofibrations are each generated by small sets $\mathcal{I}$ and $\mathcal{J}$ respectively; the factorizations come from applying Theorem 41.5 to $\mathcal{I}$ and $\mathcal{J}$. Both Quillen-**sSet** and Quillen-**Top** are cofibrantly generated.

41.3 Cofibrant, Fibrant, and Bifibrant Objects

41.3.1 Formalizing

Let $\emptyset$ and 1 denote the initial and terminal objects of $\mathcal{C}$ (existing by M5).

Definition 41.7 (Cofibrant, fibrant, bifibrant)**.** An object $X \in \mathcal{C}$ is:

- **Cofibrant** if the unique map $\emptyset \to X$ is a cofibration.
- **Fibrant** if the unique map $X \to 1$ is a fibration.
- **Bifibrant** if it is both cofibrant and fibrant.

Theorem 41.8 (Functorial replacement)**.** *There exist endofunctors $Q, R : \mathcal{C} \to \mathcal{C}$ and natural transformations $Q \xrightarrow{\sim} \mathrm{id}$ and $\mathrm{id} \xrightarrow{\sim} R$ such that for every X:*

- *QX is cofibrant and $QX \to X$ is a trivial fibration —* ***cofibrant replacement****.*
- *$X \to RX$ is a trivial cofibration and RX is fibrant —* ***fibrant replacement****.*

Proof sketch. For cofibrant replacement, factor $\emptyset \to X$ using M3 as $\emptyset \hookrightarrow QX \xrightarrow{\sim, \twoheadrightarrow} X$. For fibrant replacement, factor $X \to 1$ as $X \xrightarrow{\sim, \hookrightarrow} RX \twoheadrightarrow 1$. Functoriality of the factorizations (M3 with functorial choice via the small object argument) gives the natural transformations. □ □

Remark 41.9 (Why bifibrant?)**.** The localization $\mathrm{Ho}(\mathcal{C}) = \mathcal{C}[\mathcal{W}^{-1}]$ is built from $\mathcal{C}$'s objects, but its hom-sets between bifibrant X, Y admit the explicit description

$$\mathrm{Ho}(\mathcal{C})(X, Y) \;=\; \mathcal{C}(X, Y)/\sim,$$

where $\sim$ is the equivalence relation of **homotopy** (defined below via cylinder/path objects). Combined with bifibrant replacement (apply Q then R, or any composite producing a bifibrant object), this gives the small-hom-class construction of $\mathrm{Ho}(\mathcal{C})$.

41.3.2 Reorganizing: cylinders and homotopies

Definition 41.10 (Cylinder object, path object)**.** A **cylinder object** for X is a factorization $X \sqcup X \xrightarrow{(i_0,i_1)} \mathrm{Cyl}(X) \xrightarrow{\sim} X$ of the fold map, with (i_0, i_1) a cofibration. A **path object** for Y is a factorization $Y \xrightarrow{\sim} \mathrm{Path}(Y) \xrightarrow{(p_0,p_1)} Y \times Y$ of the diagonal, with (p_0, p_1) a fibration. Two maps $f, g : X \to Y$ are **left homotopic** if there is $H : \mathrm{Cyl}(X) \to Y$ with $Hi_0 = f$, $Hi_1 = g$; **right homotopic** if there is $H : X \to \mathrm{Path}(Y)$ with $p_0H = f$, $p_1H = g$.

Theorem 41.11 (Homotopy is an equivalence relation on bifibrant maps)**.** *If X is cofibrant and Y is fibrant, left and right homotopy coincide and define an equivalence relation on $\mathcal{C}(X, Y)$, which we denote $[X, Y]$. The composition $\mathcal{C}(X, Y) \times \mathcal{C}(Y, Z) \to \mathcal{C}(X, Z)$ descends to $[-, -]$, giving the homotopy category $\mathrm{Ho}(\mathcal{C})_{\mathrm{bif}}$ on bifibrant objects.*

41.4 The Two Foundational Examples

41.4.1 sSet with the Quillen model structure

Theorem 41.12 (Quillen model structure on **sSet**)**.** *The category* **sSet** *admits a model structure in which:*

- *Weak equivalences are maps whose realization is a weak homotopy equivalence in* **Top**.
- *Cofibrations are monomorphisms (equivalently, level-wise injections).*
- *Fibrations are* ***Kan fibrations****: maps $p : X \to Y$ with RLP against every horn inclusion $\Lambda^n_k \hookrightarrow \Delta^n$ ($n \geq 1$, $0 \leq k \leq n$).*

This is the ***Kan–Quillen model structure****. Generating cofibrations: the boundary inclusions $\partial\Delta^n \hookrightarrow \Delta^n$. Generating trivial cofibrations: the horn inclusions $\Lambda^n_k \hookrightarrow \Delta^n$.*

Remark 41.13 (Fibrant objects of Quillen-**sSet**)**.** The fibrant objects are exactly the Kan complexes (Chapter 39): $X \to 1$ having RLP against all horn inclusions is exactly the Kan condition. Every simplicial set is cofibrant (the unique map from $\emptyset$ is a monomorphism), so bifibrant simplicial sets are Kan complexes.

Where the Name Comes From

The **Kan complex** terminology was introduced for the Kan–Quillen model structure on **sSet**. Daniel Kan defined the horn-filling condition in 1958; Quillen (1967) elevated it to "the fibrancy condition for the model structure on **sSet**." The dual – cofibrations as monomorphisms – is forced by the lifting axiom plus the generating cofibrations $\partial\Delta^n \hookrightarrow \Delta^n$.

41.4.2 Top with the Quillen model structure

Theorem 41.14 (Quillen model structure on **Top**)**.** **Top** *(compactly generated weak Hausdorff) admits a model structure with:*

- *Weak equivalences: weak homotopy equivalences (isomorphisms on all π_n).*
- *Cofibrations: retracts of relative cell complexes, generated by the sphere inclusions $S^{n-1} \hookrightarrow D^n$.*
- *Fibrations:* ***Serre fibrations****: maps with RLP against $D^n \hookrightarrow D^n \times [0, 1]$.*

Every space is fibrant; cofibrant spaces are (retracts of) CW complexes.

Where the Name Comes From

Serre fibration is named for *Jean-Pierre Serre* (b. 1926, French mathematician, Fields Medal 1954). Serre's 1951 thesis introduced the notion of fibration with the homotopy lifting property restricted to maps from D^n — exactly the class characterizing the Quillen fibrations in **Top**. The corresponding cofibrations (retracts of relative cell complexes) trace back to J.H.C. Whitehead's work on CW complexes (1949).

41.4.3 The Quillen equivalence

Theorem 41.15 (Realization–singular Quillen equivalence)**.** *The adjunction $|-| \dashv \mathrm{Sing}$ of Chapter 39 is a* ***Quillen equivalence*** *between Quillen-***sSet** *and Quillen-***Top***: it descends to an equivalence of homotopy categories*

$$\mathrm{Ho}(\mathbf{sSet}) \simeq \mathrm{Ho}(\mathbf{Top}).$$

Remark 41.16 (The homotopy hypothesis)**.** Theorem 41.15 is the formal version of the **homotopy hypothesis**: the homotopy theory of spaces is equivalent to the homotopy theory of simplicial sets, hence — restricting to Kan complexes — to the homotopy theory of ∞-groupoids. Chapter 42 develops the proof via derived functors. We will use this equivalence throughout Volume IV to move between topological and simplicial language as convenient.

Where the Name Comes From

Quillen equivalence is the standard term for an adjoint pair that descends to an equivalence on homotopy categories — i.e., the strongest possible Quillen adjunction. The terminology dates to Quillen's original 1967 monograph and is preserved in the modern literature (Hovey 1999, Riehl 2014, etc.).

41.4.4 Preview: the Joyal model structure

Where the Name Comes From

The **Joyal model structure** on **sSet** is named for *André Joyal* (the same Joyal of Chapter 37's monoid theorem). Joyal developed the structure around 2000–2008, building on his earlier work on quasi-categories. The fibrant objects are quasi-cats; the model structure is the technical home for ∞-category theory in the simplicial-set framework (HTT Chapter 2; Cisinski 2019).

Remark 41.17 (Joyal's model structure)**. sSet** also admits a *different* model structure, due to Joyal, in which:

- Cofibrations are still monomorphisms (same as Kan–Quillen).
- Fibrations are **quasi-fibrations**: maps with RLP against *inner* horn inclusions $\Lambda^n_k \hookrightarrow \Delta^n$ $(0 < k < n)$.
- Weak equivalences are characterized by induced equivalences of homotopy 1-categories after fibrant replacement.

The fibrant objects are the **quasi-categories** (Chapter 43). The Joyal model structure makes **sSet** a model for ∞-categories exactly as Kan–Quillen makes it a model for ∞-groupoids.

41.5 Exercises

What is a model category?	A complete and cocomplete category with three classes of morphisms — weak equivalences $\mathcal{W}$, cofibrations $\mathcal{C}$of, fibrations $\mathcal{F}$ib — satisfying 2-out-of-3, retract closure, factorization, lifting, and completeness (M1–M5).
What is a trivial cofibration?	A cofibration that is also a weak equivalence. Equivalently: a map with LLP against all fibrations.
What is a trivial fibration?	A fibration that is also a weak equivalence. Equivalently: a map with RLP against all cofibrations.
State the lifting axiom (M4).	Cofibrations lift on the left against trivial fibrations; trivial cofibrations lift on the left against fibrations.
State the factorization axiom (M3).	Every map factors as cofib $\circ$ trivial fib, and also as trivial cofib $\circ$ fib.
Why is one of $\mathcal{W}, \mathcal{C}\mathrm{of}, \mathcal{F}\mathrm{ib}$ determined by the other two?	Lifting plus factorization: cofibrations are exactly LLP($\mathcal{W} \cap \mathcal{F}$ib), fibrations are RLP($\mathcal{W} \cap \mathcal{C}$of), and $\mathcal{W}$ is determined by either via the factorizations.
What is the left lifting property LLP($\mathcal{S}$)?	The class of maps i such that every square with i on the left and some $p \in \mathcal{S}$ on the right admits a diagonal lift.
What closure properties does LLP($\mathcal{S}$) enjoy?	Composition, pushouts, transfinite composition, retracts. These are the building blocks of "cell complex" arguments.
What is the small object argument?	Given a small set of morphisms $\mathcal{I}$ with small domains, every map factors functorially as (transfinite cell-pushout of $\mathcal{I}$) followed by (map with RLP against $\mathcal{I}$).
What is a cofibrantly generated model category?	One where $\mathcal{C}$of and $\mathcal{W} \cap \mathcal{C}$of are each generated by a small set under closure of LLP(RLP($\cdot$)). Both Quillen-**sSet** and Quillen-**Top** are cofibrantly generated.
Define cofibrant and fibrant.	X cofibrant: $\emptyset \to X$ is a cofibration. X fibrant: $X \to 1$ is a fibration. Bifibrant: both.
Why do we need replacement?	To represent arbitrary objects by bifibrant ones, on which the homotopy relation is well-behaved. Factorize $\emptyset \to X$ and $X \to 1$ to get QX and RX.
State the homotopy-category description.	If X is cofibrant and Y is fibrant, $\mathrm{Ho}(\mathcal{C})(X, Y) = \mathcal{C}(X, Y)/\sim$, where $\sim$ is the (left = right) homotopy equivalence relation.

What is the Kan–Quillen model structure on **sSet**?	Cofibrations = monomorphisms; fibrations = Kan fibrations (RLP against horn inclusions); weak equivalences = realization-weak-homotopy-equivalences. Fibrant objects = Kan complexes.
What is the Quillen model structure on **Top**?	Cofibrations = retracts of relative cell complexes (generated by $S^{n-1} \hookrightarrow D^n$); fibrations = Serre fibrations; weak equivalences = weak homotopy equivalences. Every space is fibrant; CW complexes are cofibrant.
What is the Quillen equivalence between **sSet** and **Top**?	The adjunction $\lvert - \rvert \dashv \mathrm{Sing}$ descends to an equivalence $\mathrm{Ho}(\mathbf{sSet}) \simeq \mathrm{Ho}(\mathbf{Top})$. This is the homotopy hypothesis.
What is the Joyal model structure?	Same cofibrations as Kan–Quillen; fibrations = inner-fibrations (RLP against *inner* horns only); weak equivalences different. Fibrant objects = quasi-categories (Chapter 43).
Why use cofibrant replacement before computing π_n?	Because π_n is only well-defined and homotopy-invariant on bifibrant objects. Cofibrant replacement ensures the inputs to representable functors are correctly resolved.
What's a cylinder object for X?	A factorization $X \sqcup X \hookrightarrow \mathrm{Cyl}(X) \xrightarrow{\sim} X$ of the fold map. Two maps $X \to Y$ are left-homotopic via a $\mathrm{Cyl}(X) \to Y$.
Path object dually?	$Y \xrightarrow{\sim} \mathrm{Path}(Y) \twoheadrightarrow Y \times Y$ factoring the diagonal. Right homotopies live here.
Why does the homotopy category have small hom-sets in model categories?	Because for bifibrant X, Y, hom-sets are quotients of $\mathcal{C}(X, Y)$ (small) by an equivalence relation, hence small. This solves the size issue of Gabriel–Zisman localization (Chapter 40).
Is **sSet** with the Joyal structure a model for ordinary categories?	No — for ∞-categories. Ordinary categories are the homotopy 1-category. Joyal-**sSet** retains higher coherence data (Chapter 43).
What's special about a Kan fibration?	It satisfies the homotopy lifting property for all simplicial sets, including for outer horns — modeling Serre fibrations of topological spaces.

Chapter Summary

Chapter Summary

Model category. Complete cocomplete category with three classes $\mathcal{W}, \mathcal{C}\mathrm{of}, \mathcal{F}\mathrm{ib}$ satisfying 2-out-of-3, retract closure, factorization, lifting, and completeness. Two classes determine the third via lifting properties.
Lifting and the small object argument. LLP/RLP define the classes from each other. The small object argument produces factorizations from generating sets of cofibrations.
Cofibrant/fibrant. $\emptyset \to X$ cof, $X \to 1$ fib; bifibrant when both. Replacement functors give $QX \xrightarrow{\sim} X \xrightarrow{\sim} RX$. The homotopy category between bifibrant objects is $\mathcal{C}(X, Y)/\sim$.
Two foundational examples. Kan–Quillen on **sSet** (fibrations = Kan fibrations; fibrant = Kan complexes); Quillen on **Top** (fibrations = Serre; cofibrant = CW). The adjunction $\lvert - \rvert \dashv \mathrm{Sing}$ is a Quillen equivalence — the homotopy hypothesis.
Joyal model structure preview. Same cofibrations as Kan–Quillen; fibrations only RLP against *inner* horns. Fibrant objects: quasi-categories. The home of ∞-category theory (Chapter 43).

Formal Restatement

Formal Restatement

Quillen's axioms. $(\mathcal{C}, \mathcal{W}, \mathcal{C}\mathrm{of}, \mathcal{F}\mathrm{ib})$ satisfying:

- (M1) 2-out-of-3 for $\mathcal{W}$.
- (M2) Retract closure for $\mathcal{W}, \mathcal{C}\mathrm{of}, \mathcal{F}\mathrm{ib}$.

- (M3) $\mathrm{Mor}(\mathcal{C}) = \mathcal{C}\mathrm{of} \circ (\mathcal{W} \cap \mathcal{F}\mathrm{ib}) = (\mathcal{W} \cap \mathcal{C}\mathrm{of}) \circ \mathcal{F}\mathrm{ib}$.
- (M4) $\mathcal{C}\mathrm{of} = \mathrm{LLP}(\mathcal{W} \cap \mathcal{F}\mathrm{ib})$; $\mathcal{F}\mathrm{ib} = \mathrm{RLP}(\mathcal{W} \cap \mathcal{C}\mathrm{of})$.
- (M5) $\mathcal{C}$ has all small (co)limits.

Small object argument. For $\mathcal{I}$ a small set of maps with small domains, every f factors functorially as $\mathrm{cell}(\mathcal{I}) \circ \mathrm{RLP}(\mathcal{I})$.

Replacement. $Q : \mathcal{C} \to \mathcal{C}_{\mathrm{cof}}$ with $QX \xrightarrow{\sim} X$; $R : \mathcal{C} \to \mathcal{C}_{\mathrm{fib}}$ with $X \xrightarrow{\sim} RX$. Both natural in X.

Homotopy category. $\mathrm{Ho}(\mathcal{C})(X, Y) = [QX, RY]$ where $[-, -]$ denotes left (= right) homotopy classes; this is the small-hom-class model for $\mathcal{C}[\mathcal{W}^{-1}]$.

Quillen-sSet.
- $\mathcal{W}$: realization weak homotopy equivalences.
- $\mathcal{C}\mathrm{of}$: monomorphisms; generators $\partial\Delta^n \hookrightarrow \Delta^n$.
- $\mathcal{F}\mathrm{ib}$: Kan fibrations; generators of $\mathcal{W} \cap \mathcal{C}\mathrm{of}$: $\Lambda^n_k \hookrightarrow \Delta^n$ for $0 \le k \le n$.

Quillen-Top.
- $\mathcal{W}$: weak homotopy equivalences.
- $\mathcal{C}\mathrm{of}$: retracts of relative cell complexes; generators $S^{n-1} \hookrightarrow D^n$.
- $\mathcal{F}\mathrm{ib}$: Serre fibrations; generators of $\mathcal{W} \cap \mathcal{C}\mathrm{of}$: $D^n \hookrightarrow D^n \times [0, 1]$.

Quillen equivalence. $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : \mathrm{Sing}$ is a Quillen equivalence; $\mathrm{Ho}(\mathbf{sSet}) \simeq \mathrm{Ho}(\mathbf{Top})$.

Chapter 42

Derived Functors and Quillen Equivalences

Chapter Overview

A functor between model categories rarely descends to homotopy categories directly: a typical $F : \mathcal{C} \to \mathcal{D}$ fails to preserve weak equivalences. The standard fix — **derive** F — is to precompose with cofibrant or fibrant replacement before evaluating. The resulting **total left** or **total right derived functor** $\mathbb{L}F, \mathbb{R}F : \mathrm{Ho}(\mathcal{C}) \to \mathrm{Ho}(\mathcal{D})$ is the homotopy-coherent extension.

A **Quillen adjunction** $F \dashv G$ is a pair compatible with the model structures: F preserves cofibrations and trivial cofibrations, G preserves fibrations and trivial fibrations. Such adjunctions induce **derived adjunctions** $\mathbb{L}F \dashv \mathbb{R}G$ on homotopy categories. A Quillen adjunction is a **Quillen equivalence** when the derived adjunction is an equivalence — the strongest comparison between two model categories.

This chapter develops the derived-functor machinery and applies it to prove the realization–singular Quillen equivalence (Theorem 41.41.15). The chapter closes with two previews of Volume IV's later applications: the bar construction as a derived left Kan extension along $\Delta_+ \to \mathcal{V}$ (foreshadowing ∞-monad bar resolutions of Chapter 51), and derived tensor / derived hom as the entry points to homological algebra (Chapter 54). The phase-one foundational machinery ends here; Chapter 43 turns to quasi-categories as the unifying ∞-categorical model.

42.1 Total Derived Functors

42.1.1 Informally

Most functors do not respect weak equivalences. Tensoring a chain complex with a non-flat module, for instance, does not preserve quasi-isomorphisms: if $C \xrightarrow{\sim} C'$ is a quasi-isomorphism and M is not flat, $C \otimes M$ and $C' \otimes M$ may have different homology. The **derived tensor product** $- \otimes^{\mathbb{L}} M$ fixes this by first replacing the input chain complex by a quasi-isomorphic flat complex (a *cofibrant replacement*) and then tensoring.

The pattern is universal: to derive a functor that fails to respect weak equivalences, replace inputs by cofibrant (or outputs by fibrant) objects before evaluating. The result depends only on the weak-equivalence class of the input, hence descends to $\mathrm{Ho}(\mathcal{C})$.

42.1.2 Formalizing

Where the Name Comes From

Derived functor dates to Cartan and Eilenberg's *Homological Algebra* (1956). For an additive functor F between abelian categories, the *left/right derived functors* L_nF and R^nF measure F's failure to be exact, by computing homology/cohomology of F applied to projective/injective resolutions. The term *derived* marks them as "descended from F" via the resolution machinery — they capture the deeper, derived information that F alone obscures. The Volume IV $\mathbb{L}F, \mathbb{R}F$ are the categorical generalization: derived functors as left/right Kan extensions of F along the localization γ.

Definition 42.1 (Total left and right derived functor)**.** Let $\mathcal{C}, \mathcal{D}$ be model categories and $F : \mathcal{C} \to \mathcal{D}$ a functor.

The **total left derived functor** of F, when it exists, is the left Kan extension of the composite $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{\gamma_{\mathcal{D}}} \mathrm{Ho}(\mathcal{D})$ along the localization $\gamma_{\mathcal{C}} : \mathcal{C} \to \mathrm{Ho}(\mathcal{C})$:

$$\mathbb{L}F \;=\; \mathrm{Lan}_{\gamma_{\mathcal{C}}}(\gamma_{\mathcal{D}} F).$$

The **total right derived functor** $\mathbb{R}F$ is dually the right Kan extension.

How to Say This

$\mathbb{L}F$ *(read: "the total left derived functor of F")*; $\mathbb{R}F$ *(read: "the total right derived functor of F")*. The blackboard-bold notation distinguishes the homotopical version from F itself. Some texts write LF and RF, or $F^{\mathbb{L}}$ and $F^{\mathbb{R}}$.

Theorem 42.2 (Pointwise formulas via replacement)**.** *If F sends weak equivalences between cofibrant objects to weak equivalences in $\mathcal{D}$, then $\mathbb{L}F$ exists and*

$$\mathbb{L}F(X) \;\cong\; F(QX) \quad \text{in } \mathrm{Ho}(\mathcal{D}),$$

where QX is a cofibrant replacement (Theorem 41.41.8). Dually, if F sends weak equivalences between fibrant objects to weak equivalences,

$$\mathbb{R}F(X) \;\cong\; F(RX).$$

Proof sketch. For $\mathbb{L}F$: define $\mathbb{L}F(X) = F(QX)$ on objects. Well-definedness on $\mathrm{Ho}(\mathcal{C})$: if $X \xrightarrow{\sim} X'$ is a weak equivalence, the induced $QX \xrightarrow{\sim} QX'$ between cofibrants becomes a weak equivalence under F by hypothesis, hence is an isomorphism in $\mathrm{Ho}(\mathcal{D})$. The left-Kan-extension universality follows from the universal property of cofibrant replacement: $QX \xrightarrow{\sim} X$ is initial among weak-equivalences-out-of-cofibrants into X. □ □

Where the Name Comes From

Ken Brown's lemma is named for *Kenneth S. Brown*, who formulated it in his 1973 paper "Abstract Homotopy Theory and Generalized Sheaf Cohomology." The lemma is a small but indispensable technical tool in homotopical algebra: it reduces the existence of derived functors from a strong invariance condition to a weaker (and checkable) one on trivial cofibrations.

Lemma 42.3 (Ken Brown's lemma)**.** *If $F : \mathcal{C} \to \mathcal{D}$ sends trivial cofibrations between cofibrant objects to weak equivalences, then F sends* all *weak equivalences between cofibrant objects to weak equivalences — so $\mathbb{L}F$ exists.*

Remark 42.4 (Ken Brown in practice)**.** Ken Brown's lemma reduces the hypothesis "preserves weak equivalences between cofibrants" to "preserves trivial cofibrations." That is exactly the half of the Quillen-functor condition that left adjoints often satisfy; the next section makes this systematic.

42.2 Quillen Adjunctions

42.2.1 Formalizing

Where the Name Comes From

A **Quillen adjunction** is named for *Daniel Quillen*, who introduced both the notion and the model-categorical framework in 1967. The asymmetry between "left Quillen" (preserves cof + trivial cof) and "right Quillen" (preserves fib + trivial fib) reflects the dual roles that cofibrations and fibrations play under adjunction. As Quillen put it: every adjunction between model categories is either compatible with the model structures or it isn't, and "compatible" has a small precise meaning.

Definition 42.5 (Quillen adjunction)**.** A **Quillen adjunction** between model categories $\mathcal{C}$ and $\mathcal{D}$ is an adjoint pair $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$, $F \dashv G$, such that any (equivalently, all) of the following hold:

1. F preserves cofibrations and trivial cofibrations.
2. G preserves fibrations and trivial fibrations.
3. F preserves cofibrations and G preserves fibrations.

We call F **left Quillen** and G **right Quillen**.

Theorem 42.6 (Equivalence of Quillen-adjunction conditions)**.** *For an adjunction $F \dashv G$, the three conditions in Definition 42.5 are equivalent.*

Proof sketch. (1) ⇔ (2): cofibrations are exactly LLP against trivial fibrations; by the adjunction $F \dashv G$, a square involving $F(i)$ and p has a lift iff the adjoint square involving i and $G(p)$ does. So F preserves cofibrations iff G preserves trivial fibrations. Dually for trivial cofibrations and fibrations.

(1) ⇒ (3) is trivial. (3) ⇒ (1): if F preserves cofibrations, then F preserves LLP-against-trivial-fibrations; combined with G preserving fibrations, F also preserves trivial cofibrations (via LLP-against-fibrations). □ □

Theorem 42.7 (Derived adjunction)**.** *A Quillen adjunction $F \dashv G$ induces an adjunction of total derived functors*

$$\mathbb{L}F \;\dashv\; \mathbb{R}G \quad : \quad \mathrm{Ho}(\mathcal{C}) \rightleftarrows \mathrm{Ho}(\mathcal{D}).$$

Proof sketch. By Ken Brown's lemma, F preserving trivial cofibrations between cofibrants implies $\mathbb{L}F$ exists; dually for $\mathbb{R}G$. The adjunction descends because the hom-set bijection $\mathcal{D}(FX, Y) \cong \mathcal{C}(X, GY)$ is compatible with the equivalence relation defining Ho: replacing X by QX on the left and Y by RY on the right gives

$$\mathrm{Ho}(\mathcal{D})(\mathbb{L}F(X), Y) \;\cong\; \mathrm{Ho}(\mathcal{C})(X, \mathbb{R}G(Y)). \quad \square$$

$\square$

Example 42.8 (Realization–singular is Quillen)**.** The adjunction $|-| \dashv \mathrm{Sing}$ between $\mathbf{sSet}_{\mathrm{Q}}$ and $\mathbf{Top}_{\mathrm{Q}}$ is a Quillen adjunction. Verification: $|-|$ preserves monomorphisms (cofibrations of **sSet**) since it preserves colimits and monos in **Top** are pullback-stable. It also preserves horn inclusions $\Lambda^n_k \hookrightarrow \Delta^n$ (trivial cofibrations of **sSet**), since their realizations are deformation retracts — hence trivial cofibrations in **Top**.

42.3 Quillen Equivalences

42.3.1 Formalizing

Definition 42.9 (Quillen equivalence)**.** A Quillen adjunction $F \dashv G$ is a **Quillen equivalence** if any (equivalently, all) of the following hold:

1. The derived adjunction $\mathbb{L}F \dashv \mathbb{R}G$ is an equivalence of categories on $\mathrm{Ho}(\mathcal{C}) \simeq \mathrm{Ho}(\mathcal{D})$.
2. For every cofibrant $X \in \mathcal{C}$ and fibrant $Y \in \mathcal{D}$, a map $FX \to Y$ is a weak equivalence in $\mathcal{D}$ iff its adjunct $X \to GY$ is a weak equivalence in $\mathcal{C}$.
3. For every cofibrant $X \in \mathcal{C}$, the composite $X \to GFX \xrightarrow{G(\text{fib repl})} GRFX$ is a weak equivalence in $\mathcal{C}$; and dually for Y fibrant.

Theorem 42.10 (Realization–singular Quillen equivalence)**.** *The adjunction* $|-| \dashv \mathrm{Sing}$ *between* $\mathbf{sSet}_{\mathrm{Q}}$ *and* $\mathbf{Top}_{\mathrm{Q}}$ *is a Quillen equivalence.*

Proof sketch. Use condition (2). For $X \in \mathbf{sSet}$ cofibrant (every simplicial set is) and $Y \in \mathbf{Top}$ fibrant (every space is), the question is whether $|X| \to Y$ is a weak homotopy equivalence iff $X \to \mathrm{Sing}\, Y$ is.

Both directions reduce to the unit and counit:

- Unit $\eta_X : X \to \mathrm{Sing}\,|X|$. By a theorem of Milnor, this is a weak equivalence in **sSet** for every X: $\mathrm{Sing}\,|X|$ and X have the same simplicial homotopy groups (Chapter 38's π_n definition, via inner-horn filling).
- Counit $\epsilon_Y : |\mathrm{Sing}\, Y| \to Y$. This is a weak homotopy equivalence for every space Y (Quillen's theorem), classically equivalent to Y admitting a CW approximation.

Both unit and counit being weak equivalences is condition (3); hence the adjunction is a Quillen equivalence. $\square$ $\square$

Remark 42.11 (The homotopy hypothesis, formalized)**.** Theorem 42.10 is the precise statement of the *homotopy hypothesis* we previewed in Chapter 39: simplicial sets and topological spaces present equivalent homotopy theories. From this point onward in Volume IV, we may freely move between **sSet** and **Top** at the level of homotopy categories.

42.4 Examples and Bridges

42.4.1 The bar construction as a derived left Kan extension

Example 42.12 (Derived bar)**.** For a monoid M in a strict monoidal model category $(\mathcal{V}, \otimes, I)$, the bar construction $B_\bullet M : \Delta^{\mathrm{op}} \to \mathcal{V}$ (Chapter 37 Corollary 37.18) gives a simplicial object. Its **geometric realization** $|B_\bullet M|$ — the coend $\int^{[n]} M^{\otimes(n+1)} \otimes |\Delta^n|$ — computes the derived colimit

$$\mathbb{L}\mathrm{colim}_{\Delta^{\mathrm{op}}} B_\bullet M \;\simeq\; |B_\bullet M|.$$

This is the model-categorical incarnation of "the universal property of Δ_+ derived to its homotopical version." The ∞-monad machinery of Chapter 51 makes this systematic.

42.4.2 Derived tensor and Ext

Where the Name Comes From

Tor and **Ext** are the prototypical derived functors, named in Cartan–Eilenberg (1956): *Tor* (from *torsion*) because it measures the torsion-like obstruction to flatness of tensor product, and *Ext* (from *extension*) because $\mathrm{Ext}^1(M, N)$ classifies extensions $0 \to N \to E \to M \to 0$. The categorical $\mathbb{L}\otimes$ and $\mathbb{R}\mathrm{Hom}$ are the modern ∞-categorical packaging, with H_n recovering Tor and H^n recovering Ext.

Example 42.13 (Derived tensor product)**.** Let R be a ring, M a right R-module, N a left R-module. Cofibrantly resolve M in the projective model structure on $\mathbf{Ch}(\text{Mod-}R)$ to obtain a chain complex $P_\bullet \xrightarrow{\sim} M$ of projective modules. Then

$$M \otimes^{\mathbb{L}}_R N \;=\; P_\bullet \otimes_R N,$$

the **derived tensor product**, computes the Tor groups $\mathrm{Tor}_n^R(M,N)$ as $H_n(P_\bullet \otimes_R N)$.

Example 42.14 (Derived hom)**.** Dually, $\mathbb{R}\mathrm{Hom}_R(M,N) = \mathrm{Hom}_R(P_\bullet, N)$ where $P_\bullet \xrightarrow{\sim} M$ is a projective resolution (or $N \xrightarrow{\sim} I^\bullet$ an injective resolution), with homology computing $\mathrm{Ext}_R^n(M,N) = H^n\mathbb{R}\mathrm{Hom}_R(M,N)$.

Remark 42.15 (Phase 1 closes here)**.** With derived functors and Quillen equivalences in place, the simplicial / model-categorical foundations of Volume IV are complete. The next phase (Chapters 43–48) builds the theory of quasi-categories on this foundation. The key takeaway: ∞-category theory will generalize *both* ordinary categories (via the nerve) *and* simplicial homotopy types (via Kan complexes), unifying the two pictures under a single combinatorial framework.

42.5 Exercises

Define the total left derived functor $\mathbb{L}F$.	The left Kan extension of $\gamma_{\mathcal{D}} \circ F$ along the localization $\gamma_{\mathcal{C}}$; pointwise $\mathbb{L}F(X) \simeq F(QX)$ when F preserves weak equivalences between cofibrants.
What does $\mathbb{R}F$ compute?	Total right derived functor: $\mathbb{R}F(X) \simeq F(RX)$ via fibrant replacement, when F preserves weak equivalences between fibrants.
State Ken Brown's lemma.	If F preserves trivial cofibrations between cofibrant objects, it preserves all weak equivalences between cofibrant objects.
Why is Ken Brown's lemma useful?	It reduces the existence of $\mathbb{L}F$ to checking F preserves trivial cofibrations — a much simpler condition than "preserves all weak equivalences."
Define a Quillen adjunction.	An adjunction $F \dashv G$ with F left Quillen (preserves cof and trivial cof) and G right Quillen (preserves fib and trivial fib).
Why are the three conditions in Definition 42.5 equivalent?	Cofibrations are LLP against trivial fibrations, and LLP/RLP are exchanged by the adjunction. So F preserves cof iff G preserves trivial fib, etc.
State the derived adjunction theorem.	A Quillen adjunction $F \dashv G$ induces $\mathbb{L}F \dashv \mathbb{R}G : \mathrm{Ho}(\mathcal{C}) \rightleftarrows \mathrm{Ho}(\mathcal{D})$.
Define a Quillen equivalence.	A Quillen adjunction whose derived adjunction is an equivalence of homotopy categories.
Equivalent characterization in terms of unit/counit?	$X \xrightarrow{\eta} GFX \xrightarrow{G(\sim)} GRFX$ is a weak equivalence for every cofibrant X, and the dual for fibrant Y.
Why is $\lvert - \rvert \dashv \mathrm{Sing}$ a Quillen equivalence?	Unit $X \to \mathrm{Sing}\,\lvert X\rvert$ is a weak equivalence (Milnor); counit $\lvert \mathrm{Sing}\, Y\rvert \to Y$ is a weak equivalence (Quillen / CW approximation).
Why is $\lvert - \rvert$ left Quillen?	It preserves monomorphisms (so cofibrations of **sSet**) and sends horn inclusions to deformation retracts (so trivial cofibrations).
Why is Sing right Quillen?	By adjunction with $\lvert - \rvert$ being left Quillen — or directly: $\mathrm{Sing}(p)$ is a Kan fibration iff p is a Serre fibration.
What does $\mathbb{L}F$ compute when F already preserves weak equivalences?	Just F itself, descended to $\mathrm{Ho}(\mathcal{C})$. The derivation is trivial.
When is $F : \mathcal{C} \to \mathcal{D}$ *homotopical*?	When F preserves *all* weak equivalences. Such F has $\mathbb{L}F = \mathbb{R}F = F_{\mathrm{Ho}}$.
What is the derived tensor product?	$M \otimes_R^{\mathbb{L}} N = P_\bullet \otimes_R N$, where $P_\bullet \xrightarrow{\sim} M$ is a projective resolution. H_n recovers $\mathrm{Tor}_n^R(M,N)$.
What is derived hom?	$\mathbb{R}\mathrm{Hom}_R(M,N) = \mathrm{Hom}_R(P_\bullet, N)$ for $P_\bullet$ projective resolution of M; cohomology recovers $\mathrm{Ext}_R^n(M,N)$.

Why does Tor not just equal tensor at the chain level?	Because tensoring with a non-flat module doesn't preserve quasi-isomorphisms; deriving via cofibrant replacement (projective resolution) corrects this.
What is the homotopy colimit?	hocolim $= \mathbb{L}$colim. For a simplicial diagram, $\mathrm{hocolim}_\Delta = \lvert - \rvert$ (geometric realization of the simplicial replacement).
The bar construction as a derived colimit?	$\lvert B_\bullet M\rvert \simeq \mathbb{L}\mathrm{colim}_{\Delta^{\mathrm{op}}} B_\bullet M$ — the realization of the bar simplicial object is the derived colimit.
What's a Bousfield localization?	A new model structure on the same underlying category and cofibrations, with a strictly larger class of weak equivalences. Not developed here, but central in advanced applications.
Does every Quillen adjunction yield a Quillen equivalence?	No — only those that, in addition, witness that no homotopy-theoretic information is lost. The realization–singular adjunction does; many others don't.
What is a homotopy hypothesis?	The claim that ∞-groupoids and homotopy types coincide. Theorem 42.10 is its model-categorical version: $\mathbf{sSet}_{\mathrm{Q}} \simeq_{\mathrm{Q}} \mathbf{Top}_{\mathrm{Q}}$.
What's the next phase of Volume IV?	Quasi-categories (Chapter 43) — combining the nerve perspective (Chapter 39) with the Joyal model structure (Chapter 41 preview) to give a single combinatorial model of ∞-categories.

Chapter Summary

Chapter Summary

Total derived functor. $\mathbb{L}F = \mathrm{Lan}_\gamma(\gamma_\mathcal{D} F)$ — pointwise $\mathbb{L}F(X) \simeq F(QX)$ via cofibrant replacement. Dually $\mathbb{R}F(X) \simeq F(RX)$.
Ken Brown. Preserving trivial cofibrations between cofibrants suffices for $\mathbb{L}F$ to exist. The right-handed dual gives $\mathbb{R}F$.
Quillen adjunction. $F \dashv G$ with F left Quillen (preserves cof and trivial cof) and G right Quillen (dually). The conditions are exchanged by adjunction. Induces derived adjunction $\mathbb{L}F \dashv \mathbb{R}G$ on homotopy categories.
Quillen equivalence. Derived adjunction is an equivalence. Realization–singular $\lvert - \rvert \dashv \mathrm{Sing}$ is the canonical example: simplicial sets and topological spaces are Quillen-equivalent. This is the homotopy hypothesis formalized.
Bridges. The bar construction realizes as a derived colimit; derived tensor computes Tor; derived hom computes Ext. Phase 1 foundations close here; Phase 2 (Chapter 43+) builds quasi-categories on top.

Formal Restatement

Formal Restatement

Total derived functors.
$$\mathbb{L}F = \mathrm{Lan}_{\gamma_\mathcal{C}}(\gamma_\mathcal{D} F), \qquad \mathbb{R}F = \mathrm{Ran}_{\gamma_\mathcal{C}}(\gamma_\mathcal{D} F).$$
Pointwise: $\mathbb{L}F(X) \simeq F(QX)$, $\mathbb{R}F(X) \simeq F(RX)$.

Ken Brown's lemma. If F sends trivial cofibrations between cofibrants to weak equivalences, F sends all weak equivalences between cofibrants to weak equivalences.

Quillen adjunction. $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ with F preserving $\mathcal{C}$of and $\mathcal{W} \cap \mathcal{C}$of (equivalently G preserving $\mathcal{F}$ib and $\mathcal{W} \cap \mathcal{F}$ib).

Derived adjunction. $\mathbb{L}F \dashv \mathbb{R}G : \mathrm{Ho}(\mathcal{C}) \rightleftarrows \mathrm{Ho}(\mathcal{D})$.

Quillen equivalence. Quillen adjunction whose derived adjunction is an equivalence. Equivalently: for X cofibrant, Y fibrant, $FX \to Y$ is a w.e. in $\mathcal{D}$ iff $X \to GY$ is a w.e. in $\mathcal{C}$.

Homotopy hypothesis (Theorem). $\lvert - \rvert : \mathbf{sSet}_{\mathrm{Q}} \rightleftarrows \mathbf{Top}_{\mathrm{Q}} : \mathrm{Sing}$ is a Quillen equivalence; $\mathrm{Ho}(\mathbf{sSet}) \simeq \mathrm{Ho}(\mathbf{Top})$.

Derived algebra. $\operatorname{Tor}_n^R(M, N) = H_n(M \otimes_R^{\mathbb{L}} N)$; $\operatorname{Ext}_R^n(M, N) = H^n \mathbb{R}\operatorname{Hom}_R(M, N)$. Cofibrant replacement = projective resolution; fibrant replacement on opposite side = injective resolution.

Chapter 43

Quasi-Categories: Definition and First Examples

Chapter Overview

A **quasi-category** is a simplicial set in which every inner horn admits a filler. The condition is one weakening of the nerve characterization (Theorem 39.39.9) — drop the uniqueness; keep only existence — and the resulting structure is the working model of an ∞-category that organizes Volume IV's remaining chapters.

This chapter introduces the definition, builds the dictionary between quasi-categorical and ordinary-categorical vocabulary, and works through the first examples. We meet:

- Nerves of ordinary categories (Chapter 39): quasi-categories with unique fillers.
- Kan complexes (Chapter 39): quasi-categories whose mapping spaces are all ∞-groupoids.
- Differential graded categories (locally) and topological categories (after a coherent-nerve construction): the bridge to practice.

The technical heart of the chapter is the construction of the **homotopy category** $\mathrm{ho}(\mathcal{C})$ of a quasi-category $\mathcal{C}$, an ordinary category whose objects are the 0-simplices of $\mathcal{C}$ and whose morphisms are equivalence classes of 1-simplices under a specific homotopy relation defined via 2-simplex fillers. The adjunction $\tau_1 \dashv \mathrm{N}$ of Chapter 39 restricts to a functor $\mathrm{ho} : \mathbf{QCat} \to \mathbf{Cat}$ — the first descent from the ∞-categorical to the 1-categorical world. Mapping spaces, equivalences, the Joyal model structure, joins/slices, and the fibrations of Cluster D all build on the basic combinatorics established here.

43.1 Quasi-Categories

43.1.1 Informally

A quasi-category is what one gets when "a category up to coherent homotopy" is presented combinatorially. Take a simplicial set and read its 1-simplices as morphisms, its 0-simplices as objects. The *composition* of two composable 1-simplices f, g is not a single 1-simplex but a 2-simplex whose two short edges are f, g and whose long edge is the composite. Such a 2-simplex always exists (this is exactly the inner-horn filling condition), but it is not unique — different fillers give different composites, all related by higher-dimensional simplices.

43.1.2 Formalizing

Where the Name Comes From

Quasi-category was coined by *André Joyal* around 2002, following an earlier name **weak Kan complex** introduced by *Boardman* and *Vogt* in their 1973 book *Homotopy Invariant Algebraic Structures on Topological Spaces*. "Quasi-" (Latin *quasi*, "as if, almost") marks the structure as *almost a category*: composition exists but is not strictly unique; associativity holds up to coherent homotopy rather than on the nose. The Joyal terminology has won out because it is shorter, more mnemonic, and clearly distinguishes the class from Kan complexes (which have a stronger condition, not weaker).

Where the Name Comes From

The alternative name **∞-category** reflects the modern viewpoint: a quasi-category is a categorical structure with *∞-many levels of morphisms* (objects, 1-morphisms, 2-morphisms between those, 3-morphisms, ...), where coherence is witnessed by higher-dimensional simplices rather than by axioms. The "∞" marks the unbounded tower of higher cells. The two names are interchangeable in this book; "quasi-category" emphasizes the simplicial combinatorics, "∞-category" emphasizes the categorical content.

Definition 43.1 (Quasi-category)**.** A **quasi-category** (also called an **∞-category** or, in older literature, **weak Kan complex**) is a simplicial set $\mathcal{C}$ in which every inner horn admits a filler: for $n \geq 2$ and $0 < k < n$, every map $\Lambda^n_k \to \mathcal{C}$ extends to $\Delta^n \to \mathcal{C}$.

How to Say This

$\mathcal{C}$ *(read: "a quasi-category $\mathcal{C}$")* or *(read: "an ∞-category $\mathcal{C}$")*; we use both interchangeably, preferring "quasi-category" when emphasizing the simplicial-set combinatorics and "∞-category" when emphasizing the categorical content. The category **QCat** is the full subcategory of **sSet** on quasi-categories.

Example 43.2 (Nerves of categories)**.** Every nerve $\mathrm{N}(\mathcal{D})$ of an ordinary category $\mathcal{D}$ is a quasi-category. The inner-horn filler is in fact unique (Theorem 39.39.9), so nerves are "1-truncated" quasi-categories. Conversely, a quasi-category with unique inner-horn fillers is a nerve.

Example 43.3 (Kan complexes)**.** Every Kan complex (Definition 39.39.11) is a quasi-category: inner horns are a subset of all horns. In particular, $\mathrm{Sing}(T)$ for any topological space T is a quasi-category — specifically, an ∞-groupoid (every morphism is invertible up to coherent homotopy).

Example 43.4 (Joyal model fibrant simplicial sets)**.** The fibrant objects of the Joyal model structure on **sSet** (Remark 41.41.17) are exactly the quasi-categories. This is the precise sense in which **sSet** "presents" ∞-categories: $\mathrm{Ho}(\mathbf{sSet}_{\mathrm{Joyal}})$ is the homotopy category of ∞-categories.

Remark 43.5 (Why drop uniqueness?)**.** Ordinary category theory enforces that composites are unique by axiom. In "higher-dimensional" settings — chain complexes up to chain homotopy, spaces up to coherent homotopy, derived categories with their triangulated structure — composites are unique only up to a higher-dimensional equivalence, which itself is unique up to a still-higher equivalence, and so on. Quasi-categories present this entire tower with a single combinatorial structure: the simplices of dimension ≥ 2 are the witnesses to all the higher coherences.

43.2 The Homotopy Category

43.2.1 Formalizing

A quasi-category has too much higher-dimensional data to be a category in the ordinary sense — but it has a quotient that is.

Reader's Annotation

The "homotopy category" of a quasi-category $\mathcal{C}$ is the 1-*categorical shadow* obtained by collapsing all the higher coherence data: a single 1-category with the same objects and hom-sets given by equivalence classes of 1-simplices. The map $\mathcal{C} \to \mathrm{ho}(\mathcal{C})$ is structurally analogous to "take π_0 of mapping spaces" — the construction discards all higher homotopical information and keeps only the categorical skeleton.

Definition 43.6 (Homotopy between 1-simplices)**.** Let $\mathcal{C}$ be a quasi-category and $f, g : x \to y$ two 1-simplices with the same source x and target y (i.e., $d_1 f = d_1 g = x$, $d_0 f = d_0 g = y$). We say f is **homotopic** to g, written $f \sim g$, if there exists a 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$ whose boundary is

$$d_0\sigma = g, \quad d_1\sigma = f, \quad d_2\sigma = s_0 x,$$

i.e., a 2-simplex with edges g (face opposite 0), f (face opposite 1), and the degenerate identity $s_0 x$ on x (face opposite 2).

Lemma 43.7 (Homotopy is an equivalence relation)**.** *The relation $\sim$ on 1-simplices with fixed source and target is an equivalence relation: reflexive, symmetric, and transitive.*

Proof sketch. Reflexive. Use the degenerate $s_1 f : \Delta^2 \to \mathcal{C}$ with $d_0(s_1 f) = f$, $d_1(s_1 f) = f$, $d_2(s_1 f) = s_0 x$.

Symmetric. Given a witnessing 2-simplex for $f \sim g$ together with the degenerate $s_1 g$ for $g \sim g$, fill the inner horn Λ^3_2 they form to produce a witness for $g \sim f$.

Transitive. Given witnesses for $f \sim g$ and $g \sim h$, together with $s_0 x$, fill the inner horn Λ^3_1 to produce a witness for $f \sim h$. □ □

Definition 43.8 (Homotopy category)**.** The **homotopy category** $\mathrm{ho}(\mathcal{C})$ of a quasi-category $\mathcal{C}$ is the ordinary category with:

- Objects: $\mathrm{Ob}(\mathrm{ho}(\mathcal{C})) = \mathcal{C}_0$.
- Morphisms $\mathrm{ho}(\mathcal{C})(x, y) = \mathcal{C}_1(x, y)/\sim$, equivalence classes of 1-simplices with source x, target y.
- Composition: for $[f] : x \to y$ and $[g] : y \to z$, choose 2-simplex fillers of the inner horn $\Lambda^2_1 \to \mathcal{C}$ on (f, g) and set $[g] \circ [f] = [d_1\sigma]$ where σ is any such filler.
- Identity: $[s_0 x] : x \to x$.

Theorem 43.9 (Well-definedness of ho)**.** *Composition in $\mathrm{ho}(\mathcal{C})$ does not depend on the choice of 2-simplex filler, nor on the representatives f, g within equivalence classes. Associativity and unit laws hold strictly in $\mathrm{ho}(\mathcal{C})$.*

Proof sketch. Filler-independence. Two fillers σ, σ' of the same inner horn glue into an inner horn Λ^3_1 whose filling witnesses $d_1\sigma \sim d_1\sigma'$.

Representative-independence. Replacing f by f' with $f \sim f'$ substitutes the relevant 2-simplex faces by homotopic ones; the Λ^3 horn-filling argument transports the conclusion.

Associativity. Given three composable 1-simplices, fill a Λ^4_2 horn with the relevant 2-cells as faces. The 4-simplex filler witnesses associativity.

Units. Degenerate 1-simplices s_0x are units for the composition by direct simplex-counting. □ □

Corollary 43.10 (Functor ho : **QCat** → **Cat**)**.** *The assignment* $\mathcal{C} \mapsto \mathrm{ho}(\mathcal{C})$ *extends to a functor* $\mathrm{ho} : \mathbf{QCat} \to \mathbf{Cat}$*, left adjoint to the nerve* $\mathrm{N} : \mathbf{Cat} \to \mathbf{QCat}$*.*

Proof sketch. Extend $\tau_1 \dashv \mathrm{N}$ of Theorem 39.39.8 to the full subcategory **QCat**; the adjunction restricts. $\mathrm{ho}(\mathcal{C}) = \tau_1(\mathcal{C})$ for $\mathcal{C}$ a quasi-category. □ □

Remark 43.11 ("Homotopy 1-category")**.** The functor ho flattens an ∞-category to its underlying 1-category by collapsing all higher coherence data. The kernel of this flattening is exactly the higher-categorical content that distinguishes quasi-categories from ordinary categories.

43.3 Bridges and Working Examples

43.3.1 Differential graded categories

Where the Name Comes From

A **differential graded category** (dg-category) is a category enriched over chain complexes: hom-objects are chain complexes (with differential), composition is a chain map. The dg-nerve $\mathrm{N}^{\mathrm{dg}}(\mathcal{D})$ converts a dg-category to a quasi-category by encoding the chain-complex structure via $\mathbb{E}_\infty$-multiplicative simplicial data. Hinich (1997) and Tabuada (2005) developed the modern theory; the resulting ∞-categories model triangulated/derived categories and are the standard home of homological algebra in higher form (Chapter 53).

Remark 43.12 (The dg-nerve)**.** For a differential graded (dg) category $\mathcal{D}$ over a ring k, the **dg-nerve** $\mathrm{N}^{\mathrm{dg}}(\mathcal{D})$ is a quasi-category whose n-simplices are coherent diagrams in $\mathcal{D}$ together with $\mathbb{E}_\infty$-multiplicative homotopies between their composites. This construction sends dg-categories to quasi-categories in a way compatible with the homotopy theory of dg-categories. It is the standard source of *stable* quasi-categories arising in homological algebra (Chapter 53).

43.3.2 Topological categories

Where the Name Comes From

The **coherent nerve** $\mathrm{N}^{\mathrm{coh}}$ is also called the *simplicial nerve* or *Cordier nerve*, after *Jean-Marc Cordier* (1982) who introduced an early version. Lurie's 2009 book *Higher Topos Theory* established the modern form as left adjoint to the rigidification functor $\mathfrak{C}$. The "coherent" modifier reflects that the nerve produces homotopy-coherent rather than strict diagrams.

Remark 43.13 (The coherent nerve)**.** A **topological category** (or **simplicial category**) is a category enriched in **Top** (or **sSet**). Lurie's **coherent nerve** $\mathrm{N}^{\mathrm{coh}} : \mathbf{sCat} \to \mathbf{sSet}$ sends each simplicial category to a simplicial set whose n-simplices are "coherent" chains of morphisms. The image of the coherent nerve lands in quasi-categories, and the functor descends to an equivalence of ∞-categories

$$\mathbf{sCat} \;\simeq_{\mathrm{Q}}\; \mathbf{sSet}_{\mathrm{Joyal}}$$

(a Quillen equivalence) — saying that the two definitions of "∞-category" (simplicial categories vs. quasi-categories) agree.

43.3.3 The ∞-category of spaces

Where the Name Comes From

The name **anima** for an object of $\mathcal{S}$ was popularized recently (post-2018) by *Cisinski*, *Khan*, and the broader French school. The Latin *anima* ("soul, life") was suggested to displace the historically loaded names "space," "homotopy type," and "∞-groupoid." Each prior name has connotations that don't always fit; *anima* carries the spirit of the object without the baggage. The terminology is gaining adoption in younger circles; "space" remains the dominant usage in HTT-style literature.

Example 43.14 (The ∞-category $\mathcal{S}$ of spaces (anima))**.** The **∞-category of spaces** (or **anima**) is the quasi-category

$$\mathcal{S} \;=\; \mathrm{N}^{\mathrm{coh}}(\mathbf{Kan}),$$

the coherent nerve of the simplicial category of Kan complexes (with mapping spaces $\mathrm{Map}_{\mathbf{sSet}}(X, Y)$). Equivalently, $\mathcal{S}$ is the ∞-categorical localization of **sSet** at weak equivalences. Its homotopy category $\mathrm{ho}(\mathcal{S}) \simeq \mathrm{Ho}(\mathbf{Top})$ is the classical homotopy category of spaces.

Remark 43.15 (Notation for spaces in $\mathcal{S}$)**.** We will write objects of $\mathcal{S}$ as *anima* (Lurie's term), *spaces*, or simply "∞-groupoids," depending on emphasis. The notation $\mathcal{S}$ for the ∞-category of spaces is standard in Lurie's *Higher Topos Theory* and *Kerodon*.

43.3.4 Three pictures of an ∞-category

Remark 43.16 (Three models, one homotopy theory)**.** There are at least three established combinatorial models for ∞-categories:

- *Quasi-categories* (Joyal–Lurie): simplicial sets with inner-horn fillers. Our working model in Volume IV.
- *Simplicial categories* (**sCat**): categories enriched in **sSet**.
- *Complete Segal spaces* (Rezk): bisimplicial sets satisfying completeness conditions.

All three are connected by Quillen equivalences (Bergner, Joyal–Tierney, Lurie); the choice of model is a matter of which combinatorics is most convenient. Volume IV exclusively uses quasi-categories; readers graduating to Lurie's *Higher Topos Theory* encounter all three.

43.4 Exercises

Define a quasi-category.	A simplicial set in which every inner horn $\Lambda^n_k \to \mathcal{C}$ (with $0 < k < n$) extends to $\Delta^n \to \mathcal{C}$.
What's the relation to nerves?	Nerves of ordinary categories are exactly the quasi-categories with *unique* inner-horn fillers. Dropping uniqueness gives quasi-categories proper.
Why are Kan complexes quasi-categories?	Inner horns are a subset of all horns; if every horn fills, in particular every inner one does.
What's the heuristic for inner-horn filling?	Composition: $\Lambda^2_1 \to \mathcal{C}$ is two composable 1-simplices; the filler $\Delta^2 \to \mathcal{C}$ provides a composite as the long edge, plus a 2-cell witnessing it.
When is composition unique?	Never strictly — only up to a Λ^3_1-filler-witnessed homotopy. In the nerve of an ordinary category, this homotopy is forced to be the identity by the uniqueness of fillers.
Define homotopy of 1-simplices.	$f \sim g$ iff there is a $\Delta^2 \to \mathcal{C}$ with edges f, g, and $s_0 x$ on $x = d_1 f$.
Why is $\sim$ an equivalence relation?	Reflexivity uses degenerate 2-simplices; symmetry and transitivity use Λ^3-filler arguments built from the given witnesses plus degeneracies.
Construct $\mathrm{ho}(\mathcal{C})$.	Objects $\mathcal{C}_0$, morphisms $\mathcal{C}_1/\sim$, composition by inner-horn filler (well-defined up to $\sim$), identities $[s_0 x]$.
Why is composition in $\mathrm{ho}(\mathcal{C})$ well-defined?	Two fillers of the same horn produce homotopic long edges (by a Λ^3_1 filler); replacing f, g by homotopic representatives also produces homotopic composites.
Why does associativity hold in $\mathrm{ho}(\mathcal{C})$?	Three composable 1-simplices plus their two composites give the data for a Λ^4_2-horn, whose filler is a 4-simplex witnessing associativity at the homotopy level.
State the adjunction $\mathrm{ho} \dashv \mathrm{N}$.	$\mathbf{Cat}(\mathrm{ho}(\mathcal{C}), \mathcal{D}) \cong \mathbf{QCat}(\mathcal{C}, \mathrm{N}(\mathcal{D}))$. Maps into a 1-category factor through the homotopy 1-category.
What does ho destroy?	All higher-dimensional simplices beyond their incidence with 1-simplices — i.e., all higher coherence data.
Name three models of ∞-categories.	Quasi-categories (Joyal–Lurie); simplicial categories (**sSet**-enriched); complete Segal spaces (Rezk).

Why does Volume IV use quasi-categories?	They are the most combinatorially direct: simplicial sets with a single horn-filling condition. The other models are Quillen-equivalent but require more setup.
What is the dg-nerve?	$\mathrm{N}^{\mathrm{dg}}(\mathcal{D})$: a quasi-category from a dg-category, whose simplices encode coherent chain homotopies. The source of stable ∞-categories in homological algebra.
What is the coherent nerve?	$\mathrm{N}^{\mathrm{coh}} : \mathbf{sCat} \to \mathbf{sSet}$ sending a simplicial category to a quasi-category. Lands in $\mathbf{QCat}$; descends to a Quillen equivalence $\mathbf{sCat} \simeq_{\mathrm{Q}} \mathbf{sSet}_{\mathrm{Joyal}}$.
What is $\mathcal{S}$?	The ∞-category of spaces (anima); $\mathcal{S} = \mathrm{N}^{\mathrm{coh}}(\mathbf{Kan})$. Equivalently, the ∞-localization of $\mathbf{sSet}$ at weak equivalences.
What's $\mathrm{ho}(\mathcal{S})$?	The classical homotopy category $\mathrm{Ho}(\mathbf{Top})$ of topological spaces (or equivalently, $\mathrm{Ho}(\mathbf{sSet})$).
Distinguish "∞-category" and "quasi-category."	Same thing in this book. "Quasi-category" emphasizes simplicial-set combinatorics; "∞-category" emphasizes the categorical content.
What is an "∞-groupoid"?	A quasi-category in which every morphism is an equivalence — equivalently, a Kan complex. The objects of $\mathcal{S}$.
Why doesn't quasi-category composition simply pick a chosen filler?	Functoriality would require coherent choice across all horns, and no such global choice exists in general — choosing strictly would specialize to nerves of categories.
What's the next chapter's contribution?	Mapping spaces $\mathrm{Map}_{\mathcal{C}}(x, y)$ as Kan complexes; equivalences of quasi-categories; the Joyal model structure that organizes them.
What's the philosophical takeaway?	The same simplicial set $\mathbf{sSet}$ supports two model structures: Kan–Quillen (for spaces) and Joyal (for ∞-categories). The combinatorics is shared; the homotopy theory differs in which class of morphisms is inverted.

Chapter Summary

Chapter Summary

Quasi-category. A simplicial set in which every inner horn $\Lambda^n_k \to \mathcal{C}$ $(0 < k < n)$ extends to $\Delta^n \to \mathcal{C}$. Nerves of categories: unique inner fillers. Kan complexes: all horn fillers. Quasi-categories sit between: existence-only for inner horns.
Homotopy of 1-simplices. $f \sim g$ iff there is a 2-simplex with edges $f, g, s_0 x$. The relation is an equivalence by Λ^3-horn arguments.
Homotopy category $\mathrm{ho}(\mathcal{C})$**.** Objects $\mathcal{C}_0$; morphisms $\mathcal{C}_1/\sim$; composition via inner-horn filler (well-defined modulo $\sim$). ho is left adjoint to the nerve.
Bridges. Differential-graded categories (via N^{dg}) yield stable quasi-categories; topological / simplicial categories (via $\mathrm{N}^{\mathrm{coh}}$) yield general quasi-categories. The ∞-category $\mathcal{S}$ of spaces, $\mathcal{S} = \mathrm{N}^{\mathrm{coh}}(\mathbf{Kan})$, is the universal example.
Three models. Quasi-categories (this book); simplicial categories; complete Segal spaces. All Quillen-equivalent.

Formal Restatement

Formal Restatement

Quasi-category. $\mathcal{C} \in \mathbf{sSet}$ with the property: $\forall n \geq 2, \forall 0 < k < n, \forall \alpha : \Lambda^n_k \to \mathcal{C}, \exists \tilde{\alpha} : \Delta^n \to \mathcal{C}$ extending α.

Horn-type hierarchy.
- Inner horns fill uniquely: $\mathcal{C} = \mathrm{N}(\mathcal{D})$ for some category $\mathcal{D}$.
- Inner horns fill (existence): $\mathcal{C}$ is a quasi-category.
- All horns fill: $\mathcal{C}$ is a Kan complex (∞-groupoid).

Homotopy relation. For $f, g \in \mathcal{C}_1(x, y)$: $f \sim g \iff \exists\sigma \in \mathcal{C}_2$ with $d_0\sigma = g, d_1\sigma = f, d_2\sigma = s_0x$. Equivalence relation.

Homotopy category. $\mathrm{ho}(\mathcal{C})$: $\mathrm{Ob} = \mathcal{C}_0$, $\mathrm{Hom}(x, y) = \mathcal{C}_1(x, y)/\sim$, $[g] \circ [f] = [d_1\sigma]$ for any filler $\sigma : \Delta^2 \to \mathcal{C}$ of (f, g), $\mathrm{id}_x = [s_0x]$.

Adjunction. $\mathrm{ho} : \mathbf{QCat} \rightleftarrows \mathbf{Cat} : \mathrm{N}$, $\mathrm{ho} \dashv \mathrm{N}$.

∞-category of spaces. $\mathcal{S} = \mathrm{N}^{\mathrm{coh}}(\mathbf{Kan})$; $\mathrm{ho}(\mathcal{S}) = \mathrm{Ho}(\mathbf{Top}) = \mathrm{Ho}(\mathbf{sSet})$.

Chapter 44

Mapping Spaces, Equivalences, and the Joyal Model Structure

Chapter Overview

A quasi-category $\mathcal{C}$ has, for each pair of objects $x, y \in \mathcal{C}_0$, a **mapping space** $\mathrm{Map}_{\mathcal{C}}(x, y)$ — a Kan complex (not merely a set) whose vertices are the 1-simplices from x to y, whose 1-simplices are homotopies between them, and whose higher simplices are higher homotopies. This is the ∞-categorical analogue of a hom-set, with the entire tower of coherences explicitly present.

The chapter develops three threads.

First, the construction of $\mathrm{Map}_{\mathcal{C}}(x, y)$ via slice simplicial sets, and its core property: it is always a Kan complex (every horn fills), regardless of whether $\mathcal{C}$ is.

Second, the definition of **equivalences in a quasi-category** — 1-simplices $f : x \to y$ such that $[f]$ is an isomorphism in $\mathrm{ho}(\mathcal{C})$ — and of **equivalences of quasi-categories** $F : \mathcal{C} \to \mathcal{D}$ — functors inducing an equivalence on homotopy categories *and* a weak equivalence on every mapping space.

Third, the **Joyal model structure** on **sSet** (already previewed in Chapter 41). We state Joyal's theorem characterizing it: weak equivalences are exactly the maps that become equivalences after fibrant replacement; fibrant objects are exactly the quasi-categories; cofibrations are monomorphisms. The Joyal weak equivalences differ sharply from Kan–Quillen weak equivalences — the same simplicial set supports two distinct homotopy theories. This is the source of the oft-repeated fact that "∞-groupoids and ∞-categories are different things, even though both live in **sSet**."

The chapter closes with the homotopy n-category construction, which forms a 1-category from a quasi-category by truncating away all k-cells for $k > n$.

44.1 Mapping Spaces

44.1.1 Informally

In an ordinary category, the data $\mathrm{Hom}(x, y)$ is a set: a discrete collection of morphisms with no further structure. In a quasi-category, the data $\mathrm{Map}_{\mathcal{C}}(x, y)$ is itself a space (Kan complex): a 0-simplex is a morphism $f : x \to y$; a 1-simplex is a homotopy $f \sim g$; a 2-simplex is a homotopy between homotopies; and so on. The mapping "set" has become a mapping "space." This is exactly the data lost when one passes to the homotopy category — which takes π_0 of each mapping space — but visible at the ∞-categorical level.

44.1.2 Formalizing

There are several models for the mapping space, all weakly equivalent as Kan complexes. We give the simplest definition (the **slice model**) and refer to alternatives in the Formal Restatement.

Where the Name Comes From

The term **mapping space** is from algebraic topology, where the *mapping space* $\mathrm{Map}(X, Y)$ between two topological spaces is the space of continuous maps with the compact-open topology. The ∞-categorical $\mathrm{Map}_{\mathcal{C}}(x, y)$ is the combinatorial analogue: a Kan complex encoding the homotopy types of 1-morphisms, 2-morphisms-between-them, etc. from x to y. The use of "space" (rather than "set" or "object") signals that the mapping data is a *higher-dimensional* thing, not a discrete collection.

Definition 44.1 (Slice mapping space)**.** For a quasi-category $\mathcal{C}$ and $x, y \in \mathcal{C}_0$, define the **slice mapping space** $\mathrm{Map}_{\mathcal{C}}(x, y)$ to be the simplicial set whose n-simplices are maps $\Delta^{n+1} \to \mathcal{C}$ sending the initial vertex $0 \in \Delta^{n+1}$ to x and the final vertex $n + 1 \in \Delta^{n+1}$ to y.

How to Say This

$\mathrm{Map}_{\mathcal{C}}(x, y)$ *(read: "the mapping space from x to y in $\mathcal{C}$").* Other notations: $\mathcal{C}(x, y)$ (Lurie), $\mathrm{Hom}_{\mathcal{C}}(x, y)$, or just $\mathrm{Map}(x, y)$ when $\mathcal{C}$ is understood.

Theorem 44.2 (Mapping spaces are Kan complexes)**.** *For every quasi-category $\mathcal{C}$ and every pair $x, y \in \mathcal{C}_0$, the simplicial set $\mathrm{Map}_{\mathcal{C}}(x, y)$ is a Kan complex.*

Proof sketch. A horn $\Lambda^n_k \to \mathrm{Map}_{\mathcal{C}}(x, y)$ corresponds, by the slice description, to a partial map $\Lambda^{n+1}_{k'} \to \mathcal{C}$ in which two faces are degenerate-on-x and -on-y. By the inner-horn-filling property of $\mathcal{C}$ (plus a separate combinatorial argument for outer horns of Map — which correspond to inner horns of $\mathcal{C}$ after the slice shift), the partial map extends. ☐ ☐

Remark 44.3 (Connected components of Map recover hom-sets)**.** $\pi_0 \mathrm{Map}_{\mathcal{C}}(x, y) = \mathrm{ho}(\mathcal{C})(x, y)$: connected components of the mapping space are equivalence classes of 1-simplices, i.e., morphisms in the homotopy category. Higher $\pi_n \mathrm{Map}_{\mathcal{C}}(x, y)$ encode higher homotopical information lost to ho.

44.2 Equivalences

44.2.1 Formalizing

Where the Name Comes From

Equivalence in ∞-category theory generalizes the classical notion of equivalence between objects of a category (mutual inverse arrows). For ∞-categories, equality is too strict and even "natural isomorphism" is too strict; the right notion is invertibility *up to coherent homotopy*, which requires the entire infinite tower of higher 2-cells, 3-cells, etc. to commute. The terminology and formalization is mostly due to Joyal (2002) and Lurie (HTT 2009).

Definition 44.4 (Equivalence in a quasi-category)**.** A 1-simplex $f : x \to y$ in a quasi-category $\mathcal{C}$ is an **equivalence** if any (equivalently, all) of the following hold:

1. $[f]$ is an isomorphism in $\mathrm{ho}(\mathcal{C})$.
2. There exist $g : y \to x$ and 2-simplices witnessing $g \circ f \sim \mathrm{id}_x$ and $f \circ g \sim \mathrm{id}_y$.
3. There exists a homotopy-coherent inverse: g as above, together with 3-simplices (and higher) extending the homotopies to a coherent pseudo-inverse.

How to Say This

Equivalence $f : x \xrightarrow{\sim} y$ *(read: "f is an equivalence" or "f is invertible up to coherent homotopy").* The wedge $\sim$ over the arrow signals an equivalence; sometimes $\simeq$ is used instead.

Theorem 44.5 (Characterization via inner Δ^3-extension)**.** *A 1-simplex f in a quasi-category $\mathcal{C}$ is an equivalence iff every $\Delta^1 \to \mathcal{C}$ landing on f extends to a $\Delta^0 \star \Delta^1 = \Delta^2 \to \mathcal{C}$ with all 1-edges mapping to equivalences. Equivalently, f is an equivalence iff the slice quasi-category $\mathcal{C}_{f/}$ (Chapter 45) is a Kan complex.*

Example 44.6 (Identity is an equivalence)**.** Every identity $\mathrm{id}_x = s_0 x$ is an equivalence: take $g = s_0 x$ and the constant 2-simplices $s_1 s_0 x$ as witnesses.

Definition 44.7 (Equivalence of quasi-categories)**.** A simplicial-set map $F : \mathcal{C} \to \mathcal{D}$ between quasi-categories is an **equivalence** (also called a **categorical equivalence**) if any (equivalently, all) of:

1. The induced functor $\mathrm{ho}(F) : \mathrm{ho}(\mathcal{C}) \to \mathrm{ho}(\mathcal{D})$ is an equivalence of 1-categories, and for each $x, y \in \mathcal{C}_0$, the induced map $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(Fx, Fy)$ is a weak homotopy equivalence of Kan complexes.
2. F has a pseudo-inverse G with natural equivalences $GF \simeq \mathrm{id}_{\mathcal{C}}$ and $FG \simeq \mathrm{id}_{\mathcal{D}}$.
3. F is a weak equivalence in the Joyal model structure (Theorem 44.9).

Where the Name Comes From

Whitehead's theorem (cross-ref Ch. 40) generalizes from spaces to quasi-categories: a functor is an equivalence iff essentially surjective + fully faithful (mapping-space). The analogue for spaces was J.H.C. Whitehead's 1949 theorem on CW complexes; the ∞-categorical version is due to Joyal–Tierney and Lurie.

Theorem 44.8 (Whitehead's theorem for quasi-categories)**.** *$F : \mathcal{C} \to \mathcal{D}$ is an equivalence iff:*

- *F is **essentially surjective**: every object of $\mathcal{D}$ is equivalent to one in the image of F.*
- *F is **fully faithful**: each $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(Fx, Fy)$ is a weak homotopy equivalence.*

Proof sketch. $\Rightarrow$. Essential surjectivity follows from $\mathrm{ho}(F)$ being essentially surjective; full faithfulness follows from the second hypothesis of (1).

$\Leftarrow$. Construct G object-by-object using essential surjectivity (pick an equivalent inverse image for each $y \in \mathcal{D}$), then on morphisms using full faithfulness (lift mapping spaces). Coherence of G as a quasi-functor follows from the Kan-complex structure on each Map. □ □

44.3 The Joyal Model Structure

44.3.1 Formalizing

Reader's Annotation

The Joyal model structure is the technical home of ∞-category theory in the simplicial-set framework. The same underlying category **sSet** carries two distinct model structures: *Kan–Quillen* (fibrant objects = Kan complexes = ∞-groupoids) and *Joyal* (fibrant objects = quasi-categories = ∞-categories). The same simplicial set X can be a fibrant Kan-complex but not a fibrant quasi-category, or vice versa: the two notions are genuinely different, just as ∞-groupoids and ∞-categories are different.

Theorem 44.9 (Joyal's model structure on **sSet**)**.** *The category* **sSet** *admits a model structure — the* ***Joyal model structure****, denoted* $\mathbf{sSet}_{\text{Joyal}}$ *— characterized by:*

- *Cofibrations: monomorphisms (same as Kan–Quillen).*
- *Fibrant objects: quasi-categories.*
- *Weak equivalences: maps* $F : X \to Y$ *such that for every quasi-category* $\mathcal{C}$*, the induced* $\mathrm{Map}_{\mathbf{sSet}}(Y, \mathcal{C}) \to \mathrm{Map}_{\mathbf{sSet}}(X, \mathcal{C})$ *is a weak equivalence of Kan complexes.*

The fibrations are characterized by RLP against inner-anodyne maps, where ***inner-anodyne*** *maps are generated under saturation by the inner horn inclusions* $\Lambda^n_k \hookrightarrow \Delta^n$ $(0 < k < n)$*.*

Where the Name Comes From

Isofibration is the term used for Joyal fibrations between quasi-categories. The prefix "iso-" (Greek *isos*, "equal") reflects the additional condition: an isofibration must lift not only inner horns (giving an inner fibration) but also equivalences (i.e., the inclusion $\{0\} \hookrightarrow J$ from the walking-equivalence groupoid). The combination of "inner fibrant" plus "lift-equivalences" is captured in this one word.

Where the Name Comes From

Anodyne (as in "inner anodyne map") is from Greek *anōdunos* — "free from pain, painless." Joyal's terminology: the anodyne maps are the trivial cofibrations of the Joyal model structure — they lift against every isofibration "without trouble." The metaphor: they are "painless" to lift past, in the same way an analgesic eases pain.

Remark 44.10 (Joyal fibrations vs. Kan fibrations)**.** A Joyal fibration (between quasi-categories) is sometimes called an **isofibration**. The condition is: RLP against inner horn inclusions and against the inclusion $\{0\} \hookrightarrow J$, where J is the walking equivalence (the nerve of the contractible groupoid on two objects). The latter condition lifts equivalences. By contrast, a Kan fibration is RLP against *all* horn inclusions, which is a stronger condition.

Theorem 44.11 (Joyal weak equivalences vs. Kan–Quillen weak equivalences)**.** *Let* W_{Joyal} *and* W_{KQ} *denote the classes of weak equivalences in the Joyal and Kan–Quillen model structures respectively. Then* $W_{\text{Joyal}} \subsetneq W_{\text{KQ}}$*: every Joyal weak equivalence is a Kan–Quillen weak equivalence, but not conversely.*

The discrepancy is concrete: the inclusion $\Delta^0 \hookrightarrow \Delta^1$ *is a Joyal cofibration (a monomorphism) but not a Joyal weak equivalence — it cannot be inverted while keeping all higher-categorical structure — whereas the same inclusion is a Kan–Quillen trivial cofibration (both* Δ^0 *and* Δ^1 *are contractible).*

Remark 44.12 (The two homotopy theories cohabit **sSet**)**.** The Joyal–Kan–Quillen discrepancy is the most important fact about **sSet** in ∞-category theory. The same underlying category supports two distinct homotopy theories:

- Kan–Quillen: weak equivalences = realization-weak-homotopy-equivalences; fibrant objects = Kan complexes = ∞-groupoids; $\mathrm{Ho} \simeq \mathrm{Ho}(\mathbf{Top})$.
- Joyal: weak equivalences = categorical equivalences; fibrant objects = quasi-categories = ∞-categories; $\mathrm{Ho}_{\text{Joyal}}$ is the homotopy category of ∞-categories.

The forgetful inclusion "every ∞-groupoid is an ∞-category" becomes the left Quillen functor $\mathbf{sSet}_{\text{KQ}} \hookrightarrow \mathbf{sSet}_{\text{Joyal}}$ (identity on underlying **sSet**).

44.4 Homotopy n-Category

44.4.1 Formalizing

Where the Name Comes From

The **core** $\mathcal{C}^{\mathrm{core}}$ (also written $\mathcal{C}^{\simeq}$) of a quasi-category is named by analogy with the *core* of a group or ring — the largest sub-structure of a specified kind. Here, the maximal Kan subcomplex (i.e., the maximal ∞-groupoid contained in $\mathcal{C}$). The terminology is due to Joyal.

Definition 44.13 (Homotopy n-category)**.** For $n \geq 0$ and a quasi-category $\mathcal{C}$, the **homotopy n-category** $\mathrm{h}_n\mathcal{C}$ is obtained by quotienting $\mathcal{C}$ at the level of n-simplices: two n-simplices are identified if they agree on all boundary $(n-1)$-faces and are connected by an $(n+1)$-simplex with degenerate top face.

Example 44.14 ($n = 0$: homotopy set of equivalence classes)**.** $\mathrm{h}_0\mathcal{C} = \pi_0(\mathcal{C}^{\mathrm{core}})$, where $\mathcal{C}^{\mathrm{core}} \subset \mathcal{C}$ is the **core** — the maximal Kan subcomplex (objects $= \mathcal{C}_0$, k-simplices all of whose edges are equivalences).

Example 44.15 ($n = 1$: ordinary homotopy category)**.** $\mathrm{h}_1\mathcal{C} = \mathrm{ho}(\mathcal{C})$ — the ordinary 1-category construction of Chapter 43.

Example 44.16 ($n = \infty$: identity)**.** $\mathrm{h}_\infty\mathcal{C} = \mathcal{C}$ — no truncation.

Remark 44.17 (Truncation as left adjoint)**.** h_n is left adjoint to the inclusion of n-truncated quasi-categories (those equivalent to nerves of n-categories) into **QCat**. Volume V (if written) would treat n-categorical truncations in detail; here we record only the existence and the first three cases.

44.5 Exercises

Define the slice mapping space $\mathrm{Map}_\mathcal{C}(x, y)$.	n-simplices are $\Delta^{n+1} \to \mathcal{C}$ sending $0 \mapsto x$ and $n+1 \mapsto y$.
Is $\mathrm{Map}_\mathcal{C}(x, y)$ always a Kan complex?	Yes — even when $\mathcal{C}$ is merely a quasi-category. Horn-filling for Map reduces to inner-horn filling for $\mathcal{C}$ via the slice shift.
What is $\pi_0\mathrm{Map}_\mathcal{C}(x, y)$?	$\mathrm{ho}(\mathcal{C})(x, y)$ — the hom-set in the homotopy category.
What's the higher-dimensional data of the mapping space?	1-simplices: homotopies between 1-simplices in $\mathcal{C}$; 2-simplices: homotopies between those; etc. The full tower of coherences.
Define an equivalence in $\mathcal{C}$.	A 1-simplex f such that $[f]$ is an isomorphism in $\mathrm{ho}(\mathcal{C})$. Equivalently: there's a g with $g \circ f \sim \mathrm{id}$ and $f \circ g \sim \mathrm{id}$.
Are identities equivalences?	Yes — $[s_0 x]$ is the identity of x in $\mathrm{ho}(\mathcal{C})$, which is an isomorphism.
Why is the equivalence condition really an ∞-condition?	Because invertibility up to a 2-simplex requires coherence with all higher simplices to be a genuine pseudo-inverse; the data of an equivalence is a contractible space, not just a pair (g, h_1, h_2).
Define a categorical equivalence $F : \mathcal{C} \to \mathcal{D}$.	F induces (a) an equivalence on homotopy 1-categories and (b) a weak equivalence on every mapping space.
State Whitehead's theorem for quasi-categories.	F is a categorical equivalence iff essentially surjective and fully faithful (mapping spaces send to weak equivalences).
What's the Joyal model structure?	A model structure on **sSet** with cofibrations = monomorphisms; fibrant objects = quasi-categories; weak equivalences = categorical equivalences after fibrant replacement.
What is a Joyal fibration (isofibration)?	A map of quasi-categories with RLP against inner horn inclusions and against $\{0\} \hookrightarrow J$, where J is the walking equivalence.
How does a Joyal fibration differ from a Kan fibration?	Kan: RLP against all horn inclusions. Joyal: only inner horns, plus equivalence-lifting. Every Kan fibration of quasi-categories is a Joyal fibration; not conversely.

Why are Joyal and Kan–Quillen weak equivalences different?	Joyal sees the ∞-categorical structure (must respect non-invertible morphisms); Kan–Quillen sees only the ∞-groupoidal structure (inverts everything).
Give a concrete distinguishing example.	$\Delta^0 \hookrightarrow \Delta^1$: Kan–Quillen weak equivalence (both contractible), but *not* a Joyal weak equivalence (cannot invert the morphism $0 \to 1$ while preserving categorical structure).
What does "the same simplicial set carries two model structures" mean philosophically?	The choice of which morphisms to invert is independent of the underlying combinatorics: Kan–Quillen inverts everything, Joyal only invertibles. The two homotopy theories are different in essence.
Define the core $\mathcal{C}^{\text{core}}$.	The maximal Kan subcomplex of $\mathcal{C}$: objects $= \mathcal{C}_0$; k-simplices have all 1-edges equivalences.
What does the core compute?	$\mathcal{C}^{\text{core}}$ is the ∞-groupoid of equivalences in $\mathcal{C}$: invertible morphisms, invertible 2-cells between them, etc.
What is $\mathrm{h}_n\mathcal{C}$?	The homotopy n-category — truncates higher simplices by identifying $(n+1)$-simplices with degenerate top face.
$\mathrm{h}_0\mathcal{C} = ?$	$\pi_0\mathcal{C}^{\text{core}}$: equivalence classes of objects.
$\mathrm{h}_1\mathcal{C} = ?$	$\mathrm{ho}(\mathcal{C})$: the ordinary homotopy 1-category.
Why does Lurie write $\mathrm{Map}_{\mathcal{C}}(x, y)$ rather than $\mathcal{C}(x, y)$?	To emphasize that the mapping is a *space* (Kan complex), not a set. The set $\pi_0\mathrm{Map}_{\mathcal{C}}(x, y)$ is the morphism set of $\mathrm{ho}(\mathcal{C})$.
Are there other models for mapping spaces?	Yes: pinched (left, right, two-sided) and tensored models, all weakly equivalent. The slice model is the simplest; the right-pinched model $\mathcal{C}^{\Delta^1} \times_{\mathcal{C}} \Delta^0$ at (x, y) is computationally convenient.
When does Joyal coincide with Kan–Quillen?	Restricted to the full subcategory of Kan complexes (∞-groupoids), the two model structures agree.

Chapter Summary

Chapter Summary

Mapping spaces. $\mathrm{Map}_{\mathcal{C}}(x, y) \in \mathbf{sSet}$ is a Kan complex whose n-simplices are $\Delta^{n+1} \to \mathcal{C}$ with prescribed initial/final vertices. $\pi_0\mathrm{Map}_{\mathcal{C}}(x, y) = \mathrm{ho}(\mathcal{C})(x, y)$.
Equivalences. A 1-simplex f is an equivalence iff $[f]$ is iso in ho. A functor $F : \mathcal{C} \to \mathcal{D}$ is a categorical equivalence iff $\mathrm{ho}(F)$ is an equivalence and each mapping space map is a weak equivalence. Whitehead: equivalent to essential surjectivity plus full faithfulness on mapping spaces.
Joyal model structure. Cofibrations = monos; fibrant = quasi-categories; weak equivalences = categorical equivalences (after fibrant replacement); fibrations = inner fibrations + equivalence-lifting (isofibrations). $W_{\text{Joyal}} \subsetneq W_{\text{KQ}}$: Joyal respects directionality, Kan–Quillen inverts everything.
Homotopy n-category. $\mathrm{h}_n\mathcal{C}$ truncates higher coherences: $\mathrm{h}_0 = \pi_0$ of the core, $\mathrm{h}_1 = \mathrm{ho}$, $\mathrm{h}_\infty =$ identity.

Formal Restatement

Formal Restatement

Mapping space (slice model). $\mathrm{Map}_{\mathcal{C}}(x, y)_n = \{\, \sigma : \Delta^{n+1} \to \mathcal{C} : \sigma(0) = x, \sigma(n+1) = y \,\}$.

Kan-complex theorem. For any quasi-category $\mathcal{C}$ and $x, y \in \mathcal{C}_0$, $\mathrm{Map}_{\mathcal{C}}(x, y)$ is a Kan complex; $\pi_0\mathrm{Map}_{\mathcal{C}}(x, y) = \mathrm{ho}(\mathcal{C})(x, y)$.

Equivalence (object). $f \in \mathcal{C}_1$ is an equivalence $\iff$ $[f] \in \mathrm{ho}(\mathcal{C})$ is an isomorphism.

Categorical equivalence (functor). $F : \mathcal{C} \to \mathcal{D}$ is a categorical equivalence $\iff$
- $\mathrm{ho}(F) : \mathrm{ho}(\mathcal{C}) \to \mathrm{ho}(\mathcal{D})$ is essentially surjective,

- $\mathrm{Map}_{\mathcal{C}}(x, y) \to \mathrm{Map}_{\mathcal{D}}(Fx, Fy)$ is a weak homotopy equivalence for all x, y.

Joyal model structure.
- $\mathcal{C}$of: monomorphisms.
- $\mathcal{F}$ib: isofibrations (RLP against inner horns and against $\{0\} \hookrightarrow J$).
- $\mathcal{W}$: categorical equivalences after fibrant replacement.

Fibrant objects: quasi-categories.

Homotopy n-category. $\mathrm{h}_n : \mathbf{QCat} \to \mathbf{nCat}$, left adjoint to inclusion. $\mathrm{h}_0\mathcal{C} = \pi_0(\mathcal{C}^{\mathrm{core}})$, $\mathrm{h}_1\mathcal{C} = \mathrm{ho}(\mathcal{C})$, $\mathrm{h}_\infty\mathcal{C} = \mathcal{C}$.

Chapter 45

Joins, Slices, and Limits in a Quasi-Category

Chapter Overview

∞-limits and ∞-colimits are the central universal constructions of ∞-category theory. Chapter 49 will develop them in earnest; this chapter assembles the combinatorial machinery they require: the **join** $X \star Y$ of simplicial sets, and the **slice** quasi-categories $\mathcal{C}_{/p}$ and $\mathcal{C}_{p/}$ that the join produces.

The join is a binary operation on **sSet** with $\Delta^a \star \Delta^b = \Delta^{a+b+1}$. Geometrically, it juxtaposes two simplicial sets and adds simplices for "all the new connecting edges" from one to the other. The slice $\mathcal{C}_{/p}$ over a diagram $p : K \to \mathcal{C}$ is the right adjoint of $- \star K$ at p: its n-simplices are extensions of p along $\Delta^n \star K \to \mathcal{C}$.

With slices in hand, ∞-limits and ∞-colimits admit a striking definition: $\lim p$ is the **terminal object** of $\mathcal{C}_{/p}$, and $\operatorname{colim} p$ is the **initial object** of $\mathcal{C}_{p/}$. The shift to slice quasi-categories converts universal property into "existence of a terminal/initial object," which turns out to be the cleanest formulation in the ∞-categorical setting.

The chapter closes with cofinal maps — diagrams sharing the same colimit. Joyal–Lurie's criterion is the practical workhorse: $i : K' \to K$ is cofinal iff for every object x of the target ∞-category, the fibre of i over $\mathcal{C}_{/x}$ has a contractible slice. This will be used repeatedly in Chapters 48–51.

45.1 The Join of Simplicial Sets

45.1.1 Formalizing

Where the Name Comes From

The **join** $X \star Y$ takes its name from the classical join operation in convex geometry: the join of two polytopes (or topological spaces) $A \star B$ is the polytope/space obtained by "joining every point of A to every point of B" via line segments. For simplicial complexes, $\Delta^a \star \Delta^b = \Delta^{a+b+1}$: joining an a-simplex with a b-simplex produces an $(a+b+1)$-simplex. The combinatorial join on **sSet** is the simplicial-set version of this geometric operation. The symbol $\star$ ("star") is universal; the alternative $*$ (asterisk) is also used.

Definition 45.1 (Join of simplicial sets)**.** For $X, Y \in$ **sSet**, the **join** $X \star Y$ is the simplicial set with

$$(X \star Y)_n \;=\; X_n \sqcup Y_n \sqcup \coprod_{i+j=n-1} X_i \times Y_j.$$

The face and degeneracy operators are the natural ones from X on the first piece, from Y on the second piece, and on the mixed piece $X_i \times Y_j$ by applying X-operators when the indexed face/degeneracy stays in the X-part and Y-operators when it stays in the Y-part, with appropriate "connecting" behavior at the boundary.

How to Say This

$X \star Y$ *(read: "X join Y").* The star is the standard typographic convention; sometimes $X * Y$ is written instead.

Theorem 45.2 (Join on standard simplices)**.** $\Delta^a \star \Delta^b \;\cong\; \Delta^{a+b+1}$.

Proof sketch. Δ^{a+b+1} has $a+b+2$ vertices. Label the first $a+1$ as those of Δ^a and the last $b+1$ as those of Δ^b. An n-simplex of Δ^{a+b+1} is a monotone map $[n] \to [a+b+1]$, which can be split into the X-part (image in $[a]$), the Y-part (image in $[b]$, shifted), and the mixed part (one vertex in each). The three pieces match the three summands in Definition 45.1. ☐ ☐

Where the Name Comes From

The **right cone** $X^{\triangleright}$ and **left cone** $X^{\triangleleft}$ generalize the classical *cone* on a space (a point joined to every point of X by line segments). The triangular shapes $\triangleright$ and $\triangleleft$ are visual mnemonics: the bare "$\triangleright$" faces right (cone point at the right end) and $\triangleleft$ faces left. The notation is due to Joyal.

Example 45.3 (Cones)**.** For any $X \in \mathbf{sSet}$:

- $X \star \Delta^0$ is the **right cone** on X, denoted $X^{\triangleright}$. It adjoins a single "cone point" to which every vertex of X is connected.
- $\Delta^0 \star X$ is the **left cone** on X, denoted $X^{\triangleleft}$. The cone point is the source instead of the target.

Theorem 45.4 (Functoriality of the join)**.** *The join $\star : \mathbf{sSet} \times \mathbf{sSet} \to \mathbf{sSet}$ is a functor in each variable separately. It does* not *preserve colimits in either variable, however, so does not extend to an ∞-categorical tensor product without modification.*

45.2 Slice Quasi-Categories

45.2.1 Formalizing

Where the Name Comes From

The **slice category** construction (Vol I/II) takes a category $\mathcal{C}$ and an object x to the category $\mathcal{C}_{/x}$ of "things with a map to x." In the ∞-categorical setting the construction is upgraded: $\mathcal{C}_{/p}$ for a diagram $p : K \to \mathcal{C}$ consists of "cones over p." The fundamental fact is that this slice operation is the *right adjoint* of the join construction: $- \star K \dashv (-)_{/K}$. The naming convention is inherited from Vol I/II; the join-slice adjunction generalizes the classical fact that $\mathrm{Cat}(C \star K, \mathcal{D})$ corresponds to cones over a diagram of shape K.

Definition 45.5 (Slice quasi-category over a diagram)**.** Let $\mathcal{C}$ be a simplicial set and $p : K \to \mathcal{C}$ a map (thought of as a **diagram** of shape K in $\mathcal{C}$). The **slice** $\mathcal{C}_{/p}$ is the simplicial set with

$$(\mathcal{C}_{/p})_n \;=\; \{\, \sigma : \Delta^n \star K \to \mathcal{C} : \sigma|_K = p \,\}.$$

That is, an n-simplex of $\mathcal{C}_{/p}$ is an extension of p along $K \hookrightarrow \Delta^n \star K$.

Definition 45.6 (Slice quasi-category under a diagram)**.** Dually, $\mathcal{C}_{p/}$ has n-simplices $\sigma : K \star \Delta^n \to \mathcal{C}$ with $\sigma|_K = p$.

How to Say This

$\mathcal{C}_{/p}$ *(read: "$\mathcal{C}$ over p"); $\mathcal{C}_{p/}$ (read: "$\mathcal{C}$ under p").* For the special case $K = \Delta^0$ with p picking out an object x, we write $\mathcal{C}_{/x}$ and $\mathcal{C}_{x/}$.

Theorem 45.7 (Slice quasi-categories are quasi-categories)**.** *If $\mathcal{C}$ is a quasi-category, then for any diagram $p : K \to \mathcal{C}$, the slices $\mathcal{C}_{/p}$ and $\mathcal{C}_{p/}$ are quasi-categories.*

Proof sketch. An inner horn $\Lambda^n_k \to \mathcal{C}_{/p}$ corresponds to a partial extension of p from K to $(\Lambda^n_k) \star K \to \mathcal{C}$. By the inner-horn-filling property of $\mathcal{C}$ (the relevant horn in $\mathcal{C}$ being still inner), the extension exists, giving $\Delta^n \to \mathcal{C}_{/p}$. □ □

Theorem 45.8 (Join-slice adjunction)**.** *For each simplicial set K, the functor $- \star K : \mathbf{sSet} \to \mathbf{sSet}_{K/}$ (sending X to $X \star K$ with its canonical inclusion of K) has a right adjoint $\mathcal{C} \mapsto \mathcal{C}_{/p}$ where $p : K \to \mathcal{C}$ is the structure map of an object of $\mathbf{sSet}_{K/}$:*

$$\mathbf{sSet}(X \star K, \mathcal{C}) \cong \mathbf{sSet}(X, \mathcal{C}_{/p}) \quad (\textit{over } K).$$

Dually for $- \star K$ on the left and $\mathcal{C}_{p/}$.

45.3 Limits and Colimits

45.3.1 Informally

In an ordinary category, the limit of a diagram is the universal cone over it. In a quasi-category, the analogous data is: an object $\lim p \in \mathcal{C}$, plus a cone (a map $\Delta^0 \star K \to \mathcal{C}$ extending p), plus an entire tower of higher-dimensional witnesses that "no better cone exists." The shift to slices packages all this data as: an object of $\mathcal{C}_{/p}$ — namely the limit cone — that is *terminal* in $\mathcal{C}_{/p}$.

45.3.2 Formalizing

Where the Name Comes From

Initial and **terminal** objects extend from Vol I/II's 1-categorical definitions (object with a unique map to / from every other) to the ∞-setting. "Unique" becomes "contractible space of maps" — the natural ∞-categorical version of uniqueness, by the same logic as Vol II's representability principle.

Definition 45.9 (Initial and terminal objects)**.** An object x of a quasi-category $\mathcal{D}$ is:

- **Initial** if $\mathrm{Map}_{\mathcal{D}}(x, y)$ is contractible for every $y \in \mathcal{D}_0$.

- **Terminal** if $\mathrm{Map}_{\mathcal{D}}(y, x)$ is contractible for every $y \in \mathcal{D}_0$.

Remark 45.10 (Why "contractible"?)**.** "Unique up to coherent homotopy" is the same as "contractible." In **Set**, a one-point set is the unique set such that all $\mathrm{Hom}(x, y)$ are singletons. In quasi-categories, "singleton" is replaced by "contractible Kan complex" — its weak-equivalent counterpart.

Definition 45.11 (∞-limit and ∞-colimit)**.** Let $\mathcal{C}$ be a quasi-category and $p : K \to \mathcal{C}$ a diagram.

A **limit** of p, denoted $\lim p$, is a *terminal* object of the slice $\mathcal{C}_{/p}$ — when it exists.

A **colimit** of p, denoted $\operatorname{colim} p$, is an *initial* object of $\mathcal{C}_{p/}$.

How to Say This

$\lim p$ *(read: "the limit of p")* or *(read: "the ∞-limit")*. For emphasis we sometimes write $\lim^{\infty} p$ to distinguish from the 1-categorical limit, but typically context makes this clear.

Example 45.12 (∞-products and ∞-equalizers)**.** For K a finite discrete set, an ∞-limit of $p : K \to \mathcal{C}$ is the **∞-product** of the objects $p(k)$. For $K = \Delta^1 \rightrightarrows \Delta^0$ (the walking parallel pair), it is the **∞-equalizer**.

These ∞-versions reduce to ordinary 1-categorical products and equalizers in $\mathrm{ho}(\mathcal{C})$, but the ∞-data retains all higher coherences.

Theorem 45.13 (Uniqueness of ∞-(co)limits)**.** *When $\lim p$ exists, the space of limit cones — the space of terminal objects of $\mathcal{C}_{/p}$ — is either empty or contractible.*

Proof sketch. By definition, a terminal object x of a quasi-category $\mathcal{D}$ has $\mathrm{Map}_{\mathcal{D}}(y, x)$ contractible for all y. In particular, the space of terminal objects of $\mathcal{D}$ is the limit of contractible spaces over a contractible diagram, hence contractible (or empty). □ □

Remark 45.14 (Limits in $\mathcal{S}$)**.** For the ∞-category of spaces, ∞-limits computed in $\mathcal{S}$ are homotopy limits of spaces. Standard homotopy fibre and homotopy pullback constructions in **Top** are recovered as particular cases.

45.4 Cofinality

45.4.1 Formalizing

Where the Name Comes From

Cofinal (sometimes **final**) comes from the order-theoretic notion: a subset $S \subseteq P$ of a poset is *cofinal* if every element of P is dominated by some element of S. The categorical generalization (a functor $i : K' \to K$ such that every object of K "factors through" K' in the appropriate sense) has the same intuitive content: K' captures enough of K to compute colimits correctly. The Joyal–Lurie cofinality criterion makes this precise for ∞-colimits.

Definition 45.15 (Cofinal map)**.** A map $i : K' \to K$ between simplicial sets is **cofinal** (also called **final**) if for every quasi-category $\mathcal{C}$ and every diagram $p : K \to \mathcal{C}$ such that $\operatorname{colim} p$ exists, the map $\operatorname{colim} p \to \operatorname{colim}(p \circ i)$ is an equivalence in $\mathcal{C}$. Dually, i is **coinitial** (or **initial**) if it preserves ∞-limits.

Theorem 45.16 (Joyal–Lurie criterion)**.** *A map $i : K' \to K$ is cofinal iff for every vertex $k \in K_0$, the slice $K'_{k/} := K' \times_K K_{k/}$ is weakly contractible (i.e., its realization is a contractible topological space).*

Example 45.17 (Inclusion of an initial object)**.** If K has an initial object k_0, then the inclusion $\{k_0\} \hookrightarrow K$ is cofinal. Hence

$$\operatorname{colim}_K p \;\simeq\; p(k_0).$$

Diagrams with initial objects collapse on colimits. This generalizes the fact that the colimit of a diagram with an initial vertex in **Set** is the image of that initial vertex.

Example 45.18 (Filtered cofinality)**.** If K, K' are both ∞-filtered (their nerves are filtered as quasi-categories), and $i : K' \to K$ is essentially surjective, then i is cofinal. Filtered colimits depend only on the asymptotic shape.

Remark 45.19 (Cofinality in practice)**.** Cofinality is the central tool for *computing* ∞-(co)limits: replacing a complicated diagram by a cofinal subdiagram, the latter being easier to handle. In Chapter 49 we will see how cofinality computes ∞-(co)limits via cone-cone arguments, generalizing Chapter 6's ordinary-category cone-and-limit reductions.

45.5 Exercises

Define the join $X \star Y$.	$(X \star Y)_n = X_n \sqcup Y_n \sqcup \coprod_{i+j=n-1} X_i \times Y_j$, with face and degeneracy operators piecing together the three components.

What is $\Delta^a \star \Delta^b$?	Δ^{a+b+1} — the standard simplex on the disjoint union of vertex sets.
What is the right cone $X^{\triangleright}$?	$X^{\triangleright} = X \star \Delta^0$: X with a single "cone-point" vertex added, connected to every vertex of X.
What is the left cone $X^{\triangleleft}$?	$X^{\triangleleft} = \Delta^0 \star X$: dually, with the cone point as source.
Is $\star$ associative? Symmetric?	Associative (up to natural isomorphism); not symmetric — $X \star Y \neq Y \star X$ in general (cones are directional).
Define the slice $\mathcal{C}_{/p}$.	$(\mathcal{C}_{/p})_n = \{\sigma : \Delta^n \star K \to \mathcal{C} : \sigma\|_K = p\}$; n-simplices are extensions of p from K along $K \hookrightarrow \Delta^n \star K$.
What is $\mathcal{C}_{/x}$ for x a single object?	The case $K = \Delta^0$: n-simplices are $\Delta^n \star \Delta^0 = \Delta^{n+1} \to \mathcal{C}$ with last vertex x. The slice $\mathcal{C}_{/x}$ has objects "arrows into x."
Is $\mathcal{C}_{/p}$ a quasi-category when $\mathcal{C}$ is?	Yes — inner-horn filling for the slice reduces to inner-horn filling for $\mathcal{C}$.
State the join-slice adjunction.	$\mathbf{sSet}(X \star K, \mathcal{C}) \cong \mathbf{sSet}(X, \mathcal{C}_{/p})$ over K. So $- \star K \dashv (-)_{/K}$.
Define initial and terminal objects in a quasi-category.	x is initial if $\mathrm{Map}_{\mathcal{D}}(x, y)$ is contractible for all y; terminal if $\mathrm{Map}_{\mathcal{D}}(y, x)$ is contractible for all y.
Why "contractible" rather than "one element"?	Because the ∞-categorical hom is a space, not a set. "Unique up to coherent homotopy" is the same as "contractible space of choices."
Define the ∞-limit of $p : K \to \mathcal{C}$.	A terminal object of the slice $\mathcal{C}_{/p}$.
Define the ∞-colimit.	An initial object of $\mathcal{C}_{p/}$.
Why is the limit object well-defined?	When it exists, the space of terminal objects of $\mathcal{C}_{/p}$ is contractible (or empty). So $\lim p$ is unique up to a contractible space of equivalences.
Recover ∞-products and ∞-equalizers from the definition.	∞-product $=$ ∞-limit over a finite discrete K. ∞-equalizer $=$ ∞-limit over $K = (\Delta^1 \rightrightarrows \Delta^0)$.
Define a cofinal map.	$i : K' \to K$ such that for every $p : K \to \mathcal{C}$, $\mathrm{colim}\,(p \circ i) \simeq \mathrm{colim}\, p$.
State the Joyal–Lurie cofinality criterion.	i is cofinal iff for every vertex $k \in K_0$, the fibre $K' \times_K K_{k/}$ is weakly contractible.
Show that the inclusion of an initial object is cofinal.	If k_0 is initial in K, every slice $K_{k/}$ for $k = k_0$ is K itself (contractible); the criterion holds trivially for k_0, and by initiality for other k.
Cofinality and filtered colimits?	Essentially-surjective maps between filtered quasi-categories are cofinal. Filtered colimits depend on the asymptotic shape, not specific diagram structure.
Why are cofinal maps the workhorse?	They reduce computation of complicated colimits to simpler subdiagrams. In Chapter 49 we use this to characterize all ∞-(co)limits.
What is $\mathrm{Map}_{\mathcal{C}_{/p}}(\alpha, \beta)$ for cones α, β?	The space of "arrows of cones from α to β" — combinatorially, the relevant Kan complex of extensions in $\mathcal{C}_{/p}$.
When is $\mathrm{Map}_{\mathcal{D}}(y, x)$ contractible for all y but x not the terminal object?	Never, by definition. "Terminal" just is "all Maps into x contractible."
What is the homotopy hypothesis's reflection in this chapter?	Limits in $\mathcal{S}$ (the ∞-category of spaces) recover homotopy limits in **Top**. Colimits in $\mathcal{S}$ recover homotopy colimits. The two pictures agree via the Quillen equivalence.

Chapter Summary

Chapter Summary

Join. $X \star Y$ has n-simplices the disjoint union of X_n, Y_n, and "connector" simplices $X_i \times Y_j$ with $i + j = n - 1$. $\Delta^a \star \Delta^b = \Delta^{a+b+1}$. Cones: $X^{\triangleright} = X \star \Delta^0$, $X^{\triangleleft} = \Delta^0 \star X$.
Slice. $\mathcal{C}_{/p}$ has n-simplices given by extensions $\Delta^n \star K \to \mathcal{C}$ of $p : K \to \mathcal{C}$. Adjoint to join: $- \star K \dashv (-)_{/K}$. Slices of quasi-categories are quasi-categories.
Limits and colimits. $\lim p$ is a terminal object of $\mathcal{C}_{/p}$; $\operatorname{colim} p$ is an initial object of $\mathcal{C}_{p/}$. When they exist, they are unique up to contractible space of equivalences. Recovers ordinary products, equalizers, and their homotopical analogues.
Cofinality. $i : K' \to K$ is cofinal iff $K' \times_K K_{k/}$ is contractible for every k (Joyal–Lurie). Reduces computation of $\operatorname{colim}_K p$ to that of $\operatorname{colim}_{K'}(p \circ i)$ for a simpler K'.

Formal Restatement

Formal Restatement

Join. $(X \star Y)_n = X_n \sqcup Y_n \sqcup \coprod_{i+j=n-1} X_i \times Y_j$. $\Delta^a \star \Delta^b = \Delta^{a+b+1}$.

Cones. $X^{\triangleright} = X \star \Delta^0$; $X^{\triangleleft} = \Delta^0 \star X$.

Slice. $(\mathcal{C}_{/p})_n = \{\sigma : \Delta^n \star K \to \mathcal{C} : \sigma|_K = p\}$. $(\mathcal{C}_{p/})_n = \{\sigma : K \star \Delta^n \to \mathcal{C} : \sigma|_K = p\}$.

Join-slice adjunction. $- \star K : \mathbf{sSet} \rightleftarrows \mathbf{sSet}_{K/} : (-)_{/K}$.

Initial and terminal. x initial $\iff \operatorname{Map}_{\mathcal{D}}(x, -) \simeq *$; x terminal $\iff \operatorname{Map}_{\mathcal{D}}(-, x) \simeq *$.

∞-limit and ∞-colimit. $\lim p$ = terminal object of $\mathcal{C}_{/p}$; $\operatorname{colim} p$ = initial object of $\mathcal{C}_{p/}$.

Cofinality (Joyal–Lurie). $i : K' \to K$ cofinal $\iff \forall k \in K_0$, $K' \times_K K_{k/}$ is weakly contractible. Cofinal maps preserve colim.

Chapter 46

Cartesian and coCartesian Fibrations

Chapter Overview

The Grothendieck construction in ordinary category theory packages a pseudofunctor $F : \mathcal{S}^{\mathrm{op}} \to \mathbf{Cat}$ as a single category $\int F$ together with a **fibration** $\int F \to \mathcal{S}$ whose fibres are the categories $F(s)$ and whose "transport" is the functoriality of F. The ∞-categorical version is **Cartesian** (and **coCartesian**) **fibrations**: morphisms $p : \mathcal{X} \to \mathcal{S}$ between quasi-categories satisfying a combinatorial lifting property that encodes coherent transport up the fibres.

This chapter introduces the definitions and works through the foundational examples. The technical heart is the notion of a **p-Cartesian morphism**: a 1-simplex $\alpha : x \to y$ in $\mathcal{X}$ such that restricting along α gives a homotopy equivalence on mapping spaces. Cartesian morphisms supply the transport data; Cartesian fibrations are those p in which enough Cartesian morphisms exist.

Examples organize the chapter:

- The **family fibration**: for a coCartesian fibration, the fibres $\mathcal{X}_s$ form a functor $\mathcal{S} \to \mathbf{QCat}$.
- The **arrow fibration** $\mathcal{C}^{\Delta^1} \to \mathcal{C}$ (evaluation at the source or target): target evaluation is a Cartesian fibration; source evaluation is a coCartesian fibration.
- The **universal Cartesian fibration**: a Cartesian fibration $\mathcal{U} \to \mathcal{S}$ classifying functors $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$. Existence is the content of Chapter 47's straightening theorem.

After this chapter, Chapter 47's straightening/unstraightening equivalence between Cartesian fibrations and functors $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$ closes the loop.

46.1 Cartesian Morphisms

46.1.1 Informally

Let $p : \mathcal{X} \to \mathcal{S}$ be a map of quasi-categories. A morphism $\alpha : x \to y$ in $\mathcal{X}$ is **p-Cartesian** if it presents y as "the image of x along α in a universal way relative to the base $\mathcal{S}$." Equivalently: α is p-Cartesian when any morphism $z \to y$ in $\mathcal{X}$ that lies over a factorization $p(z) \to p(x) \to p(y)$ in $\mathcal{S}$ lifts uniquely to a factorization $z \to x \to y$ in $\mathcal{X}$.

The intuition is direct from 1-categorical Grothendieck fibrations: α is the **transport** of x along $p(\alpha)$, and any other morphism into y should "factor through" α in a unique way.

46.1.2 Formalizing

Where the Name Comes From

Cartesian (as in "Cartesian morphism," "Cartesian fibration") is from *Grothendieck* via the standard categorical **Cartesian square** (= pullback square). A Cartesian morphism is one that fits into a universal pullback-style square with respect to the base; a Cartesian fibration is a family of categories with enough Cartesian morphisms to encode the contravariant action of the base. The term "Cartesian" goes back to Descartes (1596–1650) via "Cartesian product" — the original universal-product construction in set theory. Grothendieck (1960s) lifted the terminology to fibred categories; Lurie (HTT 2009) extended it to the ∞-categorical setting.

Definition 46.1 (p-Cartesian morphism)**.** Let $p : \mathcal{X} \to \mathcal{S}$ be a map of simplicial sets and $\alpha : x \to y$ a 1-simplex in $\mathcal{X}$. We say α is **p-Cartesian** if for every $n \geq 2$, every commutative diagram of solid arrows

$$\begin{tikzcd} \Lambda^n_n \arrow[r] \arrow[d] & \mathcal{X} \arrow[d, "p"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & \mathcal{S} \end{tikzcd}$$

in which the restriction of the upper map to the final edge $\Delta^{\{n-1,n\}}$ of Λ^n_n is α, admits a (dashed) lift.

How to Say This

α *(read: "a p-Cartesian morphism" or "p-Cartesian")*; the condition specifies a *lifting* property against outer horns $\Lambda^n_n \hookrightarrow \Delta^n$ that have α on the final edge. The dual coCartesian condition uses initial-edge horns Λ^n_0.

Theorem 46.2 (Equivalent characterizations)**.** *For $\alpha : x \to y$ in $\mathcal{X}$ over $f : s' \to s$ in $\mathcal{S}$, the following are equivalent:*

1. *α is p-Cartesian.*
2. *For every $z \in \mathcal{X}$, the canonical map*
$$\mathrm{Map}_{\mathcal{X}}(z, x) \longrightarrow \mathrm{Map}_{\mathcal{X}}(z, y) \times_{\mathrm{Map}_{\mathcal{S}}(p(z),p(y))} \mathrm{Map}_{\mathcal{S}}(p(z), p(x))$$
is a weak homotopy equivalence of Kan complexes.
3. *Every diagram*
$$\begin{tikzcd} z \arrow[r, dashed, "g"'] \arrow[rr, bend left] & x \arrow[r, "\alpha"] & y \end{tikzcd}$$
admits a unique (up to coherent homotopy) lift g for any compatible $z \to y$ in $\mathcal{X}$ and $p(z) \to p(x)$ in $\mathcal{S}$.

Remark 46.3 (Cartesian for ordinary categories)**.** For an ordinary functor $p : \mathbf{C} \to \mathbf{S}$ regarded as a map of nerves, the condition collapses: α is p-Cartesian iff for every $g : z \to y$ in $\mathbf{C}$ and every factorization $p(z) \to p(x) \to p(y)$ of $p(g)$ in $\mathbf{S}$, there is a unique $\bar{g} : z \to x$ making both pieces commute. This is the standard 1-categorical definition of Grothendieck-Cartesian.

46.2 Cartesian Fibrations

46.2.1 Formalizing

Where the Name Comes From

The dichotomy **Cartesian** vs. **coCartesian** reflects the direction of transport: *Cartesian* fibrations have transport *against* the base direction (over $f : s \to s'$, transport sends fibre$_{s'}$ to fibre$_s$), so the fibres form a *contravariant* functor $\mathcal{S}^{\mathrm{op}} \to \mathrm{Cat}_\infty$. *coCartesian* fibrations have transport *with* the base direction, so the fibres form a covariant functor. The dual prefix "co-" is standard category-theoretic convention.

Definition 46.4 (Cartesian fibration)**.** A map $p : \mathcal{X} \to \mathcal{S}$ of simplicial sets is a **Cartesian fibration** if:

- p is an inner fibration (RLP against inner horns; equivalently, the fibres $\mathcal{X}_s = \mathcal{X} \times_{\mathcal{S}} \{s\}$ are quasi-categories).
- For every edge $f : s' \to s$ in $\mathcal{S}$ and every $y \in \mathcal{X}$ with $p(y) = s$, there exists a p-Cartesian morphism $\alpha : x \to y$ in $\mathcal{X}$ with $p(\alpha) = f$.

Definition 46.5 (coCartesian fibration)**.** p is a **coCartesian fibration** if it is an inner fibration and for every $f : s \to s'$ and every x with $p(x) = s$, there exists a p-coCartesian morphism $\alpha : x \to y$ (defined dually) with $p(\alpha) = f$.

Example 46.6 (Trivial fibrations are Cartesian and coCartesian)**.** A trivial Kan fibration (RLP against all $\partial\Delta^n \hookrightarrow \Delta^n$) is automatically both Cartesian and coCartesian. The transport along any base edge is via the lifting property; every morphism in the total space is both Cartesian and coCartesian.

Example 46.7 (The target evaluation $\mathcal{C}^{\Delta^1} \to \mathcal{C}$)**.** For any quasi-category $\mathcal{C}$, the target evaluation $\mathrm{ev}_1 : \mathcal{C}^{\Delta^1} \to \mathcal{C}$ (which sends an arrow $f : a \to b$ to b) is a Cartesian fibration. The Cartesian lift of $g : b' \to b$ over an object $f : a \to b$ is the precomposition $g^* f = f \circ g : a \to b'$, presented as a 2-simplex.

Example 46.8 (Source evaluation $\mathcal{C}^{\Delta^1} \to \mathcal{C}$)**.** Dually, $\mathrm{ev}_0 : \mathcal{C}^{\Delta^1} \to \mathcal{C}$ (sending $f : a \to b$ to a) is a *coCartesian* fibration: along $h : a \to a'$, transport an arrow $f : a \to b$ to its postcomposition $h_* f$.

Where the Name Comes From

The **right fibration** / **left fibration** terminology denotes the directionality of the lifting condition: a right fibration has *right* lifting against outer horns Λ^n_n (final-vertex horns, on the right), giving the contravariant fibre functor; a left fibration has right lifting against Λ^n_0 (initial-vertex horns), giving the covariant fibre functor. The terminology is Joyal's.

Example 46.9 (Right fibrations)**.** A **right fibration** is a Cartesian fibration in which every morphism of $\mathcal{X}$ is p-Cartesian. Equivalently, p has RLP against all outer horns Λ^n_n — a stronger condition than RLP against just final-edge horns. Right fibrations are the ∞-categorical analogue of ordinary discrete fibrations (a slice $\mathcal{C}_{/x} \to \mathcal{C}$ is a right fibration).

46.2.2 Reorganizing: fibres are functorial

Theorem 46.10 (Functoriality of fibres)**.** *For a Cartesian fibration $p : \mathcal{X} \to \mathcal{S}$, the assignment $s \mapsto \mathcal{X}_s$ extends to a functor*
$$\mathcal{S}^{\mathrm{op}} \longrightarrow \mathcal{C}\mathrm{at}_\infty$$

(into the ∞-category of small ∞-categories), called the ***functor classified by*** *p. The transport along an edge $f : s' \to s$ in $\mathcal{S}$ is the functor $f^* : \mathcal{X}_s \to \mathcal{X}_{s'}$ sending $y \mapsto x$, where $\alpha : x \to y$ is the chosen Cartesian lift.*

Remark 46.11 (Choice issue)**.** The transport functor f^* depends on choices of Cartesian lifts. Different choices give homotopic functors, but to make the assignment *strictly* functorial in f requires the ∞-categorical straightening of Chapter 47.

46.3 Examples in the Wild

46.3.1 The arrow ∞-category

Example 46.12 (The arrow ∞-category)**.** $\mathcal{C}^{\Delta^1}$ is the ∞-category of arrows of $\mathcal{C}$. Together with $(\mathrm{ev}_0, \mathrm{ev}_1) : \mathcal{C}^{\Delta^1} \to \mathcal{C} \times \mathcal{C}$, it is the universal ∞-category over $\mathcal{C} \times \mathcal{C}$ classifying "data of an arrow."

The product map $(\mathrm{ev}_0, \mathrm{ev}_1)$ is *not* a Cartesian fibration in general — the joint conditions on source and target are too restrictive — but each component is a Cartesian fibration (target) or coCartesian fibration (source).

46.3.2 Free fibrations

Example 46.13 (Free fibration generated by a vertex)**.** For any quasi-category $\mathcal{C}$ and object $x \in \mathcal{C}_0$, the slice $\mathcal{C}_{/x} \to \mathcal{C}$ is a right fibration. It classifies the representable presheaf $\mathrm{Map}_{\mathcal{C}}(-, x) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ — a presheaf of spaces (Chapter 48).

46.3.3 Universal Cartesian fibration

Remark 46.14 (The universal example, foreshadowed)**.** There is a Cartesian fibration $\mathcal{U} \to \mathcal{C}\mathrm{at}_\infty^{\mathrm{op}}$ whose fibre over an ∞-category $\mathcal{D}$ is $\mathcal{D}$ itself, classifying the identity functor $\mathrm{id} : \mathcal{C}\mathrm{at}_\infty^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$. Every Cartesian fibration $\mathcal{X} \to \mathcal{S}$ is the pullback of $\mathcal{U}$ along the classifying functor $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$. This is the universal property of $\mathcal{C}\mathrm{at}_\infty$, made precise by the straightening theorem of Chapter 47.

46.4 Exercises

Define a p-Cartesian morphism.	A 1-simplex $\alpha : x \to y$ in $\mathcal{X}$ that lifts every outer horn Λ^n_n ending on α. Equivalently: restriction along α gives a homotopy equivalence on mapping spaces.
Why is the lifting condition specifically against Λ^n_n?	Because that horn shape leaves the final edge (α) free and demands the lift to fit a Cartesian-square diagram of $p(\alpha)$.
Define a Cartesian fibration.	An inner fibration $p : \mathcal{X} \to \mathcal{S}$ such that for every edge in $\mathcal{S}$ and every object over its target, there is a p-Cartesian morphism over that edge with that target.
Define a coCartesian fibration.	Dually: p-coCartesian morphisms exist for every edge in $\mathcal{S}$ and object over its source.
Why are Cartesian fibrations the higher analogue of Grothendieck fibrations?	Because Cartesian morphisms supply the transport data, and the totality of Cartesian lifts encodes the action of the base on the fibres — functorially up to coherent homotopy.
What's a right fibration?	A Cartesian fibration in which every morphism is p-Cartesian. Equivalently: RLP against all outer horns Λ^n_n (no constraint on which final edge).
When is $\mathrm{ev}_1 : \mathcal{C}^{\Delta^1} \to \mathcal{C}$ Cartesian?	Always, for any quasi-category $\mathcal{C}$. The Cartesian lift over $g : b' \to b$ at $f : a \to b$ is the precomposition $f \circ g$.
When is $\mathrm{ev}_0 : \mathcal{C}^{\Delta^1} \to \mathcal{C}$ coCartesian?	Always — dually. coCartesian lifts over $h : a \to a'$ at $f : a \to b$ are postcompositions $h^* f$.
Is ev_0 Cartesian?	No in general — postcomposing along an arrow in the base is the natural direction; precomposing requires inverses, which may not exist.
What's a slice as a right fibration?	$\mathcal{C}_{/x} \to \mathcal{C}$ is a right fibration for any object x of $\mathcal{C}$. It classifies the representable presheaf $\mathrm{Map}_{\mathcal{C}}(-, x)$.
What does "$\mathcal{X}_s$ is the fibre over s" mean?	$\mathcal{X}_s := \mathcal{X} \times_{\mathcal{S}} \{s\}$ — the pullback of $\mathcal{X} \to \mathcal{S}$ along $\{s\} \hookrightarrow \mathcal{S}$.

What's the functor classified by a Cartesian fibration p?	$\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$, $s \mapsto \mathcal{X}_s$. Transport along $f : s' \to s$: $\mathcal{X}_s \to \mathcal{X}_{s'}$ via Cartesian lifts.
Why is the transport functor only well-defined up to homotopy?	Because the choice of Cartesian lift over f is only unique up to a contractible space of choices. Strict functoriality requires straightening.
What is the universal Cartesian fibration?	$\mathcal{U} \to \mathcal{C}\mathrm{at}_\infty^{\mathrm{op}}$ classifying the identity functor. Every Cartesian fibration is a pullback of $\mathcal{U}$ — content of straightening.
Difference between Cartesian and inner fibration?	Inner fibrations have RLP against inner horns only (giving quasi-category fibres but no transport). Cartesian fibrations additionally have the Cartesian lift property (existence of transport data).
How does Cartesian fibration relate to right fibration?	Right fibration $\Rightarrow$ Cartesian (in particular). Right fibration: ALL morphisms are Cartesian. General Cartesian: only enough Cartesian lifts to handle base-edge transport.
What's a left fibration?	Dual of right: every morphism is p-coCartesian. Equivalently, RLP against Λ^n_0 horns.
Are right fibrations Kan fibrations?	Restricted to fibres over a Kan complex, yes — the fibrewise structure forces it. In general, no: right fibrations are RLP against Λ^n_n but not Λ^n_k for $0 < k < n$.
What does a Cartesian fibration p over Δ^1 look like?	Two fibres $\mathcal{X}_0$ and $\mathcal{X}_1$ together with a functor $\mathcal{X}_1 \to \mathcal{X}_0$ given by the Cartesian lifts of the single edge $0 \to 1$.
And over Δ^0?	A Cartesian fibration over a point is just a quasi-category $\mathcal{X}_0$ (the single fibre); the Cartesian condition is vacuous.
Why are Cartesian fibrations the right notion (vs. strict pseudofunctors)?	Because ∞-categorical functoriality is strict only up to coherent homotopy; encoding it via Cartesian lifts captures all the coherences without forcing strict commutativity.
What's the relationship between Cartesian and coCartesian for the same p?	p^{op} is Cartesian iff p is coCartesian. (Take opposites.) A self-dual map is called a *bifibration* or *biCartesian*.
What is $\mathcal{C}\mathrm{at}_\infty$?	The ∞-category of small ∞-categories. Its objects are quasi-categories; mapping spaces are spaces of functors (equivalences as 1-cells); the universal Cartesian fibration $\mathcal{U} \to \mathcal{C}\mathrm{at}_\infty^{\mathrm{op}}$ has fibres themselves.

Chapter Summary

Chapter Summary

Cartesian morphism. A 1-simplex α in $\mathcal{X}$ over $p : \mathcal{X} \to \mathcal{S}$ such that outer horn lifts $\Lambda^n_n \to \mathcal{X}$ ending on α extend to Δ^n. Equivalently, restriction along α is a weak equivalence on mapping spaces.
Cartesian and coCartesian fibrations. Inner fibrations admitting enough Cartesian (resp. coCartesian) lifts. Right fibration: every morphism Cartesian (so all of $\mathcal{X}$ lies over $\mathcal{S}$ "co-discretely"). Left fibration: every morphism coCartesian.
Functoriality of fibres. A Cartesian fibration p gives a functor $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$, $s \mapsto \mathcal{X}_s$, with transport along edges via Cartesian lifts. The choice of lifts is unique up to coherent homotopy.
Examples. Target evaluation $\mathcal{C}^{\Delta^1} \to \mathcal{C}$ (Cartesian); source evaluation (coCartesian); slices $\mathcal{C}_{/x}$ (right fibration); universal Cartesian fibration $\mathcal{U} \to \mathcal{C}\mathrm{at}_\infty^{\mathrm{op}}$ (classifying the identity).

Formal Restatement

Formal Restatement

Cartesian morphism. $\alpha : x \to y$ in $\mathcal{X}$ over p is Cartesian iff $\forall n \geq 2$, $\Lambda^n_n \to \mathcal{X}$ restricting to α on the final edge extends to Δ^n.

Equivalent characterization. α is Cartesian iff for all $z \in \mathcal{X}$:

$$\mathrm{Map}_{\mathcal{X}}(z, x) \to \mathrm{Map}_{\mathcal{X}}(z, y) \times_{\mathrm{Map}_{\mathcal{S}}(p(z),p(y))} \mathrm{Map}_{\mathcal{S}}(p(z), p(x))$$

is a weak equivalence of Kan complexes.

Cartesian fibration. $p : \mathcal{X} \to \mathcal{S}$, inner fibration, with: $\forall (f : s' \to s) \in \mathcal{S}_1$, $\forall y \in \mathcal{X}_s$, $\exists \alpha \in \mathcal{X}_1$ with $d_0\alpha = y, p(\alpha) = f, \alpha$ Cartesian.

coCartesian fibration. Dual: lift over source rather than target.

Right / left fibration. Cartesian (resp. coCartesian) with *every* morphism Cartesian (resp. coCartesian). Equivalent: RLP against Λ^n_n (resp. Λ^n_0) horns.

Universal Cartesian fibration. $\mathcal{U} \to \mathcal{C}\mathrm{at}^{\mathrm{op}}_\infty$, fibre over $\mathcal{D}$ equals $\mathcal{D}$; classifies $\mathrm{id} : \mathcal{C}\mathrm{at}^{\mathrm{op}}_\infty \to \mathcal{C}\mathrm{at}_\infty$. Every Cartesian fibration is a pullback (Chapter 47).

Chapter 47

Straightening and Unstraightening

Chapter Overview

The **straightening** and **unstraightening** constructions of Joyal and Lurie make precise the relationship previewed in Chapter 46: Cartesian fibrations over a simplicial set $\mathcal{S}$ are equivalent, in a homotopy-coherent sense, to functors $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$.

The full theorem is one of the heaviest technical results in ∞-category theory — Lurie's *Higher Topos Theory* devotes Chapter 2 to it, roughly 100 pages of dense combinatorial argument. We follow the chapter's proof-depth policy: state the theorem clearly, work the low-dimensional cases explicitly, sketch the technical strategy, and refer to HTT §2.2 for the full apparatus. The reader who has digested this chapter is then prepared to consult HTT directly.

The chapter introduces:

- **Marked simplicial sets** $\mathbf{sSet}^+$: simplicial sets together with a marked subset $M \subseteq X_1$ containing degeneracies. The marked edges are the "would-be Cartesian morphisms."
- The **Cartesian model structure** on $\mathbf{sSet}^+/\mathcal{S}$ whose fibrant objects are Cartesian fibrations over $\mathcal{S}$ (with their Cartesian edges as the marking).
- The **straightening functor** $\mathrm{St}_\mathcal{S} : \mathbf{sSet}^+/\mathcal{S} \to \mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+)$, where $\mathfrak{C}[\mathcal{S}]$ is the simplicial category rigidification of $\mathcal{S}$.
- The **unstraightening** $\mathrm{Un}_\mathcal{S}$ as right adjoint.

Theorem 47.3: $\mathrm{St}_\mathcal{S} \dashv \mathrm{Un}_\mathcal{S}$ is a Quillen equivalence between the Cartesian model structure on $\mathbf{sSet}^+/\mathcal{S}$ and the projective model structure on $\mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+)$. On homotopy categories, this gives the equivalence

$$\mathrm{Cart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}\mathrm{at}_\infty),$$

the precise statement of "Cartesian fibrations = presheaves of ∞-categories."

47.1 The Straightening Equivalence

47.1.1 Formalizing

Where the Name Comes From

Straightening and **unstraightening** are Lurie's terminology (HTT 2009, building on Joyal's earlier work). The intuition: a Cartesian fibration $\mathcal{X} \to \mathcal{S}$ has a "crooked" representation (coherent rather than strict data); the *straightening* functor converts it to a *straight* (functorial) presentation as a Cat_∞-valued functor. *Unstraightening* is the reverse operation, building a Cartesian fibration from a functor. The adjunction sits between the two presentations and is, in fact, a Quillen equivalence.

The classical 1-categorical version of straightening is the **Grothendieck construction** (1960), which converts a pseudofunctor $F : \mathcal{S}^{\mathrm{op}} \to \mathrm{Cat}$ to a fibred category $\int F \to \mathcal{S}$. Lurie's ∞-categorical version generalizes both constructions to the homotopy-coherent setting.

Where the Name Comes From

Marked simplicial set refers to the marking of certain edges. The marking marks which edges are "Cartesian" or "would-be equivalences." The terminology is descriptive (mathematical objects with extra "marks") and goes back at least to model-category literature of the 1990s.

Definition 47.1 (Marked simplicial set)**.** A **marked simplicial set** is a pair (X, M) where $X \in \mathbf{sSet}$ and $M \subseteq X_1$ is a subset containing all degenerate 1-simplices. A morphism $(X, M) \to (Y, N)$ is a simplicial-set map $X \to Y$ sending M into N. The category is $\mathbf{sSet}^+$.

How to Say This

$\mathbf{sSet}^+$ *(read: "marked simplicial sets")*; (X, M) has M as the "marked edges" or "would-be equivalences/Cartesian morphisms." For a quasi-category $\mathcal{C}$, the natural marking is its equivalences; for a Cartesian fibration $p : \mathcal{X} \to \mathcal{S}$, the natural marking on $\mathcal{X}$ is its Cartesian morphisms.

Where the Name Comes From

The **rigidification** functor $\mathfrak{C}$ converts a simplicial set into a (rigid, i.e., strict) simplicial category, in contrast to the coherent-nerve direction which goes the other way. "Rigid" here means "strict" — the resulting structure has on-the-nose composition, not just up-to-coherent-homotopy composition. The notation $\mathfrak{C}$ is German typographer's blackletter (Fraktur), used because Lurie's HTT chose Fraktur letters for simplicial-category-valued constructions; the choice is suggestive of the older, more rigid mathematical typography.

Definition 47.2 (Rigidification $\mathfrak{C}$)**.** The **rigidification** functor $\mathfrak{C} : \mathbf{sSet} \to \mathbf{sCat}$ sends a simplicial set $\mathcal{S}$ to a simplicial category whose objects are the vertices of $\mathcal{S}$ and whose mapping simplicial sets $\mathfrak{C}[\mathcal{S}](x, y)$ are specific Kan complexes built from the simplices of $\mathcal{S}$ with appropriate end-conditions.

For $\mathcal{S} = \mathrm{N}(\mathcal{D})$ the nerve of an ordinary category, $\mathfrak{C}[\mathcal{S}]$ is the trivial simplicial enrichment of $\mathcal{D}$ (mapping sets are discrete simplicial sets).

Theorem 47.3 (Straightening–Unstraightening Equivalence)**.** *For every simplicial set $\mathcal{S}$, there is a Quillen adjunction*

$$\mathrm{St}_{\mathcal{S}} : \mathbf{sSet}^+/\mathcal{S} \ \rightleftarrows\ \mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+) : \mathrm{Un}_{\mathcal{S}}$$

between the Cartesian model structure on $\mathbf{sSet}^+/\mathcal{S}$ and the projective model structure on $\mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+)$. The adjunction is a Quillen equivalence.

On homotopy categories, after restricting to fibrant objects, this gives

$$\mathrm{Cart}(\mathcal{S}) \ \simeq\ \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}at_\infty).$$

Proof strategy. The full proof spans HTT §2.2 (about 100 pages of dense combinatorics). The strategy:

Step 1: Construct $\mathrm{St}_{\mathcal{S}}$ *explicitly.* For $(\mathcal{X}, M) \to \mathcal{S}$ a marked simplicial set over $\mathcal{S}$, define $\mathrm{St}_{\mathcal{S}}(\mathcal{X}, M)(s)$ as a specific simplicial set built from the marked n-simplices over chains starting at s.

Step 2: Verify the adjunction. The right adjoint $\mathrm{Un}_{\mathcal{S}}$ has an explicit formula as a certain limit; the unit and counit are checked combinatorially.

Step 3: Prove Quillen. Use the cofibrant generation of both model structures and check that $\mathrm{St}_{\mathcal{S}}$ sends generating (trivial) cofibrations to (trivial) cofibrations.

Step 4: Prove Quillen equivalence. The hardest step. Verify that unit and counit on cofibrant–fibrant objects are weak equivalences. This uses the rigidification's good behavior and a detailed analysis of how marked simplicial sets present Cartesian fibrations.

We will work through the low-dimensional cases in §47.2 below. ☐ ☐

47.2 Low-Dimensional Warm-Up

47.2.1 Formally: $\mathcal{S} = \Delta^0$

Example 47.4 (Straightening over a point)**.** When $\mathcal{S} = \Delta^0$, $\mathfrak{C}[\Delta^0]$ is the trivial simplicial category with one object and one morphism. Then:

- $\mathbf{sSet}^+/\Delta^0 = \mathbf{sSet}^+$.
- Cartesian fibrations over Δ^0 are simply quasi-categories (with their natural marking by equivalences).
- $\mathrm{Fun}(\mathfrak{C}[\Delta^0]^{\mathrm{op}}, \mathbf{sSet}^+) = \mathbf{sSet}^+$.

The straightening is the identity on the underlying $\mathbf{sSet}$. The equivalence trivially holds.

47.2.2 Formally: $\mathcal{S} = \Delta^1$

Example 47.5 (Straightening over the walking arrow)**.** When $\mathcal{S} = \Delta^1$, $\mathfrak{C}[\Delta^1]$ is a simplicial category with two objects $0, 1$ and one non-trivial morphism $0 \to 1$ (plus identities). The mapping simplicial set $\mathfrak{C}[\Delta^1](0, 1)$ is Δ^0.

A Cartesian fibration $p : \mathcal{X} \to \Delta^1$ has fibres $\mathcal{X}_0$ and $\mathcal{X}_1$, and the Cartesian condition gives a transport functor $\phi : \mathcal{X}_1 \to \mathcal{X}_0$ (over the edge $0 \to 1$ in Δ^1). The data is captured by:

- Two quasi-categories: $\mathcal{X}_0, \mathcal{X}_1$.
- A functor: $\phi : \mathcal{X}_1 \to \mathcal{X}_0$.

A functor $\mathfrak{C}[\Delta^1]^{\mathrm{op}} \to \mathbf{sSet}^+$ is also exactly the data of: $F(1), F(0)$ (two marked simplicial sets) and a morphism $F(0 \to 1)^{\mathrm{op}} = \phi : F(1) \to F(0)$.

So St_{Δ^1} assigns $(\mathcal{X}_0, \mathcal{X}_1, \phi)$ to $\mathcal{X} \to \Delta^1$, which is the same data on both sides. The straightening is essentially the identity. □

47.2.3 Formally: general low-dimensional pattern

Remark 47.6 (Why low-dimensional examples are easy)**.** For $\mathcal{S}$ a 1-dimensional simplicial set (so $\mathfrak{C}[\mathcal{S}]$ is a trivial simplicial category), the straightening reduces to ordinary 1-categorical Grothendieck unstraightening. Coherence comes for free because all mapping simplicial sets in $\mathfrak{C}[\mathcal{S}]$ are Δ^0.

The non-trivial content of $\mathrm{St}_{\mathcal{S}}$ appears for $\mathcal{S}$ of dimension ≥ 2 (where $\mathfrak{C}[\mathcal{S}]$ has non-trivial mapping simplicial sets) and especially for $\mathcal{S}$ a general quasi-category (where the rigidification injects infinite-dimensional combinatorics from 2-simplices and above).

47.3 The Cartesian Model Structure

47.3.1 Formalizing

Theorem 47.7 (Cartesian model structure on $\mathbf{sSet}^+/\mathcal{S}$)**.** *The category $\mathbf{sSet}^+/\mathcal{S}$ admits a model structure — the* ***Cartesian model structure*** *— characterized by:*

- *Cofibrations: monomorphisms (of underlying simplicial sets).*
- *Fibrant objects: Cartesian fibrations $(\mathcal{X}, M_{\mathrm{Cart}}) \to \mathcal{S}$, marked by their Cartesian edges.*
- *Weak equivalences: maps inducing equivalences on the corresponding ∞-categories of straightened functors.*

Remark 47.8 (Two markings on a quasi-category)**.** A quasi-category $\mathcal{C}$ has two natural markings:

- Maximal marking $\mathcal{C}^\sharp$: every 1-simplex marked.
- Minimal (equivalence) marking $\mathcal{C}^\flat$: only equivalences marked.

The maximal marking corresponds to viewing $\mathcal{C}$ as a Kan complex (every edge is "an equivalence" by fiat); the minimal marking is the “honest” one for ∞-category theory. The Cartesian model structure sees them differently.

47.3.2 Reorganizing: presheaves as a model

Theorem 47.9 (Equivalence with presheaves of ∞-categories)**.** *On homotopy categories, the straightening induces an equivalence*

$$\mathrm{Ho}(\mathbf{sSet}^+/\mathcal{S})_{\mathrm{Cart}} \;\simeq\; \mathrm{Ho}\Big(\mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+)\Big)_{\mathrm{proj}}.$$

Both sides are models for the homotopy category of presheaves of ∞-categories on $\mathcal{S}$. Cartesian fibrations are an “encoded form” of such presheaves.

47.4 Computational Use

47.4.1 Formally

Example 47.10 (Pulling back classifying functors)**.** Given a functor $F : \mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$, the corresponding Cartesian fibration $\mathcal{X}_F \to \mathcal{S}$ satisfies: for every $s \in \mathcal{S}_0$, the fibre $(\mathcal{X}_F)_s \simeq F(s)$ in $\mathcal{C}\mathrm{at}_\infty$. The unstraightening $\mathrm{Un}_{\mathcal{S}}$ implements this passage.

Pullback: for $f : \mathcal{T} \to \mathcal{S}$, the composition $\mathrm{Un}_{\mathcal{T}}(F \circ f^{\mathrm{op}}) = \mathcal{X}_F \times_{\mathcal{S}} \mathcal{T}$.

Example 47.11 (Representable presheaves and slices)**.** For an object x of a quasi-category $\mathcal{C}$, the slice fibration $\mathcal{C}_{/x} \to \mathcal{C}$ is a right fibration whose classifying functor under straightening is the representable presheaf

$$\mathrm{Map}_{\mathcal{C}}(-, x) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}.$$

Right fibrations land in the subcategory of ∞-*groupoid-valued* functors $\mathcal{C}^{\mathrm{op}} \to \mathcal{S}$. The straightening gives, in particular, a precise sense in which “every right fibration is a presheaf of spaces.”

47.4.2 Reorganizing: where this goes

Remark 47.12 (Phase-2 milestone)**.** With straightening in place, Chapter 48 will state and prove the ∞-categorical Yoneda lemma in the form

$$\mathcal{C} \;\hookrightarrow\; \mathcal{P}(\mathcal{C}) := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}),$$

a fully faithful embedding into presheaves of spaces. This is the ∞-categorical avatar of the classical Yoneda embedding (Chapter 9, 15, 20, 28), and it organizes the rest of Volume IV.

47.5 Exercises

What does the straightening theorem say in one sentence?	Cartesian fibrations over $\mathcal{S}$ are equivalent (as ∞-categories) to functors $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$.
What is a marked simplicial set?	A pair (X, M) with $X \in \mathbf{sSet}$ and $M \subseteq X_1$ containing degeneracies. The marking is the would-be Cartesian/equivalence morphisms.
Why mark simplicial sets at all?	Because Cartesian fibrations and their morphisms are best handled when the Cartesian edges are explicitly tracked. The Cartesian model structure on $\mathbf{sSet}^+$ uses this marking.
What is the rigidification $\mathfrak{C}[\mathcal{S}]$?	A simplicial category with the same objects as $\mathcal{S}$, whose mapping simplicial sets are explicit Kan complexes built from $\mathcal{S}$'s simplices. Reduces to discrete enrichment when $\mathcal{S} = \mathrm{N}(\mathcal{D})$.
State the straightening adjunction.	$\mathrm{St}_{\mathcal{S}} : \mathbf{sSet}^+/\mathcal{S} \rightleftarrows \mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+) : \mathrm{Un}_{\mathcal{S}}$, Quillen equivalence.
What do the fibres of a Cartesian fibration become under straightening?	The value of the corresponding functor $\mathcal{S}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}_\infty$ at each vertex.
What's the straightening over Δ^0?	Trivially the identity: Cartesian fibrations over a point are quasi-categories, and functors from the one-object category to $\mathbf{sSet}^+$ are just $\mathbf{sSet}^+$.
What's the straightening over Δ^1?	The data of a Cartesian fibration $p : \mathcal{X} \to \Delta^1$ — two fibres $\mathcal{X}_0, \mathcal{X}_1$ and a functor $\mathcal{X}_1 \to \mathcal{X}_0$ — equals the data of a functor $\mathfrak{C}[\Delta^1]^{\mathrm{op}} \to \mathbf{sSet}^+$. Both are: "two marked simplicial sets and a morphism between them."
Why is straightening hard for higher-dimensional $\mathcal{S}$?	Because $\mathfrak{C}[\mathcal{S}]$ has non-trivial mapping simplicial sets when $\mathcal{S}$ has 2-simplices and above. The combinatorics of comparing fibrations and functors then becomes substantial.
What are the two natural markings on a quasi-category $\mathcal{C}$?	Maximal ($\mathcal{C}^\sharp$): all edges marked. Minimal ($\mathcal{C}^\flat$): only equivalences marked. The two correspond to different categorical pictures (Kan-complex view vs. honest ∞-category view).
What is the Cartesian model structure on $\mathbf{sSet}^+/\mathcal{S}$?	Cofibrations: monomorphisms; fibrant objects: Cartesian fibrations with their Cartesian markings; weak equivalences: fibrewise equivalences plus base.
How does the equivalence read at the level of homotopy categories?	$\mathrm{Cart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}\mathrm{at}_\infty)$ — the homotopy category of Cartesian fibrations over $\mathcal{S}$ is the same as the homotopy category of $\mathcal{S}^{\mathrm{op}}$-indexed diagrams of ∞-categories.
What about coCartesian fibrations?	The dual theorem: coCartesian fibrations over $\mathcal{S}$ are equivalent to functors $\mathcal{S} \to \mathcal{C}\mathrm{at}_\infty$ (covariant). Symmetric under $\mathcal{S} \leftrightarrow \mathcal{S}^{\mathrm{op}}$.
And right fibrations?	Right fibrations correspond to functors $\mathcal{S}^{\mathrm{op}} \to \mathcal{S}$ (into the ∞-category of spaces). Because right fibrations have all-Cartesian morphisms and Kan-complex fibres.
What does the slice $\mathcal{C}_{/x}$ straighten to?	The representable presheaf $\mathrm{Map}_{\mathcal{C}}(-, x) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$.
How is straightening used in practice?	To convert between fibration-form and functor-form of structured ∞-categorical data. Many constructions are easier to describe as fibrations (universal example: limits, colimits) but easier to manipulate as functors (universal example: adjoint functor existence).

What's the universal Cartesian fibration?	$\mathcal{U} \to \mathcal{C}at_\infty^{\mathrm{op}}$, whose fibre over $\mathcal{D}$ is $\mathcal{D}$. Straightens to $\mathrm{id} : \mathcal{C}at_\infty^{\mathrm{op}} \to \mathcal{C}at_\infty$.
Why is the full proof of straightening so long?	Because verifying that $\mathrm{St}_\mathcal{S}$ is a Quillen equivalence requires checking unit and counit on fibrant–cofibrant objects involve intricate combinatorial decomposition of marked simplicial sets and their relationship to the rigidification.
Where in the literature is the full proof?	Lurie, *Higher Topos Theory*, §2.2 (about 100 pages). Riehl–Verity, *Elements of ∞-Category Theory*, gives an alternative model-independent treatment.
Can one avoid straightening?	Yes — Riehl–Verity develop ∞-category theory model-independently, using comma constructions and never explicitly invoking the straightening equivalence. Practical for many purposes, but the equivalence itself is a fundamental structural fact.
Why does this chapter sketch rather than prove?	Per the Volume IV proof-depth policy (selective full proofs, otherwise sketches). The straightening proof itself fills a textbook chapter and would dilute the bridge-to-literature mission.
What chapter uses straightening crucially?	Chapter 48 (∞-Yoneda) — the embedding $\mathcal{C} \to \mathcal{P}(\mathcal{C})$ is via straightening of the slice fibrations.
What is the homotopy-categorical statement to take away?	$\mathrm{Cart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}at_\infty)$. This is the ∞-categorical Grothendieck construction.

Chapter Summary

Chapter Summary

Marked simplicial sets. $\mathbf{sSet}^+$ = pairs (X, M) with $M \subseteq X_1$ containing degeneracies; the marking encodes Cartesian/equivalence edges.
Rigidification. $\mathfrak{C} : \mathbf{sSet} \to \mathbf{sCat}$; sends a simplicial set to a simplicial category with specific Kan-complex mapping spaces. Trivial on nerves of 1-categories.
Straightening–unstraightening. For every $\mathcal{S}$, $\mathrm{St}_\mathcal{S} \dashv \mathrm{Un}_\mathcal{S}$ is a Quillen equivalence between $(\mathbf{sSet}^+/\mathcal{S})_{\mathrm{Cart}}$ and $\mathrm{Fun}(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+)_{\mathrm{proj}}$.
Homotopy-category statement.

$$\mathrm{Cart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}at_\infty).$$

"Cartesian fibrations = presheaves of ∞-categories."
Low-dimensional reduction. For $\mathcal{S}$ a nerve of a 1-category, straightening reduces to ordinary Grothendieck construction. The full ∞-categorical content appears for $\mathcal{S}$ with non-trivial mapping simplicial sets in its rigidification.
Use. Chapter 48 uses straightening to construct the ∞-Yoneda embedding $\mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C})$.

Formal Restatement

Formal Restatement

Marked simplicial sets. $\mathbf{sSet}^+$ = category of pairs (X, M), $M \supseteq \{\text{degenerate 1-simplices of } X\}$.

Rigidification. $\mathfrak{C} : \mathbf{sSet} \to \mathbf{sCat}$, the left adjoint to the simplicial nerve $\mathrm{N}^{\mathrm{coh}}$.

Cartesian model structure on $\mathbf{sSet}^+/\mathcal{S}$.

- Cofibrations: monomorphisms.
- Fibrant: Cartesian fibrations $(\mathcal{X}, M_{\mathrm{Cart}}) \to \mathcal{S}$.
- Weak equivalences: maps inducing equivalences on classifying functors.

Straightening–unstraightening (Theorem 47.3). $\mathrm{St}_\mathcal{S} \dashv \mathrm{Un}_\mathcal{S}$, Quillen equivalence:

$$\mathbf{sSet}^+/\mathcal{S} \simeq_{\mathrm{Q}} \mathrm{Fun}\big(\mathfrak{C}[\mathcal{S}]^{\mathrm{op}}, \mathbf{sSet}^+\big).$$

Homotopy-category statement. $\mathrm{Cart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{C}at_\infty)$; $\mathrm{coCart}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}, \mathcal{C}at_\infty)$; $\mathrm{Right}(\mathcal{S}) \simeq$

$\mathrm{Fun}(\mathcal{S}^{\mathrm{op}}, \mathcal{S})$; $\mathrm{Left}(\mathcal{S}) \simeq \mathrm{Fun}(\mathcal{S}, \mathcal{S})$.

Slice as representable. $\mathcal{C}_{/x} \to \mathcal{C}$ corresponds under straightening to $\mathrm{Map}_{\mathcal{C}}(-, x) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$.

Chapter 48

The ∞-Categorical Yoneda Lemma

Chapter Overview

The Yoneda lemma is the most read theorem of category theory. It has appeared four times in this book: in Chapter 9 as a hom-set bijection; in Chapter 15 as the representability principle; in Chapter 20 in enriched form; in Chapter 28 unified via ends in the formal Yoneda statement. This chapter delivers the fifth incarnation — the ∞-categorical Yoneda lemma — which subsumes all of the above as special cases.
The statement: for any (small) ∞-category $\mathcal{C}$, there is a **Yoneda embedding**

$$y : \mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C}) := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}),$$

where $\mathcal{S}$ is the ∞-category of spaces (anima). The embedding is *fully faithful*: it induces equivalences on mapping spaces. The **∞-Yoneda lemma** computes mapping spaces in $\mathcal{P}(\mathcal{C})$ out of representables:

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \;\simeq\; F(x), \quad \text{naturally in } x \in \mathcal{C} \text{ and } F \in \mathcal{P}(\mathcal{C}).$$

The structural consequence is that $\mathcal{P}(\mathcal{C})$ is the **free cocompletion** of $\mathcal{C}$ — the universal ∞-category containing $\mathcal{C}$ in which all small colimits exist.
This chapter is the milestone of Phase 2: the foundation is laid for the rest of ∞-category theory. A reader who has completed Chapter 48 can open Lurie's *Higher Topos Theory* (especially HTT §5.1, on presentable ∞-categories) or Riehl–Verity's *Elements of ∞-Category Theory* (their development of Yoneda in Chapter 5) and read fluently.

48.1 Presheaves of Spaces

48.1.1 Formalizing

Where the Name Comes From

Yoneda lemma is named for *Nobuo Yoneda* (1930–1996, Japanese mathematician). The story goes: Yoneda communicated the result to Mac Lane on a train platform during a 1954 Paris conference, and Mac Lane brought it back to the West. Yoneda himself may never have published the result in this form. The lemma has appeared in five guises in this book: Chapter 9 (ordinary, set-valued); Chapter 15 (representability); Chapter 20 (enriched); Chapter 28 (formal, via ends); and now Chapter 48 (∞-categorical). Each layer generalizes the previous; the underlying philosophy — *"an object is determined by its representable functor"* — remains constant.

Where the Name Comes From

Presheaf (cross-ref Ch. 38) is from Leray's 1946 *faisceau* / *préfaisceau* terminology in sheaf theory. A *presheaf of spaces* on $\mathcal{C}$ is a presheaf valued in $\mathcal{S}$ rather than **Set** — the ∞-categorical refinement.

Definition 48.1 (∞-category of presheaves)**.** For a small ∞-category $\mathcal{C}$, the **∞-category of presheaves of spaces** on $\mathcal{C}$ is

$$\mathcal{P}(\mathcal{C}) \;:=\; \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}),$$

the ∞-category of functors from $\mathcal{C}^{\mathrm{op}}$ to the ∞-category of spaces $\mathcal{S}$.

Concretely, an object of $\mathcal{P}(\mathcal{C})$ is a functor of quasi-categories $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$. A morphism is a natural transformation, presented as a 1-simplex of $\mathcal{P}(\mathcal{C})$.

How to Say This

$\mathcal{P}(\mathcal{C})$ *(read: "the presheaves of spaces on $\mathcal{C}$")*; sometimes $\widehat{\mathcal{C}}$ or $\mathrm{PSh}(\mathcal{C})$. The dimensional shift is significant: presheaves of *sets* (Chapter 9) become presheaves of *spaces* (Kan complexes / ∞-groupoids) in the ∞-setting.

Remark 48.2 (Why presheaves of spaces, not of categories?)**.** The natural choice for "the values of a Yoneda embedding" is the ∞-category of ∞-*groupoids* (= spaces, = $\mathcal{S}$), not the larger $\mathcal{C}\mathrm{at}_\infty$. Reason: the representable functor $\mathrm{Map}_{\mathcal{C}}(-, x) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ takes values in Kan complexes — Map is always ∞-groupoidal (Theorem 44.44.2). We embed into $\mathcal{P}(\mathcal{C})$, not into a larger functor category.

Theorem 48.3 (Properties of $\mathcal{P}(\mathcal{C})$)**.** *$\mathcal{P}(\mathcal{C})$ is:*

1. *A presentable ∞-category (Chapter 52): cocomplete, accessible, and generated under small colimits by representables.*
2. *Equipped with a Cartesian fibration to $\mathcal{C}\mathrm{at}_\infty^{\mathrm{op}}$ (the universal fibration of Chapter 46), pulled back along the inclusion $\mathcal{S} \hookrightarrow \mathcal{C}\mathrm{at}_\infty$.*
3. *Complete and cocomplete: ∞-(co)limits computed pointwise from $\mathcal{S}$.*

Example 48.4 (Presheaves on a 1-category)**.** For $\mathcal{C} = \mathrm{N}(\mathcal{D})$ the nerve of an ordinary category, $\mathcal{P}(\mathrm{N}(\mathcal{D}))$ is the ∞-categorical refinement of the ordinary presheaf category $\widehat{\mathcal{D}} = \mathbf{Set}^{\mathcal{D}^{\mathrm{op}}}$. Specifically:

- Objects of $\mathcal{P}(\mathrm{N}(\mathcal{D}))$ are functors $\mathcal{D}^{\mathrm{op}} \to \mathcal{S}$ — i.e., simplicial presheaves (presheaves of Kan complexes).
- Morphisms are natural transformations, ∞-categorically (homotopy-coherent).

$\mathrm{ho}(\mathcal{P}(\mathrm{N}(\mathcal{D})))$ recovers $\mathrm{Ho}(\widehat{\mathcal{D}})$, the homotopy category of simplicial presheaves.

48.2 The ∞-Yoneda Embedding

48.2.1 Formalizing

Reader's Annotation

The ∞-*Yoneda lemma is the central organizing theorem of ∞-category theory*, just as the ordinary Yoneda lemma is for 1-category theory. After internalizing this section, you should be able to:

- Read mapping spaces as $F(x)$ for the right F.
- Construct ∞-functors out of $\mathcal{C}$ by specifying their values on representables.
- Recognize representability as a fundamental property of "natural" functors.

The slogans of Vol II's representability principle (Ch. 15) carry across to the ∞-setting with mapping spaces in place of hom-sets and equivalences in place of bijections.

Definition 48.5 (∞-Yoneda embedding)**.** For a small ∞-category $\mathcal{C}$, the **∞-Yoneda embedding** is the functor

$$y : \mathcal{C} \longrightarrow \mathcal{P}(\mathcal{C})$$

characterized by: $y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$, the representable presheaf at x.

Theorem 48.6 (Three constructions of y)**.** *The Yoneda functor y can be constructed in three equivalent ways:*

1. ***Via straightening (Chapter 47).*** *The Cartesian fibration $\mathrm{ev}_1 : \mathcal{C}^{\Delta^1} \to \mathcal{C}$ (Example 46.46.7) restricts over each $x \in \mathcal{C}_0$ to the slice $\mathcal{C}_{/x} \to *$. Straightening this fibration gives the functor $y : \mathcal{C} \to \mathcal{P}(\mathcal{C})$, $y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$.*
2. ***Via the universal property.*** *y is the unique colimit-preserving functor $\mathcal{P}(\mathcal{C}) \to \mathcal{P}(\mathcal{C})$ restricting to the identity on $\mathcal{C}$ — i.e., y sits inside the universal property of $\mathcal{P}(\mathcal{C})$ as a free cocompletion (§48.4).*
3. ***Via cosimplicial spaces.*** *The cosimplicial space $\Delta^n \mapsto N(\mathcal{C}^{\mathrm{op}}_{/\bullet,[n]})$ — a specific cosimplicial simplicial set — Kan-extends along Yoneda to give y.*

All three are weakly equivalent in $\mathrm{Fun}(\mathcal{C}, \mathcal{P}(\mathcal{C}))$.

Proof sketch. Construction (1). The slice $\mathcal{C}_{/x}$ is a right fibration (Example 46.46.13). Its associated functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ under straightening (Chapter 47) is $\mathrm{Map}_{\mathcal{C}}(-, x)$. Varying x gives the functor $y : \mathcal{C} \to \mathcal{P}(\mathcal{C})$.

Construction (2) is more conceptual: $\mathcal{P}(\mathcal{C})$ has a universal property as the free cocompletion (Theorem 48.10 below). The Yoneda embedding is forced by this universal property to identify $\mathcal{C}$ with the generators.

Construction (3) is the most concrete and computational, generalizing Riehl–Verity's exposition. Each cosimplicial space gives a functor $\mathbf{sSet}^{\mathcal{C}^{\mathrm{op}}} \to \mathbf{sSet}$, and the specific cosimplicial $N(\mathcal{C}^{\mathrm{op}}_{/\bullet,[n]})$ recovers Yoneda. □ □

48.3 The ∞-Yoneda Lemma

48.3.1 Formalizing

Theorem 48.7 (∞-Yoneda Lemma)**.** *For any small ∞-category $\mathcal{C}$ and any $F \in \mathcal{P}(\mathcal{C})$, there is a weak equivalence of Kan complexes*

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \;\simeq\; F(x),$$

natural in $x \in \mathcal{C}$ and $F \in \mathcal{P}(\mathcal{C})$.

Proof sketch. The strategy mirrors the 1-categorical Yoneda proof (Chapter 9), but all sets become spaces.

Construct the comparison map. Given a natural transformation $\eta : y(x) \to F$, take $\eta_x(\mathrm{id}_x) \in F(x)$ — the image of the universal element. This defines $\Phi : \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \to F(x)$.

Construct the inverse. Given $\xi \in F(x)$, define $\widetilde{\xi} : y(x) \to F$ pointwise: $\widetilde{\xi}_z : \mathrm{Map}_{\mathcal{C}}(z, x) \to F(z)$ sends f to $F(f)(\xi)$. Coherence of $\widetilde{\xi}$ as a natural transformation requires ∞-categorical functoriality of F, which is built in.

Verify mutual inverse. The two compositions are: $\xi \mapsto \widetilde{\xi} \mapsto \widetilde{\xi}_x(\mathrm{id}_x) = F(\mathrm{id}_x)(\xi) \simeq \xi$ (functoriality). The other composition $\eta \mapsto \eta_x(\mathrm{id}_x) \mapsto \widetilde{\eta_x(\mathrm{id}_x)}$ recovers η by naturality. The required coherences (at 2-simplex level and above) are checked combinatorially. □ □

Corollary 48.8 (Yoneda for representables)**.** *For $x, y \in \mathcal{C}$:*

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), y(y)) \simeq \mathrm{Map}_{\mathcal{C}}(x, y).$$

Proof. Apply ∞-Yoneda to $F = y(y)$:

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), y(y)) \simeq y(y)(x) = \mathrm{Map}_{\mathcal{C}}(x, y). \quad \square$$

□

Theorem 48.9 (Yoneda embedding is fully faithful)**.** *The functor $y : \mathcal{C} \to \mathcal{P}(\mathcal{C})$ is fully faithful: $\mathrm{Map}_{\mathcal{C}}(x, y) \xrightarrow{\sim} \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), y(y))$ is a weak homotopy equivalence of Kan complexes for all x, y.*

Proof. By Corollary 48.8 the comparison map is a weak equivalence; this is exactly full faithfulness in the ∞-categorical sense (Theorem 44.44.8). □ □

48.4 Free Cocompletion

48.4.1 Formalizing

Where the Name Comes From

Free cocompletion terminology: a category $\mathcal{P}(\mathcal{C})$ "freely adjoins" all small colimits to $\mathcal{C}$, in the same sense as a *free* group adjoins all formal products to a generating set. The construction is universal among colimit-preserving extensions out of $\mathcal{C}$. The pattern goes back to the universal-property constructions of Vol I; the ∞-categorical version was made precise by Joyal and Lurie.

Theorem 48.10 (Universal property of $\mathcal{P}(\mathcal{C})$)**.** *For every cocomplete ∞-category $\mathcal{D}$, the restriction functor*

$$\mathrm{Fun}^{\mathrm{cc}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{y^*} \mathrm{Fun}(\mathcal{C}, \mathcal{D})$$

(from colimit-preserving functors $\mathcal{P}(\mathcal{C}) \to \mathcal{D}$ to functors $\mathcal{C} \to \mathcal{D}$, via precomposition with the Yoneda embedding) is an equivalence.

*In words: $\mathcal{P}(\mathcal{C})$ is the **free cocompletion** of $\mathcal{C}$. Every functor $F : \mathcal{C} \to \mathcal{D}$ to a cocomplete ∞-category extends uniquely (up to coherent homotopy) to a colimit-preserving functor $\bar{F} : \mathcal{P}(\mathcal{C}) \to \mathcal{D}$.*

Proof sketch. Construction of $\bar{F}$. Every $P \in \mathcal{P}(\mathcal{C})$ is a colimit of representables (**density theorem**, below). Set $\bar{F}(P) := \mathrm{colim}\,(F \circ p)$ where $p : \mathcal{C}_{/P} \to \mathcal{C}$ is the canonical diagram of representables.

Uniqueness of $\bar{F}$. A colimit-preserving functor is determined by its restriction to a generating set; representables generate $\mathcal{P}(\mathcal{C})$ under colimits.

Equivalence. The equivalence is established by checking fully-faithfulness and essential surjectivity of y^*, both of which follow from the density theorem. □ □

Where the Name Comes From

The **density theorem** (also: *co-Yoneda*) says representables are "dense" in $\mathcal{P}(\mathcal{C})$, in the topological sense: every object is a (canonical) colimit of representables. The terminology is borrowed from topology (a "dense" subset is one whose closure is the whole space) but the formal meaning is purely categorical. Density of representables is what makes Kan extensions *computable* (Ch. 51) — extending a functor from representables extends it everywhere.

Theorem 48.11 (Density theorem)**.** *Every $P \in \mathcal{P}(\mathcal{C})$ is canonically a colimit of representables: there is an equivalence*

$$P \simeq \mathrm{colim}_{(x,\xi) \in \mathcal{C}_{/P}}\, y(x),$$

where $\mathcal{C}_{/P}$ is the slice of $\mathcal{C}$ over P (via y), and the colimit is computed in $\mathcal{P}(\mathcal{C})$.

Remark 48.12 (Foreshadowing Chapter 49)**.** The density theorem is the cornerstone of ∞-categorical limit and colimit theory. It says: in $\mathcal{P}(\mathcal{C})$, representables form a *dense generator*. Constructing functors out of $\mathcal{P}(\mathcal{C})$ that preserve colimits is the same as specifying their values on representables — a powerful technique used throughout Chapter 49 to construct ∞-(co)limits in general ∞-categories.

48.5 Comparison with the Earlier Yonedas

48.5.1 Reorganizing

Remark 48.13 (The Yoneda quartet, revisited)**.** The ∞-Yoneda lemma (Theorem 48.7) is the common ancestor of the four earlier appearances:

- **Chapter 9 (ordinary Yoneda).** For $\mathcal{C}$ an ordinary category, $\mathcal{P}(\mathrm{N}(\mathcal{C}))$ contains $\widehat{\mathcal{C}} = \mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}$ as the full subcategory of π_0-valued presheaves (presheaves of *discrete* spaces). Restricting ∞-Yoneda to this subcategory recovers $\mathrm{Hom}(y(x), F) \cong F(x)$.
- **Chapter 15 (representability).** The ∞-statement "every presheaf of spaces is a colimit of representables" (Density theorem) is the higher version of Chapter 15's "representability principle."
- **Chapter 20 (enriched Yoneda).** For $\mathcal{V}$-enriched categories, the $\mathcal{V}$-Yoneda lemma is a special case of ∞-Yoneda by setting $\mathcal{C}$ to the ∞-category underlying $\mathcal{V}\mathrm{Cat}$.
- **Chapter 28 (formal Yoneda via ends).** The end formula $[\mathcal{C}, \mathcal{V}](y(-), F) \cong F$ in coend form generalizes to ∞-end form $\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(-), F) \simeq F$ — the precise ∞-statement we have just proved.

Remark 48.14 (Phase-2 milestone)**.** With ∞-Yoneda in hand, the reader has the central organizing theorem of ∞-category theory. Chapter 49 will use it to develop ∞-(co)limits in general ∞-categories. Chapter 50 will develop ∞-adjunctions, building on the representability that ∞-Yoneda provides. The literature (Lurie HTT, Riehl–Verity, Land) is now accessible.

48.6 Exercises

What is $\mathcal{P}(\mathcal{C})$?	$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$ — the ∞-category of presheaves of spaces on $\mathcal{C}$.
Why values in $\mathcal{S}$ and not $\mathcal{C}\mathrm{at}_\infty$?	Because $\mathrm{Map}_{\mathcal{C}}(-, x)$ takes values in Kan complexes (Theorem 44.2). For the Yoneda embedding to land in the natural target, presheaves of spaces is the right notion.
State the Yoneda embedding.	$y : \mathcal{C} \to \mathcal{P}(\mathcal{C})$, $y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$.
Three constructions of y?	Via straightening of the slice fibration; via universal property (free cocompletion); via cosimplicial-space Kan extension.
State the ∞-Yoneda lemma.	$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \simeq F(x)$, natural in x and F.
What is the comparison map in the Yoneda proof?	$\eta \mapsto \eta_x(\mathrm{id}_x)$ — evaluation at the universal element.
What is the inverse map?	$\xi \mapsto \widetilde{\xi}$ where $\widetilde{\xi}_z(f) = F(f)(\xi)$ — transporting ξ along F's contravariant action.
State the corollary for representables.	$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), y(y)) \simeq \mathrm{Map}_{\mathcal{C}}(x, y)$.
Why does the corollary imply full faithfulness of y?	Full faithfulness of an ∞-functor = weak equivalence on each mapping space (Whitehead, Theorem 44.10). The corollary is exactly this for y.
State the free cocompletion property.	$\mathrm{Fun}^{\mathrm{cc}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{y^*} \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is an equivalence — every $F : \mathcal{C} \to \mathcal{D}$ to cocomplete $\mathcal{D}$ extends uniquely to a colimit-preserving $\bar{F}$.
State the density theorem.	Every $P \in \mathcal{P}(\mathcal{C})$ is canonically the colimit of representables: $P \simeq \mathrm{colim}_{(x,\xi)} y(x)$ over $(x, \xi) \in \mathcal{C}_{/P}$.
How does Chapter 9's Yoneda fit?	For $\mathcal{C}$ a 1-category, $\mathcal{P}(\mathrm{N}(\mathcal{C}))$ contains $\widehat{\mathcal{C}}$ as the π_0-valued subcategory; restricting ∞-Yoneda there recovers Chapter 9.

How does Chapter 28's formal Yoneda fit?	The end formula $\mathrm{Nat}(y(-), F) \cong F$ becomes $\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(-), F) \simeq F$ in ∞-categorical form.
What is the categorical content of "free cocompletion"?	$\mathcal{P}(\mathcal{C})$ is universal among cocomplete ∞-categories containing $\mathcal{C}$: any other such cocomplete embedding factors through it via a colimit-preserving functor.
Why does the density theorem hold?	Because every presheaf $P : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ is determined by its values at all objects of $\mathcal{C}$ via Yoneda; the canonical colimit over $\mathcal{C}_{/P}$ reconstructs P.
What is the slice $\mathcal{C}_{/P}$ when $P \in \mathcal{P}(\mathcal{C})$?	The ∞-category of pairs (x, ξ) where $x \in \mathcal{C}$ and $\xi \in P(x)$, with morphisms in $\mathcal{C}$ compatible with P's action on ξ.
What's the relationship to Cartesian fibrations?	$\mathcal{C}_{/P} \to \mathcal{C}$ is a right fibration, classified under straightening by the presheaf P itself.
Why is $\mathcal{P}(\mathcal{C})$ presentable?	It is cocomplete (pointwise from $\mathcal{S}$), accessible (small generated under filtered colimits), and locally small. Chapter 52 develops presentability.
What property does Yoneda preserve?	y preserves all limits that exist in $\mathcal{C}$ (since representables preserve limits). It does *not* in general preserve colimits — that's exactly the failure that makes $\mathcal{P}(\mathcal{C})$ the *free* cocompletion.
When does Yoneda preserve colimits?	Never in general — that's the point of free cocompletion. It would only happen if $\mathcal{C}$ already had all colimits and they were preserved by the embedding, which is essentially never.
What does "representable presheaf" mean?	$y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$ for some $x \in \mathcal{C}$. A presheaf is representable iff it is in the essential image of y.
What's the standard reading of the Yoneda philosophy?	An object is determined by its incoming morphisms (its representable). Two objects are equivalent iff their representables are. This generalizes from **Cat** to $\mathcal{C}\mathrm{at}_\infty$ without change.
Why is this chapter the milestone?	Because with ∞-Yoneda in hand, the reader can use the foundation in any ∞-categorical reference (Lurie's HTT especially), which assumes Yoneda freely from the outset. The bridge to literature is now fully built.

Chapter Summary

Chapter Summary

Presheaves of spaces. $\mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$. Cocomplete and presentable; complete; the ∞-categorical refinement of ordinary presheaf categories.
Yoneda embedding. $y : \mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C})$, $y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$. Constructed via straightening of slice fibrations.
∞-Yoneda lemma. For all $x \in \mathcal{C}$, $F \in \mathcal{P}(\mathcal{C})$: $\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \simeq F(x)$, natural in both. Corollary: y is fully faithful.
Free cocompletion. $\mathcal{P}(\mathcal{C})$ is universal: $\mathrm{Fun}^{\mathrm{cc}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \simeq \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ for cocomplete $\mathcal{D}$.
Density theorem. Every $P \in \mathcal{P}(\mathcal{C})$ is canonically a colimit of representables.
Bridge to literature. With ∞-Yoneda established, Lurie HTT, Riehl–Verity, Cisinski, and Kerodon are now accessible. The remaining ∞-categorical theorems of Volume IV (limits, adjunctions, monads, Kan extensions, presentability, stability, operads) all derive from this foundation.

Formal Restatement

Formal Restatement

Presheaves of spaces. $\mathcal{P}(\mathcal{C}) := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$. Cocomplete and locally small; (co)limits pointwise from $\mathcal{S}$.

Yoneda embedding. $y : \mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C})$, $y(x) = \mathrm{Map}_{\mathcal{C}}(-, x)$.

∞-Yoneda lemma.

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \;\simeq\; F(x), \qquad \text{natural in } x \in \mathcal{C}, F \in \mathcal{P}(\mathcal{C}).$$

Full faithfulness. y induces weak equivalences $\mathrm{Map}_{\mathcal{C}}(x, y) \xrightarrow{\sim} \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), y(y))$.

Free cocompletion. For cocomplete $\mathcal{D}$:

$$\mathrm{Fun}^{\mathrm{cc}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{y^*} \mathrm{Fun}(\mathcal{C}, \mathcal{D})$$

is an equivalence. The unique extension of $F : \mathcal{C} \to \mathcal{D}$ is $\bar{F}(P) = \mathrm{colim}_{(x,\xi) \in \mathcal{C}_{/P}} F(x)$.

Density. Every $P \in \mathcal{P}(\mathcal{C})$ satisfies

$$P \;\simeq\; \mathrm{colim}_{(x,\xi) \in \mathcal{C}_{/P}}\, y(x).$$

Bridge. With ∞-Yoneda available, the higher-CT literature (Lurie HTT, Riehl–Verity, Cisinski, Kerodon, Land) is accessible. Phase 2 of Volume IV closes here.

Chapter 49

∞-Limits and ∞-Colimits

Chapter Overview

Chapter 45 introduced ∞-limits and ∞-colimits as terminal and initial objects of slice quasi-categories. With the ∞-Yoneda lemma (Chapter 48) now in hand, this chapter develops the theory in earnest: ∞-(co)limits as adjoints to constant-diagram functors, their representability via Yoneda, the computation rules via cofinality and via products + equalizers, and the foundational examples in $\mathcal{S}$, $\mathcal{P}(\mathcal{C})$, and $\mathcal{C}\mathrm{at}_\infty$.

Three thematic shifts from the 1-categorical version distinguish the ∞-treatment:

- Uniqueness becomes *contractibility*: the space of limit cones is contractible, not merely singleton.
- Universality is *coherent*: limits represent functors of spaces, not just sets.
- Existence comes via *homotopy* products and equalizers, which differ from their 1-categorical analogues in critical ways even when the ∞-category at hand is a nerve.

This chapter prepares all of Volume IV's remaining material: Chapter 50's adjunctions are characterized by limit preservation; Chapter 51's ∞-Kan extensions are pointwise limits over slices; Chapter 52's presentable ∞-categories require κ-filtered colimits as their defining feature; Chapter 53's stable ∞-categories use finite limits and colimits intertwined via the fibre/cofibre construction.

49.1 Limits as Adjoints

49.1.1 Formalizing

Where the Name Comes From

Limit and **colimit** are the central universal-property constructions of category theory (Vol I Ch. 6, Vol II Ch. 16). The ∞-categorical versions retain the same names and the same universal-property intuition; the differences come from replacing *set* of cones by *space* of cones (a Kan complex), and *unique* cone by *contractible space of cones*. The prefix "co-" in "colimit" is standard categorical convention for duality: a colimit in $\mathcal{C}$ is a limit in $\mathcal{C}^{\mathrm{op}}$.

Definition 49.1 (Constant-diagram functor)**.** For a quasi-category $\mathcal{C}$ and a simplicial set K, the **constant-diagram functor**

$$\mathrm{const} : \mathcal{C} \longrightarrow \mathcal{C}^K$$

sends $x \in \mathcal{C}$ to the constant functor $\Delta x : K \to \mathcal{C}$, $\Delta x(k) = x$ for all k.

Theorem 49.2 (Limits and colimits as adjoints)**.** *If every ∞-limit (resp. ∞-colimit) over K exists in $\mathcal{C}$, the assignment $p \mapsto \lim p$ (resp. $\operatorname{colim} p$) extends to a functor $\mathcal{C}^K \to \mathcal{C}$, and we have the adjunctions*

$$\operatorname{colim} : \mathcal{C}^K \rightleftarrows \mathcal{C} : \mathrm{const}, \quad \mathrm{const} : \mathcal{C} \rightleftarrows \mathcal{C}^K : \lim .$$

Proof sketch. A natural transformation $\Delta x \to p$ (where $\Delta x \in \mathcal{C}^K$ is the constant diagram on x) is exactly the data of a cone over p with apex x. By definition of the slice $\mathcal{C}_{/p}$, this corresponds to a 0-simplex of $\mathcal{C}_{/p}$ — a cone-with-apex-x. The terminal cone is the limit, so $\mathrm{Map}_{\mathcal{C}^K}(\Delta x, p) \simeq \mathrm{Map}_{\mathcal{C}}(x, \lim p)$, giving the adjunction $\mathrm{const} \dashv \lim$. Dually for $\operatorname{colim} \dashv \mathrm{const}$. □ □

Remark 49.3 (Limits via Yoneda)**.** The adjunction characterization combined with ∞-Yoneda (Chapter 48) gives the **representability characterization of limits**: $\lim p$ exists iff the functor $\mathrm{Map}_{\mathcal{C}^K}(\Delta -, p)$ is representable, in which case the representing object is $\lim p$. This is the ∞-categorical version of the Chapter 6 representability proof of limits.

49.2 Existence: Products and Equalizers

49.2.1 Formalizing

Definition 49.4 (Complete and cocomplete ∞-category)**.** A quasi-category $\mathcal{C}$ is **complete** if it admits all ∞-limits over small simplicial sets K. **Cocomplete**: dually admits all ∞-colimits. **Bicomplete**: both.

Theorem 49.5 (Limit existence via products and equalizers)**.** *A quasi-category $\mathcal{C}$ is complete iff it admits:*

1. *All small* ***∞-products****: limits over discrete simplicial sets.*
2. *All* ***∞-equalizers****: limits over the walking parallel pair $\Delta^1 \rightrightarrows \Delta^0$.*

Dually for cocompleteness: coproducts and coequalizers suffice.

Proof sketch. The strategy mirrors Chapter 6's 1-categorical proof: an arbitrary limit over K is computed as the equalizer of two maps between products indexed by simplices of K. The ∞-categorical version replaces ordinary products and equalizers by their ∞-versions; the combinatorial verification of the universal property uses inner-horn filling. □ □

Remark 49.6 (Homotopy products differ from set-theoretic products)**.** For $\mathcal{C}$ a quasi-category and $\{x_i\}$ a family of objects, the ∞-product $\prod_i x_i$ is the limit over the discrete diagram. When $\mathcal{C} = \mathcal{S}$, this is the homotopy product $\prod_i^h x_i$ — the categorical product, which for fibrant Kan complexes is the set-theoretic product. For non-fibrant input, fibrant replacement is required.

49.2.2 Reorganizing: completeness of key examples

Example 49.7 ($\mathcal{S}$ is bicomplete)**.** The ∞-category of spaces $\mathcal{S}$ is bicomplete; (co)limits are the standard homotopy (co)limits of **Top** (or $\mathbf{sSet}_{\mathrm{Q}}$ under the Quillen equivalence). Specifically:

- Products: $X \times Y$ — Cartesian product of Kan complexes.
- Equalizers: homotopy equalizer (or path object), giving the homotopy fibre when one map is trivial.
- Coproducts: $X \sqcup Y$ — disjoint union.
- Coequalizers: homotopy coequalizer; pushout when the parallel pair shares an endpoint.

Example 49.8 ($\mathcal{P}(\mathcal{C})$ is bicomplete)**.** For any small $\mathcal{C}$, $\mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$ is bicomplete. ∞-(co)limits are computed pointwise: $(\lim_K F)(c) = \lim_K F(c)$ in $\mathcal{S}$, and dually for colimits.

Every $P \in \mathcal{P}(\mathcal{C})$ is the colimit of representables (density theorem, Theorem 48.48.11). The cocompleteness of $\mathcal{P}(\mathcal{C})$ together with the Yoneda embedding is the *free cocompletion* statement of Theorem 48.48.10.

Example 49.9 ($\mathcal{C}\mathrm{at}_\infty$ is bicomplete)**.** The ∞-category of ∞-categories is bicomplete: products are Cartesian product of quasi-categories; equalizers and coequalizers via homotopy pullback/pushout; arbitrary limits and colimits computed via Theorem 49.5.

49.3 Computing Limits via Cofinality

49.3.1 Formalizing

Theorem 49.10 (Cofinality and colimits)**.** *Let $i : K' \to K$ be a cofinal map (Definition 45.45.15), let $p : K \to \mathcal{C}$ be a diagram. If $\mathrm{colim}\,(p \circ i)$ exists in $\mathcal{C}$, so does $\mathrm{colim}\,p$, and the canonical comparison $\mathrm{colim}\,(p \circ i) \to \mathrm{colim}\,p$ is an equivalence.*

Example 49.11 (Filtered colimits in $\mathcal{S}$)**.** A filtered ∞-colimit in $\mathcal{S}$ is computed by taking the underlying 1-categorical filtered colimit (in **Set**, of π_0) together with the ∞-groupoid filtered colimit of the π_n-systems. This recovers the classical fact that filtered colimits commute with finite limits in spaces.

Example 49.12 (Pushouts and pullbacks)**.** ∞-pullbacks and ∞-pushouts are computed via the diagonal of the square diagram. In $\mathcal{S}$: pullback is the homotopy pullback (Mayer–Vietoris fits); pushout is the homotopy pushout (cofibre sequences fit). Both are recovered as limits/colimits of the standard Λ^2_0 / Λ^2_2-shaped diagrams.

49.3.2 Reorganizing: preservation and reflection

Definition 49.13 (Limit preservation and reflection)**.** A functor $F : \mathcal{C} \to \mathcal{D}$ **preserves** ∞-limits of shape K if for every $p : K \to \mathcal{C}$ with limit $\lim p \in \mathcal{C}$, the image $F(\lim p)$ is the ∞-limit of $F \circ p$ in $\mathcal{D}$. F **reflects** ∞-limits if $F(\lim p) \simeq \lim F \circ p$ implies $\lim p$ exists in $\mathcal{C}$.

Reader's Annotation

The slogan *Right Adjoints Preserve Limits* (RAPL) and its dual *Left Adjoints Preserve Colimits* (LAPC) were central in Vol I Ch. 7 and Vol II Ch. 17. They generalize verbatim to the ∞-setting: limits commute with right-adjoint functors, colimits with left-adjoint functors. The proof is a one-line Yoneda calculation (Theorem 49.14 below), unchanged in spirit from its 1-categorical ancestor.

Theorem 49.14 (Right adjoints preserve ∞-limits)**.** *If $L \dashv R : \mathcal{D} \to \mathcal{C}$ is an ∞-adjunction (Chapter 50), then R preserves all ∞-limits that exist in $\mathcal{D}$. Dually, L preserves all ∞-colimits in $\mathcal{C}$.*

Proof sketch. For a diagram $p : K \to \mathcal{D}$ with limit $\lim p$ and any $c \in \mathcal{C}$:

$$\begin{aligned}
\mathrm{Map}_{\mathcal{C}}(c, R(\lim p)) &\simeq \mathrm{Map}_{\mathcal{D}}(Lc, \lim p) && \text{(adjunction)}\\
&\simeq \lim_k \mathrm{Map}_{\mathcal{D}}(Lc, p(k)) && \text{(definition of } \lim)\\
&\simeq \lim_k \mathrm{Map}_{\mathcal{C}}(c, Rp(k)) && \text{(adjunction)}\\
&\simeq \mathrm{Map}_{\mathcal{C}}(c, \lim_k Rp(k)).
\end{aligned}$$

By ∞-Yoneda, $R(\lim p) \simeq \lim R \circ p$. □ □

49.4 Filtered and Sifted Colimits

49.4.1 Formalizing

Where the Name Comes From

Filtered colimit is the ∞-categorical generalization of the classical *filtered colimit* in 1-categories (also called *directed colimit*). The term *filtered* comes from set theory and order theory: a partially ordered set is *filtered* if every finite subset has an upper bound. The pattern abstracts to categories: a category $\mathcal{K}$ is *filtered* if every finite diagram into $\mathcal{K}$ has a cocone. Filtered colimits are ubiquitous because they commute with finite limits — a property that fails for general colimits but is essential for many applications (e.g., the colimit of a directed system of modules is again a module).

Where the Name Comes From

Sifted (also: *sift*) is from the verb "to sift" — to separate using a sieve. A category $\mathcal{K}$ is *sifted* if the colimit-over-$\mathcal{K}$ functor preserves finite products (weaker than "preserves finite limits," which would be filtered). Sifted colimits include filtered colimits and reflexive coequalizers; they are exactly the colimits that play well with finite-product structure, making them central to algebraic theories (Lurie HA Ch. 5).

Definition 49.15 (Filtered ∞-category)**.** A quasi-category $\mathcal{K}$ is **filtered** (also called **∞-filtered**) if for every finite simplicial set K and diagram $p : K \to \mathcal{K}$, the slice $\mathcal{K}_{p/}$ is contractible (Kan complex with trivial homotopy groups).

Equivalent characterizations:

- Every finite diagram in $\mathcal{K}$ extends to a cocone.
- The functor $\mathrm{colim}_{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \to \mathcal{S}$ preserves finite limits.

Theorem 49.16 (Filtered colimits commute with finite limits)**.** *For a small filtered quasi-category $\mathcal{K}$ and a small finite quasi-category $\mathcal{J}$, the canonical comparison*

$$\mathrm{colim}_{\mathcal{K}} \lim_{\mathcal{J}} F \;\simeq\; \lim_{\mathcal{J}} \mathrm{colim}_{\mathcal{K}} F$$

is an equivalence, for any $F : \mathcal{K} \times \mathcal{J} \to \mathcal{S}$.

Remark 49.17 (Sifted colimits)**.** A more general class than filtered is **sifted**: $\mathcal{K}$ is sifted if $\mathrm{colim}_{\mathcal{K}}$ preserves *finite products* (rather than finite limits). Examples: filtered $\mathcal{K}$ is sifted; reflexive coequalizers (which are not filtered) are sifted. Sifted colimits play a central role in defining algebraic theories ∞-categorically; see Chapter 52 and the literature on algebraic theories in Lurie's HA.

49.5 Exercises

Define an ∞-limit.	$\lim p$ for $p : K \to \mathcal{C}$ is a terminal object of the slice $\mathcal{C}_{/p}$ — equivalently, an object $\ell \in \mathcal{C}$ together with a cone $p \to \mathrm{const}\,\ell$ that is universal.
State the limit-as-adjoint characterization.	$\mathrm{const} \dashv \lim$ as functors $\mathcal{C} \to \mathcal{C}^K \to \mathcal{C}$. When $\mathcal{C}$ has K-shaped limits.
State the representability characterization of limits.	$\lim p$ exists iff $\mathrm{Map}_{\mathcal{C}^K}(\Delta -, p)$ is representable. The representing object is $\lim p$.
When is $\mathcal{C}$ complete?	When all small ∞-limits exist, equivalently when small products and equalizers exist (Theorem 49.5).

Is $\mathcal{S}$ complete?	Yes — bicomplete. Limits are homotopy limits in $\mathbf{Top}/\mathbf{sSet}_{\mathrm{Q}}$.
Is $\mathcal{P}(\mathcal{C})$ complete?	Yes — bicomplete. Limits are computed pointwise. Cocompleteness combined with Yoneda gives the free cocompletion property.
Is $\mathcal{C}\mathrm{at}_\infty$ complete?	Yes — bicomplete. (Co)limits via Cartesian product, homotopy pullback/pushout, etc.
Why does "RAPL" generalize?	Right adjoints preserve ∞-limits. Same proof: hom-Map-bijection + commutation through Map + Yoneda recovers the right adjoint as "limits commute with $\mathrm{Map}(-, x)$."
What does "cofinal" give in computation?	Cofinal map $i : K' \to K$ allows replacing $\mathrm{colim}_K p$ with $\mathrm{colim}_{K'}(p \circ i)$ — usually a simpler diagram.
When is $\mathcal{K}$ filtered?	When every finite diagram in $\mathcal{K}$ extends to a cocone. Equivalent: $\mathrm{colim}_{\mathcal{K}}$ preserves finite limits.
Why filtered colimits commute with finite limits?	Theorem 49.16. The combinatorial reason: filtered shapes can be "thinned" to a cofinal piece that avoids the finite limit's interaction with the colimit indexing.
Sifted vs filtered?	Sifted: colim preserves finite products. Filtered: preserves finite limits. Filtered implies sifted; reflexive coequalizers are sifted but not filtered.
How is an ∞-pushout computed in $\mathcal{S}$?	As the homotopy pushout: the geometric realization of the bar-like construction $A \sqcup_C B \to$ mapping cylinder.
How are ∞-limits in functor categories computed?	Pointwise. $(\lim F)(c) = \lim F(c)$ — using completeness of the target ∞-category.
What is the diagonal/comparison map for filtered + finite?	$\mathrm{colim}_{\mathcal{K}} \lim_{\mathcal{J}} F \to \lim_{\mathcal{J}} \mathrm{colim}_{\mathcal{K}} F$. Always exists; is an equivalence when $\mathcal{K}$ is filtered and $\mathcal{J}$ is finite.
Why is the space of limit cones contractible?	Because terminal objects form a contractible space (or empty space): the data of a terminal object is unique up to a contractible space of equivalences.
What happens to limits when the diagram has an initial object?	They collapse: the limit equals the value at the initial object (by cofinality of $\{k_0\} \hookrightarrow K$).
Are filtered colimits filtered in $\mathcal{S}$ "built from sequences"?	Roughly yes: a sequential colimit $\mathrm{colim}\,(X_0 \to X_1 \to \cdots)$ is filtered. More generally, any cofinal sequential map into a filtered $\mathcal{K}$ realizes the colimit.
What's the relationship to fibrant replacement?	In $\mathcal{S}$ presented as **sSet**, ∞-(co)limits require fibrant inputs. Replacing inputs by Kan complexes (fibrant replacement) is essential before computing.
What's an ∞-equalizer?	The ∞-limit of the parallel-pair diagram $K = \Delta^1 \rightrightarrows \Delta^0$ — generalizes ordinary equalizer to allow "coherent agreement" rather than strict equality.
When does $\mathcal{C}^K$ have all limits?	Whenever $\mathcal{C}$ does, computed pointwise. The same statement for colimits.
What's the next chapter?	Ch 50: ∞-Adjunctions — the ∞-categorical analogue of Chapter 7/17/33 adjunctions, with ∞-Yoneda and limit preservation as central tools.
Why is this chapter compact compared to Vol I Ch 6?	Vol I Ch 6 introduced limits from scratch; this chapter assumes the framework (slices, Yoneda, adjunctions) and develops only what's ∞-categorically new. The bridge to literature is the goal.

Chapter Summary

Chapter Summary

∞-limits as adjoints. const $\dashv$ lim (and colim $\dashv$ const) when ∞-(co)limits exist. Representability: $\lim p$ represents $\mathrm{Map}_{\mathcal{C}^K}(\Delta -, p)$.
Existence via products + equalizers. $\mathcal{C}$ is complete iff ∞-products and ∞-equalizers exist. Standard construction transfers from Chapter 6 with ∞-versions throughout.
Foundational bicomplete examples. $\mathcal{S}$ (limits = homotopy limits); $\mathcal{P}(\mathcal{C})$ (pointwise); $\mathcal{C}\mathrm{at}_\infty$ (Cartesian + pullback/pushout).
Cofinality. Cofinal maps preserve ∞-colimits, giving a powerful computation technique. Pushouts, pullbacks, sequential colimits all admit cofinality-based simplifications.
Adjoint preservation. Right adjoints preserve ∞-limits; left adjoints preserve ∞-colimits. The proof is one calculation in Map via Yoneda.
Filtered colimits commute with finite limits. The cornerstone identity. Filtered colimits are the bread-and-butter limit-related construction in presentable ∞-categories (Chapter 52).

Formal Restatement

Formal Restatement

∞-limit. $\lim p$ = terminal of $\mathcal{C}_{/p}$; $\operatorname{colim} p$ = initial of $\mathcal{C}_{p/}$.

Adjunction. const $\dashv$ lim; colim $\dashv$ const. Both natural in K.

Representability. $\lim p \in \mathcal{C}$ iff $\mathrm{Map}_{\mathcal{C}^K}(\Delta -, p)$ is representable.

Existence. $\mathcal{C}$ complete $\iff$ has products and equalizers; cocomplete $\iff$ has coproducts and coequalizers.

RAPL/LAPC. $L \dashv R \Rightarrow R$ preserves ∞-limits, L preserves ∞-colimits.

Cofinality. $i : K' \to K$ cofinal $\Rightarrow \operatorname{colim}(p \circ i) \simeq \operatorname{colim} p$ for all p.

Filtered. $\mathcal{K}$ filtered $\iff$ every finite diagram extends to a cocone $\iff$ $\operatorname{colim}_{\mathcal{K}}$ preserves finite ∞-limits.

Filtered $\times$ Finite (Theorem 49.16).

$$\operatorname{colim}_{\mathcal{K}} \lim_{\mathcal{J}} F \simeq \lim_{\mathcal{J}} \operatorname{colim}_{\mathcal{K}} F, \quad \mathcal{K} \text{ filtered}, \mathcal{J} \text{ finite}.$$

Chapter 50

∞-Adjunctions

Chapter Overview

Adjunctions have appeared in three earlier guises:

- Chapter 7 — hom-set bijection $\mathcal{D}(LX, Y) \cong \mathcal{C}(X, RY)$.
- Chapter 17 — triangle identities; uniqueness via Yoneda; adjoint triples.
- Chapter 33 — formal account in 2-categories.

This chapter develops the fourth incarnation: the ∞-categorical version, where the hom-bijection is replaced by an equivalence of mapping spaces.

A pair of ∞-functors $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ form an **∞-adjunction** $L \dashv R$ if there is a natural equivalence

$$\mathrm{Map}_{\mathcal{D}}(LX, Y) \ \simeq\ \mathrm{Map}_{\mathcal{C}}(X, RY), \quad X \in \mathcal{C},\ Y \in \mathcal{D}.$$

The naturality is in the ∞-categorical sense: the equivalences fit together into a coherent system, witnessing arbitrarily high-dimensional compatibilities.

This chapter develops three equivalent characterizations: hom-equivalence, unit/counit data, and "relative adjunctions" via Cartesian fibrations. The latter is the model-independent definition (Riehl–Verity) and the most flexible for proofs.

Theorems: right adjoints preserve ∞-limits, left adjoints preserve ∞-colimits (Chapter 49's RAPL/LAPC). Uniqueness up to contractible choice: any two adjoints to L are equivalent via a unique-up-to-coherent-homotopy ∞-natural isomorphism. The ∞-adjoint functor theorem (presentable case) is stated in §50.4, with proof deferred to Chapter 52.

Adjunctions are the backbone of practical ∞-categorical constructions: free/forgetful adjunctions, localizations, fibre/cofibre functors, base-change pairs in derived algebraic geometry. By Chapter 52 (presentable case), the adjoint functor theorem will become as central in the ∞-setting as in the 1-setting.

50.1 Defining ∞-Adjunctions

50.1.1 Formalizing

Where the Name Comes From

Adjunction (cross-ref Vol I Ch. 7) is one of the foundational concepts of category theory, introduced by Kan (1958) under the name "adjoint functors." The word *adjoint* is from linear algebra (Hilbert space adjoint T^*: the unique operator satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$). Kan's insight was that the categorical version of this pairing — a bijection $\mathcal{D}(LX, Y) \cong \mathcal{C}(X, RY)$ — is so fundamental that it organizes a vast amount of mathematics. The ∞-version replaces the bijection by an equivalence of mapping spaces and adds an infinite tower of higher coherences. The terminology, the symbols $\dashv$ "left adjoint to," and the unit/counit decomposition are all preserved from the classical case.

Definition 50.1 (∞-Adjunction: hom-equivalence form)**.** A pair of ∞-functors $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ form an **∞-adjunction** $L \dashv R$ if there is a natural equivalence of mapping spaces

$$\mathrm{Map}_{\mathcal{D}}(LX, Y) \ \simeq\ \mathrm{Map}_{\mathcal{C}}(X, RY)$$

in the variables $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Precisely: the two functors $\mathrm{Map}_{\mathcal{D}}(L(-), -)$ and $\mathrm{Map}_{\mathcal{C}}(-, R(-))$ from $\mathcal{C}^{\mathrm{op}} \times \mathcal{D}$ to $\mathcal{S}$ are equivalent in $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}} \times \mathcal{D}, \mathcal{S})$.

How to Say This

$L \dashv R$ *(read: "L is left adjoint to R")* or *(read: "L adjoint R")*; L called the **left adjoint**, R the **right adjoint**.

Definition 50.2 (∞-Adjunction: unit/counit form)**.** Equivalently, an ∞-adjunction is a pair $L \dashv R$ together with:

- A **unit** natural transformation $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow RL$.
- A **counit** natural transformation $\epsilon : LR \Rightarrow \mathrm{id}_{\mathcal{D}}$.

satisfying the **triangle identities up to coherent homotopy**: $\epsilon L \circ L\eta \simeq \mathrm{id}_L$ and $R\epsilon \circ \eta R \simeq \mathrm{id}_R$, with all higher coherences supplied as data.

Theorem 50.3 (Equivalence of the two forms)**.** *The hom-equivalence form (Definition 50.1) and the unit/counit form (Definition 50.2) are equivalent: there is an equivalence of ∞-categories of "∞-adjunctions presented in form A" and "form B."*

Remark 50.4 (The triangle identities require coherence data)**.** In a 1-category, the triangle identities are equations between natural transformations. In an ∞-category, they are weak equivalences of 1-simplices, requiring 2-simplex witnesses; the 2-simplex compatibilities require 3-simplex witnesses; and so on. The infinite tower of coherences is exactly what distinguishes ∞-adjunctions from naive "pseudo-adjunctions."

50.2 Relative Adjunctions via Fibrations

50.2.1 Formalizing

Riehl–Verity propose a model-independent definition of ∞-adjunctions using only the language of ∞-categorical mapping spaces and fibrations.

Definition 50.5 (Adjunction via fibrations)**.** An ∞-adjunction $L \dashv R$ is the data of:

- A coCartesian fibration $\mathcal{E} \to \Delta^1$ with fibres $\mathcal{E}_0 = \mathcal{C}$ over 0 and $\mathcal{E}_1 = \mathcal{D}$ over 1.
- The transport functor $\mathcal{C} \to \mathcal{D}$ is L.

The fibration is simultaneously a Cartesian fibration iff L has a right adjoint R. The Cartesian transport (in the reverse direction) recovers R, and the bi-fibration structure encodes the entire adjunction data (including coherences).

Remark 50.6 (Why fibrations are model-independent)**.** The fibration definition uses only ∞-categorical features: existence of (co)Cartesian morphisms, fibres, transport. It works in any model of ∞-categories (quasi-cats, simplicial cats, complete Segal spaces). The hom-equivalence and unit/counit forms are quasi-categorical specializations.

50.3 Preservation Properties

50.3.1 Formalizing

Theorem 50.7 (∞-RAPL and ∞-LAPC)**.** *Let $L \dashv R : \mathcal{D} \to \mathcal{C}$ be an ∞-adjunction.*

- *R preserves all ∞-limits that exist in $\mathcal{D}$.*
- *L preserves all ∞-colimits that exist in $\mathcal{C}$.*

Proof. Proven already in Theorem 49.49.14 as Chapter 49's Map-Yoneda calculation. □ □

Remark 50.8 (The converse is not automatic)**.** Preservation of ∞-limits does *not* imply being a right adjoint without further hypotheses (e.g., presentability of $\mathcal{D}$). The adjoint functor theorem, generalized to the ∞-setting, gives a clean criterion in the presentable case (§50.4).

50.3.2 Reorganizing: uniqueness of adjoints

Theorem 50.9 (Uniqueness of adjoints)**.** *If $L_1 \dashv R$ and $L_2 \dashv R$ are two ∞-adjunctions sharing the same right adjoint, then there is an equivalence $L_1 \simeq L_2$ unique up to a contractible space of equivalences.*

Dually: given $L \dashv R_1$ and $L \dashv R_2$, $R_1 \simeq R_2$ uniquely up to contractible choice.

Proof sketch. Both L_1, L_2 co-represent the functor $Y \mapsto \mathrm{Map}_{\mathcal{C}}(-, RY)$. By ∞-Yoneda, the corepresenting object is unique up to contractible space of equivalences. □ □

Remark 50.10 (Adjoint triples)**.** A right adjoint R may itself have a further right adjoint R', giving an adjoint triple $L \dashv R \dashv R'$. This phenomenon is essentially the same as in 1-categories (Chapter 17), but generalizes: in $\mathcal{C}\mathrm{at}_\infty$ one finds chains of arbitrary length ("adjoint chains"), e.g., in the context of 6-functor formalisms or derived algebraic geometry.

50.4 Adjoint Functor Theorem (Preview)

50.4.1 Formalizing

Where the Name Comes From

The **Adjoint Functor Theorem** (AFT) appeared in Vol II Ch. 23 in its 1-categorical form (Freyd 1964): a functor between locally small categories has a left/right adjoint iff it preserves limits/colimits and satisfies a smallness "solution-set" condition. The ∞-version (Lurie HTT Ch. 5) replaces "locally small" with "presentable" and "solution-set" with "accessible." The historical lineage runs Freyd 1964 → Adámek-Rosický 1994 (presentable categories) → Lurie 2009 (∞-version).

Theorem 50.11 (∞-Adjoint Functor Theorem: presentable case)**.** *Let $\mathcal{C}$ and $\mathcal{D}$ be presentable ∞-categories (Chapter 52). A functor $R : \mathcal{D} \to \mathcal{C}$ admits a left adjoint iff R preserves all small ∞-limits and is* ***accessible****: there exists a regular cardinal κ such that R preserves κ-filtered colimits.*

Dually, $L : \mathcal{C} \to \mathcal{D}$ admits a right adjoint iff L preserves all small ∞-colimits.

Proof sketch. The proof is the central content of HTT §5.5; we defer the full apparatus to Chapter 52. Key ingredients:

- Presentable ∞-categories admit accessible localizations.
- Accessibility plus colimit preservation gives the construction of the adjoint via Kan extension along a presentation.
- Uniqueness up to contractible choice follows from ∞-Yoneda.

□ □

Remark 50.12 (Working horse of higher algebra)**.** Theorem 50.11 is the engine of much of higher algebra: existence of left adjoints to forgetful functors (giving free constructions), right adjoints to restriction functors (giving induction), all derive from this single theorem applied to presentable categories.

50.5 Examples

50.5.1 Foundational adjunctions in Volume IV

Example 50.13 ($\mathrm{ho} \dashv \mathrm{N}$)**.** The homotopy category functor $\mathrm{ho} : \mathbf{QCat} \to \mathbf{Cat}$ is left adjoint to the nerve $\mathrm{N} : \mathbf{Cat} \to \mathbf{QCat}$ (Theorem 43.43.10).

Example 50.14 ($|-| \dashv \mathrm{Sing}$)**.** Geometric realization is left adjoint to the singular complex functor (Theorem 39.39.2). The adjunction is a Quillen equivalence (Theorem 42.42.10); on ∞-category level, it becomes an equivalence $\mathcal{S} \simeq \mathcal{S}_{\mathrm{Top}}$.

Example 50.15 (Yoneda and free cocompletion)**.** For any cocomplete ∞-category $\mathcal{D}$ and small $\mathcal{C}$, the restriction $y^* : \mathrm{Fun}^{\mathrm{cc}}(\mathcal{P}(\mathcal{C}), \mathcal{D}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is an equivalence (Theorem 48.48.10); in particular, it has both adjoints. The unique colimit-preserving extension $\bar{F} : \mathcal{P}(\mathcal{C}) \to \mathcal{D}$ is the left adjoint of y^*.

50.5.2 Reorganizing: where this goes

Remark 50.16 (Adjunctions throughout Volume IV)**.** ∞-adjunctions appear in nearly every remaining chapter:

- Chapter 51: every ∞-monad arises from an ∞-adjunction; ∞-Kan extensions are universal adjunctions.
- Chapter 52: presentable ∞-categories are characterized by "accessibility plus AFT-style adjoint existence."
- Chapter 53: stable ∞-cats use adjoints to define suspension/loop functors.
- Chapter 55: ∞-operads are themselves built using adjunctions between operadic and ordinary ∞-categories.

The ∞-adjunction is the second pillar of ∞-category theory, after the ∞-Yoneda lemma.

50.6 Exercises

Define an ∞-adjunction via hom-equivalence.	$L \dashv R$ iff there's a natural equivalence $\mathrm{Map}_{\mathcal{D}}(LX, Y) \simeq \mathrm{Map}_{\mathcal{C}}(X, RY)$ as functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathcal{S}$.
What's the unit/counit form?	$\eta : \mathrm{id} \Rightarrow RL$, $\epsilon : LR \Rightarrow \mathrm{id}$, satisfying triangle identities up to coherent homotopy with infinitely many higher coherences.
What's the fibrational form?	A coCartesian fibration $\mathcal{E} \to \Delta^1$ with fibres $\mathcal{C}, \mathcal{D}$ that is also a Cartesian fibration. The transport in two directions gives L and R.
Why is the fibrational form model-independent?	Because it uses only fibration data, available in any model of ∞-categories.

State ∞-RAPL/LAPC.	$L \dashv R \Rightarrow R$ preserves ∞-limits and L preserves ∞-colimits. Theorem 49.49.14.
Why is the proof of RAPL one short calculation?	$\mathrm{Map}_{\mathcal{C}}(c, R(\lim p)) \simeq \mathrm{Map}_{\mathcal{D}}(Lc, \lim p) \simeq \lim \mathrm{Map}_{\mathcal{D}}(Lc, p) \simeq \lim \mathrm{Map}_{\mathcal{C}}(c, Rp) \simeq \mathrm{Map}_{\mathcal{C}}(c, \lim Rp)$. Yoneda concludes.
Does preserving limits imply being a right adjoint?	Only in special cases (presentability). The full ∞-AFT theorem (§50.4) says: right adjoints among accessible functors between presentable ∞-cats = limit-preserving functors.
State uniqueness of adjoints.	Two left adjoints to the same R are equivalent up to a contractible space of equivalences. Same for right adjoints.
Why uniqueness?	Both adjoints corepresent the same functor $\mathrm{Map}_{\mathcal{C}}(-, RY)$; ∞-Yoneda gives the corepresenting object up to contractible choice.
What's an adjoint triple?	$L \dashv R \dashv R'$: R has both a left adjoint L and a further right adjoint R'.
Give a foundational ∞-adjunction.	ho $\dashv$ N: $\mathbf{QCat} \rightleftarrows \mathbf{Cat}$. Going up: N embeds 1-cats into ∞-cats; going down: ho truncates to a 1-cat.
Another?	$\lvert - \rvert \dashv \mathrm{Sing}$: realization and singular complex. A Quillen equivalence; on ∞-cat level an equivalence $\mathcal{S} \simeq \mathcal{S}_{\mathrm{Top}}$.
And another?	$y^* \dashv (\bar{-})$ via Yoneda: restriction is left adjoint to free-cocompletion extension; gives the universal property of $\mathcal{P}(\mathcal{C})$.
State the ∞-AFT in the presentable case.	For presentable $\mathcal{C}, \mathcal{D}$: a functor admits a left adjoint iff it preserves all ∞-limits and is accessible. Dually for right adjoints.
What does "accessible" mean?	Preserves κ-filtered colimits for some regular cardinal κ. Together with limit preservation, characterizes right adjoints among presentable-target functors.
Where is ∞-AFT proven?	Chapter 52 (presentability) develops the necessary machinery. Lurie HTT §5.5 has the full account.
Can L be a right adjoint too?	Yes — this is called **ambidexterity**. Examples: limits and colimits commute in $\mathcal{S}$ (filtered + finite), giving the bifibrational structure.
How does this generalize Vol I Ch 7 hom-set bijection?	The hom-set bijection $\mathcal{C}(X, RY) \cong \mathcal{D}(LX, Y)$ becomes a weak equivalence of ∞-groupoids (Kan complexes) Map. Naturality is coherent.
How does it generalize Vol II Ch 17 triangle identities?	Triangle identities $(\epsilon L)(L\eta) = \mathrm{id}_L$ become weak equivalences with infinitely many higher coherences as data.
How does it generalize Vol III Ch 33 formal version?	The 2-categorical Vol III account becomes the ∞-categorical one: $\mathcal{C}\mathrm{at}_\infty$ replaces **Cat** as the ambient structure.
Are ∞-adjunctions composable?	Yes — composition of left adjoints is a left adjoint to the composition of right adjoints. The corresponding fibrations compose via Cartesian product over Δ^1.
When is an adjunction an equivalence?	When unit and counit are weak equivalences. Equivalent to: both L and R are fully faithful and essentially surjective.

What's the next chapter?	Ch 51: ∞-Monads and ∞-Kan Extensions. Monads are precisely the algebras associated to ∞-adjunctions; Kan extensions are universal adjoints. Together they complete the higher-categorical spine.

Chapter Summary

Chapter Summary

∞-Adjunction. $L \dashv R$ iff there's a natural equivalence $\mathrm{Map}_{\mathcal{D}}(LX, Y) \simeq \mathrm{Map}_{\mathcal{C}}(X, RY)$. Three forms: hom-equivalence, unit/counit-with-coherences, fibration over Δ^1.
∞-RAPL/LAPC. Right adjoints preserve ∞-limits; left adjoints preserve ∞-colimits. One-line Yoneda proof.
Uniqueness. Any two left adjoints to the same R are equivalent up to a contractible space of equivalences (∞-Yoneda).
∞-AFT (presentable case). Among accessible functors between presentable ∞-categories, "right adjoint = preserves ∞-limits." Engine of higher-algebra constructions.
Foundational examples. $\mathrm{ho} \dashv \mathrm{N}$; $|-| \dashv \mathrm{Sing}$; Yoneda's $y^* \dashv (\bar{-})$.
Role in Volume IV. Used in Chapters 51 (monads), 52 (presentability), 53 (stability), 55 (operads). The second pillar of ∞-CT after Yoneda.

Formal Restatement

Formal Restatement

∞-Adjunction. $L \dashv R$:

$$\mathrm{Map}_{\mathcal{D}}(LX, Y) \simeq \mathrm{Map}_{\mathcal{C}}(X, RY), \quad \text{natural in } X, Y.$$

Unit and counit. $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow RL$, $\epsilon : LR \Rightarrow \mathrm{id}_{\mathcal{D}}$. Triangle equivalences: $\epsilon L \circ L\eta \simeq \mathrm{id}_L$, $R\epsilon \circ \eta R \simeq \mathrm{id}_R$, plus all higher coherences.

Fibration form. $\mathcal{E} \to \Delta^1$ co- and Cartesian fibration; fibres $\mathcal{C}, \mathcal{D}$; transport in two directions gives L, R.

Preservation. $L \dashv R \Rightarrow R$ preserves ∞-limits, L preserves ∞-colimits.

Uniqueness. The space of adjoints to a given functor is either empty or contractible.

∞-AFT (presentable). For $\mathcal{C}, \mathcal{D}$ presentable and $R : \mathcal{D} \to \mathcal{C}$:

$$R \text{ has a left adjoint} \iff R \text{ preserves } \infty\text{-limits and is accessible.}$$

Dually, L has a right adjoint $\iff$ L preserves ∞-colimits.

Chapter 51

∞-Monads and ∞-Kan Extensions

Chapter Overview

The two remaining pillars of Vol III's spine — monads and Kan extensions — complete their ∞-categorical lifts here. The chapter closes Phase 3 of Volume IV (Cluster E: the higher-categorification of Vol III).

∞-Monads. Every ∞-adjunction $L \dashv R$ produces a monad RL on the source $\mathcal{C}$. Conversely, the **∞-monadicity** theorem characterizes when a right adjoint $R : \mathcal{D} \to \mathcal{C}$ exhibits $\mathcal{D}$ as the ∞-category of algebras over the monad RL. Beck's theorem of Chapter 21 generalizes to the ∞-setting, with the technical content concentrated in the existence of RL-split simplicial objects.

∞-Kan extensions. The functors $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$ of Chapter 11/18/31 lift directly. The pointwise formula reads

$$(\mathrm{Lan}_p F)(d) \;\simeq\; \mathrm{colim}_{(c,\alpha)\in \mathcal{C}^p_{/d}} F(c),$$

where the indexing ∞-category is the comma $p \downarrow d$.

The **density theorem** of Chapter 48 (every presheaf is a colimit of representables) reappears as a Kan extension identity: $\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \simeq \mathrm{id}_{\mathcal{P}(\mathcal{C})}$. **Codensity**: every functor $F : \mathcal{C} \to \mathcal{D}$ induces its codensity monad $\mathrm{Ran}_F F$, the universal monad whose algebras model F as a forgetful functor.

The Phase 3 milestone — all of Vol III's spine generalizes to ∞ — is delivered. Phase 4 (presentable, stable, higher algebra) builds the practical apparatus on this foundation.

51.1 ∞-Monads

51.1.1 Formalizing

Where the Name Comes From

Monad (cross-ref Vol I Ch. 8) is from Greek *monos* (single, alone) — the categorical version of "a single algebraic structure on the endo-functor monoid." The term was introduced by Saunders Mac Lane, displacing earlier names "triple" (Eilenberg-Moore) and "standard construction" (Godement). The ∞-monad is the same data lifted to the homotopy-coherent setting: a monoid in $\mathrm{End}(\mathcal{C})$ where the monoid axioms hold up to coherent homotopy.

Definition 51.1 (∞-monad)**.** Let $\mathcal{C}$ be a quasi-category. An **∞-monad on $\mathcal{C}$** is a monoid object in the ∞-category $\mathrm{End}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}, \mathcal{C})$ with composition as monoidal structure: an endofunctor $T : \mathcal{C} \to \mathcal{C}$, a unit $\eta : \mathrm{id} \Rightarrow T$, a multiplication $\mu : TT \Rightarrow T$, satisfying associativity and unit laws up to coherent homotopy (with infinitely many higher coherences).

Theorem 51.2 (Monad from ∞-adjunction)**.** *Every ∞-adjunction $L : \mathcal{C} \rightleftarrows \mathcal{D} : R$ gives rise to an ∞-monad RL on $\mathcal{C}$, with:*

- *Unit $\eta_{\mathrm{ad}} : \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ (the adjunction unit).*
- *Multiplication $\mu : RLRL \Rightarrow RL$ given by $R\epsilon L$, where $\epsilon : LR \Rightarrow \mathrm{id}_{\mathcal{D}}$ is the adjunction counit.*

The monad laws hold by the triangle identities of the adjunction (and all higher coherences).

Example 51.3 (Free-forgetful monads)**.** For the free-forgetful adjunction $L \dashv U$ between **Set** and groups, the monad UL on **Set** assigns $X \mapsto |F(X)|$, the underlying set of the free group on X. This is the *group monad*; algebras for it are groups.

In Chapter 52's presentable ∞-setting, the same construction classifies ∞-categories of algebraic structures up to coherent homotopy.

51.1.2 Reorganizing: algebras and the Eilenberg–Moore ∞-category

Definition 51.4 (T-algebra in a quasi-category)**.** For an ∞-monad T on $\mathcal{C}$, a **T-algebra** is an object $x \in \mathcal{C}$ together with a structure map $a : Tx \to x$ satisfying unit ($a \circ \eta_x \simeq \mathrm{id}_x$) and multiplicativity ($a \circ \mu_x \simeq a \circ Ta$), plus all higher coherences. The ∞-category of T-algebras is denoted $\mathrm{Alg}_T(\mathcal{C})$.

Theorem 51.5 (Free T-algebras)**.** *The forgetful functor $\mathrm{Alg}_T(\mathcal{C}) \to \mathcal{C}$ admits a left adjoint sending $x \in \mathcal{C}$ to the free T-algebra (Tx, μ_x). The resulting adjunction*

$$F_T : \mathcal{C} \rightleftarrows \mathrm{Alg}_T(\mathcal{C}) : U_T$$

recovers T as $U_T F_T$.

51.2 Bar Resolutions and ∞-Monadicity

51.2.1 Formalizing

Where the Name Comes From

The **bar resolution** (cross-ref Ch. 37 bar construction) goes back to Eilenberg-MacLane's 1953 paper on the homology of cyclic groups. The vertical bars in the original notation $[a_1|a_2|\cdots|a_n]$ gave the construction its name. In the monad setting, the bar resolution $\mathrm{Bar}_\bullet(L, T, x)$ is the simplicial-object version: the free T-algebra resolution of x assembled into a single simplicial diagram. Lurie's ∞-monadicity criterion uses the bar resolution to characterize when an adjunction is monadic — generalizing Beck (1967) to the ∞-categorical setting.

Definition 51.6 (Bar simplicial object)**.** For an ∞-adjunction $L \dashv R$ and an object $x \in \mathcal{C}$, the **bar resolution** of x is the simplicial object

$$\mathrm{Bar}_\bullet(L, T, x) : \Delta^{\mathrm{op}} \to \mathcal{C}, \quad n \mapsto T^{n+1}x,$$

where $T = RL$. Face maps come from μ (applied in different positions); degeneracies from η. This is Chapter 37's bar construction (Corollary 37.18) applied to the monoid T in $\mathrm{End}(\mathcal{C})$.

Theorem 51.7 (∞-monadicity, Lurie)**.** *An ∞-adjunction $L \dashv R : \mathcal{D} \to \mathcal{C}$ is **monadic** — i.e., the comparison functor $\mathcal{D} \to \mathrm{Alg}_T(\mathcal{C})$ is an equivalence — iff:*

1. *R is conservative (reflects equivalences).*
2. *R preserves **U_T-split** simplicial objects: simplicial objects whose image under R admits a split (i.e., extends to Δ_+^{op}).*

Proof strategy. This is the ∞-categorical Beck monadicity theorem, generalizing Chapter 21's Vol III result. The key step is constructing the inverse to the comparison functor via geometric realization of the bar resolution:

$$\mathrm{Alg}_T(\mathcal{C}) \to \mathcal{D}, \quad (x, a) \mapsto |\mathrm{Bar}_\bullet(L, T, x)|.$$

R-split simplicial objects give the necessary "effective epimorphism" property to recognize colimits over Δ^{op}.

The proof is HTT §6.2 / HA §4.7. Forwards: Lurie's HA Theorem 4.7.3.5 gives the precise formulation. □ □

Remark 51.8 (Applications of ∞-monadicity)**.** ∞-monadicity is the central tool for recognizing ∞-categories of algebras. Lurie uses it to identify, e.g., the ∞-category of modules over a ring spectrum, the ∞-category of comodules over a coalgebra, sheaves on a stack, and many other "categories of structured objects."

51.3 ∞-Kan Extensions

51.3.1 Formalizing

Where the Name Comes From

Kan extension (cross-ref Vol I Ch. 11, Vol II Ch. 18, Vol III Ch. 31) is named for *Daniel Kan* again (his ubiquity in ∞-category theory's foundations is unsurpassed). Kan introduced the construction in 1958: given a functor $p : \mathcal{C} \to \mathcal{E}$ and another $F : \mathcal{C} \to \mathcal{D}$, the **left Kan extension** $\mathrm{Lan}_p F$ is the "best" extension of F along p — the universal cocontinuous extension. The ∞-version preserves the universal property and adds the coherence data; the pointwise formula via comma categories carries over without change.

Definition 51.9 (∞-Kan extensions)**.** For an ∞-functor $p : \mathcal{C} \to \mathcal{E}$ and an ∞-category $\mathcal{D}$, the **left ∞-Kan extension** of $F : \mathcal{C} \to \mathcal{D}$ along p is a functor $\mathrm{Lan}_p F : \mathcal{E} \to \mathcal{D}$ together with a universal natural transformation $F \to \mathrm{Lan}_p F \circ p$.

When $\mathcal{D}$ is cocomplete, $\mathrm{Lan}_p F$ always exists and is the left adjoint to precomposition with p:

$$\mathrm{Lan}_p : \mathrm{Fun}(\mathcal{C}, \mathcal{D}) \rightleftarrows \mathrm{Fun}(\mathcal{E}, \mathcal{D}) : p^*.$$

Dually, $p^* \dashv \mathrm{Ran}_p$ gives the right Kan extension.

Theorem 51.10 (Pointwise formula)**.** *When $\mathcal{D}$ is cocomplete and the indexing ∞-categories are appropriately small,*

$$(\mathrm{Lan}_p F)(e) \;\simeq\; \mathrm{colim}_{(c,\alpha)\in p\downarrow e} F(c),$$

where $p \downarrow e$ is the comma ∞-category whose objects are pairs $(c \in \mathcal{C}, \alpha : p(c) \to e \in \mathcal{E})$.

Dually, $(\mathrm{Ran}_p F)(e) \simeq \lim_{(c,\alpha)\in e\downarrow p} F(c)$.

Theorem 51.11 (Adjoint triple)**.** *For any $p : \mathcal{C} \to \mathcal{E}$ and cocomplete $\mathcal{D}$,*

$$\mathrm{Lan}_p \;\dashv\; p^* \;\dashv\; \mathrm{Ran}_p.$$

This is the ∞-categorical version of Chapter 18's adjoint-triple theorem.

51.4 Density, Codensity, and Free Cocompletion

51.4.1 Density via Kan

Theorem 51.12 (Density as Kan extension)**.** *For the Yoneda embedding $y : \mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C})$:*

$$\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \;\simeq\; \mathrm{id}_{\mathcal{P}(\mathcal{C})}.$$

That is, every presheaf is the left Kan extension of the identity on $\mathcal{C}$ along Yoneda. Equivalently, every $P \in \mathcal{P}(\mathcal{C})$ is the colimit of representables (the density theorem, Theorem 48.48.11).

Proof sketch. Apply the pointwise formula to $\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}}$ at $P \in \mathcal{P}(\mathcal{C})$:

$$(\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}})(P) \simeq \mathrm{colim}_{(c,\xi)\in y\downarrow P}\, y(c) = \mathrm{colim}_{\mathcal{C}_{/P}}\, y(c) \simeq P.$$

The last equivalence is the density theorem. □ □

51.4.2 Codensity monads

Where the Name Comes From

Codensity monad comes from *co-* (dual to) + *density* (cross-ref Ch. 48). A functor $F : \mathcal{C} \to \mathcal{D}$ is *dense* if its left Kan extension $\mathrm{Lan}_F F$ recovers the identity on $\mathcal{P}(\mathcal{C})$ (every object is a colimit of F-images). The *codensity monad* $\mathrm{Ran}_F F$ is the right-Kan-extension analogue — the universal monad on $\mathcal{D}$ for which the unit measures the failure of F to be dense. The terminology is due to *Anders Kock* (1966), with the classical ultrafilter example traced to *Manes* (1976).

Definition 51.13 (Codensity monad)**.** For an ∞-functor $F : \mathcal{C} \to \mathcal{D}$, the **codensity monad** is the right Kan extension

$$T_F \;:=\; \mathrm{Ran}_F F : \mathcal{D} \to \mathcal{D}.$$

T_F is an ∞-monad on $\mathcal{D}$ (the unit and multiplication arise from the universal property of Ran_F).

Example 51.14 (Codensity for ultrafilters)**.** For the inclusion $F : \mathbf{FinSet} \hookrightarrow \mathbf{Set}$, the codensity monad on **Set** is the **ultrafilter monad**: $T_F(X)$ is the set of ultrafilters on X. This is the classical Manes theorem, generalized to the ∞-setting in Chapter 32 (formal theory of monads) and applicable here directly.

51.4.3 Free cocompletion

Theorem 51.15 (Free cocompletion via Kan)**.** *For a small ∞-category $\mathcal{C}$ and cocomplete $\mathcal{D}$, the unique colimit-preserving extension $\bar{F} : \mathcal{P}(\mathcal{C}) \to \mathcal{D}$ of $F : \mathcal{C} \to \mathcal{D}$ (Theorem 48.48.10) is the left Kan extension*

$$\bar{F} \;=\; \mathrm{Lan}_y F.$$

Remark 51.16 (The Volume IV spine complete)**.** With ∞-Kan extensions in place, every theorem of Vol III's spine has been lifted to the ∞-setting:

- Vol III Ch 28 (formal Yoneda) $\rightsquigarrow$ Ch 48 (∞-Yoneda).
- Vol III Ch 32 (formal monads) $\rightsquigarrow$ this chapter, §1.
- Vol III Ch 31 (general Kan extensions) $\rightsquigarrow$ this chapter, §3.
- Vol III Ch 33 (formal adjunctions) $\rightsquigarrow$ Ch 50.

The remaining Phase 4 chapters (52–57) build the practical apparatus that makes these tools usable in derived algebraic geometry, stable homotopy theory, and higher algebra.

51.5 Exercises

Define an ∞-monad.	A monoid object in $\mathrm{End}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}, \mathcal{C})$ under composition: an endofunctor T, unit $\eta : \mathrm{id} \Rightarrow T$, multiplication $\mu : T^2 \Rightarrow T$, with associativity and unit laws up to coherent homotopy.
How does an ∞-adjunction give a monad?	$L \dashv R$ gives $T = RL$ with η_{ad} as unit and $R\epsilon L$ as multiplication. The adjunction triangle identities (with coherences) give the monad laws.
What's the bar resolution?	$\mathrm{Bar}_n(L, T, x) = T^{n+1}x$, a simplicial object. Face d_i from μ at position i; degeneracies from η.
State ∞-monadicity.	$L \dashv R$ is monadic iff R is conservative and preserves U_T-split simplicial objects.
What does "conservative" mean?	R reflects equivalences: Rf an equivalence implies f an equivalence.
What's a U_T-split simplicial object?	One whose image under the forgetful U_T extends to the augmented category Δ_+^{op} — i.e., admits a coaugmentation, plus splittings, making the colimit computable termwise.
What's the comparison functor?	$\mathcal{D} \to \mathrm{Alg}_T(\mathcal{C})$, sending d to $(Rd, R\epsilon_d : RLRd \to Rd)$, the T-algebra structure inherited.
When is the comparison an equivalence?	Precisely the monadicity condition above.
Define $\mathrm{Lan}_p F$.	Left ∞-Kan extension of $F : \mathcal{C} \to \mathcal{D}$ along $p : \mathcal{C} \to \mathcal{E}$: the left adjoint to $p^* : \mathrm{Fun}(\mathcal{E}, \mathcal{D}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{D})$.
Pointwise formula?	$(\mathrm{Lan}_p F)(e) \simeq \mathrm{colim}_{(c,\alpha) \in p \downarrow e} F(c)$, where $p \downarrow e$ is the comma ∞-category.
State the adjoint triple.	$\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$, the ∞-version of Chapter 18's Vol II theorem.
What's the Volume IV density theorem in Kan form?	$\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \simeq \mathrm{id}_{\mathcal{P}(\mathcal{C})}$: every presheaf is the canonical left Kan extension of the identity on $\mathcal{C}$ along Yoneda.
Define a codensity monad.	For $F : \mathcal{C} \to \mathcal{D}$, the right Kan extension $T_F = \mathrm{Ran}_F F$. An ∞-monad on $\mathcal{D}$.
What is the codensity monad of **FinSet** $\hookrightarrow$ **Set**?	The ultrafilter monad: $T_F(X) =$ set of ultrafilters on X (Manes's theorem, generalizing to ∞).
Why does codensity often matter?	Because it captures the "inverse limit of finite approximations": $\mathrm{Ran}_F F(x) = \lim_{(c,\alpha) \in F \downarrow x} F(c)$ recovers x as a limit over its F-approximations.
State the free-cocompletion via Kan formula.	$\bar{F} = \mathrm{Lan}_y F$ for the unique colimit-preserving extension of $F : \mathcal{C} \to \mathcal{D}$ (cocomplete $\mathcal{D}$).
Why is the bar construction central?	It computes free T-algebra resolutions; geometric realization recovers the ∞-categorical version. Generalizes simplicial resolutions in homological algebra.
What's the relationship between ∞-monads and operads (Ch 55)?	Every ∞-monad arises from an operad with one color. More general operads give multi-categorical generalizations.
Where does monadicity get used?	Recognizing ∞-categories of structured objects: modules over a ring spectrum, comodules over a coalgebra, sheaves on a stack, etc.
How does Ch 51 generalize Chapter 32 (formal monads in 2-cats)?	Chapter 32 worked in 2-categories; this chapter works in ∞-categories. The data is the same (monoid in End); the difference is that "equalities" become coherent homotopies.
What does "$\mathcal{D}$ is cocomplete" buy?	Existence of Lan_p for all small p. Without cocompleteness, Kan extensions may not exist.

How are Kan extensions used in derived algebraic geometry?	Pull-back / push-forward of sheaves between spaces involves ∞-Kan extensions along the structure maps. Six-functor formalisms organize the multitude of Kan-extension-based functors.
What's the next phase?	Phase 4: Ch 52–57. Presentable ∞-cats (52); stable ∞-cats (53); triangulated and t-structures (54); ∞-operads (55); $\mathbb{E}_n$-algebras (56); A_∞/L_∞ algebras (57). The practical apparatus.

Chapter Summary

Chapter Summary

∞-Monad. Monoid in $\mathrm{End}(\mathcal{C})$ under composition. Every ∞-adjunction $L \dashv R$ produces an ∞-monad RL.
Bar resolution. $\mathrm{Bar}_n(L, T, x) = T^{n+1}x$, a simplicial object. Geometric realization computes the free T-algebra construction.
∞-monadicity. $L \dashv R$ is monadic iff R is conservative and preserves U_T-split simplicial objects.
∞-Kan extensions. $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$. Pointwise: $(\mathrm{Lan}_p F)(e) = \mathrm{colim}_{p\downarrow e} F$.
Density and Kan. $\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \simeq \mathrm{id}_{\mathcal{P}(\mathcal{C})}$ — every presheaf is a canonical left Kan extension along Yoneda.
Codensity monads. $T_F = \mathrm{Ran}_F F$, the canonical monad associated to a functor. Classical example: ultrafilter monad = codensity of **FinSet** $\hookrightarrow$ **Set**.
Phase 3 milestone. All of Vol III's spine (Yoneda, monads, adjunctions, Kan) lifts to ∞. Phase 4 builds the applied apparatus.

Formal Restatement

Formal Restatement

∞-Monad. $(T, \eta, \mu) \in \mathrm{Mon}(\mathrm{End}(\mathcal{C}))$.

Monad from adjunction. $L \dashv R \Rightarrow RL$ a monad, $\eta = \eta_{\mathrm{ad}}$, $\mu = R\epsilon L$.

Bar. $\mathrm{Bar}_\bullet(L, T, x) : \Delta^{\mathrm{op}} \to \mathcal{C}$, $\mathrm{Bar}_n = T^{n+1}x$; $d_i = \mu$ at position i; $s_i = \eta$.

∞-monadicity (Lurie). $L \dashv R$ monadic $\iff$ R conservative and preserves U_T-split simplicial objects.

∞-Kan adjoint triple. $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$.

Pointwise formula. $(\mathrm{Lan}_p F)(e) \simeq \mathrm{colim}_{p\downarrow e} F$; $(\mathrm{Ran}_p F)(e) \simeq \lim_{e\downarrow p} F$.

Density via Kan. $\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \simeq \mathrm{id}_{\mathcal{P}(\mathcal{C})}$.

Codensity monad. $T_F = \mathrm{Ran}_F F : \mathcal{D} \to \mathcal{D}$ an ∞-monad.

Free cocompletion. $\bar{F} = \mathrm{Lan}_y F$ for the unique colimit-preserving extension.

Phase 3 closed. Vol III spine (Yoneda, monads, adjunctions, Kan) fully lifted to ∞. Phase 4 develops the applied higher algebra.

Chapter 52

Presentable ∞-Categories

Chapter Overview

Presentable ∞-categories are the working class of "manageable" ∞-categories: those that are cocomplete and generated under small colimits by a small "presentation." Examples: $\mathcal{S}$ (spaces), $\mathcal{P}(\mathcal{C})$ (presheaves on a small $\mathcal{C}$), and most ∞-categories arising in derived algebraic geometry, homotopy theory, and topos theory.

This chapter develops the theory:

- **κ-filtered colimits** for a regular cardinal κ — the building blocks of accessibility.
- **Accessible ∞-categories** — those generated under κ-filtered colimits by a small set of κ-compact objects.
- **Presentable ∞-categories** — accessible + cocomplete. Equivalently, accessible localizations of $\mathcal{P}(\mathcal{C})$.

The capstone is the **∞-Adjoint Functor Theorem** for presentable ∞-categories: a functor $R : \mathcal{D} \to \mathcal{C}$ admits a left adjoint iff R preserves all small ∞-limits and is *accessible*. This is the practical engine for proving existence of ∞-categorical free constructions, base changes, sheafifications, and the like.

A reader who completes this chapter has acquired the standard "working infrastructure" assumed throughout Lurie's *Higher Topos Theory* (Chapters 5–6).

52.1 Accessibility

52.1.1 Formalizing

Where the Name Comes From

Accessible and **presentable** categories were introduced by *Jiří Adámek* and *Jiří Rosický* in their 1994 book *Locally Presentable and Accessible Categories*. The terminology: *accessible* means "reachable from a small set of generators under κ-filtered colimits" (the category is "accessible" from a small generating set); *presentable* (or *locally presentable*) means accessible + cocomplete (i.e., "every object has a small presentation in terms of generators and relations"). The ∞-version was developed by Lurie in HTT Ch. 5 with the same philosophy.

Definition 52.1 (κ-filtered)**.** For a regular cardinal κ, a quasi-category $\mathcal{K}$ is **κ-filtered** if every diagram $K \to \mathcal{K}$ of size $< \kappa$ admits a cocone in $\mathcal{K}$.

A ω-filtered ($\kappa = \aleph_0$) quasi-category is just a filtered quasi-category in the sense of Chapter 49.

Where the Name Comes From

κ-compact (also called **κ-presentable**) is from topology / set theory: a topological space is *compact* if every open cover has a finite subcover; an object of a category is *κ-compact* if every map from it into a κ-filtered colimit factors through a stage. The terminology was lifted into category theory by Gabriel and Ulmer (1971). The classical ω-compact (= compact) objects in **Top** are the finite CW-complexes — generalizing the topological compactness exactly.

Definition 52.2 (κ-compact object)**.** An object x of a cocomplete ∞-category $\mathcal{C}$ is **κ-compact** (or **κ-presentable**) if $\mathrm{Map}_{\mathcal{C}}(x, -) : \mathcal{C} \to \mathcal{S}$ preserves κ-filtered colimits.

The full subcategory of κ-compact objects is $\mathcal{C}^{\kappa} \subseteq \mathcal{C}$.

Example 52.3 (Compactness in $\mathcal{S}$)**.** An object of $\mathcal{S}$ is ω-compact iff it is equivalent to a finite CW complex (a finitely-presented Kan complex). For larger κ, κ-compact spaces are exactly Kan complexes with κ-bounded simplices.

Definition 52.4 (Accessible ∞-category)**.** A quasi-category $\mathcal{C}$ is **accessible** if there is a regular cardinal κ and a small set $\{x_i\}_{i \in I}$ of κ-compact objects such that every object of $\mathcal{C}$ is a κ-filtered colimit of the x_i.

Equivalently: $\mathcal{C}$ has κ-filtered colimits and the inclusion $\mathcal{C}^{\kappa} \hookrightarrow \mathcal{C}$ extends along κ-filtered colimits to an equivalence $\mathrm{Ind}^{\kappa}(\mathcal{C}^{\kappa}) \simeq \mathcal{C}$.

How to Say This

$\mathrm{Ind}^\kappa(\mathcal{D})$ *(read: "the κ-Ind completion of $\mathcal{D}$")*: freely add κ-filtered colimits to $\mathcal{D}$.

Theorem 52.5 (Accessibility-via-Ind)**.** *For a small ∞-category $\mathcal{D}$, the universal κ-filtered colimit completion $\mathrm{Ind}^\kappa(\mathcal{D})$ exists, is the κ-filtered-colimit closure of $\mathcal{D}$ in $\mathcal{P}(\mathcal{D})$, and is universal among accessible ∞-categories admitting $\mathcal{D}$ as their κ-compact subcategory.*

52.2 Presentable ∞-Categories

52.2.1 Formalizing

Definition 52.6 (Presentable ∞-category)**.** A quasi-category $\mathcal{C}$ is **presentable** (or **locally presentable**) if it is:

1. Cocomplete (admits all small ∞-colimits).
2. Accessible (a κ-Ind completion of a small ∞-category, for some κ).

Theorem 52.7 (Characterizations of presentability)**.** *A quasi-category $\mathcal{C}$ is presentable iff any (equivalently, all) of:*

1. *It is an **accessible localization** of some $\mathcal{P}(\mathcal{D})$ — i.e., a reflective subcategory whose reflector preserves κ-filtered colimits for some κ.*
2. *It is cocomplete and there is a small set of objects $\{x_i\}$ that generates $\mathcal{C}$ under small colimits.*
3. *There exists a small $\mathcal{D}$ and a small set of morphisms W in $\mathcal{P}(\mathcal{D})$ such that $\mathcal{C} \simeq \mathcal{P}(\mathcal{D})[W^{-1}]$ — a Bousfield-localization presentation.*

Remark 52.8 (The class Pr^L)**.** The category Pr^L has presentable ∞-categories as objects and *colimit-preserving* (left adjoint) functors as morphisms. Lurie's HTT establishes that Pr^L is itself an ∞-category, in fact a closed symmetric monoidal ∞-category under the *Lurie tensor product*. This last is the working substrate of ∞-operads (Chapter 55) and stable homotopy theory.

52.2.2 Reorganizing: foundational examples

Example 52.9 ($\mathcal{S}$ is presentable)**.** $\mathcal{S}$ is presentable: ω-compact objects are finite CW complexes, and every space is an ω-filtered colimit of finite ones. In fact, $\mathcal{S} \simeq \mathrm{Ind}^\omega(\mathcal{S}^\omega)$.

Example 52.10 ($\mathcal{P}(\mathcal{C})$ is presentable)**.** For any small $\mathcal{C}$, $\mathcal{P}(\mathcal{C})$ is presentable. ω-compact objects: presheaves with finite support (representable or finite colimits of representables). Generation: representables.

Example 52.11 (Modules over a ring)**.** Mod_R for a connective $\mathbb{E}_\infty$-ring spectrum R is presentable. ω-compact: finitely generated projective R-modules. Generation: R itself. This is the principal example of practical higher algebra.

52.3 The ∞-Adjoint Functor Theorem

52.3.1 Formalizing

Theorem 52.12 (∞-AFT, presentable case)**.** *Let $\mathcal{C}, \mathcal{D}$ be presentable ∞-categories. A functor $R : \mathcal{D} \to \mathcal{C}$ admits a left adjoint iff:*

1. *R preserves all small ∞-limits.*
2. *R is accessible: preserves κ-filtered colimits for some regular cardinal κ.*

Dually, $L : \mathcal{C} \to \mathcal{D}$ admits a right adjoint iff L preserves all small ∞-colimits.

Proof strategy. The proof is HTT §5.5.2 (Theorem 5.5.2.9). The strategy:

Existence of left adjoint. For each $y \in \mathcal{D}$, the functor $\mathrm{Map}_{\mathcal{C}}(-, Ry) \circ R : \mathcal{D} \to \mathcal{S}$ preserves limits and is accessible. By the **Adjoint Functor Theorem for $\mathcal{S}$-valued functors**, it is representable; the representing object is the value of L at the corresponding object of $\mathcal{C}$.

Uniqueness follows from ∞-Yoneda.

Necessity of conditions is Theorem 50.50.7 (RAPL) plus the observation that left adjoints to accessible functors are accessible.

The dual statement uses the symmetric structure: right adjoints exist iff left adjoints to opposite functors do. ☐ ☐

Example 52.13 (Building free objects)**.** $\mathrm{Mod}_R \to \mathcal{S}$, sending a module to its underlying space, preserves all limits and is accessible. The adjoint functor theorem produces a left adjoint $\mathcal{S} \to \mathrm{Mod}_R$: the *free R-module* functor, $S \mapsto R \otimes \Sigma^\infty_+ S$.

52.4 Bousfield Localization and Reflective Subcategories

52.4.1 Formalizing

Where the Name Comes From

Bousfield localization is named for *Aldridge Knight "Pete" Bousfield* (1941–2020), American mathematician who pioneered the construction in his 1975 paper "The Localization of Spectra with Respect to Homology." Bousfield's insight was that one can localize an entire homotopy category at a chosen class of maps without leaving the ambient category — by working with the reflective subcategory of "W-local objects." The construction is the homotopy-theoretic generalization of the Gabriel-Zisman localization (Ch. 40), and it extends naturally to the ∞-categorical setting via reflective subcategories of presentable ∞-cats.

Definition 52.14 (Reflective subcategory)**.** A full subcategory $\mathcal{D} \hookrightarrow \mathcal{C}$ is **reflective** if the inclusion has a left adjoint $L : \mathcal{C} \to \mathcal{D}$. The composition $\mathcal{C} \xrightarrow{L} \mathcal{D} \hookrightarrow \mathcal{C}$ is then an idempotent monad (a **localization**).

Theorem 52.15 (Bousfield localization)**.** *For any presentable $\mathcal{C}$ and small set W of morphisms of $\mathcal{C}$, there is a reflective subcategory $L_W\mathcal{C} \hookrightarrow \mathcal{C}$ whose objects are the* ***W-local*** *ones: x such that $\mathrm{Map}_{\mathcal{C}}(w, x)$ is an equivalence for every $w \in W$.*

The reflection $L_W : \mathcal{C} \to L_W\mathcal{C}$ inverts every morphism in W and is universal among colimit-preserving functors out of $\mathcal{C}$ that do.

Remark 52.16 (Bousfield localization and presentability)**.** Every presentable ∞-category arises (up to equivalence) as a Bousfield localization of some $\mathcal{P}(\mathcal{C})$: this is Theorem 52.7's characterization (3). In particular, the ∞-category of sheaves on a Grothendieck site is the Bousfield localization of presheaves of spaces at the covers.

52.5 Exercises

Define κ-filtered.	A quasi-category $\mathcal{K}$ is κ-filtered if every diagram of size $< \kappa$ has a cocone.
What's a κ-compact object?	$x \in \mathcal{C}$ is κ-compact if $\mathrm{Map}_{\mathcal{C}}(x, -)$ preserves κ-filtered colimits.
Define accessible.	$\mathcal{C}$ is accessible if it equals $\mathrm{Ind}^{\kappa}(\mathcal{C}^{\kappa})$ for some κ and the small subcategory $\mathcal{C}^{\kappa}$.
Define presentable.	Cocomplete and accessible. Equivalent: accessible localization of $\mathcal{P}(\mathcal{C})$ for some small $\mathcal{C}$.
What are ω-compact objects in $\mathcal{S}$?	Finite CW complexes (equivalently, finitely generated Kan complexes).
Why is $\mathcal{P}(\mathcal{C})$ presentable?	Cocomplete by construction (limits and colimits computed pointwise); accessible because representables are compact and generate under colimits.
What is Pr^L?	The ∞-category of presentable ∞-categories with colimit-preserving (left adjoint) functors as morphisms. Closed symmetric monoidal under Lurie tensor product.
State the ∞-adjoint functor theorem.	For presentable $\mathcal{C}, \mathcal{D}$: $R : \mathcal{D} \to \mathcal{C}$ has a left adjoint iff R preserves ∞-limits and is accessible. Dually for left adjoints.
Why does the AFT work?	Map-Yoneda: limit-preserving accessible $\mathcal{D} \to \mathcal{S}$ is representable; this gives the adjoint value-by-value.
What is Bousfield localization?	For presentable $\mathcal{C}$ and small $W \subseteq \mathrm{Mor}(\mathcal{C})$, the reflective subcategory $L_W\mathcal{C} \subseteq \mathcal{C}$ on W-local objects.
Define W-local.	x is W-local iff $\mathrm{Map}(w, x)$ is an equivalence of spaces for every $w \in W$.
Why is every sheaf ∞-category a Bousfield localization?	Sheaves are presheaves local with respect to covers: $\mathrm{Sh}(X) = L_{\mathrm{covers}}\mathcal{P}(\mathrm{Open}(X))$.
What is $\mathrm{Ind}^{\kappa}(\mathcal{D})$?	The free κ-filtered colimit completion of $\mathcal{D}$: representables plus all κ-filtered colimits of them.

Why do compactness and accessibility differ from 1-categories?	They don't, conceptually — the definitions are direct analogues — but in the ∞-setting "size" and "finite generation" are subtler because of the higher-dimensional structure of mapping spaces.
What's a reflective subcategory?	Full subcategory $\mathcal{D} \hookrightarrow \mathcal{C}$ with a left adjoint $L : \mathcal{C} \to \mathcal{D}$. The composition $\mathcal{C} \to \mathcal{D} \to \mathcal{C}$ is an idempotent endofunctor.
Foundational example of localization?	The localization of **sSet** at weak equivalences gives $\mathcal{S}$. The localization of $\mathcal{P}(\mathbf{Aff})$ at étale covers gives the ∞-topos of étale stacks.
How does presentability help with adjoint existence?	Presentability ensures that AFT-style "representability" arguments terminate (the small-presentation provides a stopping point for transfinite induction).
Are all left adjoints in Pr^L accessible?	Yes — accessibility of a functor reflects in/out of accessible ∞-cats, so left adjoints between presentables are automatically accessible.
What's the difference between presentable and locally presentable?	The terms are interchangeable in most modern literature; "locally presentable" emphasizes the underlying 1-categorical analogue (Adámek–Rosický).
What's a generator?	A small set $\{x_i\}$ of objects such that every object of $\mathcal{C}$ is a colimit of x_i's. Compact objects of an accessible ∞-cat form a generator.
Why is the Lurie tensor product important?	It makes Pr^L symmetric monoidal, allowing tensor products of presentable ∞-categories — central to symmetric monoidal ∞-categories (Ch 55) and ring spectra.
What's the next chapter?	Ch 53: Stable ∞-Categories. Presentability + stability gives "presentable stable ∞-categories" $= \mathrm{Pr}^L_{\mathrm{Stab}}$ — the working category for stable homotopy theory and derived algebraic geometry.
Why does presentability matter in practice?	Because nearly every ∞-category arising in modern mathematics — sheaves, modules, schemes, types — is presentable. The AFT is then a free tool.

Chapter Summary

Chapter Summary

κ-filtered and κ-compact. κ-filtered: diagrams of size $< \kappa$ have cocones. κ-compact: $\mathrm{Map}(x, -)$ preserves κ-filtered colimits.
Accessible. $\mathcal{C} \simeq \mathrm{Ind}^\kappa(\mathcal{C}^\kappa)$ for some κ and small $\mathcal{C}^\kappa$. Generated under κ-filtered colimits by a small set.
Presentable. Cocomplete + accessible. Equivalent characterizations: accessible localization of $\mathcal{P}(\mathcal{C})$; small generator under colimits; Bousfield localization $\mathcal{P}(\mathcal{C})[W^{-1}]$.
∞-AFT. For presentable $\mathcal{C}, \mathcal{D}$: R has a left adjoint iff R preserves ∞-limits and is accessible. Engine of free constructions throughout higher algebra.
Bousfield localization. For any small $W \subseteq \mathrm{Mor}(\mathcal{C})$ in a presentable $\mathcal{C}$, the W-local subcategory $L_W\mathcal{C}$ is reflective with reflector inverting W. Every presentable ∞-cat is a Bousfield localization of some $\mathcal{P}(\mathcal{C})$.
Pr^L. The ∞-category of presentable ∞-categories with colimit-preserving functors. Closed symmetric monoidal under Lurie tensor product.

Formal Restatement

Formal Restatement

Accessibility. $\mathcal{C}$ accessible $\iff$ $\mathcal{C} \simeq \mathrm{Ind}^{\kappa}(\mathcal{D})$ for some small $\mathcal{D}$ and regular κ.

Presentability. $\mathcal{C}$ presentable $\iff$ cocomplete and accessible.

Characterizations (equivalent).

- Accessible localization of $\mathcal{P}(\mathcal{D})$.
- Cocomplete with a small generating set.
- $\mathcal{P}(\mathcal{D})[W^{-1}]$ for small $\mathcal{D}, W$.

∞-AFT. For presentable $\mathcal{C}, \mathcal{D}$:

$$R : \mathcal{D} \to \mathcal{C} \text{ has left adjoint } \iff R \text{ preserves } \infty\text{-limits and is accessible.}$$

Dually, L has right adjoint $\iff$ L preserves ∞-colimits.

Bousfield localization. $\mathcal{C}$ presentable, W small $\Rightarrow$ $L_W\mathcal{C} \hookrightarrow \mathcal{C}$ reflective; L_W universal among W-inverting colimit-preserving functors out of $\mathcal{C}$.

Pr^L. The ∞-category of presentable ∞-cats with colimit-preserving functors. Closed symmetric monoidal under Lurie tensor.

Chapter 53

Stable ∞-Categories

Chapter Overview

A **stable ∞-category** is the ∞-categorical analogue of an abelian category (Chapter 13/20) and, more directly, of a triangulated category (Chapter 54). The defining property — every fibre sequence is a cofibre sequence and vice versa — captures the duality between left and right exactness that characterizes additive categories.
Equivalently, a stable ∞-category is a pointed ∞-category in which the suspension functor $\Sigma : \mathcal{C} \to \mathcal{C}$ is an equivalence. This makes the homotopy category $\mathrm{ho}(\mathcal{C})$ triangulated (Chapter 54).
The motivating examples are:

- **Spectra** Sp: the stabilization of $\mathcal{S}$, the universal stable ∞-category receiving a colimit-preserving functor from $\mathcal{S}_*$ (pointed spaces).
- **Derived categories** $\mathcal{D}(\mathcal{A})$ of an abelian category, lifted to stable ∞-categories via the dg-nerve.
- **Module categories** Mod_R for an $\mathbb{E}_\infty$-ring spectrum R.

Stable ∞-categories are the working substrate of stable homotopy theory, derived algebraic geometry, and the modern picture of homological algebra. The class $\mathrm{Pr}^L_{\mathrm{St}}$ of presentable stable ∞-categories is closed symmetric monoidal under Lurie's tensor product, and serves as the universal home for stable phenomena.

53.1 Pointed ∞-Categories

53.1.1 Formalizing

Where the Name Comes From

Pointed (as in "pointed ∞-category") is from algebraic topology, where a *pointed space* has a distinguished basepoint. A pointed ∞-category has a distinguished *zero object* (both initial and terminal). The zero object plays the role of the basepoint at the categorical level — and is exactly what's needed to define fibre/cofibre constructions.

Definition 53.1 (Pointed ∞-category)**.** An ∞-category $\mathcal{C}$ is **pointed** if it admits a **zero object**: an object $0 \in \mathcal{C}$ that is both initial and terminal. Equivalently, the unique map $\emptyset \to$ terminal in $\mathcal{C}$ is an equivalence.

Example 53.2 (Pointed examples)**. Pointed spaces $\mathcal{S}_*$:** the slice $\mathcal{S}_{*/}$ under a chosen point. Zero object: the point.

Chain complexes $\mathrm{Ch}(\mathcal{A})$: the zero complex is initial and terminal.

Stable ∞-categories: every stable ∞-cat is pointed (a defining feature).

Not pointed: $\mathcal{S}$ itself. The empty space (initial) and the point (terminal) differ.

Remark 53.3 (Stabilization and pointed)**.** Pointed-ness is a prerequisite for stability: without a zero object, the fibre/cofibre constructions are undefined. The functor $\mathcal{S} \to \mathcal{S}_*$ adjoining a disjoint basepoint is the universal way to pointify, and is left adjoint to the forgetful $\mathcal{S}_* \to \mathcal{S}$.

53.2 Fibre and Cofibre Sequences

53.2.1 Formalizing

Where the Name Comes From

Fibre and **cofibre** sequences come from algebraic topology. A *fibre sequence* $F \to E \to B$ is the long-exact-sequence data of a fibration; the fibre F is the preimage of the basepoint. A *cofibre sequence* is the dual: $A \to X \to X/A$ for a cofibration $A \hookrightarrow X$, with X/A the quotient (cofibre). The ∞-categorical versions are pullback/pushout against the zero object: $\mathrm{fib}(f) = 0 \times_y x$, $\mathrm{cofib}(f) = 0 \sqcup_x y$. The duality between fibre and cofibre is what makes *stable*

∞-categories interesting: stability means the two notions coincide.

Definition 53.4 (Fibre and cofibre)**.** Let $\mathcal{C}$ be a pointed ∞-category with a zero object 0. For a morphism $f : x \to y$:

- The **fibre** $\mathrm{fib}(f)$ is the ∞-pullback of f along $0 \to y$ (when it exists).
- The **cofibre** $\mathrm{cofib}(f)$ is the ∞-pushout of f along $x \to 0$ (when it exists).

A **fibre sequence** is a sequence $a \to b \to c$ where $a = \mathrm{fib}(b \to c)$.

A **cofibre sequence** is a sequence $a \to b \to c$ where $c = \mathrm{cofib}(a \to b)$.

Theorem 53.5 (Loop and suspension)**.** *The fibre and cofibre constructions define endofunctors $\Omega = \mathrm{fib}(0 \to -) : \mathcal{C} \to \mathcal{C}$ and $\Sigma = \mathrm{cofib}(- \to 0) : \mathcal{C} \to \mathcal{C}$ on a pointed cocomplete $\mathcal{C}$, with $\Sigma \dashv \Omega$. Called the **suspension** and **loop** functors.*

Example 53.6 (Loop and suspension in $\mathcal{S}_*$)**.** For pointed spaces $\mathcal{S}_*$:

- $\Sigma X = X \wedge S^1$ (reduced suspension).
- $\Omega X = \mathrm{Map}_*(S^1, X)$ (based loop space).

The adjunction $\Sigma \dashv \Omega$ is the classical loop-suspension adjunction.

53.3 Stable ∞-Categories

53.3.1 Formalizing

Where the Name Comes From

Stable comes from *stable homotopy theory*: a phenomenon in algebraic topology is *stable* if it persists after sufficient iterations of suspension Σ (and is hence detected by spectra rather than spaces). For example, $\pi_n(S^k)$ are unstable in general but stabilize as $k \to \infty$ to the stable homotopy groups of spheres. The categorical version: an ∞-category $\mathcal{C}$ is *stable* if $\Sigma : \mathcal{C} \to \mathcal{C}$ is an equivalence — i.e., suspension has reached a fixed point, and homotopy is "stabilized." The ∞-category of spectra Sp is the universal stable ∞-category, just as it was the universal stable phenomenon in classical algebraic topology.

Definition 53.7 (Stable ∞-category)**.** An ∞-category $\mathcal{C}$ is **stable** if it is pointed, admits all finite limits and colimits, and satisfies:

- A square is a pullback iff it is a pushout. Equivalently: every fibre sequence is a cofibre sequence and vice versa.

Theorem 53.8 (Equivalent characterizations of stability)**.** *For a pointed cocomplete $\mathcal{C}$, the following are equivalent:*

1. *$\mathcal{C}$ is stable.*
2. *The suspension functor $\Sigma : \mathcal{C} \to \mathcal{C}$ is an equivalence (with inverse Ω).*
3. *Every fibre sequence in $\mathcal{C}$ extends to a long fibre sequence $\cdots \to \Omega c \to a \to b \to c \to \Sigma a \to \cdots$ which is also a long cofibre sequence.*

Where the Name Comes From

Spectra (singular: *spectrum*) was introduced by *Edwin Spanier* and *J.H.C. Whitehead* (1953) — a *spectrum* in the topological sense is a sequence of pointed spaces $\{E_n\}$ with structure maps $\Sigma E_n \to E_{n+1}$. The name comes from the spectral-sequence machinery of algebraic topology, which spectra naturally produce. The modern ∞-category Sp is the homotopy-coherent version of this concept, with $\mathbb{S}$ the sphere spectrum being the unit object.

Example 53.9 (Spectra)**.** The ∞-category Sp of **spectra** is the universal stable ∞-category receiving a colimit-preserving functor from $\mathcal{S}_*$:

$$\Sigma^\infty : \mathcal{S}_* \to \mathrm{Sp}, \qquad \Omega^\infty : \mathrm{Sp} \to \mathcal{S}_*.$$

Equivalently, Sp is the inverse limit of the sequence

$$\cdots \to \mathcal{S}_* \xrightarrow{\Omega} \mathcal{S}_* \xrightarrow{\Omega} \mathcal{S}_*.$$

Spectra make Σ invertible by construction.

Example 53.10 (Derived ∞-categories)**.** For an abelian category $\mathcal{A}$ with enough projectives, the derived ∞-category $\mathcal{D}(\mathcal{A})$ (a refinement of Chapter 40's homotopy-category derived category) is stable. Objects are chain complexes; the suspension is shift by one.

Example 53.11 (Modules over an $\mathbb{E}_\infty$-ring)**.** For an $\mathbb{E}_\infty$-ring spectrum R, the ∞-category Mod_R of R-modules is stable. This is the natural home of higher algebra: ring spectra, structured ring spectra, Hochschild homology, K-theory, all live in stable ∞-categories.

53.3.2 Reorganizing: stability and presentability

Theorem 53.12 ($\mathrm{Pr}^L_{\mathrm{St}}$)**.** *The class $\mathrm{Pr}^L_{\mathrm{St}}$ of presentable stable ∞-categories with colimit-preserving functors is closed symmetric monoidal under Lurie's tensor product. The unit is* Sp*:*

$$\mathrm{Sp} \otimes \mathcal{C} \simeq \mathcal{C}$$

for any $\mathcal{C} \in \mathrm{Pr}^L_{\mathrm{St}}$.

Remark 53.13 (Lurie tensor product)**.** The Lurie tensor product on Pr^L is the universal way to combine two presentable ∞-categories: $\mathcal{C} \otimes \mathcal{D}$ classifies functors bilinear in $\mathcal{C}, \mathcal{D}$ that preserve colimits in each variable. Restricting to $\mathrm{Pr}^L_{\mathrm{St}}$ preserves all the structure plus stability; spectra act as the unit because $\Sigma^\infty \dashv \Omega^\infty$ exhibits stabilization as tensoring with Sp.

53.4 Exact Functors

53.4.1 Formalizing

Definition 53.14 (Exact functor)**.** A functor $F : \mathcal{C} \to \mathcal{D}$ between stable ∞-categories is **exact** if it preserves all finite limits (equivalently, finite colimits — they coincide in stable ∞-categories).

Theorem 53.15 (Exact = colimit-preserving on finite limits)**.** *For stable ∞-categories $\mathcal{C}, \mathcal{D}$, the following are equivalent:*

1. *F preserves finite limits.*
2. *F preserves finite colimits.*
3. *F preserves fibre sequences.*
4. *F preserves cofibre sequences.*

Remark 53.16 (Exactness in derived ∞-cats)**.** For $\mathcal{D}(\mathcal{A}) \to \mathcal{D}(\mathcal{B})$ exact functors, exactness coincides with the classical notion of **exact functors between abelian categories** extended to derived categories. Derived tensor and derived hom (Chapter 42) are exact in this sense.

53.5 Exercises

Define a pointed ∞-category.	One with a zero object: both initial and terminal.
Define fibre and cofibre.	$\mathrm{fib}(f)$ = pullback of $f : x \to y$ along $0 \to y$. $\mathrm{cofib}(f)$ = pushout of f along $x \to 0$.
Define suspension and loop.	$\Sigma x = \mathrm{cofib}(x \to 0)$; $\Omega x = \mathrm{fib}(0 \to x)$. Adjoint: $\Sigma \dashv \Omega$.
Define a stable ∞-category.	Pointed, with finite (co)limits, where pullback squares = pushout squares. Equivalently: Σ is an equivalence.
Why are stable ∞-cats called "stable"?	Because Σ is invertible: "susupension is no longer adding information." The homotopy theory has reached an equilibrium between past and future.
What is Sp?	The universal stable ∞-cat receiving a colimit-preserving functor from $\mathcal{S}_*$. Equivalently, $\lim(\cdots \to \mathcal{S}_* \xrightarrow{\Omega} \mathcal{S}_*)$.
Why is $\mathcal{S}$ not stable?	$\Sigma S^0 = S^1$, but ΩS^0 is the point — not an inverse. The suspension is not invertible on spaces.
Is $\mathcal{D}(\mathcal{A})$ stable?	Yes, for any abelian $\mathcal{A}$. Suspension is shift by one: $\Sigma C = C[1]$.
What is Mod_R for R an $\mathbb{E}_\infty$-ring spectrum?	The stable ∞-cat of R-module spectra. Foundational example of higher algebra; presentable and stable.
What is $\mathrm{Pr}^L_{\mathrm{St}}$?	Presentable stable ∞-cats with colimit-preserving functors. Closed symmetric monoidal under Lurie tensor; unit is Sp.
Why is Sp the unit?	Because $\mathcal{S}_* \otimes \mathcal{C} \simeq \mathcal{C}$ when $\mathcal{C}$ is pointed (since pointed-ness is the unit on $\mathcal{S}_*$-side), and stabilizing $\mathcal{S}_*$ gives Sp. Tensoring with Sp becomes the identity on stable categories.

Define an exact functor.	Preserves finite limits (equivalently, finite colimits) between stable ∞-categories.
Why do limits and colimits coincide in stable cats?	Because pullbacks = pushouts: finite limits and colimits are built from the same data.
What's a long fibre sequence?	$\cdots \to \Omega^2 c \to \Omega b \to \Omega a \to a \to b \to c \to \Sigma a \to \Sigma b \to \cdots$. Extends a single fibre sequence by iterated Ω and Σ.
Equivalent: long cofibre sequence?	Yes, the same data — in a stable ∞-category, fibre and cofibre sequences are interchangeable.
Why are stable ∞-cats the home of homological algebra?	Because the fibre/cofibre interchangeability captures additivity: ho of a stable ∞-cat is automatically triangulated (Chapter 54).
What's the relation to chain complexes?	$\mathcal{D}(\mathcal{A})$ is the principal example. Chain complexes of R-modules underly Mod_R for R a discrete ring.
Where does stable homotopy theory live?	In Sp, the ∞-category of spectra. Spectra are the generalized cohomology theories, ring spectra, and stable analogues of spaces.
Why $\mathbb{E}_\infty$-ring spectrum?	Because commutativity in Sp is coherence-up-to-everything; an $\mathbb{E}_\infty$-structure (Ch 56) provides the full coherent commutativity.
Does stability respect equivalences?	Yes — it's an invariant of the ∞-categorical equivalence class.
Can a 1-category be a stable ∞-category?	Only trivially: the zero category, or any pointed groupoid. A non-trivial stable ∞-cat necessarily has higher coherence.
What's a t-structure (preview)?	Chapter 54: a way to extract an abelian *heart* from a stable ∞-cat. Generalizes H_0 for chain complexes.
What's the homotopy hypothesis's avatar here?	The stabilization $\mathcal{S}_* \to \mathrm{Sp}$ extends the homotopy hypothesis to its stable analogue: spectra are the universal stable home for pointed homotopy types.

Chapter Summary

Chapter Summary

Pointed. $\mathcal{C}$ pointed iff has a zero object (initial = terminal). Prerequisite for fibre/cofibre.
Fibre/cofibre. $\mathrm{fib}(f)$ = pullback of f along $0 \to y$; $\mathrm{cofib}(f)$ = pushout along $x \to 0$. Σ and Ω: $\Sigma x = \mathrm{cofib}(x \to 0)$, $\Omega x = \mathrm{fib}(0 \to x)$.
Stable. Pointed with finite (co)limits, pullback squares = pushout squares. Equivalent: $\Sigma : \mathcal{C} \to \mathcal{C}$ an equivalence.
Examples. Sp (universal stable receiving $\mathcal{S}_*$); $\mathcal{D}(\mathcal{A})$ (derived ∞-cat); Mod_R for $R \in \mathrm{CAlg}^\infty(\mathrm{Sp})$.
$\mathrm{Pr}^L_{\mathrm{St}}$. Presentable stable ∞-cats with colimit-preserving functors; closed symmetric monoidal under Lurie tensor; unit Sp.
Exact functors. Between stable ∞-cats, preserve finite limits $\iff$ preserve finite colimits $\iff$ preserve fibre/cofibre sequences.

Formal Restatement

Formal Restatement

Pointed. $\mathcal{C}$ pointed: $\exists\, 0 \in \mathcal{C}$ with 0 initial and terminal.

Fibre/cofibre. $\mathrm{fib}(f) = 0 \times_y x$; $\mathrm{cofib}(f) = 0 \sqcup_x y$. $\Omega = \mathrm{fib}(0 \to -)$, $\Sigma = \mathrm{cofib}(- \to 0)$, $\Sigma \dashv \Omega$.

Stable. $\mathcal{C}$ stable $\iff$ pointed, finite (co)complete, and: square is pullback $\iff$ pushout.

Equivalent characterizations.

- Σ is an equivalence.

- Every fibre sequence is a cofibre sequence.
- $\mathcal{C}$ has a triangulated homotopy category (Chapter 54).

Foundational examples. Sp; $\mathcal{D}(\mathcal{A})$; Mod_R.

$\mathrm{Pr}^L_{\mathrm{St}}$**.** Closed symmetric monoidal under Lurie tensor; Sp unit; $\mathrm{Sp} \otimes \mathcal{C} \simeq \mathcal{C}$.

Exact. F between stable ∞-cats exact $\iff$ preserves finite limits $\iff$ preserves finite colimits.

Chapter 54

Triangulated Categories and t-Structures

Chapter Overview

The homotopy category $\mathrm{ho}(\mathcal{C})$ of a stable ∞-category $\mathcal{C}$ comes equipped with an extra layer of structure: it is a **triangulated category**, in the sense of Verdier. The triangulated structure remembers what the ∞-categorical fibre/cofibre sequences look like at the 1-categorical level: distinguished triangles $a \to b \to c \to \Sigma a$ replace the original homotopical data.
This chapter has two main themes.
First, the construction ho(stable) $\to$ triangulated. We define triangulated categories axiomatically (Verdier's axioms TR1–TR4), verify that $\mathrm{ho}(\mathcal{C})$ satisfies them, and explore the classical example: the homotopy category of chain complexes $\mathbf{K}(\mathcal{A})$ together with its triangulated structure.
Second, **t-structures**: a pair of full subcategories $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ of a stable ∞-category (or its triangulated homotopy category) satisfying conditions that ensure

- Each object decomposes into "connective and coconnective parts."
- The **heart** $\mathcal{C}^{\heartsuit} := \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$ is an abelian category.

The motivating example: the standard t-structure on $\mathcal{D}(\mathcal{A})$ has heart $\mathcal{A}$ itself, and the truncation functors recover the standard cohomology $H^n : \mathcal{D}(\mathcal{A}) \to \mathcal{A}$. Other t-structures (perverse, étale, motivic) produce different "hearts" suitable for specific applications.
Together, stable ∞-categories and t-structures supply the ∞-categorical home for homological algebra and its applications.

54.1 Triangulated Categories

54.1.1 Formalizing

Where the Name Comes From

Triangulated category was introduced by *Jean-Louis Verdier* (1935–1989) in his 1967 PhD thesis (advisor: Grothendieck). The name comes from *distinguished triangles* — sequences $a \to b \to c \to \Sigma a$ that play the role in triangulated categories that exact sequences play in abelian categories. The triangle is a tightening of the long exact sequence $\cdots \to a \to b \to c \to \Sigma a \to \cdots$ into a single rotating triple. Verdier's axiomatization captures the structure of derived categories of abelian categories abstractly; the ∞-categorical lift via stable ∞-cats was Lurie (HA 2017).

Where the Name Comes From

The **octahedron axiom** (TR4) is the most distinctive — and most delicate — axiom of triangulated categories. The name comes from the geometric shape of the diagram: given two composable morphisms $a \to b \to c$, the resulting four distinguished triangles fit into an octahedron with eight triangular faces. The axiom expresses a compatibility between the three composable cones. Verdier's original 1967 formulation; many authors have rewritten the axiom over the years (e.g., Beilinson-Bernstein-Deligne introduced a cleaner 9-diagram version).

Definition 54.1 (Triangulated category)**.** A **triangulated category** is an additive category $\mathcal{T}$ equipped with:

- A **shift** or **suspension** autoequivalence $\Sigma : \mathcal{T} \xrightarrow{\sim} \mathcal{T}$.
- A class of **distinguished triangles** of the form $a \to b \to c \to \Sigma a$,

satisfying Verdier's axioms **TR1**–**TR4**:

TR1 (Identity) For every a, the triangle $a \xrightarrow{\mathrm{id}} a \to 0 \to \Sigma a$ is distinguished. Distinguished triangles are closed under isomorphism.

TR2 (Rotation) $a \to b \to c \to \Sigma a$ is distinguished iff $b \to c \to \Sigma a \xrightarrow{-\Sigma f} \Sigma b$ is.

TR3 (Morphisms of triangles) Given two distinguished triangles and a morphism between their first two terms, there is a (not necessarily unique) extension to a morphism between the full triangles.

TR4 (Octahedron axiom) Given two composable morphisms $a \to b \to c$, the resulting distinguished triangles fit into an "octahedron" of compatible distinguished triangles.

How to Say This

Distinguished triangle *(read: "distinguished triangle" or "exact triangle")*. The notation $a \to b \to c \to \Sigma a$ implicitly carries the data of all three connecting morphisms.

54.1.2 Reorganizing: where triangulated comes from

Theorem 54.2 (ho of stable is triangulated)**.** *For any stable ∞-category $\mathcal{C}$, the homotopy* 1*-category* ho$(\mathcal{C})$ *is naturally triangulated. The shift is induced by Σ; the distinguished triangles are the images in* ho *of the (co)fibre sequences in $\mathcal{C}$.*

Proof strategy. Verifying TR1–TR4 amounts to constructing simplices in $\mathcal{C}$:

- TR1: the trivial fibre sequence $a \xrightarrow{\text{id}} a \to 0$ provides the identity triangle.
- TR2 (rotation) is the iterated cofibre construction.
- TR3 comes from inner-horn filling in $\mathcal{C}$.
- TR4 (octahedron) corresponds to a specific 3-simplex construction in $\mathcal{C}$ encoding compatibility of two cofibres.

All steps use the stability axiom (pullbacks = pushouts) at the ∞-categorical level. □ □

Example 54.3 ($\mathbf{K}(\mathcal{A})$ and $\mathcal{D}(\mathcal{A})$)**.** For an abelian category $\mathcal{A}$:

- $\mathbf{K}(\mathcal{A})$ (chain complexes modulo chain homotopy) is triangulated: shift by one, mapping cone gives distinguished triangles.
- $\mathcal{D}(\mathcal{A})$ (the derived category) is triangulated: the same structure, descending through localization at quasi-isomorphisms.

$\mathcal{D}(\mathcal{A}) =$ ho of the stable ∞-category $\mathcal{D}^\infty(\mathcal{A})$.

54.2 t-Structures

54.2.1 Formalizing

Where the Name Comes From

t-structure was introduced by *Beilinson, Bernstein, and Deligne* (*Faisceaux Pervers*, 1982) in their development of perverse sheaves. The letter "t" originally stood for *topology* — BBD's t-structures generalize the Postnikov truncation of topological spaces, where the bounded-above and bounded-below subcategories correspond to "everything above" and "everything below" a given degree. The *heart* of a t-structure (the intersection of the two parts) is an abelian category — a remarkable recovery of abelian structure from triangulated data. The terminology has stuck despite the more modern view that t-structures generalize *many* kinds of truncations (Postnikov, étale, perverse, etc.).

Definition 54.4 (t-structure)**.** A **t-structure** on a stable ∞-category $\mathcal{C}$ is a pair of full subcategories $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ of $\mathcal{C}$ satisfying:

1. $\Sigma\mathcal{C}_{\geq 0} \subseteq \mathcal{C}_{\geq 0}$ and $\Omega\mathcal{C}_{\leq 0} \subseteq \mathcal{C}_{\leq 0}$.
2. For $x \in \mathcal{C}_{\geq 0}$ and $y \in \Sigma^{-1}\mathcal{C}_{\leq 0}$: $\mathrm{Map}_{\mathcal{C}}(x, y) \simeq 0$ (mapping space is contractible).
3. Every $z \in \mathcal{C}$ fits into a fibre sequence $\tau_{\geq 0} z \to z \to \tau_{\leq -1} z$ with $\tau_{\geq 0} z \in \mathcal{C}_{\geq 0}$ and $\tau_{\leq -1} z \in \mathcal{C}_{\leq -1} = \Sigma^{-1}\mathcal{C}_{\leq 0}$.

We define $\mathcal{C}_{\geq n} = \Sigma^n \mathcal{C}_{\geq 0}$ and $\mathcal{C}_{\leq n} = \Sigma^n \mathcal{C}_{\leq 0}$.

How to Say This

$\mathcal{C}_{\geq 0}$ *(read: "the connective part of $\mathcal{C}$")* or *(read: "the ≥ 0-truncated part")*. $\mathcal{C}_{\leq 0}$: *(read: "coconnective")*. $\tau_{\geq 0}, \tau_{\leq 0}$ are **truncation functors**.

Where the Name Comes From

The **heart** $\mathcal{C}^\heartsuit$ of a t-structure is the abelian category at the "center" of the triangulated structure — the intersection of the connective and coconnective parts. The terminology is BBD (1982): the heart is the "heart" of the structure, the central part from which everything else can be reconstructed via cohomology functors. The symbol ♡ (heart suit) is universal across the literature.

Theorem 54.5 (Heart of a t-structure)**.** *The intersection $\mathcal{C}^\heartsuit := \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$ is an* abelian category*. It is closed under finite limits and finite colimits in $\mathcal{C}$, but not closed under suspension (which takes $\mathcal{C}^\heartsuit$ outside itself).*

Example 54.6 (Standard t-structure on $\mathcal{D}(\mathcal{A})$)**.** For an abelian category $\mathcal{A}$, the derived ∞-category $\mathcal{D}(\mathcal{A})$ has a canonical t-structure:

- $\mathcal{D}(\mathcal{A})_{\geq 0}$: chain complexes whose $H^n = 0$ for $n < 0$.
- $\mathcal{D}(\mathcal{A})_{\leq 0}$: chain complexes whose $H^n = 0$ for $n > 0$.
- Heart: $\mathcal{D}(\mathcal{A})^\heartsuit \simeq \mathcal{A}$ via the cohomology functor H^0.

54.2.2 Reorganizing: cohomology functors

Definition 54.7 (Cohomology functor)**.** The **n-th cohomology functor** of a t-structure is

$$H^n : \mathcal{C} \to \mathcal{C}^\heartsuit, \quad H^n(x) = \tau_{\leq 0}\tau_{\geq 0}(\Omega^n x).$$

H^n is exact in the sense that it sends fibre/cofibre sequences in $\mathcal{C}$ to long exact sequences in $\mathcal{C}^\heartsuit$.

Theorem 54.8 (Long exact sequence in cohomology)**.** *A fibre sequence $a \to b \to c$ in $\mathcal{C}$ induces a long exact sequence in the heart $\mathcal{C}^\heartsuit$:*

$$\cdots \to H^{n-1}(c) \to H^n(a) \to H^n(b) \to H^n(c) \to H^{n+1}(a) \to \cdots.$$

54.3 Other t-Structures

54.3.1 Formally: variants

Example 54.9 (Perverse t-structure)**.** On the derived category of constructible sheaves on a topological space, the **perverse t-structure** (Beilinson–Bernstein–Deligne) has distinct connective and coconnective parts from the standard one. Its heart consists of **perverse sheaves**, an abelian category playing a central role in geometric representation theory and the geometric Langlands program.

Example 54.10 (Étale t-structure)**.** On the derived ∞-category of étale sheaves on a scheme, a different t-structure (the standard one for étale cohomology) has heart the category of étale sheaves of abelian groups.

Example 54.11 (Postnikov t-structure on spectra)**.** On Sp, the **Postnikov t-structure** has connective spectra ($\pi_n = 0$ for $n < 0$) and coconnective spectra ($\pi_n = 0$ for $n > 0$). Heart: **Ab**, the category of abelian groups, via Eilenberg–MacLane spectra.

54.4 Exercises

Define a triangulated category.	An additive category with a shift autoequivalence Σ and a class of distinguished triangles, satisfying Verdier's TR1–TR4.
State TR2 (rotation).	$a \to b \to c \to \Sigma a$ distinguished iff $b \to c \to \Sigma a \xrightarrow{-\Sigma f} \Sigma b$ distinguished — rotating triangles preserves the distinguished property up to a sign.
State TR4 (octahedron).	Given $a \to b \to c$ as composable morphisms, the resulting four distinguished triangles fit into an octahedron of compatible morphisms.
State the theorem connecting stable ∞-cats and triangulated categories.	For any stable ∞-cat $\mathcal{C}$, $\mathrm{ho}(\mathcal{C})$ is canonically triangulated. Distinguished triangles are images of fibre/cofibre sequences in $\mathcal{C}$.
Why does $\mathrm{ho}(\mathcal{C})$ acquire triangulated structure?	Because stability of $\mathcal{C}$ provides the symmetry between fibre and cofibre sequences (TR2); inner-horn filling provides morphism extensions (TR3); and a specific 3-simplex realizes octahedron (TR4).
What's $\mathbf{K}(\mathcal{A})$?	The category of chain complexes modulo chain homotopy — the most classical example of a triangulated category. Distinguished triangles arise from mapping cones.
Define a t-structure.	A pair $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ of subcategories satisfying closure under suspension/loop, mapping-space orthogonality, and a truncation triangle for every object.

Define the heart.	$\mathcal{C}^\heartsuit = \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$ — an abelian category.
What's the standard t-structure on $\mathcal{D}(\mathcal{A})$?	$\mathcal{D}(\mathcal{A})_{\geq 0}$ = complexes with $H^n = 0$ for $n < 0$; $\mathcal{D}(\mathcal{A})_{\leq 0}$ = complexes with $H^n = 0$ for $n > 0$. Heart: $\mathcal{A}$.
What's the cohomology functor?	$H^n(x) = \tau_{\leq 0}\tau_{\geq 0}(\Omega^n x)$. Takes a stable object to its n-th cohomology in the heart.
State the long exact sequence in cohomology.	For a fibre sequence $a \to b \to c$: a LES $\cdots \to H^{n-1}(c) \to H^n(a) \to H^n(b) \to H^n(c) \to \cdots$ in the heart.
What's the Postnikov t-structure on Sp?	Connective spectra (positive π_n only) vs coconnective (negative π_n only). Heart: **Ab** via Eilenberg–MacLane spectra.
What's a perverse t-structure?	A non-standard t-structure on derived categories of constructible sheaves whose heart is perverse sheaves. Central to geometric representation theory.
Why don't all stable ∞-categories have natural t-structures?	Because t-structures encode a specific choice of "positive/negative" decomposition that not every stable category admits canonically. Spectra have one; abstract triangulated categories may not.
Is the heart of a t-structure abelian?	Yes (Theorem 54.5). The conditions on the t-structure guarantee exactness of finite limits and colimits within the heart.
Are stable ∞-categories "more structured" than triangulated ones?	Yes — stable ∞-cats retain higher-coherence data lost in passing to ho. Triangulated categories are the 1-truncation; some constructions (e.g., octahedron) become non-canonical at the 1-level.
Why is the octahedron axiom delicate?	Because the choice of morphisms in TR4 is non-unique at the triangulated level; only the ∞-categorical lift makes it functorial. This is a source of incompleteness in classical triangulated theory.
What's the relation to derived functors of Chapter 42?	Derived functors are exact functors between stable ∞-cats (or triangulated cats). Tor and Ext (Ch 42) compute cohomology in standard t-structures.
What does Riehl–Verity / Lurie use stable ∞-cats for primarily?	Stable homotopy theory, derived algebraic geometry, modular representation theory, and the modern theory of categorified algebra (e.g., 2-categorical structures arising from stable diagram homotopy).
What's Verdier's theorem on triangulated functors?	Functors between triangulated categories preserving shifts and distinguished triangles correspond, ∞-categorically, to exact functors. So ho of an exact functor of stable ∞-cats is a triangulated functor.
Is every triangulated category ho of a stable ∞-cat?	Not obviously: there exist triangulated categories with no known stable ∞-categorical lift ("ghost-classified" triangulated categories). It's a famous open problem in higher algebra.
What's a t-exact functor?	A functor between stable ∞-cats with t-structures that preserves the connective and coconnective parts: $F(\mathcal{C}_{\geq 0}) \subseteq \mathcal{D}_{\geq 0}$ and similarly for ≤ 0.
What's the next chapter?	Ch 55: ∞-Operads. Moving from stable to higher-algebraic structures: the operadic language for organizing multilinear and commutative-up-to-coherent operations.

Chapter Summary

Chapter Summary

Triangulated category. Additive category with shift Σ and distinguished triangles, satisfying Verdier's TR1–TR4 (identity, rotation, morphisms-extend, octahedron).
ho(stable) **is triangulated.** Every stable ∞-cat's homotopy category is triangulated, with distinguished triangles from fibre/cofibre sequences.
Classical examples. $\mathbf{K}(\mathcal{A})$, $\mathcal{D}(\mathcal{A})$: chain complexes modulo chain homotopy, and the derived category.
t-structure. Pair $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ with closure, orthogonality, truncation. Encodes a "positive/negative" decomposition. Heart $\mathcal{C}^{\heartsuit}$ is abelian.
Cohomology. $H^n : \mathcal{C} \to \mathcal{C}^{\heartsuit}$ takes stable objects to abelian cohomology. Sends fibre sequences to long exact sequences.
Variants. Standard t-structure on $\mathcal{D}(\mathcal{A})$ with heart $\mathcal{A}$; Postnikov on Sp with heart **Ab**; perverse for constructible sheaves; étale for étale sheaves.

Formal Restatement

Formal Restatement

Triangulated category. $(\mathcal{T}, \Sigma, \text{distinguished triangles})$ satisfying TR1–TR4.

ho-**stable.** $\mathcal{C}$ stable $\Rightarrow$ ho$(\mathcal{C})$ canonically triangulated.

t-structure. $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ on a stable $\mathcal{C}$:

- $\Sigma\mathcal{C}_{\geq 0} \subseteq \mathcal{C}_{\geq 0}$, $\Omega\mathcal{C}_{\leq 0} \subseteq \mathcal{C}_{\leq 0}$.
- $x \in \mathcal{C}_{\geq 0}, y \in \mathcal{C}_{\leq -1} \Rightarrow \mathrm{Map}(x, y) = 0$.
- $\forall z, \exists$ fibre seq $\tau_{\geq 0}z \to z \to \tau_{\leq -1}z$ with $\tau_{\geq 0}z \in \mathcal{C}_{\geq 0}$, $\tau_{\leq -1}z \in \mathcal{C}_{\leq -1}$.

Heart. $\mathcal{C}^{\heartsuit} = \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$: abelian.

Cohomology. $H^n = \tau_{\leq 0}\tau_{\geq 0}\Omega^n : \mathcal{C} \to \mathcal{C}^{\heartsuit}$. LES from fibre sequences.

Examples. $\mathcal{D}(\mathcal{A})^{\heartsuit} = \mathcal{A}$; $\mathrm{Sp}^{\heartsuit} = \mathbf{Ab}$ (Postnikov); perverse and étale variants.

Chapter 55

∞-Operads

Chapter Overview

An **operad** is a coherent description of "algebraic operations of varying arity" — a collection of n-ary operations $\{\mathcal{O}(n)\}$, together with composition rules that say how to plug operations into operations. Classical examples: the **associative operad** Assoc, whose algebras are associative algebras; the **commutative operad** Comm, whose algebras are commutative algebras; the **Lie operad** Lie, whose algebras are Lie algebras.

The **∞-operad** is the higher-categorical refinement, in which all the operad data is coherently homotopical. This chapter develops the Lurie definition: an ∞-operad is a coCartesian fibration $\mathcal{O}^{\otimes} \to \mathrm{N}(\mathrm{Fin}_*)$ over the nerve of pointed finite sets, satisfying a specific Segal-style condition that encodes the operadic structure.

Closely related: **symmetric monoidal ∞-categories** — ∞-categories with a coherent commutative tensor product — are the same as ∞-operads whose fibres are themselves ∞-categories.

The chapter introduces:

- Classical operads as warm-up.
- The category Fin_* of pointed finite sets.
- Lurie's coCartesian-fibration definition.
- Algebras over an ∞-operad, and the principal examples (commutative, associative, Lie).

Chapters 56–57 specialize to the most important operadic structures: $\mathbb{E}_n$-algebras (little disks), A_∞-algebras (Stasheff associahedra), and L_∞-algebras (the Lie analogue).

55.1 Classical Operads

55.1.1 Formalizing

Where the Name Comes From

Operad was coined by *Peter May* (1972) as a portmanteau of *operations* + *monad*: an operad generalizes the structure of a monoid (one binary operation) to handle *operations of multiple arities* simultaneously (binary, ternary, n-ary). May's original motivation was to formalize the structure of iterated loop spaces, but operads have since become a unifying language for algebraic structures across mathematics. The classical (set-valued) operad has Σ_n-actions on each $\mathcal{O}(n)$ and substitution maps; the ∞-categorical version (Lurie) packages the same structure as a coCartesian fibration.

Definition 55.1 (Operad in a symmetric monoidal category)**.** An **operad** in a symmetric monoidal category $(\mathcal{V}, \otimes, I)$ consists of:

- For each $n \geq 0$, an object $\mathcal{O}(n) \in \mathcal{V}$ of **n-ary operations**, equipped with a Σ_n-action.
- A distinguished **identity** $\eta : I \to \mathcal{O}(1)$.
- For each $n, k_1, \dots, k_n$, a **composition** map

$$\gamma : \mathcal{O}(n) \otimes \mathcal{O}(k_1) \otimes \cdots \otimes \mathcal{O}(k_n) \to \mathcal{O}(k_1 + \cdots + k_n)$$

  that "plugs n operations of arities $k_1, \dots, k_n$ into one operation of arity n."

Subject to associativity (of nested composition), unit (the identity is a unit for γ), and equivariance (Σ_n-action compatible with γ) axioms.

How to Say This

$\mathcal{O}(n)$ *(read: "the operad's n-ary operations")*. γ *(read: "operadic composition" or "substitution")*; pictorially, a tree of operations being grafted into one.

Example 55.2 (Associative operad Assoc)**.** Assoc(n) = the set of total orderings on $\{1, \ldots, n\}$ (or, equivalently, Σ_n regarded with regular action). Composition: nested orderings via concatenation. Algebras over Assoc in **Set**: associative monoids.

Example 55.3 (Commutative operad Comm)**.** Comm(n) = singleton (one element), for every n. Composition trivial. Algebras: commutative monoids.

Example 55.4 (Lie operad Lie)**.** Lie(n) in **Ch**(k-vec): $(n-1)!$-dimensional in degree 0, supported on **Lyndon basis** of bracketings. Algebras: Lie algebras.

55.2 Fin$_*$ and Symmetric Monoidal ∞-Categories

55.2.1 Formalizing

Where the Name Comes From

Fin$_*$ is the category of pointed finite sets, introduced by *Graeme Segal* (1974) in his work on classifying spaces of Γ-spaces. Segal's insight: a symmetric monoidal category can be encoded as a functor Fin$_* \to \mathcal{C}$ satisfying a coherence condition (the "Segal condition"). Lurie's ∞-operad framework lifts this to the coCartesian-fibration setting. The basepoint $*$ in each $\langle n\rangle$ represents the "output" or "trivial slot" in an operation; non-basepoint elements are "inputs."

Definition 55.5 (Fin$_*$)**.** Fin$_*$ is the category of pointed finite sets. Objects are $\langle n\rangle = \{*, 1, 2, \ldots, n\}$ with $*$ the basepoint. Morphisms are basepoint-preserving functions. We typically write N(Fin$_*$) for its nerve in **sSet**.

Definition 55.6 (Symmetric monoidal ∞-category, à la Lurie)**.** A **symmetric monoidal ∞-category** is a coCartesian fibration $\mathcal{C}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ such that for each $\langle n\rangle$, the canonical map

$$\mathcal{C}^\otimes_{\langle n\rangle} \to \mathcal{C}^{\otimes\ n}_{\langle 1\rangle}$$

(induced by the n "inclusion" maps $\langle n\rangle \to \langle 1\rangle$ collapsing all but one position) is an equivalence.

How to Say This

A symmetric monoidal ∞-category *(read: "a symmetric monoidal ∞-category")* is the data of *the entire fibration* — "$\mathcal{C}^\otimes$" with the additional structure encoded as the fibration. The underlying ∞-category $\mathcal{C} = \mathcal{C}^\otimes_{\langle 1\rangle}$ is just the value at the "one-element" pointed set.

Theorem 55.7 (Equivalence with classical symmetric monoidal ∞-categories)**.** *The Lurie definition agrees with the symmetric monoidal ∞-category defined via tensor functors $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ with all coherences (associator, unitor, symmetry, and all higher Mac Lane–Stasheff data). The fibrational form packages all coherences into a single combinatorial object.*

55.3 ∞-Operads

55.3.1 Formalizing

Definition 55.8 (∞-operad, Lurie)**.** An **∞-operad** is a functor $\mathcal{O}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ between simplicial sets such that:

1. It is an inner fibration.
2. Certain edges in Fin$_*$ — the **inert** maps (those that are bijective on non-base points after restriction) — admit coCartesian lifts.
3. For each $\langle n\rangle$, the Segal-type map
$$\mathcal{O}^\otimes_{\langle n\rangle} \to \prod_{i=1}^{n} \mathcal{O}^\otimes_{\langle 1\rangle}$$
is an equivalence (it factors over inert maps).

Remark 55.9 (Why Fin$_*$ rather than Fin?)**.** The basepoint corresponds to "outputs that disappear"; the basepoint-preserving structure captures the operadic distinction between inputs (non-basepoint elements) and outputs (the basepoint, in this encoding). This is Segal's original idea, generalized to the ∞-categorical setting.

Definition 55.10 (Algebra over an ∞-operad)**.** For an ∞-operad $\mathcal{O}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ and a symmetric monoidal ∞-category $\mathcal{C}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$, an **$\mathcal{O}$-algebra in $\mathcal{C}$** is a Fin$_*$-functor $\mathcal{O}^\otimes \to \mathcal{C}^\otimes$ preserving inert coCartesian morphisms.

The ∞-category of $\mathcal{O}$-algebras in $\mathcal{C}$ is $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$.

55.4 Foundational ∞-Operads

55.4.1 Formally

Example 55.11 (Commutative ∞-operad)**.** Comm $=$ N(Fin$_*$) itself. Algebras in $\mathcal{C}^{\otimes}$ are **$\mathbb{E}_\infty$-algebras** — commutative algebras up to coherent homotopy.

Example 55.12 (Associative ∞-operad)**.** Assoc: the ∞-operad of the nerve of the cyclic category Λ projected onto Fin$_*$. Algebras: **$\mathbb{E}_1$-algebras** — associative algebras up to coherent homotopy. Equivalent definition: algebras over the operad of trees.

Example 55.13 ($\mathbb{E}_n$-operad (preview to Chapter 56))**.** For $n \geq 1$, the **little n-disks operad** $\mathbb{E}_n$ encodes operations parametrized by configurations of n-disks. Algebras over $\mathbb{E}_n$ in Sp (or any symmetric monoidal ∞-category) are the **$\mathbb{E}_n$-algebras** of higher algebra. Chapter 56 develops them in detail.

Limiting cases: $\mathbb{E}_1 =$ Assoc (associative); $\mathbb{E}_\infty =$ Comm (commutative).

Remark 55.14 (Lie ∞-operad and L_∞-algebras)**.** The Lie ∞-operad Lie$_\infty$ gives **L_∞-algebras**, the strong-homotopy version of Lie algebras. Chapter 57 develops them together with A_∞-algebras (the strong-homotopy associative version of $\mathbb{E}_1$-algebras).

55.5 Exercises

Define a classical operad.	Collection of arity-n operations $\mathcal{O}(n)$ with Σ_n-actions, identity, and composition maps, satisfying associativity, unit, and equivariance.
What is the associative operad?	Assoc(n) = set of total orderings on $\{1,\dots,n\}$. Algebras: associative monoids.
What is the commutative operad?	Comm(n) = singleton for all n. Algebras: commutative monoids.
What is the Lie operad?	Lie(n) has dimension $(n-1)!$ in degree 0, spanned by the Lyndon basis. Algebras: Lie algebras.
Define Fin$_*$.	Category of pointed finite sets $\langle n\rangle = \{*, 1, \dots, n\}$ with basepoint-preserving functions as morphisms.
Why pointed finite sets, not finite sets?	The basepoint encodes "outputs that disappear" under the operadic interpretation; it distinguishes inputs from outputs.
Define a symmetric monoidal ∞-category (Lurie style).	A coCartesian fibration $\mathcal{C}^{\otimes} \to \mathrm{N}(\mathrm{Fin}_*)$ with $\mathcal{C}^{\otimes}_{\langle n\rangle} \simeq \mathcal{C}^{\otimes\, n}_{\langle 1\rangle}$ via the Segal map.
What does the Segal condition do?	It ensures the fibre over $\langle n\rangle$ is the n-fold "product" of the fibre over $\langle 1\rangle$ — encoding the tensor product as a higher-coherence operation.
Define an ∞-operad.	Inner fibration $\mathcal{O}^{\otimes} \to \mathrm{N}(\mathrm{Fin}_*)$ with inert-edge coCartesian lifts and a Segal-type condition.
What's an inert map?	A map of pointed finite sets that is bijective on non-basepoint elements after restriction. "Doesn't multiply elements."
Define an algebra over an ∞-operad.	A Fin$_*$-functor $\mathcal{O}^{\otimes} \to \mathcal{C}^{\otimes}$ preserving inert coCartesian morphisms.
What's an $\mathbb{E}_\infty$-algebra?	An algebra over the commutative ∞-operad Comm. Coherently commutative algebra in a symmetric monoidal ∞-category.
What's an $\mathbb{E}_1$-algebra?	Algebra over Assoc $= \mathbb{E}_1$. Coherently associative (but not necessarily commutative) algebra.
What's an $\mathbb{E}_n$-algebra?	Algebra over the little n-disks operad. Between fully commutative and fully associative; interpolates by the codimension of configuration space.

What's the homotopy hypothesis for operads?	Operads in $\mathcal{S}$ classify "operadic structures up to homotopy" — exactly the ∞-operads we've defined.
Why is the Lurie definition fibrational rather than enriched?	Because the fibration captures all coherences uniformly, avoiding the need to specify higher associators, symmetries, etc. separately. It's a Grothendieck-style packaging.
What's a symmetric monoidal functor in this framework?	A coCartesian-preserving morphism over $\mathrm{N}(\mathrm{Fin}_*)$: a functor $\mathcal{C}^\otimes \to \mathcal{D}^\otimes$ commuting with the projections and preserving coCartesian lifts.
How does this generalize Chapter 19 (monoidal categories)?	Chapter 19 worked in 1-categories; here in ∞-categories. All Mac Lane coherences (pentagon, hexagon, etc.) plus higher coherences are encoded uniformly via Fin_*.
What's Lurie's tensor product on Pr^L?	The closed symmetric monoidal structure on presentable ∞-categories. Unit: $\mathcal{S}$. Restricting to $\mathrm{Pr}^L_{\mathrm{St}}$: unit becomes Sp.
Where does Chapter 56 take this?	Detailed analysis of $\mathbb{E}_n$-algebras, the little disks operads, and their algebras. Connects to factorization homology and configuration spaces.
Where does Chapter 57?	A_∞ and L_∞ algebras: strong-homotopy variants, with explicit operadic descriptions via Stasheff associahedra and the Koszul dual of Lie.
Why doesn't this chapter prove the equivalence with the Stasheff definition?	Because the equivalence requires extensive combinatorial work (HA Chapter 2–3). We follow proof-depth policy: state the equivalence, sketch the strategy, point to Lurie's HA for details.
What's an "operadic algebra" versus an "operad"?	An operad is a system of operations (definition); an algebra is an object equipped with a representation of that system. Comm-operad: definition. Comm-algebra: a commutative monoid.
What does it mean to talk about Sp-modules or Mod_R?	Sp is a symmetric monoidal ∞-cat under smash product; modules over an $\mathbb{E}_\infty$-ring R in Sp are Mod_R-objects, themselves a symmetric monoidal ∞-cat.

Chapter Summary

Chapter Summary

Classical operads. $\{\mathcal{O}(n)\}$ with Σ_n-action, identity, and substitution γ. Examples: Assoc, Comm, Lie.
Fin_*. The category of pointed finite sets; $\mathrm{N}(\mathrm{Fin}_*)$ as a simplicial set is the indexing simplicial set for ∞-operads.
Symmetric monoidal ∞-category. CoCartesian fibration $\mathcal{C}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ satisfying the Segal condition. Encodes the symmetric monoidal structure plus all higher coherences.
∞-operad. Inner fibration $\mathcal{O}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ with inert-edge lifts and Segal condition. Algebras: Fin_*-functors preserving inert coCartesian.
Foundational examples. Comm ($\mathbb{E}_\infty$); Assoc $= \mathbb{E}_1$; $\mathbb{E}_n$ for $1 \leq n \leq \infty$; Lie and Lie_∞.
Algebras live everywhere. Modules, monoids, sheaves of rings, ring spectra — all are operadic algebras in various symmetric monoidal ∞-categories.

Formal Restatement

Formal Restatement

Classical operad. $\{\mathcal{O}(n)\}_{n\geq 0}$ with Σ_n-action, $\eta : I \to \mathcal{O}(1)$, γ, satisfying classical axioms.

Fin_*. Pointed finite sets, basepoint-preserving morphisms.

Symmetric monoidal ∞-cat. $\mathcal{C}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ coCartesian fibration, Segal: $\mathcal{C}^\otimes_{\langle n\rangle} \simeq \mathcal{C}^{\otimes\,n}_{\langle 1\rangle}$.

∞-Operad. $\mathcal{O}^\otimes \to \mathrm{N}(\mathrm{Fin}_*)$ inner fibration; inert-edge coCartesian lifts; Segal condition over inerts.

Algebra over $\mathcal{O}$. Fin_*-functor $\mathcal{O}^{\otimes} \to \mathcal{C}^{\otimes}$ preserving inert coCartesian morphisms.

Foundational. $\mathrm{Comm} = \mathbb{E}_\infty$; $\mathrm{Assoc} = \mathbb{E}_1$; $\mathbb{E}_n$ for general n; $\mathrm{Lie}, \mathrm{Lie}_\infty$.

Chapter 56

$\mathbb{E}_n$-Algebras and the Little Disks Operads

Chapter Overview

The **little n-disks operad** $\mathbb{E}_n$ is the most important family of operads in higher algebra. Its n-th component $\mathbb{E}_n(k)$ is the space of configurations of k disjoint n-disks inside the unit n-disk. Algebras over $\mathbb{E}_n$ in a symmetric monoidal ∞-category — **$\mathbb{E}_n$-algebras** — interpolate between fully associative ($\mathbb{E}_1$) and fully commutative ($\mathbb{E}_\infty$) structures, indexed by codimension.

This chapter introduces:

- Configuration spaces $\mathrm{Conf}_k(\mathbb{R}^n)$ and the action of Σ_k.
- The little disks operad $\mathbb{E}_n$ in Top (or $\mathcal{S}$ after fibrant replacement).
- $\mathbb{E}_n$-algebras in symmetric monoidal ∞-categories.
- The endpoints: $\mathbb{E}_1 = \mathrm{Assoc}$, $\mathbb{E}_\infty = \mathrm{Comm}$.
- **Dunn additivity**: $\mathbb{E}_m \otimes \mathbb{E}_n \simeq \mathbb{E}_{m+n}$ — the operadic tensor product of little disks interpolates dimensions additively.

The applications cited (not developed in detail): factorization homology, commutative ring spectra, structured ring spectra, the cohomology of loop spaces (May's recognition theorem). The reader who has completed this chapter can open Lurie's *Higher Algebra* Chapter 5 directly.

56.1 Configuration Spaces

56.1.1 Formalizing

Definition 56.1 (Configuration space)**.** For $n \geq 0$ and $k \geq 0$, the **configuration space** is

$$\mathrm{Conf}_k(\mathbb{R}^n) \;=\; \Big\{(x_1, \ldots, x_k) \in (\mathbb{R}^n)^k : x_i \neq x_j \text{ for } i \neq j\Big\}.$$

The symmetric group Σ_k acts on $\mathrm{Conf}_k(\mathbb{R}^n)$ by permuting coordinates.

Example 56.2 (Low-dimensional examples)**.** $\mathrm{Conf}_2(\mathbb{R}) = \{(x, y) \in \mathbb{R}^2 : x \neq y\}$ has two connected components (separated by $x = y$). Modulo Σ_2, one component.

$\mathrm{Conf}_2(\mathbb{R}^2) \simeq \mathbb{R}^4 \setminus \mathrm{diag}$ has homotopy type of S^1 (the angle between two points). Quotient by Σ_2: real projective line, $\mathbb{RP}^1 \simeq S^1$.

$\mathrm{Conf}_k(\mathbb{R}^\infty)$ is contractible for every k — the infinite-dimensional configuration space "has room for everything."

56.2 The Little n-Disks Operad $\mathbb{E}_n$

56.2.1 Formalizing

Where the Name Comes From

Little disks operad was introduced by *J. Michael Boardman* and *Rainer Vogt* (1973) in their book *Homotopy Invariant Algebraic Structures on Topological Spaces*. The original operad-of-little-cubes models the structure of n-fold loop spaces: the space of k disjoint n-cubes embedded in a larger n-cube parameterizes "k-fold operations" that can be performed on an n-fold loop space. Boardman–Vogt's recognition theorem characterizes n-fold loop spaces as algebras over the $\mathbb{E}_n$ operad.

The notation $\mathbb{E}_n$ is due to *Peter May*: "E" for *everything* (an $\mathbb{E}_\infty$-algebra has *everything* commutative, including all higher homotopies), with the subscript n recording the dimension at which commutativity is enforced. The naming convention has become universal in higher algebra.

Definition 56.3 (Little disks $\mathbb{E}_n$)**.** For $1 \leq n \leq \infty$, the **little n-disks operad** $\mathbb{E}_n$ has:

- $\mathbb{E}_n(k)$ = the space of **rectilinear embeddings** of k disjoint n-disks into the unit n-disk. (A rectilinear embedding scales and translates.)
- Σ_k-action by permuting the disks.
- Composition: nesting embeddings.

We have $\mathbb{E}_n(k) \simeq \mathrm{Conf}_k(\mathbb{R}^n)$ (the configuration of disk centers, weakly equivalently).

How to Say This

$\mathbb{E}_n$ *(read: "E-sub-n" or "the little n-disks operad")*; also written $\mathbb{E}_n$ or E_n depending on the typographic convention. $\mathbb{E}_\infty$ is the colimit as $n \to \infty$.

Theorem 56.4 ($\mathbb{E}_n$ as an ∞-operad)**.** *$\mathbb{E}_n$ has a canonical structure of an ∞-operad in $\mathcal{S}$ (the ∞-category of spaces), with its k-th component being the (weak homotopy type of the) configuration space.*

56.3 $\mathbb{E}_n$-Algebras and the Endpoints

56.3.1 Formalizing

Definition 56.5 ($\mathbb{E}_n$-algebra)**.** For a symmetric monoidal ∞-category $\mathcal{C}^{\otimes}$, an **$\mathbb{E}_n$-algebra in $\mathcal{C}$** is an algebra over the ∞-operad $\mathbb{E}_n$:

$$\mathrm{Alg}_{\mathbb{E}_n}(\mathcal{C}) \;=\; \mathrm{Fun}^{\mathrm{inert}}_{\mathrm{Fin}_*}(\mathbb{E}_n^{\otimes}, \mathcal{C}^{\otimes}).$$

Theorem 56.6 (Endpoints of the $\mathbb{E}_n$-tower)**.**

- *$\mathbb{E}_1$-algebras in $\mathcal{C}^{\otimes}$ are* ***associative monoids in $\mathcal{C}^{\otimes}$*** *(the coherent version of associative algebras).*
- *$\mathbb{E}_\infty$-algebras in $\mathcal{C}^{\otimes}$ are* ***commutative algebras in $\mathcal{C}^{\otimes}$*** *(coherent commutative algebras).*

For each $1 \le n \le \infty$, $\mathbb{E}_n$ interpolates: $\mathbb{E}_n$-algebras have n-disk-parametrized commutativity, weaker than full commutativity for $n < \infty$ and weaker than associativity for $n = 0$ (which trivializes).

Example 56.7 (Ring spectra)**.** $\mathbb{E}_n$-algebras in Sp are **$\mathbb{E}_n$-ring spectra**. Most prominent cases:

- $\mathbb{E}_1$-ring spectrum = associative ring spectrum (Lurie's "$\mathbb{A}_\infty$-ring spectrum").
- $\mathbb{E}_\infty$-ring spectrum = commutative ring spectrum. Underlies most modern stable homotopy theory: $\mathbb{S}$ (the sphere spectrum), $H\mathbb{Z}$, K-theory KU, complex cobordism MU, all are $\mathbb{E}_\infty$.

Example 56.8 ($\mathbb{E}_2$ in topology: braided monoids)**.** $\mathbb{E}_2$-algebras in **Set** (or $\mathcal{S}$): braided monoidal structures. The $\mathbb{E}_2$-operad's configuration spaces have non-trivial π_1 (braids), encoding the data of *half-twists* between commuting elements.

56.4 Dunn Additivity

56.4.1 Formalizing

Where the Name Comes From

Dunn additivity is named for *Gerald Dunn*, who proved the theorem in his 1988 paper "Tensor product of operads and iterated loop spaces." The result is striking: the operadic tensor product $\mathbb{E}_m \otimes \mathbb{E}_n \simeq \mathbb{E}_{m+n}$ shows that codimensions *add* under the tensor product of little-disks operads. The Eckmann-Hilton argument is the $m = n = 1$ special case: two compatible monoid structures on a set are equivalent to a single commutative monoid (i.e., $\mathbb{E}_1 \otimes \mathbb{E}_1 \simeq \mathbb{E}_2$, weakened in the 1-categorical case to commutativity).

Theorem 56.9 (Dunn additivity)**.** *For any $m, n \ge 1$, there is an equivalence of ∞-operads*

$$\mathbb{E}_m \otimes \mathbb{E}_n \;\simeq\; \mathbb{E}_{m+n},$$

where $\otimes$ is the Boardman–Vogt tensor product of ∞-operads.

Equivalently: $\mathbb{E}_{m+n}$-algebras are the same as $\mathbb{E}_m$-algebras whose underlying multiplication is itself $\mathbb{E}_n$-coherent — codimensions add.

Example 56.10 ($\mathbb{E}_2 = \mathbb{E}_1 \otimes \mathbb{E}_1$)**.** A doubly-monoidal structure (an $\mathbb{E}_1$-algebra in $\mathbb{E}_1$-algebras) is exactly an $\mathbb{E}_2$-algebra. This is the ∞-categorical version of the Eckmann–Hilton argument (Chapter 19): two compatible monoid structures form a braided monoid.

Example 56.11 (Limit case)**.** $\mathbb{E}_\infty = \mathrm{colim}_n \mathbb{E}_n$, achieved by stabilizing the configuration spaces. Dunn additivity gives $\mathbb{E}_\infty \otimes \mathbb{E}_n \simeq \mathbb{E}_\infty$ — adding any finite n to infinity remains infinity.

56.5 Exercises

Define $\mathrm{Conf}_k(\mathbb{R}^n)$.	k-tuples of distinct points in $\mathbb{R}^n$. Σ_k acts by permutation.
What is $\mathrm{Conf}_2(\mathbb{R}^2)$ homotopy-equivalent to?	S^1 (the angle between the two points, modulo translations and scalings).
What is $\mathrm{Conf}_k(\mathbb{R}^\infty)$?	Contractible — every configuration can be "spread out" in the infinite-dimensional space.
Define the little n-disks operad $\mathbb{E}_n$.	$\mathbb{E}_n(k)$ = rectilinear embeddings of k disjoint n-disks into the unit n-disk. Equivalently (up to weak equivalence), $\mathrm{Conf}_k(\mathbb{R}^n)$ with Σ_k-action.
What's $\mathbb{E}_\infty$?	$\mathrm{colim}_n \mathbb{E}_n$: the limit as $n \to \infty$. All configurations become equivalent because $\mathrm{Conf}_k(\mathbb{R}^\infty)$ is contractible.
Define an $\mathbb{E}_n$-algebra.	Algebra over the $\mathbb{E}_n$-operad in a symmetric monoidal ∞-category. Concretely: an object with n-disk-parametrized multiplication.
What's an $\mathbb{E}_1$-algebra?	Associative monoid (up to coherent homotopy). The most familiar coherent algebra structure.
What's an $\mathbb{E}_\infty$-algebra?	Commutative monoid (coherently). All higher coherences for commutativity are explicit data.
Why do $\mathbb{E}_n$-structures interpolate between associativity and commutativity?	Because configuration spaces $\mathrm{Conf}_k(\mathbb{R}^n)$ have increasingly trivial fundamental groups as n grows: π_1 = braid group for $n = 2$, becomes simpler for $n \geq 3$.
Give the principal example in Sp.	$\mathbb{E}_\infty$-ring spectra: $\mathbb{S}$ (sphere), $H\mathbb{Z}$, KU, MU. All foundational ring spectra of stable homotopy theory.
What's an $\mathbb{E}_2$-algebra in **Set**?	A braided monoidal set: commutative-up-to-braid-twist; rare in concrete sets, common in higher categorical settings.
State Dunn additivity.	$\mathbb{E}_m \otimes \mathbb{E}_n \simeq \mathbb{E}_{m+n}$. Tensor product of little-disks operads adds dimensions.
Why is Dunn additivity natural?	Because configurations in $\mathbb{R}^{m+n}$ "decompose" as configurations in $\mathbb{R}^m$ times configurations in $\mathbb{R}^n$ (in the limit of nested compatibility).
What does "$\mathbb{E}_m$-algebra in $\mathbb{E}_n$-algebras" produce?	By Dunn: an $\mathbb{E}_{m+n}$-algebra. Eckmann–Hilton in disguise: doubly-monoidal = braided = $\mathbb{E}_2$.
Give a non-trivial example of Dunn at work.	$\mathbb{E}_3$ as $\mathbb{E}_1 \otimes \mathbb{E}_2$: three commuting monoid structures, with the inner two compatible via a braid. Higher coherence between three operations.
Are all $\mathbb{E}_\infty$-rings (in Sp) commutative?	Coherently commutative, yes. Strict commutativity is rare and typically forced by being concentrated in degree 0; coherent commutativity allows higher data without being strict.
What's factorization homology?	An $\mathbb{E}_n$-algebra invariant for n-manifolds: $\int_M A$ for M an n-manifold and A an $\mathbb{E}_n$-algebra. Generalizes Hochschild homology ($n = 1$, S^1).
What's May's recognition theorem?	An $\mathbb{E}_n$-algebra in $\mathcal{S}_*$ is equivalent to an n-fold loop space $\Omega^n X$ for some X. The classical recognition theorem, generalized to all n and to all symmetric monoidal ∞-cats.
What's the $\mathbb{E}_n$-center?	For an $\mathbb{E}_n$-algebra A, the center $Z_{\mathbb{E}_n}(A)$ is the largest $\mathbb{E}_{n+1}$-algebra mapping to A. Generalizes the classical center of an associative algebra.

Where does this chapter take us?	Foundations of higher algebra à la Lurie's HA Chapter 5. Reader is prepared to encounter $\mathbb{E}_n$-algebras in actual contexts: ring spectra, derived geometry, factorization homology.
What's the next chapter?	Ch 57: A_∞ and L_∞ algebras. Strong-homotopy associative (A_∞, equivalent to $\mathbb{E}_1$) and Lie (L_∞, Koszul dual of A_∞) algebras, with explicit operadic presentations.
Are $\mathbb{E}_n$-operads the only operads in higher algebra?	No — Lie$_\infty$, A_∞, C_∞, and many bespoke operads play key roles. The $\mathbb{E}_n$-family is foundational because of Dunn additivity.
What's the symmetric counterpart of $\mathbb{E}_n$?	$\mathbb{E}_\infty$ for fully symmetric (commutative). The Σ_k-action on $\mathbb{E}_n(k)$ becomes increasingly homotopy-trivial as n grows.

Chapter Summary

Chapter Summary

Configuration spaces. $\mathrm{Conf}_k(\mathbb{R}^n)$: k-tuples of distinct points in $\mathbb{R}^n$. Foundational object; $\mathbb{E}_n(k) \simeq \mathrm{Conf}_k(\mathbb{R}^n)$ as Σ_k-spaces.
Little disks $\mathbb{E}_n$. ∞-operad whose k-th component is the configuration of k disjoint n-disks in the unit n-disk.
$\mathbb{E}_n$-algebra. An algebra over $\mathbb{E}_n$ in a symmetric monoidal ∞-category. $\mathbb{E}_1$ = associative; $\mathbb{E}_\infty$ = commutative; intermediate n interpolate.
Foundational examples. In Sp: $\mathbb{S}, H\mathbb{Z}, KU, MU$ are all $\mathbb{E}_\infty$-ring spectra. $\mathbb{E}_2$ in $\mathcal{S}$: braided monoidal spaces (e.g., 2-fold loop spaces).
Dunn additivity. $\mathbb{E}_m \otimes \mathbb{E}_n \simeq \mathbb{E}_{m+n}$. Codimensions add under operadic tensor product. Eckmann–Hilton as a special case.
Bridge to applications. Factorization homology, ring spectra, May's recognition theorem, $\mathbb{E}_n$-center, $\mathbb{E}_n$-Hochschild homology.

Formal Restatement

Formal Restatement

Configuration space. $\mathrm{Conf}_k(\mathbb{R}^n) = \{(x_1, \dots, x_k) \in (\mathbb{R}^n)^k : x_i \neq x_j\}$.

$\mathbb{E}_n$-operad. $\mathbb{E}_n(k) \simeq \mathrm{Conf}_k(\mathbb{R}^n)$ as Σ_k-space; composition by nesting.

$\mathbb{E}_n$-algebra. $\mathrm{Alg}_{\mathbb{E}_n}(\mathcal{C})$: ∞-functor $\mathbb{E}_n^\otimes \to \mathcal{C}^\otimes$ preserving inert coCartesian.

Endpoints. $\mathbb{E}_1 = \mathrm{Assoc}$; $\mathbb{E}_\infty = \mathrm{Comm}$.

Dunn additivity. $\mathbb{E}_m \otimes \mathbb{E}_n \simeq \mathbb{E}_{m+n}$ for $m, n \geq 0$.

Examples. $\mathbb{E}_\infty$-ring spectra: $\mathbb{S}, H\mathbb{Z}, KU, MU$; $\mathbb{E}_2$ in $\mathcal{S}$: braided ∞-monoids; $\mathbb{E}_n$-algebras in Sp for general n.

Chapter 57

A_∞- and L_∞-Algebras

Chapter Overview

A_∞- and L_∞-algebras are the strong-homotopy variants of associative and Lie algebras. Where an associative algebra has a single multiplication $m_2 : V \otimes V \to V$ satisfying $m_2(m_2 \otimes 1) = m_2(1 \otimes m_2)$ exactly, an A_∞-algebra has higher operations $m_n : V^{\otimes n} \to V$ for every $n \geq 2$, with associativity witnessed by an infinite tower of constraints involving the m_n's.

The chapter develops:

- A_∞-algebras as graded vector spaces with operations m_n of degree $2 - n$, satisfying Stasheff's relations.
- **Stasheff associahedra** K_n: polytopes encoding the higher associativity.
- Equivalence with $\mathbb{E}_1$-algebras in chain complexes / spectra.
- L_∞-algebras as the Lie analogue: symmetric brackets $l_n : V^{\wedge n} \to V$ with Jacobi-up-to-higher-coherence.
- **Bar/cobar duality** and the Koszul-dual relationship $A_\infty \leftrightarrow L_\infty^{\mathrm{cog}}$.

These algebras are the workhorses of homotopical and deformation theory: deformation theory of associative algebras via A_∞; Maurer-Cartan theory and BV formalism via L_∞; rational homotopy theory; the modern picture of derived categories.

Chapter 57 closes Cluster G (higher algebra). Phase 5 (Ch 58–60) opens to derived algebraic geometry, homotopy type theory, and the synthesis of Volume IV.

57.1 A_∞-Algebras

57.1.1 Formalizing

Where the Name Comes From

A_∞-algebra was introduced by *Jim Stasheff* in his 1963 thesis (Princeton, advisor: John Milnor). The name reads "A infinity algebra" or "A-sub-infinity algebra" and stands for *associative up to coherent homotopy of all orders* (the infinitely many higher coherences encoded in the operations m_n). Stasheff's original motivation was to characterize the homotopy associativity of H-spaces; the operadic interpretation as algebras over the topological associahedra-operad came later. The notation has spawned a whole family: A_∞ (associative), L_∞ (Lie), C_∞ (commutative), E_∞ (Stasheff's original "everything"-symmetric).

Definition 57.1 (A_∞-algebra)**.** An **A_∞-algebra** over a field k is a $\mathbb{Z}$-graded k-vector space $V = \bigoplus_n V^n$ equipped with multilinear operations

$$m_n : V^{\otimes n} \to V, \quad n \geq 1,$$

each of degree $2 - n$, satisfying the **Stasheff identities**: for each $n \geq 1$,

$$\sum_{r+s+t=n,\ s\geq 1} (-1)^{r+st}\, m_{r+1+t}(\mathbf{1}^{\otimes r} \otimes m_s \otimes \mathbf{1}^{\otimes t}) \;=\; 0.$$

How to Say This

A_∞-algebra *(read: "A-infinity algebra" or "strong-homotopy associative algebra")*; m_n *(read: "the n-th A_∞-operation")*. The notation A_∞ comes from "associative up to homotopy of all orders."

Example 57.2 (Low-order Stasheff identities)**.** **$n = 1$:** $m_1 m_1 = 0$. So m_1 is a degree-1 differential.

$n = 2$: $m_1 m_2 + m_2(m_1 \otimes 1) + m_2(1 \otimes m_1) = 0$. Leibniz: m_1 is a derivation for m_2.

$n = 3$: $m_2(m_2 \otimes 1) - m_2(1 \otimes m_2) = m_1 m_3 + m_3(m_1 \otimes 1 \otimes 1) + m_3(1 \otimes m_1 \otimes 1) + m_3(1 \otimes 1 \otimes m_1)$. Associativity of m_2 holds modulo m_3-corrections — the higher operation m_3 witnesses non-associativity.

Theorem 57.3 ($A_\infty = \mathbb{E}_1$ in chain complexes)**.** *The category of A_∞-algebras over k is equivalent (as an ∞-category) to the category of $\mathbb{E}_1$-algebras in $\mathbf{Ch}(k)$ (chain complexes of k-vector spaces with their natural symmetric monoidal structure). Equivalently: an A_∞-algebra is exactly a homotopy associative algebra in $\mathbf{Ch}(k)$.*

57.2 Stasheff Associahedra

57.2.1 Formalizing

Where the Name Comes From

The **Stasheff associahedra** K_n are convex polytopes named for *Jim Stasheff*, who introduced them in his 1963 PhD thesis as the topological structure encoding higher associativity. The n-th associahedron K_n is a polytope of dimension $n-2$ whose vertices correspond to the C_{n-1} Catalan-number-many parenthesizations of an n-fold product. Tamari (1962) studied the combinatorics of these polytopes independently — the *Tamari lattice* of bracketings is the 1-skeleton of an associahedron. The associahedra organize the higher coherence data of A_∞-algebras: K_4 = Mac Lane's pentagon, K_5 = a 3-dimensional polytope, etc.

Definition 57.4 (Associahedron K_n)**.** The **n-th Stasheff associahedron** K_n ($n \geq 2$) is a convex polytope of dimension $n-2$ whose vertices are in bijection with the ways to fully parenthesize a product of n symbols. Edges correspond to single re-bracketings; higher-dimensional faces encode higher-order associativities.

Example 57.5 (Small associahedra)**.**

- K_2 = point: one way to parenthesize ab.
- K_3 = interval: two parenthesizations $(ab)c$ and $a(bc)$, with an edge between them.
- K_4 = pentagon: five parenthesizations of $abcd$ (the Mac Lane pentagon, Chapter 19), with edges along 5 re-bracketings.
- K_5 = a 3-dimensional polytope with 14 vertices.

Theorem 57.6 (A_∞-operad via associahedra)**.** *The chain operad whose n-th component is the cellular chain complex of K_n is exactly the A_∞-operad. Algebras over this operad in $\mathbf{Ch}(k)$ are A_∞-algebras.*

The topological operad whose n-th component is K_n (with appropriate gluing) is weakly equivalent to $\mathbb{E}_1$.

Remark 57.7 (Mac Lane pentagon revisited)**.** The pentagon of Mac Lane coherence (Chapter 19) is exactly the boundary of K_4. Mac Lane's theorem says the pentagon commutes; the Stasheff picture says the pentagon is the natural witness for higher associativity. In an A_∞-algebra, the pentagon is replaced by 3-dimensional K_5, then K_6, and so on — the higher associahedra encode all higher coherences.

57.3 L_∞-Algebras

57.3.1 Formalizing

Where the Name Comes From

L_∞-algebra ("L infinity") was introduced by *Tom Lada* and *Jim Stasheff* (1992), generalizing A_∞-algebras to the *Lie* setting: a graded vector space with symmetric brackets l_n satisfying a generalized Jacobi identity up to coherent homotopy. L_∞-algebras play the central role in derived deformation theory (every formal moduli problem is governed by an L_∞-algebra, Lurie HA / Pridham). The parallel naming with A_∞ is intentional; the two are Koszul-dual operads.

Definition 57.8 (L_∞-algebra)**.** An **L_∞-algebra** over k is a $\mathbb{Z}$-graded vector space V with symmetric (in a graded sense) operations

$$l_n : V^{\wedge n} \to V, \quad n \geq 1,$$

each of degree $2-n$, satisfying the **generalized Jacobi identities**:

$$\sum_{i+j=n+1} \sum_{\sigma \in \mathrm{Sh}(i,n-i)} (-1)^\sigma \chi(\sigma)\, l_j(l_i(x_{\sigma(1)}, \dots, x_{\sigma(i)}), x_{\sigma(i+1)}, \dots, x_{\sigma(n)}) \;=\; 0,$$

where $\mathrm{Sh}(i, n-i)$ are unshuffles and $\chi(\sigma)$ is the Koszul sign.

How to Say This

L_∞-algebra *(read: "L-infinity algebra" or "strong-homotopy Lie algebra")*; l_n *(read: "the n-th L_∞-bracket").*

Example 57.9 (Low-order L_∞-identities)**.** $n = 1$**:** $l_1 l_1 = 0$. Differential.

$n = 2$**:** $l_1 l_2 + l_2(l_1 \otimes 1) + l_2(1 \otimes l_1) = 0$ (with appropriate Koszul signs). Leibniz.

$n = 3$**:** The graded Jacobi identity holds up to a l_1-coboundary involving l_3. The bracket l_2 is Jacobi-up-to-homotopy.

Theorem 57.10 (L_∞ and rational homotopy)**.** *Rational homotopy types of simply connected spaces are equivalent to L_∞-algebras over $\mathbb{Q}$ (Sullivan, Quillen). The L_∞-structure on $\pi_*(X) \otimes \mathbb{Q}$ encodes the rational homotopy type completely.*

57.4 Bar/Cobar Duality and Koszul Duality

57.4.1 Formalizing

Definition 57.11 (Bar and cobar)**.** For an A_∞-algebra A, the **bar construction** BA is the free coassociative coalgebra $T(sA)$ on the suspension sA, with differential b obtained from the operations m_n via a specific formula. Dually, for a coassociative coalgebra C (with differential), the **cobar construction** ΩC is the free associative algebra on the desuspension $s^{-1}C$, with differential from the coproduct.

$B \dashv \Omega$ is an adjunction; for connected A_∞-algebras, $\Omega BA \xrightarrow{\sim} A$ is a quasi-isomorphism.

Where the Name Comes From

Koszul duality is named for *Jean-Louis Koszul* (1921–2018, French mathematician). Koszul (1950) studied the homology of Lie algebras via a complex now called the *Koszul complex*, exhibiting a remarkable duality between associative and Lie structures at the level of homology. The modern operadic formulation — $\mathrm{Assoc}^! \simeq \mathrm{Lie}$ — was made precise by Ginzburg and Kapranov (1994); the chain-level bar/cobar implementation is due to Quillen, Adams, and others stretching back to the 1960s.

Theorem 57.12 (Koszul duality, $A_\infty \leftrightarrow L_\infty^{\mathrm{coalg}}$)**.** *There is a Quillen-equivalent pair of adjoint functors connecting A_∞-algebras and L_∞-coalgebras, induced by the bar/cobar adjunction. This is the **Koszul duality** for the associative and Lie operads:*

$$\mathrm{Assoc}^! \simeq \mathrm{Lie}, \quad \mathrm{Lie}^! \simeq \mathrm{Assoc}.$$

Where the Name Comes From

Maurer-Cartan equations are named for *Ludwig Maurer* (1859–1927, German mathematician) and *Élie Cartan* (1869–1951, French mathematician). The classical Maurer-Cartan equation $d\omega + \frac{1}{2}[\omega, \omega] = 0$ for a Lie-algebra-valued 1-form on a Lie group dates to the late 19th century (Maurer 1879, Cartan 1880s) and is central to differential geometry. The L_∞-version $\sum_n \frac{1}{n!} l_n(x^{\otimes n}) = 0$ extends the classical equation to infinitely many brackets — exactly the data of a solution to a formal moduli problem (Lurie HA Ch. 1).

Remark 57.13 (Maurer–Cartan in deformation theory)**.** L_∞-algebras govern derived deformation theory: a deformation problem over k corresponds to a Spec-formal moduli problem, which by Lurie's theorem (HA 1.3) is equivalent to an L_∞-algebra. Solutions to the deformation problem are **Maurer–Cartan elements** $x \in V^1$ satisfying $l_1 x + \frac{1}{2} l_2(x, x) + \frac{1}{6} l_3(x, x, x) + \cdots = 0$.

57.5 Exercises

Define an A_∞-algebra.	Graded vector space V with operations $m_n : V^{\otimes n} \to V$ of degree $2 - n$ satisfying the Stasheff identities.
What does the $n = 1$ Stasheff identity say?	$m_1^2 = 0$. m_1 is a differential.
What does the $n = 3$ Stasheff identity say?	m_2 is associative up to a coboundary involving m_3. The m_3-term is the explicit non-associativity correction.
What's the operadic statement?	A_∞-algebras = algebras over an operadic resolution of Assoc via Stasheff associahedra K_n. Equivalently, $\mathbb{E}_1$-algebras in $\mathbf{Ch}(k)$.
Define the n-th associahedron K_n.	A convex polytope of dimension $n - 2$ whose vertices are full parenthesizations of n symbols. Edges = single re-bracketings; faces = higher coherences.
What is K_4?	The Mac Lane pentagon: 5 parenthesizations of *abcd*, with 5 edges (re-bracketings).
Define an L_∞-algebra.	Graded vector space V with symmetric brackets $l_n : V^{\wedge n} \to V$ of degree $2 - n$, satisfying generalized Jacobi identities.
What does the $n = 3$ L_∞-identity say?	Jacobi for l_2 holds up to a coboundary involving l_3. Strong-homotopy Jacobi.
State Koszul duality.	$\mathrm{Assoc}^! \simeq \mathrm{Lie}$ and $\mathrm{Lie}^! \simeq \mathrm{Assoc}$. Bar/cobar adjunction is the implementation.

What's a Maurer–Cartan element?	$x \in V^1$ with $\sum_n \frac{1}{n!} l_n(x,\dots,x) = 0$. Solutions to deformation problems in the L_∞-formalism.
Where does L_∞ appear in deformation theory?	Every formal moduli problem over k is equivalent to an L_∞-algebra; solutions to the moduli problem correspond to Maurer–Cartan elements modulo gauge.
Why use A_∞ rather than just associative?	Because in derived/homotopical settings, exact associativity is too restrictive; A_∞ relaxes to associativity-up-to-coherent-homotopy, capturing the natural categories of dg-categories and ring spectra.
Equivalence with $\mathbb{E}_1$?	Yes: A_∞-algebras = $\mathbb{E}_1$-algebras in $\mathbf{Ch}(k)$. The two definitions encode the same structure: coherently associative algebra.
What's the bar construction for an A_∞-algebra?	$BA = T(sA)$ with differential b from the m_n. A free coassociative coalgebra realization.
What's the cobar?	Right adjoint Ω to B: for a coalgebra C, $\Omega C = T(s^{-1}C)$ with differential from the coproduct. $\Omega BA \to A$ is a quasi-isomorphism.
Why are A_∞ and L_∞ Koszul dual?	Because the operads Assoc and Lie are Koszul dual as quadratic operads. The bar construction implements the duality at the algebra level.
What's the homotopy hypothesis here?	L_∞ (over $\mathbb{Q}$) classifies rational homotopy types of simply connected spaces: $\pi_* X \otimes \mathbb{Q}$ is an L_∞-algebra completely determining $X_\mathbb{Q}$.
What's an A_∞-category?	Same as an A_∞-algebra but multi-object: a graded set of objects, hom-complexes with A_∞-composition. Foundational in Fukaya theory (symplectic topology).
Why do dg-categories sometimes suffice?	Because every A_∞-category has a quasi-equivalent dg-category model — but the A_∞-form is more natural for many constructions (mapping cones, twisted complexes).
What's a strict A_∞-algebra?	One with $m_n = 0$ for $n \geq 3$: an honest associative dg-algebra with differential m_1 and product m_2.
Where does this chapter close?	Cluster G (higher algebra) ends. Cluster H (Ch 58–60): derived algebraic geometry, homotopy type theory, and the higher landscape.
What does a reader of this chapter know?	Both $\mathbb{E}_n$-algebras (Ch 56) and their concrete realizations (A_∞, L_∞). Foundation of higher algebra is laid; Lurie's HA Chapter 5 onward is now accessible.
What's the next phase?	Phase 5: Ch 58 (derived alg geom preview), Ch 59 (HoTT preview), Ch 60 (the higher landscape synthesis).

Chapter Summary

Chapter Summary

A_∞-algebra. Graded V with $m_n : V^{\otimes n} \to V$ of degree $2-n$ satisfying Stasheff identities. Equivalent to $\mathbb{E}_1$-algebra in $\mathbf{Ch}(k)$.
Stasheff associahedra K_n. Polytopes of dim $n-2$ encoding higher associativity. K_4 = Mac Lane pentagon. K_n's chain complex is the A_∞-operad.
L_∞-algebra. Graded V with symmetric brackets $l_n : V^{\wedge n} \to V$ of degree $2-n$ satisfying generalized Jacobi. Equivalent to Lie$_\infty$-algebras in $\mathbf{Ch}(k)$.
Bar/cobar duality. $B \dashv \Omega$ between A_∞-algebras and connected coassociative dg-coalgebras. Recovers Koszul duality $\mathrm{Assoc}^! = \mathrm{Lie}$.

Deformation theory. Maurer-Cartan elements of an L_∞-algebra $\mathfrak{g}$ classify deformations of the corresponding formal moduli problem.
Phase 4 closes. Higher algebra foundations complete: $\mathbb{E}_n$, A_∞, L_∞, presentable, stable. Phase 5 synthesizes and previews ∞-categorical applications.

Formal Restatement

Formal Restatement

A_∞-algebra. $(V, \{m_n\}_{n\geq 1})$: $m_n : V^{\otimes n} \to V$, $|m_n| = 2 - n$; Stasheff identities for all n.

Associahedra. K_n convex polytope, $\dim K_n = n - 2$; K_4 = Mac Lane pentagon.

L_∞-algebra. $(V, \{l_n\}_{n\geq 1})$: $l_n : V^{\wedge n} \to V$ (graded symmetric), $|l_n| = 2 - n$; generalized Jacobi.

Equivalence with $\mathbb{E}_n$. $A_\infty \simeq \mathbb{E}_1$ in $\mathbf{Ch}(k)$; $L_\infty \simeq \mathrm{Lie}_\infty$.

Koszul duality. $\mathrm{Assoc}^! = \mathrm{Lie}$; $\mathrm{Lie}^! = \mathrm{Assoc}$; bar/cobar adjunction implements at the algebra level.

Maurer–Cartan. $x \in V^1$ with $\sum_n \frac{1}{n!} l_n(x^{\otimes n}) = 0$; classifies formal moduli problems over $\mathfrak{g}$.

Cluster G complete. Higher algebra foundations laid; HA Chapter 5+ now accessible.

Chapter 58

Toward Derived Algebraic Geometry

Chapter Overview

Derived algebraic geometry (DAG) is the application of ∞-category theory to algebraic geometry: schemes, stacks, and moduli problems all promoted to coherent-homotopy versions over $\mathbb{E}_\infty$-rings or simplicial commutative rings. The result captures phenomena that classical algebraic geometry can only describe imperfectly — virtual fundamental classes, derived intersection numbers, deformation theory in characteristic-p — and provides the right setting for modern moduli problems in physics and number theory.

This chapter is a *preview*. We define the basic objects (spectral schemes, derived schemes, formal moduli problems) and state the principal theorems, but full development is beyond the scope of Volume IV — these would be the content of an entire volume of their own, modeled on Lurie's *Higher Algebra* and *Higher Topos Theory*.

The chapter introduces:

- Spectral and derived schemes via $\mathbb{E}_\infty$-rings (resp. simplicial commutative rings).
- The cotangent complex.
- Formal moduli problems and Lurie's classification theorem (moduli $\sim$ L_∞-algebras).
- Stacks: ∞-categorical generalization of sheaves of groupoids.
- Reading recommendations: Toën–Vezzosi, Lurie's HAG II, recent survey papers.

After Chapter 58, the reader should be able to open Toën–Vezzosi or the relevant chapters of HA and follow.

58.1 Spectral and Derived Schemes

58.1.1 Formalizing

Where the Name Comes From

Derived algebraic geometry (DAG) was developed in two parallel streams: *Bertrand Toën* and *Gabriele Vezzosi* ($\sim$2002–2008, "Homotopical Algebraic Geometry I, II") working over simplicial commutative rings, and *Jacob Lurie* (HTT 2009, HA 2017, SAG ongoing) working over $\mathbb{E}_\infty$-ring spectra. Over $\mathbb{Q}$ the two streams agree; in positive characteristic the spectral approach has better behavior. The "derived" modifier signals that the framework naturally handles derived intersections, virtual classes, and the deformation-theoretic phenomena that classical algebraic geometry can only approximate.

Definition 58.1 (Spectral and derived rings)**.** There are two principal approaches to the "derived" generalization of commutative rings:

- **Spectral** approach: commutative rings replaced by **$\mathbb{E}_\infty$-ring spectra**, the category $\mathrm{CAlg}(\mathrm{Sp})$. Foundational example: $\mathbb{S}$, the sphere spectrum; $H\mathbb{Z}$ is the Eilenberg–MacLane $\mathbb{E}_\infty$-ring of integers.
- **Derived** approach: commutative rings replaced by **simplicial commutative rings** sCRing, the opposite of derived affine schemes.

Over $\mathbb{Q}$, the two approaches agree; for arithmetic applications (positive characteristic), spectral methods are preferred.

Definition 58.2 (Spectral scheme)**.** A **spectral scheme** is an ∞-category-valued sheaf $\mathcal{X}$ on the (small) Zariski site of $\mathbb{E}_\infty$-rings — equivalently, an object of $\mathrm{Sh}(\mathrm{CAlg}^{\mathrm{op}})$ valued in $\mathcal{S}$ and locally affine.

A **derived scheme** is the simplicial-commutative-ring analogue.

Example 58.3 ($\operatorname{Spec}\mathbb{S}$)**.** $\operatorname{Spec}\mathbb{S}$ is the spectral scheme associated to the sphere spectrum. It is the "unit" object: every spectral scheme has a unique morphism to $\operatorname{Spec}\mathbb{S}$.

58.2 The Cotangent Complex

58.2.1 Formalizing

Where the Name Comes From

The **cotangent complex** was introduced by *Daniel Quillen* (1970) and developed extensively by *Luc Illusie* (1971, *Complexe Cotangent et Déformations*). The motivation was algebraic-geometric deformation theory: the cotangent complex $\mathbb{L}_{B/A}$ classifies infinitesimal deformations and obstructions of a ring morphism $A \to B$. The name *cotangent* reflects that $\Omega^1_{B/A}$ (Kähler differentials) is the cotangent module — and $\mathbb{L}_{B/A}$ is its derived analogue.

Definition 58.4 (Cotangent complex)**.** For an $\mathbb{E}_\infty$-ring map $A \to B$, the **cotangent complex** $L_{B/A} \in \mathrm{Mod}_B$ is the left derived functor of Ω^1 (Kähler differentials), evaluated in the ∞-categorical setting:

$$L_{B/A} \;=\; \mathbb{L}_{B/A}(\mathbb{L}\Omega^1).$$

$L_{B/A}$ controls deformation theory and obstruction theory for the morphism $A \to B$.

Theorem 58.5 (Properties of the cotangent complex)**.**

- *For $A \to B$ a smooth morphism of ordinary rings, $L_{B/A} = \Omega^1_{B/A}$ concentrated in degree zero (classical limit).*
- *For $A \to B$ a closed immersion of derived schemes, $L_{B/A}$ = conormal complex shifted by one.*
- *Fundamental triangle: $L_{C/A} \to L_{C/B} \to L_{B/A}[1]$ for composable $A \to B \to C$.*

Remark 58.6 (Why derived?)**.** The classical Kähler differentials are only well-defined for smooth morphisms; for general morphisms, derived information is required to capture the deformation theory. The cotangent complex is the canonical "derived" answer.

58.3 Formal Moduli Problems

58.3.1 Formalizing

Where the Name Comes From

Formal moduli problem terminology dates to *Michael Schlessinger*'s 1968 paper "Functors of Artin Rings," which set up the formal framework for studying infinitesimal deformations. "Formal" because the problem is studied over *formal* (= infinitesimal) artin rings, capturing the local-to-first-order behavior of moduli problems. Lurie's classification theorem (FMPs equivalent to L_∞-algebras in char 0) is the modern higher-categorical generalization, with the "moduli space" replaced by the underlying L_∞-algebra (Maurer-Cartan elements as deformations).

Definition 58.7 (Formal moduli problem)**.** A **formal moduli problem** over a field k is a functor $F : \mathrm{Art}_k^{\mathrm{aug}} \to \mathcal{S}_*$ from augmented Artin local k-algebras (in derived sense) to pointed spaces, such that:

1. $F(k) = \mathrm{pt}$ (the point).
2. F preserves certain pullback squares (a Schlessinger-type condition).

Theorem 58.8 (Lurie's classification of formal moduli problems)**.** *For a field k of characteristic zero, the ∞-category of formal moduli problems over k is equivalent to the ∞-category of L_∞-algebras over k:*

$$\mathrm{FMP}_k \;\simeq\; \mathrm{Alg}_{\mathrm{Lie}_\infty}(\mathrm{Mod}_k).$$

*In positive characteristic, the corresponding theorem replaces L_∞-algebras with **partition Lie algebras**.*

Remark 58.9 (Maurer–Cartan and moduli)**.** By Theorem 58.8, solutions to a formal moduli problem F over a Artin ring A correspond to Maurer–Cartan elements of the associated L_∞-algebra (Chapter 57). Deformation theory is thus controlled by L_∞-algebras.

58.4 Stacks

58.4.1 Formalizing

Definition 58.10 (∞-stack)**.** For a site $(\mathcal{C}, J)$ — a Grothendieck topology on a small ∞-category — an **∞-stack** is a sheaf $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ satisfying the descent condition: for every covering family $\{U_i \to U\}_J$,

$$F(U) \;\simeq\; \lim\Big(\prod_i F(U_i) \rightrightarrows \prod_{ij} F(U_{ij}) \;\to\; \cdots\Big).$$

The ∞-category of ∞-stacks is $\mathrm{Sh}^\infty(\mathcal{C}, J)$.

Example 58.11 (Foundational ∞-stacks)**.** ***BG* as an ∞-stack:** the classifying ∞-stack of a group scheme G, with $BG(R) = \mathrm{Map}_*(\mathrm{Spec}\,R, BG)$ the ∞-groupoid of G-torsors on $\mathrm{Spec}\,R$.

Moduli of elliptic curves $\mathcal{M}_{\mathrm{ell}}$: an ∞-stack whose points classify elliptic curves up to isomorphism; the derived enhancement records automorphism data.

Spec $\mathbb{S}$**:** the affine ∞-stack of the sphere spectrum, the universal example.

Remark 58.12 (Why ∞-stacks?)**.** Classical stacks (Deligne–Mumford, Artin) are 1-truncated ∞-stacks — sheaves of groupoids rather than of spaces. ∞-stacks add the ability to encode *higher automorphisms*, essential when working with derived categories, perfect complexes, or moduli of higher-categorical structures.

58.5 Reading Recommendations

58.5.1 Formally

Remark 58.13 (Where to go from here)**.** For derived algebraic geometry proper, the canonical references are:

- Toën–Vezzosi, *Homotopical Algebraic Geometry* (HAG I, II) — the foundational text developing DAG over simplicial commutative rings.
- Lurie, *Higher Algebra* (Chapters 7–8) — DAG over $\mathbb{E}_\infty$-rings.
- Lurie, *Spectral Algebraic Geometry* (online book) — the spectral approach in dedicated form.
- Khan, *Lectures on derived and spectral algebraic geometry* — modern survey.

Key applications and current research:

- Gaitsgory–Rozenblyum, *A study in derived algebraic geometry* — derived Langlands.
- Pantev–Toën–Vaquié–Vezzosi, *Shifted symplectic structures* — derived symplectic geometry.
- Recent work on p-adic Hodge theory, prismatic cohomology, condensed mathematics — all build on derived foundations.

58.6 Exercises

What is the goal of derived algebraic geometry?	To extend algebraic geometry to allow $\mathbb{E}_\infty$-ring spectra (or simplicial commutative rings) as "coordinate rings," capturing derived intersection numbers and higher coherence data.
Two approaches?	Spectral (over $\mathbb{E}_\infty$-rings); derived (over simplicial commutative rings). Over $\mathbb{Q}$, equivalent.
What is CAlg(Sp)?	The ∞-category of $\mathbb{E}_\infty$-ring spectra. The home for spectral DAG.
What is the cotangent complex?	$L_{B/A} = \mathbb{L}\Omega^1$ for an $\mathbb{E}_\infty$-ring map $A \to B$. Controls deformation theory.
Smooth case for the cotangent complex?	For a smooth ordinary morphism, $L_{B/A} = \Omega^1_{B/A}$ concentrated in degree zero.
State the fundamental triangle.	$L_{C/A} \to L_{C/B} \to L_{B/A}[1]$ for composable $A \to B \to C$, in Mod_C.
What's a formal moduli problem?	A functor $F : \mathrm{Art}_k^{\mathrm{aug}} \to \mathcal{S}_*$ on augmented Artin local k-algebras with $F(k) = \mathrm{pt}$ and certain pullback preservation.
State Lurie's classification.	$\mathrm{FMP}_k \simeq \mathrm{Alg}_{\mathrm{Lie}_\infty}(\mathrm{Mod}_k)$ in characteristic zero. Formal moduli = L_∞-algebras.
Why L_∞?	Because the natural infinitesimal structure of a derived deformation is governed by a tangent complex with an L_∞-structure (the higher analogue of a Lie algebra of vector fields).
What's an ∞-stack?	A sheaf $\mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ on a site, satisfying ∞-categorical descent: gluing along covers is a homotopy limit.
What's BG?	Classifying ∞-stack of a group scheme G: $BG(R) =$ G-torsors on Spec R, as an ∞-groupoid.

What's the moduli stack of elliptic curves?	$\mathcal{M}_{\mathrm{ell}}$: ∞-stack whose points classify elliptic curves. Derived enhancement captures automorphism groups and higher structure.
Why ∞-stacks rather than 1-stacks?	Because higher automorphisms (e.g., perfect complexes, derived categories of modules) require the full ∞-categorical data; classical stacks lose information.
What's the moduli of perfect complexes?	An ∞-stack Perf whose points are perfect complexes on a scheme. Classical version is a 2-stack (objects up to automorphisms up to automorphisms of automorphisms); ∞-stack captures all higher.
What's the role of derived intersections?	Derived intersection of subschemes is naturally a derived scheme; the cotangent complex encodes virtual intersection numbers, generalizing classical excess intersection formulas.
What's prismatic cohomology?	A modern p-adic cohomology theory introduced by Bhatt–Scholze, building on derived algebraic geometry to handle integral p-adic phenomena.
What's condensed mathematics?	Scholze and Clausen's framework for analytical geometry using a derived/∞-categorical foundation for topological spaces and rings.
What's shifted symplectic geometry?	Pantev–Toën–Vaquié–Vezzosi's extension of symplectic geometry to derived stacks; shifted symplectic forms generalize the classical symplectic form.
Is DAG developed in this chapter?	No — only previewed. Full development requires its own volume; see Lurie, Toën–Vezzosi, Khan.
What does this chapter prepare the reader for?	Toën–Vezzosi's HAG II, Lurie's Spectral Algebraic Geometry, current literature on derived/spectral geometry.
What's the next chapter?	Ch 59: Toward Homotopy Type Theory. The intersection of ∞-category theory and dependent type theory.
Where do DAG and HoTT meet?	In Lurie's vision of ∞-topoi as classifying ∞-toposes for homotopy types; the univalence axiom of HoTT corresponds to the Yoneda philosophy at the level of higher categories.

Chapter Summary

Chapter Summary

Two approaches. Spectral (over $\mathbb{E}_\infty$-rings) vs. derived (over simplicial commutative rings); equivalent over $\mathbb{Q}$.
Spectral/derived schemes. ∞-stack-valued sheaves on the Zariski/étale site of $\mathbb{E}_\infty$-rings or sCRings; locally affine.
Cotangent complex. $L_{B/A}$ controls deformation theory. Smooth limit: Ω^1; closed immersion: shifted conormal. Fundamental triangle for compositions.
Formal moduli (Lurie). $\mathrm{FMP}_k \simeq \mathrm{Alg}_{\mathrm{Lie}_\infty}(\mathrm{Mod}_k)$ in char 0; partition Lie algebras in char p. Deformations are Maurer–Cartan elements.
∞-stacks. Sheaves on a site valued in $\mathcal{S}$, with ∞-categorical descent. Generalize classical stacks by encoding higher automorphisms.
Preview only. Full DAG is the content of its own volume. Reading: Toën–Vezzosi, Lurie's HA/SAG, Khan.

Formal Restatement

Formal Restatement

Derived rings. CAlg(Sp) (spectral) or sCRing (derived).

Spectral/derived scheme. ∞-stack on the small Zariski site of $\mathbb{E}_\infty$-rings or sCRings, locally affine.

Cotangent complex. $L_{B/A} = \mathbb{L}\Omega^1$ for $A \to B$ in CAlg(Sp). Smooth: $= \Omega^1_{B/A}[0]$. Closed immersion: shifted conormal complex.

Formal moduli problem. $F : \mathrm{Art}_k^{\mathrm{aug}} \to \mathcal{S}_*$, $F(k) = \mathrm{pt}$, Schlessinger pullback.

Lurie's classification. $\mathrm{FMP}_k \simeq \mathrm{Alg}_{\mathrm{Lie}_\infty}(\mathrm{Mod}_k)$ (char 0); $\mathrm{FMP}_k \simeq \mathrm{PartLie}_k$ (char p).

∞-stack. Sheaf $\mathcal{C}^{\mathrm{op}} \to \mathcal{S}$ on a site satisfying ∞-categorical descent.

Chapter 59

Toward Homotopy Type Theory

Chapter Overview

Homotopy Type Theory (HoTT) is the intersection of ∞-category theory and dependent type theory. Its central insight is the **univalence axiom**: equivalent types are equal — formally, $(A \simeq B) \simeq (A = B)$ as types. Univalence is the type-theoretic avatar of the Yoneda philosophy, and the resulting framework supplies a foundational language for mathematics in which:

- Spaces (more accurately: ∞-groupoids) are primitive rather than derived from sets.
- Identity is structural: two structures with all the same properties are formally equal.
- Constructive proofs and computer-checkable verification become natural.

This chapter is a preview, parallel to Chapter 58 on DAG. We introduce the basics — dependent types, identity types, the univalence axiom, ∞-topos models — and gesture at where HoTT is heading: the **univalent foundations** program (Voevodsky), modal type theories (synthetic homotopy theory), and applications in algebraic topology and constructive mathematics.

Full development would require a dedicated volume; the canonical references are *The HoTT Book* (Univalent Foundations Project, 2013) and Riehl–Shulman, *Synthetic theory of ∞-categories inside Homotopy Type Theory*. After this chapter, the reader can engage those works.

59.1 Dependent Type Theory: A Refresher

59.1.1 Informally

A type theory has *types* $A, B, \ldots$ and *terms* (also called *elements*) of those types, written $a : A$ to mean "a is a term of type A." Function types $A \to B$ have terms that are functions; the **lambda calculus** provides the syntax.

Dependent type theory allows types to depend on terms: $B(x)$ might be a type whose definition depends on $x : A$. Two key constructions:

- **Dependent function type** $\prod_{x:A} B(x)$: functions taking $x : A$ and producing a term of $B(x)$.
- **Dependent pair type** $\sum_{x:A} B(x)$: pairs (x, y) with $x : A$ and $y : B(x)$.

These generalize $A \to B$ (when B is constant) and $A \times B$ (when B is constant).

59.1.2 Formalizing

Where the Name Comes From

Type theory dates to *Bertrand Russell*'s 1903 attempt to resolve set-theoretic paradoxes by stratifying mathematical objects into a hierarchy of types. The modern computational version is *Alonzo Church*'s simply-typed lambda calculus (1940); the *dependent* extension is due to *Per Martin-Löf* (1972), adding types that depend on terms. The Curry-Howard-Lambek correspondence (see Vol I Ch. 46 / Vol V) identifies type-theoretic proofs with categorical morphisms, with dependent types corresponding to indexed products/coproducts. The ∞-categorical Martin-Löf type theory (Univalent Foundations, HoTT) is the modern synthesis.

Definition 59.1 (Dependent type theory, informally)**.** A **dependent type theory** consists of:

- A judgment of typing: $x : A \vdash B\,\text{type}$.
- Function types $\prod_{x:A} B(x)$ with lambda-abstraction and application.
- Sigma types $\sum_{x:A} B(x)$ with pairing and projections.
- Universe types $\mathcal{U}, \mathcal{U}_1, \mathcal{U}_2, \ldots$ with $\mathcal{U}_i : \mathcal{U}_{i+1}$ (avoiding Russell's paradox).
- Various other type formers: inductive types, $\mathbb{N}, \mathbb{B}$, etc.

Remark 59.2 (Curry–Howard–Lambek)**.** The classical **Curry–Howard–Lambek correspondence** (Chapter 46 / FP volume) identifies:

- Types $\leftrightarrow$ propositions $\leftrightarrow$ objects in a CCC.
- Terms $\leftrightarrow$ proofs $\leftrightarrow$ morphisms.
- $A \to B \leftrightarrow$ implication $\leftrightarrow$ exponential.
- $A \times B \leftrightarrow$ conjunction $\leftrightarrow$ product.
- $A + B \leftrightarrow$ disjunction $\leftrightarrow$ coproduct.

Dependent type theory extends this to higher-categorical settings: $\prod, \sum$ correspond to indexed product/coproduct = dependent adjoints.

59.2 Identity Types and Path Types

59.2.1 Formalizing

Where the Name Comes From

Identity type was introduced by *Per Martin-Löf* (1972) as part of his intuitionistic type theory. The classical view: $a =_A b$ is a type whose inhabitants are *proofs of equality* between a and b. The radical move of HoTT (Awodey-Warren 2009, Voevodsky $\sim$ 2010): take seriously the structure of identity types as ∞*-groupoids*. The path interpretation — $a = b$ is the mapping space $\mathrm{Path}(a, b)$ — connects type theory to homotopy theory in a way that has reshaped foundations.

Definition 59.3 (Identity type)**.** For terms $a, b : A$, the **identity type** $a =_A b$ (or simply $a = b$) is a type whose terms witness a and b being identical.

- Introduction: $\mathrm{refl}_a : a = a$ for every $a : A$.
- Induction (J-rule): properties holding on refl extend to all of $a = b$.

Remark 59.4 (Path interpretation)**.** In the homotopy-theoretic interpretation, A is an ∞-groupoid (Kan complex), terms $a, b : A$ are vertices, and $a = b$ is the *mapping space* $\mathrm{Path}_A(a, b)$. refl_a is the constant path at a.

Iterating: $\mathrm{refl}_a = \mathrm{refl}_a$ is the type of 2-paths (homotopies between paths); and so on. The infinite tower of identity types is exactly the infinite tower of higher cells in an ∞-groupoid.

59.3 The Univalence Axiom

59.3.1 Formalizing

Definition 59.5 (Equivalence of types)**.** For types A, B, the type $A \simeq B$ of **equivalences** from A to B consists of functions $f : A \to B$ together with proofs that f has both a left and a right homotopy inverse.

(The technical detail: equivalences are functions whose fibres are contractible, equivalent to "invertible up to coherent homotopy.")

Where the Name Comes From

The **univalence axiom** was introduced by *Vladimir Voevodsky* (2010) as part of his *Univalent Foundations* program. Voevodsky (1966–2017) was a Fields Medalist (2002) primarily known for motivic cohomology, and in his final decade turned to foundations of mathematics in HoTT. The univalence axiom — that equivalent types are equal — is the formal expression of the structural identity principle: "isomorphic structures should be treated as identical." The HoTT Book (Univalent Foundations Project 2013) is the canonical reference.

Theorem 59.6 (Univalence axiom, Voevodsky)**.** *For all types $A, B : \mathcal{U}$, the canonical map*

$$\mathrm{idtoeq} : (A = B) \to (A \simeq B)$$

is itself an equivalence. In particular, ***equivalent types are equal****.*

Remark 59.7 (Univalence as Yoneda)**.** Univalence is the type-theoretic analogue of the ∞-categorical Yoneda philosophy: "an object is determined by what maps in." In HoTT: "a type is determined by what is equivalent to it."

At the model level: HoTT is interpreted in ∞-toposes, where the universe $\mathcal{U}$ classifies ∞-groupoids; univalence corresponds to the (small-object) representability of the universe fibration.

Example 59.8 (Function extensionality from univalence)**.** A consequence of univalence: **function extensionality**. If $f, g : A \to B$ have $(x : A) \vdash f(x) = g(x)$, then $f = g$.

Function extensionality is not provable in vanilla dependent type theory but follows from univalence. This is one of the deep payoffs.

59.4 ∞-Topos Models

59.4.1 Formalizing

Where the Name Comes From

∞-topos extends *Grothendieck topos* (1957–1963) to the homotopy-coherent setting. A classical Grothendieck topos is a category of sheaves on a site; an ∞-topos is a category of ∞-sheaves on an ∞-site, with the sheaf condition phrased in ∞-categorical descent. The development is largely due to *Lurie* (HTT Ch. 6), building on *Charles Rezk*, *Bertrand Toën*, and others. The semantic claim: ∞-toposes are exactly the categories of mathematical structures definable in HoTT.

Definition 59.9 (∞-topos)**.** An **∞-topos** is an accessible left-exact localization of a presheaf ∞-category $\mathcal{P}(\mathcal{C})$. Equivalently: an ∞-category $\mathcal{X}$ satisfying Giraud's axioms in the ∞-categorical setting (colimit-presentable plus universality properties).

Theorem 59.10 (Infty-topos models of HoTT)**.** *Every ∞-topos $\mathcal{X}$ provides a model for HoTT: types are objects of $\mathcal{X}$; dependent types are morphisms; identity types are diagonals; univalence holds because $\mathcal{X}$'s universe classifies ∞-groupoid-valued sheaves.*

The principal example is $\mathcal{S}$ itself, which models HoTT trivially (types $=$ ∞-groupoids).

Remark 59.11 (Lurie's HTT and HoTT)**.** The model-theoretic foundations of ∞-toposes (Lurie, HTT §6) and the type-theoretic foundations of HoTT (Univalent Foundations) are remarkably compatible: ∞-toposes give the semantics that HoTT specifies syntactically. The synthetic and analytic pictures meet here.

59.5 Reading Recommendations

Remark 59.12 (Where to go)**.** Foundational references for HoTT:

- *The HoTT Book* (**Univalent Foundations Project**, 2013): the canonical introduction. Available at `homotopytypetheory.org`.
- Riehl–Shulman, *A type theory for synthetic ∞-categories*: ∞-category theory *within* HoTT — a more advanced treatment.
- Voevodsky's *Univalent Foundations Project* writings.
- Awodey–Warren, *Homotopy theoretic models of identity types*: the original semantic insight.

For HoTT applied to algebraic topology:

- Brunerie's PhD thesis on $\pi_4(S^3)$.
- Buchholtz–van Doorn–Rijke, *Higher groups in homotopy type theory.*
- Various papers on synthetic homotopy theory.

59.6 Exercises

What is dependent type theory?	A type theory where types can depend on terms: $x : A \vdash B(x)$ type. Includes dependent function types $\prod_{x:A} B(x)$ and dependent pair types $\sum_{x:A} B(x)$.
What is the Curry-Howard-Lambek correspondence?	Types $\leftrightarrow$ propositions $\leftrightarrow$ objects in a CCC. Terms $\leftrightarrow$ proofs $\leftrightarrow$ morphisms. Function types $\leftrightarrow$ exponentials. Connects logic, type theory, and category theory.
Define the identity type.	For $a, b : A$, $a =_A b$ is a type with $\mathrm{refl}_a : a = a$ as canonical inhabitant. Induction principle: properties on refl extend.
What's the path interpretation?	A is an ∞-groupoid; $a = b$ is the mapping space $\mathrm{Path}_A(a, b)$. Iterating gives the full tower of higher cells.
Define equivalence of types.	$A \simeq B$ is the type of equivalences: functions with contractible homotopy fibres. Equivalent to "invertible up to coherent homotopy."
State the univalence axiom.	$\mathrm{idtoeq} : (A = B) \to (A \simeq B)$ is itself an equivalence. Equivalent types are equal.

What does univalence "do"?	It identifies type-level equivalence with type-level equality, so reasoning about equivalent structures is the same as reasoning about equal ones. Reverses the classical "identity" problem of foundations.
What's function extensionality?	$(\forall x, f(x) = g(x)) \Rightarrow f = g$. Provable from univalence; independent of vanilla dependent type theory.
How does univalence relate to Yoneda?	Both say: objects determined by relationships. Yoneda: object = representable functor. Univalence: type = equivalence class under $\simeq$. Same philosophical content, different formal setup.
Define an ∞-topos.	An accessible left-exact localization of $\mathcal{P}(\mathcal{C})$ for small $\mathcal{C}$. Equivalent: ∞-cat satisfying ∞-Giraud's axioms.
What's the principal ∞-topos model of HoTT?	$\mathcal{S}$ itself: types = ∞-groupoids, types-of-types = universe in $\mathcal{S}$, identity types = path spaces.
Why is $\mathcal{S}$ an ∞-topos?	$\mathcal{S} = \mathcal{P}(\mathrm{pt}) = \mathrm{Fun}(\mathrm{pt}^{\mathrm{op}}, \mathcal{S}) = \mathcal{S}$ trivially; it satisfies the ∞-Giraud axioms.
What's an ∞-topos's universe?	A subobject classifier in the ∞-categorical sense: an object $\mathcal{U}$ such that morphisms $X \to \mathcal{U}$ classify (small) subobjects of X. Generalizes Set's universe to ∞-groupoids.
Why does univalence hold in ∞-topos models?	Because the universe's classifying property includes equivalences: a small subobject up to equivalence is the same as a small subobject up to equality.
What is synthetic homotopy theory?	Doing homotopy theory inside HoTT, without recourse to topological spaces or simplicial sets. Computations of $\pi_n(S^k)$ can be done synthetically.
Has $\pi_4(S^3)$ been computed in HoTT?	Yes — Guillaume Brunerie's PhD thesis (2016) gives a constructive proof that $\pi_4(S^3) = \mathbb{Z}/2$. Verified by computer.
What's higher inductive types (HITs)?	Type formers that introduce both points and higher-dimensional paths. Examples: $\mathbb{S}^1$ as a HIT with one point and one path-loop; sphere S^n via iterated HITs.
Why is HoTT a candidate foundation?	Because it captures higher-categorical phenomena natively, supports computer verification, and makes structural identity ($\simeq = =$) a primitive. Voevodsky's Univalent Foundations project.
Is HoTT consistent?	Yes, in the sense that it has set-theoretic models (∞-toposes). Like ZFC, consistency requires a meta-theory.
Where does HoTT meet ∞-category theory?	They're two sides of the same coin: HoTT is the internal language of ∞-toposes; ∞-category theory is the external/semantic theory of HoTT.
What about modal HoTT?	Adds modal operators (necessity, possibility) to type theory, used to model synthetic differential geometry, cohesion, and physical theories.
What's the next (final) chapter?	Ch 60: The Higher Landscape — synthesizing Vol IV, mapping its chapters to standard literature, and previewing a potential Vol V on applications.
What's Vol V (if written) supposed to be?	The fulfillment of the original v0.5 promise: connections to other branches of mathematics. Algebra (derived/triangulated), topology (spectra), logic (toposes), CS (type theory). All deferred to "after Vol IV."

Chapter Summary

Chapter Summary

Dependent type theory. Types depending on terms; $\prod_{x:A} B(x)$ and $\sum_{x:A} B(x)$ as dependent function and pair types. Curry-Howard-Lambek extended to higher categories.
Identity types. $a =_A b$ as the type of identifications. Path interpretation: $a = b$ is the mapping space in the ∞-groupoid A.
Univalence. $(A = B) \simeq (A \simeq B)$. Equivalent types are equal. The structural identity principle made formal. Function extensionality follows.
∞-topos models. Every ∞-topos models HoTT; $\mathcal{S}$ is the principal example. Synthetic and semantic foundations agree.
Reading. The HoTT Book; Riehl-Shulman; Voevodsky's Univalent Foundations; Brunerie on $\pi_4(S^3)$.
Vol IV synthesis closes Chapter 60. The original v0.5 promise (connections to other math fields) becomes potential Vol V.

Formal Restatement

Formal Restatement

Dependent type theory. Types $A : \mathcal{U}$; dependent types $x : A \vdash B(x) : \mathcal{U}$; $\prod_{x:A} B(x)$ and $\sum_{x:A} B(x)$.

Identity type. $a, b : A \Rightarrow (a = b) : \mathcal{U}$; $\mathrm{refl}_a : a = a$; J-rule eliminator.

Equivalence. $A \simeq B = \sum_{f:A\to B} \mathrm{isEquiv}(f)$.

Univalence axiom. $\mathrm{idtoeq}_{A,B} : (A = B) \to (A \simeq B)$ is an equivalence.

∞-topos. Accessible left-exact localization of $\mathcal{P}(\mathcal{C})$; equivalent: ∞-Giraud's axioms.

Model theorem. Every ∞-topos models univalent dependent type theory. $\mathcal{S}$ is the principal example.

Chapter 60

The Higher Landscape

Chapter Overview

We have arrived. Volume I built category theory from objects and arrows through the six-stage cycle. Volume II reorganized that material around representability and adjointness. Volume III completed the formal spine with ends, weighted limits, the formal Yoneda lemma, and the 2-categorical theories of monads and adjunctions. Volume IV took the final step into higher category theory: simplicial foundations, quasi-categories, the ∞-Yoneda lemma, ∞-categorical universal constructions, presentable and stable ∞-categories, ∞-operads, $\mathbb{E}_n$-algebras, A_∞- and L_∞-algebras, and previews of derived algebraic geometry and homotopy type theory.
This final chapter synthesizes Volume IV's twenty-three preceding chapters, maps them to the standard higher-CT literature, and points forward to a potential Volume V — the fulfillment of the original v0.5 promise: connections of category theory to other branches of mathematics.

Reader's Annotation

This chapter is intentionally retrospective rather than introducing new material. By this point, every term used in Vol IV has been introduced with etymology and pedagogy in its home chapter; here we synthesize and assemble the reading-list map to the existing higher-CT literature. The chapter has no new *nameorigin* callouts — but every named result and term cross-references its home chapter where the naming origin was established.

60.1 Volume IV in One Picture

60.1.1 The Phases

Volume IV's twenty-three preceding chapters arranged themselves in five phases.

Phase 1: Simplicial and Homotopical Foundations (Ch 37–42). The simplex category Δ, simplicial sets **sSet**, geometric realization and the nerve, Kan complexes; localization at weak equivalences, Quillen's model categories, derived functors, the realization–singular Quillen equivalence (the *homotopy hypothesis* at the level of model categories).

Phase 2: Quasi-Categories and ∞-Yoneda (Ch 43–48). Quasi-categories as simplicial sets with inner-horn filling, mapping spaces and equivalences, the Joyal model structure on **sSet** that makes quasi-categories the fibrant objects, joins/slices/limits, Cartesian fibrations and straightening, and the ∞-*Yoneda lemma* — the milestone delivering the working foundation of ∞-category theory.

Phase 3: Universal Constructions, Revisited (Ch 49–51). ∞-limits and ∞-colimits as adjoints to constant-diagram functors, ∞-adjunctions with their three equivalent characterizations, ∞-monads (Beck's monadicity in ∞-form) and ∞-Kan extensions. Volume III's spine is fully lifted.

Phase 4: Presentable Stability and Higher Algebra (Ch 52–57). Presentable ∞-categories with the ∞-AFT, stable ∞-categories, triangulated categories and t-structures, ∞-operads, $\mathbb{E}_n$-algebras and Dunn additivity, A_∞- and L_∞-algebras with bar/cobar Koszul duality. The practical apparatus of higher algebra.

Phase 5: Doorways (Ch 58–60). Derived algebraic geometry (preview), homotopy type theory (preview), and this synthesis.

60.1.2 The Picture

Remark 60.1. Imagine the four-floor tower of Volume III, now with a fifth floor.

Floor 1 is Volume I: objects, morphisms, basic universal properties. Floor 2 is Volume II: the same concepts reorganized around representability and adjointness. Floor 3 is Volume III: the formal spine — ends, weighted limits, 2-categorical theory.

Floor 4 — Volume IV — has five rooms:

- A simplicial room (Phase 1): Δ, **sSet**, homotopical algebra.
- A quasi-categorical room (Phase 2): the working model.
- A universal-construction room (Phase 3): limits, adjunctions, monads, Kan extensions, all ∞-categorical.
- A higher-algebra room (Phase 4): presentable, stable, operadic.
- A doorway hall (Phase 5): pointers to DAG, HoTT, and beyond.

From the doorway hall, the corridor opens to a fifth floor that is *not built here*: the applications floor. Algebra (derived categories, triangulated categories in practice), topology (spectra, stable homotopy theory in detail), logic (toposes, classifying spaces), computer science (categorical semantics, models of dependent type theory, functional programming). These rooms — the entire fifth floor — are Volume V, never written here.

60.2 The Four Pillars

60.2.1 Reorganizing

Volume IV's central content can be summarized as four theorems, each the ∞-categorical extension of a Volume III theorem:

Pillar 1: The ∞-Yoneda Lemma (Chapter 48).

$$\mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}), \quad \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \simeq F(x).$$

Every ∞-category embeds fully faithfully in its presheaf ∞-category, which is the universal cocomplete extension.

Pillar 2: ∞-Limits and Colimits (Chapter 49). const $\dashv$ lim; colim $\dashv$ const. Completeness is captured by products and equalizers, cocompleteness by coproducts and coequalizers. Filtered colimits commute with finite limits. Right adjoints preserve ∞-limits.

Pillar 3: ∞-Adjunctions (Chapter 50). $L \dashv R$ presented as hom-equivalence, unit/counit, or fibration over Δ^1. Uniqueness up to contractible space. The ∞-AFT (Chapter 52): in the presentable case, "right adjoint = preserves limits and is accessible."

Pillar 4: ∞-Monads and ∞-Kan Extensions (Chapter 51). Every ∞-adjunction gives an ∞-monad; ∞-monadicity (Lurie) characterizes when an adjunction recovers from its monad. The adjoint triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$; pointwise formula; density theorem as $\mathrm{Lan}_y \mathrm{id} = \mathrm{id}$; codensity monads.

60.2.2 The unifying view

Remark 60.2 (A theme)**.** At each level of the book, the same universal property recurs in a more abstract form:

- Vol I: hom-set bijection (Chapter 7 adjunctions).
- Vol II: representability (Chapter 17 adjunctions deeper).
- Vol III: 2-categorical universal arrow (Chapter 33 formal adjunctions).
- Vol IV: mapping-space equivalence (Chapter 50 ∞-adjunctions).

At each level, the data is the same; only the dimensional precision changes. The same is true for every Yoneda incarnation, every monad incarnation, every Kan-extension incarnation. The progression is the project: a single concept articulated with finer and finer coherence requirements.

60.3 Reading-List Map

60.3.1 Volume IV $\to$ The Higher-CT Canon

A reader who completes Volume IV can open the standard references for higher category theory. The map below shows where this volume's chapters correspond to specific sections in those references.

Lurie, *Higher Topos Theory* (HTT).

- Ch 38–39 $\to$ HTT §1.1 (simplicial preliminaries).
- Ch 43–45 $\to$ HTT §1.2 (quasi-categories, slices, mapping spaces).

- Ch 46–47 → HTT §2.2–2.4 (fibrations, straightening).
- Ch 48 → HTT §5.1.3 (Yoneda).
- Ch 49 → HTT §4.4 (limits and colimits).
- Ch 50 → HTT §5.2 (adjunctions).
- Ch 52 → HTT §5.5 (presentable ∞-categories).

Lurie, *Higher Algebra* (HA).

- Ch 51 (monads) → HA §4.7.
- Ch 53 (stable ∞-cats) → HA §1.1–1.4.
- Ch 54 (t-structures) → HA §1.2.
- Ch 55 (∞-operads) → HA §2.1–2.3.
- Ch 56 ($\mathbb{E}_n$-algebras) → HA §5.1–5.5.
- Ch 57 (A_∞, L_∞) → HA §4.1–4.5.

Riehl–Verity, *Elements of ∞-Category Theory*. A model-independent treatment. After Ch 48, this can be read essentially linearly. Chapters of RV cover Vol IV Ch 43–51 from a different angle (using 2-categorical comma constructions). Especially valuable for the analytic style; complements Lurie's combinatorial approach.

Cisinski, *Higher Categories and Homotopical Algebra*. The third major treatment, emphasizing the model-categorical underpinnings. Especially valuable for understanding the relationship between $\mathbf{sSet}_{\text{Joyal}}$ and the homotopy theory of ∞-categories.

Kerodon (Lurie's online textbook). The current most accessible exposition. Frequently updated; topics arranged for individual study. Covers most of Vol IV's material with many more examples.

Land, *Introduction to Infinity-Categories*. A shorter, more accessible introduction. Vol IV Ch 43–51 plus Ch 52 correspond to most of Land's chapters; Land is particularly suitable for a reader who finds Lurie's style daunting.

60.3.2 Specialized references

Remark 60.3. For specific topics in Volume IV:

- Stable ∞-cats / triangulated (Ch 53–54): Lurie HA §1; Krause, *Homological theory of representations*; also Neeman's classical *Triangulated categories*.
- ∞-operads (Ch 55–56): Lurie HA §2–5; Heuts–Moerdijk, *Simplicial and Dendroidal Homotopy Theory*.
- A_∞, L_∞ algebras (Ch 57): Keller, *Introduction to A_∞-algebras and modules*; Loday–Vallette, *Algebraic Operads*.
- Derived algebraic geometry (Ch 58): Toën–Vezzosi, HAG I/II; Lurie SAG.
- Homotopy type theory (Ch 59): The HoTT Book; Riehl–Shulman.

60.4 What Remains: Toward Volume V

60.4.1 Formally

The original v0.5 directive read:

> "Full formal understanding of category theory, its complete suite of theorems, and then its connections to other branches of mathematics — in that order."

Volumes I–III delivered category theory; Volume IV delivered higher category theory. The *connections* promise — the original "then" — remains as a potential Volume V.

60.4.2 Volume V outline (hypothetical)

Algebra.

- Derived categories of modules: $\mathcal{D}(R\text{-Mod})$, triangulated functors, derived tensor and Hom.
- Group cohomology via classifying ∞-stacks.
- Representation theory in stable ∞-categories; geometric Langlands.

Topology.

- Spectra in detail: ring spectra, structured ring spectra, TMF, tmf.

- Stable homotopy groups of spheres: Adams spectral sequence in ∞-categorical form.
- Cohomology theories: K-theory, cobordism, motivic cohomology.

Logic.

- Classifying topoi and geometric morphisms.
- Internal language of ∞-toposes (using Ch 59).
- Categorical model theory; categorical semantics for first-order logic.

Computer Science.

- Categorical semantics of programming languages.
- Functional programming via monads and applicative functors.
- Dependent type systems in proof assistants (Agda, Lean, Coq).
- Profunctor optics and lens theory.

60.4.3 Reorganizing: the project's arc

Remark 60.4 (The four-volume arc)**.** The book has progressed through four definite stages:

- Vol I: *learn* category theory.
- Vol II: *reorganize* it around its principles.
- Vol III: *formalize* it via the formal spine.
- Vol IV: *ascend* to higher category theory.

The fifth stage (*apply* to other domains) would complete the classical four-element pedagogy: from the abstract heart outward to the world.

But that is for another time, another writer, another volume. The elementary category theory promised at the start of Volume I is now fully delivered, in its higher-categorical form. The book closes here.

60.5 The Last Theorem

60.5.1 Formally

Remark 60.5 (Vol IV's central thesis)**.** The universal constructions of category theory — limits, colimits, adjunctions, monads, Kan extensions, Yoneda — all generalize to ∞-categories in a coherent way. The technical content of Volume IV is the verification that each classical theorem has a unique-up-to-contractible-homotopy ∞-categorical analogue. The philosophical content is that this is unavoidable: as soon as one admits coherent homotopical structure, the entire apparatus of higher category theory follows.

The reader is now equipped to:

- Open Lurie's *Higher Topos Theory* at any chapter and follow.
- Read modern algebraic and arithmetic geometry written in derived / spectral language.
- Engage with homotopy type theory and its applications.
- Pursue research in any field touching ∞-category theory.

60.6 Exercises

What are Vol IV's five phases?	(1) Simplicial/homotopical foundations (Ch 37–42); (2) quasi-categories and ∞-Yoneda (Ch 43–48); (3) universal constructions (Ch 49–51); (4) presentable, stable, higher algebra (Ch 52–57); (5) DAG, HoTT, synthesis (Ch 58–60).
What are the four pillars?	∞-Yoneda (Ch 48); ∞-(co)limits (Ch 49); ∞-adjunctions (Ch 50); ∞-monads and ∞-Kan (Ch 51).
Where in HTT does Vol IV Ch 48 map?	HTT §5.1.3 (Yoneda lemma for quasi-categories).
Where in HA does Vol IV Ch 56 map?	HA §5.1–5.5 ($\mathbb{E}_n$-algebras and the little disks operads).

What complementary perspective does Riehl-Verity provide?	Model-independent ∞-category theory via 2-categorical comma constructions. An analytic alternative to Lurie's combinatorial approach.
What does Land's book offer?	A shorter, more accessible introduction to ∞-categories. Suitable as an alternative-pace entry point after Vol IV.
What's the original goal of the book?	"Full formal understanding of category theory, its complete suite of theorems, and connections to other branches of mathematics — in that order."
What remains for a hypothetical Vol V?	The "connections to other branches" promise: algebra (derived/triangulated), topology (spectra), logic (toposes), computer science (categorical semantics).
What's the unifying view across volumes?	Each universal construction appears at increasing levels of abstraction. The data is invariant; the dimensional precision changes.
Why is the ∞-Yoneda lemma the central theorem?	Because it organizes the entire apparatus: limits, colimits, adjunctions, monads, Kan extensions all derive from representability via Yoneda.
What's the relationship between Lurie HTT and Riehl-Verity?	They model ∞-categories differently — Lurie via simplicial sets, Riehl-Verity via 2-categorical structures — but prove the same theorems. Volume IV uses Lurie's quasi-category model.
Why don't we develop straightening in detail?	Because the proof spans about 100 pages of dense combinatorics (HTT §2.2). Our chapter sketches the strategy and works low-dimensional examples; readers who need the full proof can consult HTT directly.
Why don't we prove ∞-AFT in detail?	Same reason: the proof requires the full machinery of accessible ∞-categories developed in HTT §5.5. We state the theorem cleanly and forward-point.
What does a Vol IV reader know that a Vol III reader doesn't?	∞-categorical analogues of every Vol III theorem; the simplicial-set framework; the higher algebra of operads; the bridge to Lurie's HTT and HA.
What does a Vol IV reader still not know?	Detailed model-categorical homotopy theory (Hovey); operadic combinatorics (Loday-Vallette); applications to specific fields (Vol V territory).
Is Vol IV self-contained?	Mostly. Some proofs (straightening, AFT, monadicity) are sketched rather than completed; the reader is directed to Lurie for the technical apparatus.
Why use quasi-categories rather than another model?	Combinatorial directness (single condition: inner horns fill), match with Lurie/Riehl-Verity/Kerodon literature, and accessibility from the simplicial-set foundation built in Phase 1.
What's the connection to homotopy type theory?	The two are dual: ∞-category theory provides semantics for HoTT (every ∞-topos models HoTT), while HoTT provides a synthetic language for ∞-toposes.
What's the homotopy hypothesis?	∞-groupoids $\simeq$ homotopy types of spaces. Established by the realization-singular Quillen equivalence (Ch 42). Foundational throughout Vol IV.
What's the most important ∞-category in Vol IV?	$\mathcal{S}$, the ∞-category of spaces. The universal example, used in nearly every theorem. Its homotopy 1-category is $\mathrm{Ho}(\mathbf{Top})$.

What field is most transformed by ∞-categories?	Algebraic geometry — DAG has reorganized large portions of the field. Also stable homotopy theory (Lurie's HA), representation theory (geometric Langlands), and homotopy type theory (foundations).
What does the reader who finishes Vol IV do next?	Open Lurie HTT, Riehl-Verity, Cisinski, or Land — any of the standard ∞-category references — with a strong foundation. Or open Toën-Vezzosi for DAG, or the HoTT Book for type theory.
Where does the book end?	Here, with a doorway to a fifth floor that was not built. Volume V, if written, would deliver the original v0.5 "connections to other branches" promise. Until then: the elementary category theory promised at the start is now fully delivered.

Chapter Summary

Chapter Summary

Vol IV's five phases. Simplicial foundations; quasi-categories and ∞-Yoneda; universal constructions; presentable / stable / higher algebra; doorways.
The four pillars. ∞-Yoneda (Ch 48), ∞-(co)limits (Ch 49), ∞-adjunctions (Ch 50), ∞-monads and Kan (Ch 51). Each is the higher-categorical version of a Vol III theorem.
Reading-list map. Vol IV chapters map to specific sections of Lurie's HTT, HA, SAG; Riehl-Verity; Cisinski; Kerodon; Land. The reader is equipped to engage any of these directly.
What's not here. Detailed proofs of straightening, ∞-AFT, ∞-monadicity — sketched, with forward pointers. Full DAG, full HoTT — previewed, but require dedicated volumes.
Toward Vol V. The original v0.5 promise — connections to other branches of mathematics — is preserved as a potential Vol V. Algebra, topology, logic, computer science. Not written here.
The arc. Vol I: learn. Vol II: reorganize. Vol III: formalize. Vol IV: ascend. Vol V (potential): apply. Elementary category theory in its higher-categorical form is now fully delivered.

Formal Restatement

Formal Restatement

Volume IV's central content. The universal constructions of ∞-category theory:

- $\mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$; full faithfulness; $\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(y(x), F) \simeq F(x)$ (Yoneda).
- const $\dashv$ lim; colim $\dashv$ const. ∞-bicompleteness of $\mathcal{S}$, $\mathcal{P}(\mathcal{C})$, $\mathcal{C}\mathrm{at}_\infty$.
- $L \dashv R$: Map-equivalence form, unit/counit with coherences, fibration over Δ^1. Uniqueness, RAPL/LAPC.
- Monad from adjunction: $T = RL$, η_{ad}, $\mu = R\epsilon L$. Monadicity = conservative + R-split preservation.
- Adjoint triple $\mathrm{Lan}_p \dashv p^* \dashv \mathrm{Ran}_p$; pointwise formula $\mathrm{Lan}_p F(e) = \mathrm{colim}_{p\downarrow e} F$.
- Density: $\mathrm{Lan}_y \mathrm{id}_{\mathcal{C}} \simeq \mathrm{id}_{\mathcal{P}(\mathcal{C})}$. Free cocompletion via Kan.

Higher algebra. $\mathbb{E}_n$ interpolating $\mathbb{E}_1 = \mathrm{Assoc}$ and $\mathbb{E}_\infty = \mathrm{Comm}$; Dunn additivity. A_∞, L_∞ as strong-homotopy variants; Koszul duality $\mathrm{Assoc}^! = \mathrm{Lie}$.

Presentable + stable. Pr^L closed symmetric monoidal (Lurie tensor); unit $\mathcal{S}$. $\mathrm{Pr}^L_{\mathrm{St}}$: unit Sp. ∞-AFT: right adjoint $\iff$ limit-preserving + accessible.

Connections (Vol V). Algebra, topology, logic, computer science — deferred to a potential fifth volume, not written here.

The book closes. Elementary category theory, in its higher-categorical form, is now fully delivered.